CHAPTER

ATA 34

NAVIGATION

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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CONTENTS

CONTENTS 2
NAVIGATION, GENERAL 4
FLIGHT ENVIRONMENT DATA 8
AIR DATA SYSTEM 18
INTEGRATED STANDBY INSTRUMENT (ISI) 69
ATTITUDE AND DIRECTION 75
ATTITUDE AND HEADING REFERENCE SYSTEM (AHRS) 83
STANDBY ATTITUDE AND HEADING INSTRUMENTS 132
EXTERNAL PROTECTION SYSTEM 138
WEATHER RADAR SYSTEM (PRIMUS 660) 140
ENHANCED GROUND PROXIMITY WARNING SYSTEM (EGPWS) 157
TRAFFIC ALERTING AND COLLISION AVOIDANCE SYSTEM (TCAS) 231
RADIO ALTIMETER (HONEYWELL) 275
VHF NAVIGATION SYSTEMS 288
AUTOMATIC DIRECTION FINDING (ADF) 340
DISTANCE MEASURING EQUIPMENT (DME) 365
AIR TRAFFIC CONTROL TRANSPONDER 379
FLIGHT MANAGEMENT SYSTEM − UNS−1C/1E/1EW 398

NAVIGATION, GENERAL

Introduction

The navigation system has different sub−systems to supply other systems with the data that follows:

- Pitot and static air pressures
- Pitch and bank attitude and stabilized heading
- Microwave Landing System (MLS) azimuth and elevation
- Weather, ground proximity, location and threat status of other aircraft, and Above Ground Level (AGL) altitude
- Radio navigation
- Flight management.

General Description

Figure 34- 1 NAVIGATION SYSTEM BLOCK DIAGRAM

The navigation system has the sub−systems that follow:

- Air Data System
- Attitude and Heading System
- Microwave Landing System
- External Protection System
- VHF Navigation
- Automatic Direction Finding (ADF)
- Distance Measuring Equipment (DME)
- Air Traffic Control Transponder
- Global Positioning System
- Flight Management System (FMS).

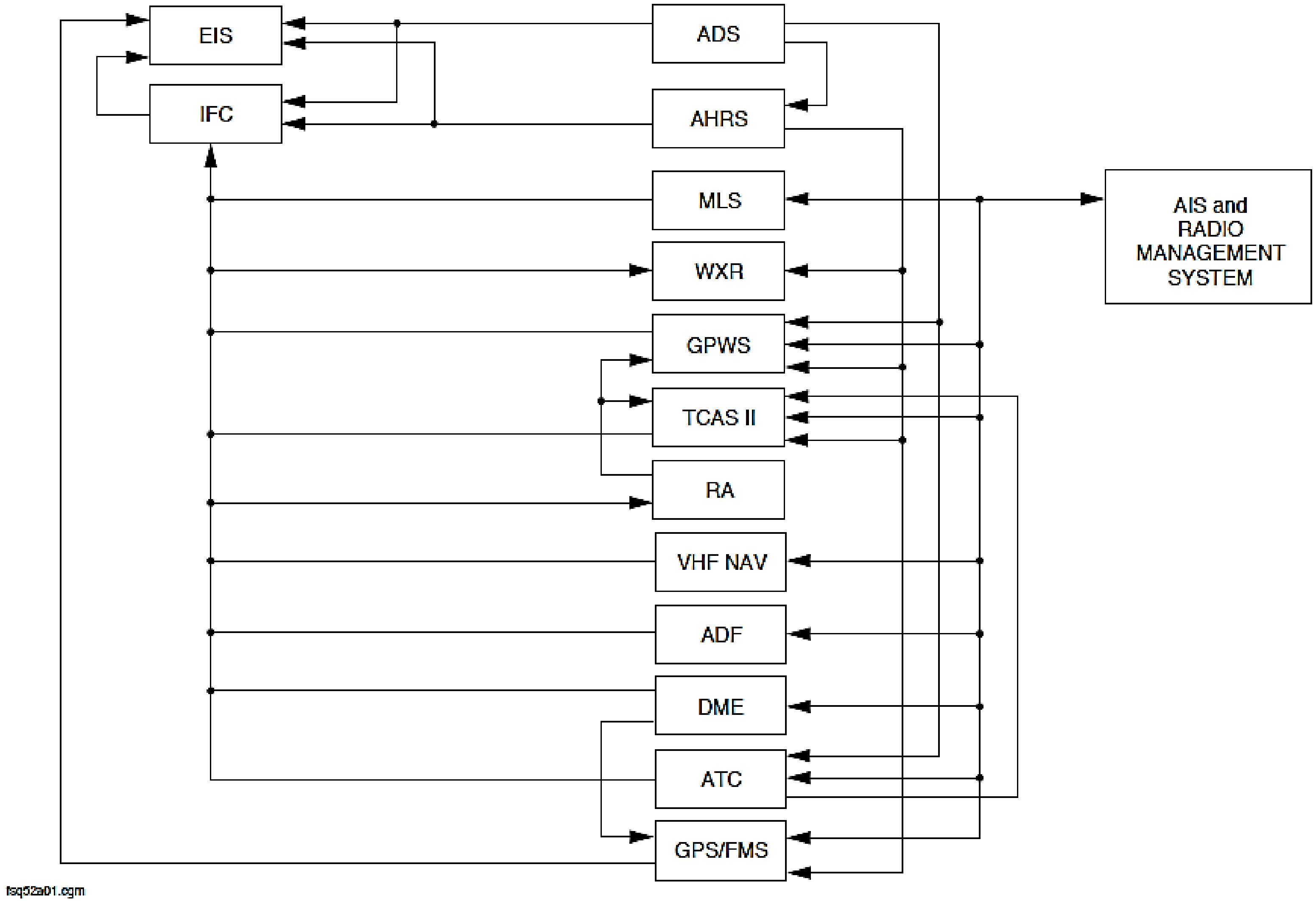


Figure 34- 1 NAVIGATION SYSTEM BLOCK DIAGRAM

Air Data System

The Air Data System (ADS) supplies air data to the Electronic Instrument System (EIS) for indication and to other aircraft systems.

The air data parameters that are shown to the pilots and used for flight guidance are critical parameters. For this reason, the ADS has the three isolated air data systems, two primary and one standby channel

Attitude and Heading System

The Attitude and Heading System (AHRS) supplies attitude and heading data to the Electronic Instrument System (EIS) indication and to other aircraft systems.

The attitude and heading parameters that are shown to the pilots and used for flight guidance are critical parameters. For this reason, the AHRS has three isolated attitude and heading data systems, two primary and one standby channel.

External Protection System

The External Protection System (EPS) has different sub−systems to give visual and aural indications that relate to the conditions that follow:

- Weather
- Ground proximity
- Location and threat status of other aircraft
- Above Ground Level (AGL) altitude.

It also gives other aircraft and ground stations with data relating to the aircraft identification and location.

The external protection system has the sub−systems that follow:

- Weather Radar Systems
- Ground Proximity Warning System (GPWS)
- Traffic Alerting and Collision Avoidance System II (TCAS II)
- Radio Altimeter.

Very High Frequency Navigation

The Very High Frequency Navigation (VHF NAV) system receives radio signals to give the navigation data that follows:

- VHF Omni−range (VOR)
- Instrument Landing System (ILS).

VHF Omni−range (VOR): The Very High Frequency Omni−range (VOR) ground stations form part of the Victor airways navigation system. The VOR gives guidance on any track to or from the station.

Instrument Landing System (ILS): The Instrument Landing System (ILS) is an integrated system that gives the approach flight path to landing on a runway.

Automatic Direction Finding

The Automatic Direction Finding (ADF) system shows the magnetic bearing to a fixed Non−Directional Beacon (NDB).

Distance Measuring Equipment

The airborne Distance Measuring Equipment (DME) measures the line of sight slant range from the aircraft to the DME ground station. It uses the slant distance data to calculate the ground speed and the time−to−go parameters.

Mode S Transponder

The Mode S Transponder system transmits a signal to identify the aircraft and its altitude parameters to Air Traffic Control (ATC) ground stations and to other aircraft equipped with Traffic Alert and Collision Avoidance System (TCAS).

Global Positioning System

The Flight Management System (FMS), Global Positioning System (GPS), and VORTAC Position Unit (VPU) are completely self–contained within a single Multi–Functional Control Display Unit (MCDU). It is located in the forward left side of the flight compartment console.

Flight Management System (FMS)

The FMS along with other interfacing equipment in the aircraft provides the following functions:

- Calculating the aircraft's position and velocity
- Guidance for the Automatic Flight Control System (AFCS)
- Managing the aircraft's flight path from origin to destination
- Tuning communication and navigation radios
- Make fuel management computations

The FMS assists the pilots by automating many routine tasks and manual computations.

FLIGHT ENVIRONMENT DATA

Introduction

The Air Data System (ADS) supplies air data to the Electronic Instrument System (EIS) for indication and to other aircraft systems.

General Description

The air data parameters that are shown to the pilots and used for flight guidance are critical parameters. For this reason, the ADS has the three isolated air data systems, two primary and one standby channel.

Figure 34- 2 AIR DATA SYSTEM BLOCK DIAGRAM, INPUTS
Figure 34- 3 PRIMARY AIR DATA SYSTEM BLOCK DIAGRAM

Each primary channel has one Air Data Unit (ADU), pitot−static probe, and temperature probe to do the functions that follow:

- Measure static and pitot pressures
- Receive air temperature data
- Receive barometric corrections
- Calculate air data parameters
- Supplies air data parameters to other systems through ARINC 429 buses to other systems.

The static air pressure sensed at each ADU is the average of the static pressures from the two pitot−static probes. Each ADU also senses pitot pressure from its related pitot−static probe and temperature from its related temperature probe.

The Index Control Panels (ICPs) are used to set the barometric correction for its related ADU.

Figure 34- 4 STANDBY AIR DATA INSTRUMENTS BLOCK DIAGRAM

The standby air data instruments function independently of the primary channel Air Data Units (ADU1, ADU2) to show the data that follows:

- Altitude
- Air speed
- Barometric correction settings.

The Air Data System (ADS) has the components that follow:

- Unit, Air Data
- Probes, Pitot−Static
- Probes, Static Temperature
- Indicator, standby altitude
- Indicator, airspeed−standby
- Probe, standby pitot−static.

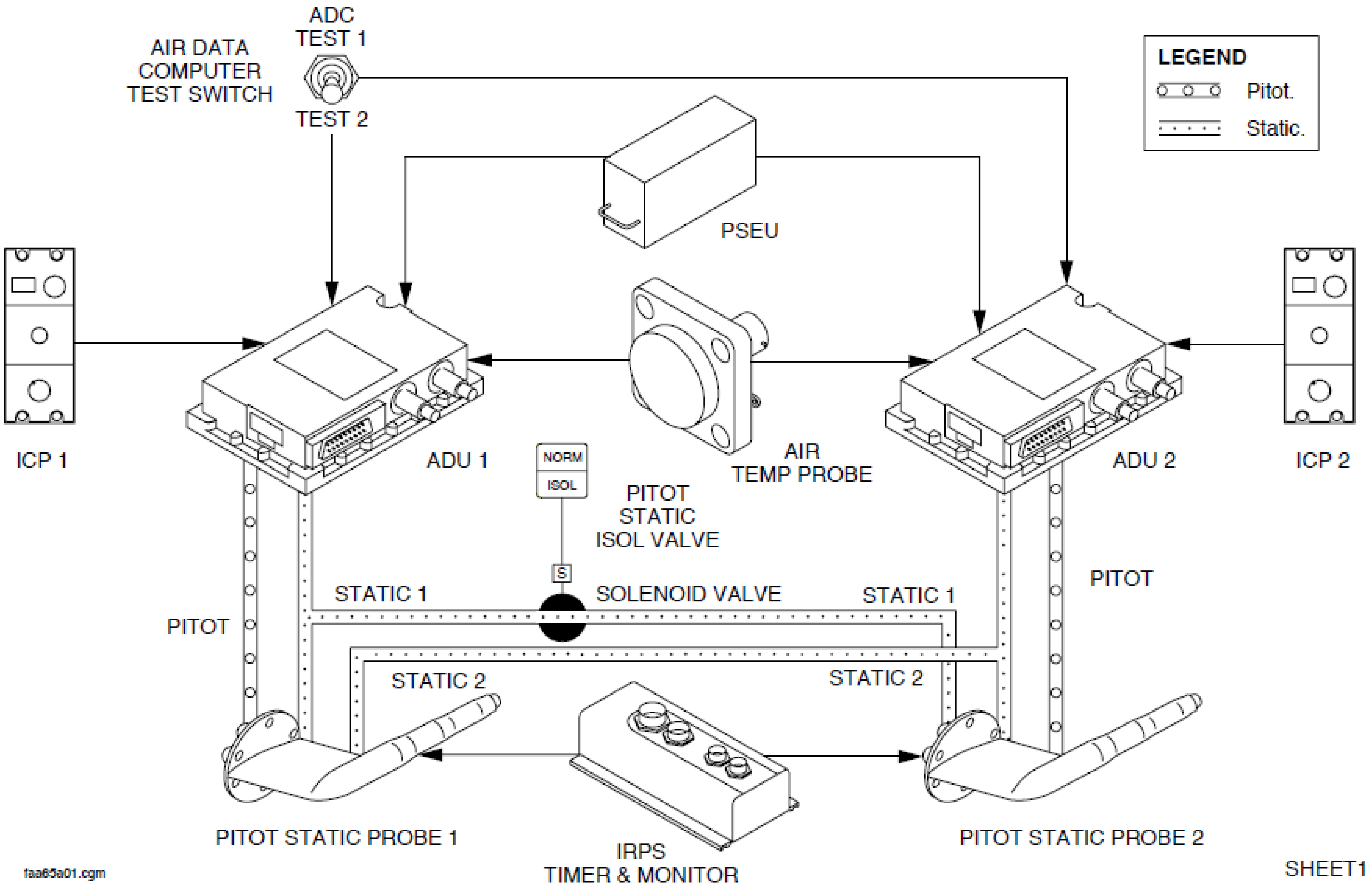


Figure 34- 2 AIR DATA SYSTEM BLOCK DIAGRAM, INPUTS

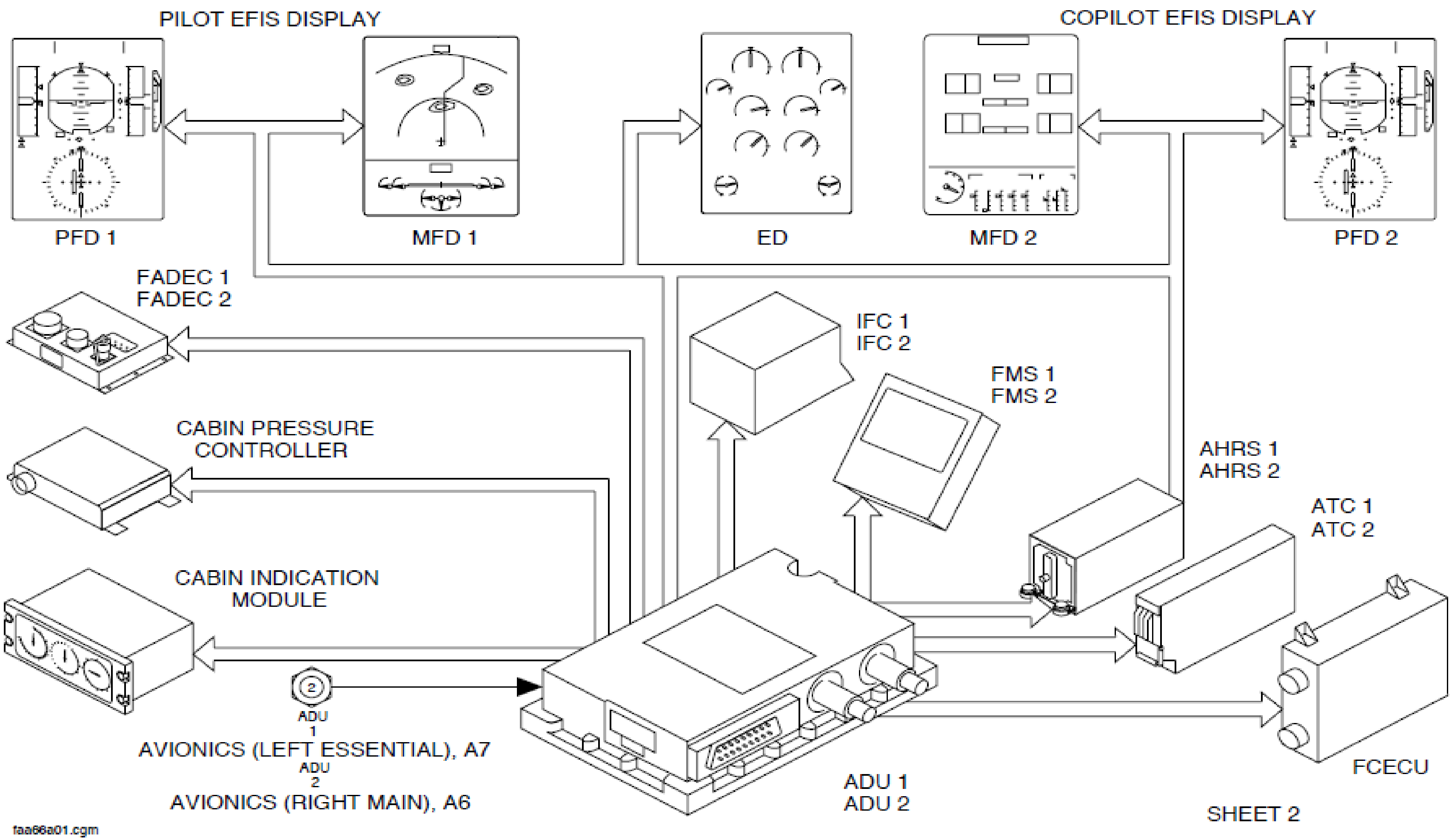


Figure 34- 3 PRIMARY AIR DATA SYSTEM BLOCK DIAGRAM

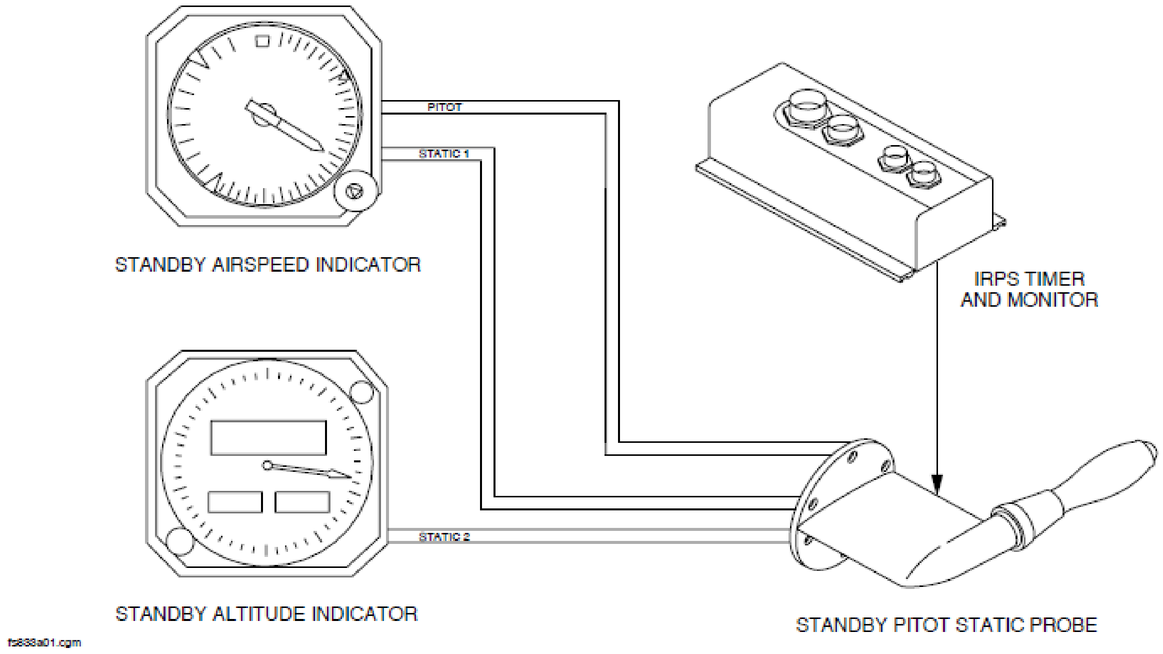


Figure 34- 4 STANDBY AIR DATA INSTRUMENTS BLOCK DIAGRAM

Air Data Unit

Figure 34- 5 AIR DATA UNIT LOCATOR

The Air Data Unit (ADU) weighs 6.12 lb (2.78 kg). It has subassemblies mechanically packaged into a 2 in. (51 mm) high, 6 in. (152 mm) wide by 7.9 in. (201 mm) long case.

It has quick disconnect pitot and static pneumatic connectors.

Both ADUs are installed in the flight compartment on the aft bulkhead using four mounting screws and have the pneumatic connectors oriented downwards to prevent entry of moisture into the units.

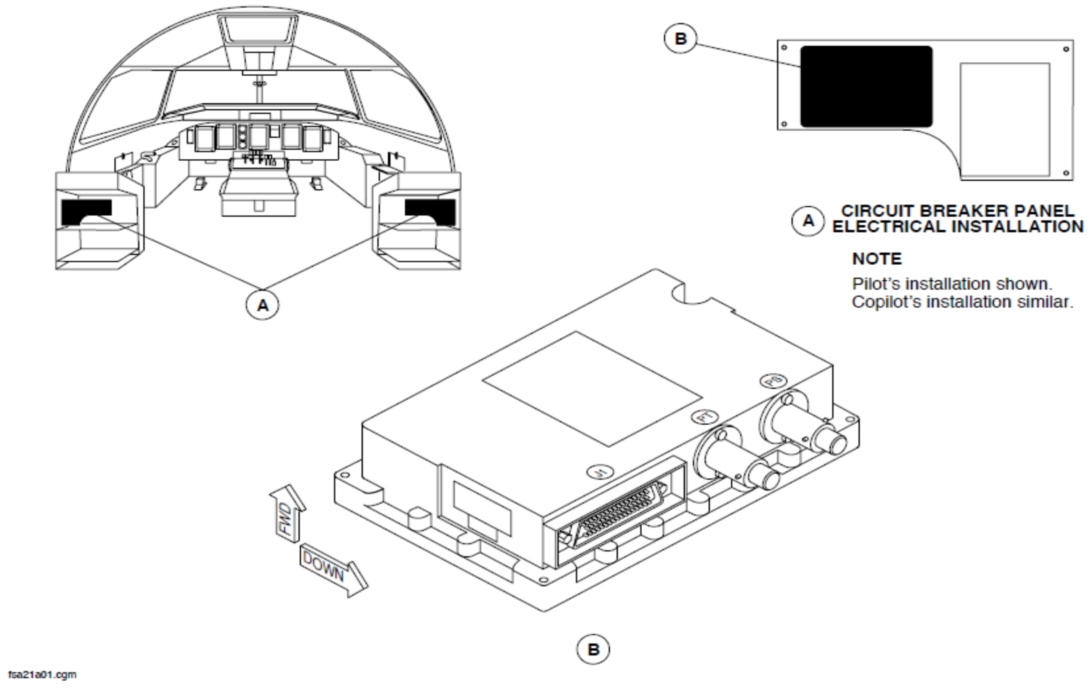


Figure 34- 5 AIR DATA UNIT LOCATOR

Probes, Pitot–Static

Figure 34- 6 PITOT–STATIC PROBES LOCATOR

The two primary heated Pitot–Static probes are installed symmetrically on the left and right sides of the fuselage.

The base plates have electrical and pressure connectors. Locating pins on the plate give an accurate angular orientation of the probe with the aircraft axis.

The primary probes are oriented 9 degrees downward relative to the horizontal reference axis of the aircraft. The angle of the probe tube relative to the mast gives a 5 degree angle in or out between the tube line and the tangential plane of the fuselage skin.

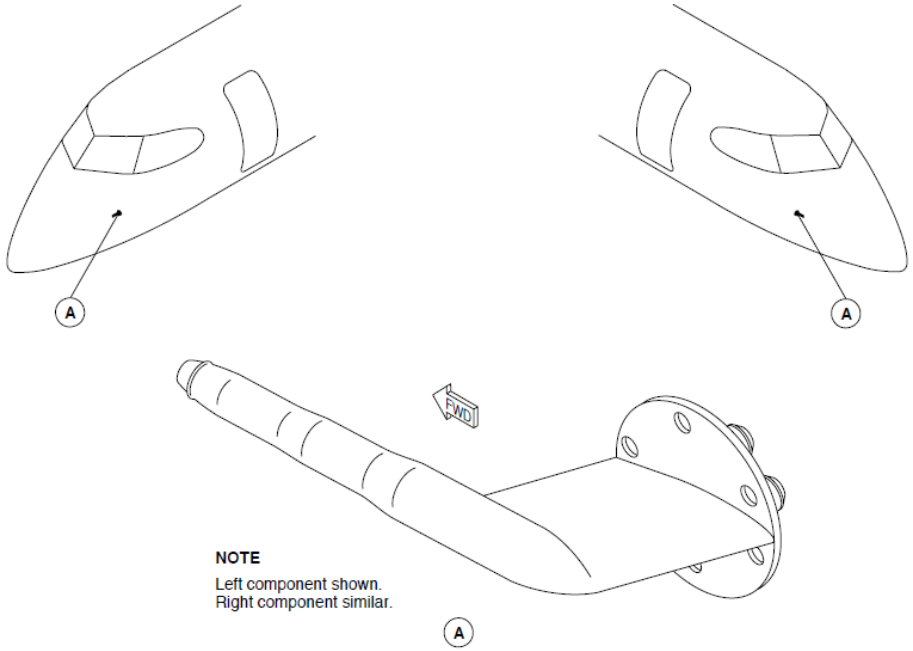


Figure 34- 6 PITOT–STATIC PROBES LOCATOR

Probe, Static Temperature

Figure 34- 7 STATIC TEMPERATURE PROBE LOCATOR

The static temperature probe weighs 0.25 pounds and is attached flush to the fuselage below the left wing root by four screws.

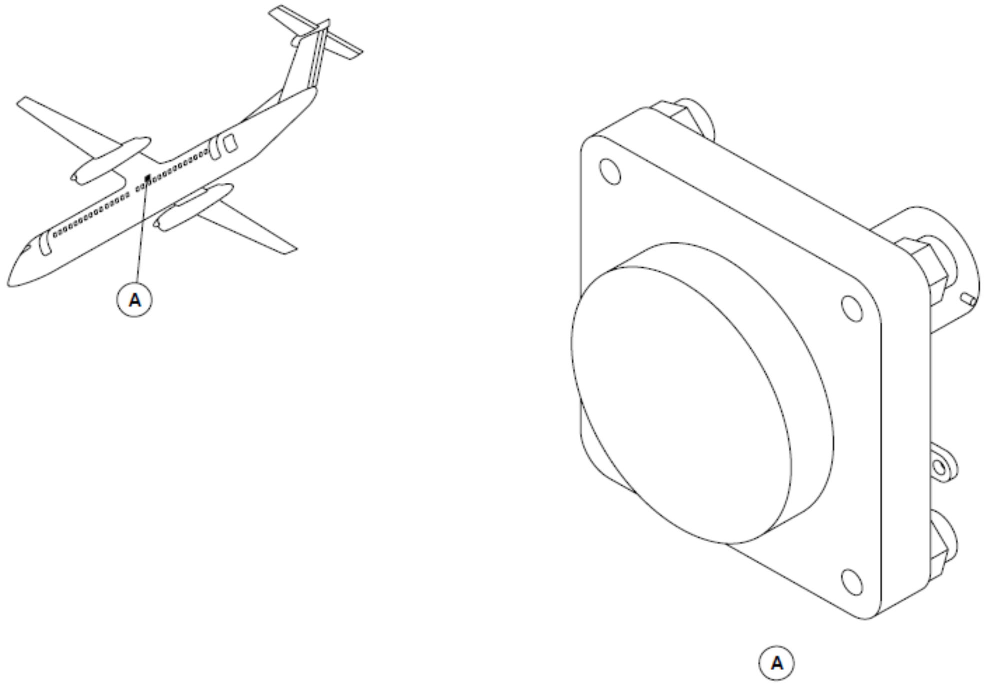


Figure 34- 7 STATIC TEMPERATURE PROBE LOCATOR

Standby Altitude Indicator

Figure 34- 8 STANDBY ALTITUDE INDICATOR LOCATOR

The standby altitude indicator weighs 2.2 lb (1 kg). It is located in the instrument panel between the pilot’s Multi−Function Display (MFD1) and the Engine Display (ED) below the standby airspeed indicator in view of the two pilots.

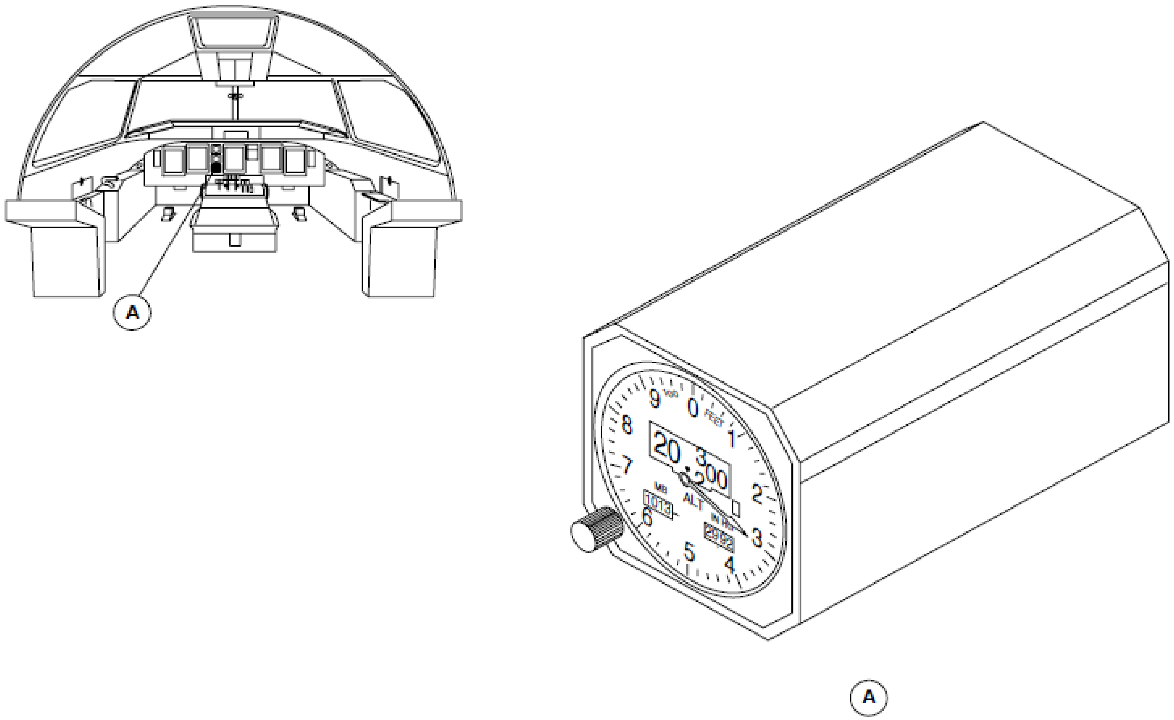


Figure 34- 8 STANDBY ALTITUDE INDICATOR LOCATOR

Standby Airspeed Indicator

Figure 34- 9 STANDBY AIRSPEED INDICATOR, LOCATOR

The standby altitude indicator weighs 1.1 lb. (0.5 kg). It is located in the instrument panel between the pilot's Multi−Function Display (MFD1) and the Engine Display (ED) above the standby altitude indicator in view of the two pilots.

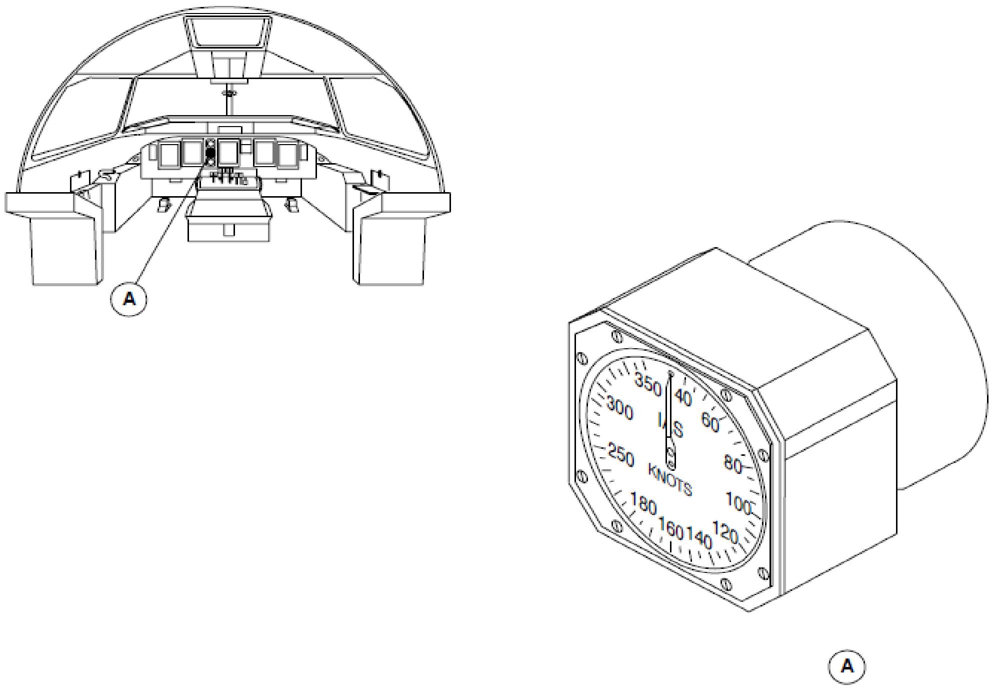


Figure 34- 9 STANDBY AIRSPEED INDICATOR, LOCATOR

Probe, Standby Pitot−Static

Figure 34- 10 STANDBY PITOT−STATIC PROBE, LOCATOR

The standby pitot−static probe is installed on the right side of the forward fuselage section.

The base plates have electrical and pressure connectors. Locating pins on the plate give an accurate angular orientation of the probe with the aircraft axis.

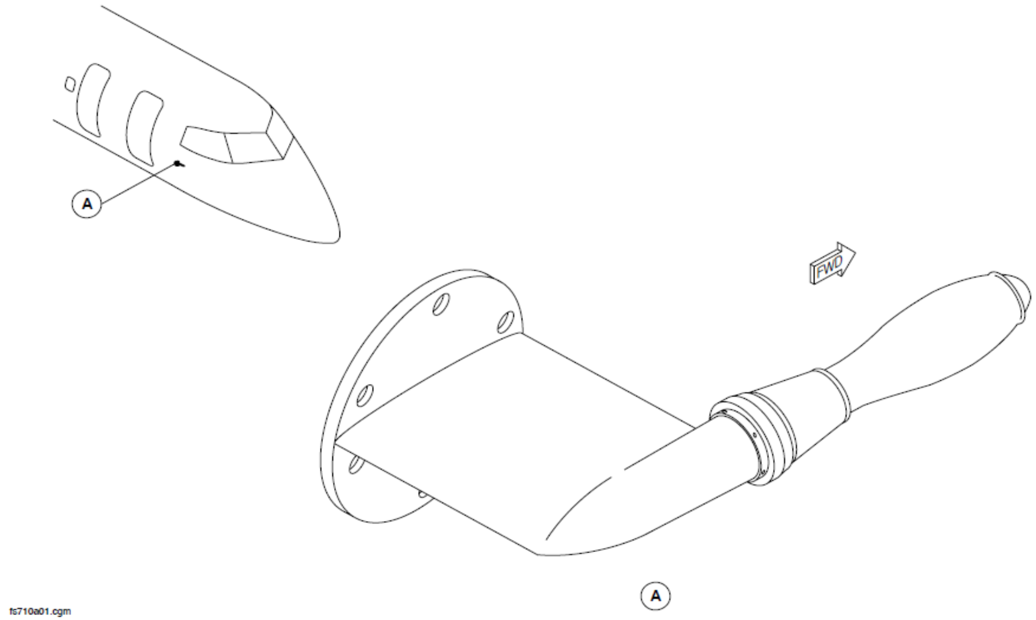


Figure 34- 10 STANDBY PITOT−STATIC PROBE, LOCATOR

AIR DATA SYSTEM

Introduction

The primary Air Data System (ADS) senses and converts static pressure, pitot (impact) pressure, and static air temperature to electronic data for use by the systems that follow:

- Cabin Pressure Control
- Flight Guidance
- Flight Controls
- Indication
- Vmo/Mmo Warning
- Navigation
- Powerplant Operation.

General Description

Figure 34- 11 AIR DATA SYSTEM BLOCK DIAGRAM, INPUTS
Figure 34- 12 PRIMARY AIR DATA SYSTEM BLOCK DIAGRAM

The two primary channel Air Data Computers (ADC1, ADC2) operate independently to calculate the data that follows:

- Altitude
- Air speed
- Vertical speed
- Temperature
- Barometric corrections.

The ADC operates in the modes that follow:

- Operational
- Initiated Test
- Maintenance.

The primary Air Data System (ADS) has the components that follow:

- Computer, Air Data
- Probes, Pitot–Static
- Probe, Static Temperature
- Valve, pitot–static isolation
- Panel, pitot–static isolation valve.

The Air Data Computers (ADC1, ADC2) supply data to the systems that follow:

- Cabin Pressure Controller, (CPC)
- Cabin Indication Module (CIM)
- Flight Guidance Module (FGM1, FGM2)
- Flight Controls Electronic Control Unit (FCECU1, FCECU2)
- Stall Protection Modules (SPM1, SPM2)
- Input/Output Processors (IOP1, IOP2)
- Primary Flight Displays (PFD1, PFD2)
- Multi–Functional Displays (MFD1, MFD2) Attitude and Heading Reference Units (AHRS1, AHRS2)
- ATC Transponder (ATC1, ATC2)
- Flight Management System (FMS1, FMS2)
- Full Authority Digital Engine Control (FADEC1, FADEC2).

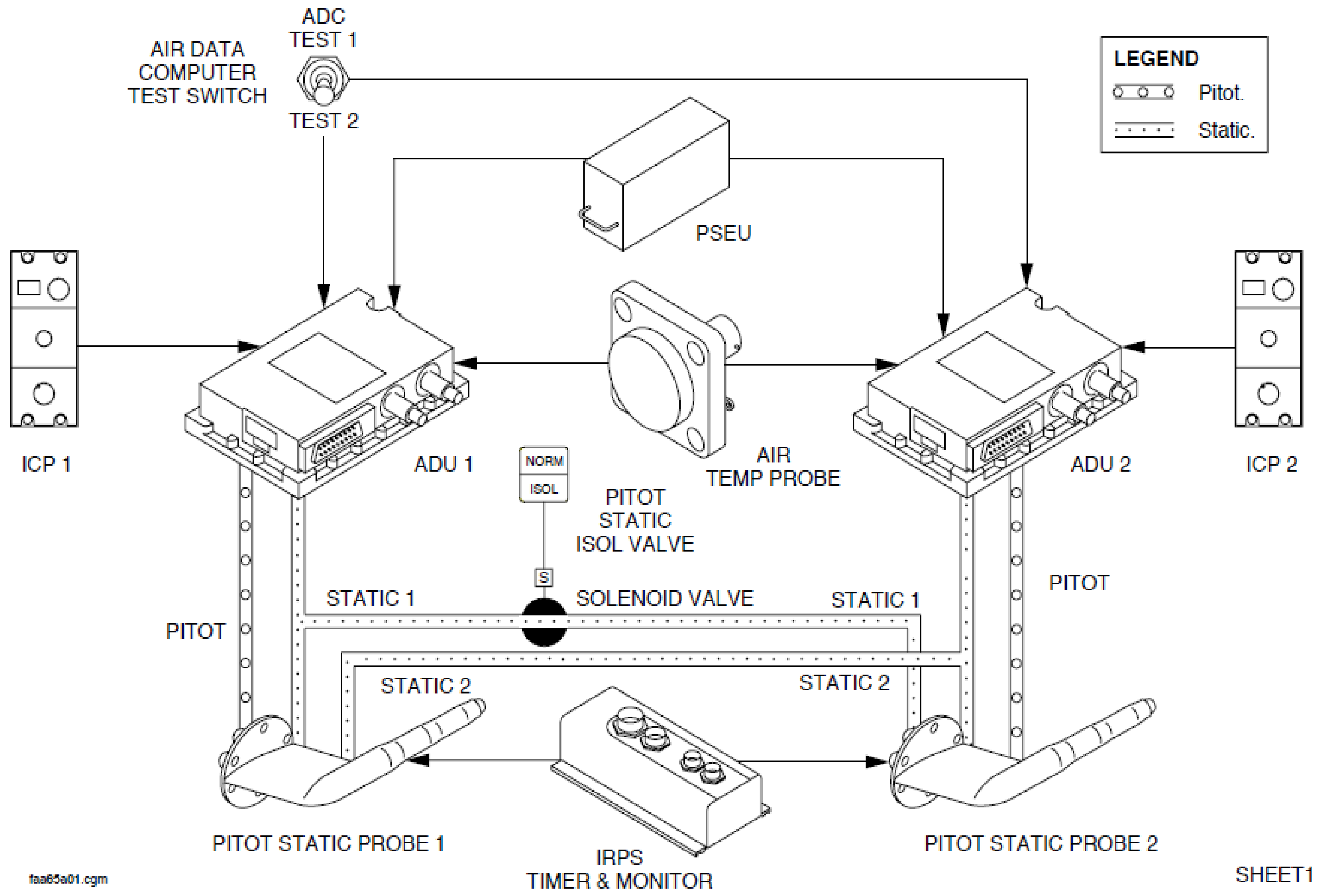


Figure 34- 11 AIR DATA SYSTEM BLOCK DIAGRAM, INPUTS

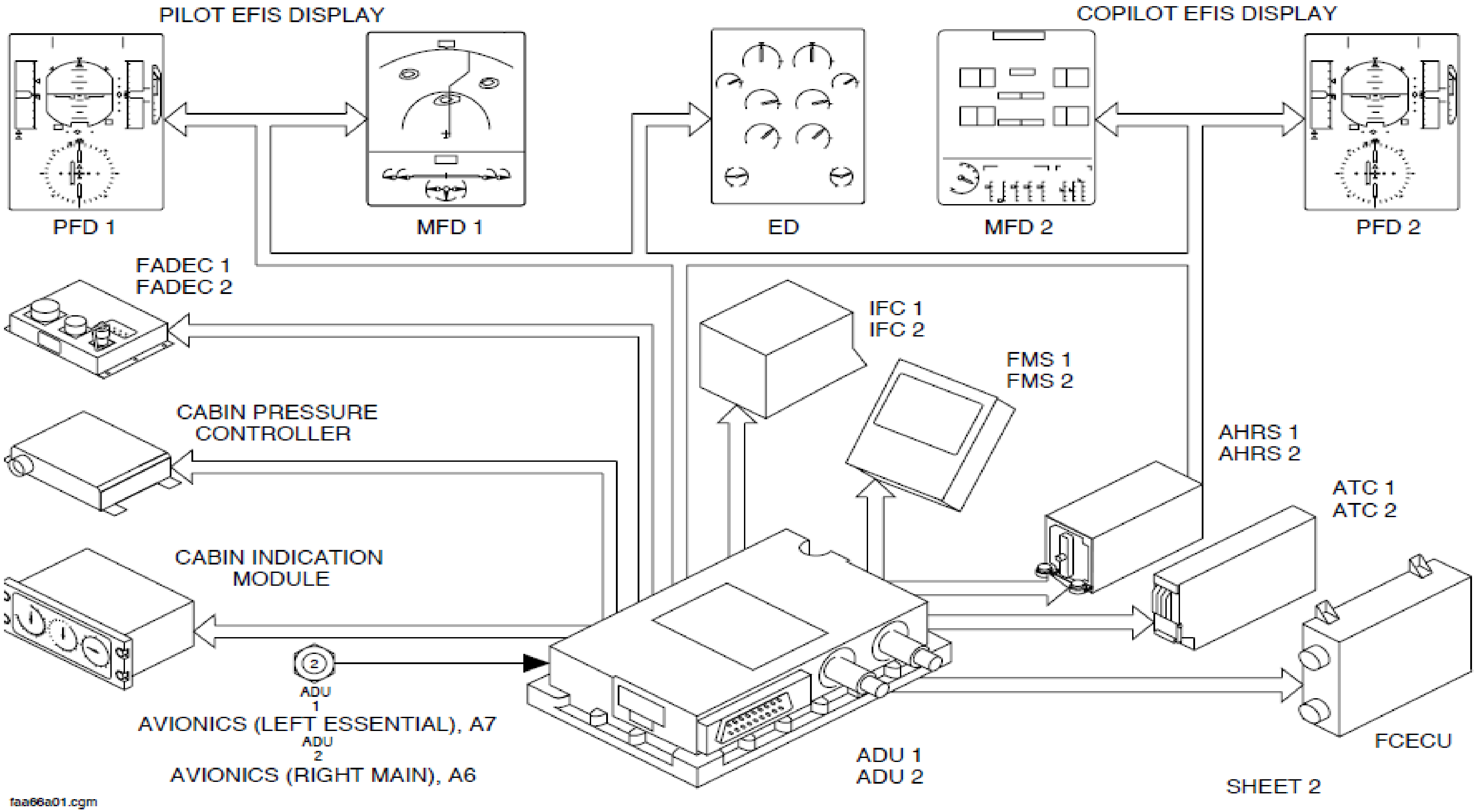


Figure 34- 12 PRIMARY AIR DATA SYSTEM BLOCK DIAGRAM

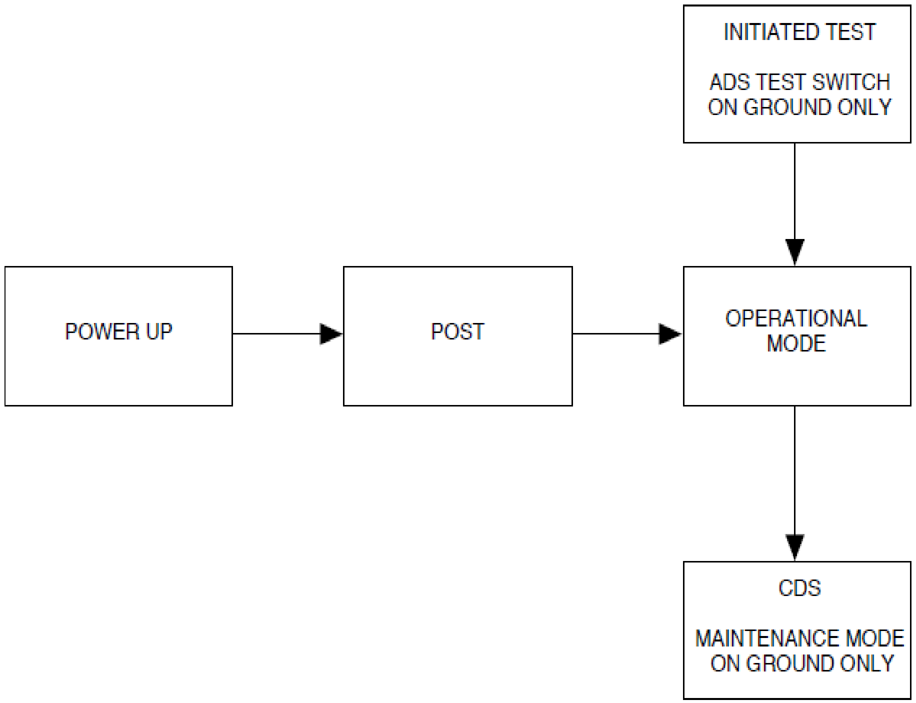


Figure 34- 13 ADC MODES OF OPERATION

Detailed Description

Figure 34- 13 ADC MODES OF OPERATION

The Air Data Computers (ADC1, ADC2) operate in the modes that follow:

- Power−On Self-Test (POST)
- Line Operational
- Initiated Test
- Maintenance mode.

Power−On Self-Test (POST): This mode automatically continues for 2 seconds after power up.

The ADCs does the self−tests that include:

- CPU self−tests
- Memory check sum
- Test of peripheral circuits
- Test of I/O ports
- Indication of test in progress and successful completion.

NOTE

The air data output parameters are invalid during the power−up tests.

The ADC calculates a failure warning if it senses a hardware failure.

When the ADC POST mode is completed, all ADC parameters are available.

Line Operational: The Air Data Computers (ADC1, ADC2) do the Primary channel operate as follows:

- Measure static and total pressures
- Measure outside air temperature
- Calculate barometric correction
- Sense probe heater failure data
- Calculate air data parameters for flight compartment indication and used by the air−conditioning, flight guidance, flight control, navigation and engine systems.

The Air Data Computers (ADC1, ADC2) calculate the air data parameters that follow:

- Static pressure (ps)
- Impact pressure (dp)
- Barometric correction in hPa and in InHg (Bcor)
- Barometric altitude corrected by barometric correction (Hc)
- Altitude rate (Vz)
- Calibrated airspeed (CAS)
- True airspeed (TAS)
- Maximum velocity in operation (VMO)
- Mach number (M)
- Static Air Temperature (SAT).

Pitot pressure from each pitot−static probe is supplied through tubing to its related ADC and averaged static pressure is supplied through tubing and a shutoff valve to the two ADCs.

A bird strike at the right side of the forward fuselage section can cause damage to the copilot's primary pitot−static probe and to the secondary pitot−static probe.

All air data indications can possibly become incorrect because the ADC1 static tubing is connected through a shutoff valve to pitot−static probe No. 2.

The shutoff valve is a usually open valve that is electrically controlled. The NORM/ISOL pushbutton switch on the pitot static isolation control panel is pushed to close the shutoff valve.

The pitot−static probe tubes have an encased heating element with a self-regulating power consumption feature to prevent ice accumulation on the pressure probes and help eliminate erroneous total and static measurements. Each pitot−static probe heater is controlled by a PITOT STATIC toggle switch on the ice protection panel.

The static temperature probe has two resistor sensors. Each sensor is connected to one ADC in order to supply the aircraft system with two independent temperature values. The Indicated Outside Air Temperature (IOAT) measured by the two sensors is affected by airspeed. A recovery factor is set in each ADC and is used to calculate and compare, using the IOAT, the Static Air Temperature (SAT) and Total Air Temperature (TAT).

The ADC receives barometric correction data from a rotary switch on its related Index Control Panel (ICP) to calculate the barometric correction and the barometric altitude.

The barometric correction value is shown on the PFD altitude digital readout in Millibars (MB) or Inches of Mercury (In Hg).

On aircraft with the Millibars (MB) barometric correction value (823SO00017): The MB barometric correction indication is set by ADC pin programming logic.

When the ADC is not energized, the barometric selection value is saved in ADC Non Volatile Memory (NVM).

The trend of the altitude calculated by the ADCs can be incorrect when the aircraft is dynamically pitched. For indication, the barometric corrected altitude data from the ADC are dampened by the Attitude and Heading Reference Units (AHRS) to show barometric inertial altitude. If the AHRU malfunctions or an opposite EFIS ATT/HDG SOURCE reversion is set, barometric corrected altitude is shown on the related PFD.

NOTE

The EIS has no indications to tell the pilots that barometric corrected altitude is shown.

The barometric altitude rate data from the ADCs are supplied through the AHRUs to make the indications instantaneous. If the related ADC malfunctions, the altitude rate data from the other ADC is automatically used to keep the IVSI indication in view.

The calibrated airspeed is supplied to the display units where it is shown as indicated airspeed.

The Air Data Computers (ADCs) calculate Non Computed Data (NCD) if a primary parameter, static pressure (ps), impact pressure (dp), or Outside Air Temperature (OAT), is sensed out of range.

NOTE

The airspeed indication is calculated as NCD when the airspeed is less than 30 kt, but, when the aircraft is stationary and on the ground, a Weight−On−Wheels (WOW) signal from the proximity sensor electronics unit will cause the Electronic Instruments System (EIS) to show 30 KIAS and 0 KTAS.

Initiated Test Mode: The ADC pre−flight test switch is used to check Vmo Warning Tone Generator (WTG) and the ADC outputs.

Maintenance Mode: The Built In Test Equipment (BITE) uses the Central Diagnostic System (CDS) to give the condition of the component. It saves faults in a Non-Volatile Memory (NVM) for reporting to line and shop maintenance.

The BITE allows aircraft maintenance personnel to:

- Do fault isolation and return to service testing after completing maintenance actions
- Access failure reports from last or previous flight legs
- Get the avionics status report
- Get the part number of an LRU.

The BITE mode monitors the condition of the component as follows:

- Power−On Self-Test (POST)
- Continuous monitoring.

Power−On Self-Test (POST): The POST checks the condition of the component at power−up or after a long power interruption of more than 200 milliseconds.

Continuous Monitoring: The continuous monitoring checks the status of the component in flight. It records faults in a Non-Volatile Memory (NVM) for later troubleshooting using the Central Diagnostic System (CDS).

The maintenance mode is selected from one or the other Audio and Radio Control Display Units (ARCDUs) during maintenance operations. The ADCs perform the tests when they receive a maintenance discrete command from IOM1 while the air speed is less than 50 kt (93 km/h) and the aircraft is on the ground.

Figure 34- 14 ESCP EFIS ADC SOURCE REVERSION PAGE 1

The EFIS ADC SOURCE reversion selector on the ESID Control Panel (ESCP) is set to the NORM position to show ADC data on the related Primary Flight Displays (PFD1, PFD2).

NOTE

The ADC data is supplied through its related AHRU.

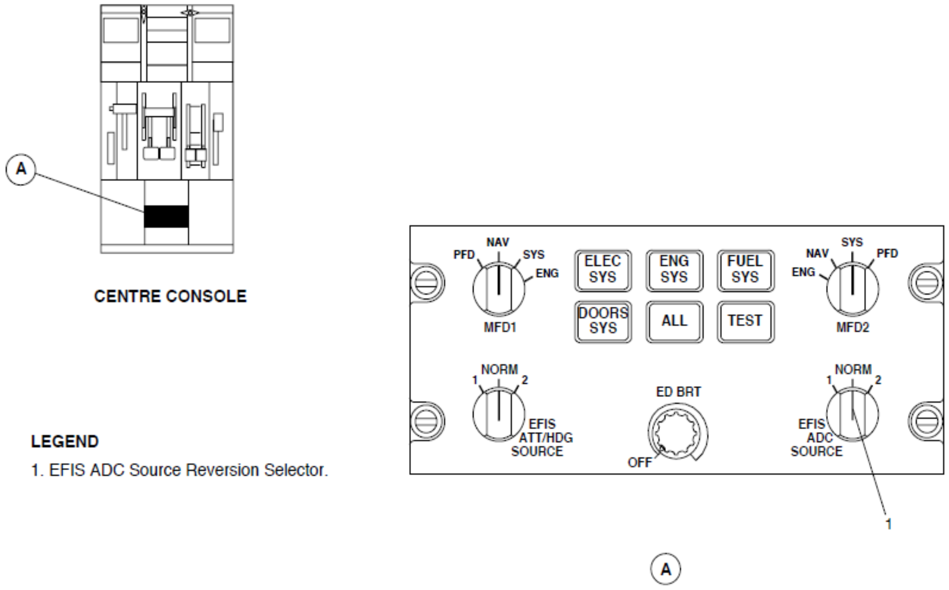


Figure 34- 14 ESCP EFIS ADC SOURCE REVERSION PAGE 1

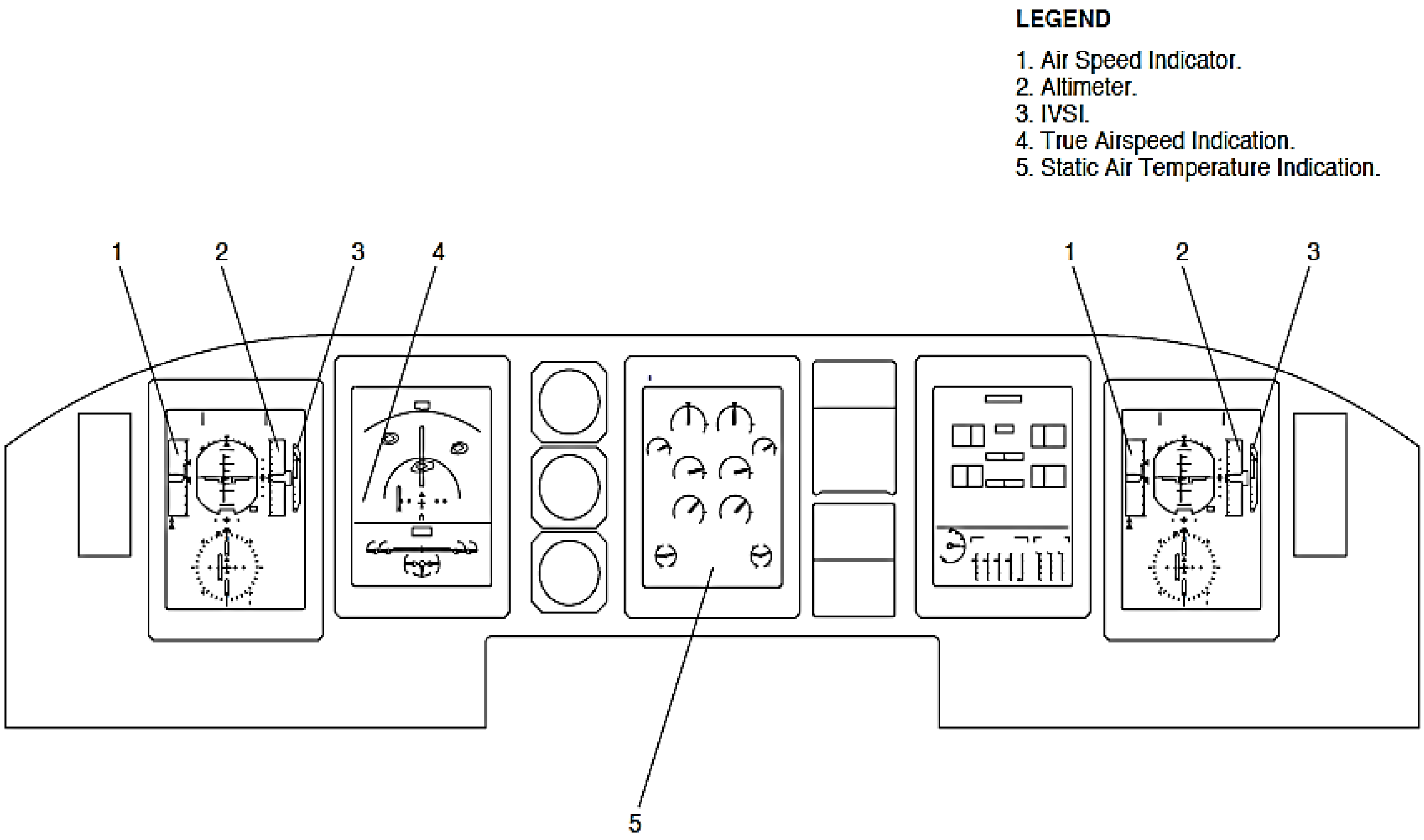


Figure 34- 15 ELECTRONIC INSTRUMENT SYSTEM (EIS) INDICATIONS

Figure 34- 15 ELECTRONIC INSTRUMENT SYSTEM (EIS) INDICATIONS

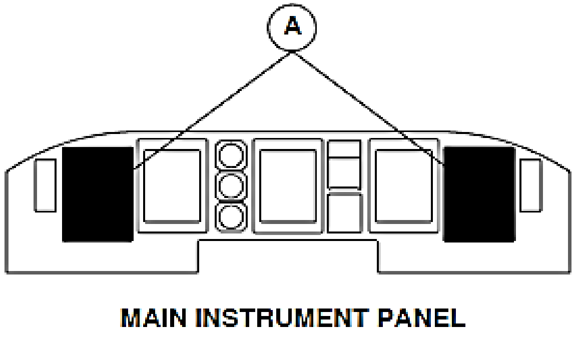
The Electronic Instrument System (EIS) shows the Air Data Computer (ADC1, ADC2) parameters that follow:

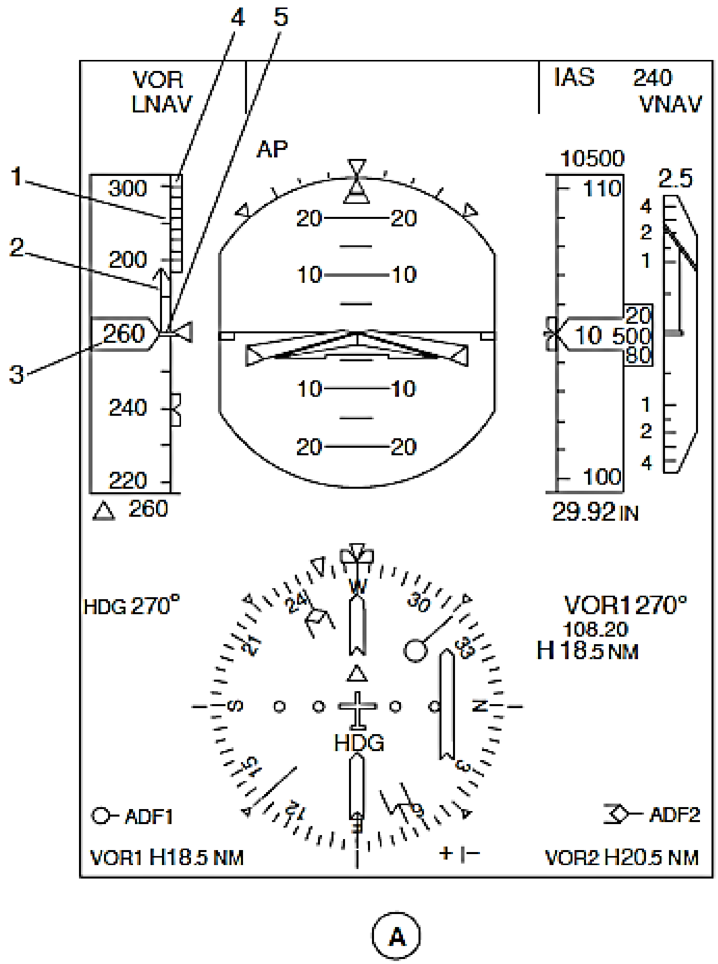
- Indicated Air Speed (IAS1, IAS2)
- Altitude (ALT1, ALT2)
- True Airspeed (TAS1, TAS2)
- Minimum Descent Altitude (MDA)
- Static Air Temperature (SAT).

Indicated Air Speed (IAS): The Indicated Air Speed (IAS) indicator shows the IAS of the selected Air Data Computer (ADC1, ADC2). It has a vertical scale with an IAS reference line and digital readout. The IAS vertical scale and reference line give an analogue indication.

The Primary Flight Display shows the Indicated Air Speed (IAS) indicator components that follow:

- IAS scale
- IAS digital readout
- IAS reference line
- VMO speed cue
- Predicted airspeed trend
- IAS mismatch flag
- IAS failure flag.





LEGEND

1. IAS Scale.
2. Predicted Airspeed Trend.
3. IAS Digital Readout.
4. VMO Speed Cue.
5. IAS Reference Line.

Figure 34- 16 PFD INDICATED AIR SPEED INDICATIONS

Figure 34- 16 PFD INDICATED AIR SPEED INDICATIONS

IAS Scale: The IAS scale has a speed range from 30 to 500 kts. It is a black filled rectangular vertical tape with scale with marks every 10 kts and numbers every 20 kts. The tape moves up for decreasing airspeeds and down for increasing airspeeds to show the IAS value.

IAS Digital Readout: The IAS value is shown through a window in the middle of the IAS indicator. The IAS indicator also shows range of values 42 kts more and less than the value shown in the window.

The IAS digital readout has white numbers that change to red when the IAS is more than the Maximum Operating Speed (VMO). The display changes back to white when the IAS decreases 2 at less than the VMO.

The numbers also change to red when the IAS is less than the low speed cue. The display changes back to white when the IAS increases 2 kts.   more than the low speed cue.

IAS Reference Line: The IAS indicator has a white line adjacent to the window to point to the current IAS.

VMO Speed Cue: The VMO speed cue gives indication of the Maximum Operating Speed (VMO) along the IAS scale from 30 to 400 kts. It is a red and black band that starts with a red edge at the VMO value and extends to the top of the scale.

The VMO speed cue is out of view when the IAS is 42 kts less than VMO.

NOTE

When VMO is invalid, the ADC will also send an invalid IAS signal to make the PFD show an IAS failure flag.

Predicted Airspeed Trend: The predicted airspeed trend gives an analog indication of the predicted airspeed in 10 seconds while the aircraft is airborne. It is a white vertical line with an arrow that starts at IAS reference line and points to the predicted airspeed.

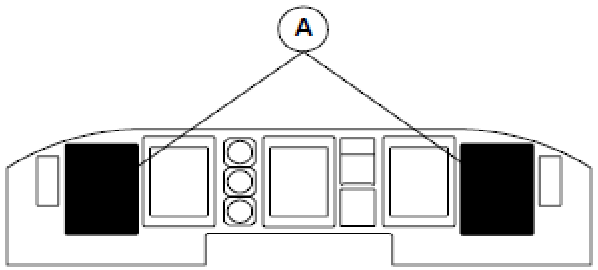
If the airspeed trend is more than the shown IAS scale, part of the indication goes out of view.

When the difference between the actual IAS and the predicted airspeed value is less than 1 kt or when the IAS parameter is not correct, the indication goes out of view

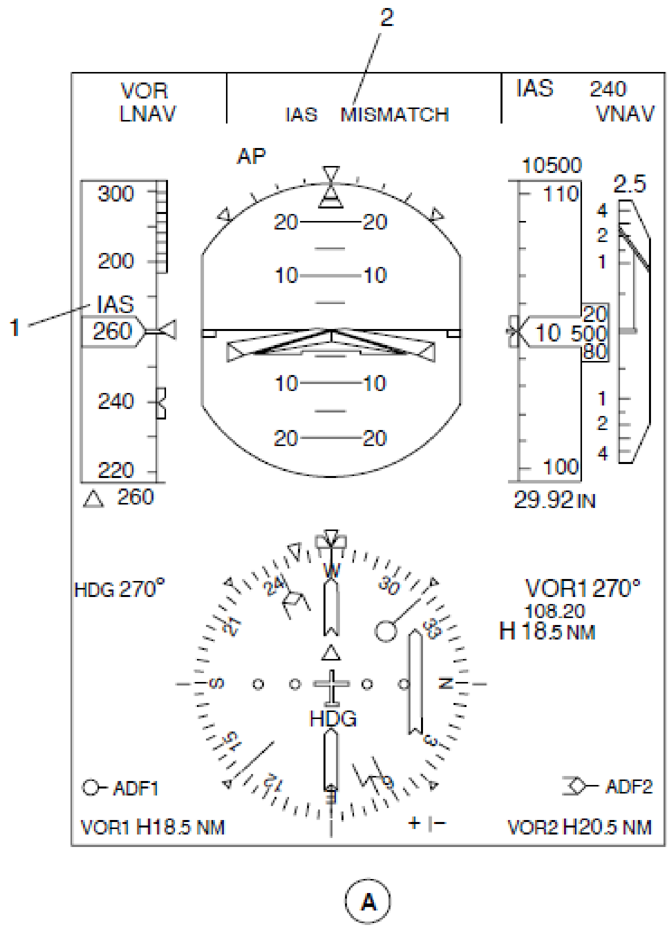
Figure 34- 17 PFD INDICATED AIR SPEED MISMATCH INDICATION

IAS Mismatch Flag: The Flight Data Processing System (FDPS) senses signal differences between the IAS1 and the IAS2. The two IAS indicators show a yellow IAS flag above its IAS digital readout and the two Flight Mode Annunciators (FMA) show a yellow IAS MISMATCH message in the centre row of the centre column. The indications flash for five seconds when they come into view and then go steady.

The airspeed mismatch threshold is equal to 10 kt.



MAIN INSTRUMENT PANEL



LEGEND

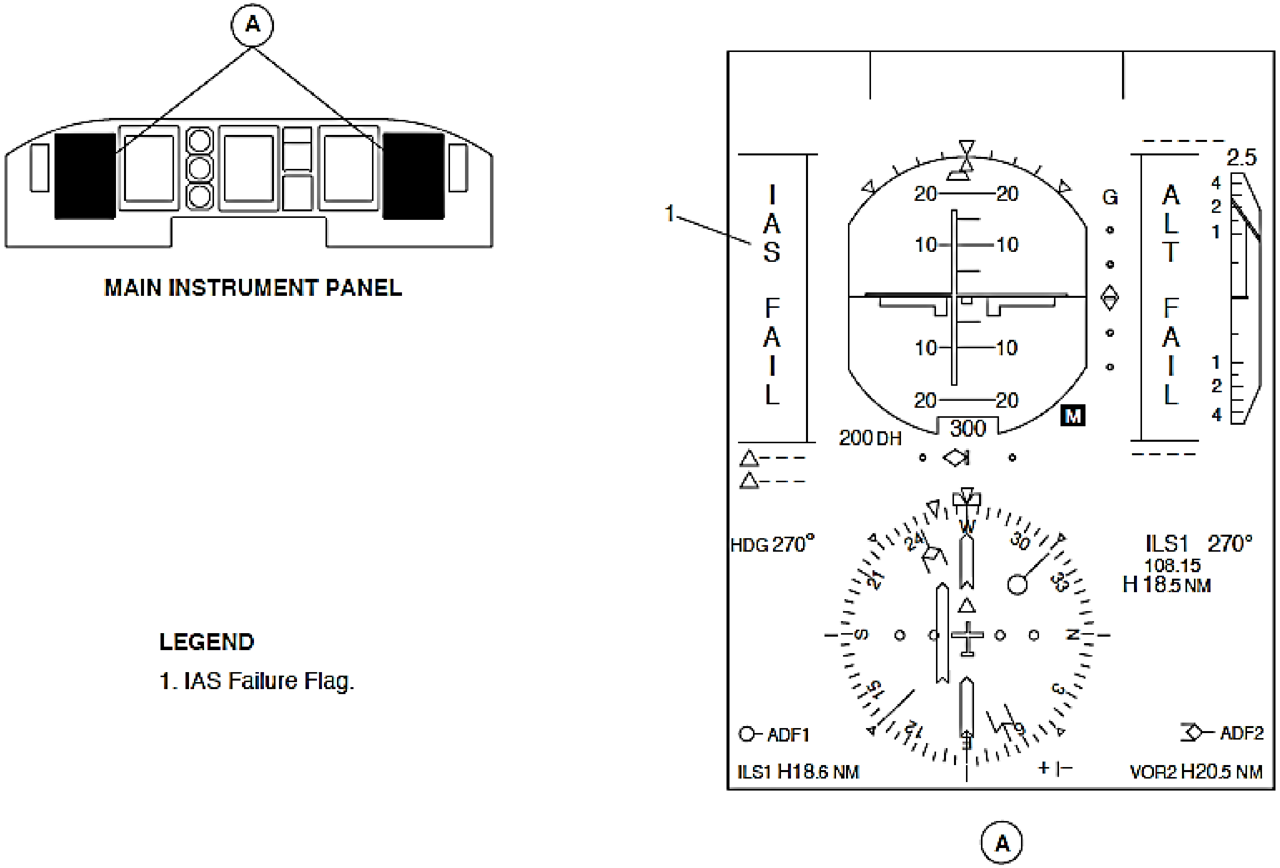
1. Mismatch Flag.
2. IAS Mismatch Message.

Figure 34- 17 PFD INDICATED AIR SPEED MISMATCH INDICATION

Figure 34- 18 PFD INDICATED AIR SPEED FAILURE FLAG

IAS Failure Flag: When the airspeed parameter malfunctions, an IAS failure flag comes into view as a red IAS FAIL message in a white rectangle.

Speed bugs are shown along the edge of the IAS scale to reference V (V1, VR, and V2) speeds



LEGEND

1. IAS Failure Flag.

Figure 34- 18 PFD INDICATED AIR SPEED FAILURE FLAG

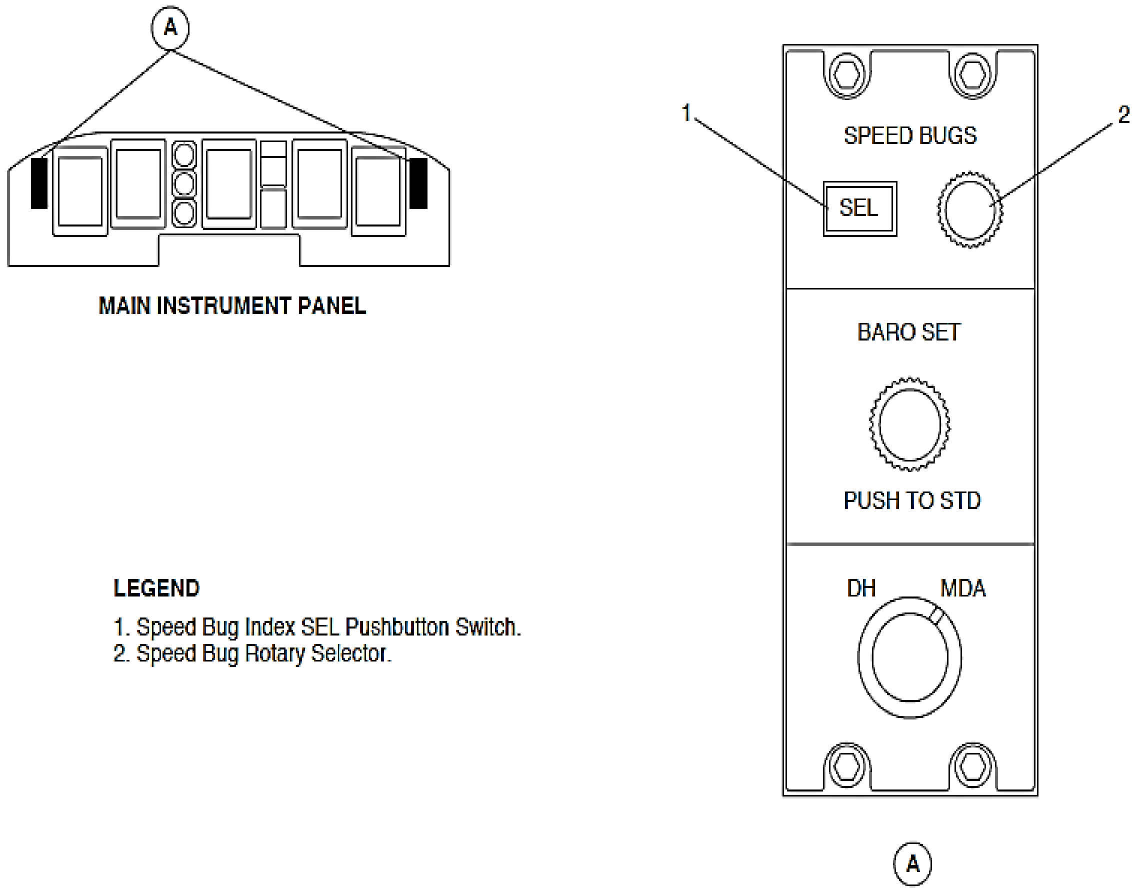


Figure 34- 19 INDEX CONTROL PANEL (ICP), SPEED BUG SELECTION

Figure 34- 19 INDEX CONTROL PANEL (ICP), SPEED BUG SELECTION

The SPEED BUG SEL pushbutton switch located on the Index Control Panel (ICP1, ICP2) is pushed to set the speed bugs that follow:

- V1, take off decision speed
- VR, rotation speed
- V2, take off safety speed
- No. 1 (Solid)
- No. 2 (Outline).

The SPEED BUG SEL pushbutton switch is pushed to set speed bugs V1, VR, and V2 when the aircraft is on the ground and speed bugs No. 1 and No.2 when the aircraft is airborne.

The SPEED BUG SEL pushbutton switch is pushed to change speed bug V1or No. 1. It is pushed again to set V1 or No. 1 and to change VR or No. 2. The SPEED BUG SEL pushbutton switch is pushed again to set V2. Each speed bug is also set five seconds after the last SPEED BUG SEL pushbutton switch or SPEED BUG rotary selection.

The SPEED BUG rotary selector is turned to change the speed bug value.

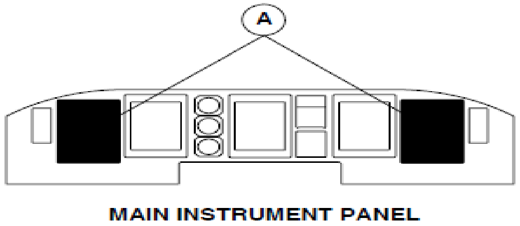
Figure 34- 20 PFD SPEED BUG INDICATIONS

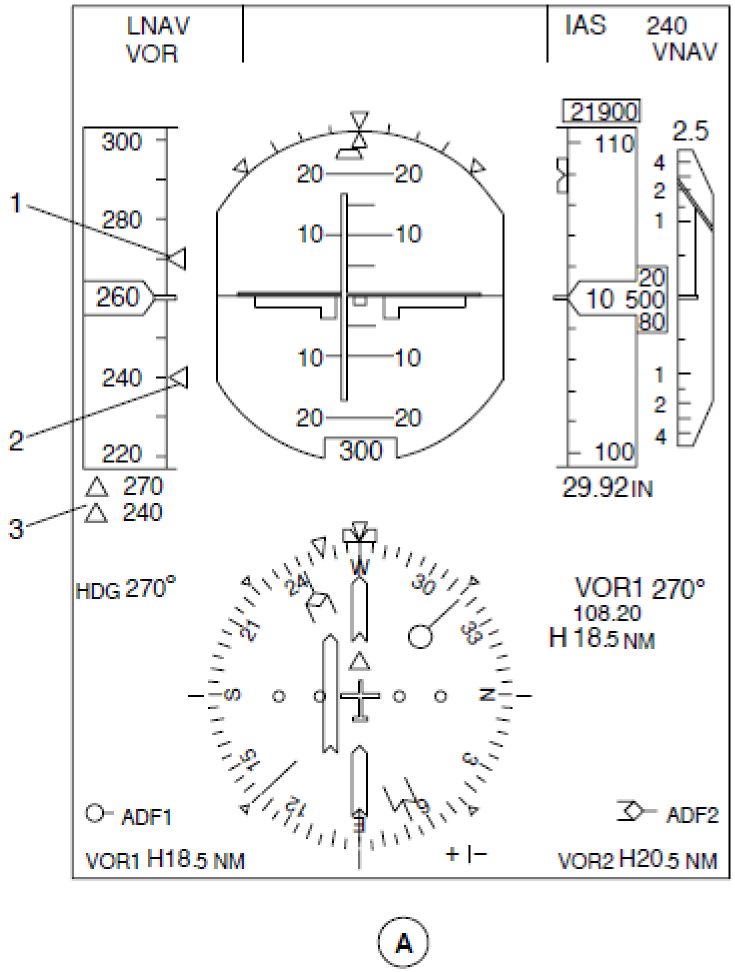
The cyan bug moves along the IAS scale from minimum 50 kt value to 400 kt, and is parked at this position for higher values. When the index value is lower than 50 kt or when not correct, it is removed from the indication.

Each index bug has a digital readout shown in cyan characters to give a digital value for the Index Control Panel (ICP1, ICP2) selection. It shows the selection in 1kt increments from 50 kts to 400 kts. The indication is out of view when the selection is less than 50 kts. Each digital value has a bug reminder.

NOTE

The speed bug is not an autopilot target and is only used as an advisory indication.





LEGEND

1. IAS Index Bug No. 1.
2. IAS Index Bug No. 2.
3. IAS Index Bug Digital Readouts.

Figure 34- 20 PFD SPEED BUG INDICATIONS

Figure 34- 21 INDEX CONTROL PANEL (ICP), BAROMETRIC CORRECTION SELECTION

The BARO SET knob located on the Index Control Panel (ICP1, ICP2) is turned to supply the related Air Data Computers (ADC1, ADC2) with a barometric correction.

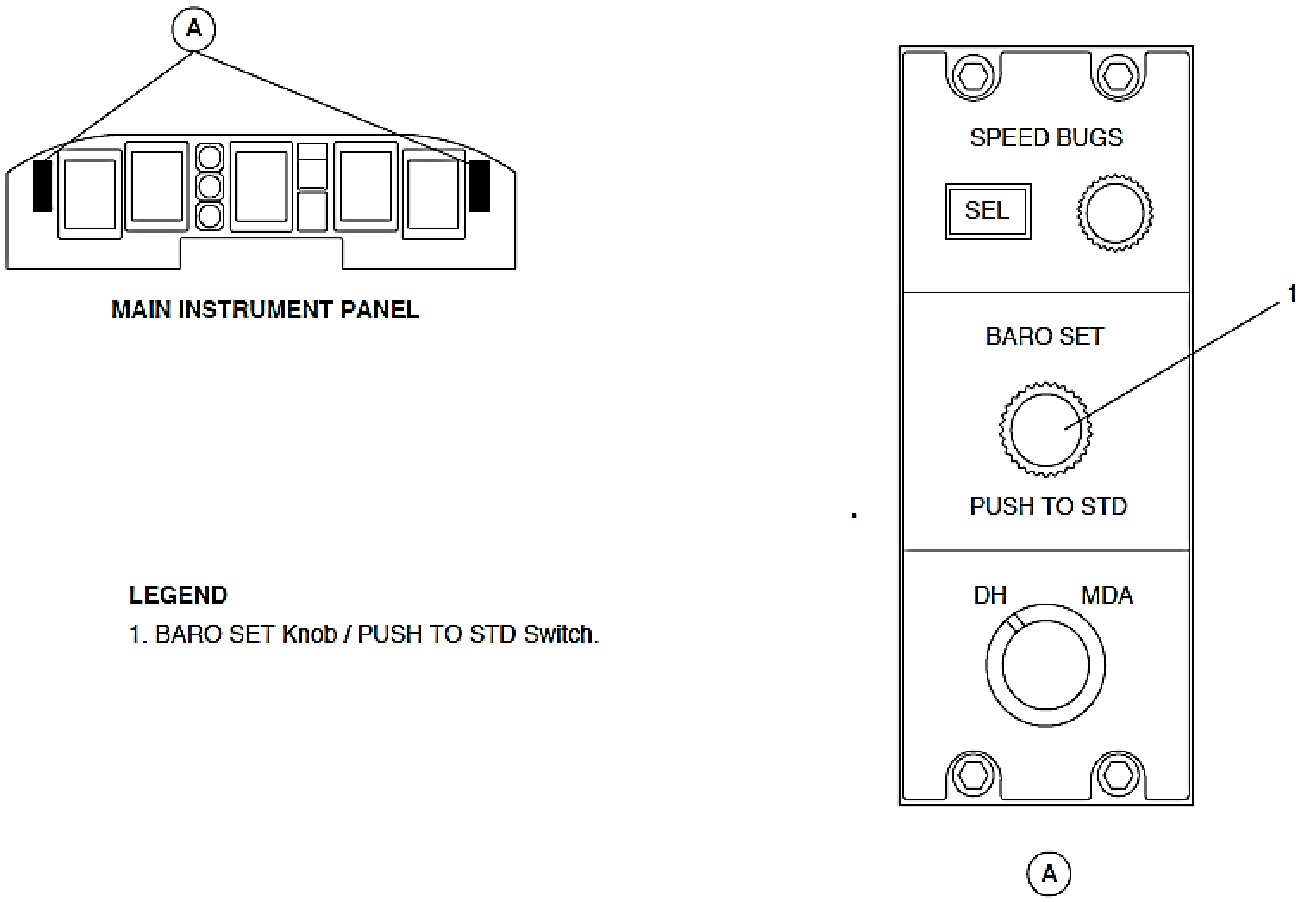
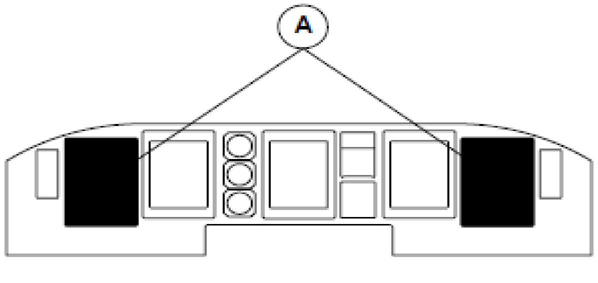
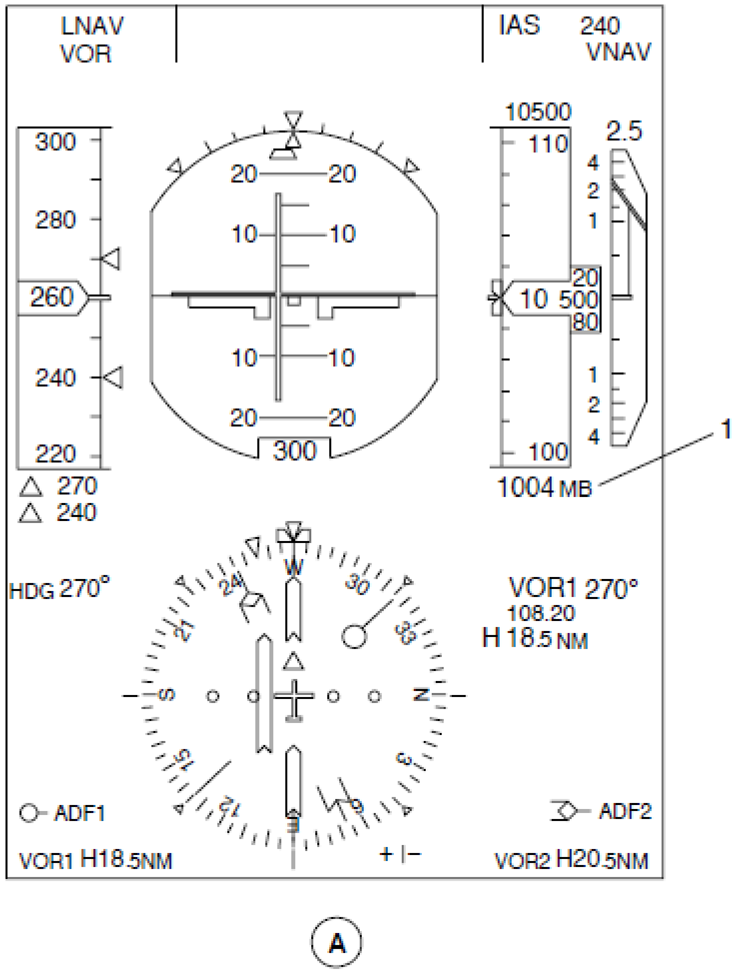


Figure 34- 21 INDEX CONTROL PANEL (ICP), BAROMETRIC CORRECTION SELECTION

Figure 34- 22 PFD BAROMETRIC CORRECTION INDICATION

The barometric correction range is 745 to 1050 mbar (22 to 31 In Hg). The Related Primary Flight Displays (PFD1 and PFD2) show the barometric correction indication below the altitude indicator with four digits shown in cyan followed by a Millibars (MB) label in white fonts. It is shown in Millibars (MB) or in Hg, as set by pin programming logic on the ADC. If a power interruption occurs, the barometric selection value is saved by the ADC for use when restarting.





LEGEND

1. Baro Setting Digital Readout.

Figure 34- 22 PFD BAROMETRIC CORRECTION INDICATION

Figure 34- 23 PFD ALTITUDE INDICATOR INDICATION

Altitude Indicator: The altitude indicator shows the barometric altitude of the selected Air Data Computer (ADC1, ADC2). It has a vertical scale with an IAS reference line and digital readout. The IAS vertical scale and reference line give an analogue indication.

The Primary Flight Display shows the altitude indicator parameters that follow:

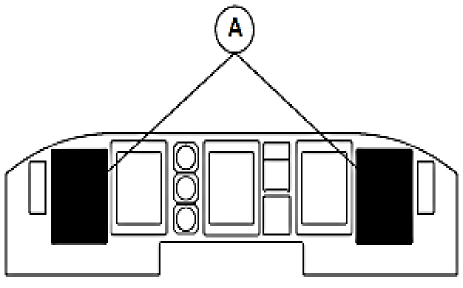
- Altitude scale
- Altitude digital readout
- Altitude reference line
- Altitude mismatch flag
- Altitude failure flag.

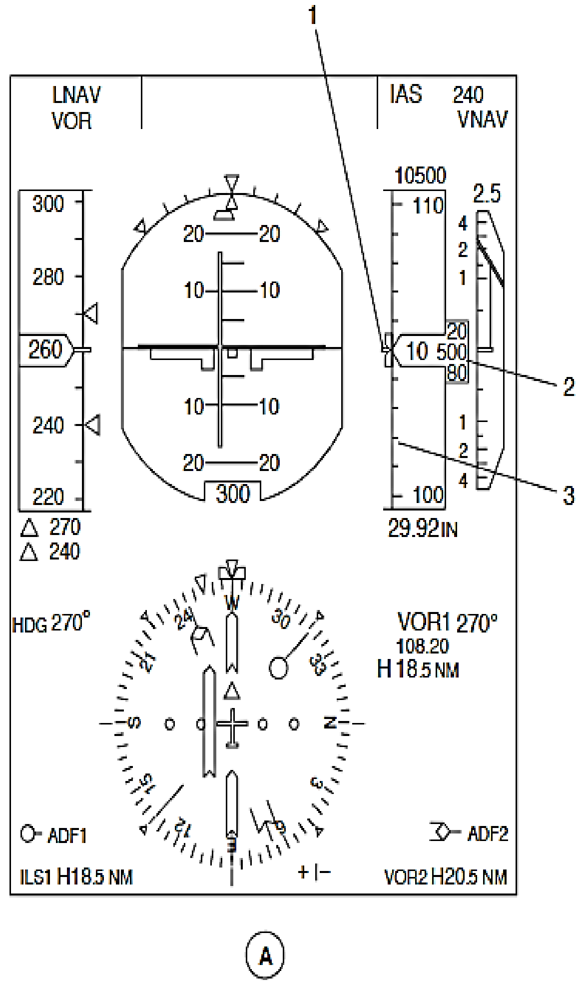
Altitude scale: The altitude scale has a range from −990 ft. to 50000 ft. It is a black filled rectangular vertical tape with scale with marks every 100 ft. and numbers every 500 ft. The tape moves up for decreasing altitudes and down for increasing altitudes.

For negative altitudes, a white NEG label is shown vertically between the 0 ft. and −500 ft., −500 and −1000 ft., and −1000 ft. and −1500 ft. marks.

Altitude digital readout: The altitude value is shown through a T shaped window in the middle of the altitude indicator as a rolling drum indication in 20 ft. increments. The altitude indicator also shows a range of values 550 ft. more and less than the value shown in the window.

Altitude reference line: The altitude indicator has a white line adjacent to the window to point to the current altitude.





LEGEND

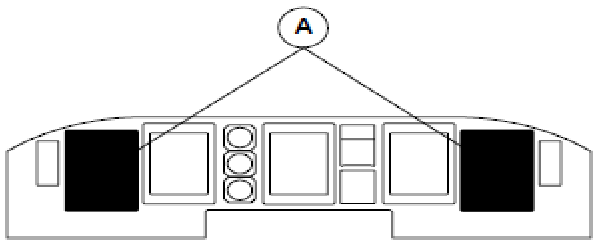
1. Altitude Reference Line.
2. Altitude Digital Readout.
3. Altitude Scale.

Figure 34- 23 PFD ALTITUDE INDICATOR INDICATION

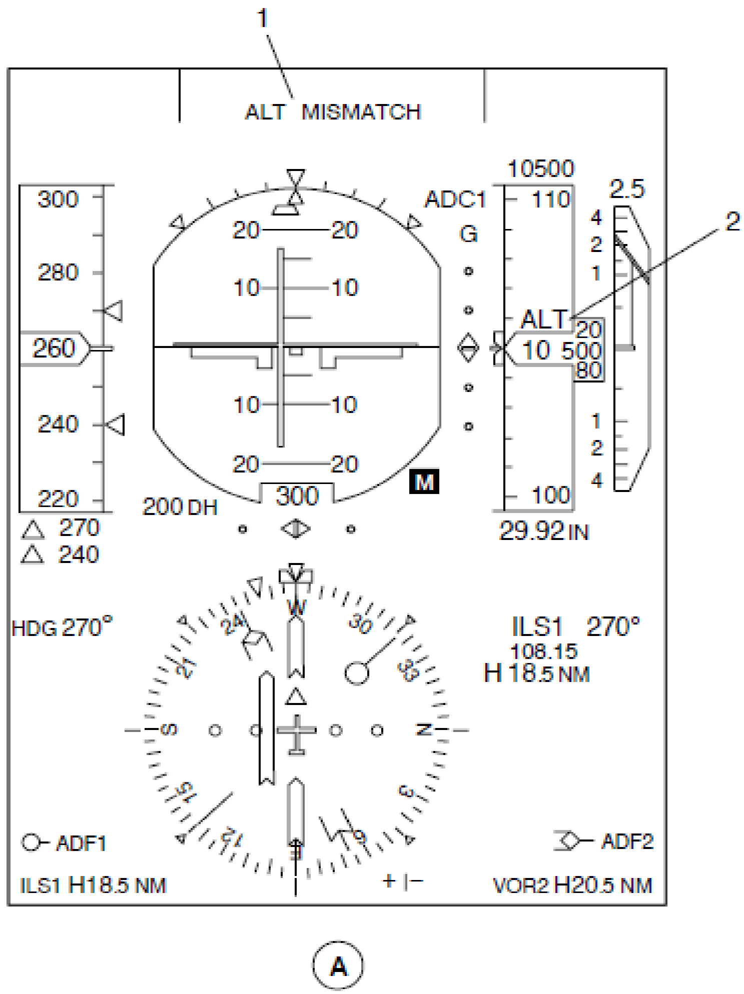
Figure 34- 24 PFD ALTITUDE INDICATOR MISMATCH INDICATION

Altitude Mismatch Flag: The Flight Data Processing System (FDPS) senses signal differences between the ALT1 and the ALT2. The two altitude indicators show a yellow ALT flag above its altitude digital readout and the two Flight Mode Annunciators (FMA) show a yellow ALT MISMATCH message in the centre row of the centre column. The indications flash for five seconds when they come into view and then go steady.

The altitude mismatch threshold increases from 60 ft. (18.3 m) on the ground to 180 ft. (55 m) at 27,000 ft. (8230 m).



MAIN INSTRUMENT PANEL



LEGEND

1. Mismatch Message.
2. Altitude Mismatch Flag.

Figure 34- 24 PFD ALTITUDE INDICATOR MISMATCH INDICATION

Figure 34- 25 PFD ALTITUDE INDICATOR FAILURE FLAG

Altitude Failure Flag: When the altitude indicator parameter malfunctions, an altitude indicator failure flag comes into view as a red ALT FAIL message in a white rectangle.

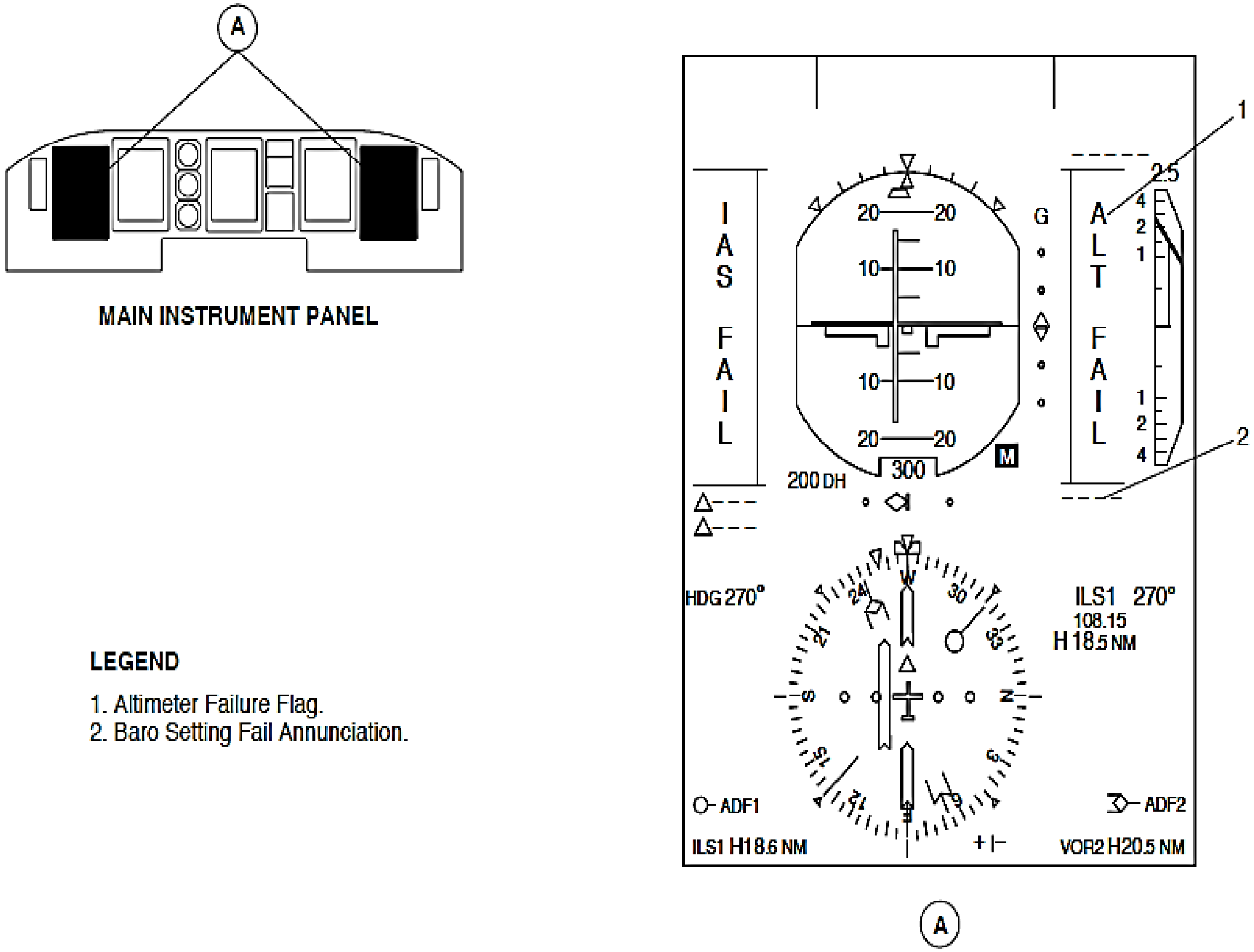


Figure 34- 25 PFD ALTITUDE INDICATOR FAILURE FLAG

Figure 34- 26 PFD METRIC SECONDARY ALTITUDE INDICATION SELECTION

On aircraft with the metric secondary altitude indications (839CH00016 OR SB84−31−74 OR SB84−31−93): When the PFD ALTIMETER UNITS toggle on the PFD altimeter unit control panel is set to the FT+M position, the altitude indicator will continue to show altitude in feet and it will also show a metric indication in meters.

On aircraft with SB84−31−44 OR SB84−31−84 incorporated, the PFD altimeter unit control panel is not installed.

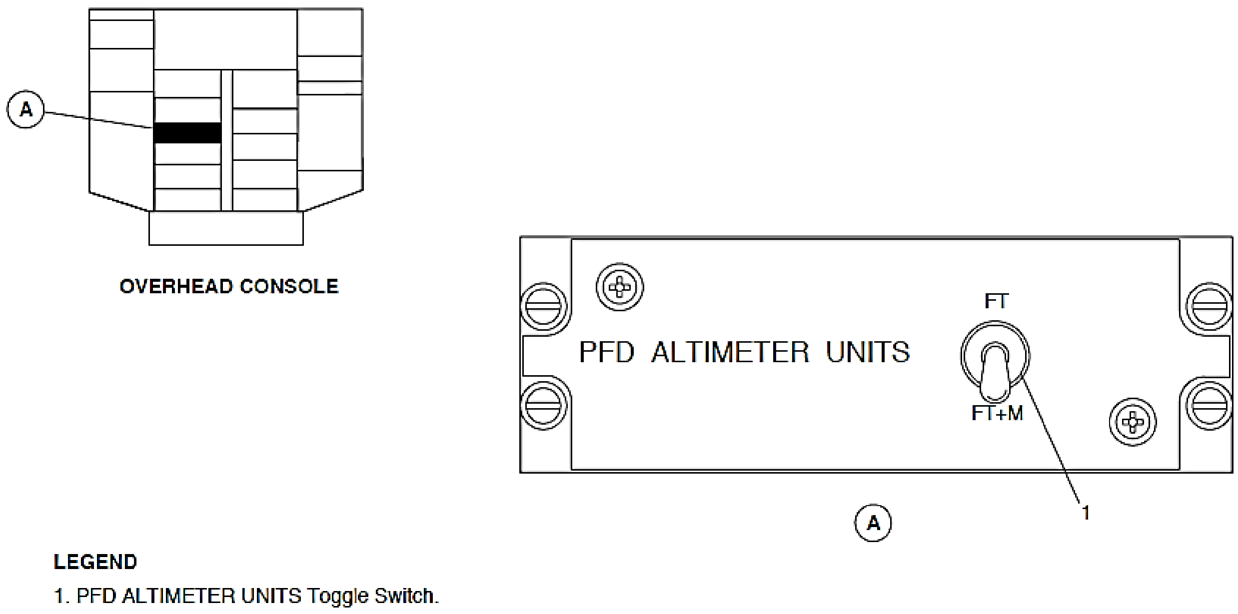


Figure 34- 26 PFD METRIC SECONDARY ALTITUDE INDICATION SELECTION

Figure 34- 27 839CH00016): PFD METRIC **ALTITUDE INDICATION**

The metric indication shows the aircraft's current altitude in 5 m increments from negative 500 m to 15900 m. It has white numbers in white box with an M label. When the metric altitude indication malfunctions, the 5 numbers are replaced with 5 white dashes.

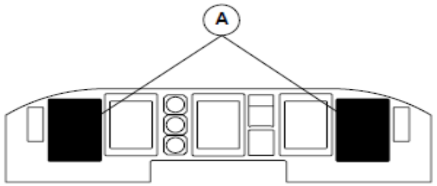
NOTE

The metric altitude indication is supplied through its related IOP. If it malfunctions, the indication is supplied through the opposite (IOP).

When the metric altitude indication is in view, the pre–selected altitude readout is also shown in meters. The range of the altitude readout is between 0 and 30500 m.

NOTE

The metric altitude readout indication changes in 100 ft. increments but is rounded up to match the 5 m metric altitude indication.



MAIN INSTRUMENT PANEL

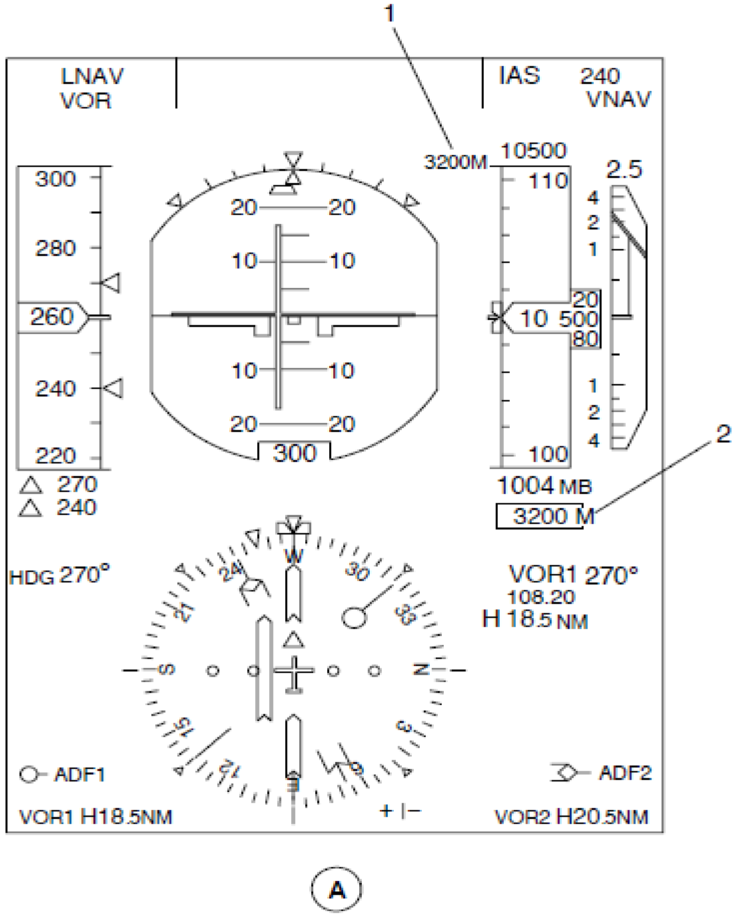


Figure 34- 27 839CH00016): PFD METRIC ALTITUDE INDICATION

LEGEND

1. Metric Altitude Digital Readout.
2. Metric Preselect Altitude Digital Readout.

Figure 34- 28 ICP MDA SELECTION

Minimum Descent Altitude (MDA): The MDA is a specified altitude that is referenced to sea level for non−precision approaches. A descent below the MDA cannot be made until a visual reference to continue the approach to land is made.

The DH/MDA rotary switch on the Index Control Panel (ICP1, ICP2) is set to MDA to let the DH/MDA knob change the MDA value. The DH/MDA knob is then turned to set a MDA for minimum descent altitude calculations.

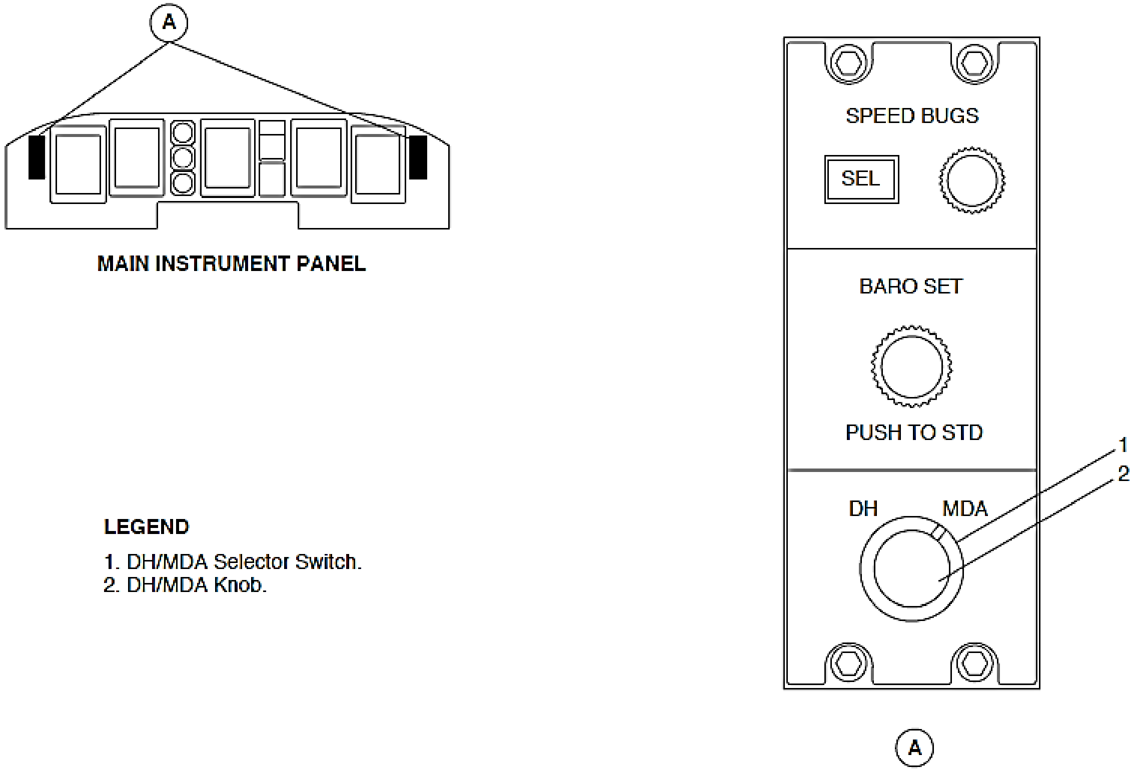
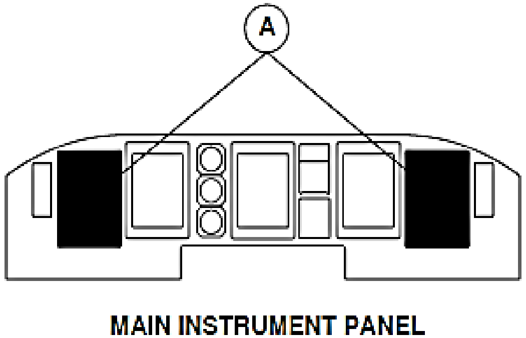
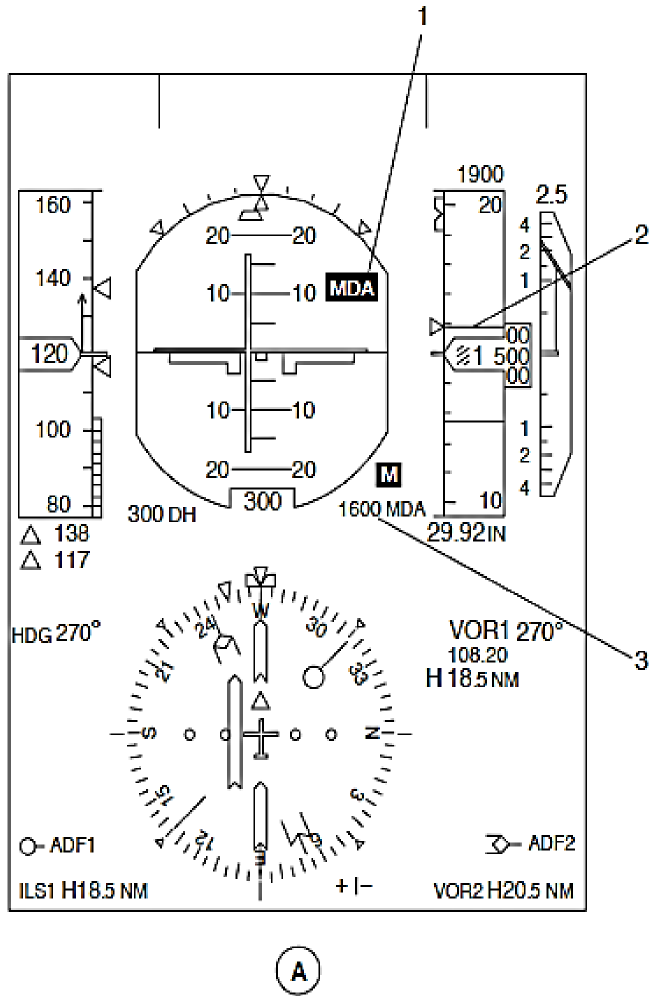


Figure 34- 28 ICP MDA SELECTION





LEGEND

1. MDA Alert Annunciator.
2. MDA Index Bug.
3. MDA Digital Readout.

Figure 34- 29 EIS MDA INDICATION

Figure 34- 29 EIS MDA INDICATION

The MDA indication has the components that follow:

- Digital readouts
- Index bug
- Alert annunciation.

MDA Digital Readouts: The bottom right part of the attitude indicator shows the selection as cyan numbers followed by a white MDA label. It shows the altitude selection in ten−feet increments from 0 through 19990 ft. When MDA setting is less than 0 ft., all of the indication goes out of view.

MDA Index Bug: When the MDA is set to a value more than zero, a cyan analogue MDA index bug on the altitude indicator also comes into view to show the selected MDA value.

When aircraft is at the selected MDA, the MDA index bug changes to a yellow indication that flashes for the first 3 seconds then goes steady.

MDA Alert Annunciation: An MDA alert annunciation shows that the aircraft is going down to the Minimum Descent Altitude (MDA).

This indication is armed while the aircraft's barometric corrected altitude is 100 ft. more than the set MDA and ground. A Weight On Wheels (WOW) condition causes the indication to disarm to make sure that the MDA alert annunciation is out of view during take-off and after landing.

When the MDA alert annunciation is armed and the aircraft altitude Is less than the set MDA value, a yellow MDA alert annunciation that flashes for the first 3 seconds then goes steady is shown in the attitude sphere.

The indication stays in view while the barometric corrected altitude is less than 100 ft. above the set MDA and ground to keep the MDA alert annunciation in view during a flight level hold at the set MDA.

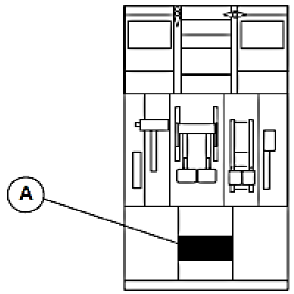
Five white dashes replace the numbers and the MDA alert annunciation goes out of view when a Primary Flight Display (PFD1, PFD2) does not receive valid data from its related Input/Output Processor (IOP1, IOP2).

Figure 34- 30 ESCP EFIS ADC SOURCE REVERSION PAGE 2

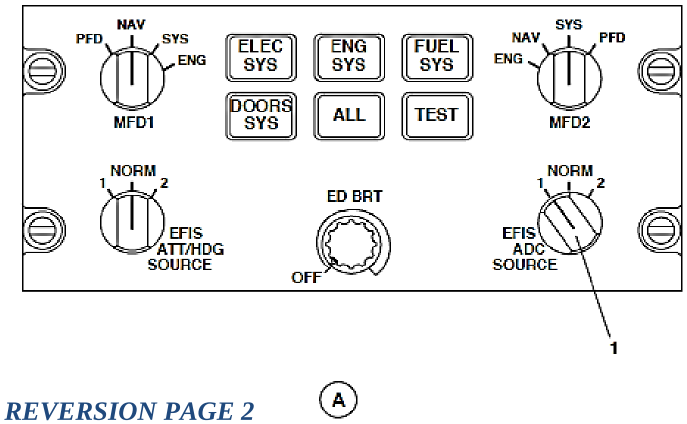
When the EFIS ADC SOURCE reversion selector on the ESID Control Panel (ESCP) is set to the 1 or 2 position, the two Primary Flight Displays (PFD1, PFD2) will show barometric inertial altitude from the selected ADC (ADC1 or ADC2) through its related AHRU (AHRU1 and AHRU2).

NOTE

When the EFIS ATT/HDG SOURCE reversion selector on the ESID Control Panel (ESCP) is set to the 1 or 2 position, the related PFD will show barometric inertial altitude from the selected ADC and AHRU and the other PFD will show barometric corrected altitude data from its related ADC.



CENTRE CONSOLE

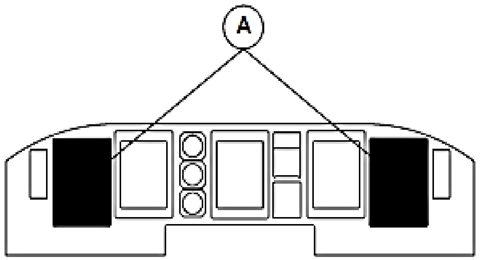


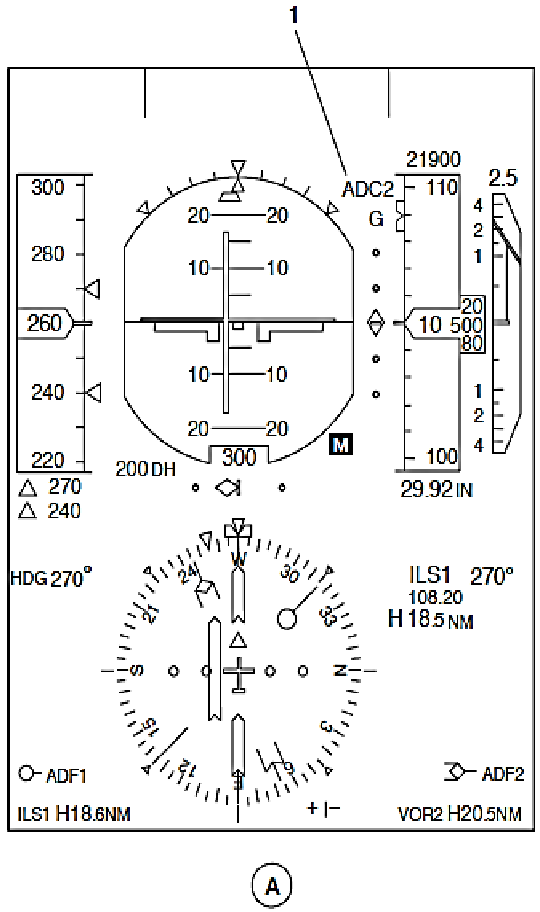
LEGEND
1. EFIS ADC Source Reversion Selector.

Figure 34- 30 ESCP EFIS ADC SOURCE REVERSION PAGE 2

Figure 34- 31 PFD EFIS ADC SOURCE REVERSION INDICATION

When the two Primary Flight Displays (PFD1, PFD2) are showing the same air data source, an air data source annunciator is shown in yellow fonts.





LEGEND

1. Air Data Source Annunciator.

Figure 34- 31 PFD EFIS ADC SOURCE REVERSION INDICATION

Figure 34- 32 ESCP MFD REVERSION SELECTION

The MFD1 and MFD2 reversion switches located on the ESID Control Panel (ESCP) are used to show the navigation page on the multi–functional displays.

The navigation page is usually shown on the upper part of MFD1 and permanent data is shown on the bottom.

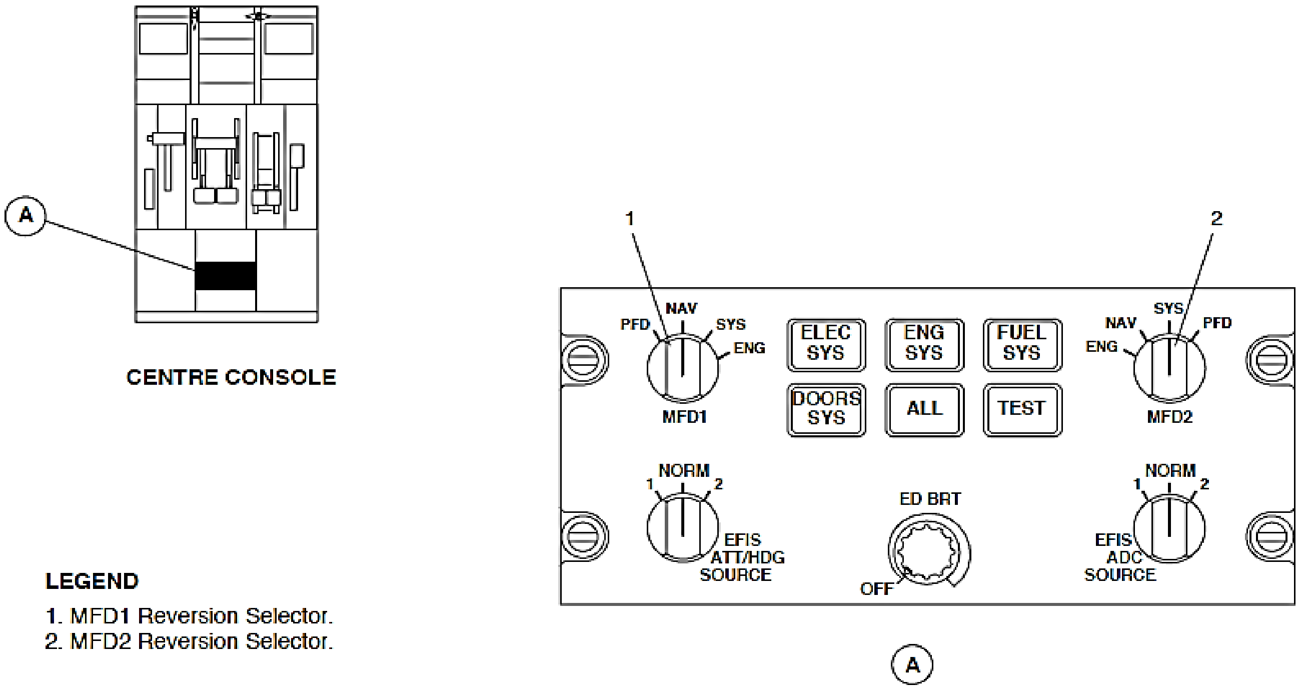
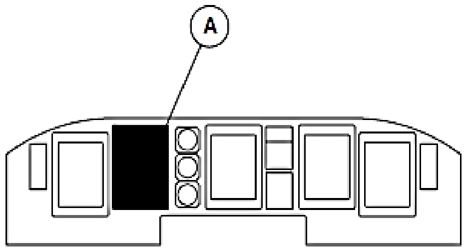


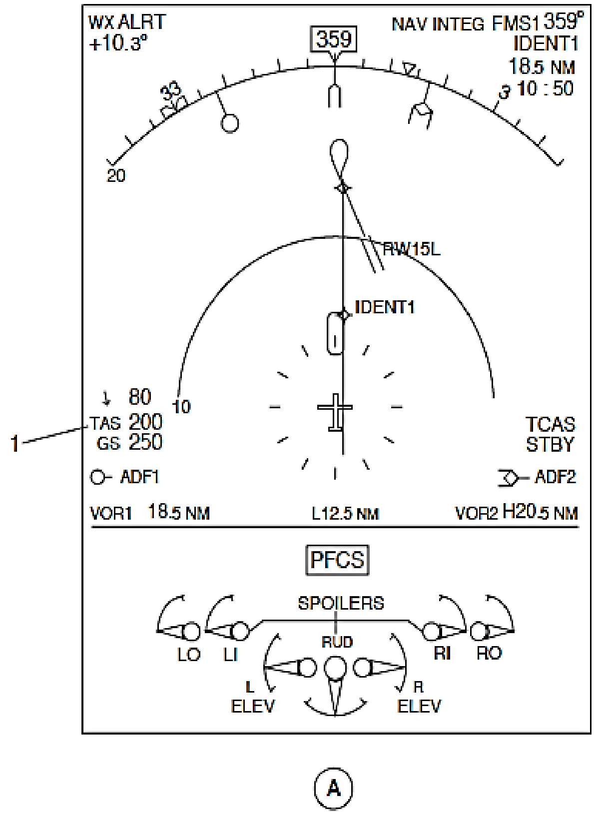
Figure 34- 32 ESCP MFD REVERSION SELECTION

Figure 34- 33 PFD TRUE AIR SPEED (TAS) INDICATION

The navigation page shows the TAS digital value in white numbers from 9 to 999 kt in 1 kt increments with a TAS label.



MAIN INSTRUMENT PANEL



LEGEND

1. True Air Speed Digital Value.

Figure 34- 33 PFD TRUE AIR SPEED (TAS) INDICATION

Figure 34- 34 PFD TRUE AIR SPEED (TAS) MALFUNCTION

When the TAS data malfunctions, the 3 digits are replaced by 3 white dashes.

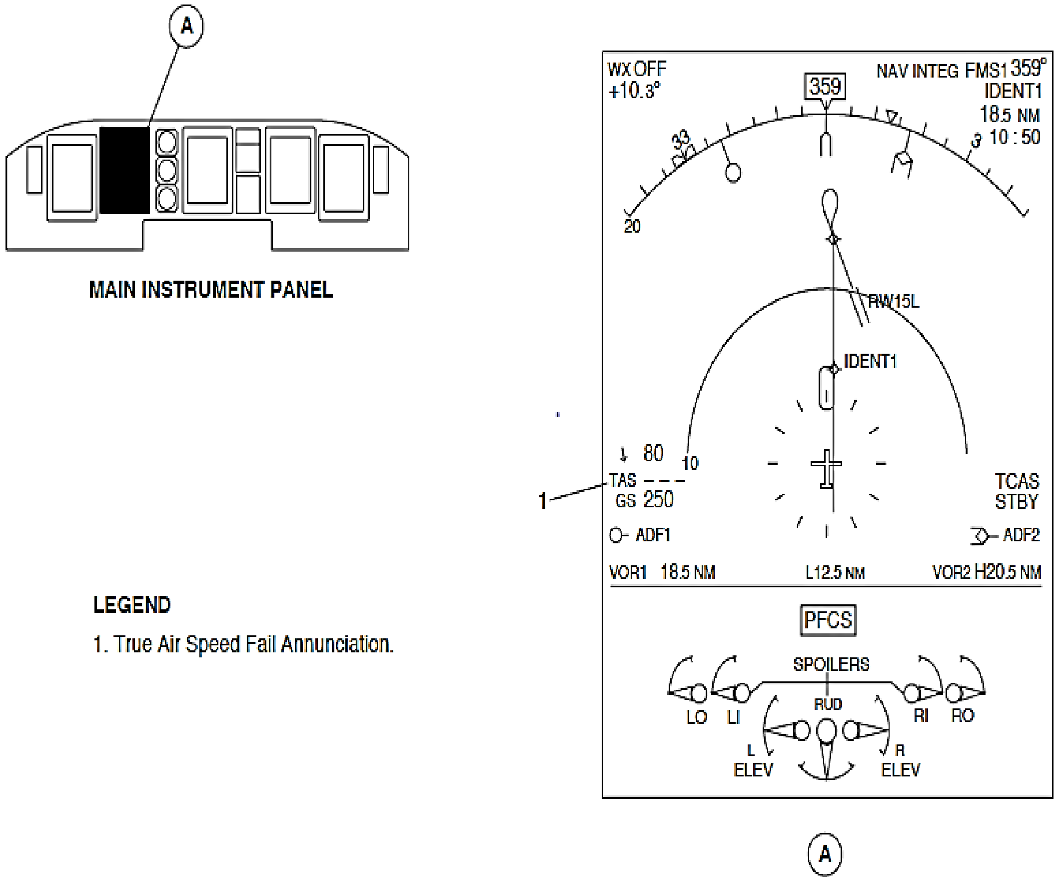


Figure 34- 34 PFD TRUE AIR SPEED (TAS) MALFUNCTION

Figure 34- 35 ED STATIC AIR TEMPERATURE (SAT) INDICATION

The Engine Display (ED) shows the Static Air Temperature (SAT) digital display in white digits with a + or − symbol and a °C label to the right of the digits.

The ED shows SAT from ADC1. When an EFIS ADC SOURCE is set to 2, the ED will show the SAT parameter from ADC2.

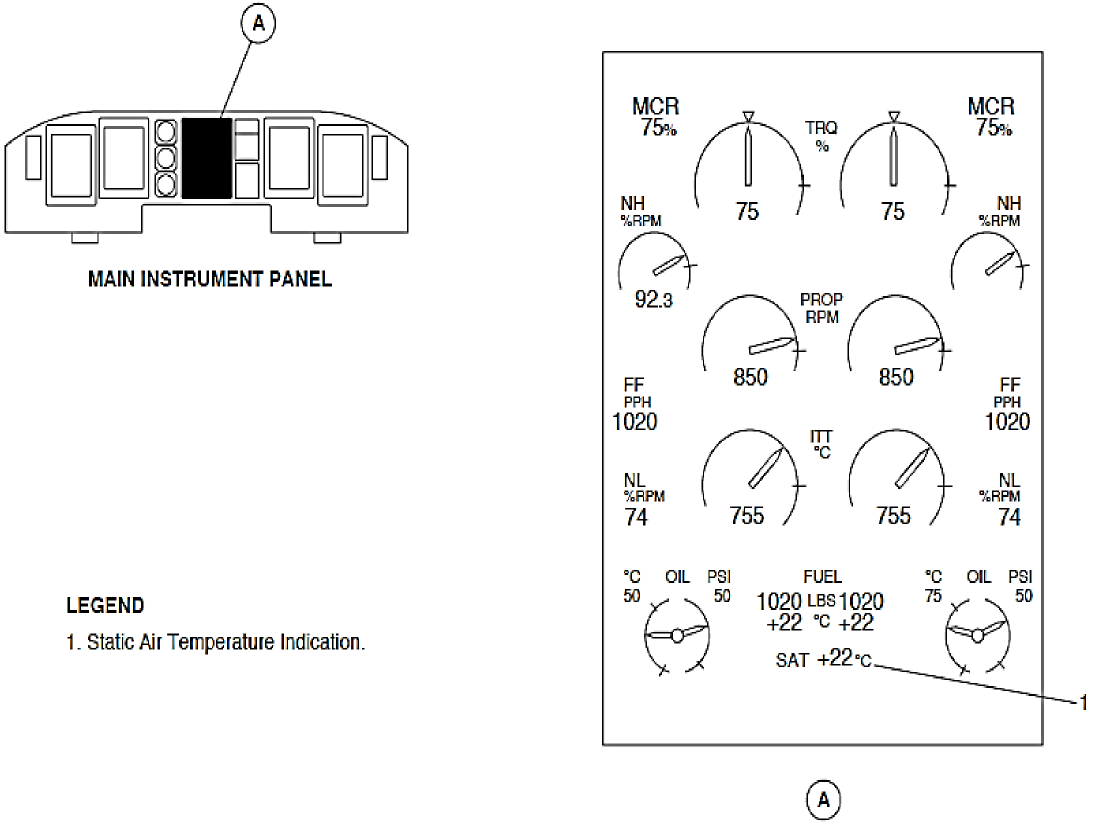
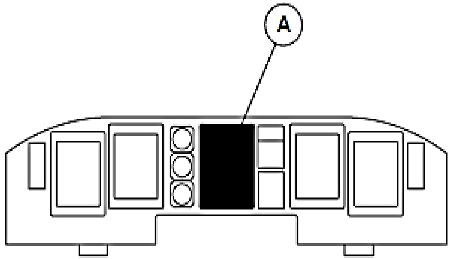


Figure 34- 35 ED STATIC AIR TEMPERATURE (SAT) INDICATION

Figure 34- 36 ED STATIC AIR TEMPERATURE (SAT) MALFUNCTION

When the SAT data malfunctions, the 2 digits are replaced by 2 white dashes.



MAIN INSTRUMENT PANEL

LEGEND

1. Static Air Temperature Fail Annunciation.

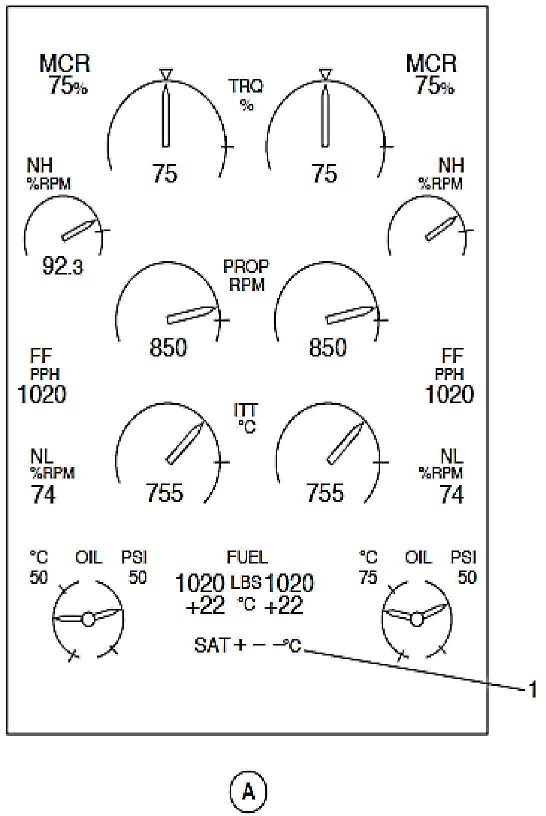
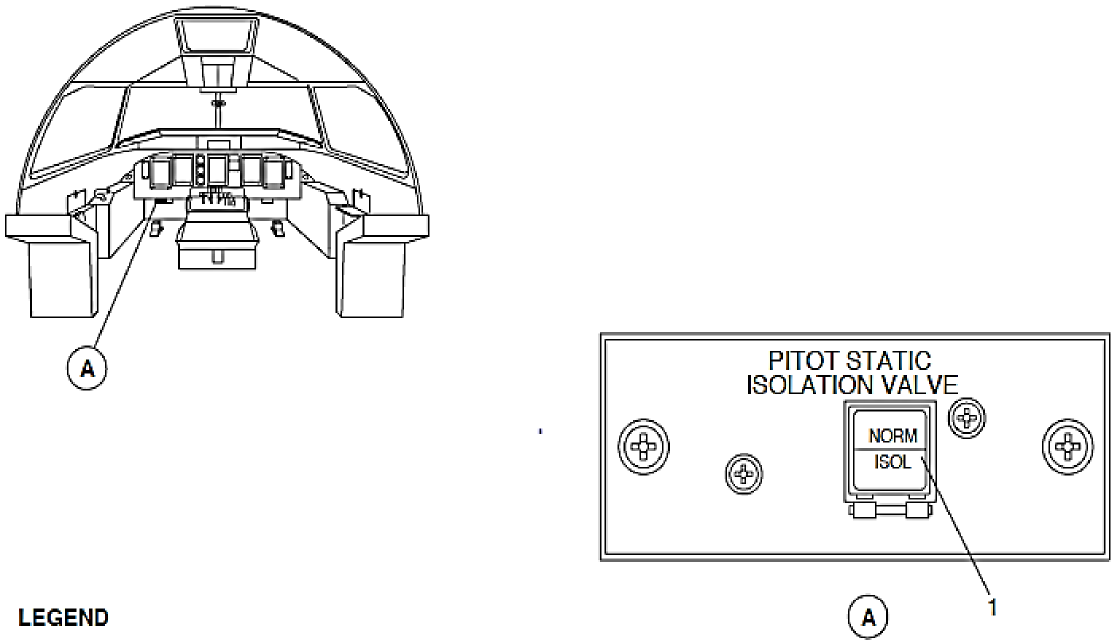


Figure 34- 36 ED STATIC AIR TEMPERATURE (SAT) MALFUNCTION

Figure 34- 37 PITOT STATIC ISOLATION SELECTION

The NORM/ISOL annunciator switch on the pitot static isolation control panel is pushed to isolate ADC1 from the pitot–static probe No. 2 static source.

The green NORM indication goes out and the amber ISOL comes on to show that the shutoff valve has moved to the closed position.



LEGEND

1. Pitot Static Isolation Valve Pushbutton Switch.

Figure 34- 37 PITOT STATIC ISOLATION SELECTION

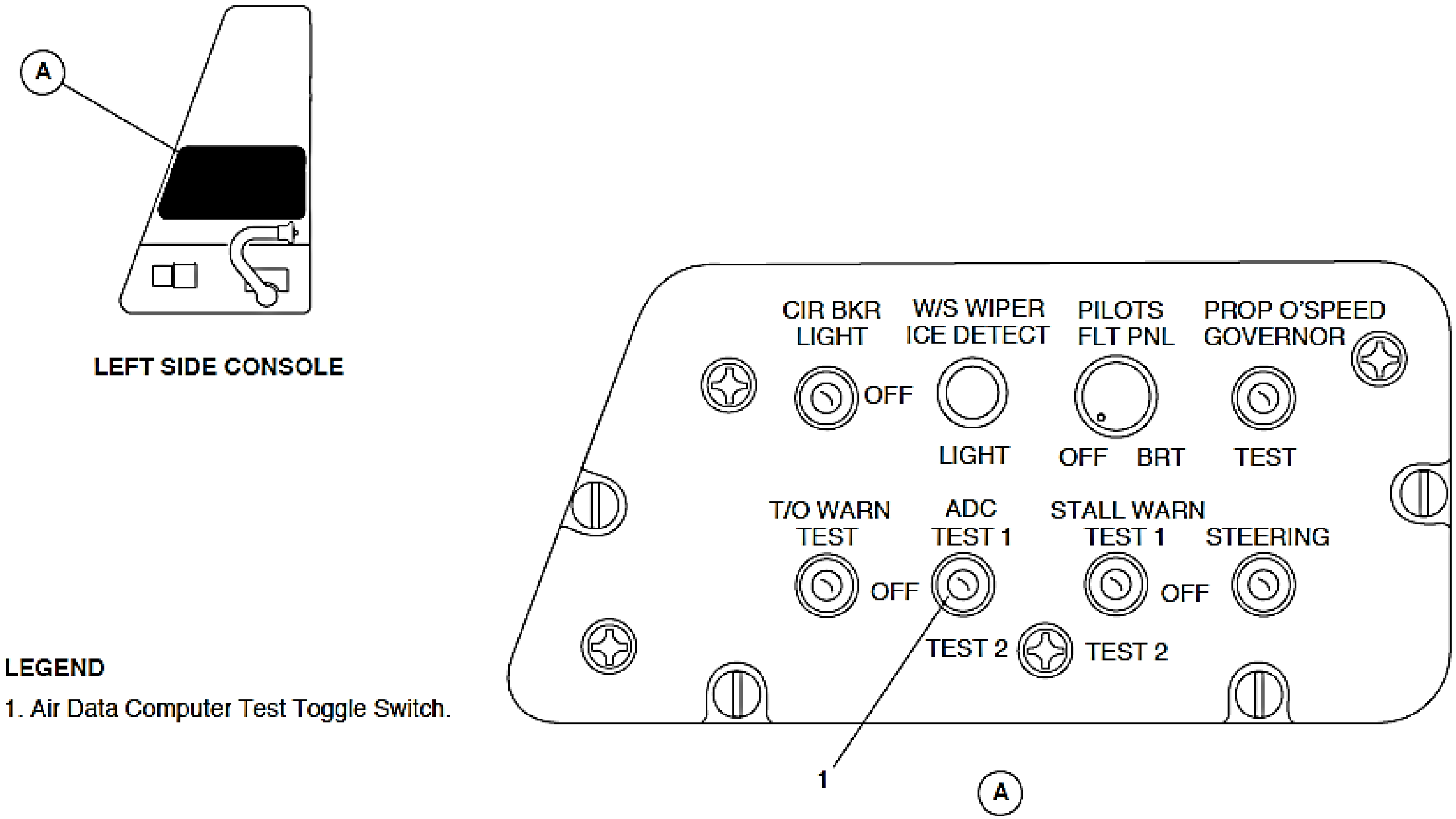


Figure 34- 38 ADC PRE−FLIGHT TEST SWITCH

Figure 34- 38 ADC PRE–FLIGHT TEST SWITCH

The ADC pre–flight test switch on the pilot's side console is set to TEST 1 or TEST 2 to check Vmo Warning Tone Generator (WTG) and the related ADC interfaces.

The Vmo warning sounds 5 seconds after making the selection for aural warning confirmation.

The specific values of air data parameters are shown in the table that follows:

PARAMETER	OUTPUT VALUES
Static Pressure	571.8125 mbar
Impact Pressure	137.875 mbar
Standard Altitude	15,000 ft
Baro correction	990 mbar 29.23 In Hg
Baro corrected altitude	14360 ft (4377 km/h)
Vertical speed	0 ft/min
Calibrated Airspeed	285 KIAS (528 km/h)

PARAMETER	OUTPUT VALUES
Mach	0.564
True Air Speed	353.125 kt (654.41 km/h)
VMO	280 KIAS (519 km/h)
Static Air Temp.	–15 °C
Total Air Temp	+1.5 °C

The PITCH TRIM, ELEVATOR FEEL, and AVIONICS caution lights also come on.

The aircraft left essential and right main busses supply 28 Vdc electrical power through 2 A circuit breakers to the Air Data Computers (ADC1, ADC2). The circuit breakers are located on the 28 Vdc avionics circuit breaker panel in position A7 and A6.

The temperature probe has 2 sensing elements. Each element receives an excitation current from its related ADC to measure its resistance. It is less than 0.5 mA to limit the self–heating error.

Each ADC receives total pressure (pt) supplied by its related pitot–static probe and a static pressure (ps) value obtained as a pneumatic average between static pressures from the two pitot–static probes.

In addition to the data received from the pneumatic interfaces, each ADC receives discrete data from the systems that follow:

- An electrical signal from its related temperature sensor to calculate the Indicated Outside Air Temperature (IOAT)
- Barometric correction data from the related Index Control Panel (ICP)
- Test discrete command from a 3 position ADC TEST switch on the pilot's side console
- Initiated Built−In−Test (IBIT) discrete command from the Centralized Diagnostic System (CDS)
- Pitot−static probe heating failure discrete from the Ice and Rain Protection System (IRPS)
- Air data source manual reversion command from the ESID Control Panel (ESCP)
- Weight−On−Wheels discrete from the Proximity Sensor Electronics Unit (PSEU).

The air data parameters that follow are controlled using ARINC 429 low speed busses (12.0 to 14.5 kilobytes per second):

PARAMETER	SOURCE
Static Pressure	FADEC1, FADEC2
Impact Pressure	FADEC1, FADEC2
Pressure Altitude	FGM1, FGM2 IOP1, IOP2 ATC1, ATC2 FMS CPC CIM

PARAMETER	SOURCE
Barometric Correction	PFD1, PFD2 MFD1, PFD2 FDR CPC
Barometric Corrected Altitude	PFD1, PFD2 MFD1, MFD2 FGM1, FGM2 AHRS1, AHRS2 ATC1, ATC2 FMS
Altitude Rate	FMS CPC CIM
Calibrated Air Speed	PFD1, PFD2 MFD1, MFD2 FGM1, FGM2 SPM1, SPM2 IOP1, IOP2 FMS FADEC1, FADEC2 FCECU1, FCECU2
Maximum Velocity in Operation	PFD1, PFD2 MFD1, MFD2 FGM1, FGM2 FMS
VMO Warning	IOP1, IOP2
Mach	SPM1, SPM2 FMS

PARAMETER	SOURCE
True Air Speed	PFD1, PFD2 MFD1, MFD2 FGM1, FGM2 SPM1, SPM2 AHRS1, AHRS2 IOP1, IOP2 FMS FADEC1, FADEC2 CIM
Static Air Temperature	MFD1, MFD2 ED IOP1, IOP2 FMS FADEC1, FADEC2
Total Air Temperature	IOP1, IOP2 FMS

Each ADC supplies air data parameters through 5 identical ARINC 429 low speed busses for use by critical systems and other avionics subsystems .For non−avionics non−critical systems, air data parameters are available on two General Purpose Data Busses (GPDB) from the IOP modules.

Each ADC also transmits to each IOM a Vmo/Mmo warning discrete.

The components interfaced with ADC ARINC 429 outputs are listed as follows:

For ADC 1:

ARINC 1	ARINC 2	ARINC 3	ARINC 4	ARINC 5
MFD1 PFD1	FGM1 SPM1 FCECU1 AHRS1 IOP1	FADEC1 Channel A FADEC2 Channel A	MFD2 PFD2 FGM2 SPM2 AHRS2	IOP2 ATC1 ATC2 CPC CIM

For ADC 2:

ARINC 1	ARINC 2	ARINC 3	ARINC 4	ARINC 5
MFD2 PFD2	FGM2 SPM2 AHRS2 FCECU2 IOP2	FADEC1, Channel B FADEC2, Channel B	MFD1 PFD1 FGM1 SPM1 AHRS2	IOP1 ATC1 ATC2 CPC CIM

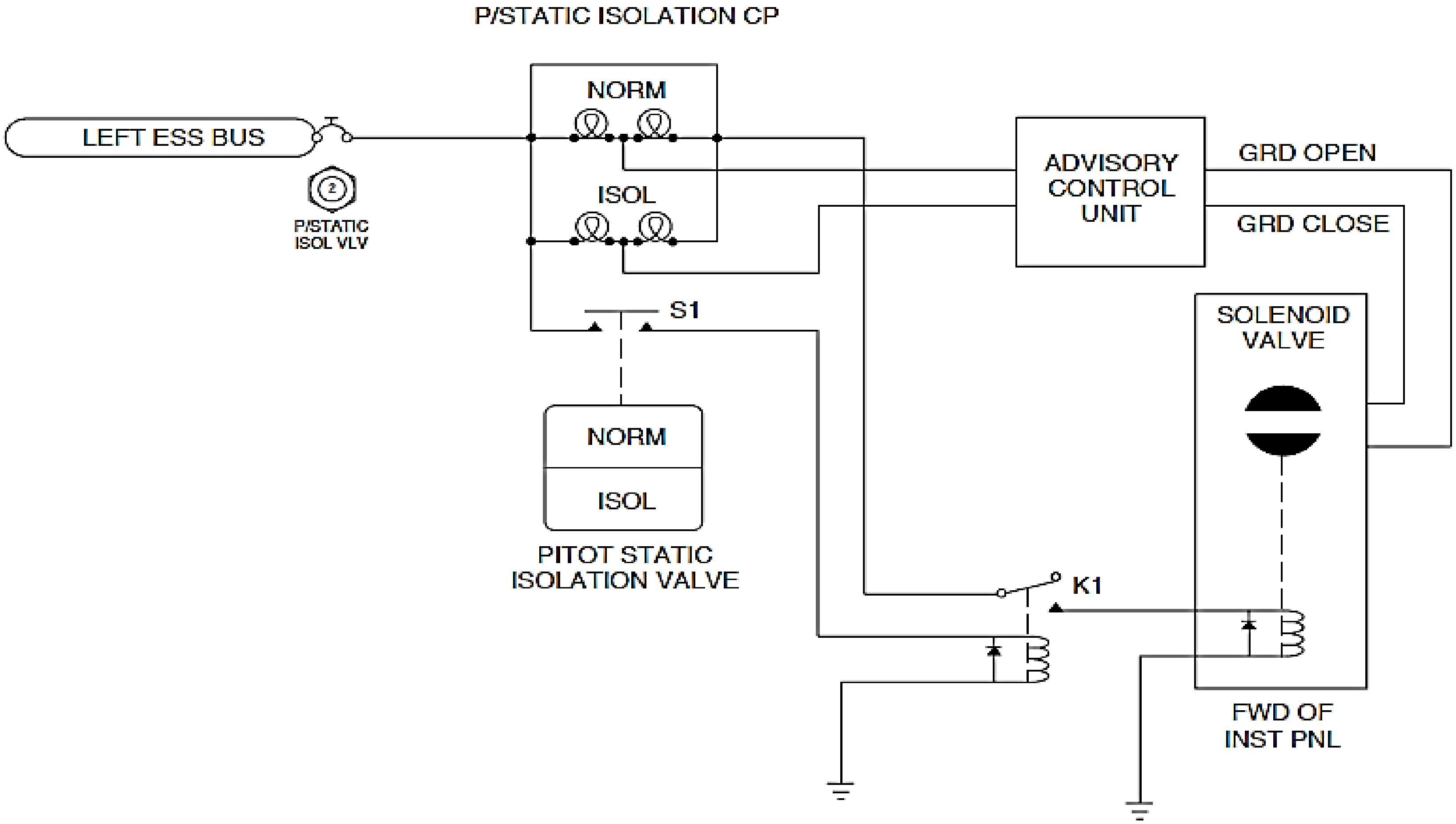


Figure 34- 39 SOLENOID SHUTOFF VALVE SIMPLIFIED SCHEMATIC

Figure 34- 39 SOLENOID SHUTOFF VALVE SIMPLIFIED SCHEMATIC

An electrically controlled solenoid shutoff valve is used to isolate ADC1 from the pitot−static probe No. 2 static source.

An annunciator switch located near the pilot in the flight compartment give manual control of the shutoff valve.

Electrical power is supplied from the 28 Vdc left essential bus through a flight compartment annunciator switch and relay to let the same electrical power source energize the shutoff valve.

When the NORM/ISOL annunciator switch is pushed, the 28 Vdc left essential bus energizes a relay to let the 28 Vdc left essential bus supply electrical power through relay contacts to the shutoff valve. The shutoff valve is stays closed while it is energized.

When the solenoid valve is in the open position, it supplies different signals through the Advisory Control Computer (ACU) to make the NORM lights in the annunciator switch come on and the ISOL lights go out.

If the solenoid valve is in the closed position, it supplies different signals through the Advisory Control Computer (ACU) to make the ISOL lights in the annunciator switch come on and the NORM lights go out.

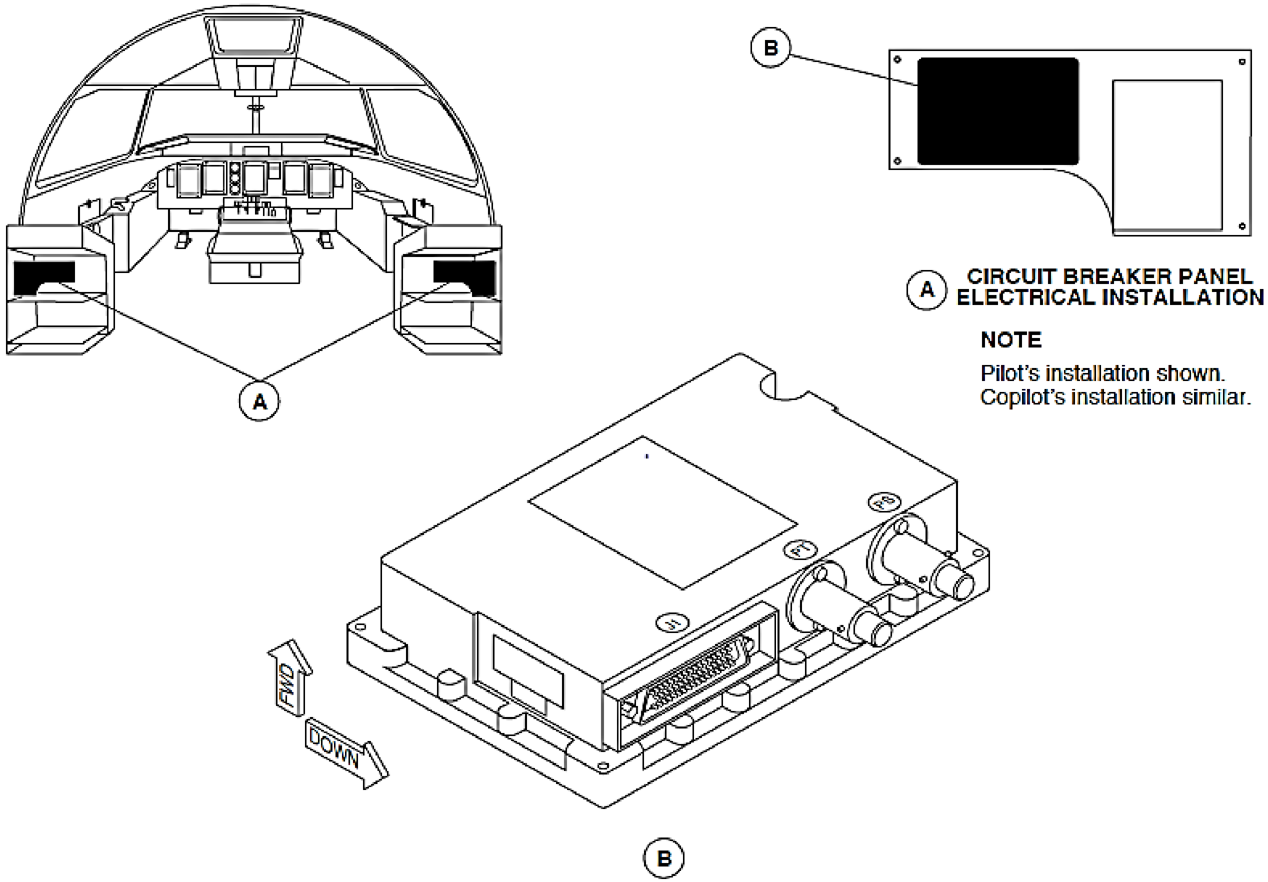


Figure 34- 40 AIR DATA UNIT LOCATOR

Computer, Air Data

Figure 34- 40 AIR DATA UNIT LOCATOR

The ADC measures static pressure (ps) and total pressure (pt) supplied from the primary pressure probes and compensates these pressures for probe position errors. The ADC also measures the temperature probe resistance to calculate Outside Air Temperature (OAT). From these primary parameters and with the acquired barometric setting information transmitted by the Index Control Panel (ICP), the ADC does the air data calculations.

The ADC is a Line Replaceable Unit (LRU) that has the parts that follow:

- An electrical board with two silicon pressure sensors P90,a CPU, analog and discrete inputs/outputs, ARINC 429 interface, power supply and electrical interface with a connector and EMC protection
- A mechanical case that secures the pressure ports

The Air Data Computer (ADC) weighs 6.12 lb. (2.78 kg). It has subassemblies mechanically packaged into a 2 in. (51 mm) high, 6 in. (152 mm) wide by 7.9 in. (201 mm) long case.

It has quick disconnect pitot and static pneumatic connectors. The two ADCs are installed in the flight compartment on the aft bulkhead using four mounting screws and have the pneumatic connectors oriented downwards to prevent entry of moisture into the units.

Figure 34- 41 AIR DATA COMPUTER INTERNAL BLOCK DIAGRAM

The ADC transmits the calculated air data parameters on 5 ARINC 429 low speed busses.

The validity of these parameters is dependent on the results of internal monitoring performed during power−up tests and continuous BIT. Interfaces with the Centralized Diagnostic System (CDS) lets any failure sensed by the ADC monitoring to be recorded during operations and lets control of a BIT in order to get ADC status.

A test mode enables confirmation of correct interface of ADC with the flight compartment displays.

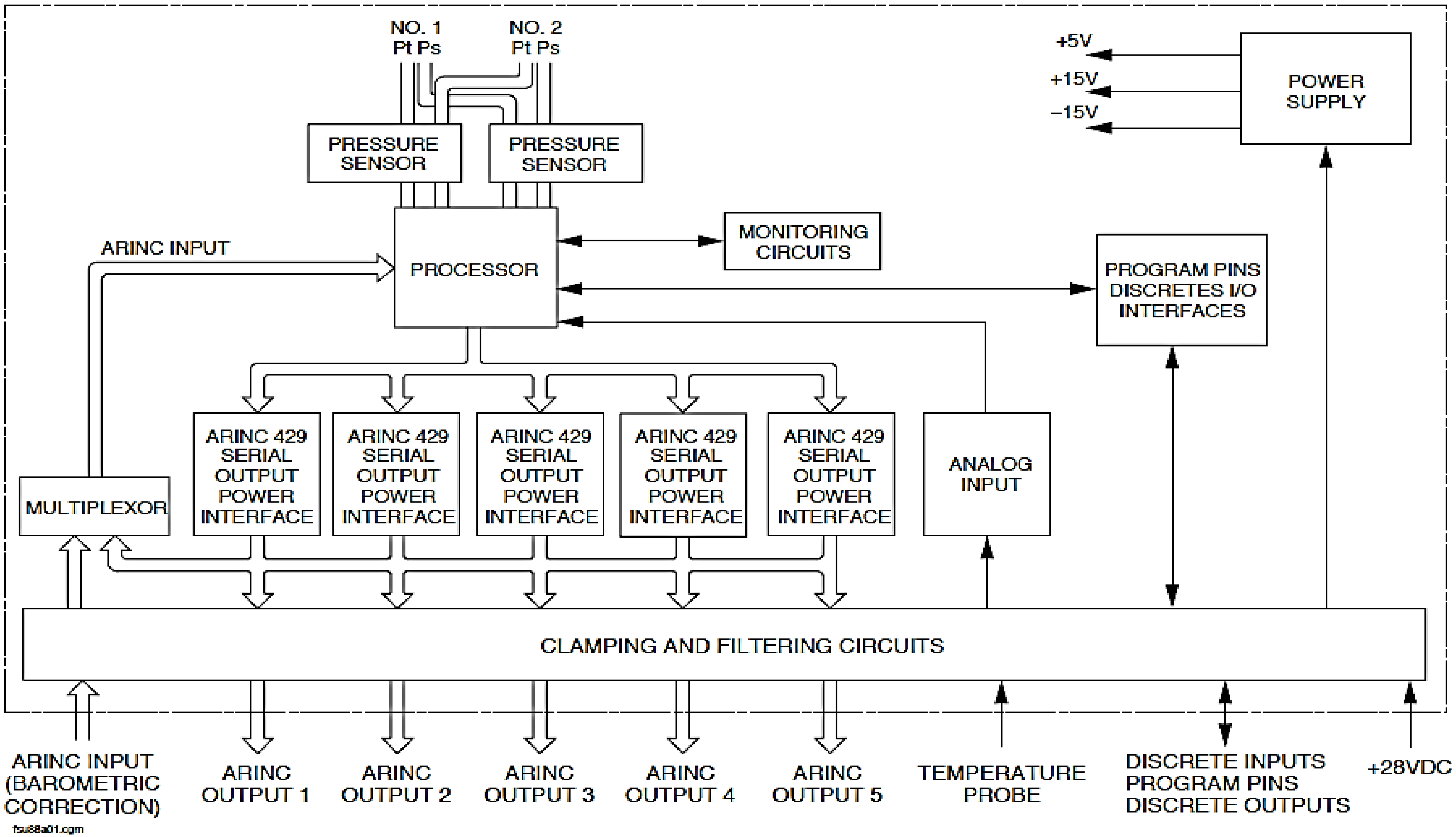


Figure 34- 41 AIR DATA COMPUTER INTERNAL BLOCK DIAGRAM

Probes, Pitot−Static

The two primary pitot−static probes are designed to accurately sense pitot and static pressures over the complete operating range of the aircraft.

An opening located at the nose of the probe senses pitot pressure. The shape of the opening is designed to reduce the sensitivity due to inclination of the tube axis to the effected varying airflow angles. This results in a total pressure measurement constant through the entire AOA and sideslip envelope. Static pressure ports are located on the side of each probe tube.

The primary probes have one total pressure connector and two static pressure connectors (for pneumatic connection with each ADC).

To prevent ice accumulation on pressure ports, probes have a single coaxial−type electrical heating element with a self−regulating power consumption feature.

Moisture accumulation in the tube is also eliminated by the use of drainholes for the total pressure circuit. The static pressure circuit is self−drained due to the locations of the static ports on the tube. The two primary heated Pitot−Static probes are installed symmetrically on the left and right sides of the fuselage.

The base plates have electrical and pressure connectors. Locating pins on the plate give an accurate angular orientation of the probe with the aircraft axis.

The primary probes are oriented 9 degrees downward in respect to the horizontal reference axis of the aircraft. The angle of the probe tube in respect with the mast gives a 5 degree angle in or out between the tube line and the tangential plane of the fuselage skin.

These probes are connected to aluminum tubing by flexible hoses. Two T−pipes, located roughly equal distance from the two primary probes, gives the pneumatic averaging of left and right static pressures delivered to each ADC. Tubing installation provides moisture drainage due to a tubing slope always greater than 3.5 degrees with drains holes installed at the lower point of each pipe. Six drains are installed.

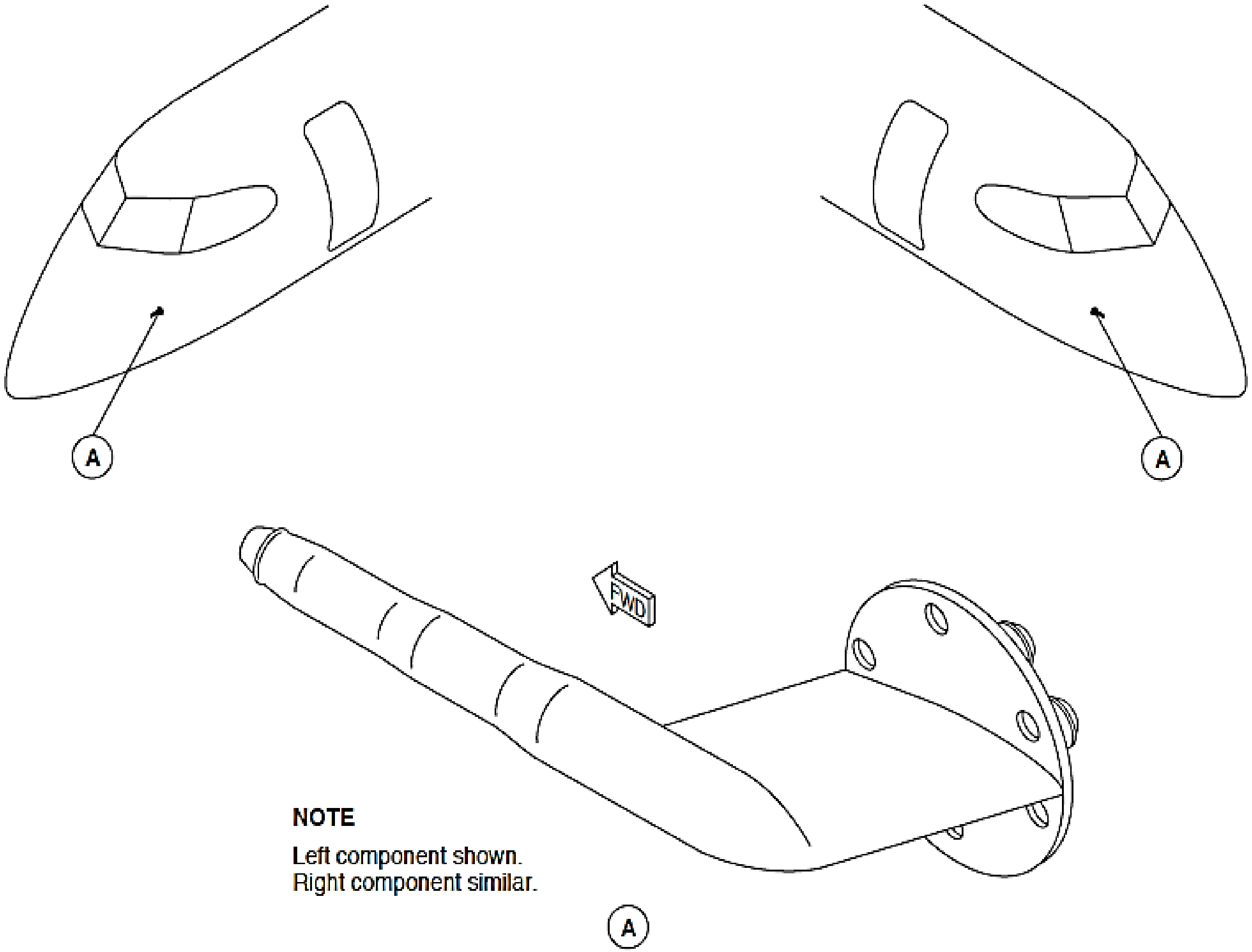


Figure 34- 42 PITOT−STATIC PROBES LOCATOR

Probe, Static Temperature

Figure 34- 43 STATIC TEMPERATURE PROBE LOCATOR

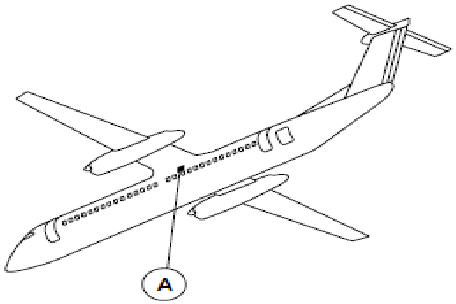
The static temperature probe measures the surrounding aircraft Indicated Outside Air Temperature (IOAT). It is a flush−attached electrical resistance transmitter with temperature resistance properties.

The IOAT measured by the two sensors is affected by airspeed, a recovery factor, (a value programmed into the ADC to change IOAT data to SAT) is stored in each ADC and is used to calculate from IOAT the Static Air Temperature (SAT) and the True Air Temperature (TAT).

The probe has two isolated platinum temperature sensing elements that have a 0 °C base resistance of 500 ohms, and a thermal coefficient of resistance of 0.00385 ohms/degree C.

The accuracy of the resistance over the –50 to +50 °C range is ± 1 ohm. This resistance error is equivalent to a temperature accuracy of 0.52 °C within this range.

The static temperature probe weighs 0.25 pounds and is attached flush to the fuselage below the left wing root by four screws.



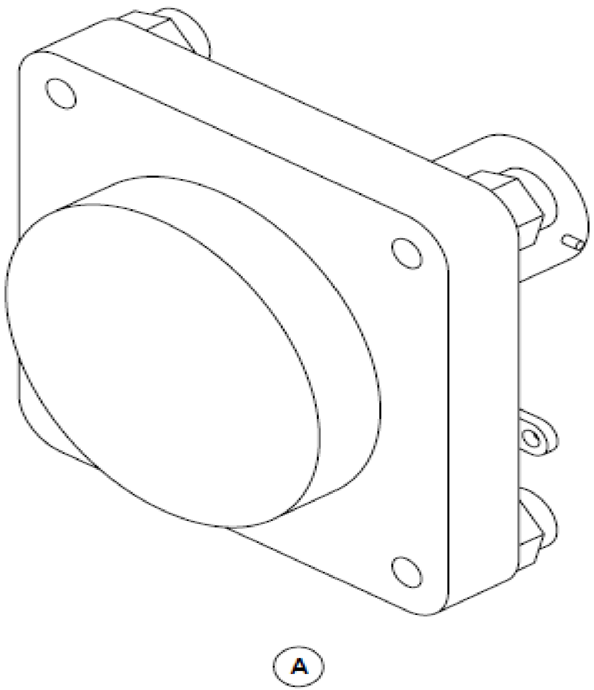


Figure 34- 43 STATIC TEMPERATURE PROBE LOCATOR

Valve, pitot−static isolation

Figure 34- 44 PITOT/STATIC ISOLATION VALVE

The pitot−static isolation valve provides physical separation between the two pitot−static detection systems. The valve is actuated with the NORM/ISOL annunciator switch.

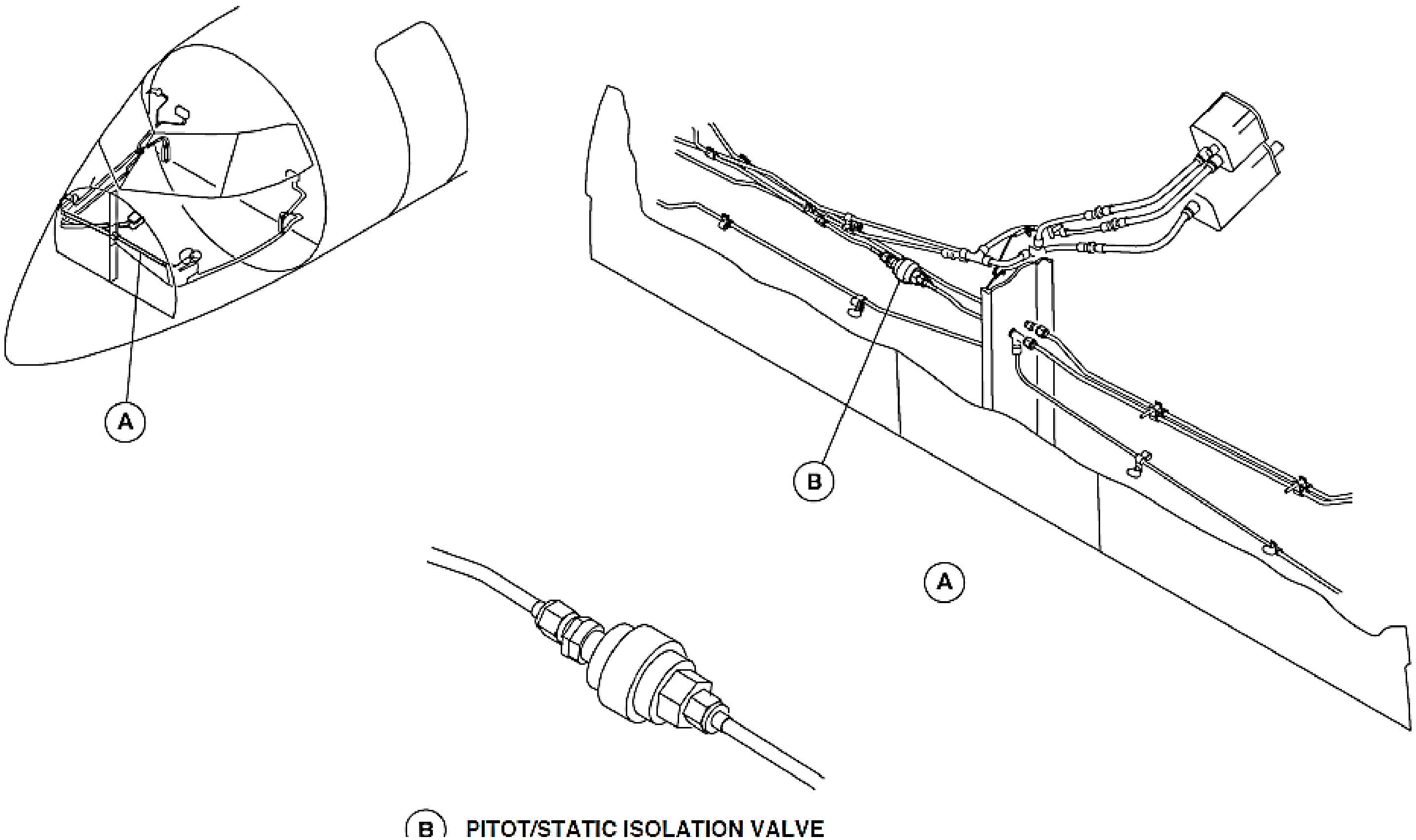


Figure 34- 44 PITOT/STATIC ISOLATION VALVE

Panel, pitot–static isolation valve

Figure 34- 45 PITOT–STATIC ISOLATION VALVE PANEL LOCATOR

The pitot–static isolation valve panel has a NORM/ISOL annunciator switch. It is pushed to energize the shutoff valve.

The NORM/ISOL annunciator switch is a guarded annunciator switch. A transparent switch cover prevents accidental selection of the switch.

The pitot–static isolation valve panel is attached to the bottom left part of the instrument panel with two mounting screws. A bonding wire connected to the chassis makes a ground continuity connection between the control panel and the aircraft structure.

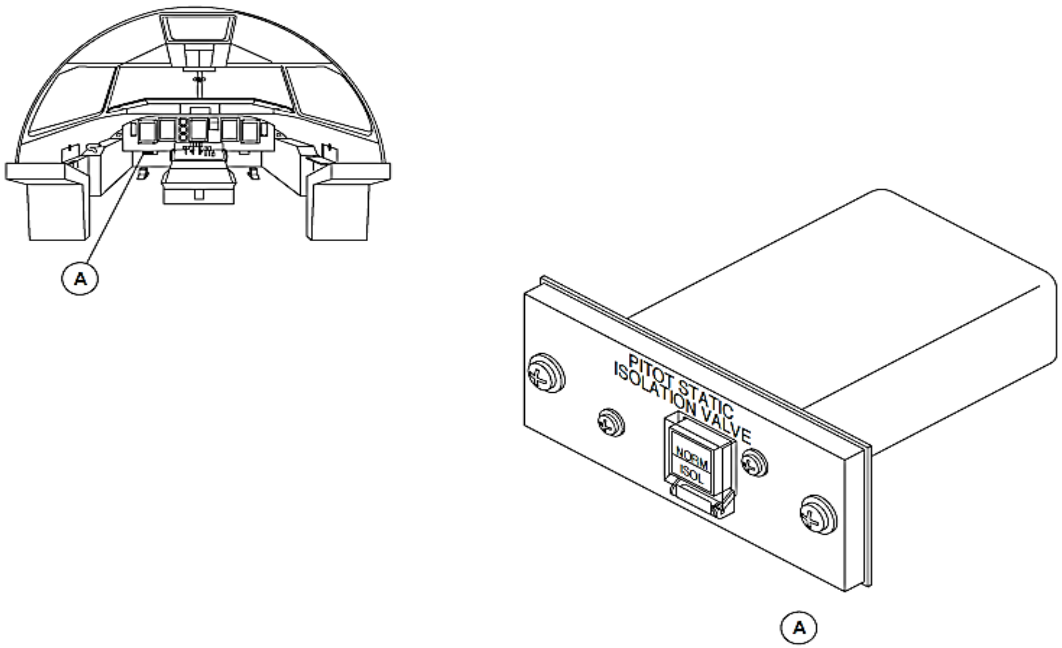


Figure 34- 45 PITOT–STATIC ISOLATION VALVE PANEL LOCATOR

Water accumulators, pitot/static system

Figure 34- 46 WATER ACCUMULATOR
Figure 34- 47 WATER ACCUMULATORS, PITOT/STATIC SYSTEM

For the aircraft with the SB84−34−59 or ModSum 4−113488 incorporated, there are eight pitot/static water accumulators installed, which includes six primary pitot/static water accumulators and two standby pitot/static water accumulators.

For the aircraft with the SB84−34−156 or ModSum 4−113815 incorporated, there are nine pitot/static water accumulators installed, which includes six primary pitot/static water accumulators, two standby pitot/static water accumulators and one additional standby pitot water accumulator.

Figure 34- 46 WATER ACCUMULATOR

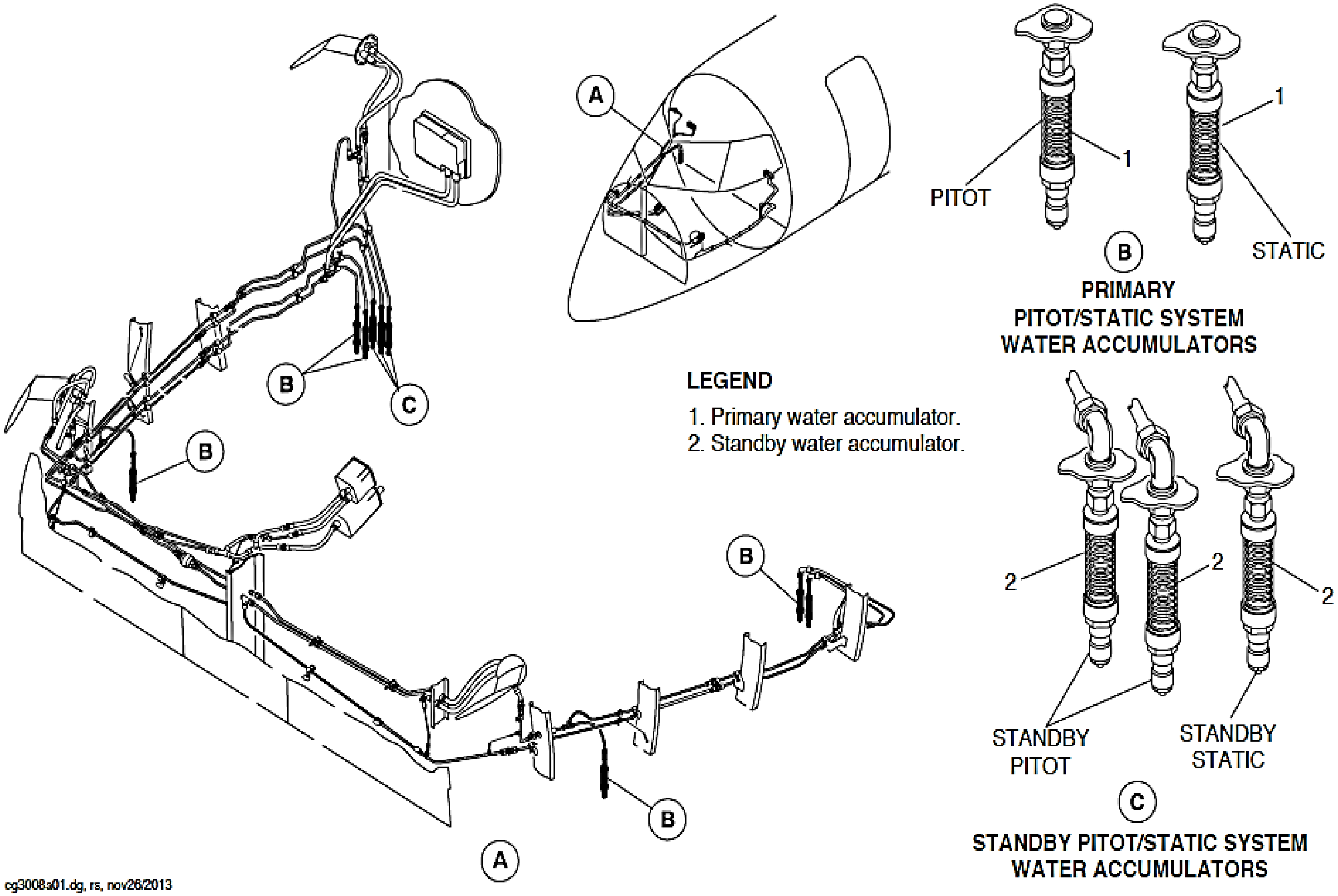


Figure 34- 47 WATER ACCUMULATORS, PITOT/STATIC SYSTEM

INTEGRATED STANDBY INSTRUMENT (ISI)

Introduction

The Integrated Standby Instrument (ISI) replaces the current mechanical standby instruments and performs their function within a single electronic unit. The ISI system provides the following information to the pilots:

- Attitude
- Altitude
- Airspeed

The ISI system operates independently and does not interface with any other systems.

General Description

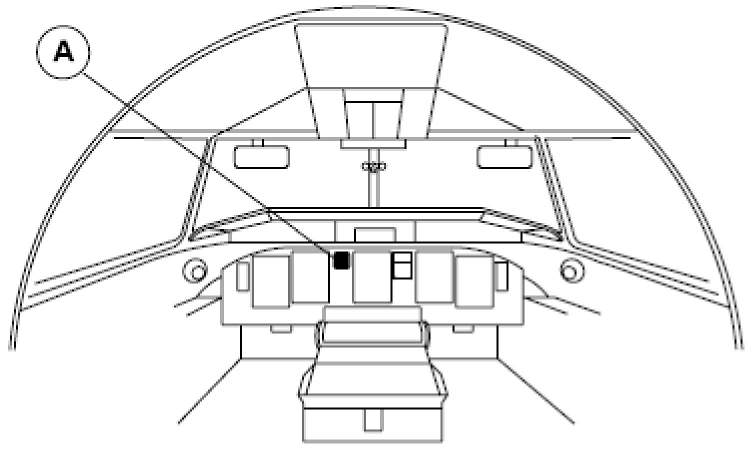
Figure 34- 48 INTEGRATED STANDBY INSTRUMENT (ISI)

The ISI is located on the instrument panel between Multi−Function Display 1 (MFD1) and the Engine Display (ED). It replaces the following mechanical standby instruments:

- Standby Attitude Direction Indicator
- Standby Altimeter
- Standby Airspeed Indicator

The ISI displays all three standby instruments on a single High Resolution Active Matrix Liquid Crystal Display (LCD) using Vertical Tape Symbology for the Indicated Airspeed (IAS) and the Altimeter. It also provides standby display for:

- Vmo (Maximum Operating Speed)
- Side−slip



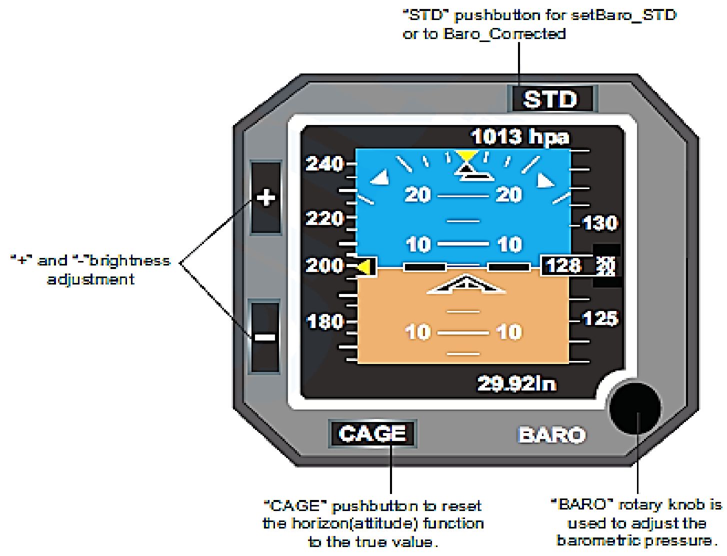


Figure 34- 48 INTEGRATED STANDBY INSTRUMENT (ISI)

Detailed Description

Figure 34- 49 INTEGRATED STANDBY INSTRUMENT (ISI) BLOCK DIAGRAM

As a standby instrument, the external requirements are limited to:

- Standby Pitot Probe
- Standby Static Probe
- Battery Bus
- Bezel Lighting

The ISI is connected to the standby pitot–static probe. The probe is in a location with minimal distorted airflow and is pointed directly into the airstream. It has an integral electrical heating element to prevent ice accumulation. The electrical power to the heater is controlled by the Timer Monitor Unit (TMU) of ice and rain protection system.

The ISI utilizes internal sensors and outside static air pressure to calculate altitude and airspeed. An Inertial Measurement Unit computes rotations and linear accelerations along the aircraft axes for the attitude function.

The aircraft right essential bus supplies 28 VDC electrical power through a 25 ampere circuit breaker to power the standby pitot–static probe. The circuit breaker is located on the 28 VDC right circuit breaker panel in position L7.

The aircraft left battery bus supplies 28 VDC electrical power through a 3 ampere circuit breaker to power the ISI. The circuit breaker is located on the 28 VDC left circuit breaker panel in position H1.

The bezel lighting (5 VDC) is embedded in the front face structure (pushbuttons) of the ISI and is controlled from the pilot's side console.

The Integrated Standby Instrument (ISI) operational features are located on the front face of the instrument and consist of:

- Control Panel Push Buttons
- Standby Attitude Display
- Standby Airspeed Display
- Standby Altitude Display

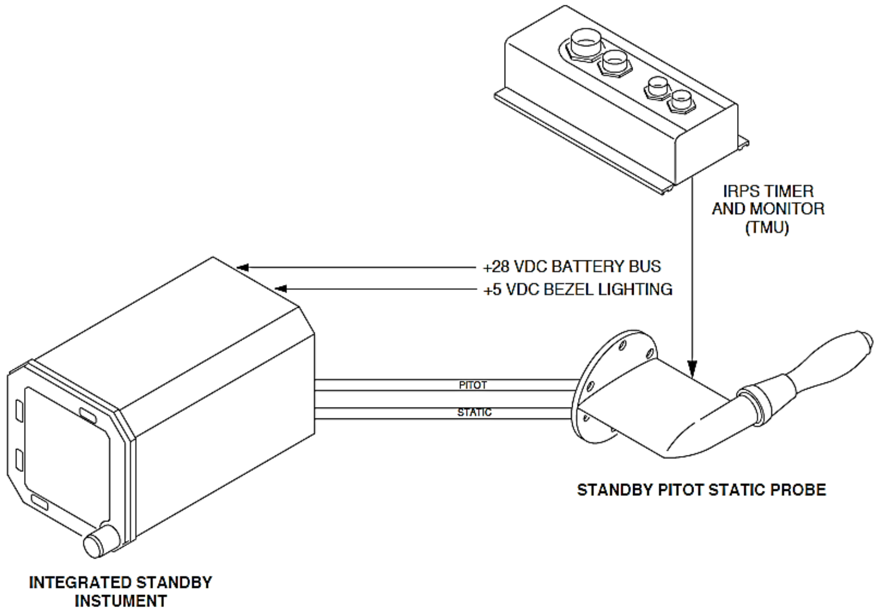


Figure 34- 49 INTEGRATED STANDBY INSTRUMENT (ISI) BLOCK DIAGRAM

Control Panel

Figure 34- 50 INTEGRATED STANDBY INSTRUMENT (ISI) CONTROL PANEL

The ISI front face is comprised of four illuminated pushbuttons and one rotary knob surrounding the LCD. The adjustments include:

- STD Pushbutton
- “+” Pushbutton
- “–” Pushbutton
- CAGE Pushbutton
- BARO Rotary Knob

The “STD” (Standard) pushbutton is used to reset barometric correction to the standard value.

The “+” pushbutton manually increases LCD brightness.

The “–” pushbutton manually decreases LCD brightness.

The “BARO” (Barometric) rotary knob enables selection of baro correction in ’mb’ (millibars) or ’in Hg’ (inches of mercury) and is used to control the barometric correction scale. A clockwise rotation increases the barometric value while a counterclockwise rotation decreases it.

The “CAGE” pushbutton resets the horizontal function to zero when depressed for more than two seconds. It will also cause a CAGE warning flag to appear in the top right of display. It is used to cancel errors induced in the attitude display following unusual aircraft maneuvering or an upset.

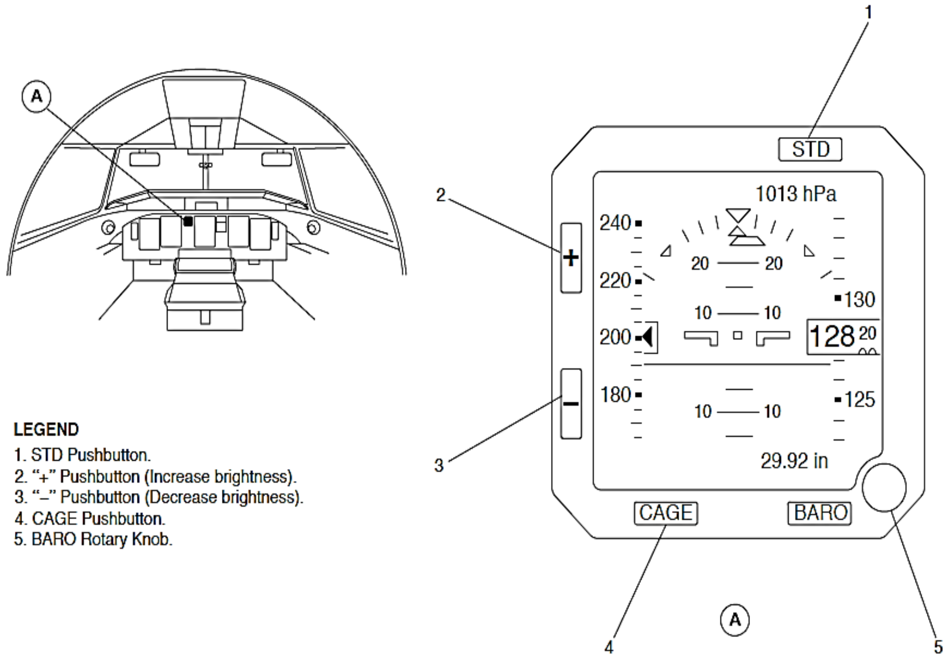


Figure 34- 50 INTEGRATED STANDBY INSTRUMENT (ISI) CONTROL PANEL

Standby Attitude Display

Figure 34- 51 INTEGRATED STANDBY INSTRUMENT (ISI) ATTITUDE DISPLAY

The standby attitude function supplies an alternate display of airplane attitude utilizing three internal quartz gyros sensors and two silicon accelerometers. It is powered by the battery bus.

The Standby Attitude Display shows the attitude information that follows:

- Airplane Symbol (white)
- Index Pointer and Roll Scale (white)
- Side−slip Indicator
- Pitch scale (white)
- Attitude Sphere and Horizon Line
- ATT Flag

The airplane symbol is a fixed airplane reference against the attitude sphere and shows the amount of airplane pitch and roll. An optional single cue airplane symbol replaces the standard airplane symbol.

The amount of roll is determined by an incremental roll scale (0°, 10°, 20°, 30°, 45° and 60°) and moves relative to a fixed index pointer.

The side−slip indicator shows lateral acceleration left or right to a maximum of 0.14g.

The pitch scale is shown on the attitude sphere and is incremented (0°, 5°, 10°, 15°, 20°, 25°, 30°, 40°, 60° and 90°). In the event of excessive pitch angles (greater than 40° pitch up or 30° pitch down), red chevrons will come into view and point to horizon.

The attitude sphere has a sky permanent sector in blue above a ground permanent sector in brown. A white horizon line divides these two sectors. The sphere moves to show pitch and roll.

An ’ATT’ (Attitude) flag indicates an invalid display due to failure of the attitude function as detected by internal monitors.

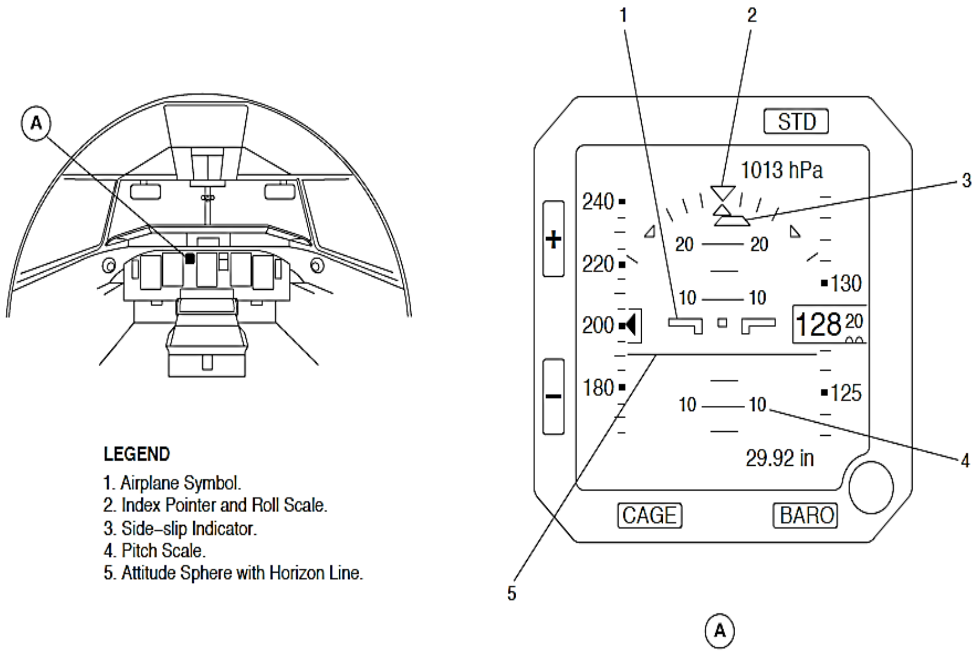


Figure 34- 51 INTEGRATED STANDBY INSTRUMENT (ISI) ATTITUDE DISPLAY

Standby Airspeed Display

Figure 34- 52 INTEGRATED STANDBY INSTRUMENT (ISI) AIRSPEED DISPLAY

The standby airspeed function supplies an alternate display of aircraft indicated airspeed in vertical tape format. It receives data from static and pitot pressure sensors utilizing static and pitot pressure supplied from the Standby Pitot Static Probe. Data is converted into digital format allowing the ISI to compute the corresponding conventional indicated airspeed. The STBY system is independent of the Air Data Computer (ADC) primary system.

The Standby Airspeed Display shows the airspeed information that follows:

- Vmo (red limit)
- Airspeed Pointer and Index (white)
- Airspeed Scale (white)
- IAS Failure Flag

The Vmo (Maximum Operating Speed) is displayed as a red tape in the left airspeed display area.

The airspeed pointer and index indicates airspeed. The airspeed ranges from 40 knots to 520 knots.

The airspeed scale is graduated every 5 knots between 40 knots and 250 knots. Above 250 knots, the scale is graduated every 10 knots up to 520 knots.

The IAS (Indicated Air Speed) failure flag replaces the airspeed tape and pointer in the event of an airspeed function failure as detected by internal monitors.

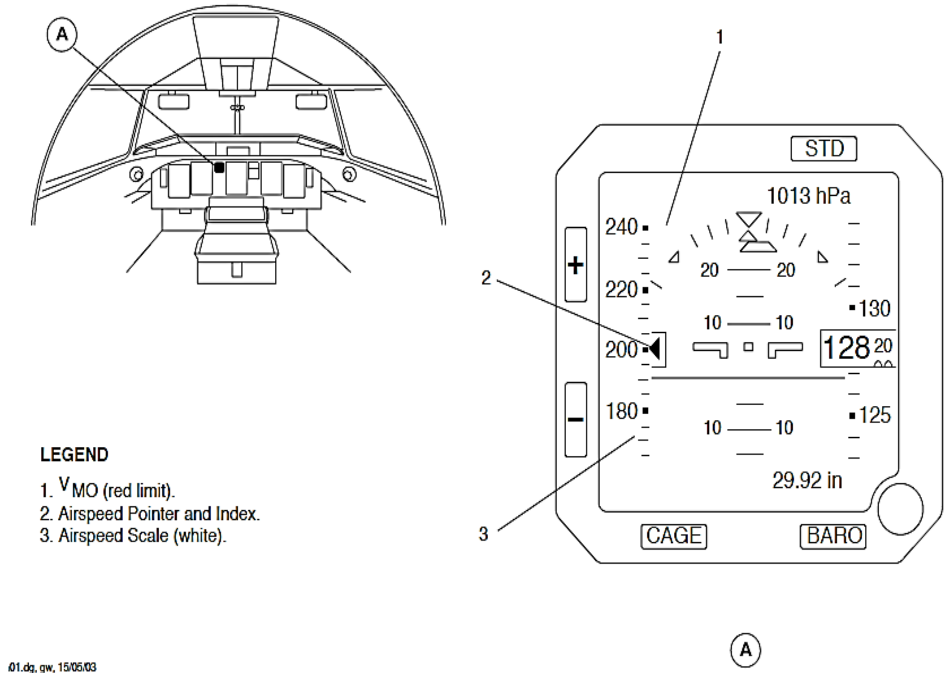


Figure 34- 52 INTEGRATED STANDBY INSTRUMENT (ISI) AIRSPEED DISPLAY

Standby Altitude Display

Figure 34- 53 INTEGRATED STANDBY INSTRUMENT (ISI) ALTITUDE DISPLAY

The standby barometric altimeter function supplies an alternate display of barometrical corrected altitude in vertical tape format. The ISI standby altimeter function receives data from the static pressure sensor utilizing static pressure supplied from the Standby Pitot static Probe. Data is converted into digital format allowing the ISI to compute the static pressure.

The Standby Altitude Display shows the altitude information that follows:

- Barometric Correction (hPa)
- Altitude Scale (white)
- Altitude Counter (white)
- Baro Correction (in Hg)
- ALT Flag
- SSEC Flag

The barometric correction displayed in the upper right quadrant of the LCD is set by the BARO rotary knob. The barometric correction ranges from 740 hPa to 1100 hPa.

The altitude scale is located on the right side of the display. It is comprised of a moving tape incremented every 100 feet and identified every 500 feet. The altitude tape range is −2000 feet to +50,000 feet.

The altitude counter appears as a digital readout right of center. It displays current altitude digitally in 20 foot increments. White hatch marks are displayed below 10,000 feet. An ’N’ is displayed in place of the ten thousands digit for negative altitudes.

The barometric correction displayed in the lower right quadrant of the LCD is set by the BARO rotary knob. The barometric correction ranges from 21.85 in Hg to 32.48 in Hg.

An optional ALT/METRIC switch on the overhead panel allows the barometric correction readout to be displayed in either meters or feet. An ’ALT’ (Altitude) flag replaces the altitude tape in event of failure of the altitude function as detected by internal monitors.

An ’SSEC’ (Static Source Error Correction) flag is displayed when SSEC corrections are no longer available for altitude computation.

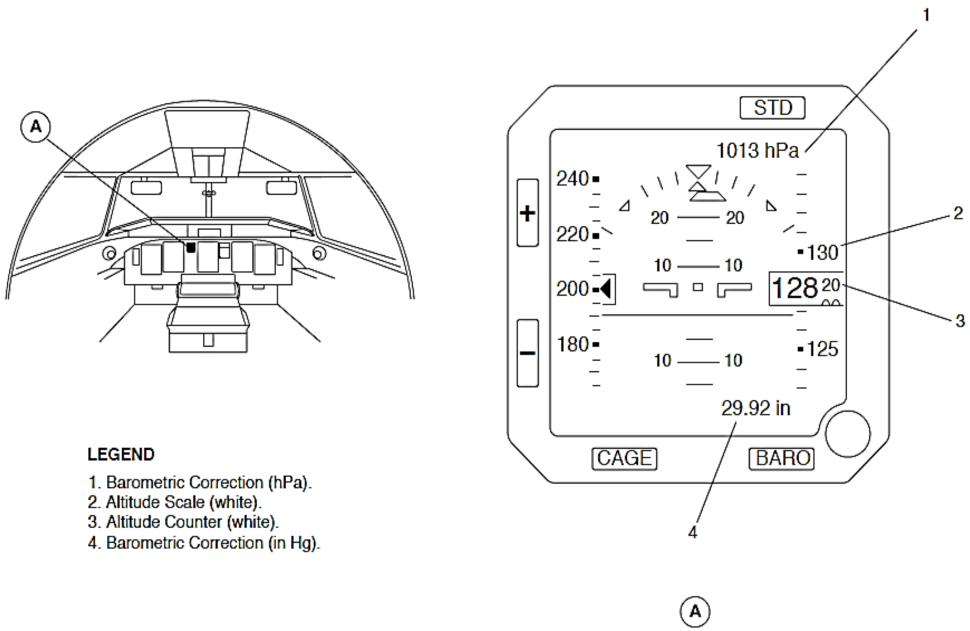


Figure 34- 53 INTEGRATED STANDBY INSTRUMENT (ISI) ALTITUDE DISPLAY

ATTITUDE AND DIRECTION

Introduction

The Attitude and Heading System (AHRS) supplies attitude and heading data to the Electronic Instrument System (EIS) indication and to other aircraft systems.

General Description

The attitude and heading parameters that are shown to the pilots and used for flight guidance are critical parameters. For this reason, the AHRS has three isolated attitude and heading data systems, two primary and one standby channel.

Figure 34- 54 ATTITUDE AND HEADING REFERENCE SYSTEM BLOCK DIAGRAM

Each primary channel has one Attitude and Heading Reference Unit (AHRU), flux valve, and AHRS control panel to the parameters that follow:

- Attitude
- Angular rates
- Lateral accelerations
- Vertical speed
- Magnetic heading.

The Attitude Heading Reference Units (AHRU1, AHRU2) function in the modes that follow:

- Power−On Self-Test (POST)
- Operational
- Maintenance
- Magnetic Compensation
- Magnetic Unit Calibration.

The standby attitude and heading instruments function independently of the primary channel Attitude and Heading Reference Units (AHRU1, AHR2) to show the data that follows:

- pitch and roll attitude
- Magnetic heading.

The Attitude and Heading Reference System (AHRS) has the components that follow:

- Units, Attitude and Heading Reference
- Valves, Flux
- Panel, AHRS Control.
- Module, Remote Memory
- Indicator, Standby Horizon
- Compass, Standby.

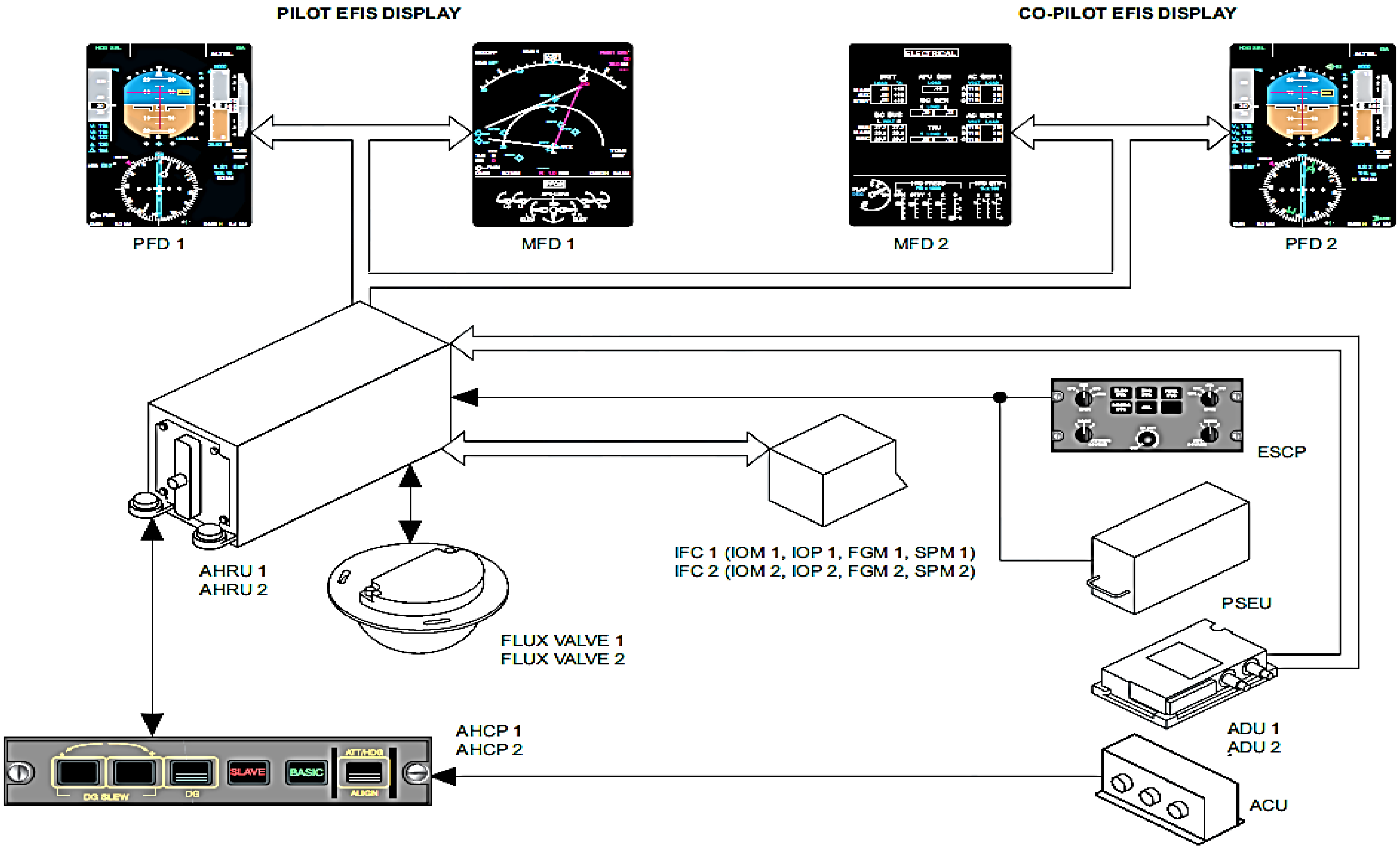


Figure 34- 54 ATTITUDE AND HEADING REFERENCE SYSTEM BLOCK DIAGRAM

Attitude and Heading Reference Units

Figure 34- 55 ATTITUDE AND HEADING REFERENCE UNITS LOCATOR

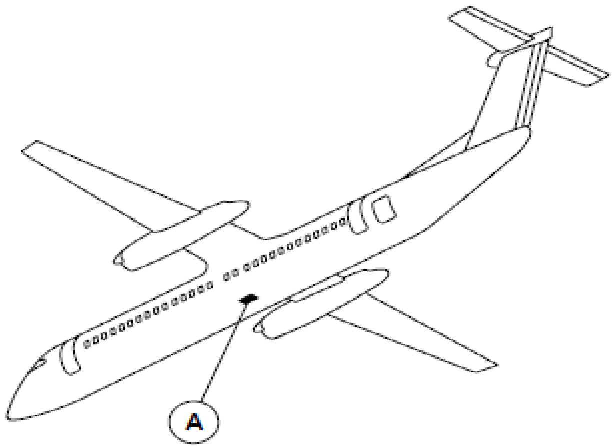
The front panel has the aircraft interface connector and the HIRF and lightning protection filter. It gives an interface with the Input/Output and power supply modules.

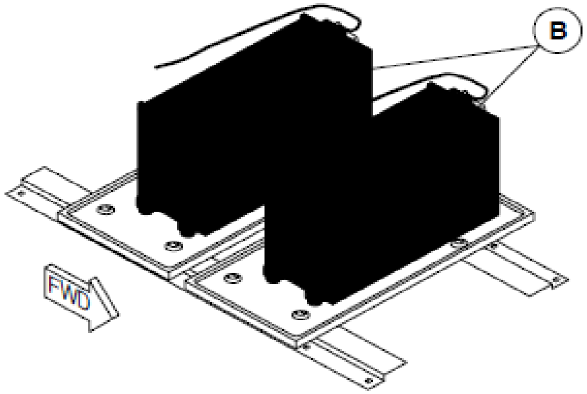
The AHRU weighs 7.7 lb. (3.49 kg). It is 4.2 in. (106.7 mm) wide, 4.5 in. (114 mm) high and 11.4 in. (290 mm) long.

The Attitude Heading Reference Units (AHRU1, AHRU2) are attached to a shock−mount with four screws.

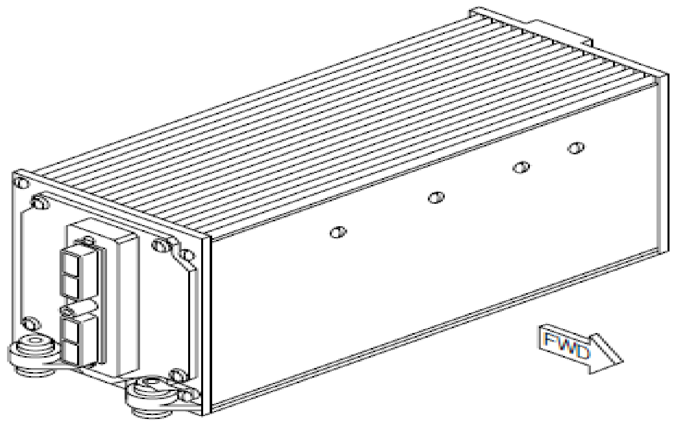
The two AHRUs have shock mounts that attach to the wing plane below the floor within the Center of Gravity (CG) limits. The AHRU is installed in a mounting tray in a lateral orientation.

Shims are installed below the AHRU mounting trays to give the correct physical alignment with aircraft attitude. The seat rails are the longitudinal and lateral references.





A AHRS INSTALLATION



B

Figure 34- 55 ATTITUDE AND HEADING REFERENCE UNITS LOCATOR

Flux Valves

Figure 34- 56 FLUX VALVES LOCATOR

The flux valve is a metallic, non−magnetic, half−spherical component. It weighs 0.9 lb. (0.41 kg).

Flux valves (MDU 1, MDU 2) are installed in the wing tips to minimize magnetic disturbances. The magnetic field at the wing tip is the most stable.

All flux valves (MDU 1, MDU 2) mounting hardware and fasteners located within an 18−inch radius are made of low magnetic permeability material.

The reference axes of the magnetic detector align with the reference axes of the aircraft.

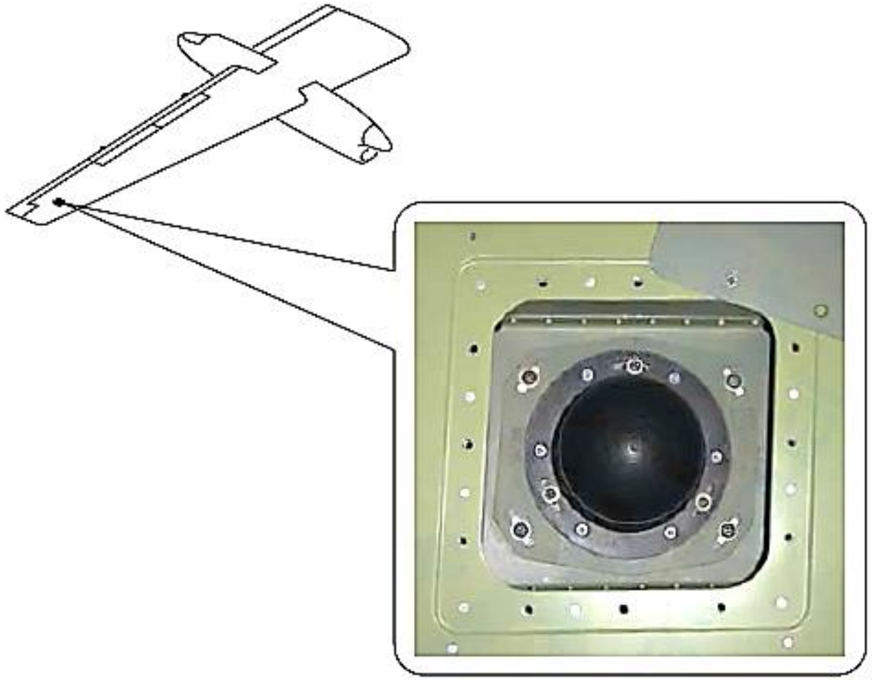


Figure 34- 56 FLUX VALVES LOCATOR

AHRS Control Panel

Figure 34- 57 ATTITUDE AND HEADING REFERENCE UNIT CONTROL PANELS LOCATOR

The AHRS Control Panel (AHCP) weighs 0.5 lb. (0.23 kg). The AHRS Control Panel (AHCP) is attached to the center−console with four DZUS fasteners.

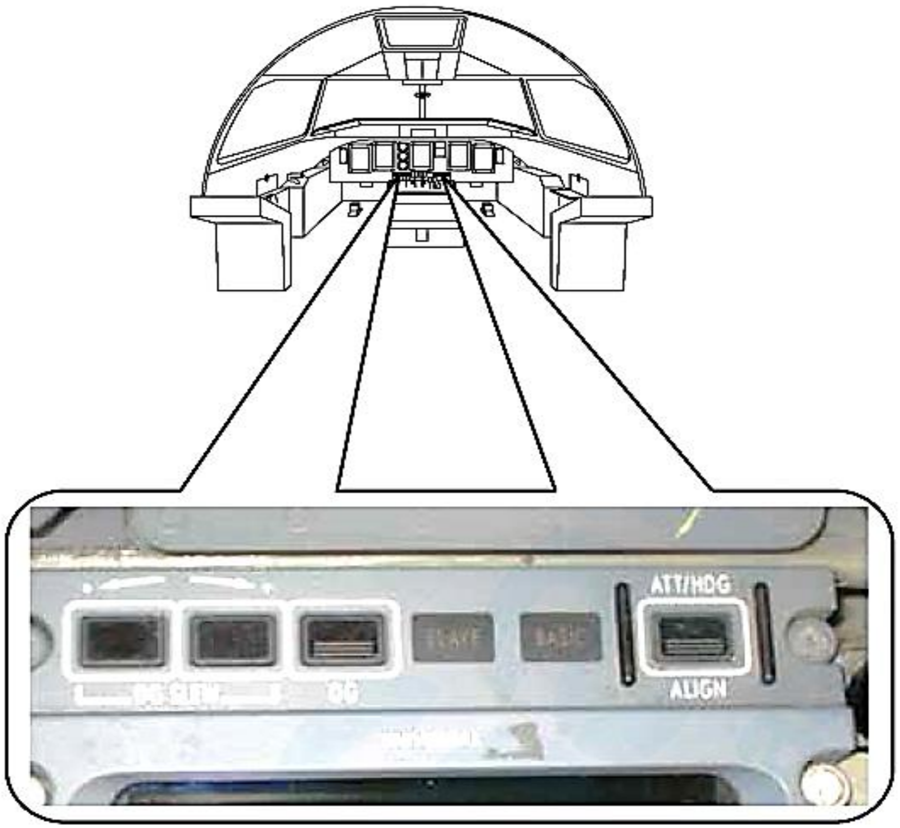


Figure 34- 57 ATTITUDE AND HEADING REFERENCE UNIT CONTROL PANELS LOCATOR

Remote Memory Module

Figure 34- 58 REMOTE MEMORY MODULES LOCATOR

The Remote Memory Modules (RMMs) connect to their related AHRUs. Two screws secure it in place. The RMMs physically attach to the aircraft by a lanyard. This makes sure that it only connects to its related Attitude Heading Reference Unit (AHRU1, AHRU2). The RMM stays in the aircraft when the Attitude Heading Reference Units (AHRU 1, AHRU 2) are removed for maintenance.

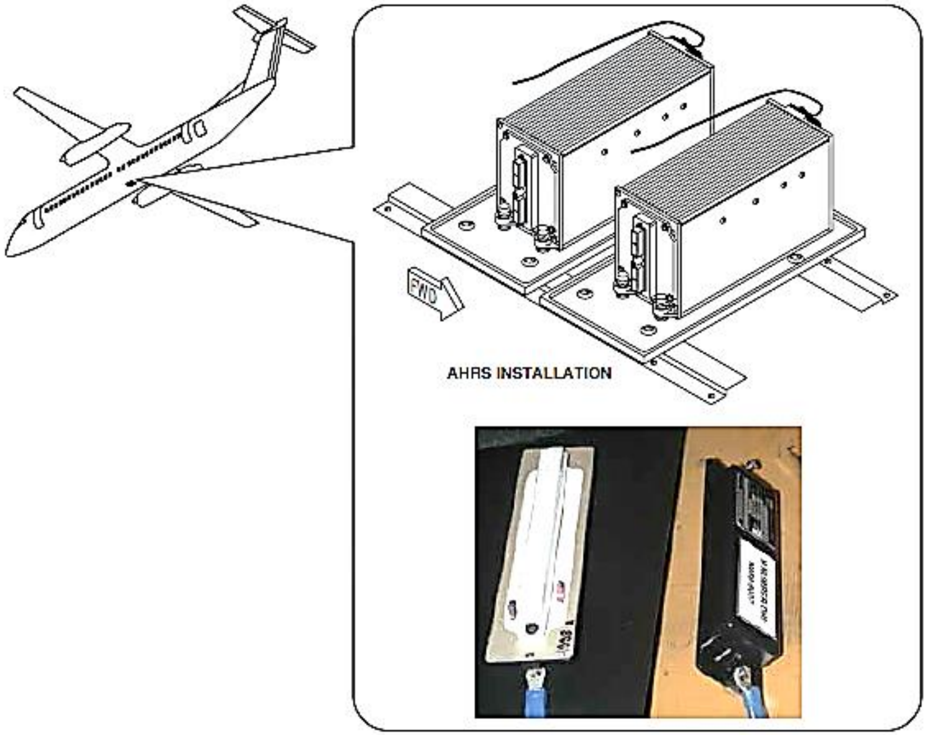


Figure 34- 58 REMOTE MEMORY MODULES LOCATOR

Standby Horizon Indicator

Figure 34- 59 STANDBY HORIZON INDICATOR LOCATOR

The standby horizon indicator is located on the instrument panel between Multi−Functional Display 1 (MFD 1) and the Engine Display (ED) above the standby airspeed indicator in view of the two pilots. The standby horizon indication is mechanically adapted to the 19.5 degrees of instrument panel tilt.

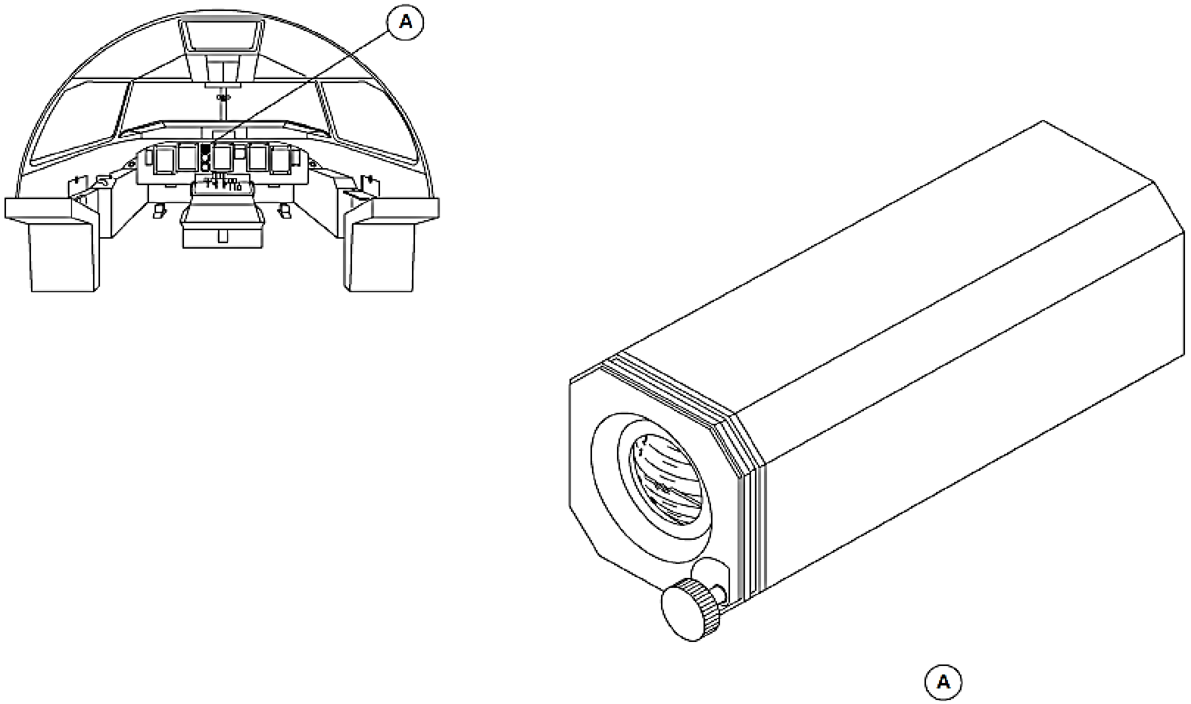


Figure 34- 59 STANDBY HORIZON INDICATOR LOCATOR

Standby Compass

Figure 34- 60 STANDBY COMPASS LOCATOR

The standby compass is installed with non−magnetic screws to the center structural support of the windshield directly below the Caution and Warning Panel (CWP).

Figure 34- 60 STANDBY COMPASS LOCATOR

ATTITUDE AND HEADING REFERENCE SYSTEM (AHRS)

Introduction

The Attitude and Heading Reference System supplies primary attitude and heading information to the systems that follow:

- Electronic Flight Instrumentation System (EFIS)
- Auto Flight Control System (AFCS)
- Other aircraft systems.

General Description

Figure 34- 61 ATTITUDE AND HEADING REFERENCE SYSTEM BLOCK DIAGRAM
Figure 34- 62 ATTITUDE AND HEADING REFERENCE SYSTEM BLOCK DIAGRAM, POWER

The two primary channel Attitude and Heading Reference Systems (AHRS 1, AHRS 2) channels supply the attitude and heading information that follows:

- Attitude
- Angular rates
- Lateral accelerations
- Vertical speed
- Magnetic heading.

The Attitude Heading Reference Units (AHRU 1, AHRU 2) function in the modes that follow:

- Power−On Self-Test (POST)
- Operational
- Maintenance
- Magnetic compensation
- Flux Valve calibration.

Each Attitude and Heading Reference System (AHRS 1, AHRS 2) has the components that follow:

- Units, Attitude and Heading Reference
- Valves, Flux
- Panel, AHRS Control.
- Module, Remote Memory.

Each Attitude and Heading Reference System (AHRS 1, AHRS 2) receives digital air data parameters from the two Air Data Units (ADU 1, ADU 2) and discrete inputs from the following systems:

- ESID Control Panel (ESCP 1, ESCP 2)
- Proximity Sensor Electronics Unit (PSEU)
- Central Diagnostic System (CDS).

The Attitude and Heading Reference Unit (AHRU) supplies specific attitude and heading parameters directly to the systems that follow:

- Auto Flight Control System (AFCS)
- Stall Protection System (SPS)
- Electronic Flight Instrumentation System (EIS).

The AHRU also supplies specific attitude and heading data to the systems that follow through the two Integrated Flight Cabinets (IFC 1, IFC 2):

- Flight Data Recorder (FDR)
- Flight Data Processing System (FDPS)

- Weather Radar (WXR)
- Ground Proximity Warning System (GPWS)
- Traffic Collision Avoidance System (TCAS)
- Central Diagnostic System (CDS).

The General Purpose Data Busses (GPDB) are used to supply attitude and heading information to the systems that follow:

- Flight Control System (FCS)
- Flight Management System (FMS).

A connection to the Portable Maintenance Access Terminal (PMAT) is used to download failure data.

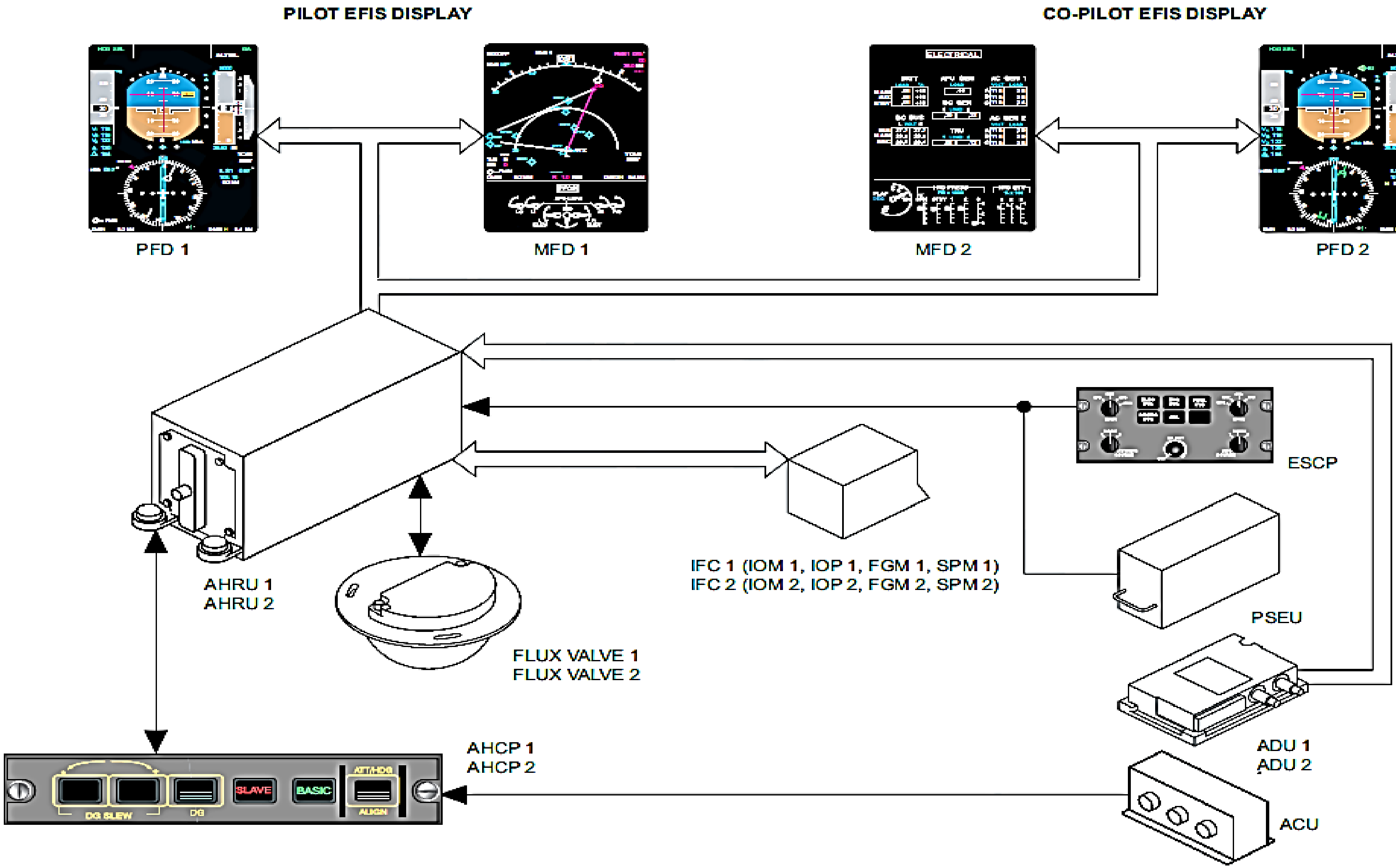


Figure 34- 61 ATTITUDE AND HEADING REFERENCE SYSTEM BLOCK DIAGRAM

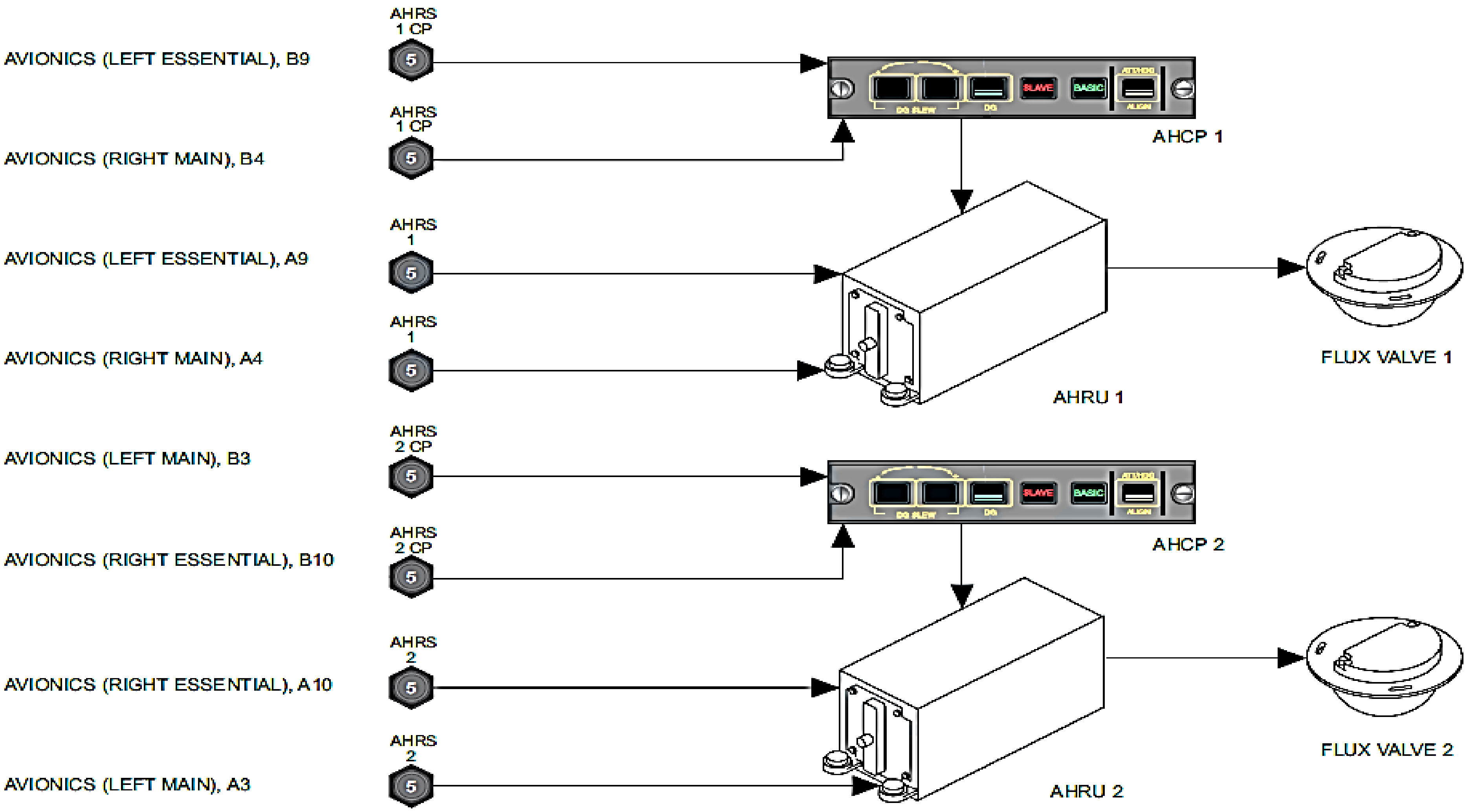


Figure 34- 62 ATTITUDE AND HEADING REFERENCE SYSTEM BLOCK DIAGRAM, POWER

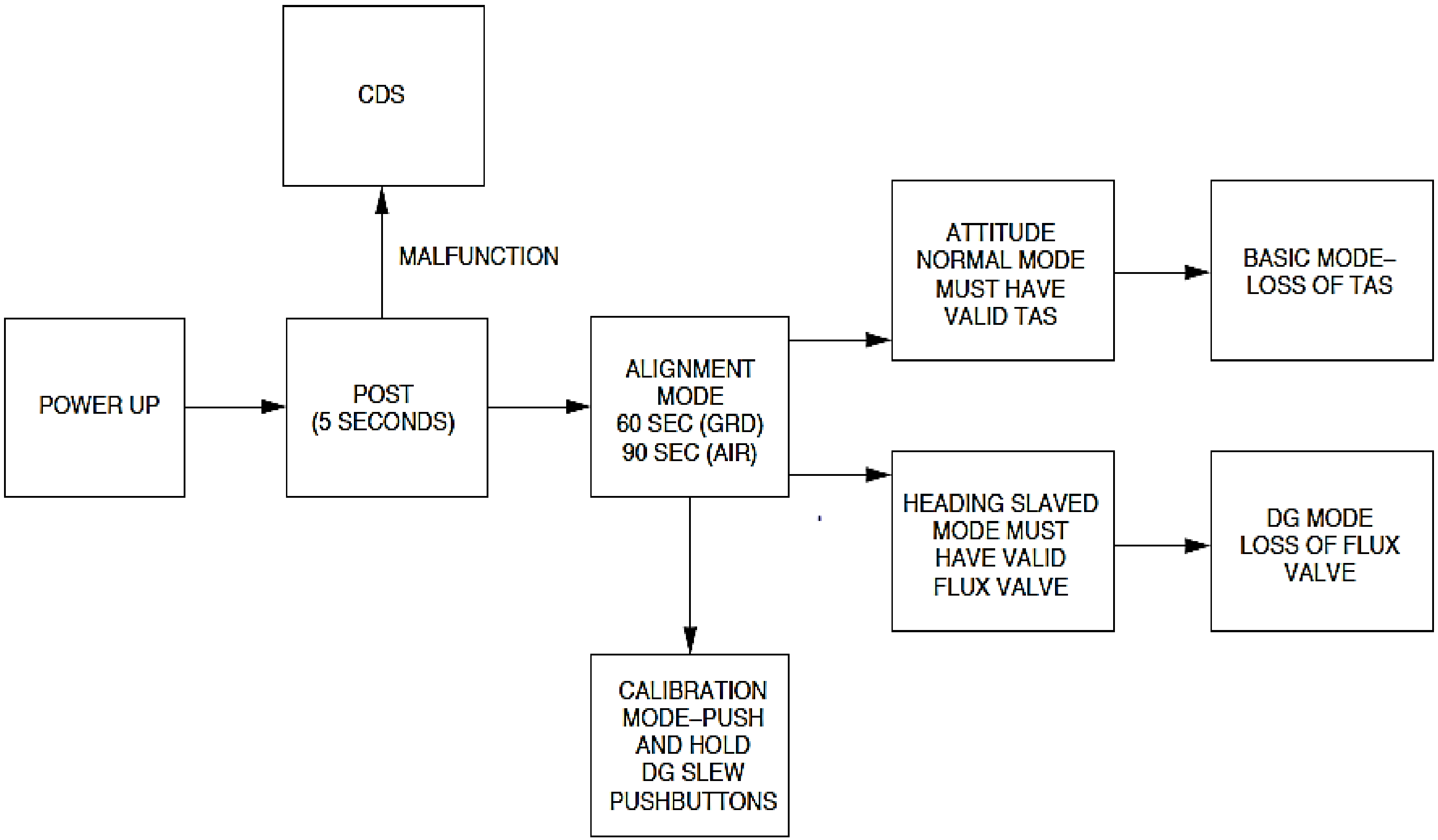


Figure 34- 63 AHRS MODES OF OPERATION

Detailed Description

Figure 34- 63 AHRS MODES OF OPERATION

The Attitude Heading Reference Units (AHRU 1, AHRU 2) function in the modes that follow:

- Power−On Self-Test (POST)
- Operational
- Maintenance
- Magnetic compensation
- Flux Valve calibration.

Power−On Self-Test (POST): The Power−On Self-Test (POST) mode automatically starts and continues for 5 seconds after power−up, during which time the AHRUs do the self−tests that follow:

- Memory check sum
- Test of peripheral circuits
- Test of I/O ports
- Indication of test in progress and successful completion.

During the self−test, the output data from AHRU 1, AHRU 2 cannot be used.

If the self−test senses a malfunction, it is sent to the Central Diagnostic System (CDS) for later analysis during maintenance functions.

Operational Mode: The operational mode has the modes that follow:

- Alignment
- Normal
- Basic
- Directional Gyro.

Alignment Mode: The alignment mode automatically starts at the end of the Power−On Self-Test (POST) mode if no malfunctions are found.

AHRU 1 and AHRU 2 calculate the parameters during the alignment mode that follow:

- Attitude and Vertical speed using Air Data Unit (ADU 1, ADU 2) altitude reference
- Heading using flux valve measurements
- Attitude using accelerations representative of gravity.

A re−alignment mode may be set using the AHRU Control Panels (AHCP 1, AHCP 2) when the aircraft is on the ground or in flight. The aircraft must be in a straight and level attitude when a re−alignment mode selection is made in flight.

AHRU 1, AHRU 2 parameters are available 20 seconds after the re−alignment mode starts.

When the re−alignment mode is completed, the AHRU operates in the primary modes that follow:

- Normal mode for attitude
- Slaved mode for heading.

Normal Mode: The AHRU will start the normal mode automatically following the POST, if the air data parameters are available that follow:

- True Air Speed (TAS)
- Barometric altitude (Zb).

Each Attitude and Heading Reference Unit (AHRU) receives air data parameters from both Air Data Units (ADU 1, ADU 2) through two data busses.

Normally, the AHRU uses its related ADU's parameters. If the related parameter is not correct, a manual ADU reversion is initiated. In this case, the other ADU is used to supply True Air Speed (TAS) and barometric altitude (Zb) parameters.

The Attitude Heading Reference Unit (AHRU) monitors its on side ADU input. If it fails completely, the AHRU automatically reverts to use the other ADU.

True Air Speed (TAS) from the ADU is used to compensate for acceleration induced attitude error. Barometric altitude is used to calculate vertical speed data for long term reference. In the short term, acceleration measurements provide vertical speed variations.

In the normal mode, if the TAS parameter is not available, the basic mode is automatically started and the AHRU functions similarly to a Vertical Gyro (VG). A green light on the AHRS Control Panel will come on to indicate the Basic mode.

Slaved Mode: When functioning in the slaved mode for heading, the AHRU slaves its inertial heading with analog magnetic heading information calculated from analog signals from the flux valve. It uses data from the MDU to calculate the magnetic heading and compensate for heading error caused by directional gyro drift.

The AHRU monitors the MDU. It causes the EIS to show a heading failure flag when the magnetic heading input from the MDU malfunctions. The apparent gravity during dynamic flight phases affects the MDU measurements. Due to the pendulous nature of the MDU, the AHRU internal logic automatically limits heading performance during accelerations.

When a heading error occurs while in the slave mode, the error may be removed by a fast selection of the DG mode and then back to the SLAVED mode. The DG push button switch on the AHRS Control Panel is pushed to start the DG mode and then is pushed again to return the system to the heading mode.

A fast selection and de−selection of the DG mode immediately causes the system to reinitialize to the MDU heading and has no effect on attitude.

In the slaved mode, if an MDU malfunctions, the PFD will show a heading failure flag and a red SLAVE annunciation on the AHRS Control Panel will come on. When the AHRU is set to DG mode, the MDU data is removed from the heading calculation and a DG heading indication is shown on the PFD. The SLAVE annunciation remains on.

The AHRU is set to the DG mode when the flux valve malfunctions or degrades when flying close to the magnetic poles.

When the DG mode is set, the Attitude and Heading Reference Units (AHRU 1, AHRU 2) do not receive magnetic data from the flux valves.

The heading is manually corrected by a plus or minus selection using the Attitude and Heading Control Panels (AHCP 1, AHCP 2). Two rates of correction speed are available. The low speed functions at 1 degree per second, and fast speed at 10 degrees per second.

Maintenance Mode: The Built In Test Equipment (BITE) uses the Central Diagnostic System (CDS) to give the condition of the component. It saves faults in a Non-Volatile Memory (NVM) for reporting to line and shop maintenance.

The maintenance mode is selected from one or the other ARCDU during maintenance operations. The AHRU does a test when it receives a maintenance discrete command through the IOM 1 while the air speed is less than 50 knots (93 km/h) and the aircraft is on the ground.

The Central Diagnostic System (CDS) sends a command directly to each primary Attitude Heading Reference Unit (AHRU 1, AHRU 2) to do a Power−On−Self−test (POST) that completes in less than 10 seconds. All sensed failures are sent in real time to the Central Diagnostic System (CDS).

The Built In Test Equipment (BITE) modes monitor the condition of the component as follows:

- Power−On Self-Test (POST)
- Continuous Monitoring.

Power−On Self-Test (POST): The Power−On Self-Test (POST) checks the condition of the component at power−up or after a long power interruption, more than 200 milliseconds.

Continuous Monitoring: The continuous monitoring checks the status of the component in flight. It records faults in a Non-Volatile Memory (NVM) for later troubleshooting using the Central Diagnostic System (CDS).

When powered, the Attitude Heading Reference Unit (AHRU 1, AHRU 2) does the different types of monitoring that follow:

- Processors and memories (program memory, calibration sensor memory, RAM memory)
- Power supply monitoring
- Sensor, FOG and accelerometer, failure monitoring and measurement range monitoring
- Internal temperature monitoring
- Flux−valve measurement monitoring, magnetic field module
- ARINC inputs and outputs.

Magnetic Compensation Mode: The magnetic compensation mode functions on the ground only. It senses magnetic heading error induced by the flux valve and aircraft heading.

The Attitude and Heading Reference Unit (AHRU) compares the magnetic heading field measurements supplied by its related flux valve and the magnetic reference values. The magnetic reference value is supplied to AHRU 1 and AHRU 2 through the Portable Maintenance Access Terminal (PMAT).

Flux Valve Calibration Mode: The flux valve calibration mode is used to set and store magnetic heading corrections.

The conditions that follow start the flux valve calibration mode:

- The aircraft is on the ground
- The ALIGN and DG push buttons switches on the AHRS Control Panel (AHCP) are pushed at the same time for three seconds.

NOTE

If the calibration mode is set, a Weight−Off−Wheels (WOFW) state causes the flux valve calibration mode to automatically stop.

The red SLAVE annunciator light on the AHRS Control Panel (AHCP) flashes throughout the calibration mode. In addition, the Electronic Horizontal Situation Indicator (EHSI) area of the Primary Flight Display (PFD) shows an amber HDG CALIBRATION message.

Figure 34- 64 ESCP, EFIS AHRS SOURCE REVERSION

The EFIS ATT/HDG SOURCE reversion selector is set to the NORM position on the ESID Control Panel (ESCP) to show data from their related Attitude and Heading Reference Units (AHRU 1, AHRU 2) on the Primary Flight Displays (PFD 1, PFD 2).

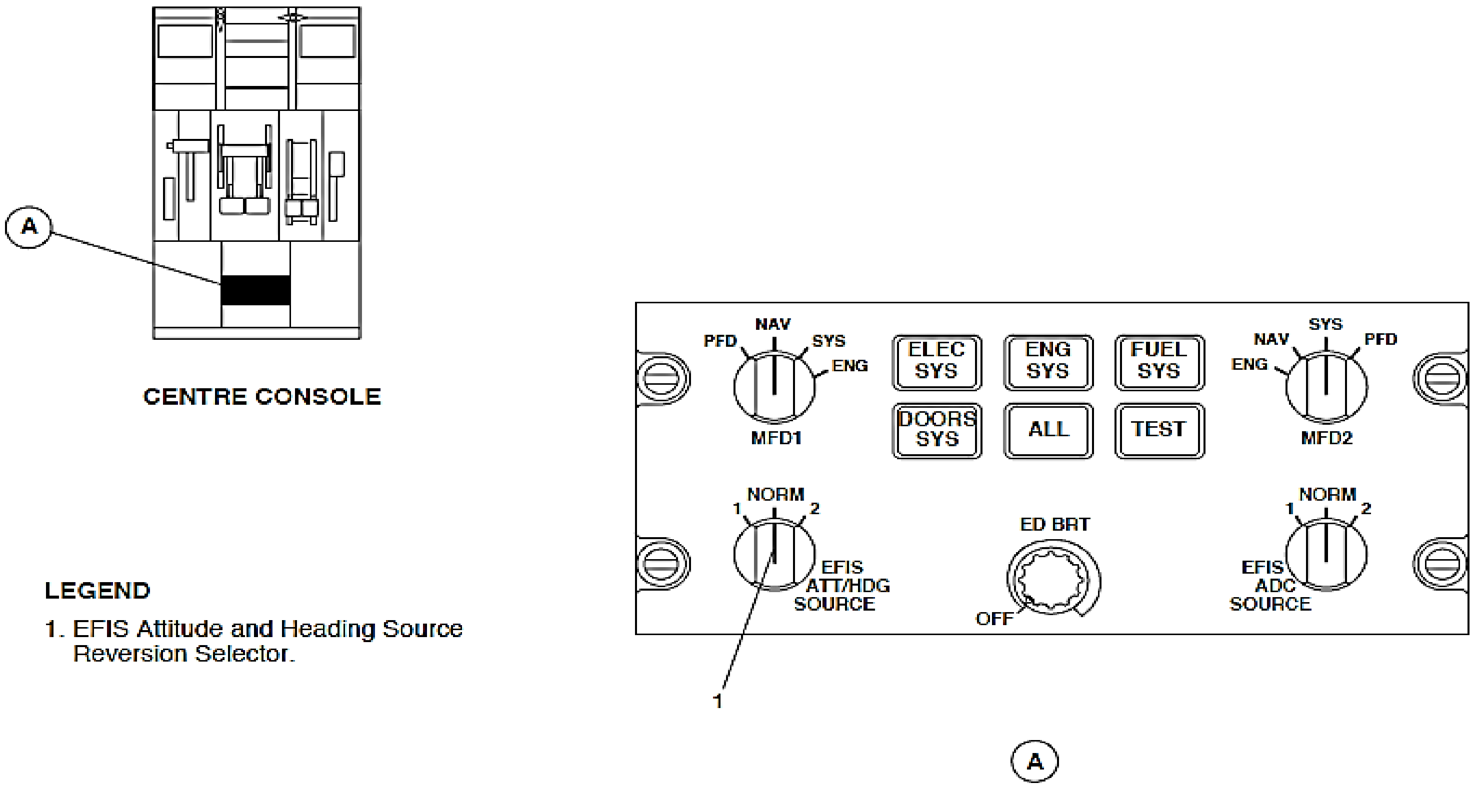


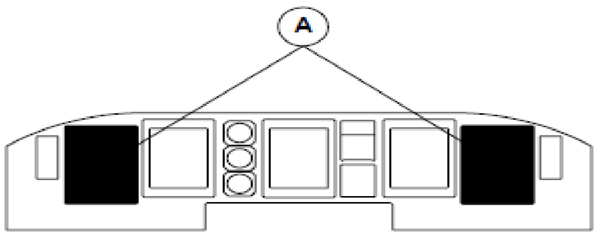
Figure 34- 64 ESCP, EFIS AHRS SOURCE REVERSION

Figure 34- 65 EIS AHRU INDICATIONS

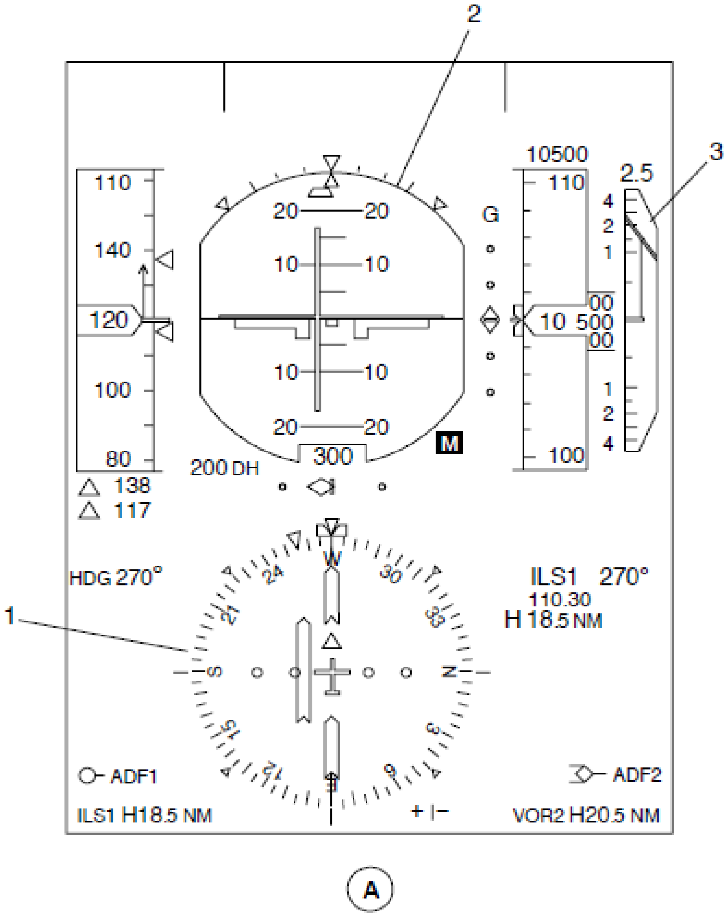
The Electronic Instrument System (EIS) shows the Attitude and Heading Reference Units (AHRU 1, AHRU 2) parameters that follow:

- Electronic Attitude Direction Indicator (EADI 1, EADI 2)
- Electronic Horizontal Situation Indicator (EHSI 1, EHSI 2)
- Inertial Vertical Speed Indicator (IVSI 1, IVSI 2).

Electronic Attitude Direction Indicator (EADI 1, EADI 2): The Electronic Attitude Direction Indicator (EADI 1, EADI 2) is used to show the aircraft's pitch, bank, and slip/skid against a representation of the earth and sky.

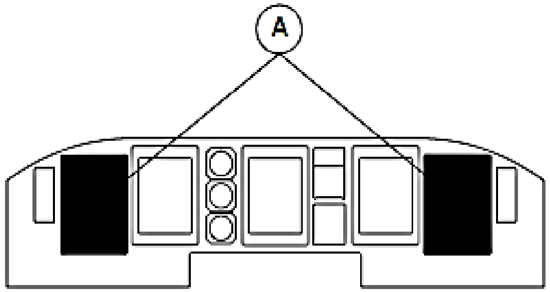


MAIN INSTRUMENT PANEL



LEGEND
1. EHSI.
2. EADI.
3. IVSI.

Figure 34- 65 EIS AHRU INDICATIONS



MAIN INSTRUMENT PANEL

LEGEND

1. Airplane Symbol.
2. Attitude Sphere.
3. Roll Pointer and Scale.
4. Slip/Skid Indicator.
5. Pitch Scale.

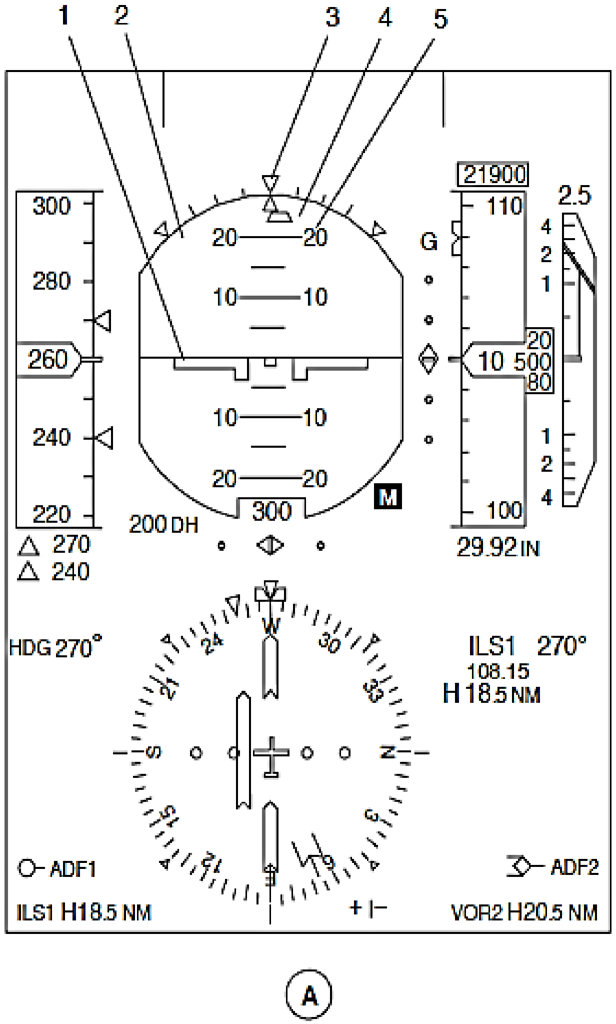


Figure 34- 66 PFD, ATTITUDE INDICATIONS

Figure 34- 66 PFD, ATTITUDE INDICATIONS

The Primary Flight Display (PFD) shows the Electronic Attitude Direction Indicator (EADI) components that follow:

- Attitude sphere
- Pitch scale
- Roll pointer and scale
- Airplane symbol
- Slip/skid indicator
- Mismatch flags
- Attitude failure flag.

Attitude Sphere: The attitude sphere has a sky permanent sector in blue above a ground permanent sector in brown and a white horizon line to divide the sectors. The attitude sphere moves to show pitch and roll.

Pitch Scale: The pitch scale is shown on the attitude sphere in 5 degree increments from 0 to 30 degrees. There are numbers show on the two sides of the pitch scale at 10, 20, 30, 40, 60, and 90 degree increments for the pitch up and pitch down attitudes.

The pitch scale has red chevrons that point towards the pitch scale to help recover from an unusual attitude.

Roll pointer and scale: The roll scale is shown on the attitude sphere in 0, 10, 20, 30, 45, and 60 degree increments at the top of the attitude sphere. A 60 degree roll mark comes into view when the roll is more than 30 degrees.

The roll pointer is located at the top of the attitude sphere, above the slip/skid indicator to show the amount of aircraft roll. It moves with the horizon line to measure aircraft roll.

Airplane Symbol: The attitude indication has an attitude sphere that moves relative to a fixed airplane symbol to show the amount of aircraft pitch.

Slip/Skid Indicator: The Slip/Skid Indicator shows the lateral acceleration of the aircraft and is a trapezoid symbol. It turns with the roll pointer and moves to left or right to show the intensity of lateral acceleration.

The maximum deflection indication shows a 0.14 gravity (g) lateral acceleration. The slip skid Indicator is shown to the left when the aircraft is slipping to the right and to the right when the aircraft is slipping to the left.

The Slip/Skid Indicator goes out of view when the aircraft roll is more than 60 degrees or the related AHRU has malfunctioned.

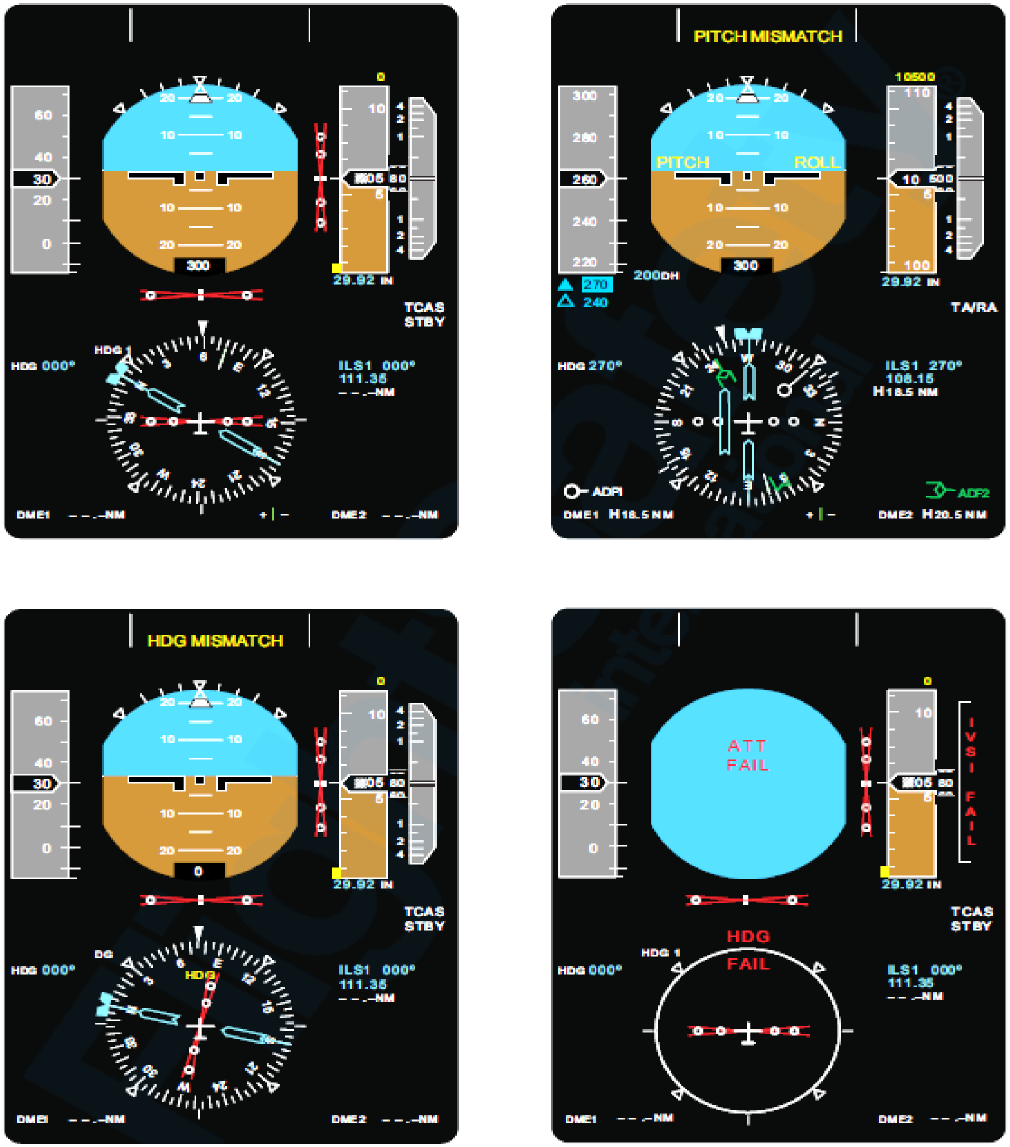


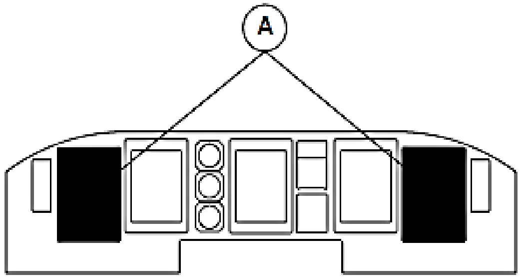
Figure 34- 67 PRIMARY FLIGHT DISPLAY - EADI INDICATIONS

Figure 34- 67 PRIMARY FLIGHT DISPLAY - EADI INDICATIONS

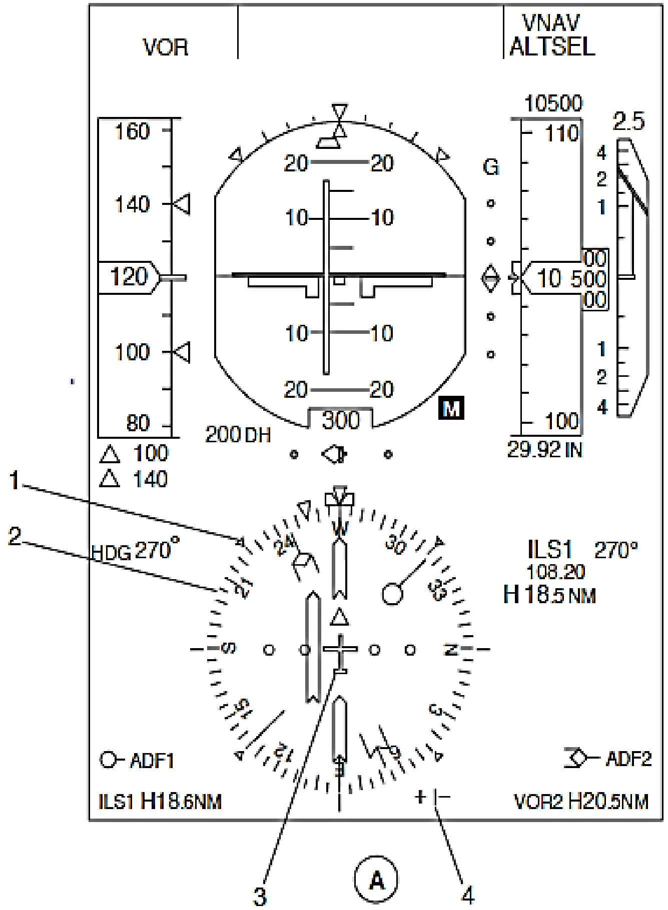
Mismatch Flags: The Flight Data Processing System (FDPS) senses signal differences between the AHRU 1 and AHRU 2 pitch and roll parameters. Both Flight Mode Annunciators (FMA) located at the top of the Primary Flight Displays (PFD 1, PFD 2) show yellow PITCH MISMATCH or ROLL MISMATCH messages in the center row of the center column. If a pitch and roll mismatch condition occur at the same time, the pitch mismatch message will be shown. It has a higher indication priority than the roll mismatch message. The indications flash for five seconds when they come into view and then go steady.

The attitude mismatch threshold is 3 degrees.

Attitude Failure Flag: When the attitude parameters malfunction, the roll scale, attitude sphere, and airplane symbol are replaced by a flag that comes into view as a red ATT FAIL message inside a white circle



MAIN INSTRUMENT PANEL



LEGEND

1. Fixed Markings.
2. Rotating Heading Dial.
3. Actual Heading Marker and Aircraft Symbol.
4. Slaving Error Annunciator.

Figure 34- 68 PFD HEADING INDICATIONS

Electronic Horizontal Situation Indicator (EHSI): The Electronic Horizontal Situation Indicator (EHSI 1, EHSI 2) is used to show the aircraft heading and the slaving error.

Figure 34- 68 PFD HEADING INDICATIONS

The Primary Flight Display (PFD) shows the Electronic Horizontal Situation Indicator EHSI components that follow:

- Rotating heading dial
- Fixed markings
- Actual heading marker and aircraft symbol
- Slaving error annunciator
- Heading mismatch flag
- Heading failure flag.

Rotating heading dial: The rotating heading dial is a full compass rose indication with markings at 5 degree intervals and numbers at 30 degree intervals except North, East, South, and West. The North, East, South, and West heading are shown as N, E, S, and W.

Fixed markings: The fixed markers are shown around the rotating heading dial at 45 degree interval.

Actual heading marker and aircraft symbol: A fixed white triangle actual heading marker is shown at the top of the EHSI with a fixed aircraft symbol in the center of the rotating heading dial to give a lubber line indication.

Slaving Error Annunciator: The compass slaving error annunciator shows the difference between the Attitude and Heading Reference Unit (AHRU 1, AHRU 2) heading and its related flux valve (MDU 1, MDU 2) heading.

A white vertical pointer that moves between a green + and – symbol is shown near the rotating heading dial indication. The maximum indication is plus or minus 4 degrees.

When the data from the AHRU is not correct, the pointer and scale are removed.

Figure 34- 69 PFD, HEADING MISMATCH INDICATION

Mismatch Flags: The Flight Data Processing System (FDPS) senses signal differences between the AHRU 1 and AHRU 2 heading parameters. Both Flight Mode Annunciators (FMA) located at the top of the Primary Flight Displays (PFD 1, PFD 2) show yellow HDG MISMATCH messages in the center row of the center column. If an attitude and heading mismatch condition occurs at the same time, the related attitude mismatch message will be shown. It has a higher indication priority than the heading mismatch message. The indications flash for five seconds when they come into view and then go steady. The heading mismatch threshold is 8 degrees.

The AHRU deactivates the flux slaving when the aircraft roll angle is greater than 7 degrees, pitch angle is greater than 15 degrees or during specified horizontal acceleration. For this reason certain aircraft maneuvers, such as extended, continuous turns may cause a “HDG MISMATCH” message to be displayed on the PFD. This is considered normal as long as the “HDG MISMATCH” message is removed within 5 minutes of returning to straight and level flight.

The AHRU deactivates the flux valve slaving on the occurrences of any of the following conditions (with hysteresis):

- The roll angle is above 7 degrees and the roll angle must return to below 6 degrees to clear this condition.
- The pitch angle is above 15 degrees and it must return to below 14 degrees to clear this condition.
- Any individual component of horizontal acceleration is above 70 mg and must return to below 60 mg.
- Filtered intensity (time constant = 5 seconds) of horizontal acceleration is above 23 mg and must return to below 20 mg.

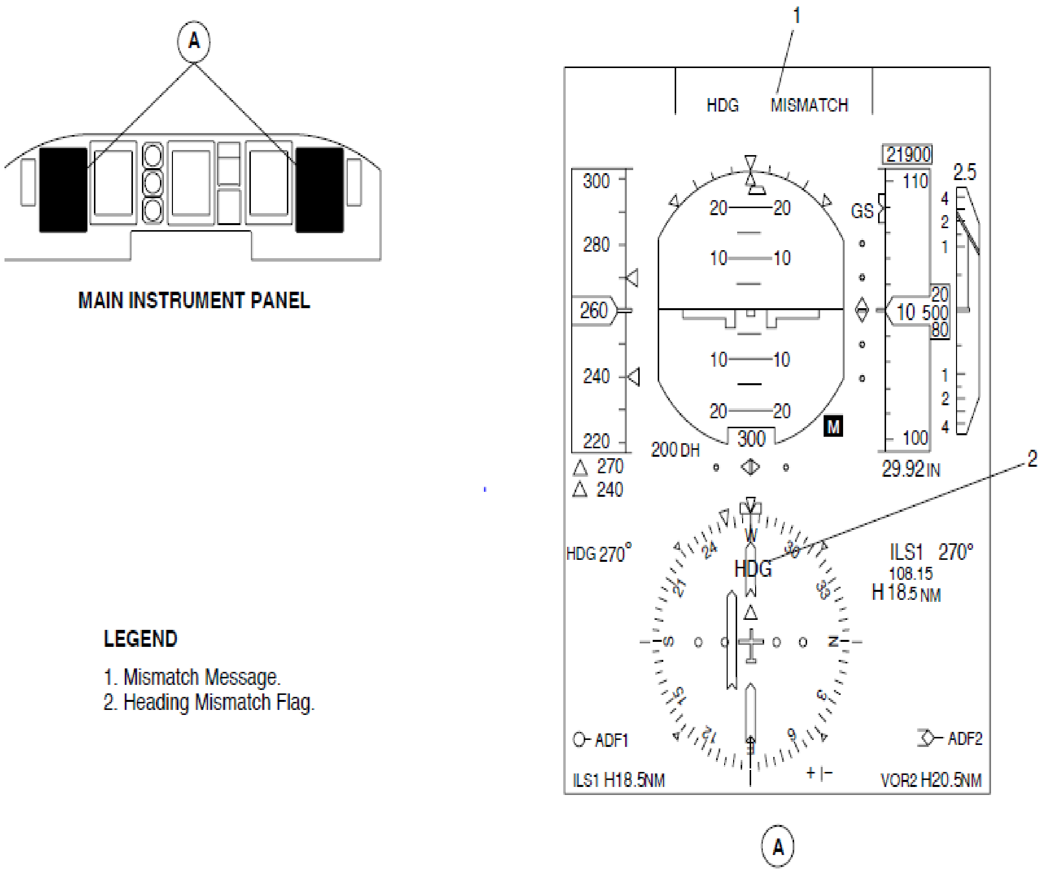
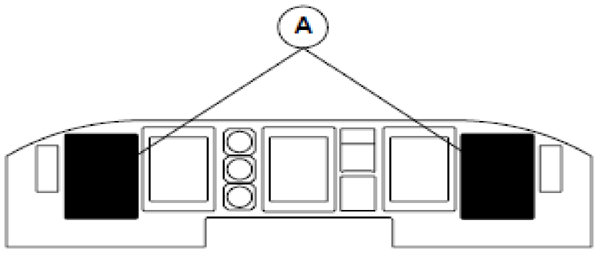
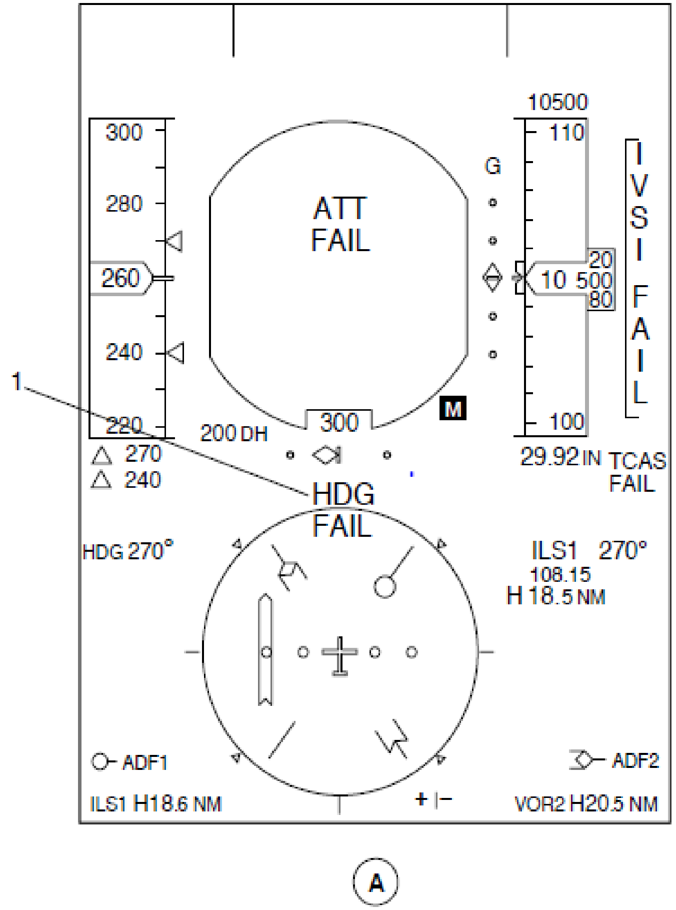


Figure 34- 69 PFD, HEADING MISMATCH INDICATION

Figure 34- 70 PFD HEADING FAILURE FLAG

Heading Failure Flag: When the heading parameters malfunction, the rotating heading dial, aircraft symbol, and slaving error annunciator are replaced by a flag that comes into view as a red HDG FAIL message inside a white circle. The fixed markings remain in view.

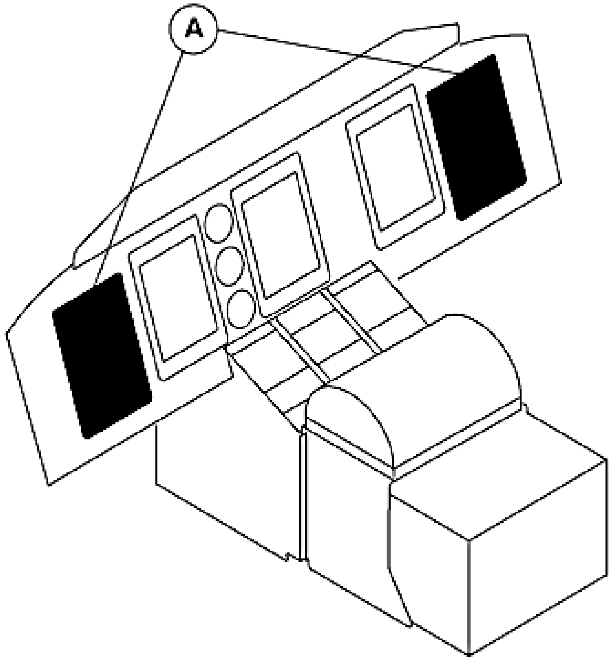


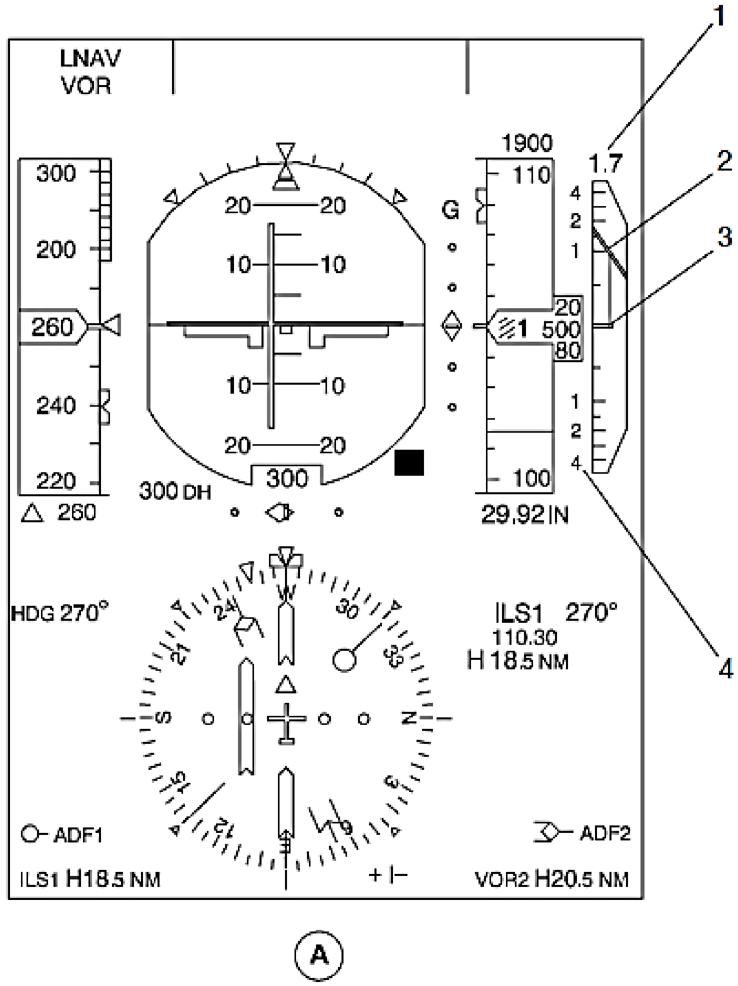


LEGEND

1. Heading Failure Flag.

Figure 34- 70 PFD HEADING FAILURE FLAG





LEGEND

1. IVSI digital readout.
2. IVSI pointer.
3. IVSI zero reference line.
4. IVSI scale.

Figure 34- 71 PFD, INERTIAL VERTICAL SPEED INDICATOR INDICATIONS

Figure 34- 71 PFD, INERTIAL VERTICAL SPEED INDICATOR INDICATIONS

Inertial Vertical Speed Indicator (IVSI 1, IVSI 2): The Inertial Vertical Speed Indicator (IVSI 1, IVSI 2) shows the aircraft inertial vertical speed.

The Primary Flight Display (PFD) shows the Inertial Vertical Speed Indicator (IVSI) components that follow:

- Scale
- Pointer needle
- Digital readout
- Failure flag.

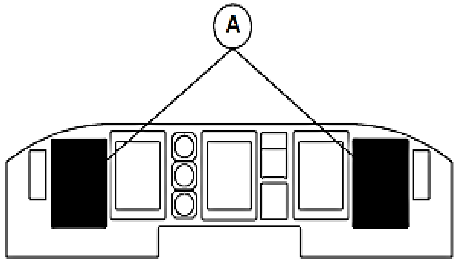
Scale: The IVSI 1 scale shows vertical speed from −4500 to +4500 feet per minute. It has a compressed scale that is in 100 feet per second increments for values between −1000 feet per minute and +1000 feet per minute. The scale expands to show increments at 500 feet per minute between −1000 and −4500 feet per minute and between +1000 and +4500 feet per minute. The markings for 1000, 2000, and 4000 feet per minute are shown as 1, 2, and 4.

Pointer Needle: The Pointer Needle points to the current inertial vertical speed of the aircraft along a fixed scale from −4500 to +4500 feet per minute.

Digital Readout: The digital readout can show the inertial vertical speed value in white numbers from −9900 to 9900 feet per minute. It shows a compressed value that is in 50 feet per minute increments from −1000 through +1000. It changes to an expanded value that is in 100 feet per minute increments between −1000 and −9900 feet per minute and between +1000 and +9900 feet per minute.

Figure 34- 72 PFD, INERTIAL VERTICAL SPEED INDICATOR FAILURE FLAG

IVSI Failure Flag: When the IVSI parameters malfunction, the scale, pointer needle, and digital readout are replaced by a flag that comes into view as a red IVSI FAIL message inside a white rectangle.



MAIN INSTRUMENT PANEL

LEGEND

1. IVSI Failure Flag.

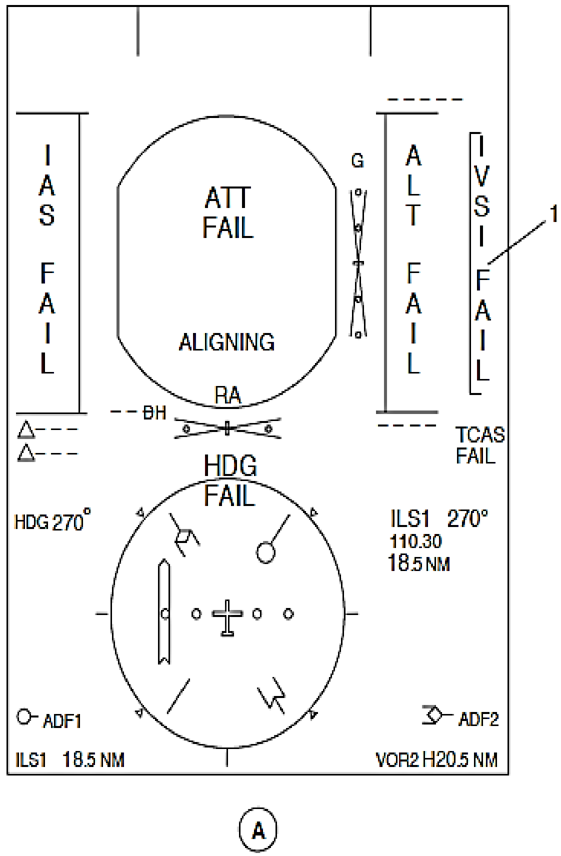
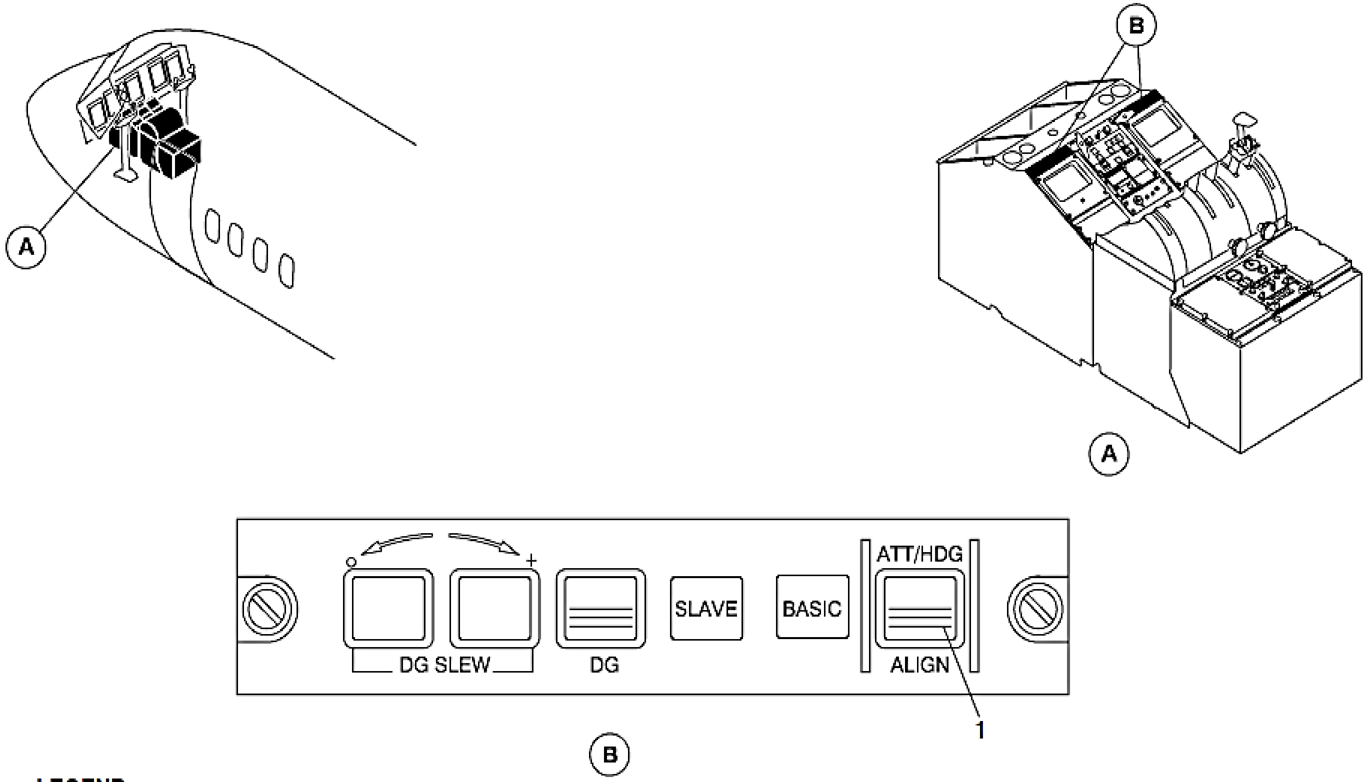


Figure 34- 72 PFD, INERTIAL VERTICAL SPEED INDICATOR FAILURE FLAG

Figure 34- 73 AHCP, ALIGNMENT MODE INDICATIONS

The ATT/HDG ALIGN annunciator switch located on the AHRS Control Panel (AHCP) is pushed to manually start the alignment mode.



LEGEND

1. ALIGN annunciator switch.

Figure 34- 73 AHCP, ALIGNMENT MODE INDICATIONS

Figure 34- 74 PFD, ALIGNMENT MODE INDICATIONS

An amber bar in the ATT/HDG ALIGN annunciator switch comes to show the alignment mode and the two Electronic Attitude Direction Indicators (EADI 1, EADI 2) show an ALIGNING message. The message comes into view in yellow fonts that flash for five seconds and then change to steady.

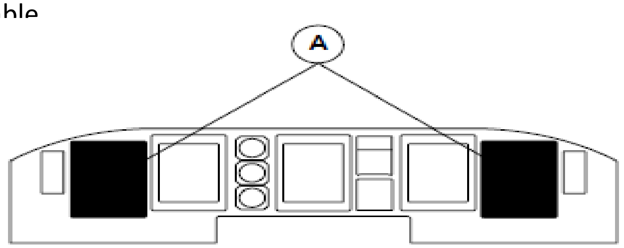
The alignment mode may be set when the aircraft is on the ground or in–flight.

NOTE

The aircraft must be in a straight and level attitude when making an alignment mode selection in–flight.

The alignment mode continues for 60 seconds on the ground and 90 seconds in–flight. The Attitude Heading Reference System (AHRS 1, AHRS 2) parameters are available 20 seconds after starting the alignment mode.

The normal mode automatically starts at the end of the alignment mode if the external inputs are available. It calculates attitude parameters when external air data inputs are available and slaved heading when the external MDU input is available.



MAIN INSTRUMENT PANEL

LEGEND

1. AHRS Aligning Annunciator.

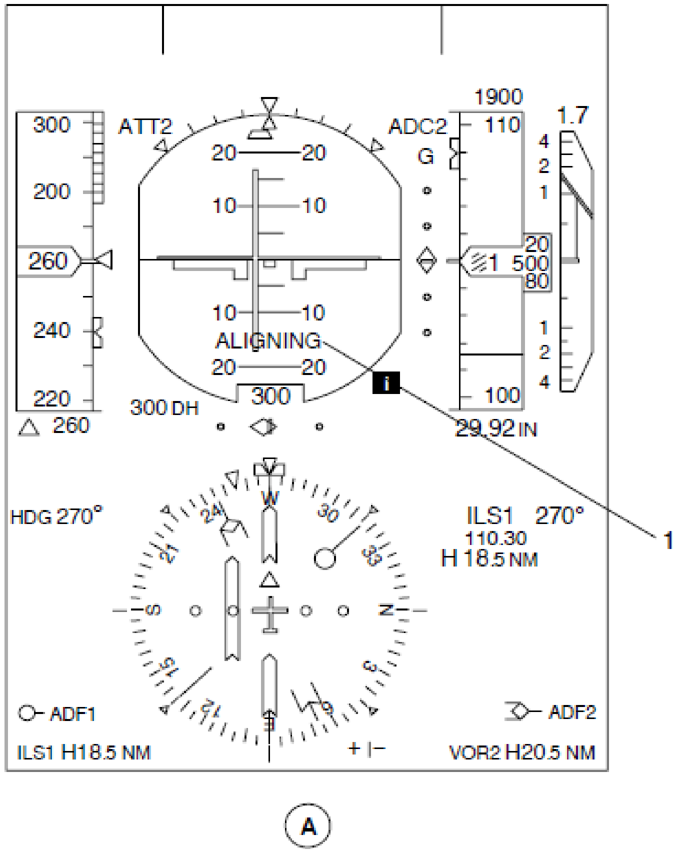


Figure 34- 74 PFD, ALIGNMENT MODE INDICATIONS

Figure 34- 75 AHCP, BASIC MODE INDICATION

If the TAS parameter is not available, a green light on the AHCPI will come on to show the automatic reversion to the basic mode.

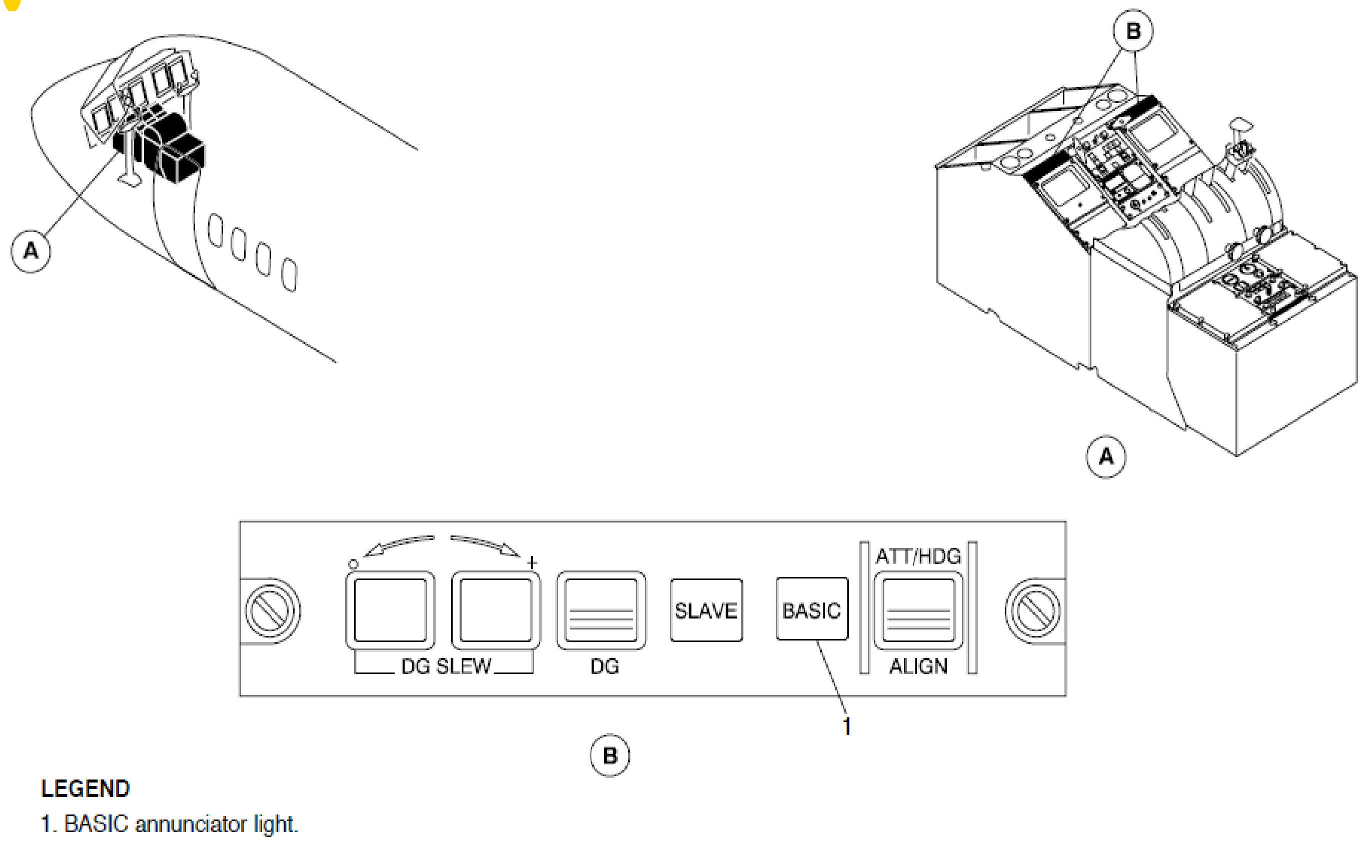
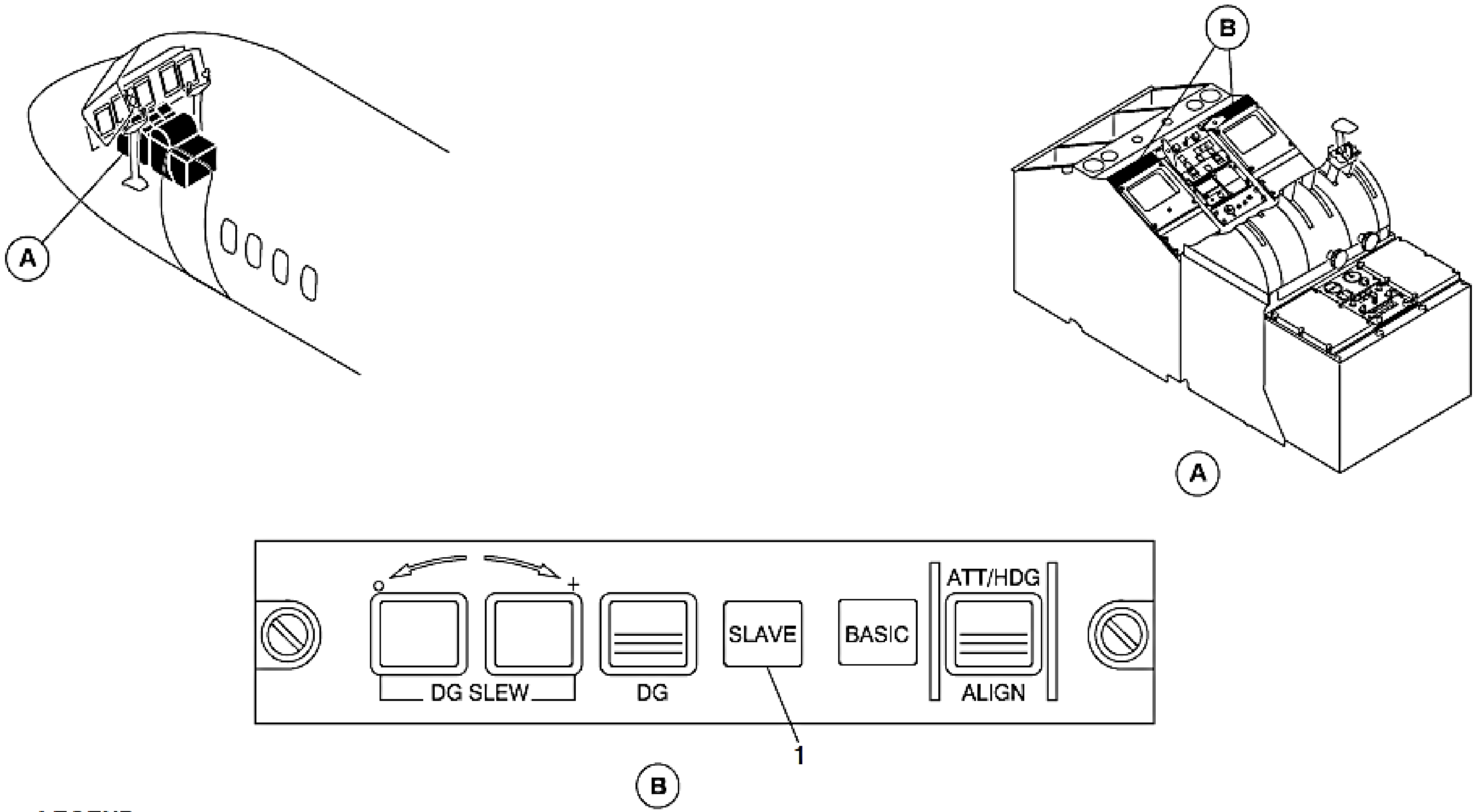


Figure 34- 75 AHCP, BASIC MODE INDICATION

Figure 34- 76 AHCP, SLAVE INDICATION

The SLAVE light on the AHCP comes on to show a flux valve malfunction.



LEGEND

1. SLAVE annunciator light.

Figure 34- 76 AHCP, SLAVE INDICATION

Figure 34- 77 AHCP, DG MODE SELECTION

The DG switch on the AHCP is pushed to manually set the DG Mode. A green bar in the annunciator switch comes on.

The + or − DG SLEW switches located on the Attitude and Heading Control Panel (AHCP 1, AHCP 2) are pushed to correct the rotating heading dial for gyro drift. The rotating heading dial turns in a clockwise direction while the + DG SLEW switch is pushed, and it moves in a counterclockwise when the − DG SLEW switch is pushed. If one slew switch or the other is initially pushed and held, the rotating heading dial turns at the low speed rate (1 degree per second), and if held for more than 3 seconds, it will then turn at the high−speed rate (10 degrees per second).

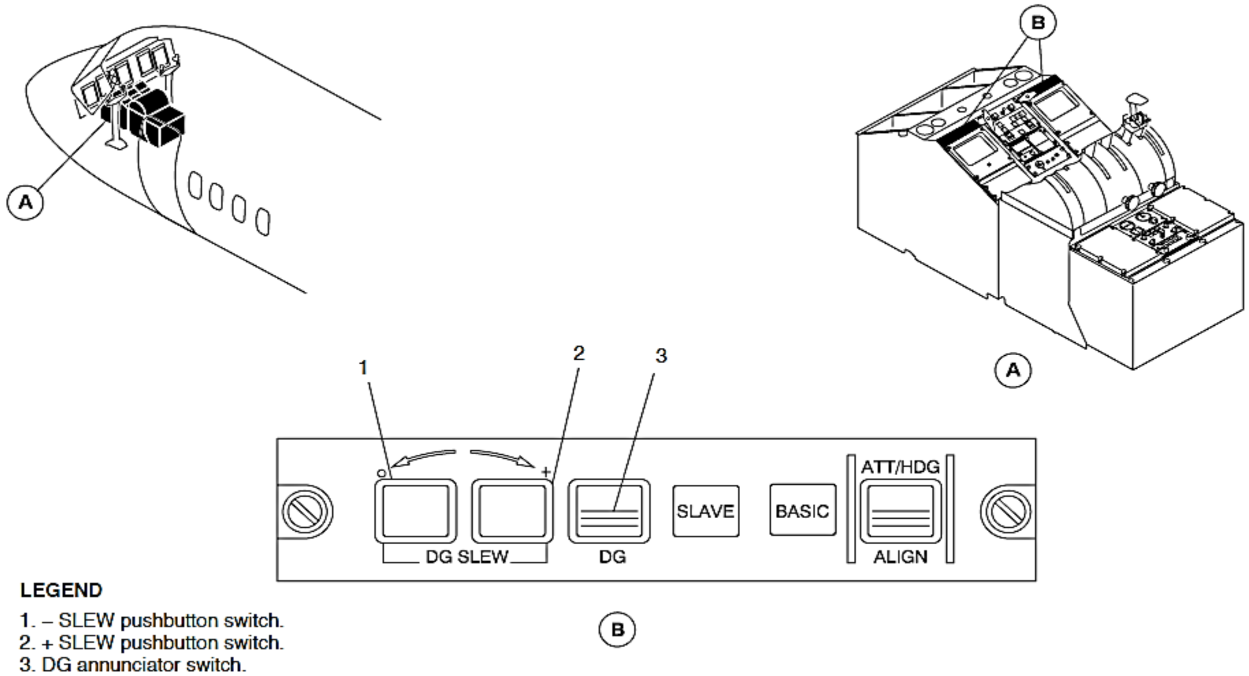
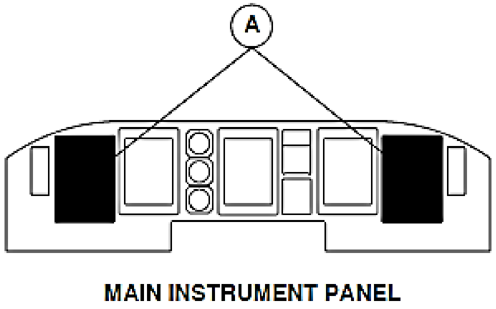


Figure 34- 77 AHCP, DG MODE SELECTION

Figure 34- 78 PFD, DG MODE INDICATION

The EHSI shows a DG heading source annunciation and the slaving error indication goes out of view to show the selection.



LEGEND

1. DG Mode Source Annunciation.

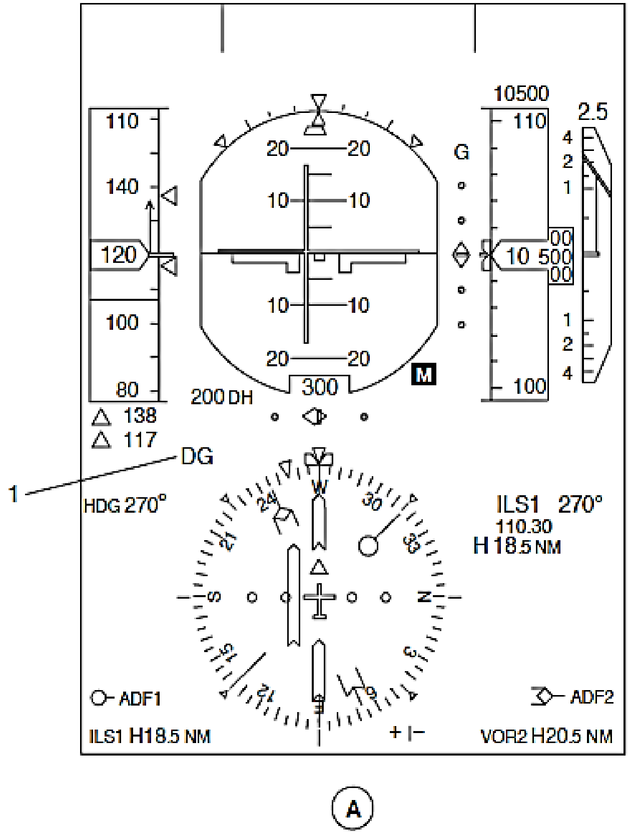


Figure 34- 78 PFD, DG MODE INDICATION

Figure 34- 79 AHCP, CALIBRATION MODE SELECTION

The flux valve calibration mode is set when the ALIGN and DG SLEW switches on the AHRS Control Panel (AHCP 1, AHCP 2) are pushed for three seconds at the same time while the aircraft is on the ground.

The flux valve calibration mode automatically cancels at take−off when the Weight On Wheels (WOW) condition changes from ground to flight.

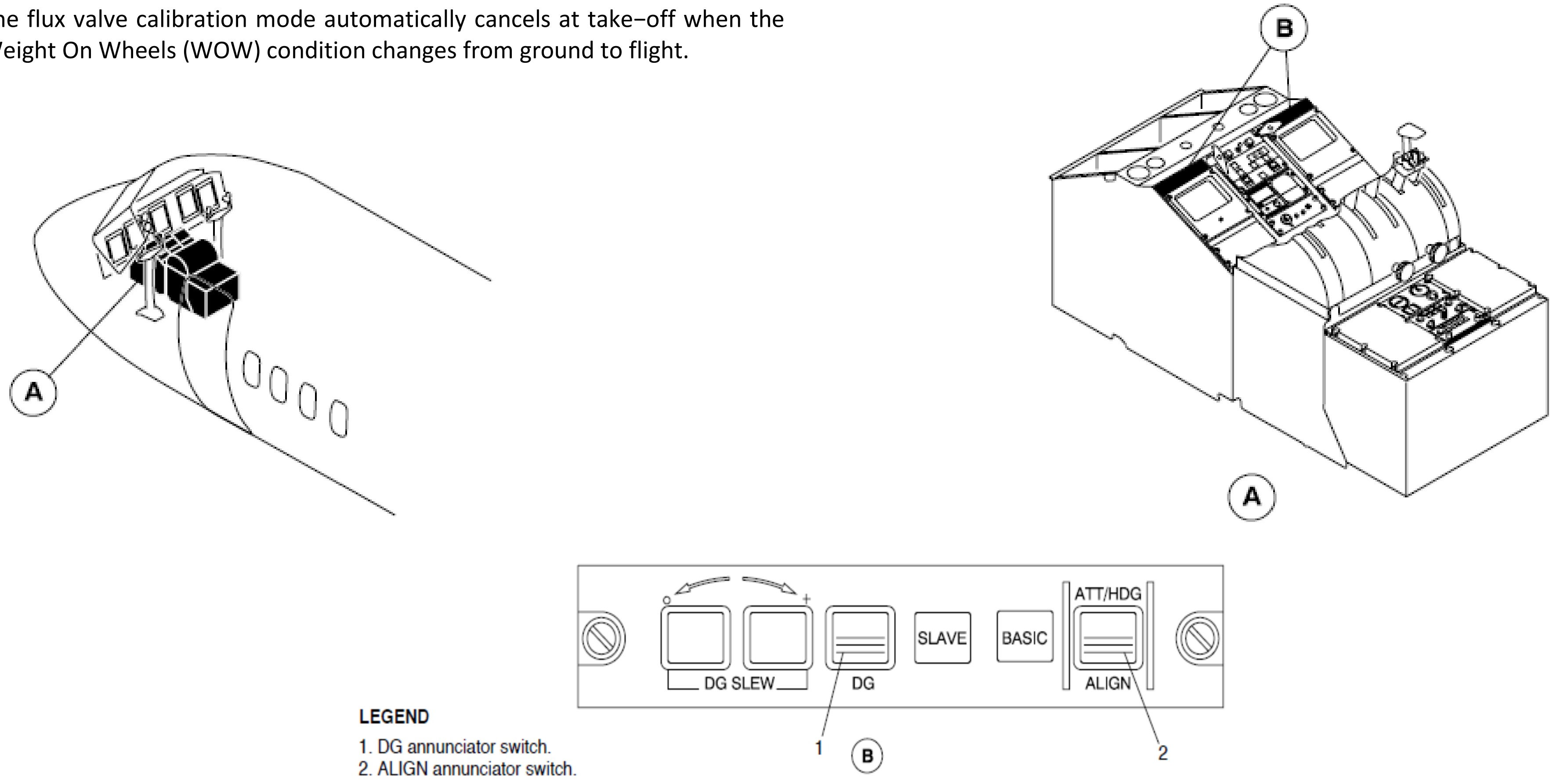


Figure 34- 79 AHCP, CALIBRATION MODE SELECTION

Figure 34- 80 PFD, CALIBRATION MODE INDICATION

The PFD and MFD EHSIs show a HDG CALIBRATION message. The MFD actual heading digital value and reference indication changes to show the heading value with four numbers. The red SLAVE annunciator light on the AHCP flashes during the calibration mode.

The flux valve calibration mode starts with an alignment phase. The ALIGN light comes on and the alignment phase continues for 5 minutes.

Heading error measurements are made when the ALIGN annunciator light goes out.

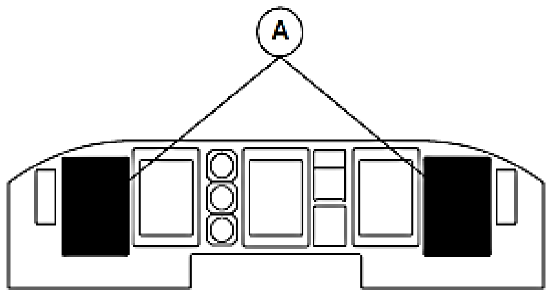
In the measurement phase, the aircraft is positioned on a northerly heading. Sight measurements are made at 45 degree intervals through 360 degrees. The AHRU is supplied with the measured reference heading data for each interval and the DG SLEW push buttons switches are used to match the rotating heading dial indication on the EHSI with the sight measurement. The DG SLEW switches are pushed at the same time to put the heading value into the AHRU.

The red SLAVE annunciator light flash rate increases to indicate a difference between the measured value and the value supplied by flux valve. The red SLAVE annunciator light goes back to the normal flash rate when the update is completed. It continues for less than 60 seconds.

The aircraft is positioned on a 30 degree heading. Eight more sight measurements are made at 45 degree intervals. The Attitude and Heading Reference Unit (AHRU) is supplied with the measured reference heading data for each interval.

The red SLAVE annunciator light stays on to show a satisfactory completion of the last measurement. The ALIGN and DG SLEW switches located on the AHCP are pushed at the same time for 3 seconds to complete the procedure.

A new alignment mode automatically initiates for 60 seconds. The Attitude and Heading Reference System (AHRS) circuit breakers are cycled to stop the calibration procedure.



MAIN INSTRUMENT PANEL

LEGEND

1. Calibration Message.

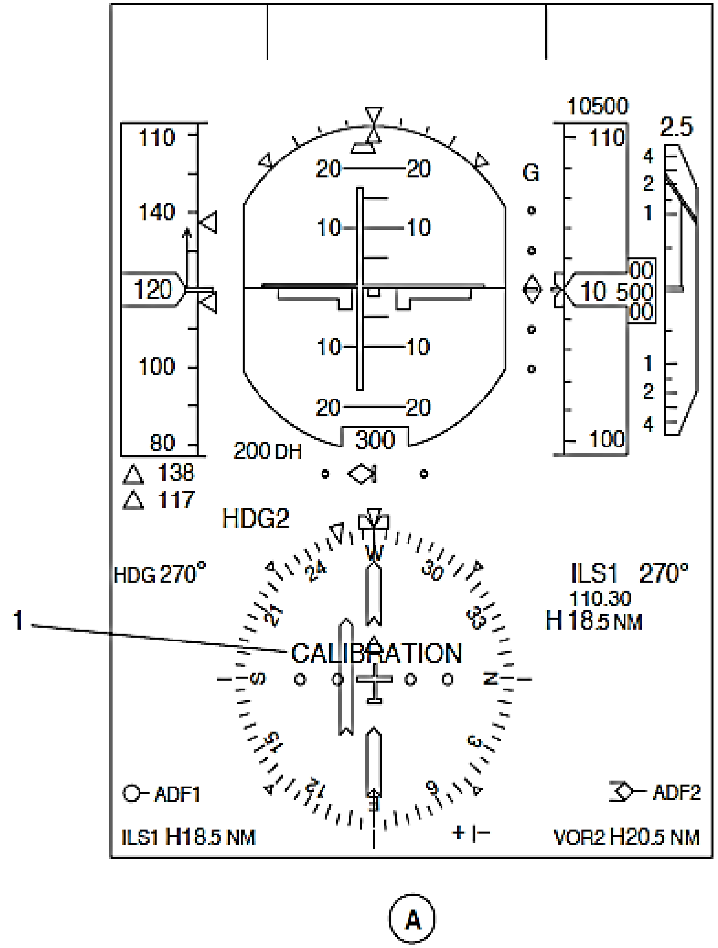
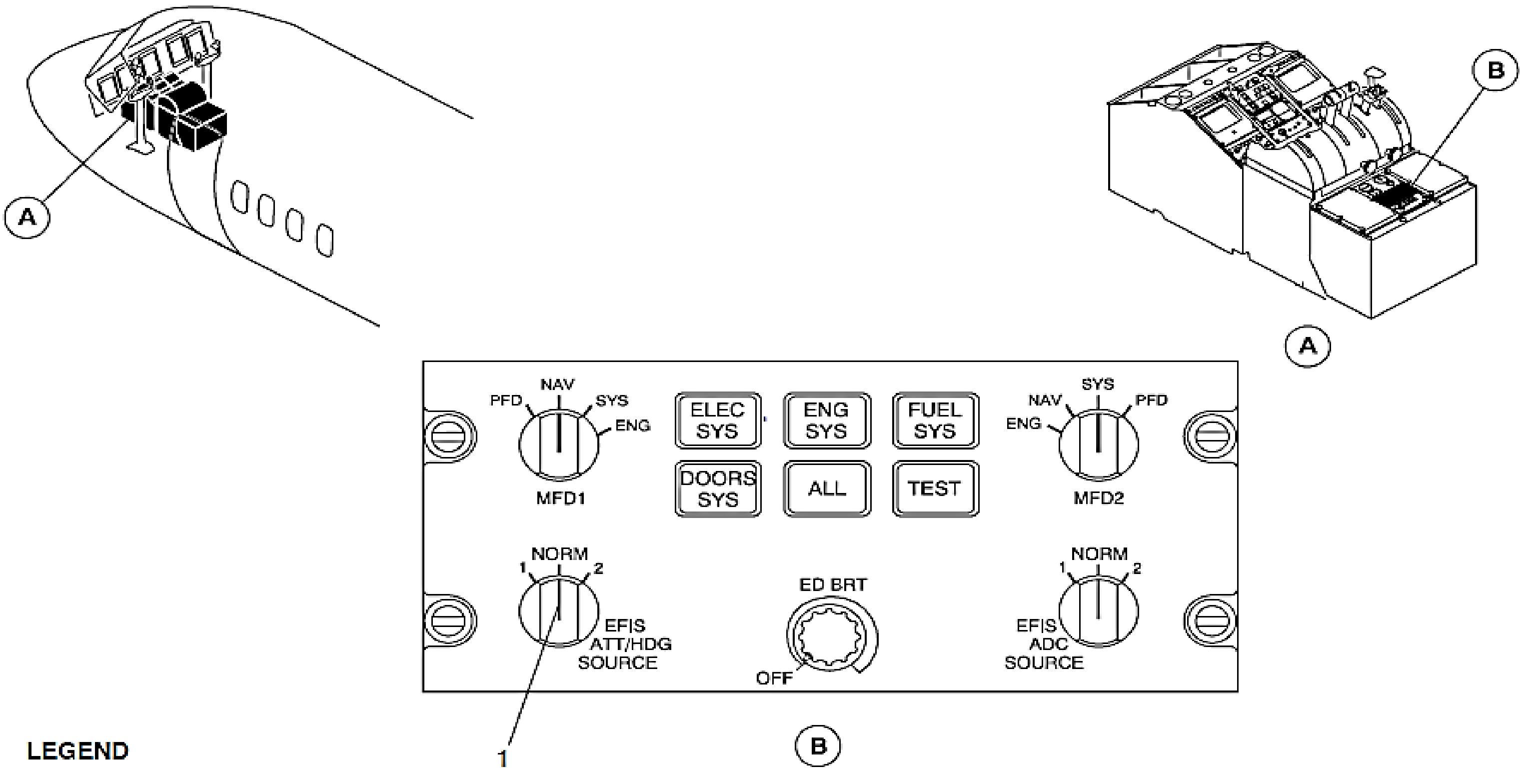


Figure 34- 80 PFD, CALIBRATION MODE INDICATION

FIGURE 34- 81 ESCP, EFIS AHRS SOURCE REVERSION SELECTION

When the EFIS ATT/HDG SOURCE reversion selector is set to 1 or 2 on the ESID Control Panel (ESCP), both Primary Flight Displays (PFD 1, PFD 2) will show data from the selected Attitude and Heading Reference Unit (AHRU 1, AHRU 2).



LEGEND

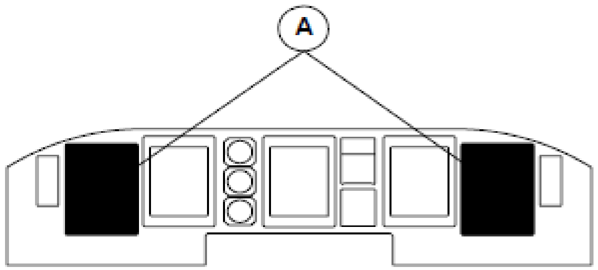
1. EFIS attitude and heading source reversion selector.

FIGURE 34- 81 ESCP, EFIS AHRS SOURCE REVERSION SELECTION

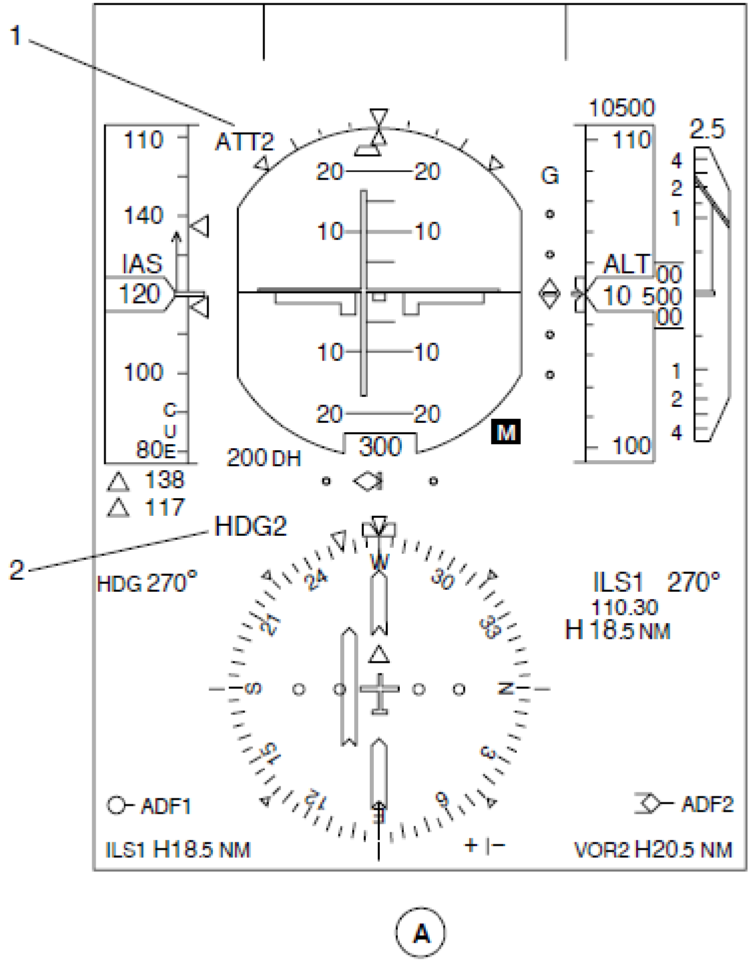
FIGURE 34- 82 PFD, EFIS ATT/HDG SOURCE REVERSION INDICATION

When the two PFDs are showing the same attitude and heading source, an attitude and heading source annunciator is shown in yellow fonts that follow:

- ATT 1, ATT 2
- HDG 1, HDG 2
- DG 1, DG 2.



MAIN INSTRUMENT PANEL



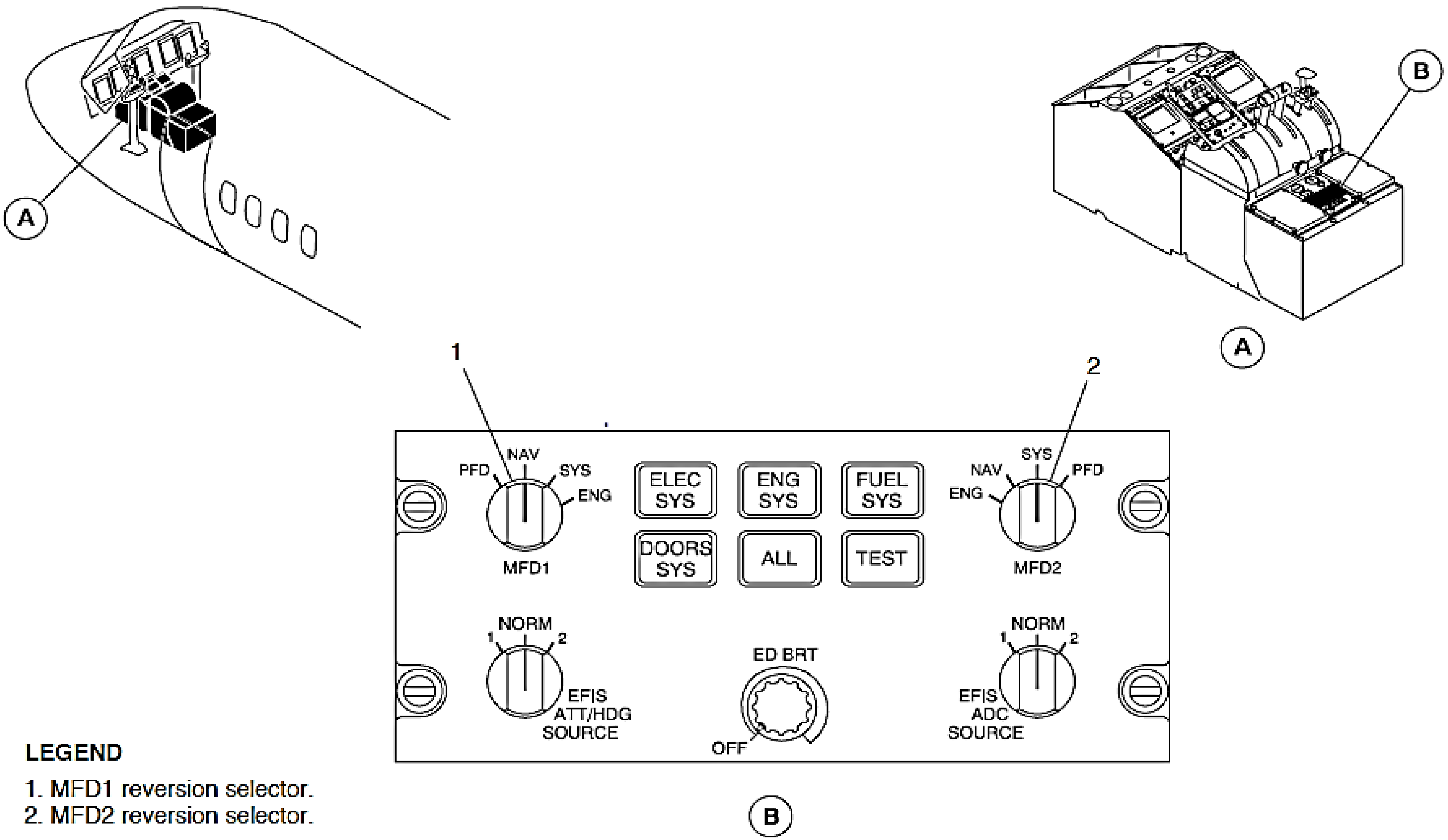
LEGEND

1. Attitude Source Annunciation.
2. Heading Source Annunciation.

FIGURE 34- 82 PFD, EFIS ATT/HDG SOURCE REVERSION INDICATION

Figure 34- 83 ESCP, MFD REVERSION SELECTION

The MFD 1 and MFD 2 Reversion switches located on the ESID Control Panel (ESCP) are used to select the Navigation Page.



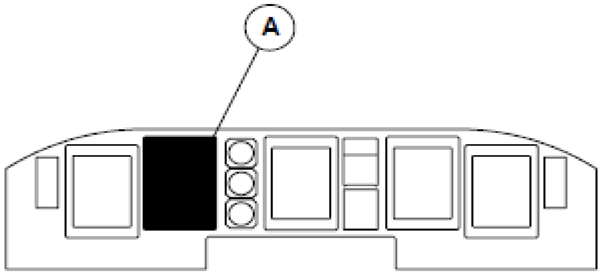
LEGEND

1. MFD1 reversion selector.
2. MFD2 reversion selector.

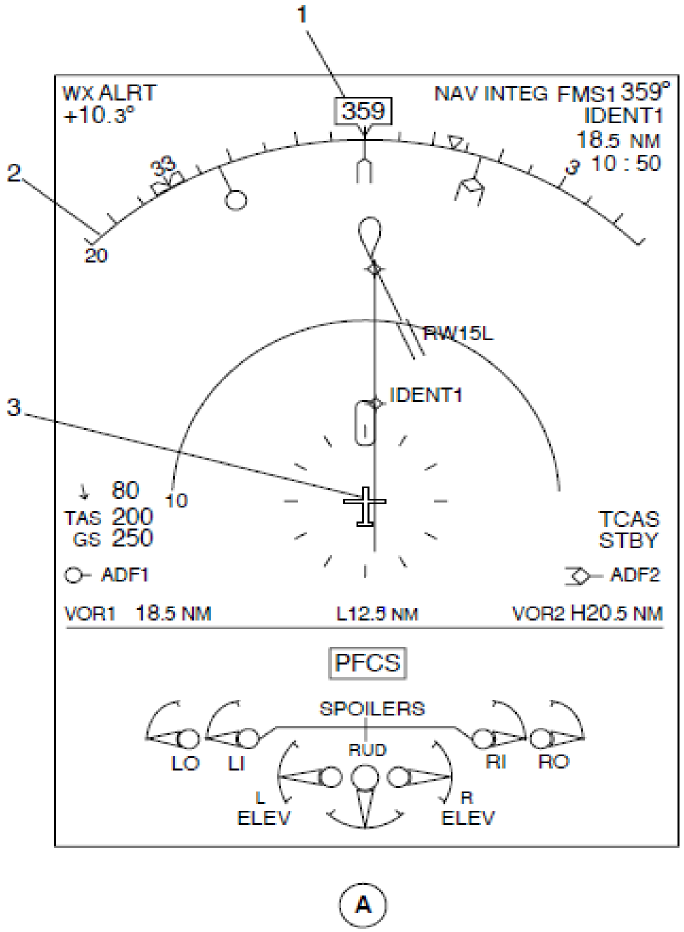
Figure 34- 83 ESCP, MFD REVERSION SELECTION

Figure 34- 84 MFD, ARC FORMAT INDICATION

The Navigation Page is usually shown on the upper part of MFD1 and permanent data is shown on the bottom. The Navigation Page comes into view in the map FMS mode format. The map mode shows the same heading parameters as its related Primary Flight Display (PFD 1, PFD 2) indication on a 90 degree heading arc.



MAIN INSTRUMENT PANEL



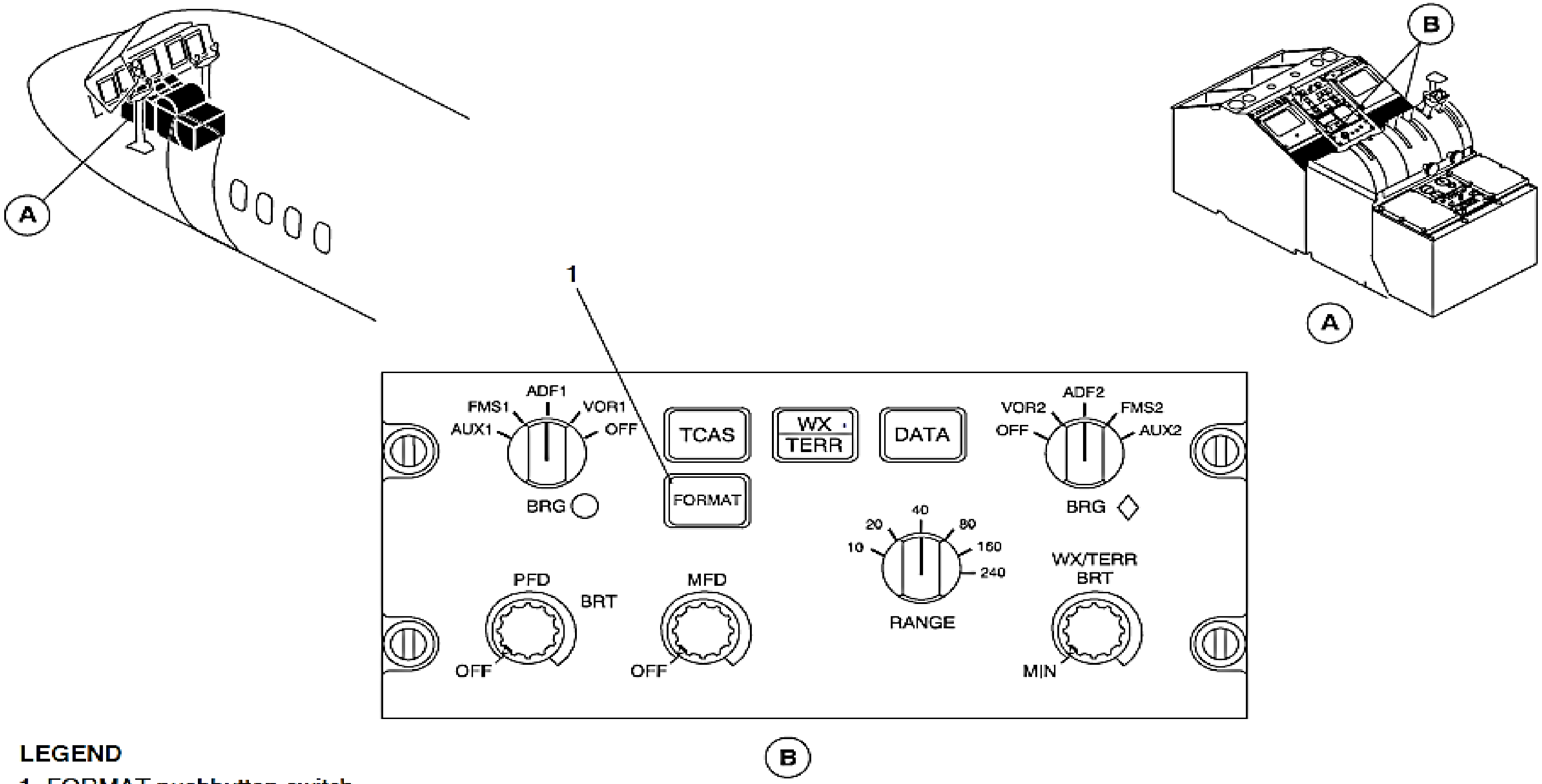
LEGEND

1. Actual Heading Digital Value and Reference.
2. Heading Scale in Arc Format.
3. Aircraft Symbol.

Figure 34- 84 MFD, ARC FORMAT INDICATION

Figure 34- 85 ESCP, PLAN FORMAT SELECTION

The FORMAT pushbutton switch located on the EFIS Control Panel (EFCP) is pushed for more than one second (long push) to select the plan FMS mode format.



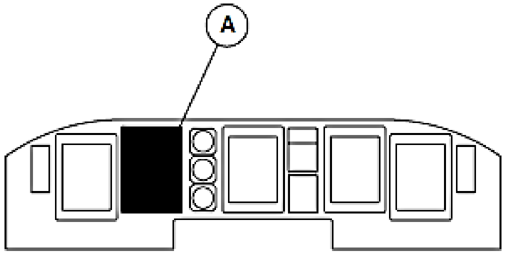
LEGEND

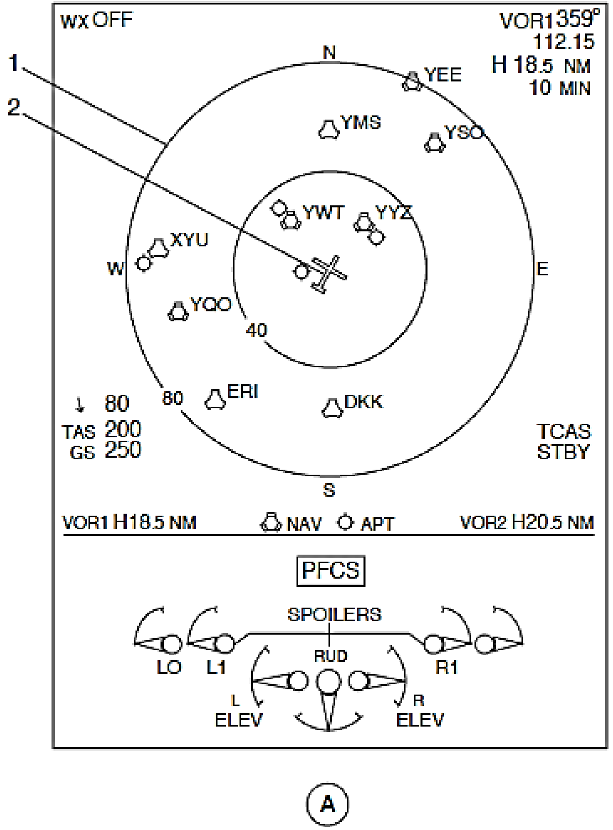
1. FORMAT pushbutton switch.

Figure 34- 85 ESCP, PLAN FORMAT SELECTION

Figure 34- 86 MFD, PLAN FORMAT INDICATION

The plan FMS mode format has a white aircraft symbol that turns about its center in the center of a fixed full map indication. When the heading parameter is not available, it is removed from view.





LEGEND

1. Fixed Heading Indication.
2. Aircraft Symbol.

Figure 34- 86 MFD, PLAN FORMAT INDICATION

The Left Main and Essential Busses supply 28 Vdc electrical power through four 5 A circuit breakers to the AHRS 1 components. The circuit breakers are located in positions A4, A9, B4, and B9 on the avionics circuit breaker panel.

The right main and essential buses supply 28 Vdc power through four 5 A circuit breakers to the AHRS 2 components. The circuit breakers are located in positions A3, A10, B3, and B10 on the avionics circuit breaker panel.

Table 34- 1 ELECTRICAL POWER INTERFACES

COMPONENT	LEFT MAIN BUS	LEFT ESS BUS	RIGHT MAIN BUS	RIGHT ESS BUS
AHRU 1	x	x		
AHCP 1	x	x		
AHRU 2			x	x
AHCP 2			x	x

The AHRUs receive digital air data parameters from the two ADUs and discrete inputs from the systems that follow:

Table 34- 2 AHRU INPUT INTERFACES

PARAMETER	SOURCE
ATT/HDG Source Reversion	ESCP 1, ESCP 2
Weight On Wheels	PSEU
Mode Commands	AHCP 1, AHCP 2
IBIT	CDS

The Attitude and Heading Reference Unit (AHRU) supplies specific attitude and heading parameters directly to the systems that follow:

Table 34- 3 AHRU OUTPUT INTERFACES

PARAMETER	SOURCE
Bank Angle	AFCS 1, AFCS 2 SPS EFIS 1, EFIS 2
Pitch Angle	AFCS 1, AFCS 2 SPS EFIS 1, EFIS 2
Body Linear Accelerations, x	AFCS 1, AFCS 2
Body Linear Accelerations, y	AFCS 1, AFCS 2
Body Linear Accelerations, z	AFCS 1, AFCS 2 SPS
Body Linear Accelerations, j	EFIS 1, EFIS 2
Body Angular Rates	AFCS 1, AFCS 2 SPS
Heading	AFCS 1, AFCS 2 EFIS 1, EFIS 2
Compass Synchro Error	EFIS 1, EFIS 2
Baro Inertial Vertical Speed	AFCS 1, AFCS 2 EFIS 1, EFIS 2

It supplies specific attitude and heading data to the systems that follow through the Integrated Flight Cabinets (IFC):

Table 34- 4 AHRU OUTPUT INTERFACES, IFCS

PARAMETER	SOURCE
Bank Angle	FDR FDPS 1, FDP 2 WXR ATC 1, ATC 2
Pitch Angle	FDR FDPS 1, FDPS 2 WXR ATC 1, ATC 2
Body Linear Accelerations, x	FDR FDPS 1, FDPS 2
Body Linear Accelerations, y	FDR FDPS 1, FDPS 2
Body Linear Accelerations, z	FDR FDPS 1 FDPS 2 FMS FCECU 1, FCECU 2
Body Angular Rates	FDPS 1, FDPS 2 FMS
Heading	FDR FDPS 1, FDPS 2

A General Purpose Data Busses (GPDB) supply attitude and heading information to the systems that follow:

Table 34- 5 AHRU OUTPUT INTERFACES, GPDB

PARAMETER	SOURCE
Body Linear Acceleration, z	FMS 1, FMS 2 FCECU 1, FCECU 2
Body Angular Rates	FMS 1, FMS 2

The AHRU uses four ARINC 429 High−Speed data busses to supply specific attitude and heading parameters to the systems that follow:

Table 34- 6 AHRU 1

ARINC 1	ARINC 2	ARINC 3	ARINC 4
MFD 1 PFD 1	FGM 1 SPM 1 CDS FDR FDPS 1	PFD 2 MFD 2 FGM 2 SPM 2	FDPS 2

Table 34- 7 AHRU 2

ARINC 1	ARINC 2	ARINC 3	ARINC 4
PFD 2 MFD 2	FGM 2 SPM 2 FDPS 2	PFD 1 MFD 1 FGM 1 SPM 1	FDPS 2 CDS FDR FDPS 1

The Input/Output Processor (IOP 1, IOP 2), Stall Protection Modules (SPM 1, SPM 2), and Flight Guidance Modules (FGM 1, FGM 2) are located inside its related Integrated Flight Cabinet (IFC 1, IFC 2).

Units, Attitude and Heading Reference

Figure 34- 88 ATTITUDE AND HEADING REFERENCE UNITS LOCATOR
Figure 34- 87 ATTITUDE AND HEADING REFERENCE UNIT

The Attitude Heading Reference Unit (AHRU) uses fiber optic gyros and accelerometers to measure angular rates and accelerations in three body axes.

It weighs 7.7 lb. (3.49 kg). It is 4.2 in. (106.7 mm) wide, 4.5 in. (114 mm) high and 11.4 in. (290 mm) long.

The case has cooling fins to dissipate heat.

The Attitude Heading Reference Unit (AHRU 1, AHRU 2) is attached to a shock−mount with four screws.

The two AHRUs have shock mounts that attach to the wing plane below the floor within the Center of Gravity (CG) limits. The AHRU is installed in a mounting tray in a lateral orientation. This prevents possible damage to the two Attitude Heading Reference Units (AHRU 1, AHRU 2) in case of engine rotor−burst. The installation is specified by the configuration software.

Shims are installed below the AHRU mounting trays to give the correct physical alignment with aircraft attitude. The seat rails are the longitudinal and lateral references.

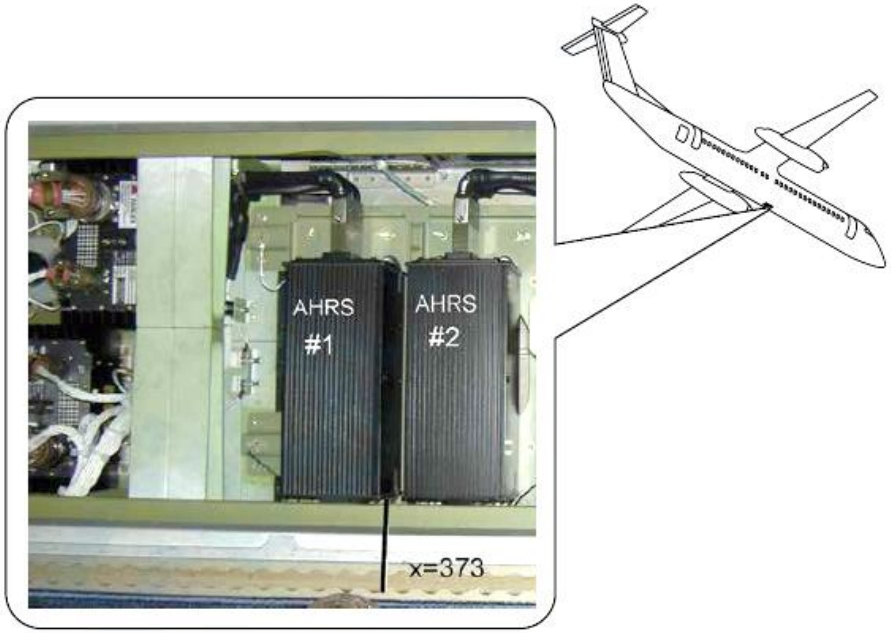


Figure 34- 87 ATTITUDE AND HEADING REFERENCE UNIT

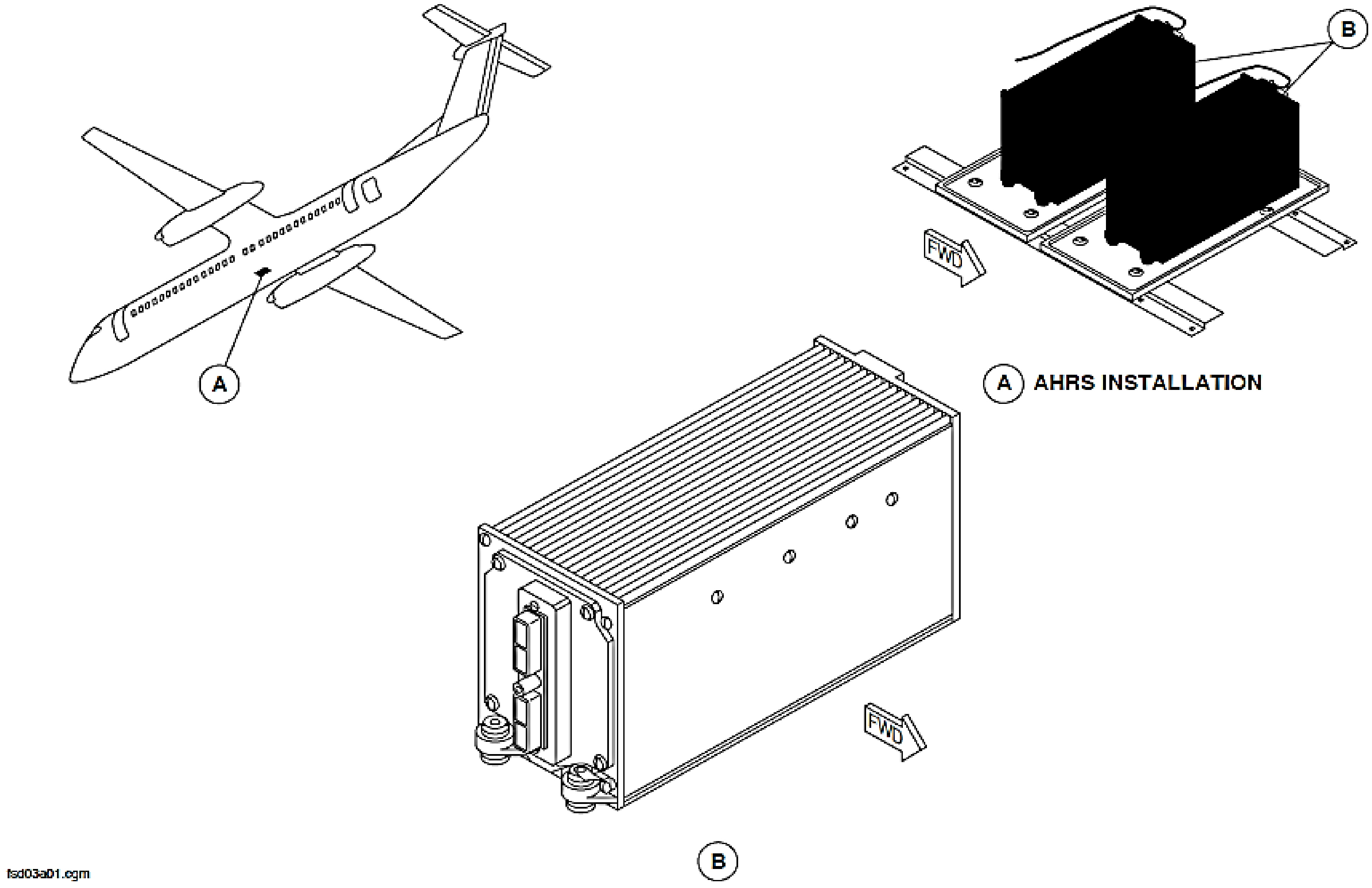


Figure 34- 88 ATTITUDE AND HEADING REFERENCE UNITS LOCATOR

Figure 34- 89 ATTITUDE AND HEADING REFERENCE INTERNAL BLOCK DIAGRAM

The Attitude and Heading Reference Unit (AHRU) has the three electronic modules that follow, installed in a rigid tubular structure:

- Sensor
- Input/Output
- Power Supply.

Sensor Module: The Sensor Module has the components that follow:

- Three−axis Fiber Optic Gyros (FOG)
- Three accelerometers
- Main processor board
- Flux−valve electronic interface board

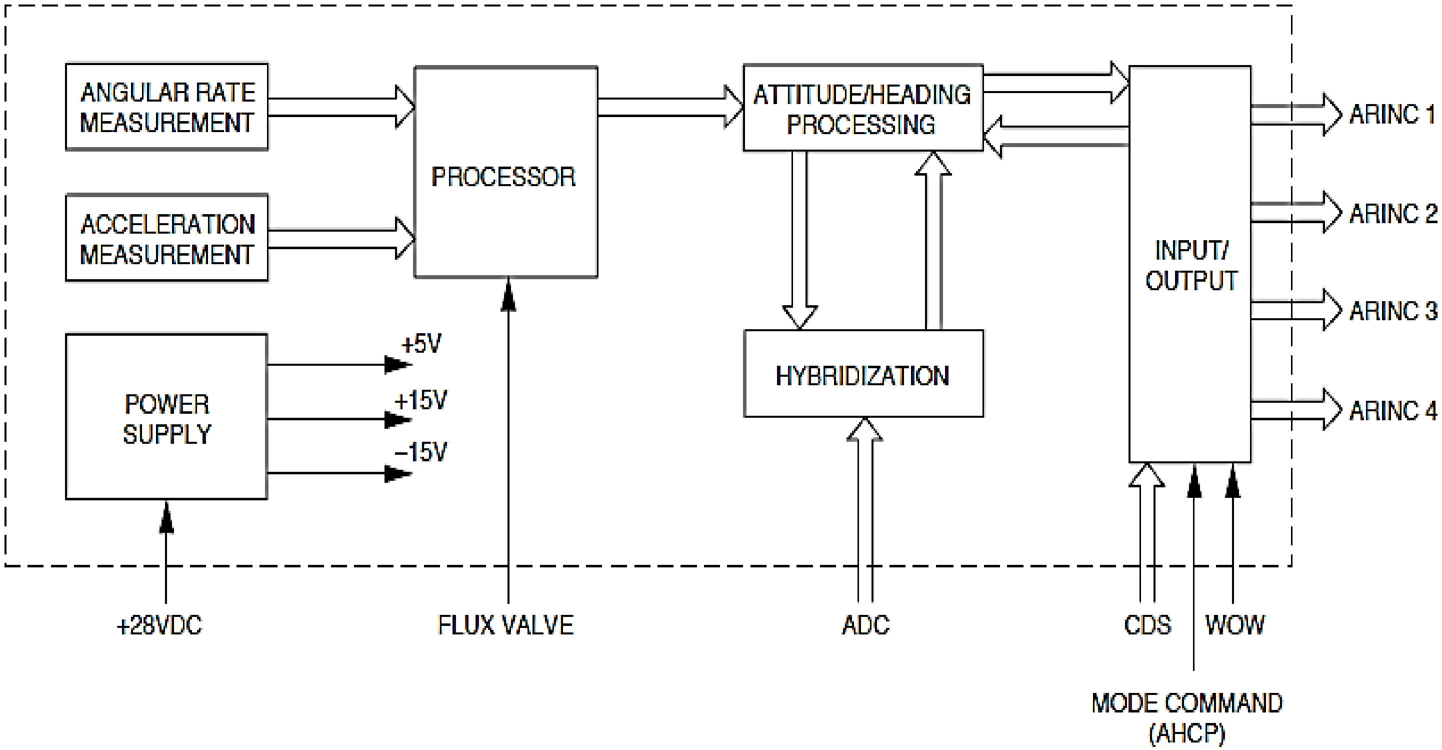


Figure 34- 89 ATTITUDE AND HEADING REFERENCE INTERNAL BLOCK DIAGRAM

Valves, Flux

Figure 34- 90 FLUX VALVES
Figure 34- 91 FLUX VALVES LOCATOR

The flux valve is a two–axis sensor. It measures the horizontal component of the earth's magnetic field.

The flux valve is a metallic, non–magnetic, half–spherical component. It weighs 0.9 lb. (0.41 kg).

One flux valve is installed in each wing tip to minimize magnetic disturbances. The magnetic field at the wing tip is most stable.

The flux valves have reference axes that are aligned with the reference axes of the aircraft.

All flux valves mounting hardware and fasteners located within an 18–inch radius are made of low magnetic permeability material.

The flux valve is attached to a support plate. Three elliptical slots and a white index mark on the support plate lets the flux valve align with the longitudinal aircraft reference axis. Its accuracy is better than ±3 degrees.

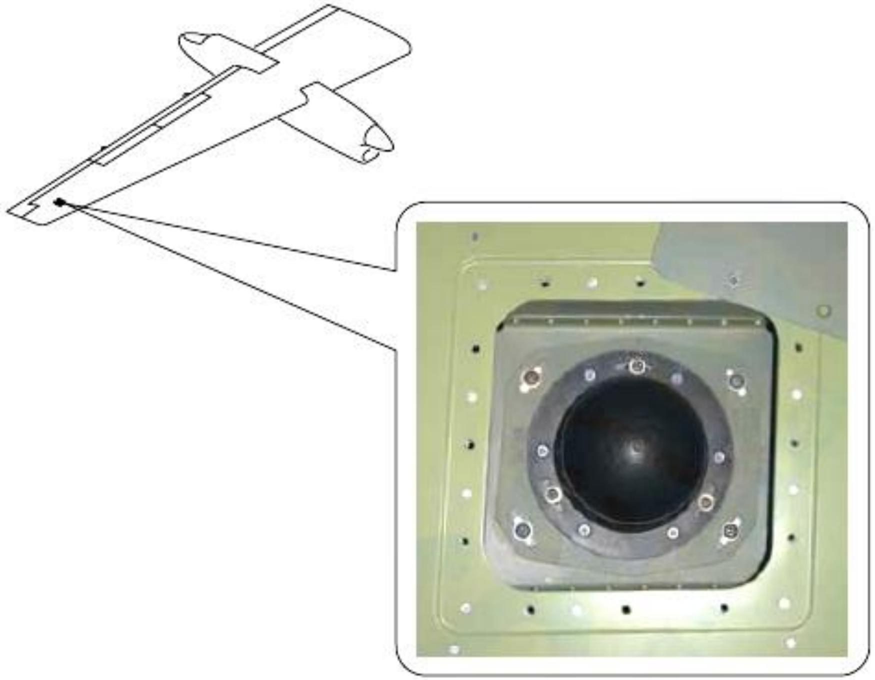


Figure 34- 90 FLUX VALVES

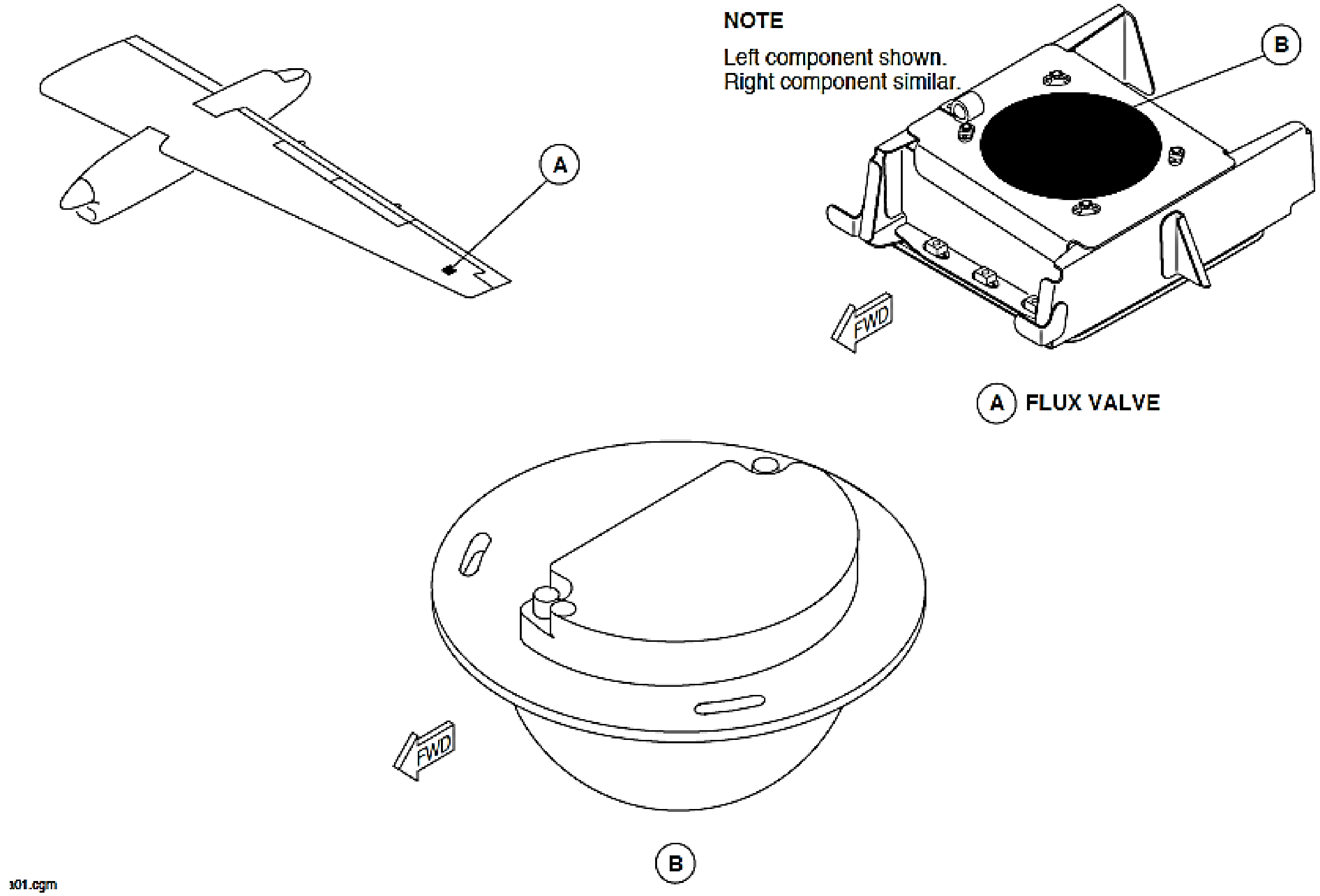


Figure 34- 91 FLUX VALVES LOCATOR

Panel, AHRS Control

Figure 34- 92 AHRS CONTROL PANEL
Figure 34- 93 ATTITUDE AND HEADING REFERENCE UNIT CONTROL PANELS LOCATOR

The AHCP is used to set the Attitude Heading Reference System (AHRS) modes and show its status.

The AHCP weighs 0.5 lb. (0.23 kg). It is attached to the forward part of the center−console with four DZUS fasteners.

The AHRS Control Panel (AHCP) has the assemblies that follow:

- Front Face
- Interface Board Module
- Outer Box.

The AHRS Control Panel (AHCP 1, AHCP 2) has the annunciators and switches that follow:

- ATT/HDG ALIGN annunciator switch
- DG annunciator switches
- DG SLEW push buttons switches
- BASIC annunciator
- SLAVE annunciator.

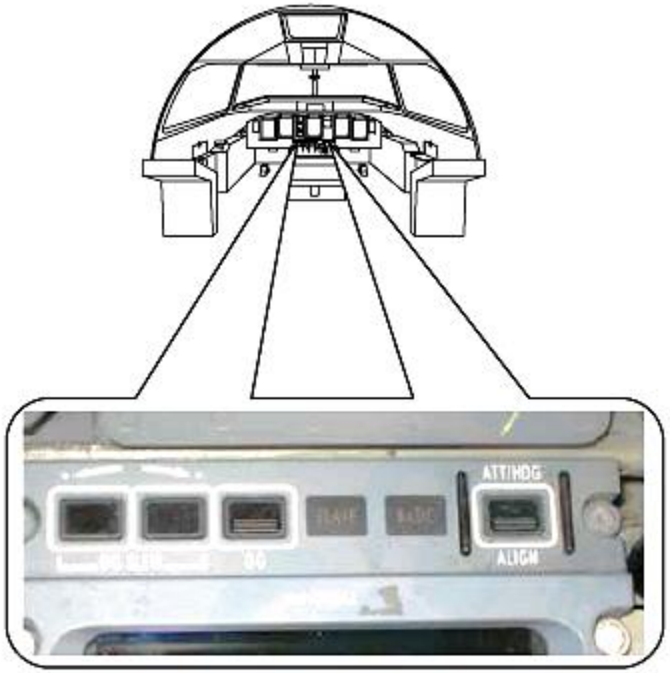


Figure 34- 92 AHRS CONTROL PANEL

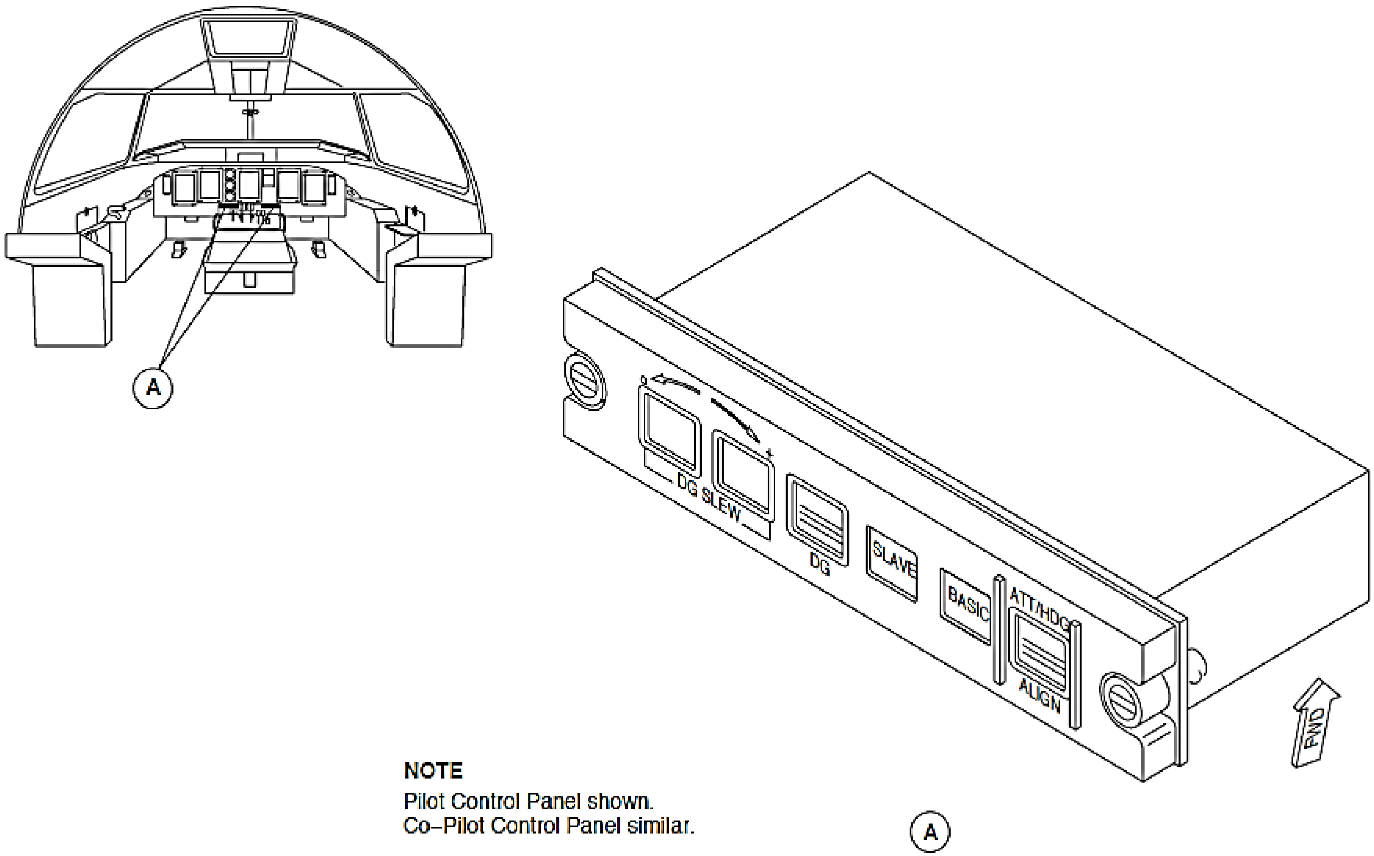


Figure 34- 93 ATTITUDE AND HEADING REFERENCE UNIT CONTROL PANELS LOCATOR

Module, Remote Memory

Figure 34- 94 REMOTE MEMORY MODULES
Figure 34- 95 REMOTE MEMORY MODULES LOCATOR

The Remote Memory Module (RMM) stores the AHRS configuration parameters for the aircraft configuration that follows:

- Magnetic compensation (usually different for each aircraft)
- Location of the AHRU in the aircraft
- Mechanical misalignment of AHRU reference axes relating to the aircraft reference axes
- The permanent orientation of the AHRU in the aircraft.

The AHRU supplies the magnetic compensation parameters to the RMM through a bi−directional RS232 serial data bus. An EEPROM in the RMM stores the magnetic compensation parameters.

The position of a single jumper at the RMM gives the pilot or copilot location in the aircraft.

The AHRU mechanical alignment make sure that the needed system accuracy without additional software compensation. The RMM installation is initially set with no compensation.

The AHRU energizes the RMM. It has no other interface.

The RMM connects to its related AHRU and is secured with two screws. The RMM is also attached to the aircraft by a lanyard to make sure that it can only be connected to its related AHRU. The RMM stays in the aircraft when the AHRU is removed for maintenance.

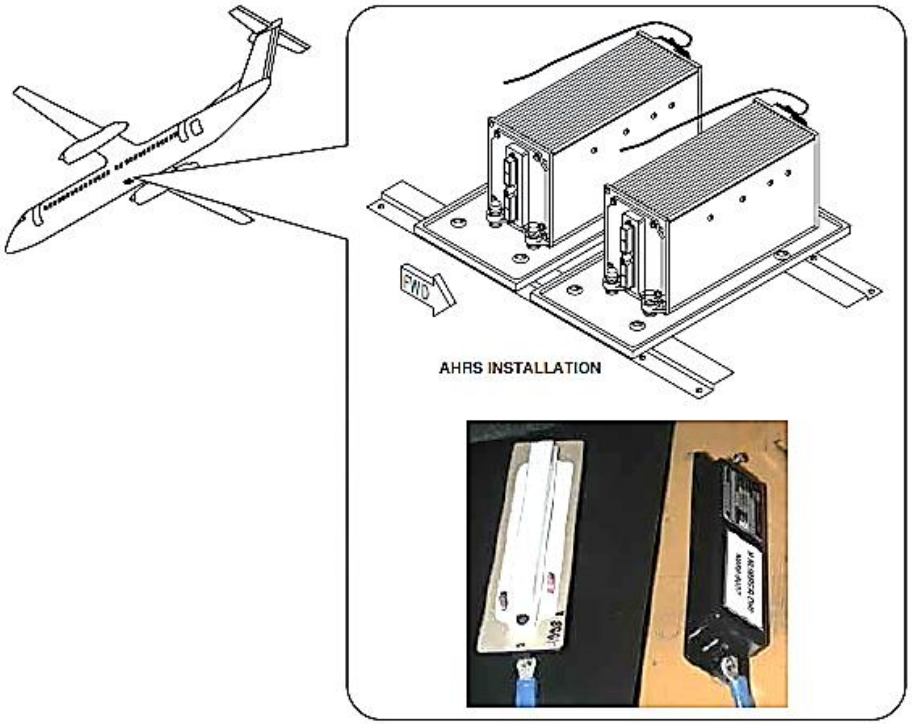


Figure 34- 94 REMOTE MEMORY MODULES

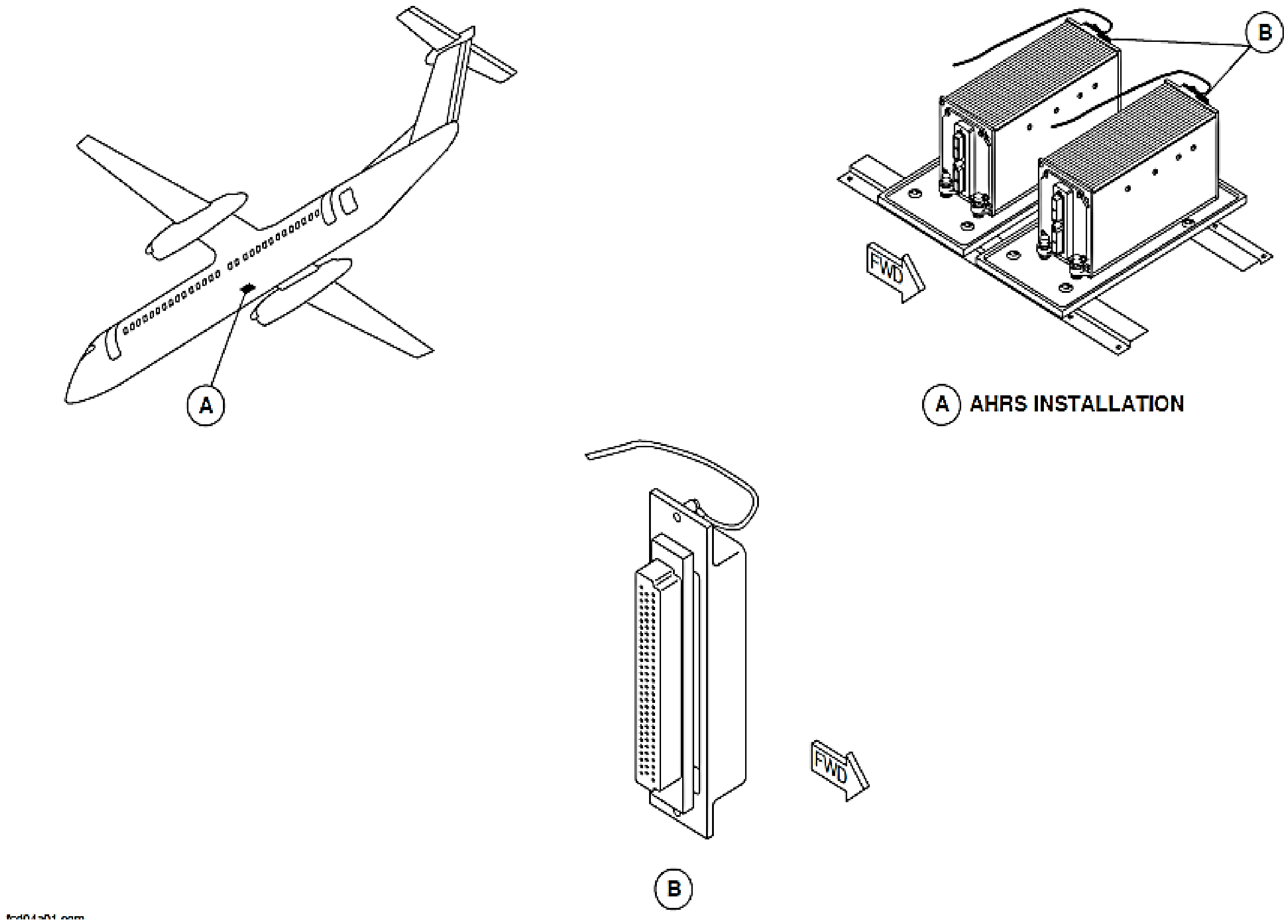


Figure 34- 95 REMOTE MEMORY MODULES LOCATOR

STANDBY ATTITUDE AND HEADING INSTRUMENTS

Introduction

The standby attitude and heading reference system shows attitude and heading information.

General Description

The standby instruments give pitch, roll, and heading indications.

A standby horizon is used to give pitch and roll attitude indications from an electro−mechanical gyroscope.

A standby magnetic compass is used to give a magnetic heading indication.

The standby units operate independently and do not interface with any other systems.

The Standby Attitude and Heading Instrument system has the components that follow:

- Indicator, Standby Horizon
- Compass, Standby.

Detailed Description

The standby horizon indicator has a vertical gyroscope that spins to remain fixed in space. It remains stable with respect to the movements of the aircraft to give an attitude indication. The standby horizon indicator has an attitude sphere with a pitch scale and roll index that moves relative to a fixed airplane symbol and roll scale.

Figure 34- 96 STANDBY HORIZON INDICATOR

The standby horizon indicator shows the components that follow:

- Attitude sphere
- Pitch scale
- Index pointer and roll scale
- Airplane symbol
- Attitude failure flag.

Attitude Sphere: The attitude sphere has a sky permanent sector in blue above a ground permanent sector in brown and a white horizon line to divide the sectors. The attitude sphere moves to show pitch and roll.

Pitch Scale: The pitch scale is shown on the attitude sphere in 5 degree increments from 0 to ±80 degrees.

Index Pointer and Roll Scale: A blue index pointer moves against a white scale that has 10 degree graduations between ±60 degrees.

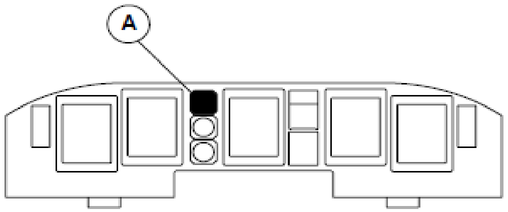
Airplane Symbol: The orange airplane symbol is a fixed aircraft reference against the attitude sphere shows the amount of aircraft pitch and roll.

Attitude Failure Flag: The instrument monitors its gyroscope for speed and its power supply for overcurrent and overheating. The indicator fail warning flag comes into view when the gyroscope speed is less that 18,000 revolutions per minute (rpm) or when a power supply malfunction is sensed. The fail warning indication is an orange flag with white cross−hatches.

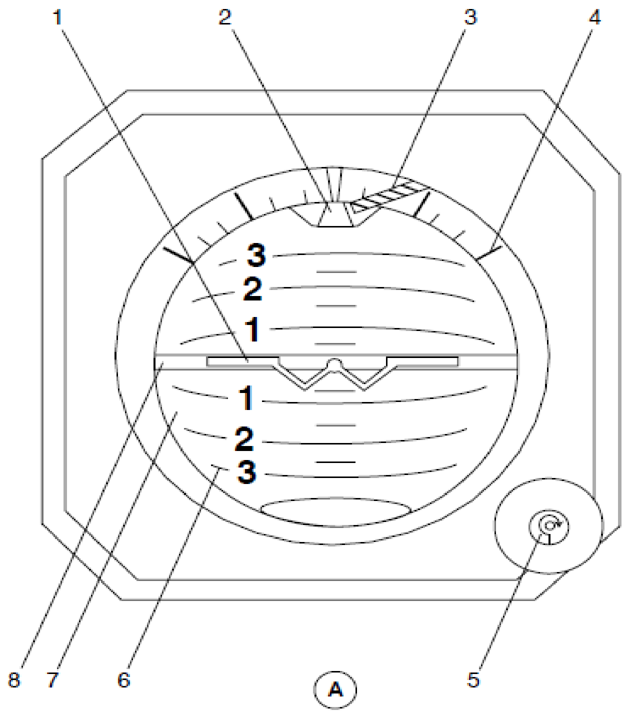
The fast erection mechanism is pulled and slowly released to stabilize the gyroscope in the vertical position after power up. The knob is located at the lower right hand corner of the instrument.

The left and right essential busses supply 28 VDC electrical power through two 3 A circuit breakers to the standby horizon indicator.

The circuit breakers are located in positions H2 on the left circuit breaker panel and A11on the avionics circuit breaker panel. The standby horizon indicator uses 1.5 A at power up and it decreases to less than 0.3 A after 3 minutes.



MAIN INSTRUMENT PANEL



LEGEND

1. Aircraft Symbol.
2. Index Pointer.
3. Attitude Failure Flag.
4. Roll Scale.
5. Fast Erect Knob.
6. Pitch Scale.
7. Attitude Sphere.
8. Horizon Line.

Figure 34- 96 STANDBY HORIZON INDICATOR

Figure 34- 97 STANDBY COMPASS

The standby compass shows the heading of the aircraft. It uses a compass card that is free to turn against a fixed lubber line. The compass card is marked in 10 degree graduations with a digital value every 30 degrees.

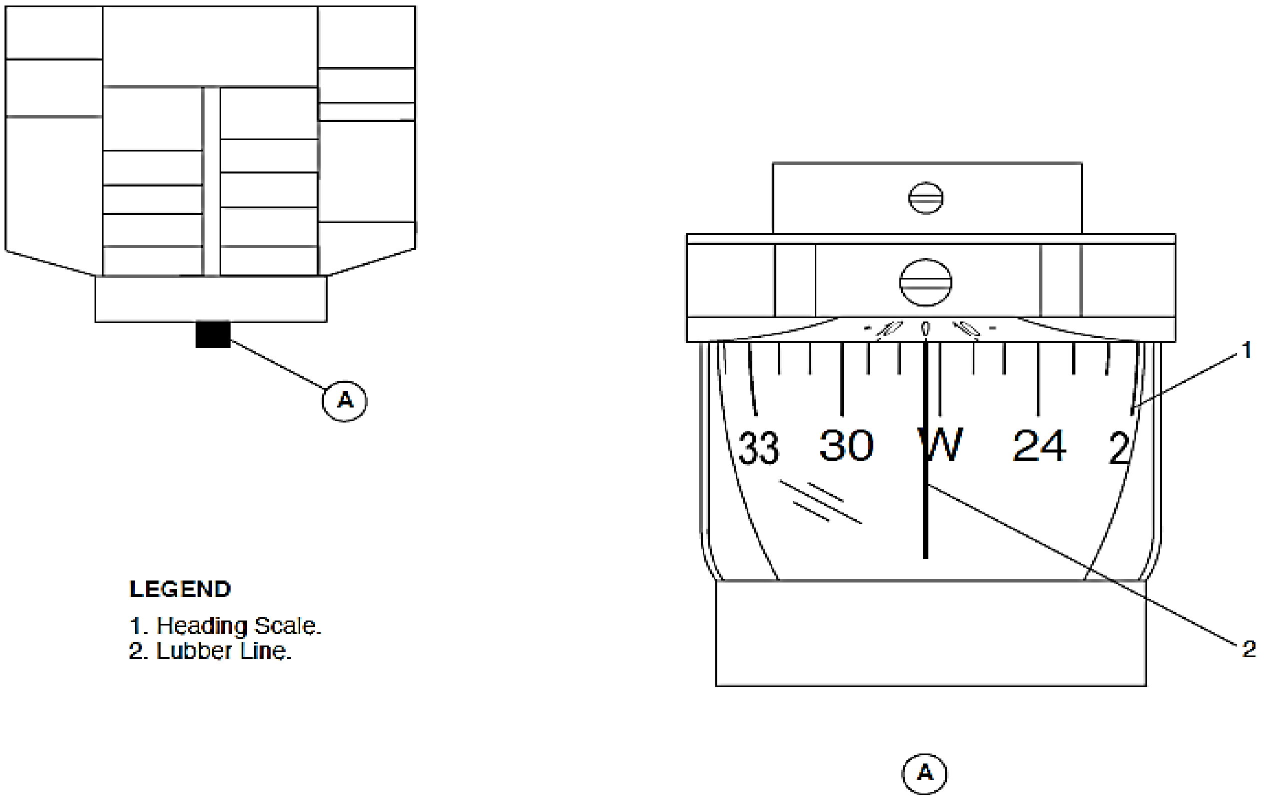


Figure 34- 97 STANDBY COMPASS

Indicator, Standby Horizon

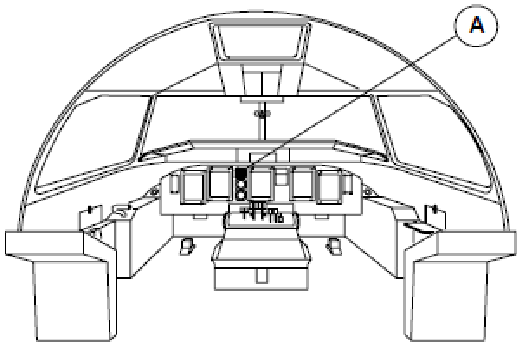
Figure 34- 98 STANDBY HORIZON INDICATOR LOCATOR

The standby horizon indicator has a gimbal mechanism that supports the vertical gyroscope. A three−phase asynchronous motor supplies the rotating movement. A static inverter powered by 28 VDC supplies the motor with 400 Hz VAC.

A mechanical erection mechanism keeps the gyroscope in the vertical axis position. It stops when a significant horizontal acceleration is sensed.

The standby horizon indicator has a glass bezel on its front panel and a black case to protect the mechanisms and electronic circuitry in it. A fast erection is located at the lower part of the front panel to erect the gyroscope and to adjust aircraft symbol.

The standby horizon indicator is located on the instrument panel between Multi−Functional Display 1 (MFD 1) and the Engine Display (ED) above the standby airspeed indicator in view of the two pilots. The standby horizon indication is mechanically adapted to the 19.5 degrees of instrument panel tilt.



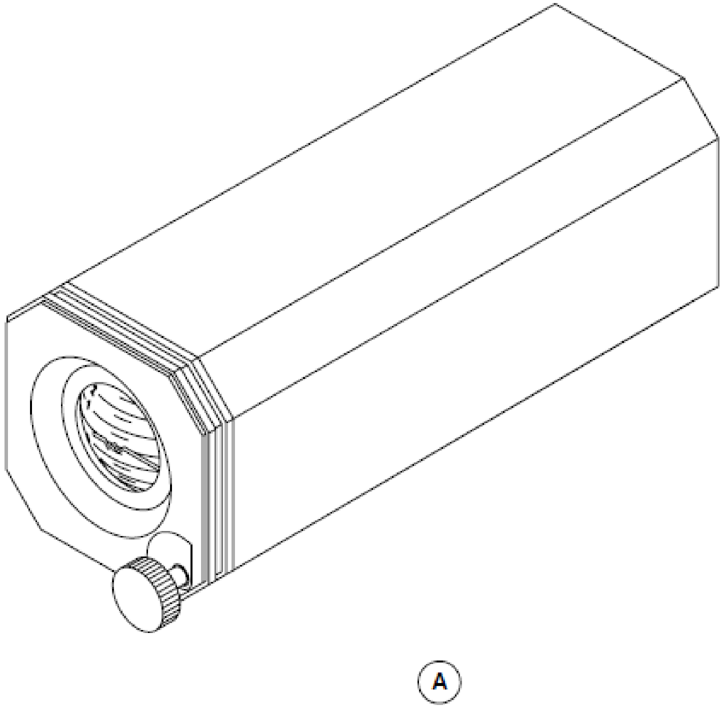


Figure 34- 98 STANDBY HORIZON INDICATOR LOCATOR

Compass, Standby

Figure 34- 99 STANDBY COMPASS LOCATOR

The standby compass gives a third source of magnetic heading. It is used as a reference when the two primary systems malfunction or when the two primary systems are calculating different heading values.

The compass has the assemblies that follow:

- Bowl
- Compensator unit
- Support plate.

Bowl: The semi–spherical bowl is transparent and filled with a stabilizing fluid in which the compass card can rotate. The compass card has directional magnets and graduations to show magnetic heading.

Compensator Unit: The deviation compensation unit is attached to the baseplate above the bowl. It has adjustable north/south and east/west magnets to compensate for the magnetic indication errors caused by magnetism around it. A related screw is turned to adjust the magnets. A white mark is shown above north/south compensator and a red mark is shown above the east/west compensator to show the "zero effect" positions. The compensation can be adjusted to 45 ±15 degrees. A cover protects the adjustment screws.

Support Plate: The bowl is attached to the support plate. A scale in degrees is engraved on the support plate near the heading mark. The bowl can be turned as part of the compensation procedure to change the heading indication. A screw is tightened in the support plate to hold the bowl in place after compensation.

The standby compass is installed with non–magnetic screws to the center structural support of the windshield directly below the Caution and Warning Panel (CWP).

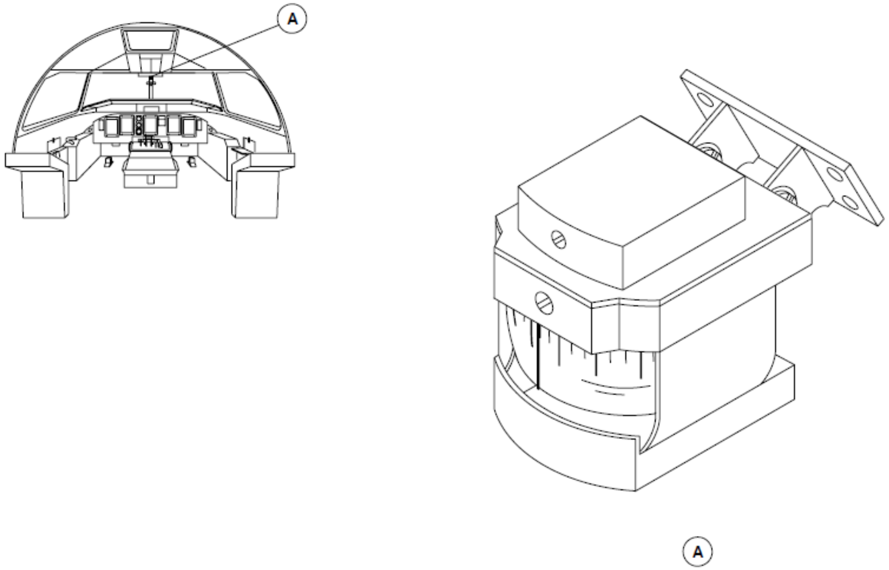


Figure 34- 99 STANDBY COMPASS LOCATOR

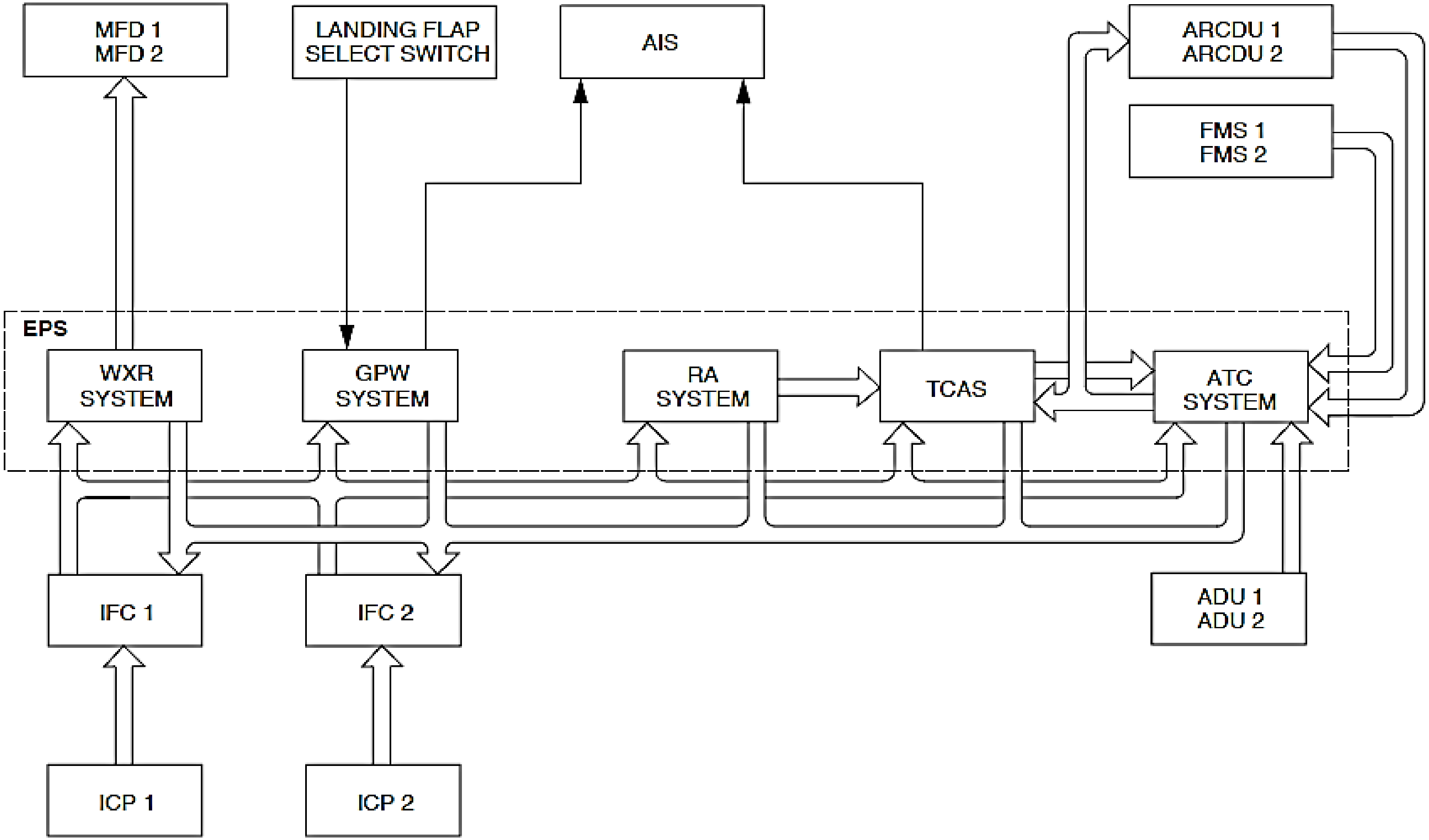


Figure 34- 100 EXTERNAL PROTECTION SYSTEM BLOCK DIAGRAM

EXTERNAL PROTECTION SYSTEM

Introduction

The External Protection System (EPS) has different sub−systems to give visual and aural indications that relate to the conditions that follow:

- Weather
- Ground proximity
- Location and threat status of other aircraft
- Above Ground Level (AGL) altitude.

It also gives other aircraft and ground stations with data relating to the aircraft identification and location.

General Description

Figure 34- 100 EXTERNAL PROTECTION SYSTEM BLOCK DIAGRAM

Each sub−system calculates the operational data to give the necessary visual and aural indications.

The EPS does the functions that follow:

- Monitors the weather conditions and gives mapping capability
- Gives visual and aural indications that related to ground proximity.
- Gives the capability for other aircraft and ground stations to identify the aircraft and locate its position
- Gives visual and aural indications that relate to the location of other aircraft and coordinate and annunciate collision avoidance maneuvers with other aircraft
- Gives AGL altitude data that is used during approach and landing conditions.

The external protection system has the sub−systems that follow:

- Weather Radar Systems
- Ground Proximity Warning System (GPWS)
- Traffic Alerting and Collision Avoidance System II (TCAS II)
- Radio Altimeter.

Weather Radar Systems

The weather radar system helps the pilot to avoid thunderstorms and associated turbulence. When selected, it supplies a continuous visual color display of the weather conditions to the Multi−Function Displays (MFD 1 and MFD 2).

Ground Proximity Warning System (GPWS)

The Ground Proximity Warning System (GPWS) gives aural and visual indications of possible Controlled Flight into Terrain (CFIT).

Traffic Alerting and Collision Avoidance System II (TCAS II)

The Traffic Collision Avoidance System II (TCAS II) gives aural and visual indications when the aircraft's position and flight path relative to an intruding aircraft may cause a potentially dangerous condition.

Radio Altimeter

The Radar Altimeter (RA) system gives Above Ground Level (AGL) altitude data for indication and to other systems.

It does Decision Height (DH) calculations during the approach phase of the flight

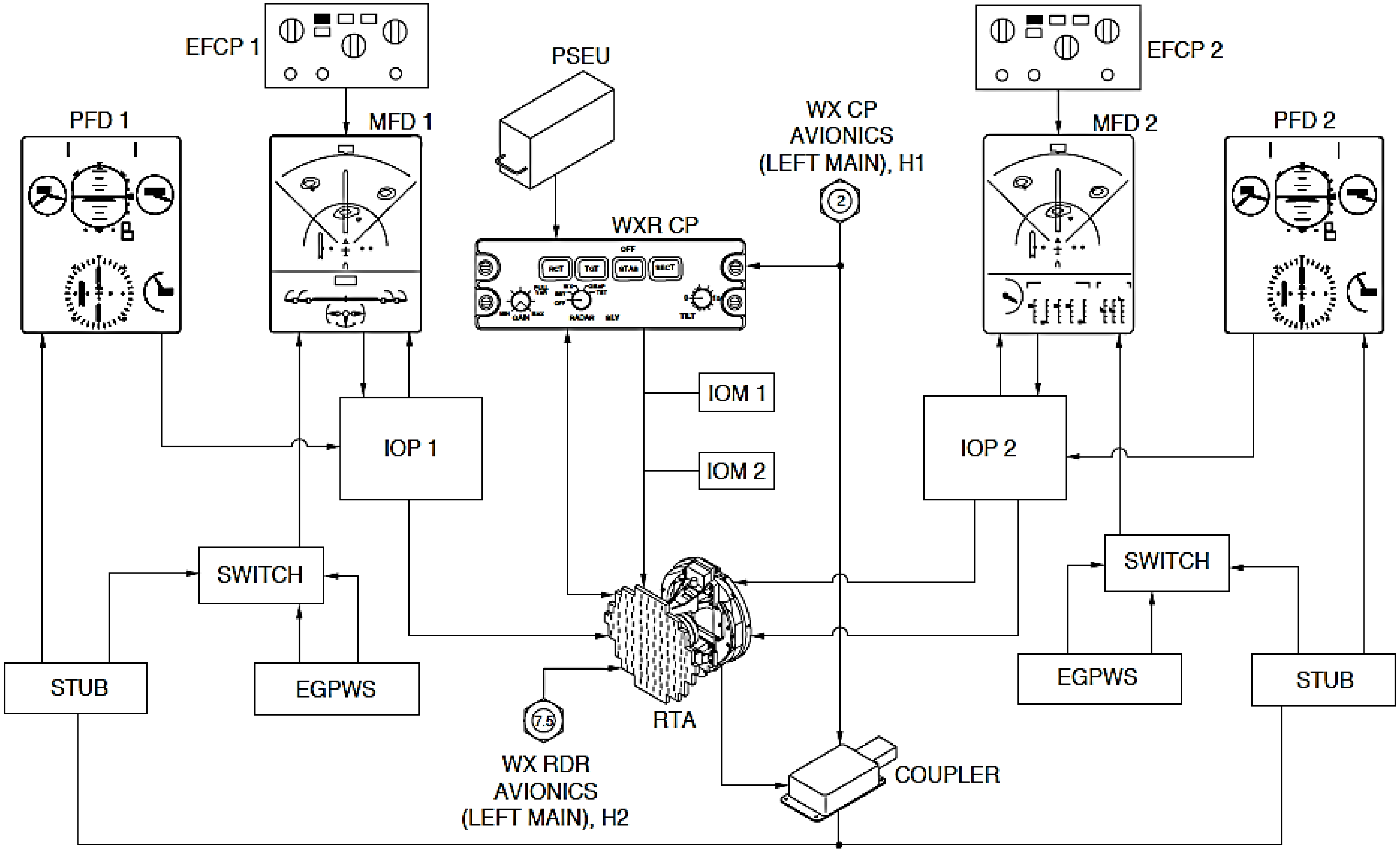


Figure 34- 101 WEATHER RADAR SYSTEM BLOCK DIAGRAM (PRIMUS 660)

WEATHER RADAR SYSTEM (PRIMUS 660)

Introduction

The Primus 660 Weather Radar (WXR) system helps the pilot to avoid thunderstorms and associated turbulence. When selected, it supplies a continuous visual color display of the weather conditions to the Multi-Function Displays (MFD 1 and MFD 2).

General Description

Figure 34- 101 WEATHER RADAR SYSTEM BLOCK DIAGRAM (PRIMUS 660)

The Primus 660 weather radar system is a light weight X−band digital radar with alpha numeric assigned for weather detection and ground mapping. The radar system also supplies the flight crew with operational features such as range selection, ground mapping, target alert, and fault annunciations.

The Primus 660 digital weather radar system detects and locates various types of storms within a 300 nm range ahead of the aircraft, and within an arc of approximately 120 degrees, and provides the pilot with a visual indication of their turbulence content based on the color of the display.

The weather radar system is operated by a control panel on the center console. The weather radar system has the components that follow:

- Receiver Transmitter
- Antenna, Weather Radar
- Panel, Radar Control
- Coupler.

The Primus 660 weather radar system has a forced standby feature that inhibits the transmitter on the ground to eliminate the X−band microwave radiation hazard.

Detailed Description

Radar is a distance measuring system using the principle of radio echoing. The term RADAR is an acronym for Radio Detecting and Ranging. The transmitter sends microwave energy in the form of pulses. These pulses are then transmitted to the antenna and focused into a radar beam. The radar beam is focused and transmitted by the antenna in such a way that it is most intense in the center of the beam and decreases in intensity near the edge. The same antenna is used to transmit and receive radar signals. When a pulse intercepts a target, the pulse is reflected as an echo, or return signal, back to the antenna. From the antenna, the returned signal is transmitted to the receiver transmitter unit. The echoes, or returned signals, are shown on the MFDs.

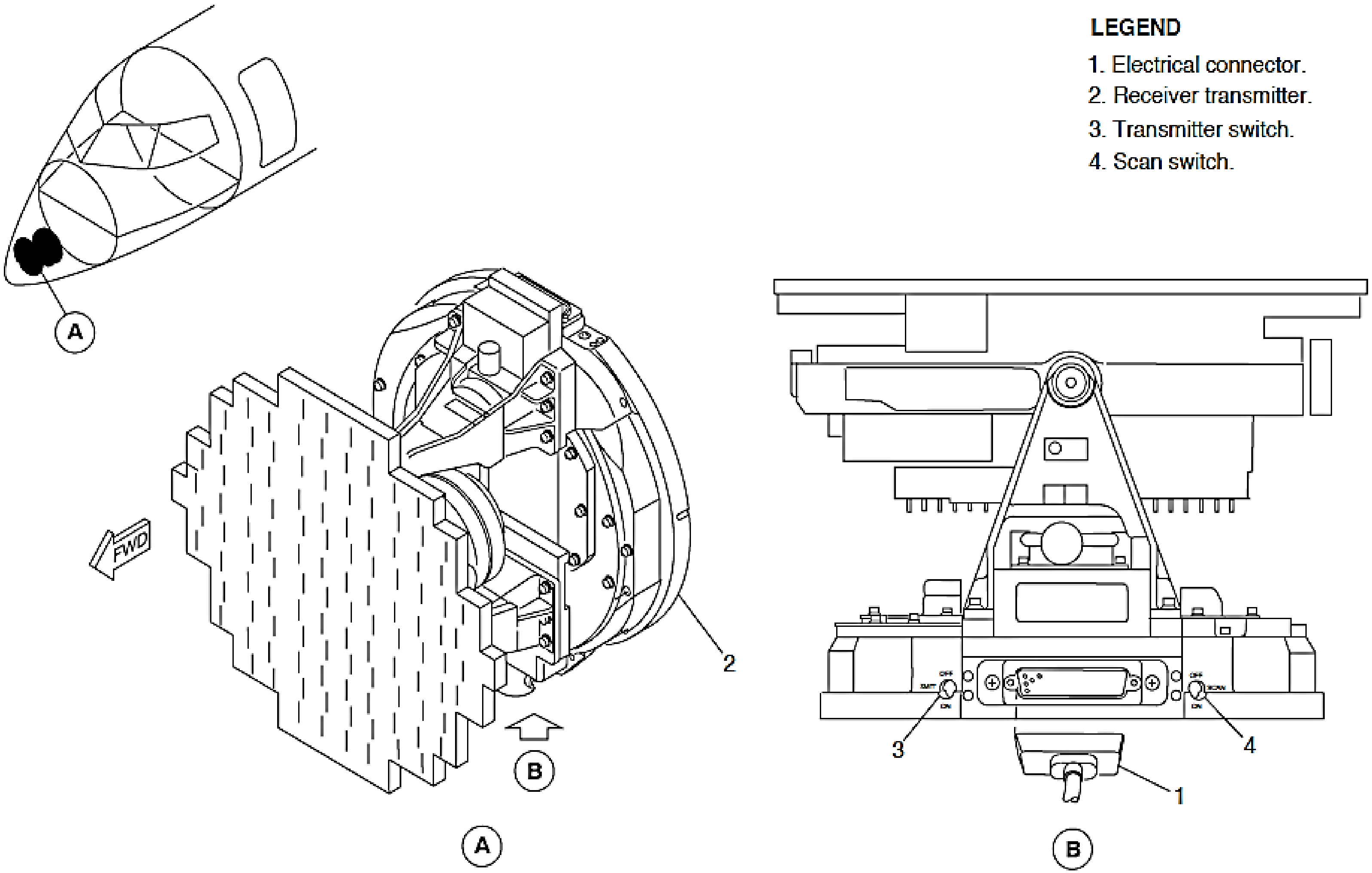


Figure 34- 102 RECEIVER TRANSMITTER (PRIMUS 660)

Receiver Transmitter Antenna (RTA)

Figure 34- 102 RECEIVER TRANSMITTER (PRIMUS 660)

The RTA is an integrated unit that incorporates a transmitter, receiver, and antenna into a single unit. The RTA is cantilever−mounted on the forward side of the bulkhead and protected by the nose cone radome. The bottom of the receiver transmitter antenna also contains maintenance related switches and an electrical connection is provided by a multi−pin connector.

The receiver transmitter antenna operates in the frequency range of 9350 MHz to 9400 MHz with a nominal magnetron power output of 10 kilowatts. The transmitter section delivers short, high−power pulses of radio−frequency energy to the antenna, which radiates the energy as a narrow beam in a direction determined by the antenna scan and tilt positions.

Pulses reflected back to the antenna by weather or ground targets are applied to the receiver section of the receiver transmitter antenna, where they are converted to an intermediate frequency, amplified, processed and applied to the indicator as video signals. Other interconnections provide sync pulses and display information to the indicator. The receiver transmitter antenna receives control information from the indicator through a Serial Control Interface (SCI) data bus.

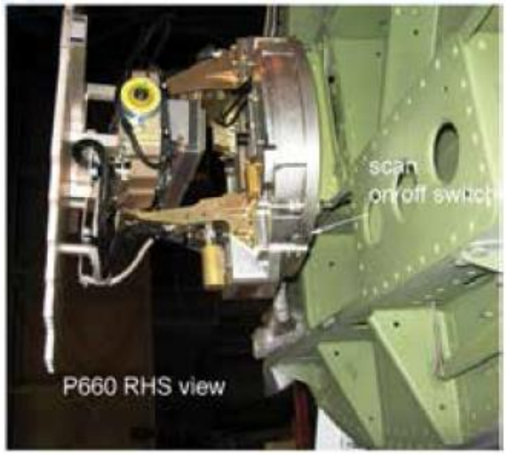
Separate outputs from the receiver provide color video, x/y coordinates, Rho−theta and digital display information to each of the EFIS system, for conversion to the display format required by the EHSI instruments.

The antenna is an 18 in. (45.72 cm) diameter flat plate phased array combining receive and transmit signals of the radar system. The RTA is mounted on the pedestal assembly which provides antenna movement in the vertical and horizontal planes. The receiver transmitter antenna contains the electromechanical drive components and feedback circuits for antenna rotation.

Logic drive signals from the receiver−transmitter operate the azimuth motor to sweep the antenna horizontally right and left for a total sweep of 60 or 120 degrees, as set by the control on the control panel. The radar scan rate is 24 sweeps per minute for the 60 degree sweep, and 12 sweeps per minute for the 120 degree sweep.

The receiver transmitter antenna circuits include antenna stabilization, data processing and timing, and regulated and unregulated dc voltage power supplies for operation of the radar system. In addition to the transmitting and receiving functions, the receiver transmitter antenna contains self−test circuits.

Vertical adjustment of the antenna is actuated by combined signals from the tilt control, signals from the vertical gyro output of the AHRS system and the airspeed and altitude signal from the air data computer. These signals permit the radar antenna to be tilted up or down 30 degrees. In normal flight, the antenna is automatically maintained in a vertical plane when the tilt control is selected to zero, or at a selected tilt angle, by servo and stabilization circuits, referenced to the AHRS pitch and roll outputs.



Multi-Functional Display

Figure 34- 103 MFD- WEATHER RADAR INDICATIONS

The MFD provides the visual displays of weather and ground mapping targets.

Each digital display cell can be black or one of four colors, dependent on the intensity of the returned signal. In weather detection mode, target returns are displayed at one of five video levels (0, 1, 2, 3, and 4). Zero is represented by a black screen because of weak or no returns, and levels 1, 2, 3, and 4 are represented by green, yellow, red, and magenta respectively to show progressively stronger returns. In ground mapping mode, video levels of increasing reflectivity are displayed as black, green, yellow, and magenta.

A 60 degree or 120 degree sector ahead of the aircraft can be shown, selectable by the SECT push−button on the Radar Control Panel (RCP). The distance of the displayed targets from the aircraft is indicated by five range marks. In addition to range, the relative position of the target in azimuth may be indicated by azimuth markers which, when selected by the AZ switch, are spaced at 30 degree intervals across the display. The antenna tilt angle, operating mode, range selection and other operating parameters are also displayed.

The radar has a forced standby function (FSTBY) a signal from the PSEU (weight on wheels) forces the radar in a standby mode on landing.

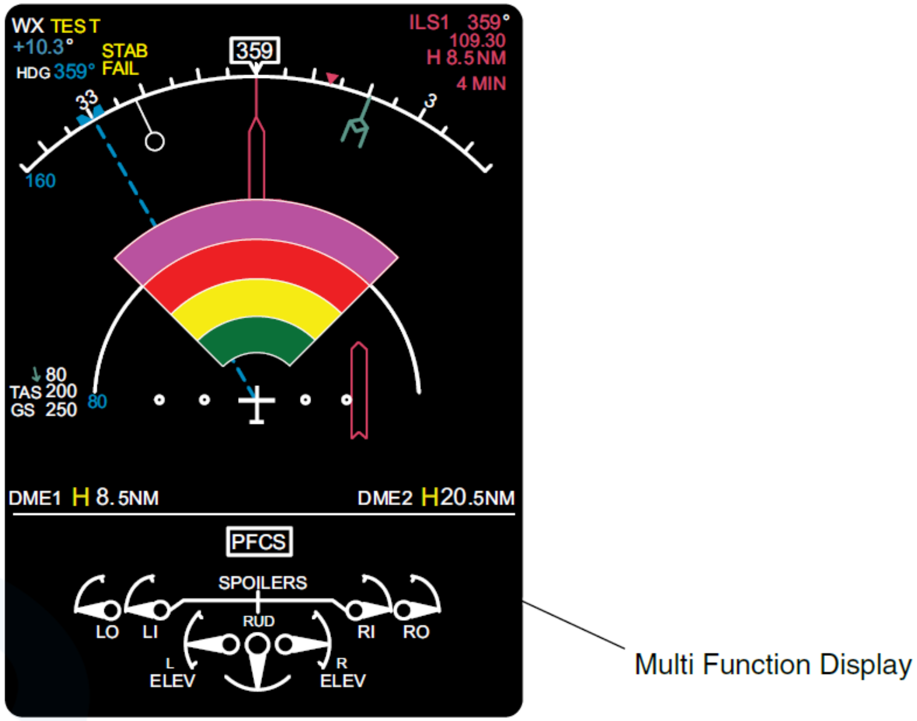


Figure 34- 103 MFD- WEATHER RADAR INDICATIONS

Weather Radar Coupler

Figure 34- 104 WEATHER RADAR COUPLER
Figure 34- 105 WEATHER RADAR COUPLER (PRIMUS 660)

The coupler is a transformer and is located in the wiring harness. It routes display data from the single receiver/transmitter to the two MFDs. Thus, the coupler acts as an interface between the weather radar and the MFD over the ARINC 453/708 bus.

The A453/708 frame transmitted by the PRIMUS 660 radar contains an additional bit before the syncro start bit of the frame. This additional bit prevents the MFD from decoding the ARINC 453/708 frame. Thus, no radar data are displayed on the EFIS when the EFIS is directly connected to the radar.

The coupler fetches the data frames, detects the synchro start bit and removes the additional data that was present before the synchro bit. The coupler does not alter the other portion of the input data frame transmitted by the radar.

The coupler has a continuous BIT capability that verifies its operation. If the coupler detects a failure or if no ARINC 453/708 frames are detected at the coupler input, a red LED located on the top of the unit will illuminate.

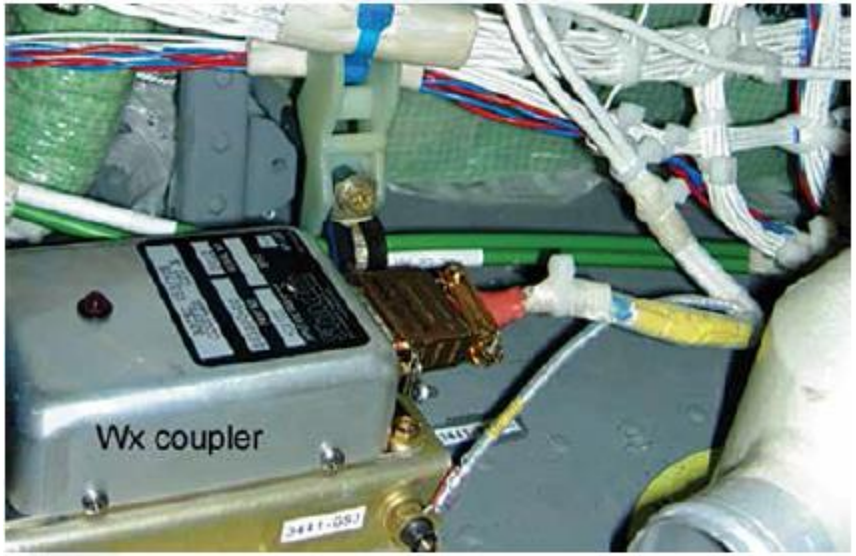


Figure 34- 104 WEATHER RADAR COUPLER

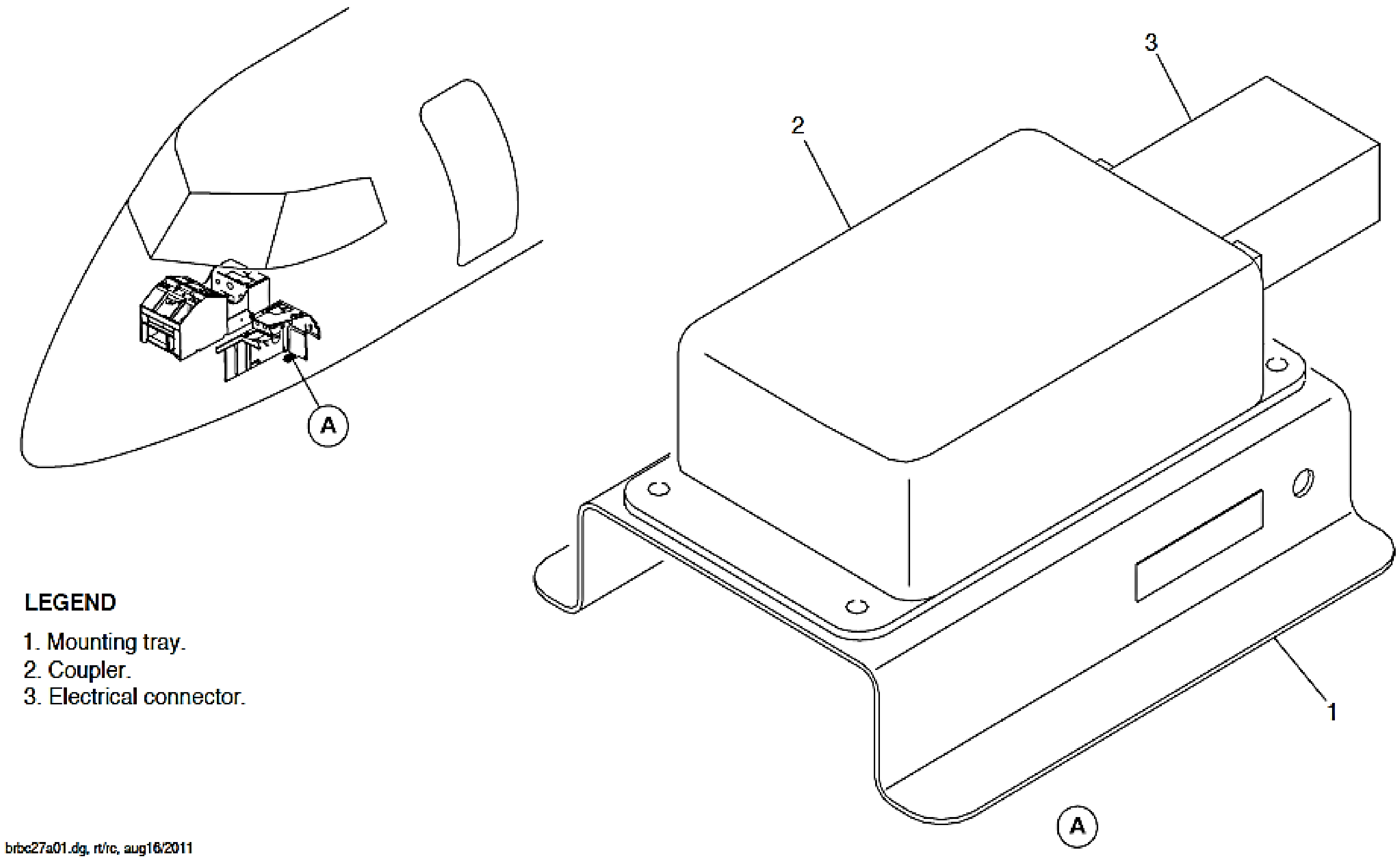


Figure 34- 105 WEATHER RADAR COUPLER (PRIMUS 660)

Power Supply

The RTA is powered from the left main bus, through a 7.5 A circuit breaker (WX RDR). The WX RDR CB is located on the 28 Vdc avionics circuit breaker panel at position H2. The RCP and the coupler is powered from the aircraft left main bus, through a 2 A circuit breaker (WX CP). The WX CP CB is located on the 28 Vdc avionics circuit breaker panel at position H1.

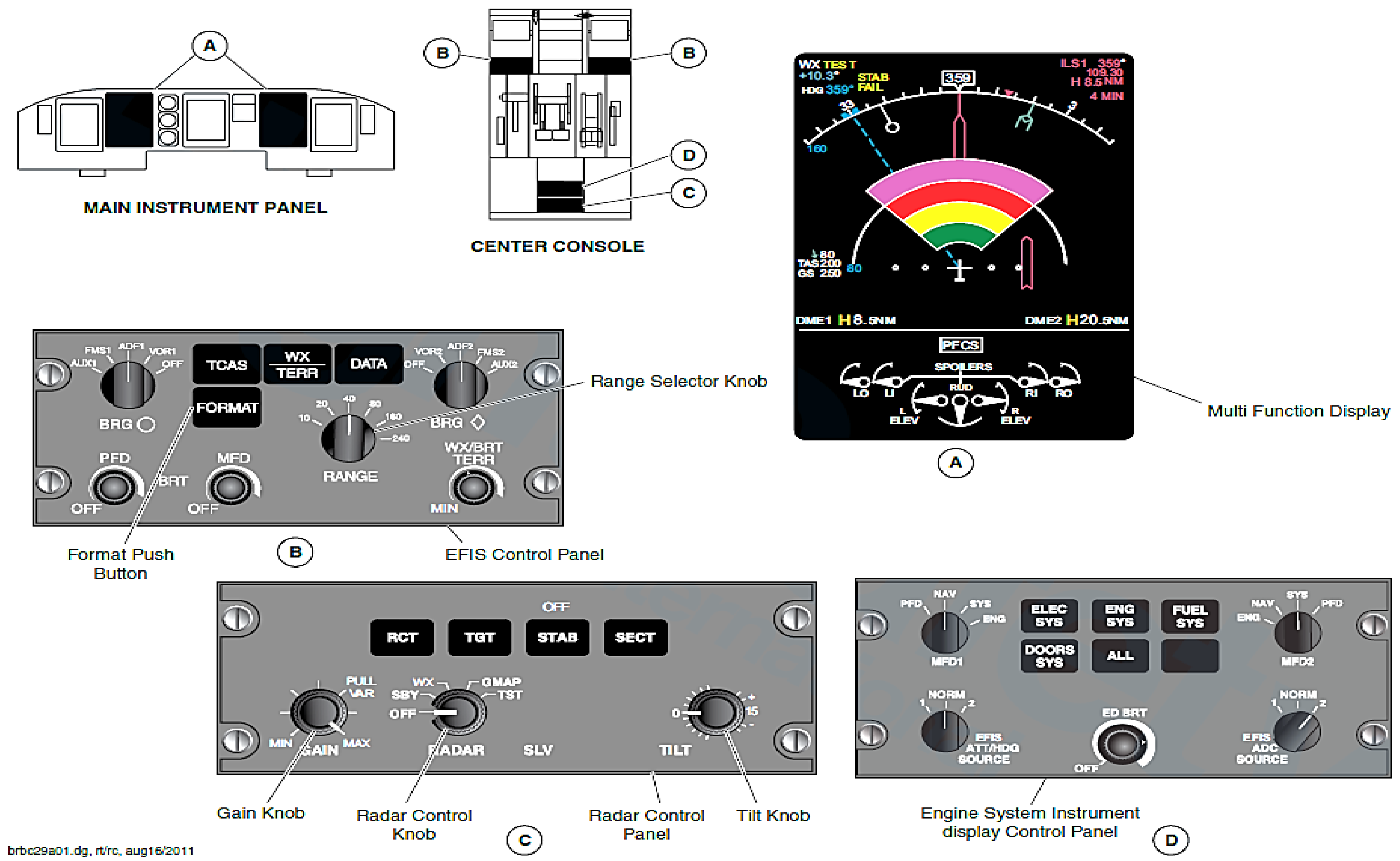


Figure 34- 106 WEATHER RADAR SYSTEM (PRIMUS 660)

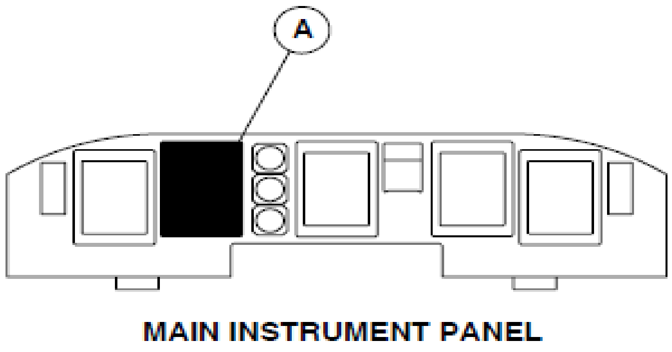
Operation and Operating Controls

Operation of the radar system and the display is controlled by four push–buttons and three rotary potentiometer switches, located on the RCP. The display and various indications are directly related to these controls.

OFF

Figure 34- 107 WEATHER RADAR OFF MODE INDICATION

- Selection of the WXR system in OFF mode is made by the control panel rotary switch. This position turns the radar system off.
- No radar transmissions will occur when the OFF mode is selected and WX OFF will be annunciated in white text on each MFD. When the weather radar is set to the OFF mode, the transmitter/receiver and the antenna scan are inhibited with the stowed in a tilt–up position.



LEGEND

1. Weather Radar Off Mode Indication.

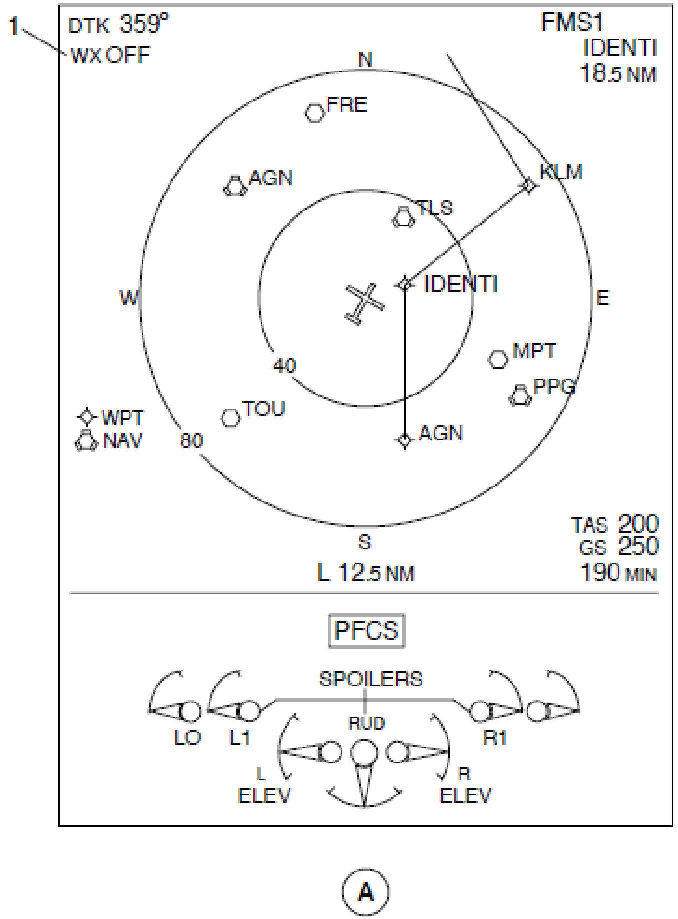


Figure 34- 107 WEATHER RADAR OFF MODE INDICATION

For power up, SBY or TST mode is selected. When power is applied, WAIT (in white) is annunciated in the upper left corner of the MFD for 40 to 100 seconds while the magnetron warms up.

Forced Standby (FSBY)

- The FSBY is an automatic non–selectable safety mode that inhibits the radar system when the aircraft is on the ground. When airborne, the system switches from the forced standby mode to the mode set on the function switch.
- In the forced standby mode, the transmitter and the antenna scan are inhibited, the display is erased from the indicator and the FSBY legend is displayed.
- To override the forced standby mode, the STAB push–button must be pushed four times, in less than three seconds. The system then switches from the forced standby mode and operates normally.

SBY (Standby)

- This position puts the radar system in standby, a ready state with the antenna stopped, the transmitter inhibited, and the display memory erased. A white WX STBY message is displayed. If STBY is set before the RTA warm–up period is over (about 100 seconds), a white WAIT is displayed in the mode field. When the warm–up period is over, the system automatically displays STBY on the MFD.
- The transmitter/receiver electronics within the RTA is powered and RTA mode selection and status data is transmitted to each MFD by the RTA data bus output. No radar transmissions will occur, when the STBY mode is selected. WX STBY will be annunciated in white text on each MFD if the radar image display is selected by the related EFCP WX push–button switch. When the SBY mode is selected, the RTA remains active and the antenna will be stowed in a tilt–up position.

WX (Weather)

- This mode puts the radar system in the weather mode of operation. The weather mode is used during storm seeking. The radar displays a target storm based on the back–scatter intensity received from the rainfall in the storm.
- Since intensity varies in each storm cell, the digital map–form colors of the display also vary. Areas of heaviest rainfall will show in magenta, the next lower level in red, the next lower level in yellow, the next lower level in green, and the least or no rainfall in black. The radar colors, rainfall rates and the intensity levels (in dBZ) are shown in Table 34-8 .

Table 34- 8 RADAR COLORS, RAINFALL RATES AND DBZ LEVELS

DISPLAY LEVEL	RAINFALL RATE (MM/HR)	RAINFALL RATE (IN.)	DBZ	CALIBRATED RANGE (NM) FOR 18 IN. ANTENNA IN 200 NM RANGE
4 Magenta	>50	>1.92	>50	290
3 Red	12–50	0.45–1.92	40–50	170
2 Yellow	4–12	0.14–0.45	32–40	116
1 Green	1–4	0.04–0.14	23–32	75
0 Black	<1	<0.04	<23	–

GMAP (Ground Mapping Mode)

- This mode puts the radar system in the ground mapping mode. The message WX GMAP is displayed in white on the MFD. When in this mode, the amount of reflected RF from ground surfaces is displayed in magenta, yellow, cyan or black from the most reflective to the least reflective surfaces. The pilot can therefore identify coastlines, hilly or mountainous regions, cities, or even large structures.
- In this Mode, RCT and TGT cannot be set.

TST (Test Mode)

Figure 34- 108 WEATHER RADAR TEST MODE INDICATION

- This position puts the radar system in the test mode. Transmitter output power can be radiated. When TEST mode is selected, the message WX is displayed in white, TEST in yellow, and the test pattern is shown in multi colored arc display. When the TEXT FAULT feature is enabled, the status text messages for the crew use, fault codes, transmit status, maintenance messages, fault types and strap codes are displayed.
- In this mode, the radar is in a non−transmitting stage.

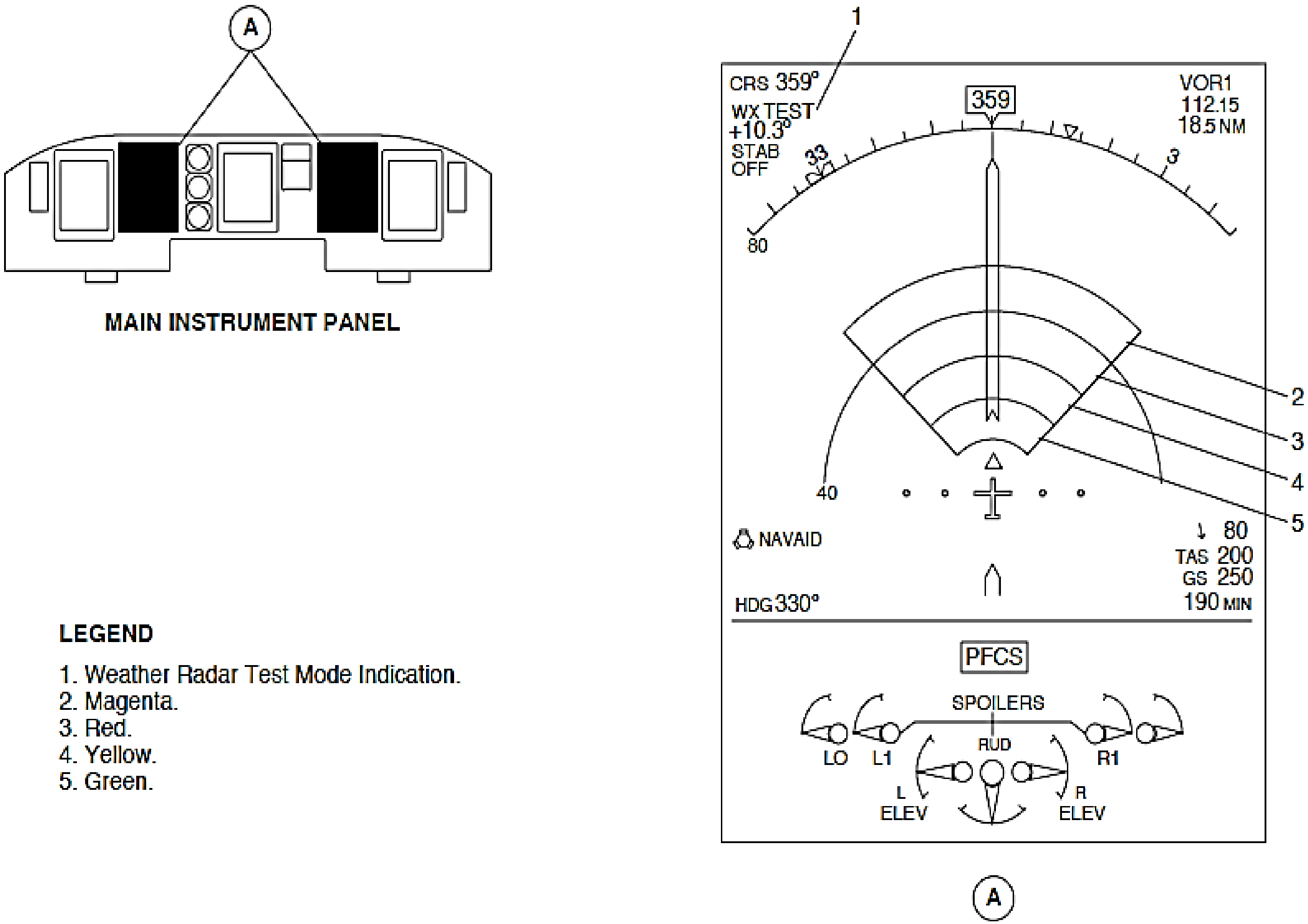


Figure 34- 108 WEATHER RADAR TEST MODE INDICATION

TILT

- This is a rotary control that is used to set the tilt angle of the antenna beam with relation to the aircraft horizontal attitude. Clockwise (cw) rotation tilts the antenna upward 0 to 15.9 degrees, counter−clockwise (ccw) rotation tilts the antenna downward 0 to −15.9 degrees. The most used range is between +5 and −5 degrees and is expanded for ease of operation. Tilt angle is displayed in cyan in 0.5 degrees increments.

GAIN

- This is a rotary push−pull control that is used to adjust the sensitivity of the radar receiver. When the knob is pulled out, variable gain is activated and the message GAIN MAIN is annunciated in white on the screen. When the knob is pushed in, the preset calibrated gain mode is activated and the gain of the receiver cannot be adjusted by rotation of the GAIN control. Calibrated gain is the normal mode for weather avoidance. In the GMAP mode the calibrated gain can also be used to resolve nearby strong target signals while ground mapping.
- Sensitivity increases with a clockwise rotation, the receiver gain is increased to the maximum when the gain control is at the full clockwise position. Setting RCT or TGT forces the system into calibrated gain.

RCT (Rain Echo Attenuation Compensation Mode)

- This is a momentary−contact alternate action push−button that turns on or off the rain echo attenuation compensation mode. Pushing the RCT switch enables the Rain Echo Attenuation Compensation Technique (REACT) circuitry. RCT mode can only be set with the weather mode and a white WX RCT is displayed on the upper left corner of the MFD. RCT mode is active in all ranges. A subsequent push on the RCT switch disables the function.

- The RCT function adjust the calibration of the radar receiver automatically to compensate the attenuation loses. The adjustment is done by measuring of the signals and deducing from the attenuation through the target, then using this information to adjust the calibration appropriately. This adjustment is continuously done on each radar azimuth radial. When the maximum value to which the sensitivity can be set due to inherent noise generated in the receiver is reached, the REACT compensation ceases. At this point, a cyan field is displayed to indicate no further compensation is possible. As the cyan field annunciates at the end of the calibrated gain, any target that occurs in the cyan area is displayed in magenta (very strong).

TGT (Target Alert)

- This is a momentary−contact alternate action push−button that turns on or off the target alert feature. When set, the message TGT is displayed in black box continuously flashing in magenta in the top centre of the radar map area. The target mode (WX TGT) is displayed in white in the upper left corner of the MFD. The TGT feature advises the pilot of any potentially hazardous targets directly in front of the aircraft outside the selected range.
- The target alert can be set in all ranges.
- In order to declare the target alert, the target must have specific depth and range characteristics, which are shown in

Table 34- 9 TARGET ALERT ACTIVE RANGE

SELECTED RANGE (NM)	TARGET DEPTH (NM)	TARGET RANGE (NM)
10	5	10–60
20	5	20–65
40	5	40–85
80	5	80–125
160	5	160–204
240	5	240–270

STAB Selector

- The radar is normally attitude stabilized. It automatically compensates for pitch and roll maneuvers. Attitude stabilization can be turned off by pushing the STAB control knob. The STAB OFF annunciator is displayed on the screen. Stabilization can be turned on when the STAB control knob is pushed in.
- The STAB control is also used to override the forced standby mode. When the STAB control knob is pushed four times in three seconds the system will switches from the forced standby mode and operates normally. The knob is also used to operate the hidden modes for in−flight analysis.

SECT (Scan Sector)

- This is a momentary−contact alternate action push−button that is used to set the antenna scan sector angle of either 120 or 60 degrees. There is a 12 scan per minute for the full 120 degrees scan sector and for a faster update a 24 scan per minute for the 60 degrees scan sector. Antenna sweep and display automatically adjusts to the set position.

EFIS (Electronic Flight Instrument System) Control Panel

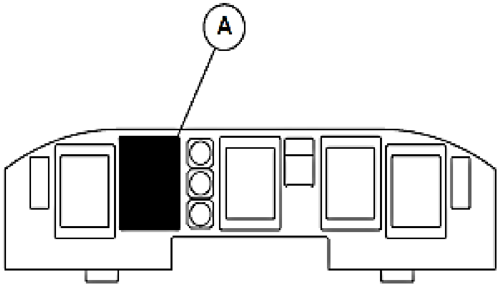
Figure 34- 106 WEATHER RADAR SYSTEM (PRIMUS 660)

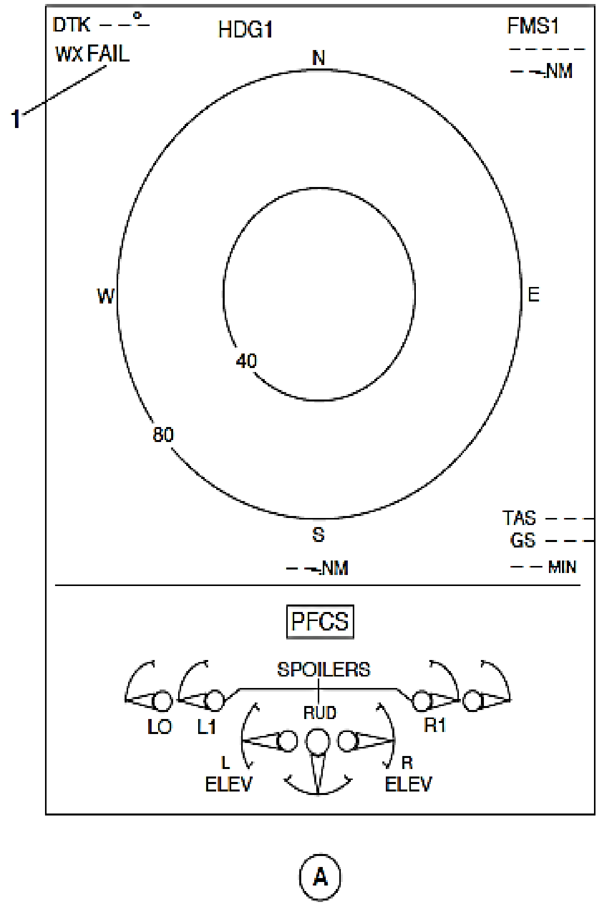
- Some of the WXR system related controls (i.e. WXR display brightness, range selection, arc/full navigation display mode selection) are controlled by the EFIS and are accessible to each flight crew member by their related EFIS Control Panel (EFCP).
- The EFCP has front−panel push buttons, rotary switches and controls. The EFCP provides radar range selection (RANGE selector) and the option of selecting the display of weather or terrain (or neither) (WX/TERR push−button). The WX/TERR push−button controls display of the radar image or the EGPWS terrain image in all formats except PLAN. The default function is WX (weather radar) in ON position with TERR (terrain) in the pop−up mode.

Fault Monitors

Figure 34- 109 WEATHER RADAR FAIL ANNUNCIATION

- Critical functions in the receiver transmitter antenna is continuously monitored for faults. The detected faults and the brief text messages are displayed in the test pattern area on the MFD.
- All faults are stored in a nonvolatile memory. If the word FAIL shows on the display, in the ON mode, this means that a failure message is been presented on the maintenance page.
- The fault display format is shown on the MFDs and on the EFIS.





LEGEND

1. Weather Radar Fail Annunciation.

Figure 34- 109 WEATHER RADAR FAIL ANNUNCIATION

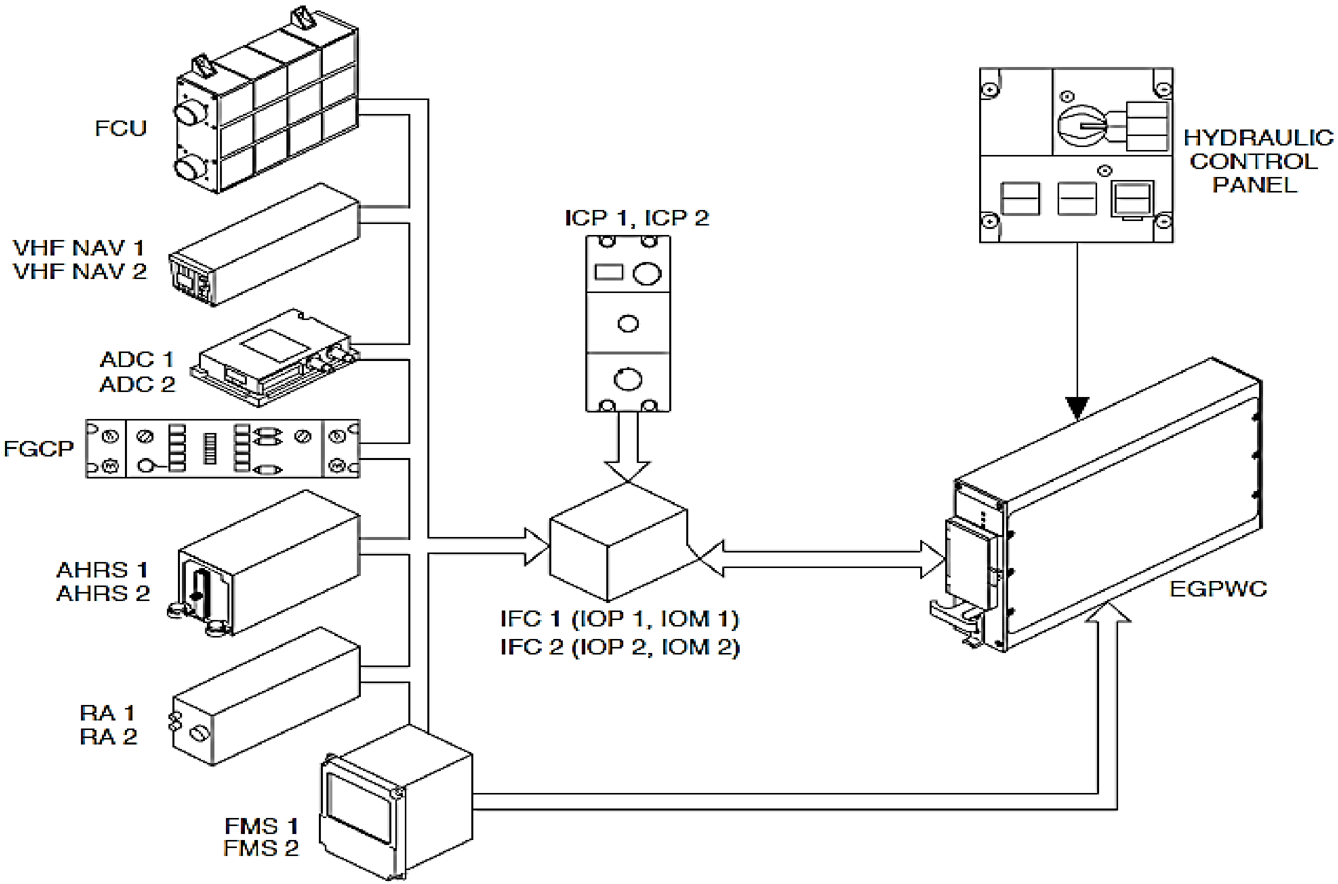


Figure 34- 110 EGPWS BLOCK DIAGRAM, INPUTS

ENHANCED GROUND PROXIMITY WARNING SYSTEM (EGPWS)

Purpose

The Enhanced Ground Proximity Warning System (EGPWS) gives aural and visual indications of possible Controlled Flight into Terrain (CFIT).

General Description

Figure 34- 110 EGPWS BLOCK DIAGRAM, INPUTS
Figure 34- 111 EGPWS BLOCK DIAGRAM, OUTPUTS

The EGPWS has the units that follow:

- EGPWS computer
- FLAP OVERRIDE annunciator switch
- PULL UP and GS annunciator switch
- INHIBIT annunciator switch
- Switching relay.

The EGPWS gives different types of indications that follow:

- Basic Ground Proximity Warning
- Terrain Clearance Floor Alerting
- Terrain Alerting
- Terrain Awareness Display Mode.

Basic Ground Proximity Warning: The EGPWS makes the basic GPWS calculations that follows:

- Mode 1 – Excessive descent rate
- Mode 2 – Excessive closure rate to terrain
- Mode 3 – Descent after takeoff
- Mode 4 – Insufficient terrain clearance
- Mode 5 – Excessive descent below the glideslope
- Mode 6 – Advisory callouts.

Mode 1 – Excessive Descent Rate: The excessive descent rate mode gives audio and visual indications for excessive descent rates into terrain. A “SINK RATE” message sounds and the PULL UP annunciator light comes on. The sensitivity of the mode changes with glideslope position and when the GPWS FLAP OVERRIDE annunciator switch is set.

On aircraft with 803SO90034 installed, the “SINK RATE” message is inhibited when the steep approach mode is selected.

Mode 2 – Excessive Closure Rate To Terrain: The excessive closure rate to terrain mode gives audio and visual indications for dangerously high terrain closure rates. The excessive closure rate to terrain mode has two sub−modes that follow:

- Mode 2A – This mode is active when the flaps are not set to the landing position.
- Mode 2B – This mode is active when the flaps are set to the landing position or while on the glideslope.

A “TERRAIN, TERRAIN” message sounds and the PULL UP annunciator light comes on. If the high terrain closure rate continues, the “TERRAIN, TERRAIN message changes to “PULL UP” and the PULL UP annunciator light stays on.

Mode 3 – Descent after Takeoff: The descent after takeoff mode gives audio and visual indications for excessive altitude loss after takeoff or go around while the aircraft is close to the ground and the flaps and landing gear is not set in the landing configuration.

A “DON’T SINK, DON’T” message sounds and the PULL UP annunciator light comes on.

The sensitivity of the mode changes when the GPWS FLAP OVERRIDE annunciator switch is set.

Mode 4 – Insufficient Terrain Clearance: The insufficient terrain clearance mode gives audio and visual indications for dangerous terrain clearance relative to flight condition, height above the ground, and speed. The insufficient terrain clearance mode has three sub−modes that follow:

- Mode 4A – This mode is active during cruise and approach with the landing gear up. A “TOO LOW TERRAIN” or “TOO LOW GEAR” message sounds depending on airspeed and the PULL UP annunciator light comes on.
- Mode 4B – This mode is active during and approach with the landing gear down and flaps up. A “TOO LOW TERRAIN” or “TOO LOW FLAPS” message sounds depending on airspeed and the PULL UP annunciator light comes on.
- Mode 4C – This mode is active during takeoff when the landing gear or flaps are up. A “TOO LOW TERRAIN” message sounds and the PULL UP annunciator light comes on.

Mode 5 – Excessive Descent Below the Glideslope: The excessive descent below the glideslope mode gives audio and visual indications for descends below the glideslope on front−course Instrument Landing System (ILS) approaches. Two levels of audio indications (soft or hard) are given depending on the deviation from the glideslope. A “GLIDESLOPE” message sounds and the BELOW G/S annunciator light comes on.

On aircraft with ModSum 4−903538 installed, the EGPWS computer is equipped with the Localizer Performance with Vertical Guidance (LPV) glide path deviation monitoring capability.

The Mode 5 of the EGPWS computer provides two levels of alerting when the aircraft flight path descends below the glideslope on the front−course ILS and LPV approaches.

Mode 6 – Advisory Callouts: The advisory callouts mode gives audio indications for the excessive bank angles, approaching minimums/minimums, and altitude awareness.

Terrain Clearance Floor (TCF) Alerting: The terrain clearance floor alerting gives more protection. It makes an increasing terrain clearance area around the airport runway to give CFIT protection when the mode 4 – insufficient terrain clearance mode give limited or no protection. The TCF mode is active during takeoff, cruise, and approach. It helps the basic mode 4 – insufficient terrain clearance mode by giving an insufficient terrain indication even if the aircraft is in a landing configuration. To make the TCF alerts calculations, the EGPWS uses the data that follows:

- Current aircraft location
- Designation runway center point
- Above Ground Level (AGL) altitude.

The TCF also has a Runway Field Clearance Floor (RFCF) alert to give short landing protection for runways that are significantly higher than the relative terrain. To make the RFCF alerts calculations, the EGPWS uses the data that follows:

- Current aircraft location
- Designation runway center point
- Above Ground Level (AGL) altitude
- Geometric or Above Sea Level (ASL) altitude.

NOTE

The EGPWS has a runway database that contains data records for all hard surface airport runways in the world 3500 ft. or more for coverage in the terrain database.

The TCF mode causes a “TOO LOW TERRAIN” message to sound and the PULL UP annunciator lights to come on.

Terrain Alerting: The terrain alerting mode gives audio and visual indications when the aircraft position relative to the local database−catalogued senses a terrain or obstacle (when and where available) threat.

The terrain alerting mode causes a “CAUTION TERRAIN” or “CAUTION OBSTACLE” message to sound and the PULL UP annunciator light to come on. If the condition continues to worsen, the “TERRAIN, TERRAIN, PULL UP” “OBSTACLE, OBSTACLE, PULL UP” message sounds and the PULL UP annunciator light stays on. The Multi−functional Display units (MFD1, MFD2) also automatically show local terrain forward of the aircraft when a terrain alert condition is sensed.

Terrain Awareness Display Mode: In the terrain awareness display mode, the background display is calculated from the airplane altitude with respect to the terrain data in the digital elevation matrix overlays.

The terrain awareness display modes are of two types:

- Standard display mode
- Peaks display mode.

Standard Display Mode − It provides a graphical plan−view image of the surrounding terrain as varying density patterns of green, yellow, and red.

Peaks Display Mode − It lets the terrain below the aircraft to be viewed during all flight conditions. At altitudes safely above the terrain for the set display range, the highest and lowest elevations are shown relative to the aircraft altitude. The peaks display mode has two elevation numbers to show the highest and lowest terrain currently being shown. Only one elevation number is shown when the aircraft is flying in the areas with no differences in terrain elevations (water or flat terrain).

The EGPWS has red PULL UP and amber BELOW G/S annunciator switches to show indications for the different conditions. The PULL UP indication comes on for the conditions that follow:

- Excessive descent rate
- Excessive closure rate to terrain
- Descent after takeoff
- Insufficient terrain clearance
- TCF alerting− Terrain alerting.

The light in the BELOW G/S annunciator switch comes on during an excessive descent below the glideslope condition. The PULL UP annunciator switch is also used to test the GPWS and the BELOW G/S annunciator switch is used to mute the excessive descent below the glideslope condition.

The different EGPWS sounds are supplied through the Remote Control Audio Unit (RCAU) to the headphones and speakers as follows:

- “SINK RATE, SINK RATE”

- "PULL UP"
- "TERRAIN, TERRAIN"
- "OBSTACLE, OBSTACLE"
- "CAUTION TERRAIN"
- "CAUTION OBSTACLE"
- "TOO LOW TERRAIN"
- "TOO LOW GEAR"
- "TOO LOW FLAPS"
- "DON'T SINK"
- "GLIDESLOPE".

The GPWS FLAP OVERRIDE annunciator switch is pushed to override unwanted indications when landing with zero or partial flaps and cancel glideslope indications.

The GPWS LANDING FLAPS switch on the HYDRAULIC control panel sets the landing flap value that is used for its calculation.

The EGPWS indications are automatically cancelled by the conditions that follow:

- Stall
- EGPWS malfunction is sensed.

The EGPWC supplies a signal to the Caution and Warning Panel (CWP) to make the GPWS caution light come on when the system has sensed a malfunction.

Basic Ground Proximity Warning System and Altitude Awareness Call–Out: The EGPWC makes the basic GPWS and Altitude Awareness Call–Out calculations. It uses data received from the units that follow through the two Input/Output Modules (IOM1, IOM2) and Input/Output Processors (IOP1, IOP2) that are located in their related Integrated Flight Cabinet (IFC1, IFC2):

- Proximity Sensor Electronics Unit (PSEU)
- Flap Position Indicator Unit (FPIU)
- Air Data Computers (ADC1, ADC2)
- Attitude and Heading Reference System (AHRS1, AHRS2)
- Radio Altimeters (RA1, RA2)
- VHF Navigation Receivers (VHF NAV1, VHF NAV2)
- Index Control Panels (ICP1, ICP2)
- Flight Guidance Control Panel.

The Enhanced Ground Proximity Warning System (EGPWS) supplies data directly to the systems that follow:

- Caution and Warning Lights Panel
- Advisory Control Unit (ACU)
- Audio Integrating System (AIS)
- Stall Protection System (SPS).

The GPWS Landing Flap Select Panel supplies flaps position data to the EGPWS.

A related IOP1−EGPWS, IOP2−EGPWS bus supplies the data that follows:

- Vertical speed
- Air speed
- Glideslope deviation, glideslope enable
- Bank angle
- Above ground level altitude radio altitude
- Radio altitude validity
- Flap setting
- Landing gear selection
- Decision Height (DH) selection
- Back course selection
- Stall indication.

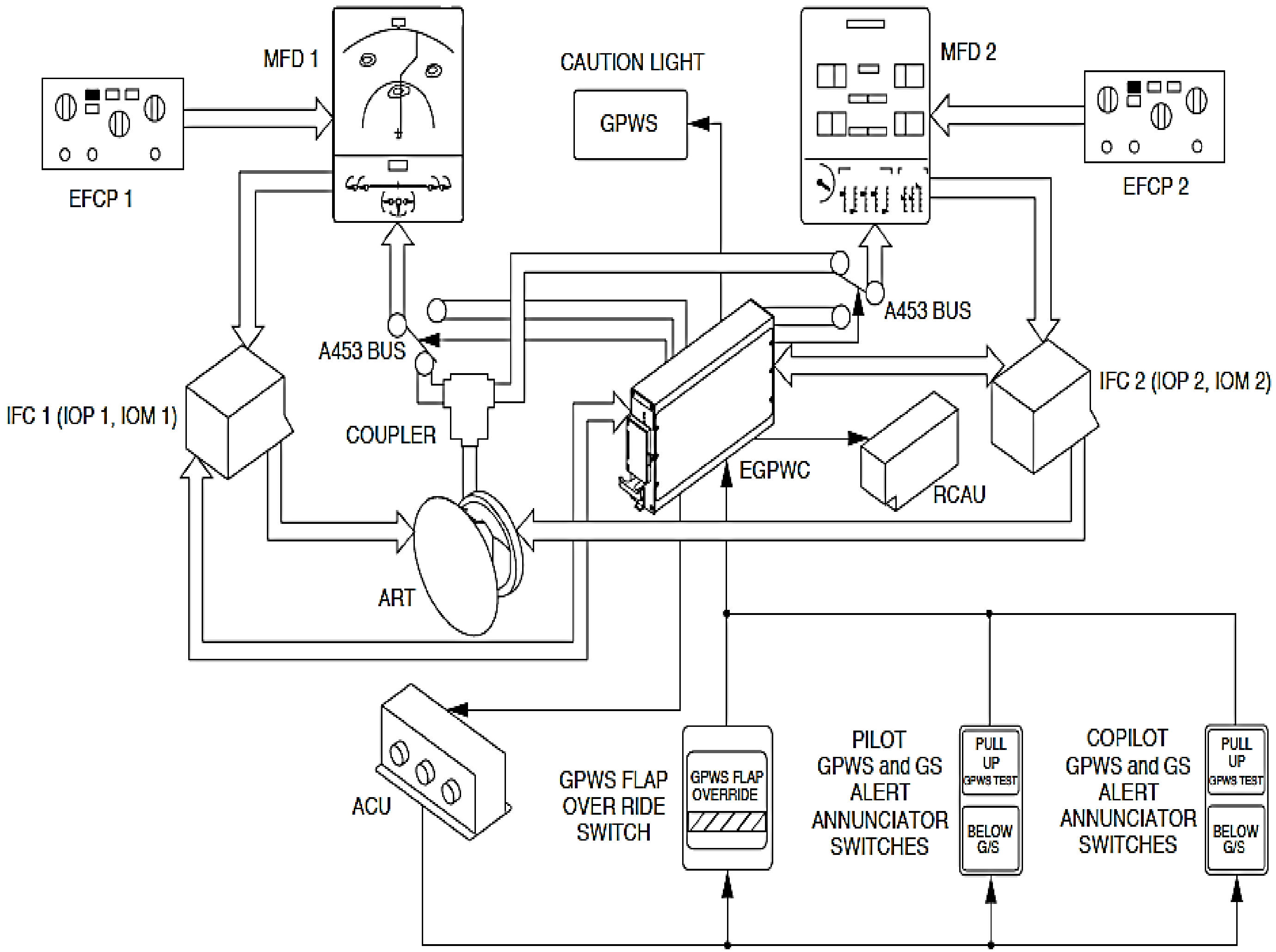


Figure 34- 111 EGPWS BLOCK DIAGRAM, OUTPUTS

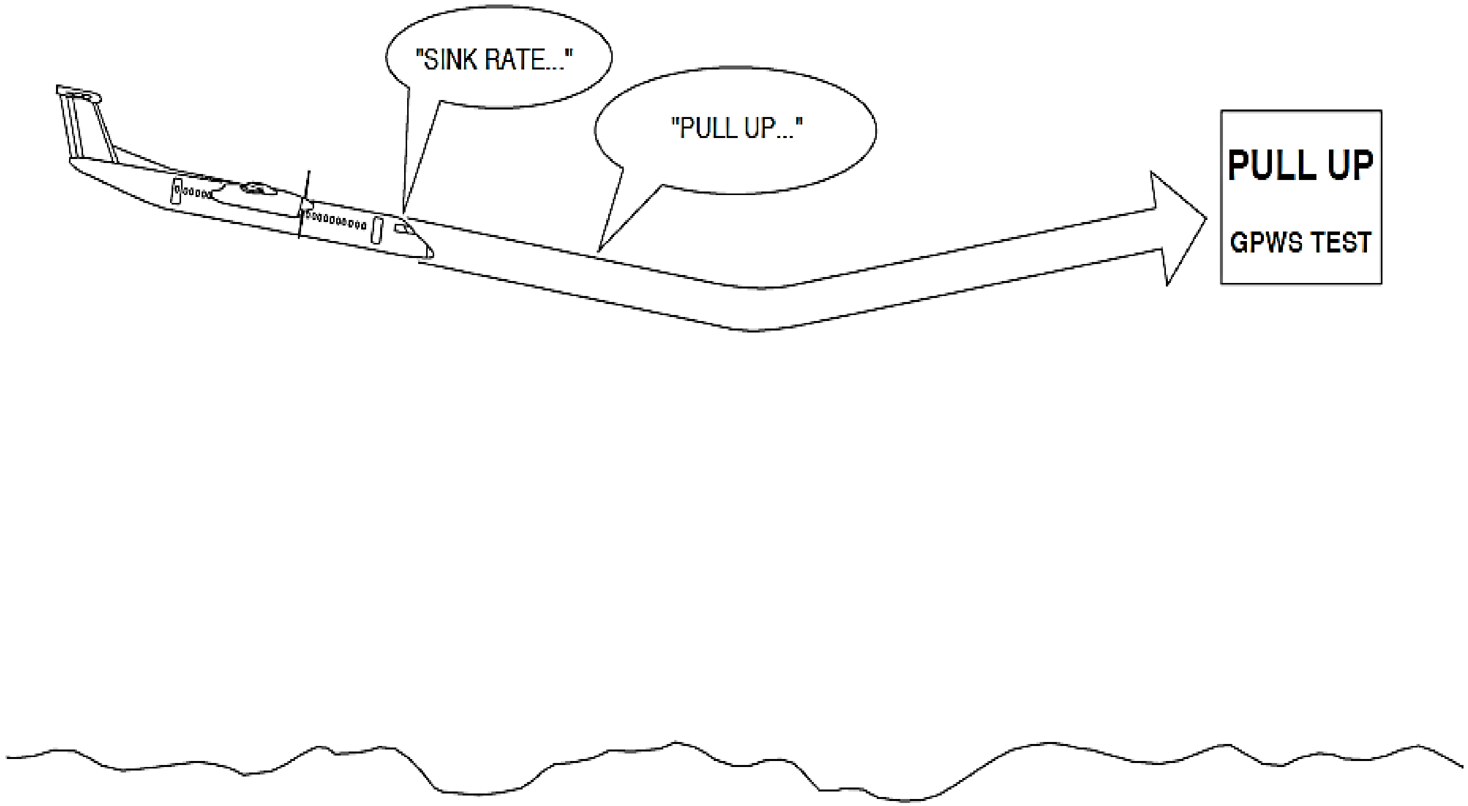


Figure 34- 112 EGPWS MODE 1, EXCESSIVE DESCENT RATE INDICATIONS

Detailed Description: Mode 1, Excessive Descent Rate:

Figure 34- 112 EGPWS MODE 1, EXCESSIVE DESCENT RATE INDICATIONS
Figure 34- 113 EGPWS MODE 1, EXCESSIVE DESCENT RATE

The Mode 1 condition gives indications for excessive rates of descent relative to the aircraft height above the terrain. The indications are given well ahead of the possible impact with the ground. The Ground Proximity Warning Computer (GPWC) calculates the possible flight into terrain from the Radio Altitude (RA) and Air Data Unit (ADU) Barometric Decent Rate (FPM) data.

The Mode 1 condition causes the "SINK RATE" and "PULL UP" aural indications to sound and the PULL UP annunciator lights come on.

The "SINK RATE" aural indication repeats every three seconds while the aircraft is in the initial indication area. As the aircraft goes into the second indication area that is closer to the terrain, a "PULL UP" aural indication sounds and not the "SINK RATE" aural indication.

An ILS glideslope selection changes the indication area to minimize incorrect warnings.

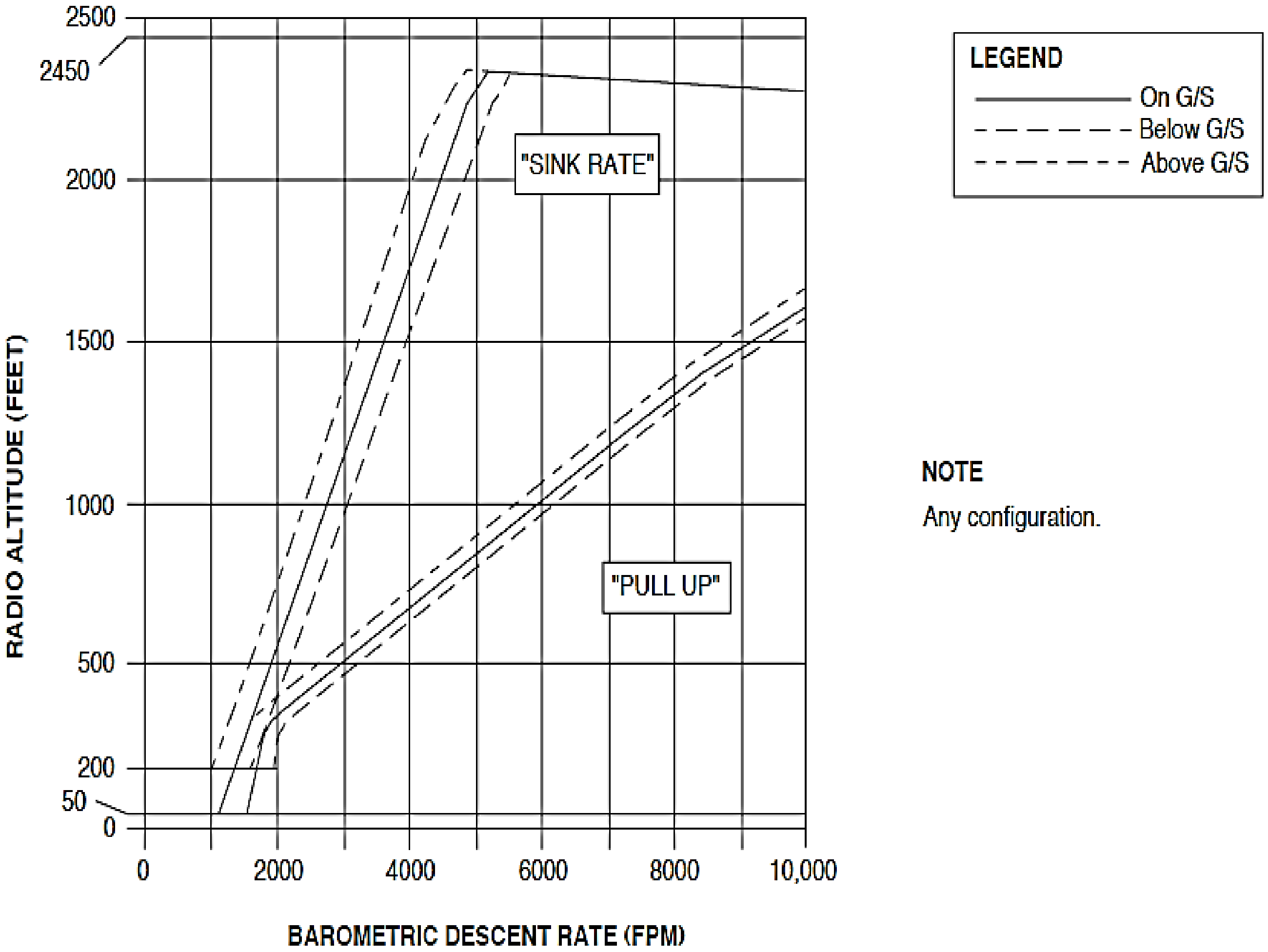


Figure 34- 113 EGPWS MODE 1, EXCESSIVE DESCENT RATE

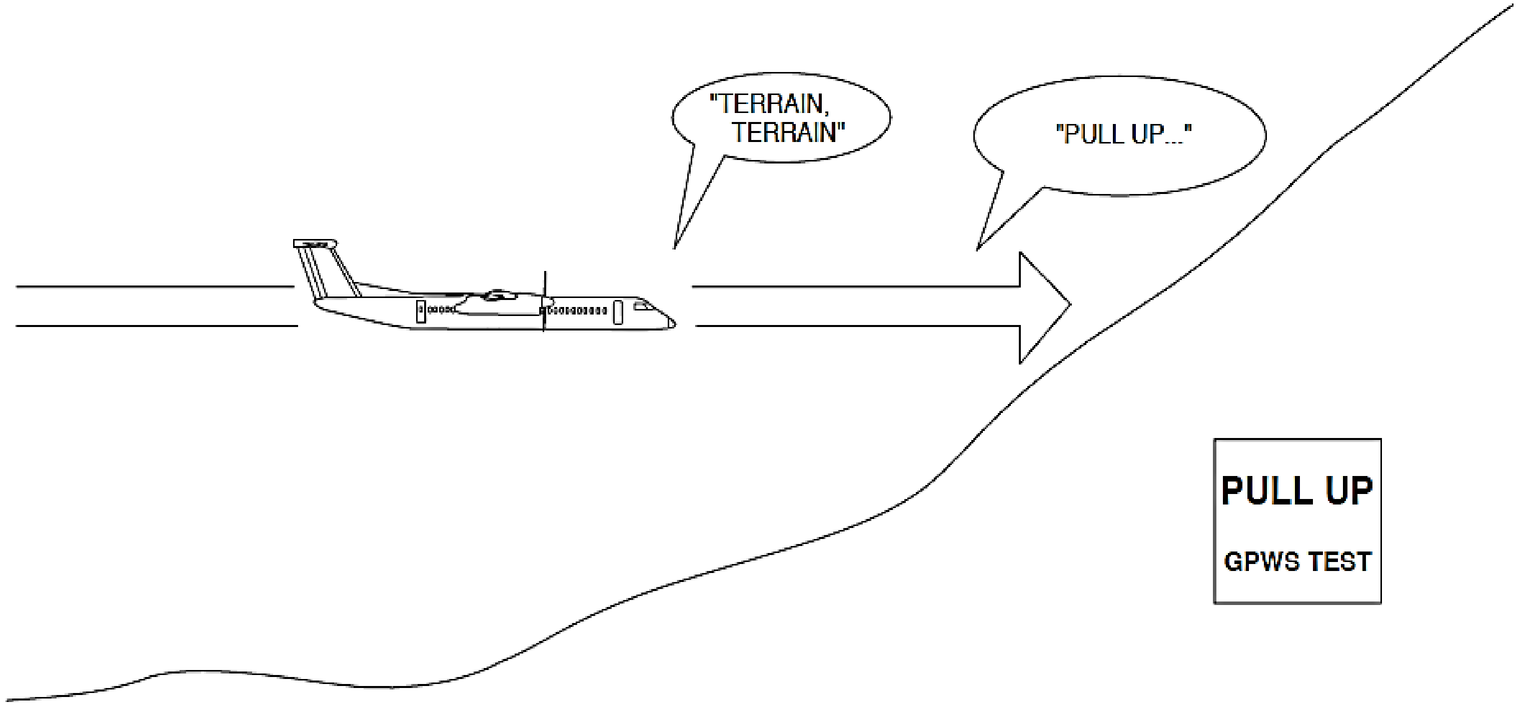


Figure 34- 114 EGPWS MODE 2, EXCESSIVE CLOSURE RATE INDICATIONS

Mode 2, Excessive Closure Rate to Terrain:

Figure 34- 114 EGPWS MODE 2, EXCESSIVE CLOSURE RATE INDICATIONS

The Mode 2 condition gives indications when terrain below the aircraft is rising dangerously fast. The Ground Proximity Warning Computer (GPWC) gives the indications well ahead of the projected collision with terrain.

The Ground Proximity Warning Computer (GPWC) monitors the conditions that follow to give the Mode 2 indication:

- AGL altitude
- Air speed
- Terrain closure rate.

The Mode 2 condition causes the "TERRAIN, TERRAIN" and "PULL UP" aural indications to sound and the PULL UP annunciator lights come on.

Mode 2 operates in the two sub–modes that follow:

- Mode 2A
- Mode 2B.

Mode 2A:

Figure 34- 115 EGPWS MODE 2A, EXCESSIVE CLOSURE RATE

The Mode 2A "PULL UP" aural indication will stop when the aircraft goes out of the warning area. The PULL UP annunciators will stay on until any condition occurs that follow:

- The barometric altitude increase is more than 300 feet
- An up acceleration that is a combination of barometric altitude gain and radio altitude, and time.

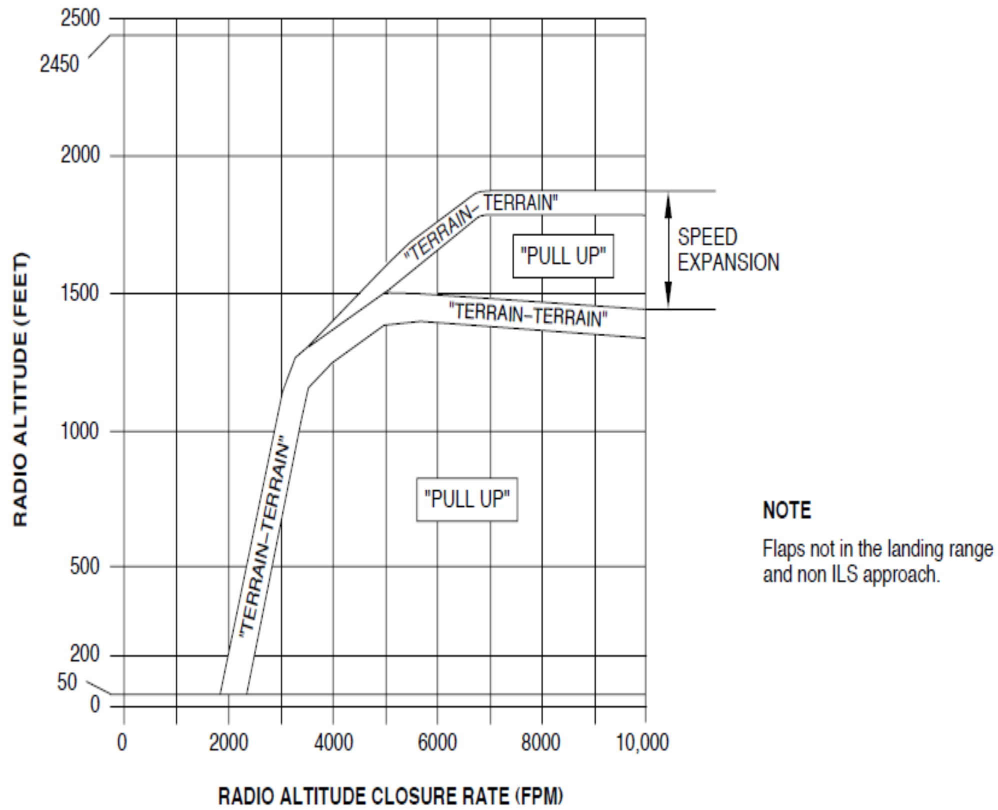


Figure 34- 115 EGPWS MODE 2A, EXCESSIVE CLOSURE RATE

Mode 2B:

Figure 34- 116 EGPWS MODE 2B, EXCESSIVE CLOSURE RATE

Mode 2B functions when any one of the conditions is met that follows:

- The flaps are in the landing position
- The GPWS flap override switch is selected
- The aircraft is on an ILS or MLS approach and the aircraft is not more than 1.3 dots below the glidepath
- The glideslope cancel feature is not selected.

When the landing gear is down, and the flaps are down or the GPWS flap override switch is set, a repetitive "TERRAIN, TERRAIN" aural indication will sound. If the landing gear is not down, a repetitive normal "PULL UP" aural indication will sound when the "TERRAIN, TERRAIN" aural indication is complete. When the aircraft goes out of the warning area, the aura indications will stop and the PULL UP annunciators go out.

NOTE

Selecting the GPWS FLAP OVERRIDE annunciator switch cancels the Mode 2 warnings. This lets the aircraft maneuver near terrain when approaching an airport.

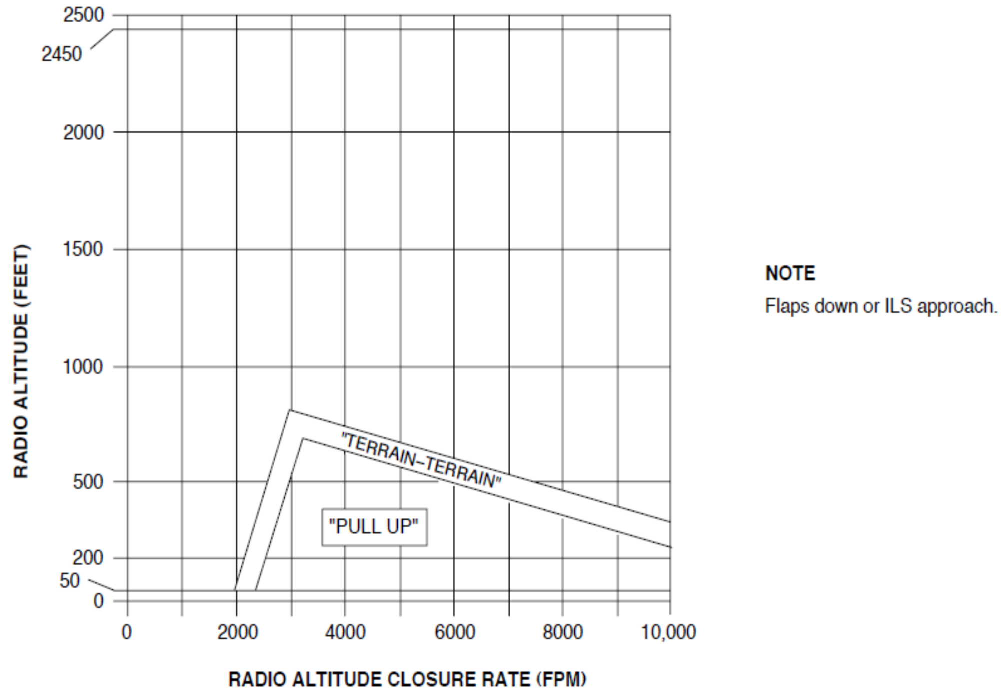


Figure 34- 116 EGPWS MODE 2B, EXCESSIVE CLOSURE RATE

Mode 3, Descent After Takeoff:

Figure 34- 117 EGPWS MODE 3, DESCENT AFTER TAKEOFF INDICATIONS
Figure 34- 118 EGPWS MODE 3, DESCENT AFTER TAKEOFF

Mode 3 functions during a takeoff or missed Approach when the conditions that follow are met:

- Increase in airspeed
- 50 feet Above Ground Level (AGL) altitude
- Increase in barometric altitude
- Landing gear retraction.

The Mode 3 gives an indication to show too much altitude decrease after takeoff or after a missed approach. A barometric altitude decrease of approximately 10 percent of the Above Ground Level (AGL) altitude starts the Mode 3 indication.

The Mode 3 condition sounds a "DON'T SINK" aural indication and the PULL UP annunciator lights come on. An increased rate of climb cancels the "DON'T SINK" aural indications and the PULL UP annunciator lights go out.

When the Above Ground Level (AGL) altitude is more than 925 feet for 17 seconds, the Mode 3 condition is cancelled.

Aircraft speed, flap and landing gear position are used to calculate a warning area below the aircraft.

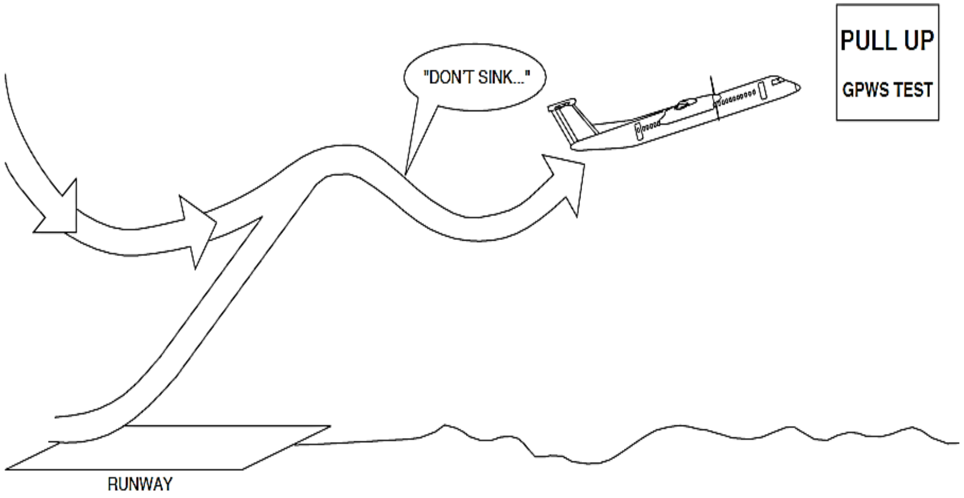


Figure 34- 117 EGPWS MODE 3, DESCENT AFTER TAKEOFF INDICATIONS

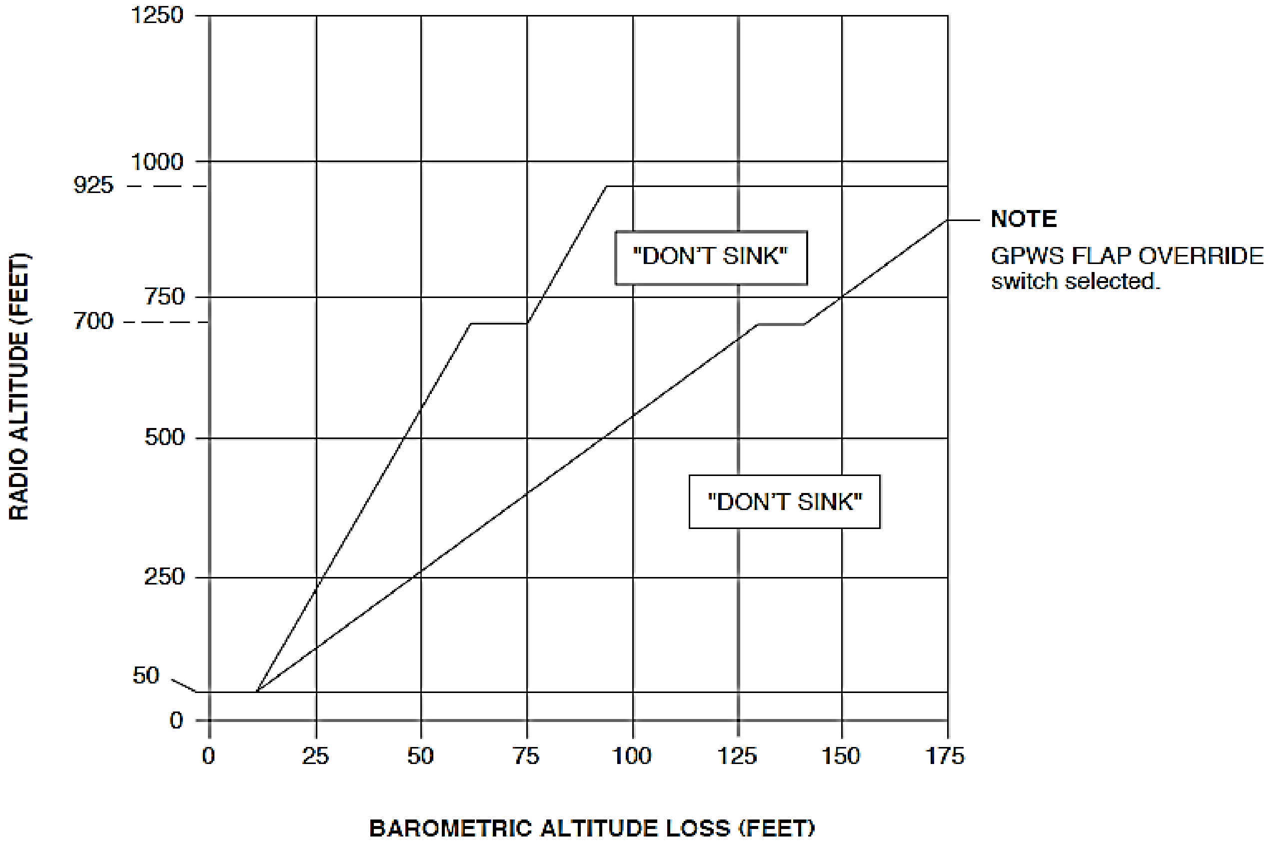


Figure 34- 118 EGPWS MODE 3, DESCENT AFTER TAKEOFF

Mode 4, Insufficient Terrain Clearance:

The Mode 4 condition gives an indication for insufficient terrain clearance during the phases of flight that follow:

- Cruise
- Approach
- Descent
- Climbout.

The Mode 4 condition gives indications when the flight path is too low for Mode 2 excessive closure rates with terrain or Mode 1 excessive descent rate calculations. It uses the data that follow:

- AGL altitude
- Airspeed
- Flight phase.

The Mode 4 condition functions in the modes that follow:

- Mode 4A
- Mode 4B
- Mode 4C.

Mode 4A:

Figure 34- 119 EGPWS MODE 4A, INSUFFICIENT TERRAIN CLEARANCE INDICATIONS
Figure 34- 120 EGPWS MODE 4A, INSUFFICIENT TERRAIN CLEARANCE

When the aircraft is in a configuration with the gear and flaps selected up, the EGPWS calculates a "floor" below the aircraft to show insufficient terrain clearance.

When the air speed is more than 250 knots and the AGL altitude is less than 1000 feet, the "TOO LOW TERRAIN" aural indication sounds and the PULL UP annunciator lights come on.

Lower AGL altitudes are used to give the same indication at airspeeds 190 to 250 knots.

At speeds less than 190 knots and the AGL altitude is less than 500 feet, the "TOO LOW GEAR" aural indication sounds and the PULL UP annunciator lights come on.

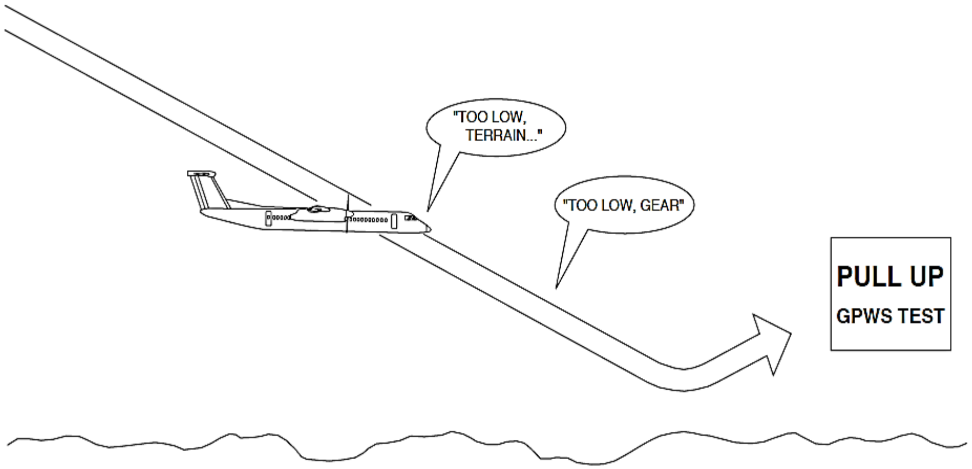
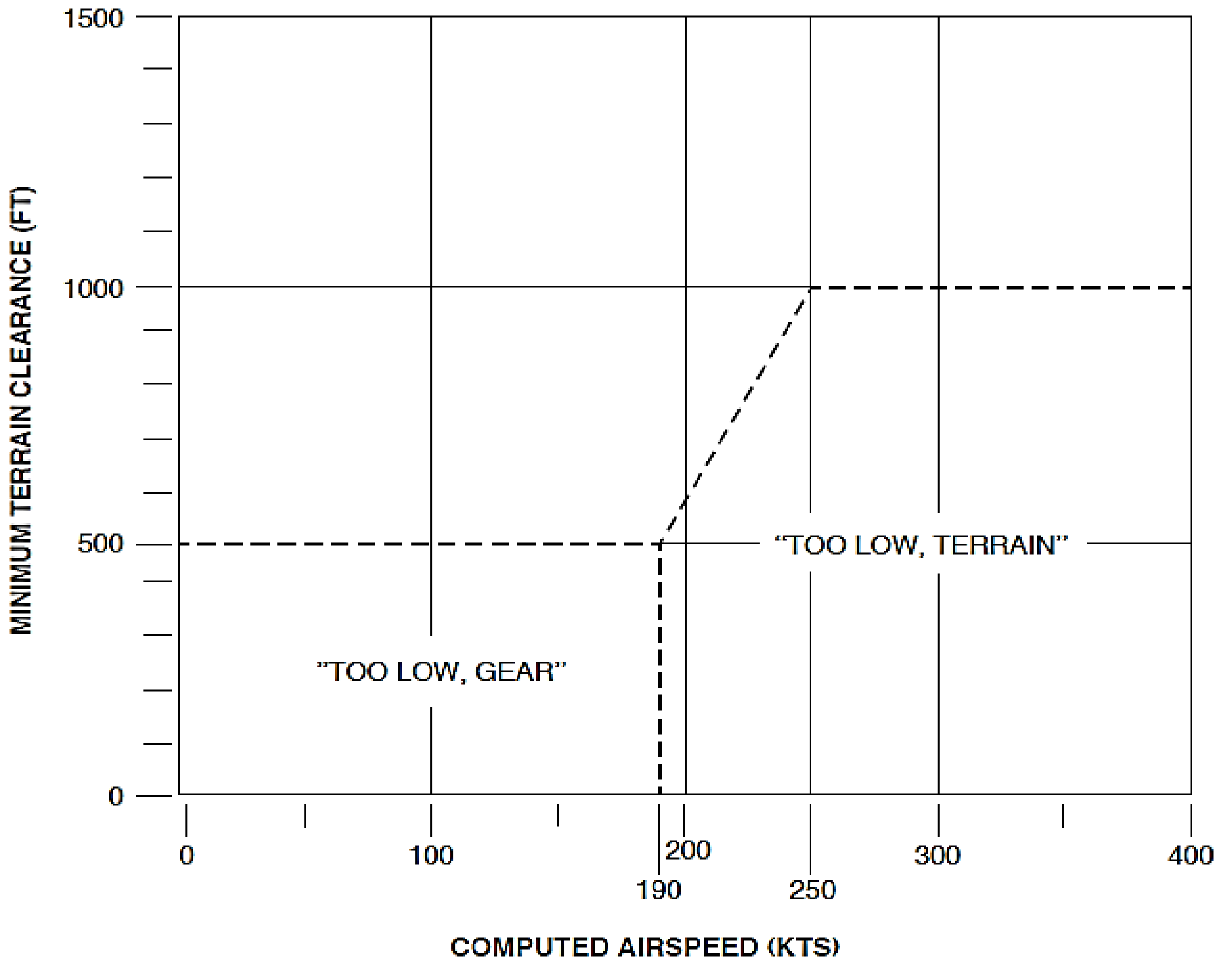


Figure 34- 119 EGPWS MODE 4A, INSUFFICIENT TERRAIN CLEARANCE INDICATIONS



NOTE
Gear up, Flaps up.

Figure 34- 120 EGPWS MODE 4A, INSUFFICIENT TERRAIN CLEARANCE

Mode 4B:

Figure 34- 121 EGPWS MODE 4B, INSUFFICIENT TERRAIN CLEARANCE INDICATIONS
Figure 34- 122 EGPWS MODE 4B, INSUFFICIENT TERRAIN CLEARANCE

Mode 4B operates during cruise and approach when the landing gear is down, the flaps not in the landing position and the EGPWS flap override switch is not set.

NOTE

The Landing Flap Selector Switch (LFSS) supplies the EGPWS with data relating to the specified landing flap position.

When the air speed is less than 250 knots and the AGL altitude is less than 750 feet, the "TOO LOW TERRAIN" aural indication sounds and the PULL UP annunciator lights come on. Lower AGL altitudes are used to give the same indication at airspeeds 148 to 225 knots.

During the initial approach, at speeds less than 148 knots and the AGL altitude is less than 170 feet, the "TOO LOW FLAPS" aural indication sounds and the PULL UP annunciator lights come on.

When the flap override annunciator switch is set, the EGPWS will not give any indications for the flaps up landing.

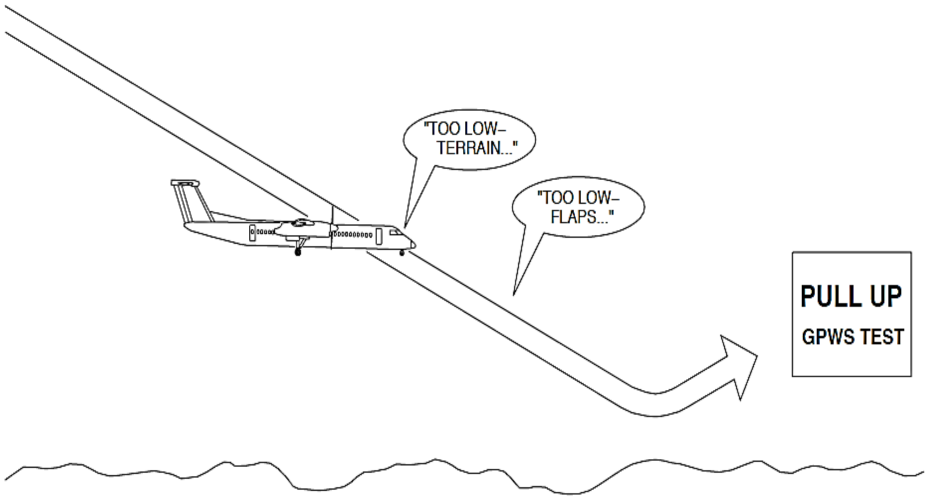
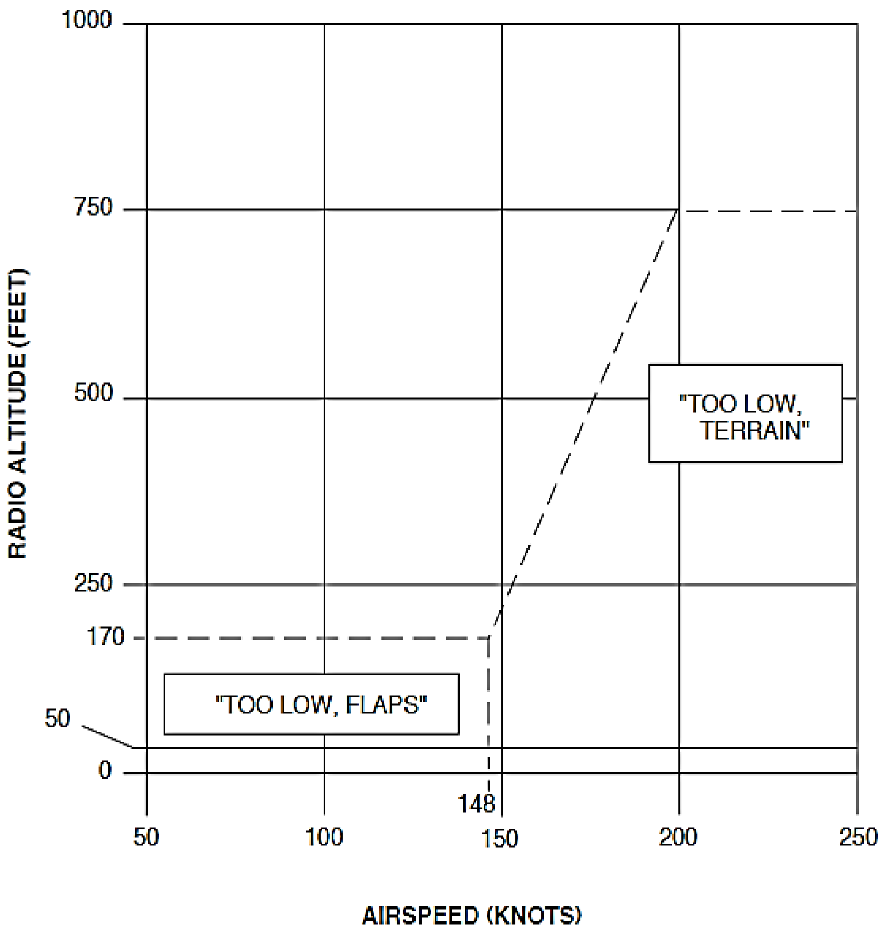


Figure 34- 121 EGPWS MODE 4B, INSUFFICIENT TERRAIN CLEARANCE INDICATIONS



NOTE

Gear down and flaps not in the landing range.

Figure 34- 122 EGPWS MODE 4B, INSUFFICIENT TERRAIN CLEARANCE

Mode 4C:

Figure 34- 123 EGPWS MODE 4C, INSUFFICIENT TERRAIN CLEARANCE INDICATIONS
FIGURE 34- 124 EGPWS MODE 4C, INSUFFICIENT TERRAIN CLEARANCE

This sub−mode is active during a takeoff or go−around. The Mode 4C gives a minimum terrain clearance that increases with the Above Ground Level (AGL) altitude.

When the AGL altitude is more than 100 feet after takeoff or 200 feet when doing a go−around, a warning floor is calculated below the aircraft. It is 75 percent of the highest ACL altitude. The Mode 4C continues to operate until the approach mode is activated or the AGL altitude goes below 30 feet.

The conditions that follow, cause the "TOO LOW TERRAIN" aural indication to sound and the PULL UP annunciator lights to come on:

- The aircraft descents after takeoff
- The terrain below the aircraft rises at a steeper slope than when the aircraft is climbing.

The warning will continue until the aircraft has sufficient clearance from the terrain.

NOTE

Warnings from Modes 4A, 4B, and 4C cannot occur at the same time.

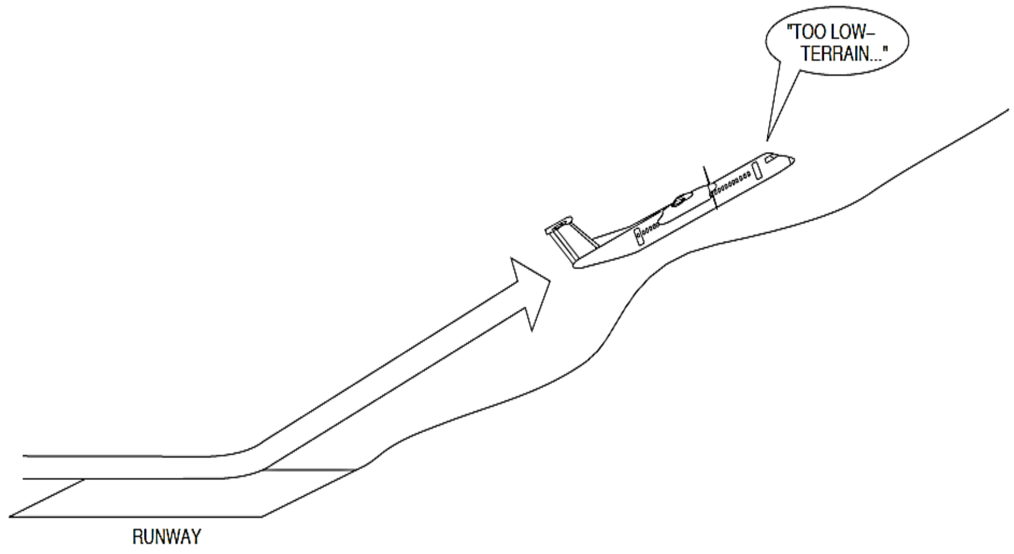


Figure 34- 123 EGPWS MODE 4C, INSUFFICIENT TERRAIN CLEARANCE INDICATIONS

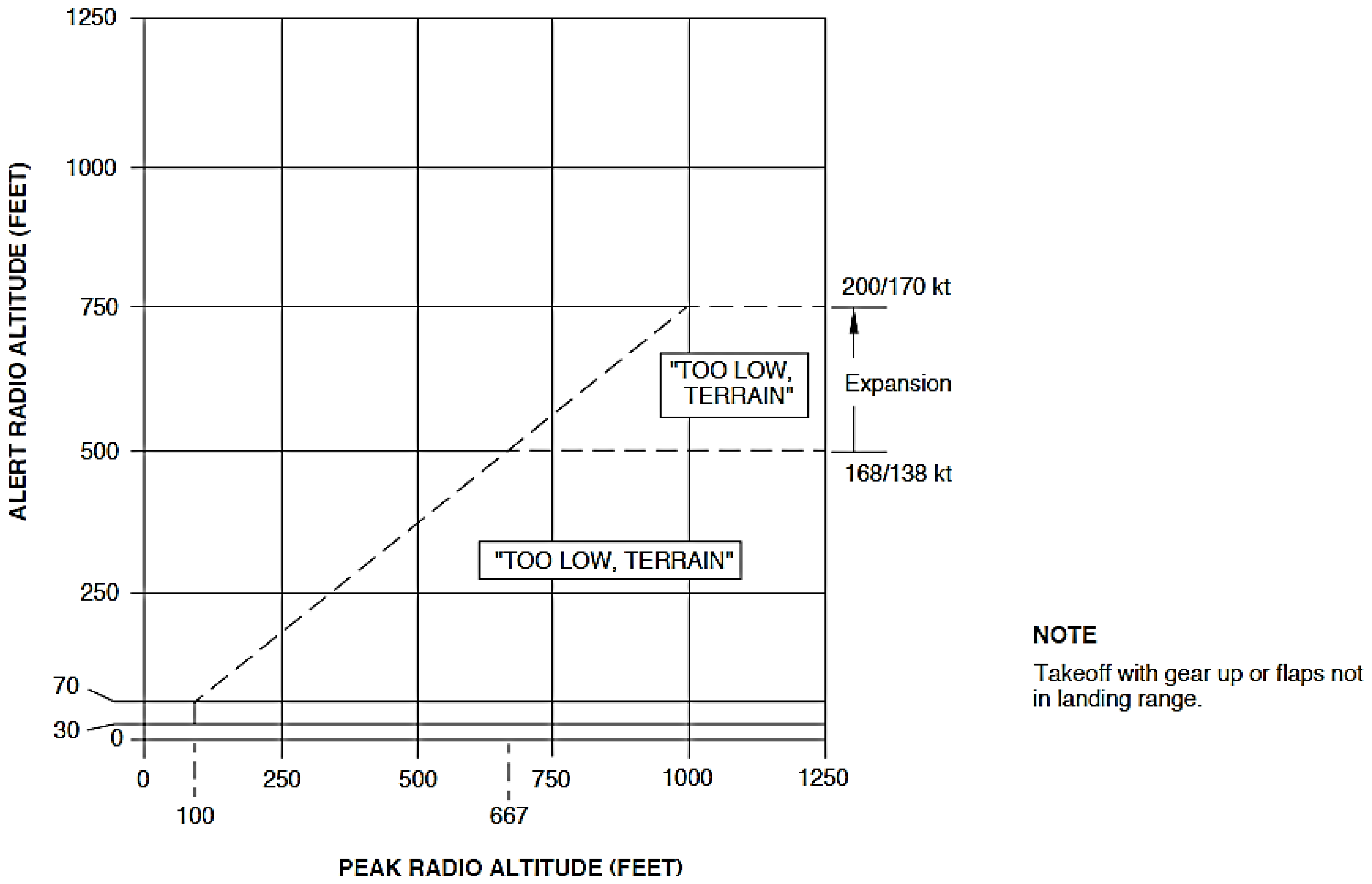


FIGURE 34- 124 EGPWS MODE 4C, INSUFFICIENT TERRAIN CLEARANCE

Mode 5, Excessive Descent Below Glideslope:

Figure 34- 125 EGPWS MODE 5, DESCENT BELOW GLIDESLOPE INDICATIONS
Figure 34- 126 EGPWS MODE 5, DESCENT BELOW GLIDESLOPE

The Mode 5 operates when the EGPWS senses the conditions that follow:

- An ILS frequency is set
- The landing gear is down
- The AGL altitude is less than 925 feet
- The aircraft is below the glidepath
- Glideslope cancel is not set.

The Mode 5 has two aural indication areas that relate to the deviation from the glidepath. Each uses a different aural indication volume level. The low volume level indication is specified as the soft indication and the louder indication is named the hard indication. The hard indication sounds two times as loud as a soft indication. The time between aural indications change if on the glideslope deviation and AGL altitude. The time between the aural indications decreases as the danger of the Mode 5 condition increases.

When the aircraft is below the glideslope and into the soft indicating area, the "GLIDESLOPE" aural indication sounds and the amber BELOW GS annunciator switches come on. The volume of the initial "GLIDESLOPE" aural indication sounds is less than the other EGPWS aural indications. While the aircraft is below the glideslope, the aural repetition rate increases as the AGL altitude decreases. If the aircraft goes into the hard indication area, the audio volume level increases to that of the other aural indications.

When the AGL altitude is less than 150 feet, more glideslope deviation is necessary to cause the indications to come on.

The Mode 5 condition cancels when the AGL altitude is less than 50 feet or by a BELOW G/S annunciator switch selection.

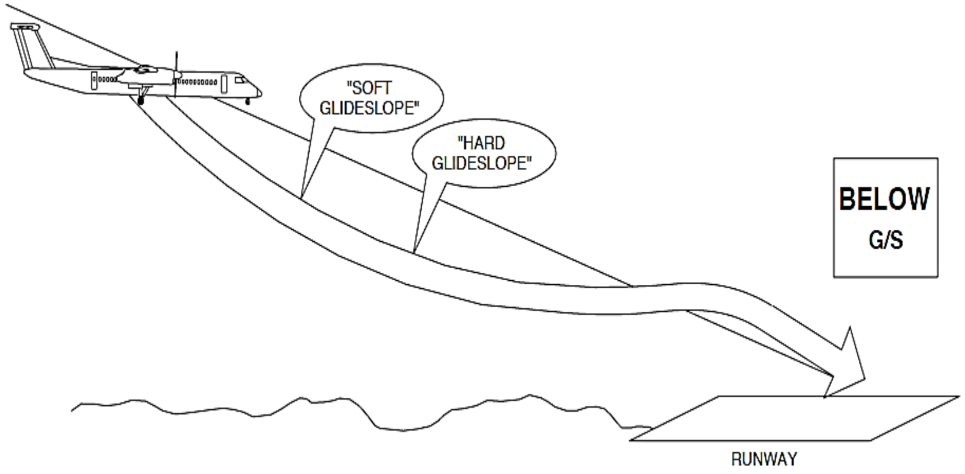


Figure 34- 125 EGPWS MODE 5, DESCENT BELOW GLIDESLOPE INDICATIONS

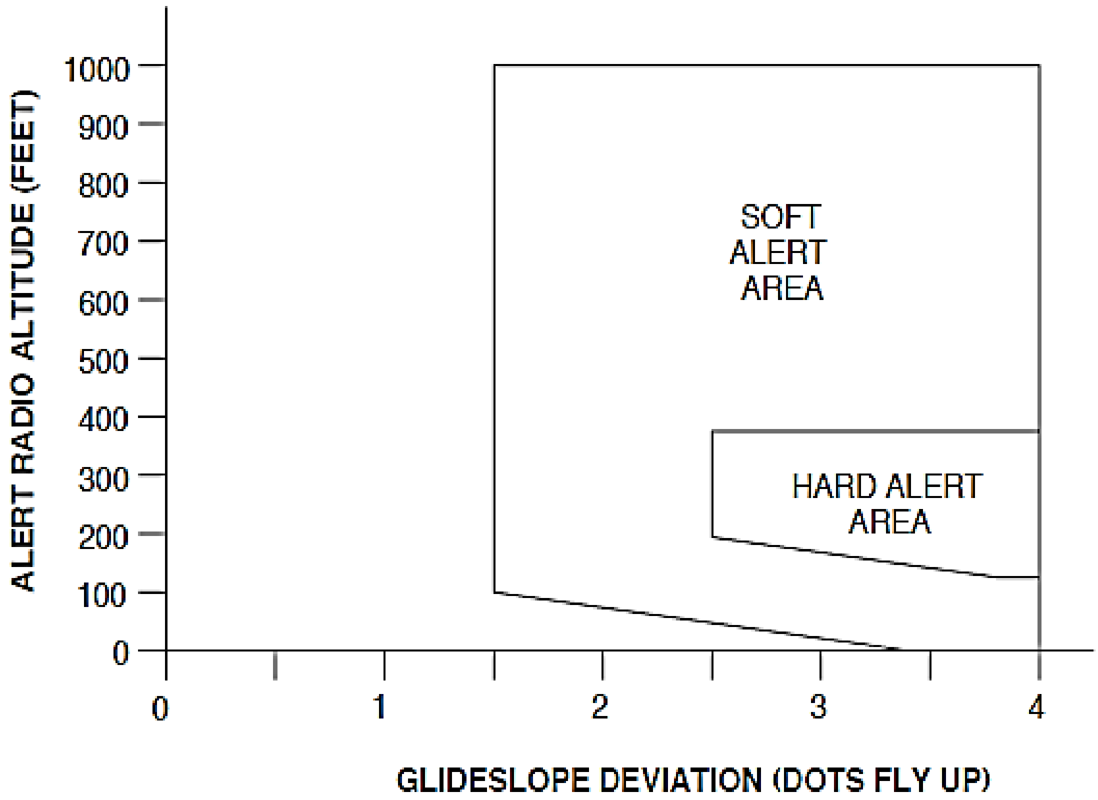


Figure 34- 126 EGPWS MODE 5, DESCENT BELOW GLIDESLOPE

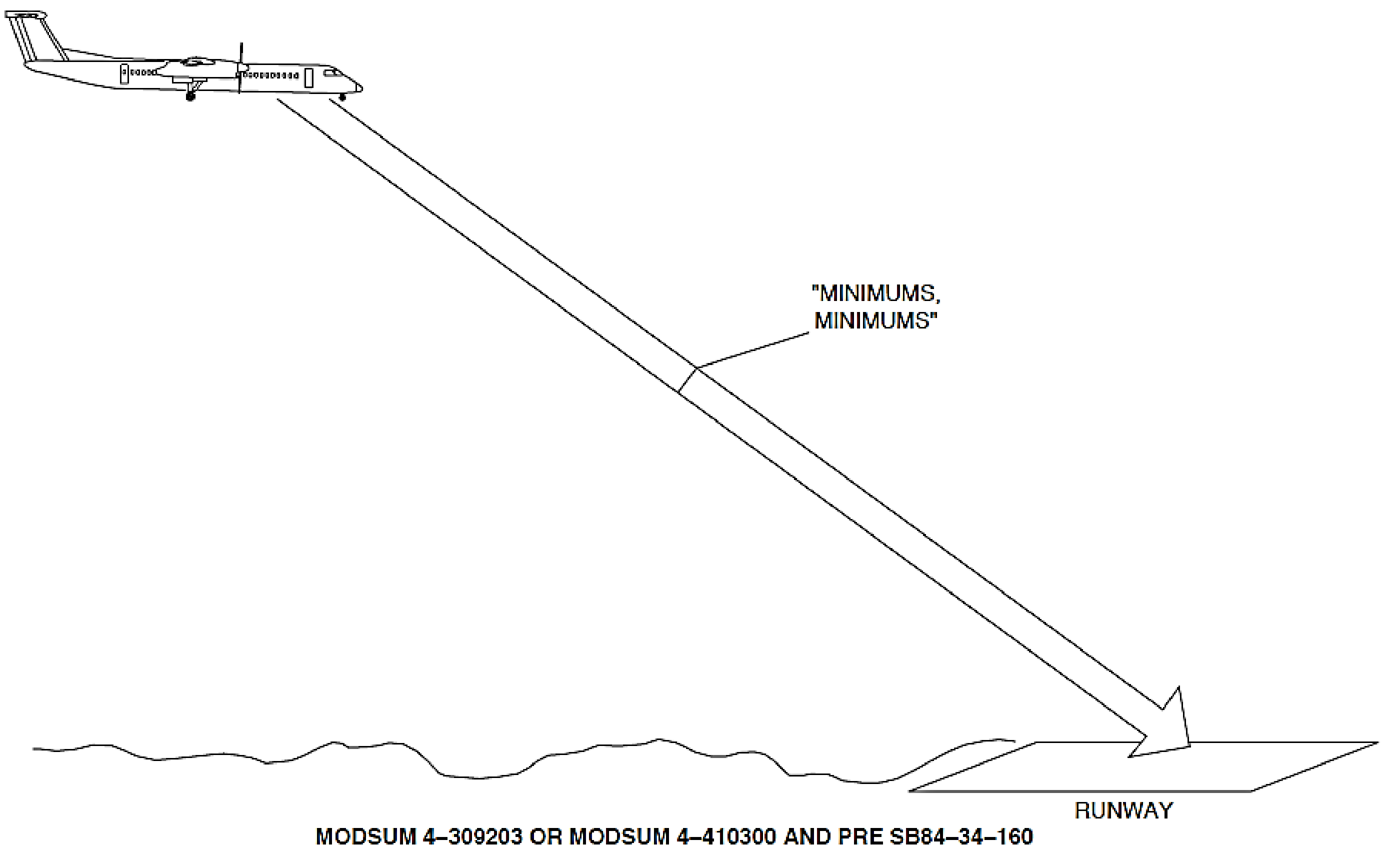


Figure 34- 127 EGPWS MODE 6, DECISION HEIGHT CALLOUTS-1

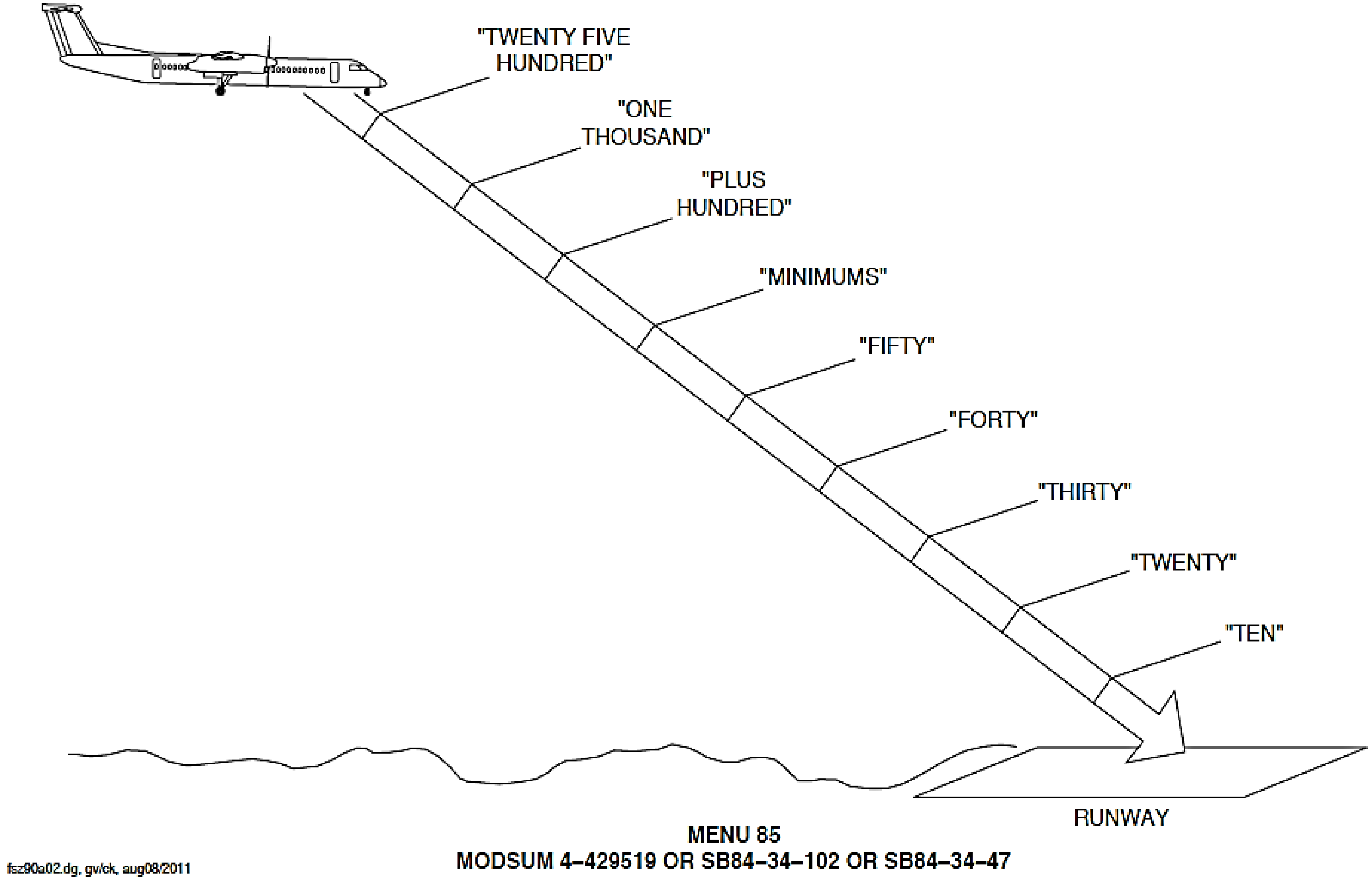
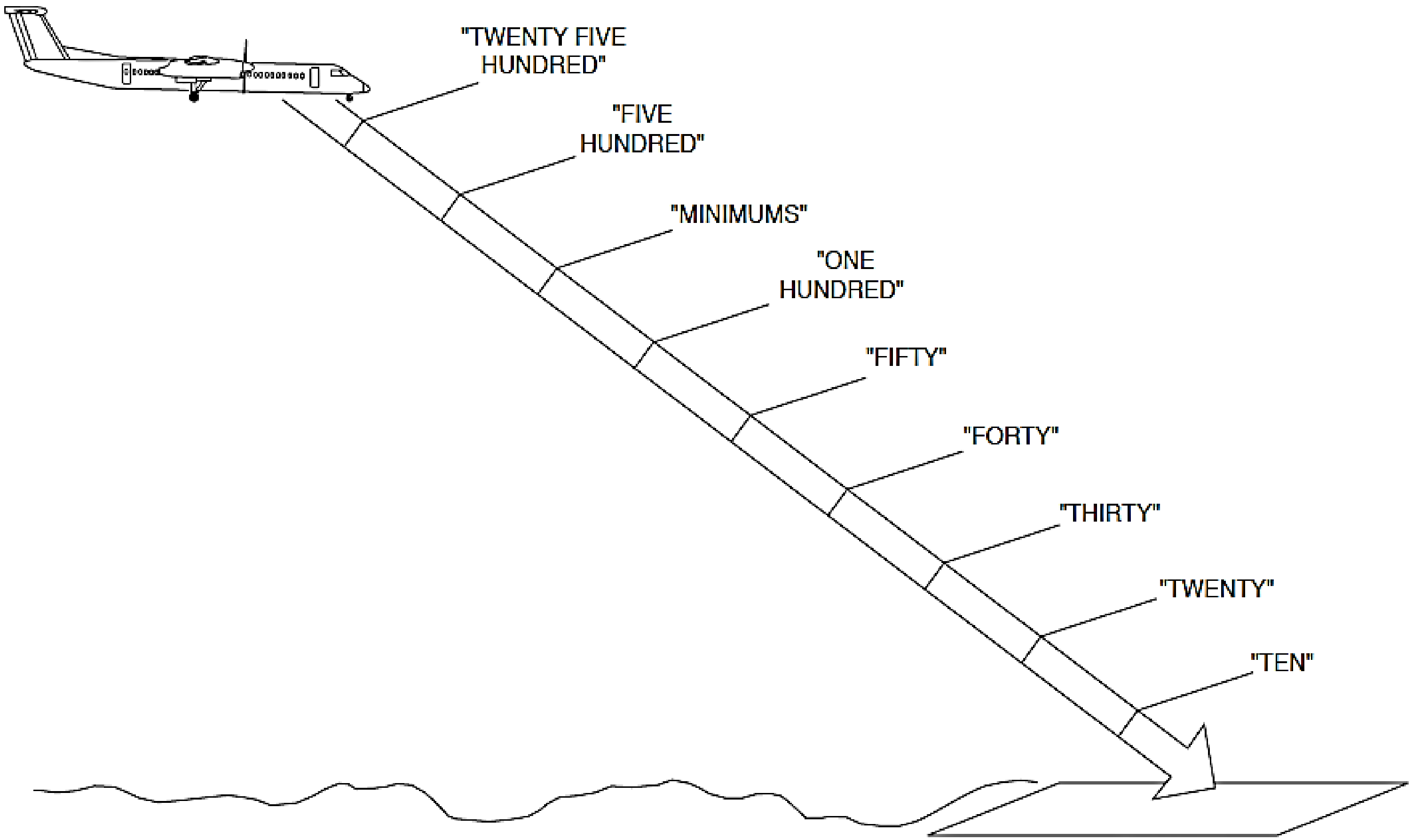


Figure 34- 128 EGPWS MODE 6, DECISION HEIGHT CALLOUTS-2



MENU 34
MODSUM 4–457043 OR SB84–34–115 OR SB84–34–36 OR SB84–34–83 OR SB84–34–111 OR SB84–34–137 OR POST SB84–34–160

FIGURE 34- 129 EGPWS MODE 6, DECISION HEIGHT CALLOUTS-3

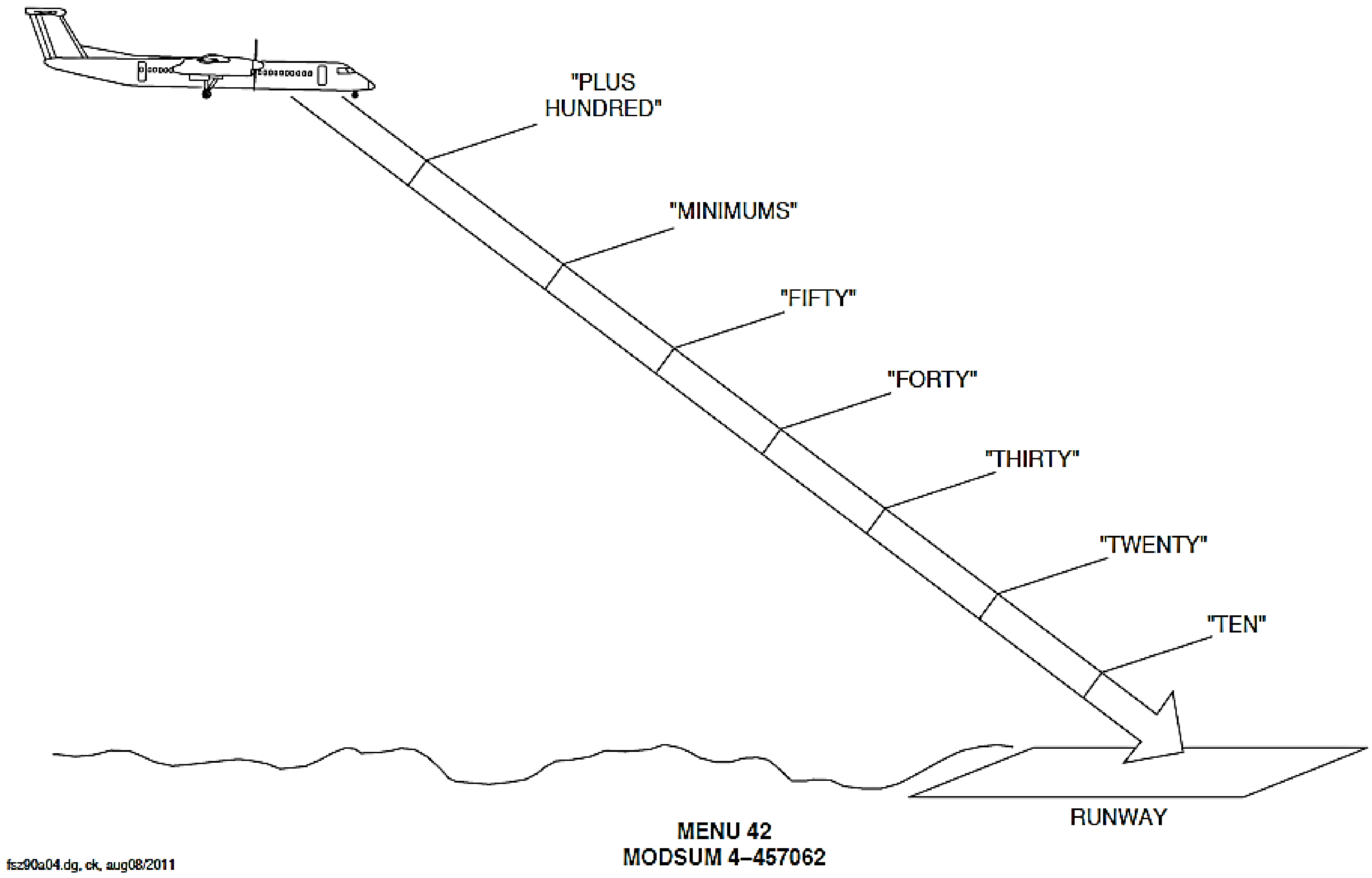


FIGURE 34- 130 EGPWS MODE 6, DECISION HEIGHT CALLOUTS-4

Mode 6, Advisory Callouts:

The mode 6 condition gives indications for excessive bank angles while maneuvering close to the runway.

Figure 34- 127 EGPWS MODE 6, DECISION HEIGHT CALLOUTS-1
Figure 34- 128 EGPWS MODE 6, DECISION HEIGHT CALLOUTS-2
FIGURE 34- 129 EGPWS MODE 6, DECISION HEIGHT CALLOUTS-3
FIGURE 34- 130 EGPWS MODE 6, DECISION HEIGHT CALLOUTS-4

On aircraft with ModSum 4−309203 or 4−410300 (without SB84−34−160) incorporated, the EGPWC gives the advisory callouts that follow:

Table 34- 10 BASIC

CALLOUT	DESCRIPTION
"MINIMUMS, MINIMUMS"	At descent below decision height setting

On aircraft with ModSum 4−429519 or SB84−34−102 or SB84−34−47 incorporated, the EGPWC gives the advisory callouts that follow:

Table 34- 11 MENU 85

CALLOUT	DESCRIPTION
"TWENTY FIVE HUNDRED"	At descent below 2500 ft.
"ONE THOUSAND"	At descent below 1000 ft.
"PLUS HUNDRED"	At descent below decision height setting plus 100 ft.
"MINIMUMS"	At descent below decision height setting
"FIFTY"	At descent below 50 ft.
"FORTY"	At descent below 40 ft.
"THIRTY	At descent below 30 ft.
"TWENTY"	At descent below 20 ft.
"TEN"	At descent below 10 ft.

NOTE

In a test scenario, when you push and hold the GPWS test switch, the 'PLUS HUNDRED' and 'MINIMUMS' callouts will be annunciated first. The actual sequence of these annunciations during the flight will depend on the Decision Height selection.

On aircraft with ModSum 4−457043 or SB84−34−115 or SB84−34−36 or SB84−34−83 or SB84−34−111 or SB84−34−137 or SB84−34−160 incorporated, the EGPWC gives the advisory callouts that follow:

Table 34- 12 MENU 34

CALLOUT	DESCRIPTION
“TWENTY FIVE HUNDRED”	At descent below 2500 ft.
“FIVE HUNDRED”	At descent below 500 ft.
“MINIMUMS”	At descent below decision height setting
“ONE HUNDRED”	At descent below 100 ft.

CALLOUT	DESCRIPTION
“FIFTY”	At descent below 50 ft.
“FORTY”	At descent below 40 ft.
“THIRTY	At descent below 30 ft.
“TWENTY”	At descent below 20 ft.
“TEN”	At descent below 10 ft.

NOTE

In a test scenario, when you push and hold the GPWS test switch, the ’MINIMUMS’ callout will be annunciated first. The actual sequence of this annunciation during the flight will depend on the Decision Height selection.

On aircraft with ModSum 4−457062 incorporated, the EGPWC gives the advisory callouts that follow:

Table 34- 13 MENU 42

CALLOUT	DESCRIPTION
“PLUS HUNDRED”	At descent below decision height setting plus 100 ft.
“MINIMUMS”	At descent below decision height setting
“FIFTY”	At descent below 50 ft.

CALLOUT	DESCRIPTION
“FORTY”	At descent below 40 ft.
“THIRTY	At descent below 30 ft.
“TWENTY”	At descent below 20 ft.
“TEN”	At descent below 10 ft.

Figure 34- 131 EGPWS MODE 6, EXCESSIVE BANK ANGLE CALLOUT

A "BANK ANGLE" aural indication gives indications for excessive bank angles relative to the AGL altitude.

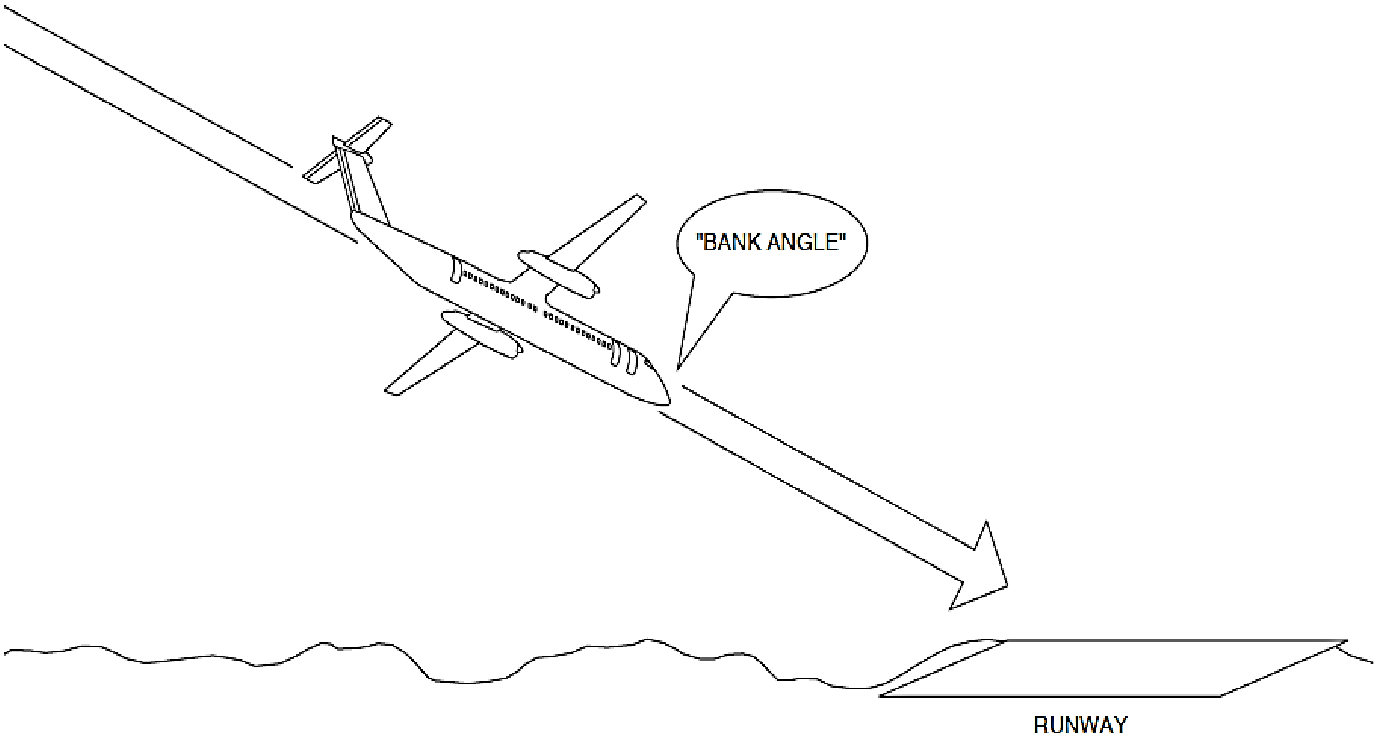


Figure 34- 131 EGPWS MODE 6, EXCESSIVE BANK ANGLE CALLOUT

Figure 34- 132 EGPWS MODE 6, EXCESSIVE BANK ANGLE

The limit of the indication decreases from 50 degrees at 190 feet AGL altitude to 15 degrees near ground level.

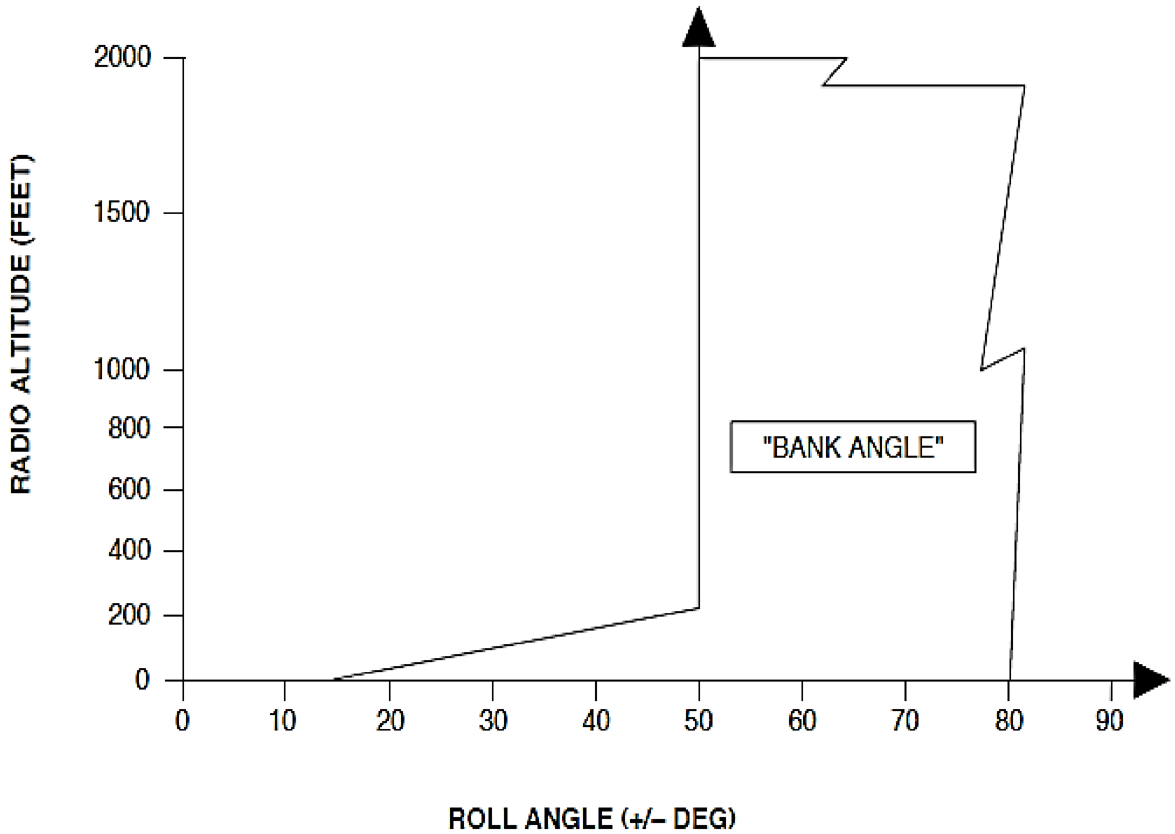


Figure 34- 132 EGPWS MODE 6, EXCESSIVE BANK ANGLE

Terrain Clearance Floor (TCF) Alerting:

Figure 34- 133 TERRAIN CLEARANCE FLOOR INDICATIONS

A TCF is calculated to give more protection to the basic GPWS modes.

It gives an indication for insufficient terrain clearance during the phases of flight that follow:

- Take Off
- Cruise
- Approach
- Descent
- Climbout.

The EGPWS calculates an increasing terrain clearance area around an airport runway that is related to the distance from the runway.

The TCF is calculated with the data that follow:

- Current aircraft location
- Nearest runway center point position
- AGL altitude.

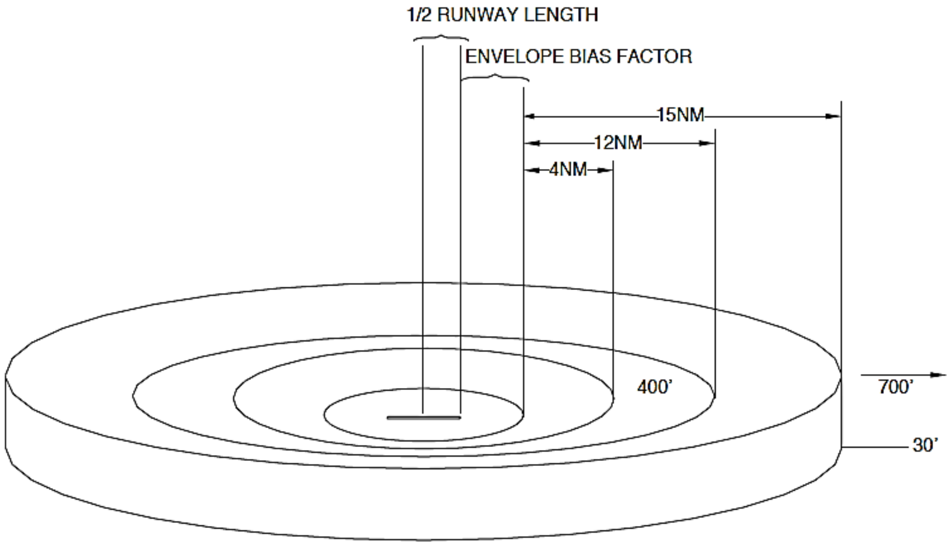


Figure 34- 133 TERRAIN CLEARANCE FLOOR INDICATIONS

Figure 34- 134 TERRAIN CLEARANCE FLOOR

If the aircraft goes below the TCF a "TOO LOW TERRAIN" aural indication sounds once and the PULL UP annunciator lights come on. The "TOO LOW TERRAIN" aural indication sounds again when AGL decreases 20 percent.

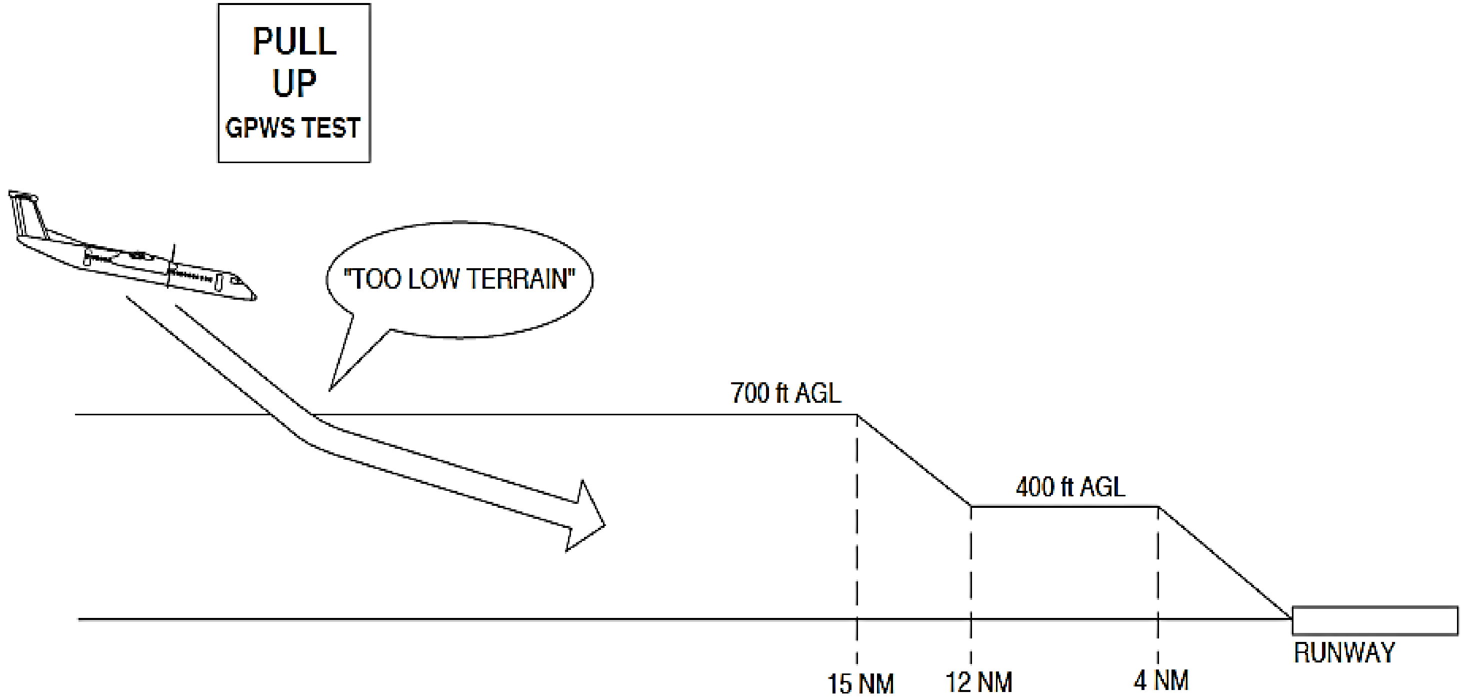


Figure 34- 134 TERRAIN CLEARANCE FLOOR

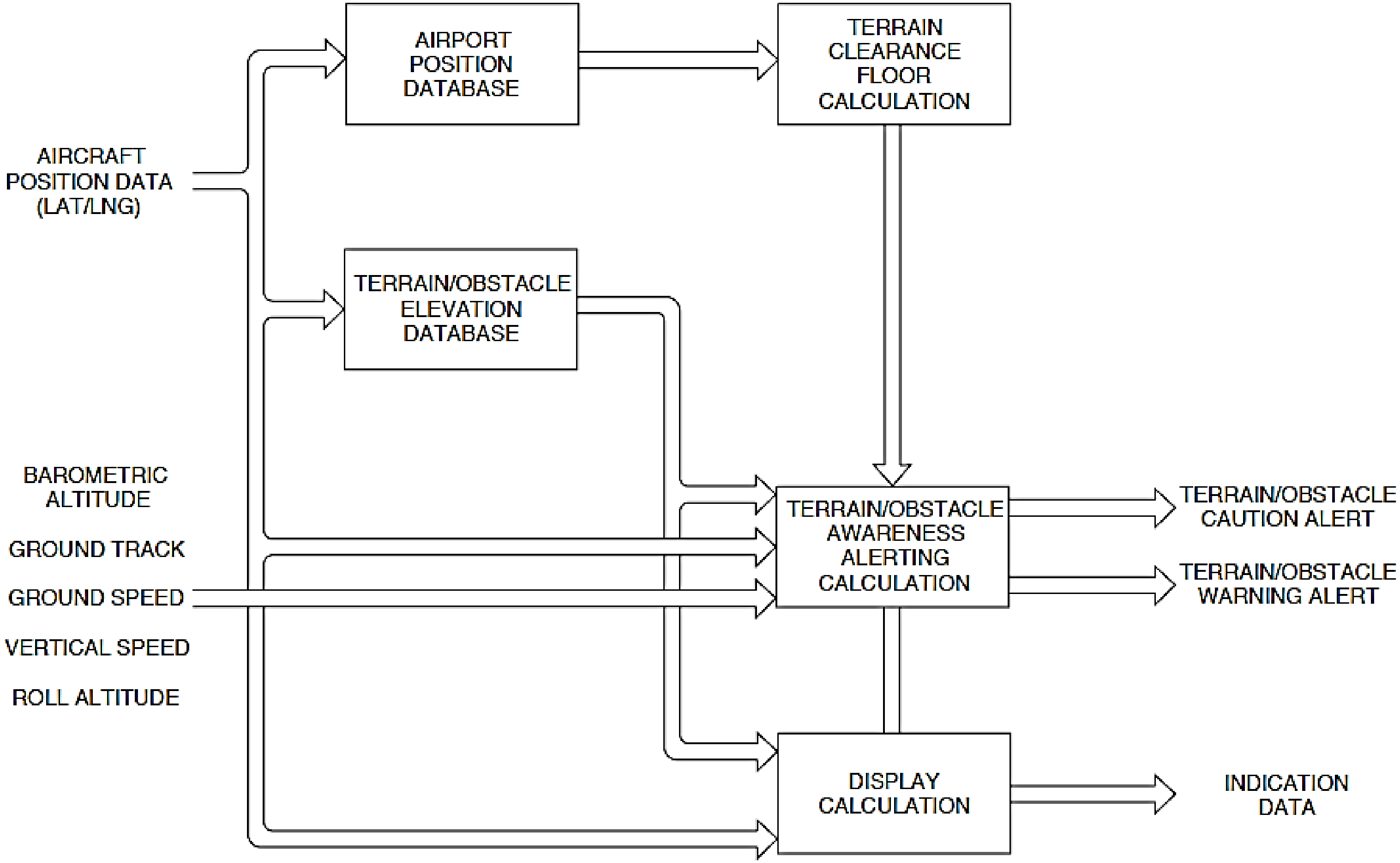


Figure 34- 135 TERRAIN ALERTING CALCULATION

Terrain Alerting:

Figure 34- 135 TERRAIN ALERTING CALCULATION

The EGPWS calculates potential conflicts between the aircraft flight path and terrain or obstacles to give an Electronic Instruments System (EIS) indication.

The terrain is calculated with the data that follow:

- Aircraft position
- Aircraft altitude
- Ground track
- Ground Speed
- Vertical Speed
- Roll attitude
- Terrain Clearance Floor
- Terrain and obstacle database.

Terrain Caution and Warning Alert Level:

Figure 34- 136 TERRAIN CAUTION AND WARNING AREAS

A terrain clearance area forward of the aircraft is continuously calculated. A terrain alert is calculated if the terrain clearance area does not agree with terrain elevation data in the terrain database.

Two alert areas are calculated as follows:

- Terrain caution alert level
- Terrain warning alert level.

If the aircraft goes into the terrain caution alert area, a "CAUTION TERRAIN, CAUTION TERRAIN" aural indication sounds and the EIS shows the conflicting terrain in yellow.

If the aircraft goes into the terrain warning alert area, a "TERRAIN, TERRAIN, PULL UP" aural indication sounds and the EIS shows the conflicting terrain in red.

NOTE

An obstacle alert has the same EIS indications but the aural indication is "CAUTION OBSTACLE, CAUTION OBSTACLE" or "OBSTACLE, OBSTACLE, PULL UP".

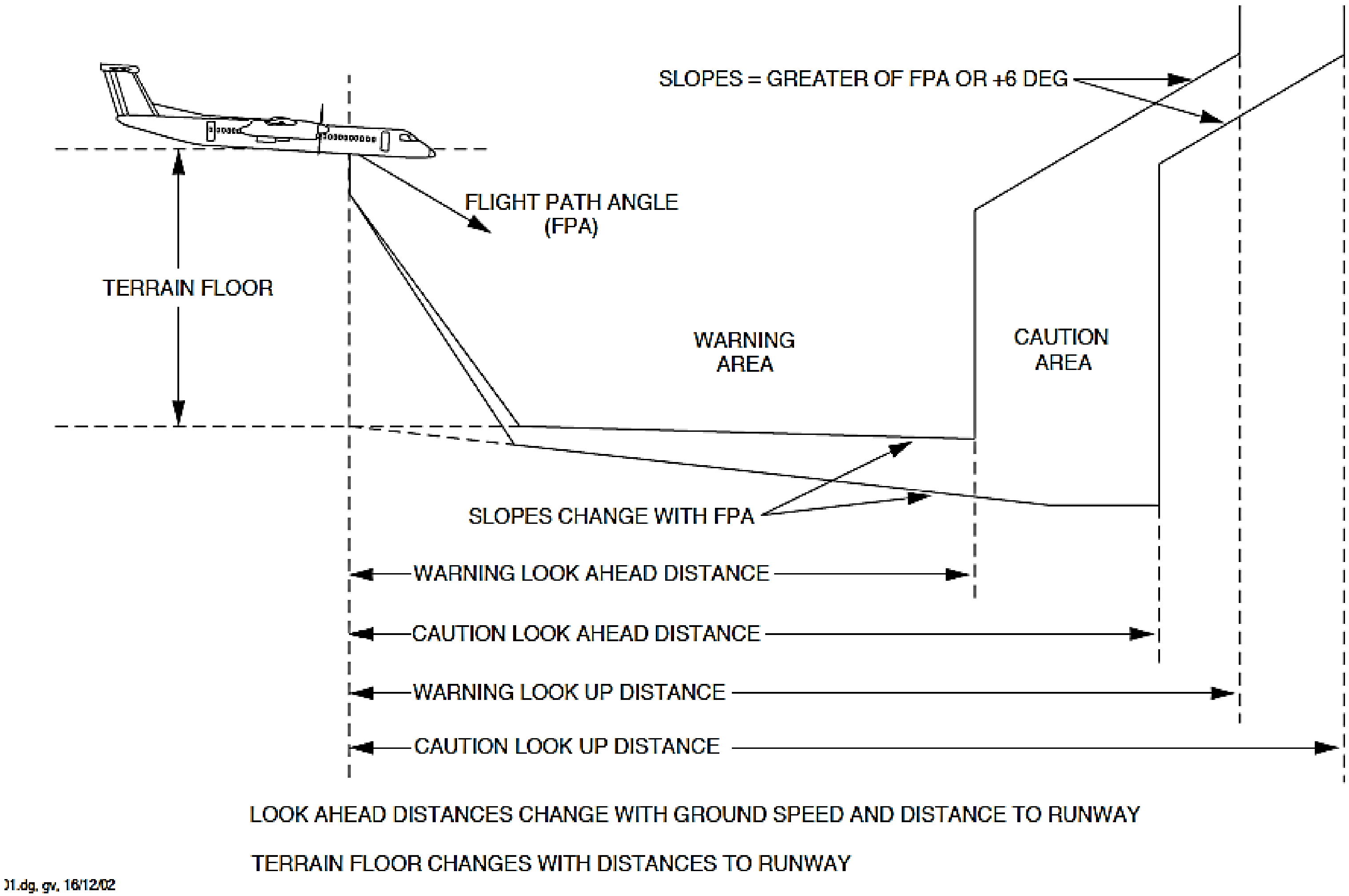


Figure 34- 136 TERRAIN CAUTION AND WARNING AREAS

Terrain Awareness Display Modes

The terrain awareness alerting and display function maintains a background display of the local terrain forward of the airplane for the optional flight compartment display. In the event of terrain or obstacle caution or warning conditions, an aural alert and lamp outputs are triggered. The background image is then enhanced to highlight the related terrain or obstacle threats forward of the airplane. There are two types of terrain awareness display modes as follows:

- Standard Display Mode
- Peaks Display Mode.

Standard and Peaks Display Mode:

Figure 34- 137 PEAKS DISPLAY, AIRCRAFT ABOVE TERRAIN
Figure 34- 138 PEAKS DISPLAY, AIRCRAFT ADJACENT TO TERRAIN
Figure 34- 139 STANDARD DISPLAY

The standard display mode lets the EIS show the terrain using different color patterns to indicate the vertical displacement between the terrain elevation and the aircraft altitude.

The red and yellow dot patterns indicate the terrain near or above the aircraft altitude. The solid yellow and red colors indicate alert and warning areas relative to the flight path of the aircraft. The medium and low density green display patterns indicate terrain that is below the aircraft and within 2000 feet of the aircraft altitude. The terrain more than 2000 feet below the aircraft is not displayed and the terrain display is typically blank during the en−route portion of the flight.

The peaks display mode lets the EIS show terrain around or below the aircraft. At altitudes that are safely above all terrain for the set indication range, the terrain is shown independently of aircraft altitude.

The highest and lowest elevations are shown to increase situational awareness.

Terrain more than 2000 feet below the aircraft is not shown. The peaks display mode gives density patterns and level thresholds relative to the terrain elevations in the indication area.

The peaks display mode has two elevation numbers to show the highest and lowest terrain that is currently shown. The elevation numbers shows terrain in hundreds of feet above sea level, Mean Sea Level (MSL). The terrain elevation numbers are displayed with the highest terrain number on top, and the lowest terrain number below it. The terrain numbers are shown in the same color as their terrain color pattern. Only one elevation number is shown when flying above flat terrain, water, or the aircraft more than 2000 feet above the highest terrain.

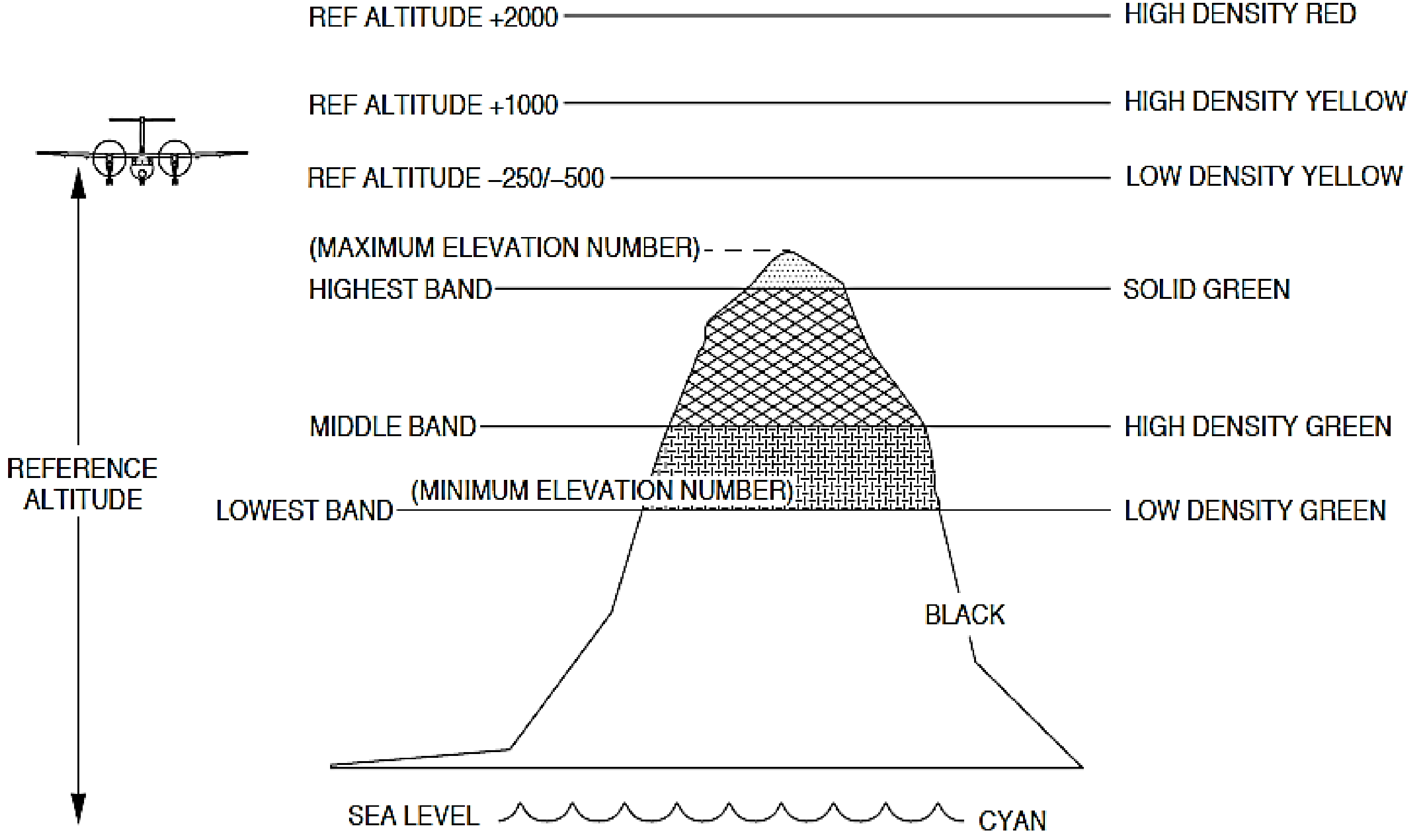


Figure 34- 137 PEAKS DISPLAY, AIRCRAFT ABOVE TERRAIN

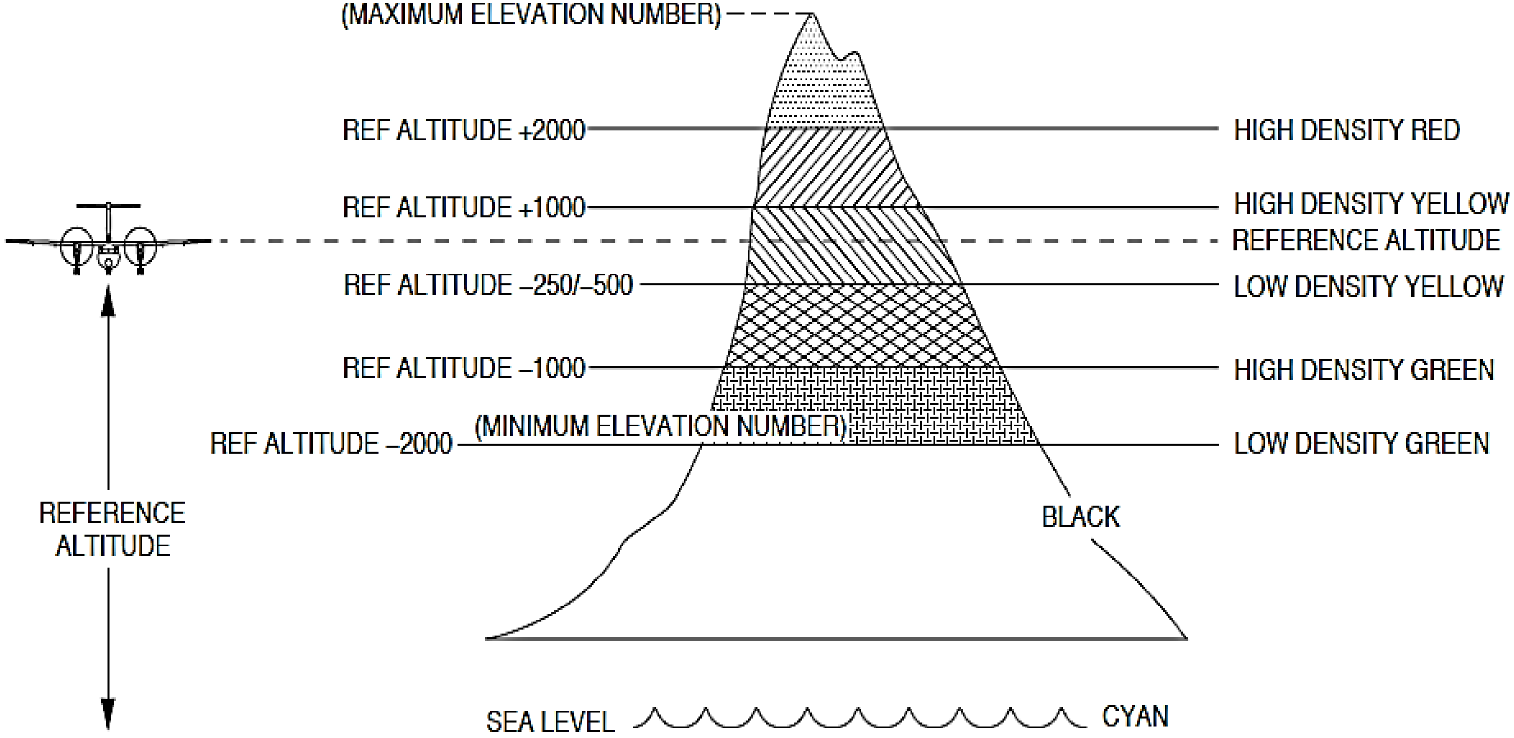


Figure 34- 138 PEAKS DISPLAY, AIRCRAFT ADJACENT TO TERRAIN

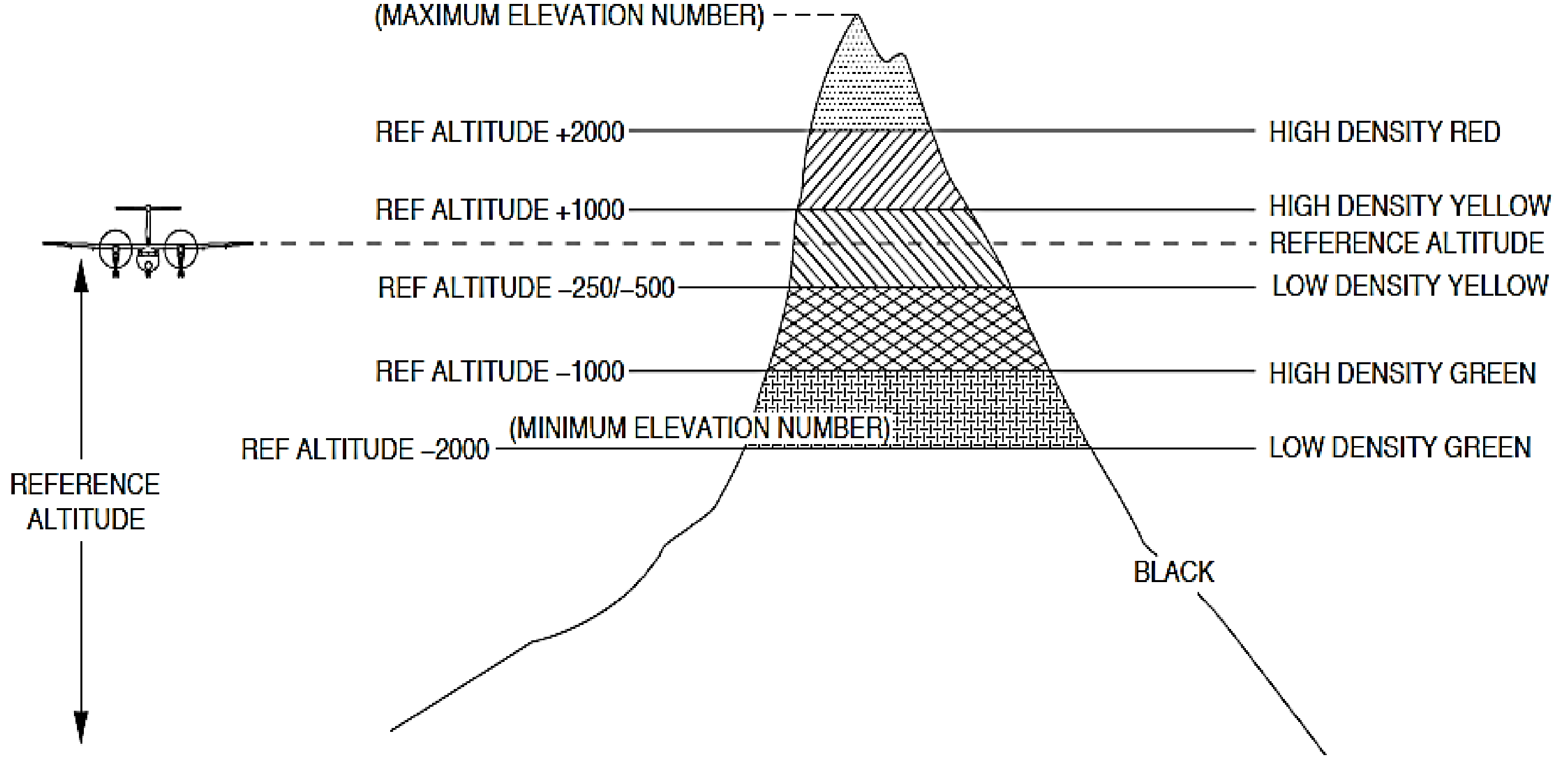


Figure 34- 139 STANDARD DISPLAY

The EGPWS supplies terrain indication data to the EIS Multi−Function Displays (MFD1, MFD2) for the standard and peaks display modes as follows:

COLOR	TERRAIN ELEVATION
Solid Red	Terrain warning
Solid Yellow	Terrain caution
High density red dots	More than 2000 ft. above aircraft
High density yellow dots	Between 1000 and 2000 ft. above aircraft

COLOR	TERRAIN ELEVATION
Black	No terrain/obstacles
Cyan	Water at sea level elevation (0 feet MSL) (for peaks display mode only)
Magenta	Unknown terrain. No terrain data in the database for the magenta area shown

COLOR	TERRAIN ELEVATION
Medium density yellow dots	Between 500 below to 1000 ft. above the aircraft
Medium density yellow dots	Landing gear down – Between 250 below to 1000 ft. above the aircraft
Solid Green	Terrain/obstacle that is the highest elevation band within 500 below the aircraft. It is shown only when no red or yellow terrain are in the range of the indication.
Solid Green	Landing gear down – Terrain/obstacle that is the highest elevation band within 250 below the aircraft. It is shown only when no red or yellow terrain are in the range of the indication.
High density green dots	Between 500 and 1000 ft. below the aircraft
High density green dots	Landing gear down – Between 250 and 1000 ft. below the aircraft
High density green dots	Terrain/obstacle that is the middle elevation band when no red or yellow terrain are in the range of the indication (for peaks display mode only).
Light density green dots	Between 1000 and 2000 ft. below aircraft
Light density green dots	Terrain/obstacle that is the lower elevation band when no red or yellow terrain are in the range of the indication (for peaks display mode only).

The reference altitude is calculated down from the actual aircraft altitude to give a thirty second advance indication of terrain when descending more than 1000 fpm.

Enhanced Ground Proximity Warning Computer (EGPWC):

Figure 34- 140 EGPWC
Figure 34- 141 COMPUTER, EGPWS LOCATOR

The Enhanced Ground Proximity Warning Computer (EGPWC) receives different system inputs and makes calculations to give aural and visual indications when the boundaries of an alerting area is exceeded. Terrain display data is supplied to the Electronic Instrument System (EIS) for indication.

The Enhanced Ground Proximity Warning Computer (EGPWC) is installed in the wardrobe on a shelf at station X−15.1, Y−21.1 and Z+161.2. It weighs 7 pounds (3.18 Kg) and is packaged in an ARINC rectangular aluminum chassis.

The EGPWC is cooled by free air circulation around the unit.

On aircraft with the Modsum 4−903303 incorporated, the EGPWC has a front panel with the following:

- Status indicators
- Door
- Self−test switch
- Headphone jack
- USB interface
- Two RJ45 Ethernet connectors
- Upload/download status indicators
- Test connector

Status Indicators: The unit's front panel has two status Light Emitting Diodes (LEDs) to show EGPWS system external fault and computer status. The External Fault LED will come on in yellow when the EGPWC has an external fault. The COMPUTER STATUS LED will come on in red when the EGPWC operates correctly and will turn to red when the EGPWC has an internal fault.

Door: A door gives access to the self−test switch, headphone jack,
USB interface, status indicators, and ethernet connectors.

Self−Test Switch and Headphone Jack: A "PRESS TO SELF−TEST" switch gives same test as the flight compartment selection. A headphone jack lets the operator listen to the test.

USB Interface, Ethernet connectors and Data−load Status Indicators: The USB Interface and Ethernet connectors are used to upload or download internal EGPWC data. Software and database is uploaded and history data is downloaded.

Test connector: There is one D−sub high−density 26−position connector on the front panel of EGPWC. The test connector can be connected to a portable PC to receive and control internal data. The front panel test connector is reserved for Honeywell use.

Figure 34- 140 EGPWC

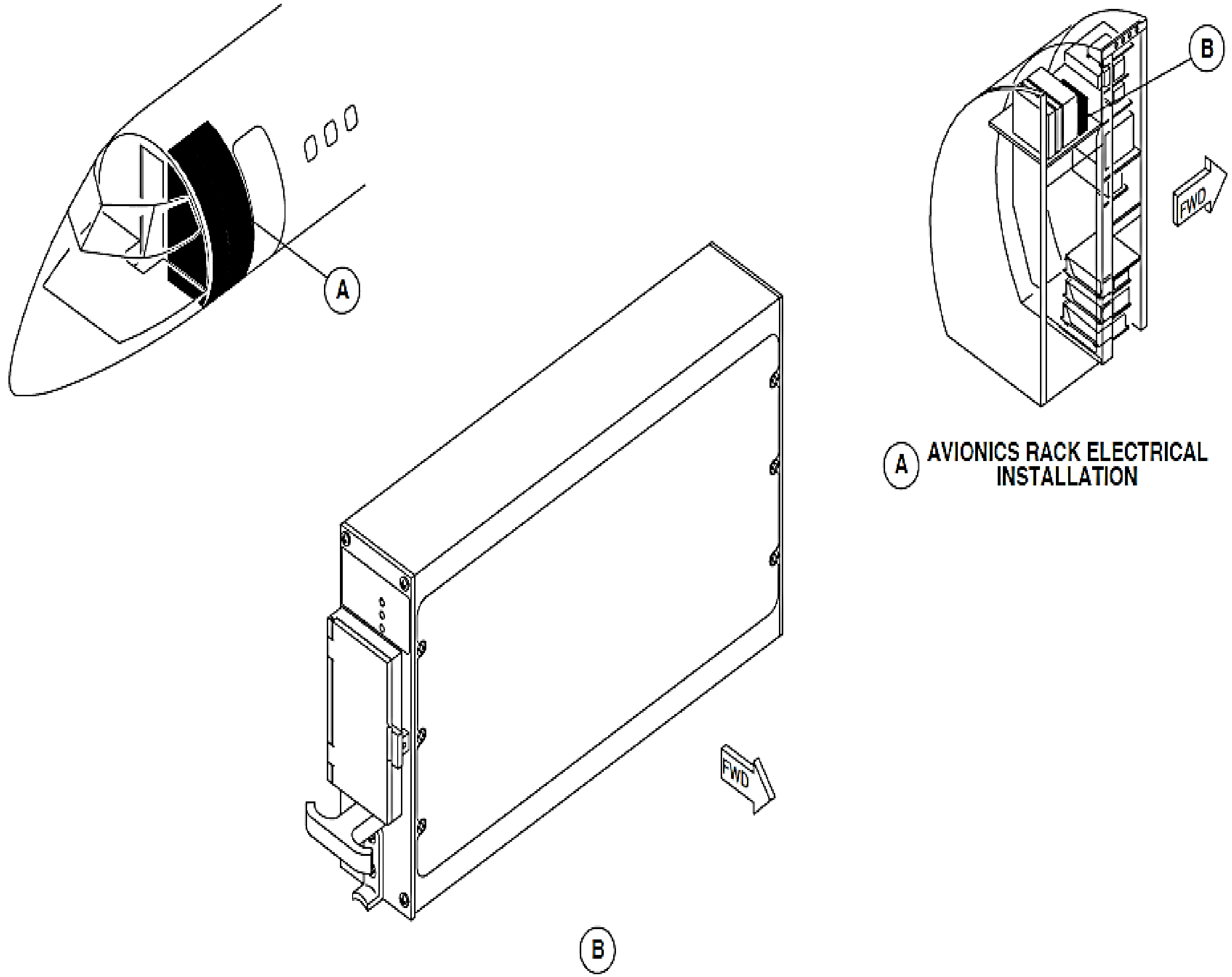


Figure 34- 141 COMPUTER, EGPWS LOCATOR

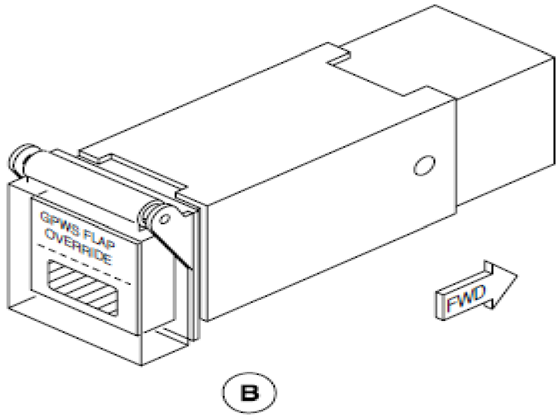
GPWS FLAP OVERRIDE Annunciator Switch:

Figure 34- 142 SWITCH, ANNUNCIATOR–FLAP OVERRIDE LOCATOR

The GPWS FLAP OVERRIDE annunciator switch is attached to pilot's left side console. DZUS fasteners and a related bonding wire directly connect to the console to make ground continuity between the side panel and console.

The GPWS FLAP OVERRIDE switch is a guarded annunciator switch. A transparent switch cover prevents accidental selection of the switch.

The GPWS FLAP OVERRIDE switch is pushed to change the indication modes. A black and yellow crosshatch indication in the switch comes on to show the manual selection to the GPWS FLAP OVERRIDE feature.



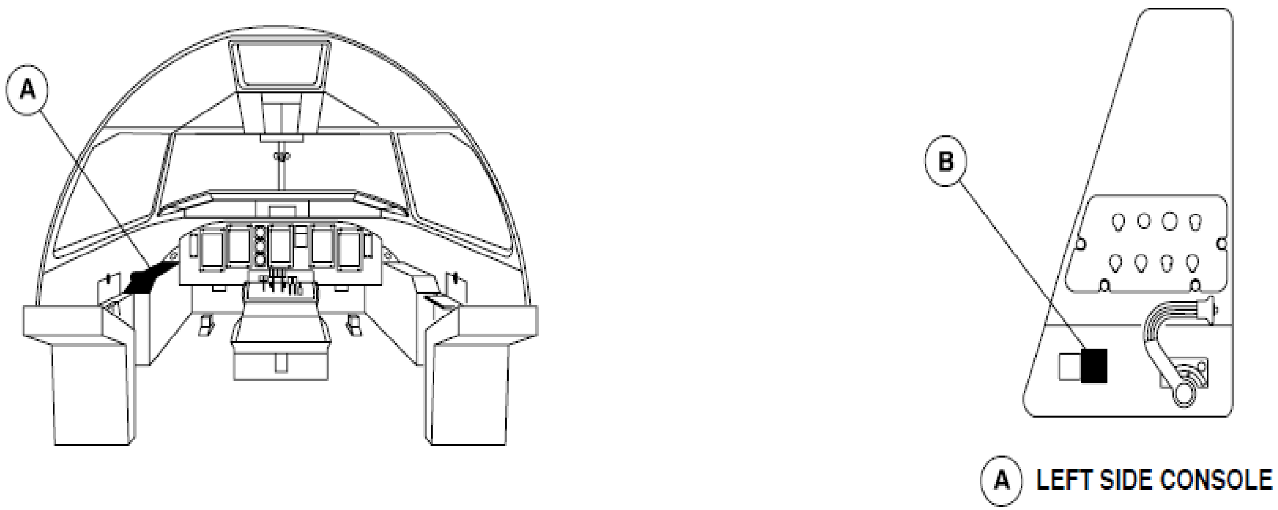


Figure 34- 142 SWITCH, ANNUNCIATOR–FLAP OVERRIDE LOCATOR

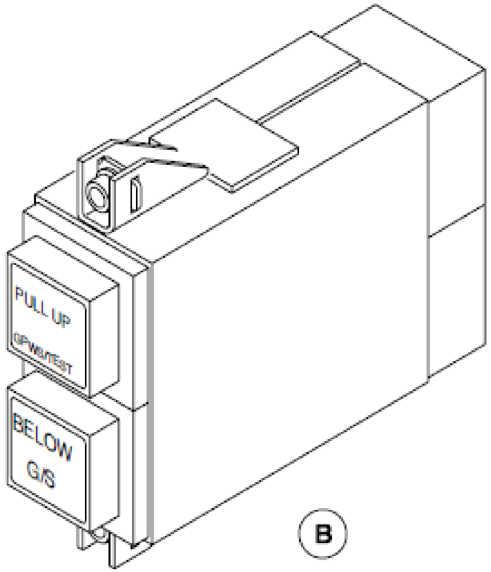
PULL UP and BELOW G/S Annunciator Switches:

Figure 34- 143 PULL UP AND BELOW G/S ANNUNCIATOR SWITCH LOCATOR

The PULL UP and BELOW G/S annunciator switches are vertically connected as a one assembly. Two PULL UP and BELOW G/S annunciator switches are attached to the glareshield panel assembly.

The PULL UP and BELOW G/S annunciator switches give the operation that follow:

- PULL UP indication
- GPWS TEST selection
- BELOW G/S indication
- BELOW G/S aural indication cancellation.



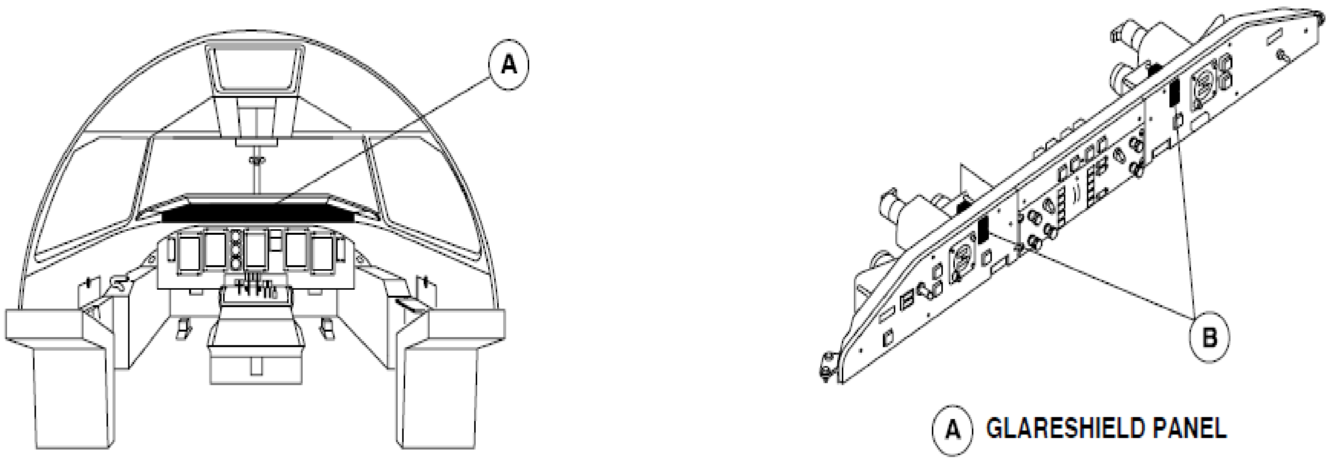


Figure 34- 143 PULL UP AND BELOW G/S ANNUNCIATOR SWITCH LOCATOR

INHIBIT Annunciator Switch:

Figure 34- 144 INHIBIT ANNUNCIATOR SWITCH LOCATOR

Two INHIBIT annunciator switch are attached to the glareshield panel assembly. Each switchlight is located in the area of view of each pilot.

The INHIBIT annunciator switches come on when one TERRAIN INHIBIT annunciator switch or the other is pushed to prevent terrain indication on the Multi−Functional Displays (MFD1, MFD2) navigation pages.

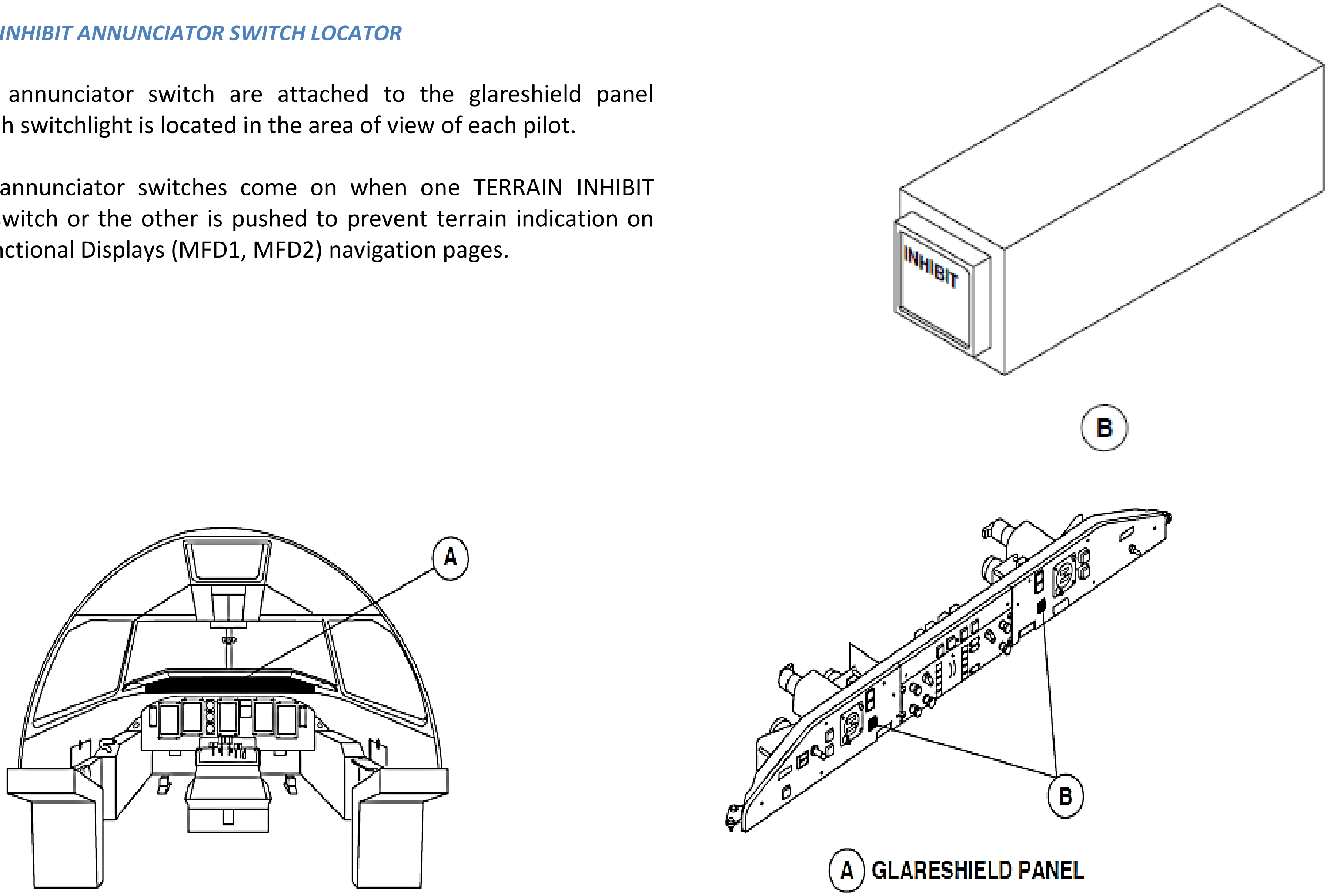


Figure 34- 144 INHIBIT ANNUNCIATOR SWITCH LOCATOR

Controls and Indications

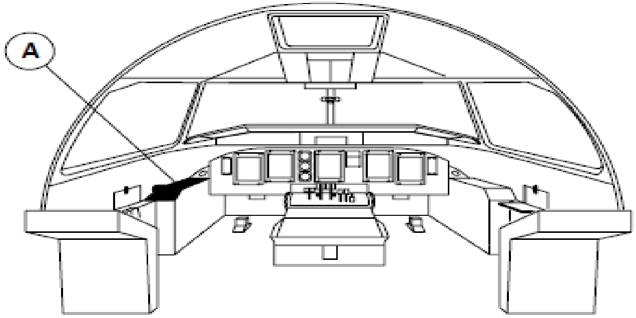
Figure 34- 145 GPWS FLAP OVERRIDE ANNUNCIATOR SWITCH

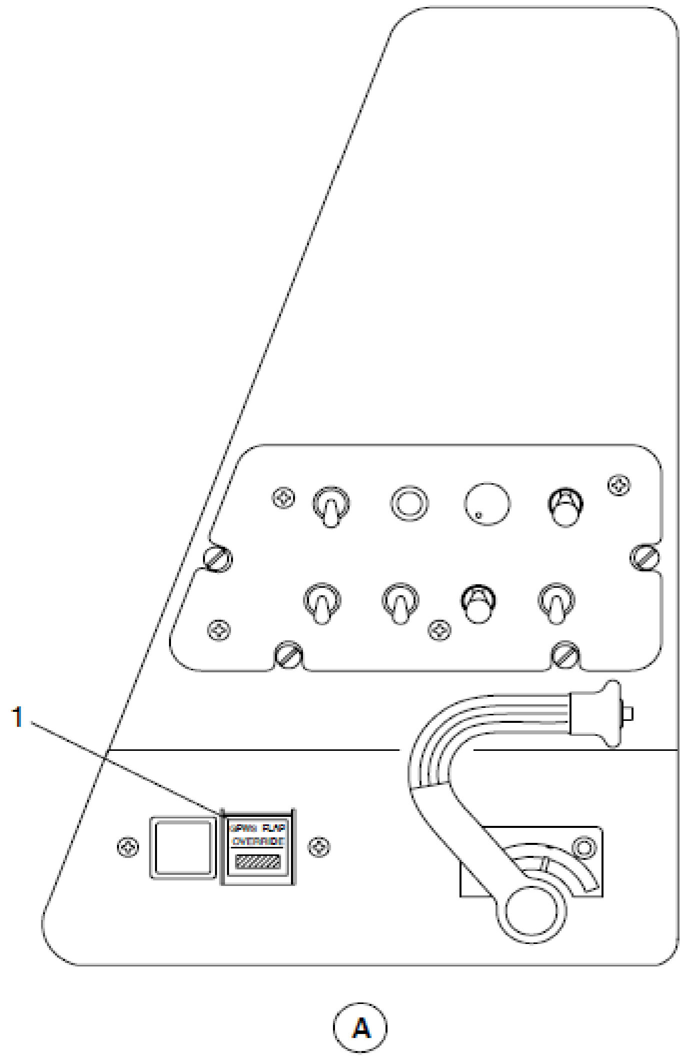
The FLAP OVERRIDE annunciator switch located on the pilot’s side console is pushed to prevent aural indications when a landing is made with zero or partial flaps extended.

A black and yellow crosshatch indication in the switch comes on to show the manual selection of the GPWS FLAP OVERRIDE feature.

The switch is pushed again to manually cancel the GPWS FLAP OVERRIDE feature.

When the Above Ground Level (AGL) is less than 50 feet, the override feature automatica





LEGEND

1. GPWS Flap Override Annunciator Switch.

Figure 34- 145 GPWS FLAP OVERRIDE ANNUNCIATOR SWITCH

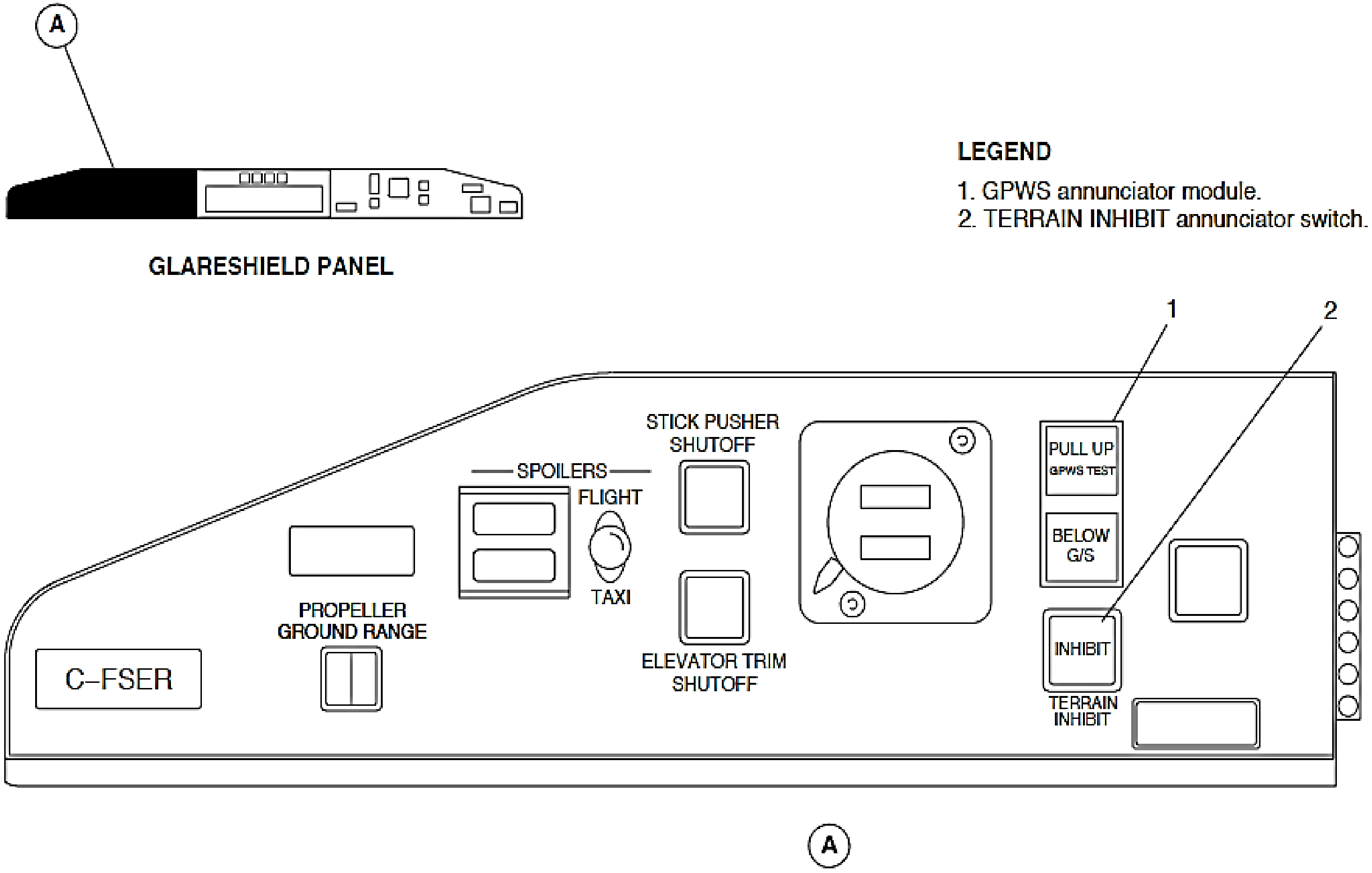


Figure 34- 146 PULL UP, BELOW G/S AND INHIBIT ANNUNCIATOR SWITCHES

Figure 34- 146 PULL UP, BELOW G/S AND INHIBIT ANNUNCIATOR SWITCHES

A red PULL UP annunciator switch located on the glareshield panel assembly comes on when the Enhanced Ground Proximity Warning System (EGPWS) calculates any of the warnings that follow:

PULL UP ANNUNCIATOR	VOICE
Mode 1, Excessive descent rate	"SINK RATE, SINK RATE", "PULL UP"
Mode 2, Excessive closure rate to terrain	"TERRAIN, TERRAIN", "PULL UP"
Mode 3, Descent after takeoff	"DON'T SINK, DON'T SINK"
Mode 4, Insufficient terrain clearance	"TOO LOW TERRAIN", "TOO LOW GEAR", "TOO LOW FLAPS"

PULL UP ANNUNCIATOR	VOICE
Terrain Clearance Floor (TCF) alerting	"TOO LOW TERRAIN"
Terrain alerting	"CAUTION TERRAIN", "CAUTION OBSTACLE", "WARNING TERRAIN, PULL UP", " WARNING OBSTACLE, PULL UP"

The PULL UP lights stay on while the aircraft is in the warning area.

An amber PULL UP annunciator switch located on the glareshield panel assembly comes on when the Enhanced Ground Proximity Warning System (EGPWS) calculates the condition that follows:

BELOW G/S ANNUNCIATOR	VOICE
Excessive descent below the Glideslope (GS)	Soft "GLIDESLOPE", Hard "GLIDESLOPE"

One BELOW G/S annunciator switch or the other is pushed for 1 second or longer to cancel the voice indication. The amber light stays on to show that the mode was manually cancelled.

The mode automatically stops when any of these conditions are set:

- The Above Ground Level (AGL) is less than 50 feet
- The Above Ground Level (AGL) is more than 1900 feet
- The ILS frequency is de–selected.

One TERRAIN INHIBIT annunciator switch or the other is pushed to prevent MFD terrain indications. The INHIBIT indication in the annunciator switch comes on and the two MFDs show a TERRAIN INHIBIT message.

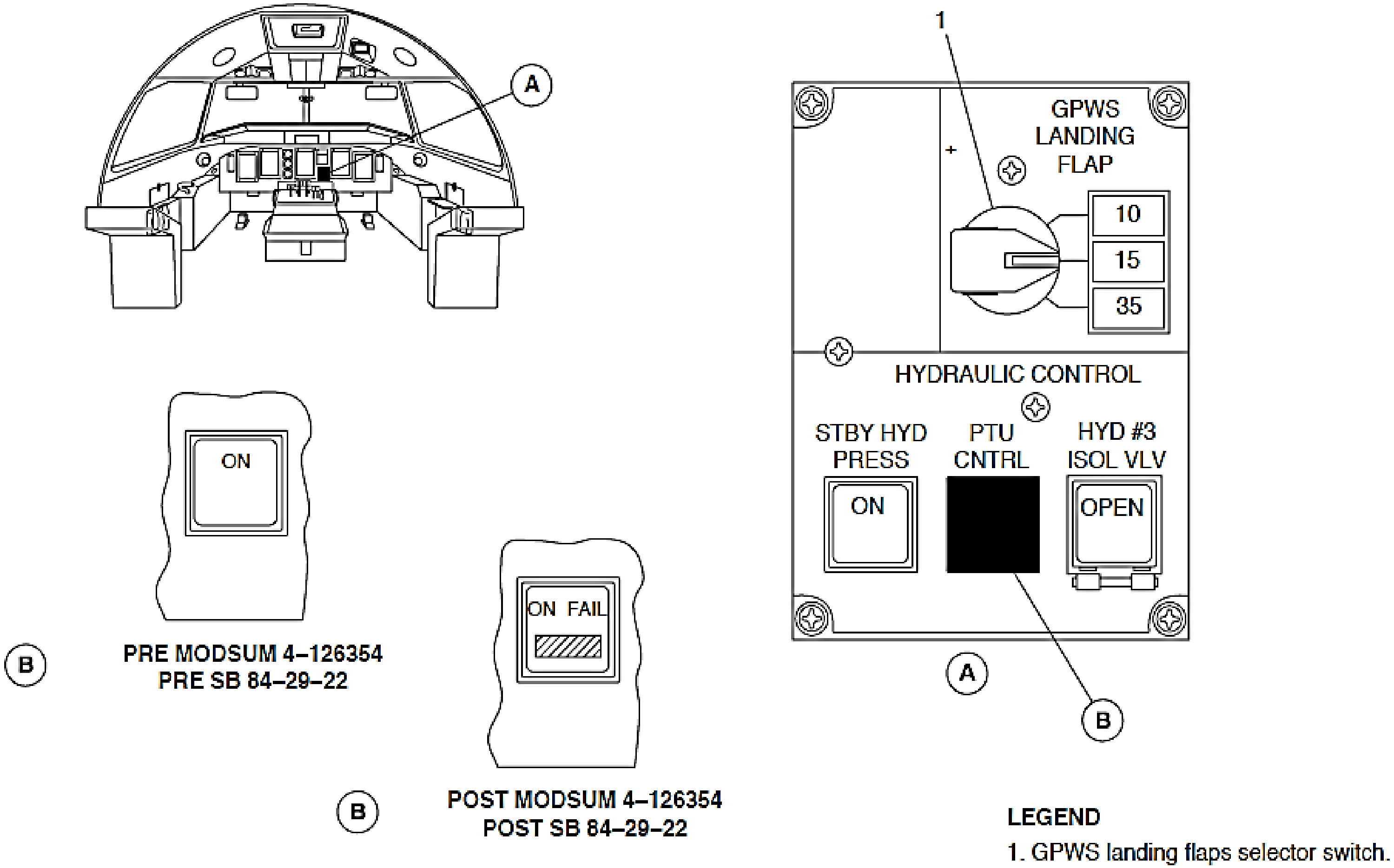


Figure 34- 147 GPWS LANDING FLAPS SELECTOR SWITCH

Figure 34- 147 GPWS LANDING FLAPS SELECTOR SWITCH
Figure 34- 148 GPWS LANDING FLAPS SELECTOR SWITCH (803SO90034)

The GPWS LANDING FLAP selector switch on the hydraulic control panel is set to one of three landing flap positions for the insufficient terrain clearance mode as follows:

- 10 degrees
- 15 degrees
- 35 degrees.

A related indication on the hydraulic control panel comes on to show the landing flap position selection.

The aircraft flaps select lever is set at the same GPWS LANDING FLAP selector switch selection.

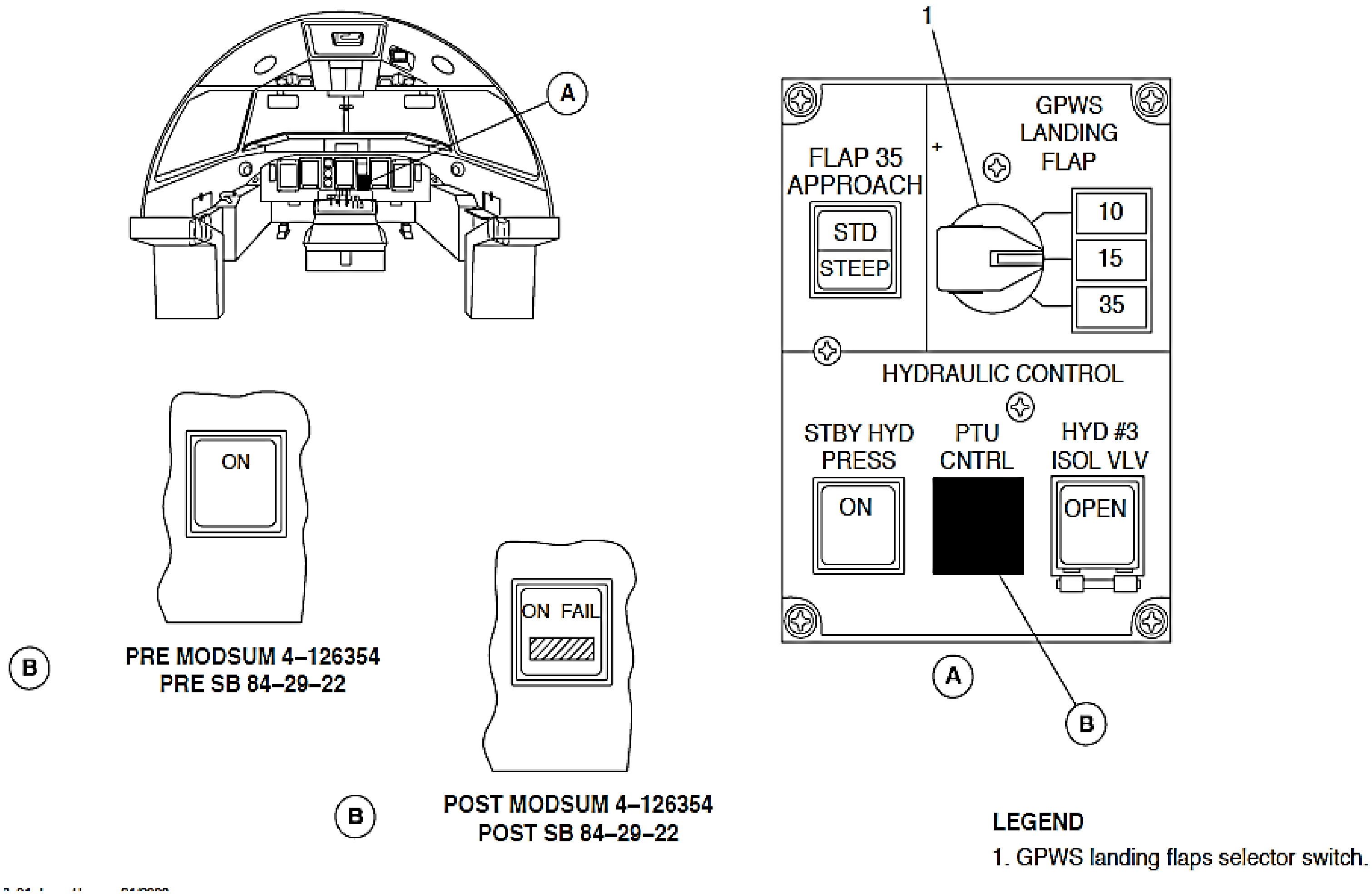
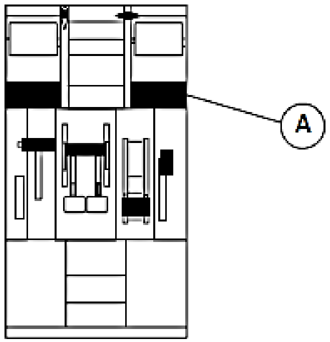
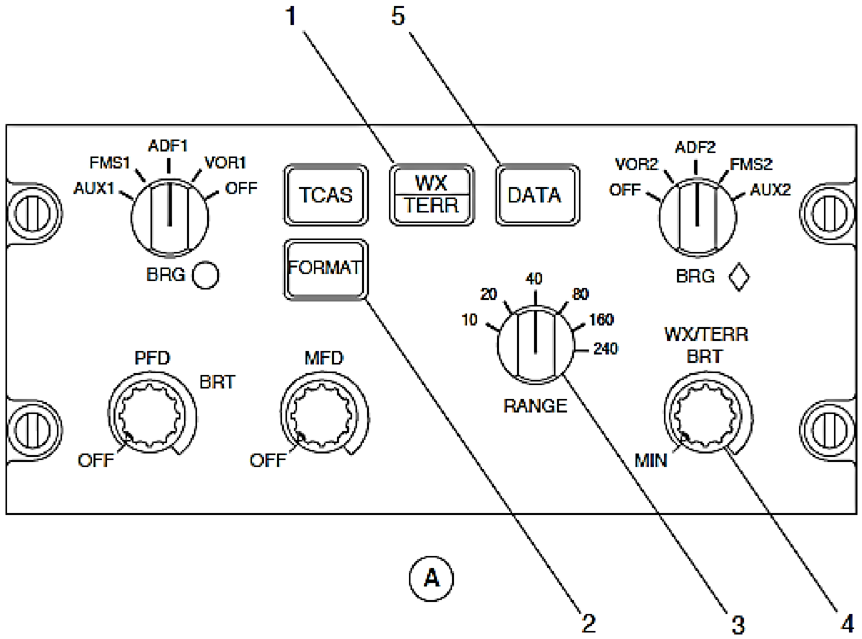


Figure 34- 148 GPWS LANDING FLAPS SELECTOR SWITCH (803SO90034)



CENTRE CONSOLE



LEGEND

1. Weather radar/terrain pushbutton switch.
2. FORMAT pushbutton switch.
3. Weather radar display range selector knob.
4. Weather radar display brightness knob.
5. DATA pushbutton switch.

Figure 34- 149 EFIS CONTROL PANEL, WEATHER RADAR CONTROLS

Figure 34- 149 EFIS CONTROL PANEL, WEATHER RADAR CONTROLS

The EFIS Control Panel (EFCP) has pushbutton and rotary switches and knobs to control the WXR and EGPWS systems indications that follow:

- WX/TERR pushbutton switch
- FORMAT pushbutton switch
- RANGE selector knob
- WX/TERR BRT knob.

WX/TERR Pushbutton Switch: A short WX/TERR pushbutton switch selection changes the MFD navigation page weather radar/EGPWS terrain automatic pop–up indication to a continuous EGPWS terrain indication. One more short WX/TERR pushbutton switch selection changes the MFD navigation page indication back to a weather radar/EGPWS terrain automatic pop–up indication.

A long WX/TERR pushbutton switch selection causes the weather radar and EGPWS terrain indications to go out of view. One more long WX/TERR pushbutton switch selection causes the weather radar and EGPWS terrain indications to come into view again.

If the navigation page is set to show a weather radar indication (arc or map format) and an EGPWS terrain alert is sensed, the terrain indication automatically comes into view. If the navigation page is set to the plan format, an EGPWS terrain alert causes it to change the arc or map format. An EGPWS terrain alert always causes the range indication to change to 10 NM. An AUTO RANGE message below the outer range digital mark shows that the range indication does not relate to the range selection on the EFCP.

NOTE

An EGPWS terrain alert condition prevents the EFCP switch selections that follow:

- WX/TERR pushbutton
- RANGE rotary switch
- FORMAT pushbutton
- DATA pushbutton.

After the EGPWS terrain alert condition, the 10 nmi range indications stays in view. The WX/TERR, TCAS, FORMAT, DATA, or RANGE switch is set to reset the indication.

If the EGPWS terrain indication is set while the aircraft is on the ground for more than 30 seconds and the WX radar is transmitting (WX ON, WX ALRT or WX GMAP), the indication changes to WX.

The MFD defaults to the WX indication at power–up. If the EFCP malfunctions, the last selection stays set unless the MFD is set to a different indication or to off.

NOTE

A WX/TERR pushbutton switch selection has no effect when the navigation page is not in view.

WX/TER BRT knob: The WXBRT knob is turned to adjust the weather radar image brightness.

RANGE selector knob: The RANGE selector knob is turned to change the weather radar and EGPWS range indications on the navigation page. The WXR and EGPWS system selectable ranges are 10, 20, 40, 80, 160 or 240 nmi.

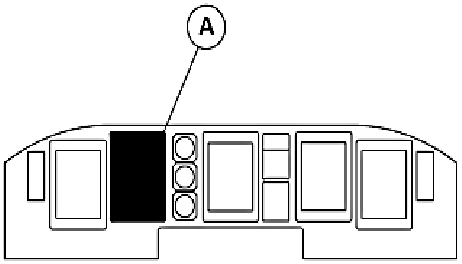
FORMAT Pushbutton Switch: A short FORMAT pushbutton switch selection changes the MFD navigation page indication from an FMS map format to a VOR/ILS arc format. One more short FORMAT pushbutton switch selection changes the MFD navigation page to show an MLS arc format, if the MLS is installed. One more short FORMAT pushbutton switch selection changes the MFD navigation page indication back to an FMS map format. A long FORMAT pushbutton switch selection changes to the MFD navigation page indication from an FMS map or VOR/ILS arc format to a plan format.

NOTE

The MFD always shows the related navigation source except when only one source is installed in the aircraft.

DATA pushbutton: Alternate action on this push−button enables the optional FMS Data selection on MFD NAV page. A DATA pushbutton selection displays on the NAV page the 10 nearest navigation aids as derived from the FMS data base. One more DATA pushbutton selection changes the NAV page to show a presentation of optional navigation aids plus the 10 nearest airports as derived from the FMS data base. One more DATA pushbutton selection removes all options (default selection).

A Terrain Awareness Display (TAD) gives an indication of the surrounding terrain in different density of green, yellow and red dot patterns. The indication is shown on the navigation display page from the aircraft altitude relative to the terrain data in the EGPWS computer.



LEGEND

1. Terrain Image.
2. Terrain Display Mask.
3. AUTO RANGE Message.
4. Terrain Mode Message.
5. Terrain Message.
6. Terrain Peaks Annunciation.

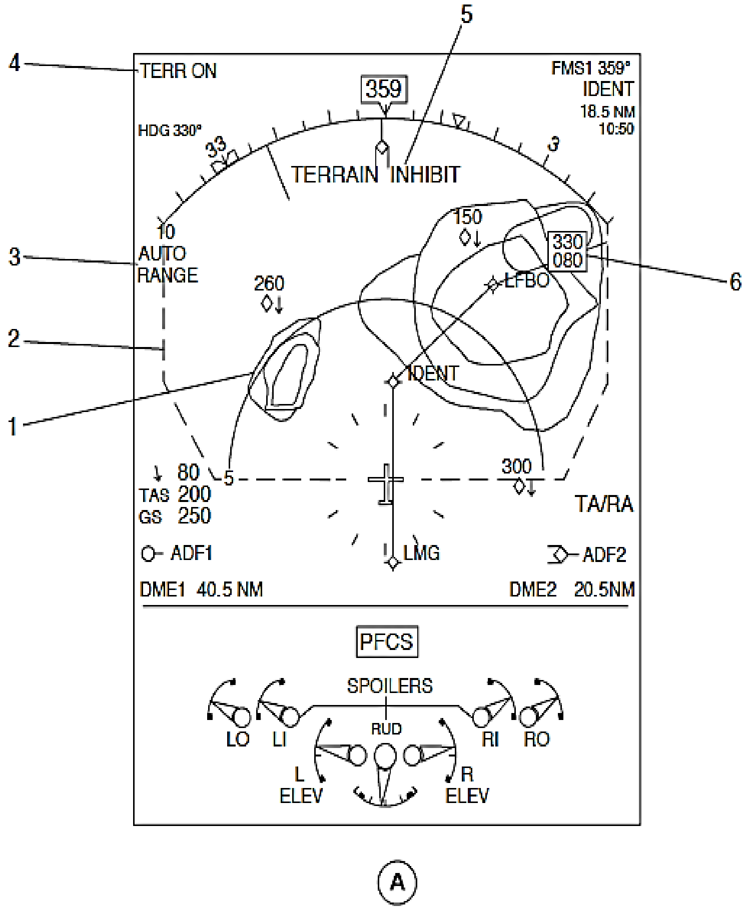


Figure 34- 150 MULTI−FUNCTION DISPLAY (MFD1, MFD2), EGPWS INDICATIONS

Figure 34- 150 MULTI–FUNCTION DISPLAY (MFD1, MFD2), EGPWS INDICATIONS

The MFD navigation pages (ARC or MAP format) give the terrain indications that follow:

- Image
- Display mask
- Auto range message
- Mode messages
- Messages
- Peaks annunciation.

Terrain Display Mask: When the terrain display is set or during a terrain alert, two white dashed radials come into view to show the lateral limits of the terrain display mask. The heading arc scale is the top limit of the indication.

Auto Range Message: A terrain alert condition will cause the 10 nmi range and terrain images to come into view with a white AUTO RANGE message below the outer range digital mark.

Terrain Mode Messages: When the terrain display is set or during a terrain alert, the upper left part of MFD navigation pages show the terrain modes messages that follow:

- TERR ON
- TERR TEST.

TERR ON Terrain Mode Message: A white TERR ON terrain mode message is shown while the terrain image is in view.

TERR TEST Terrain Mode Message: A white and yellow TERR TEST terrain mode message is shown while the test indication is in view.

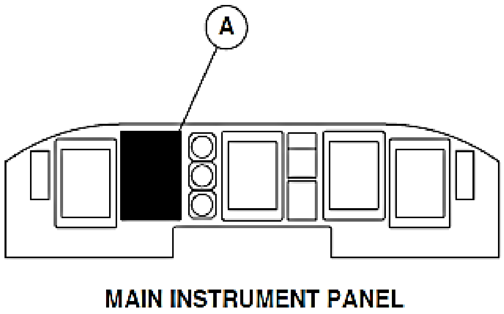
Terrain Messages: The EGPWS has terrain related messages that are shown in the upper part of the navigation display when the terrain mode is set that follow:

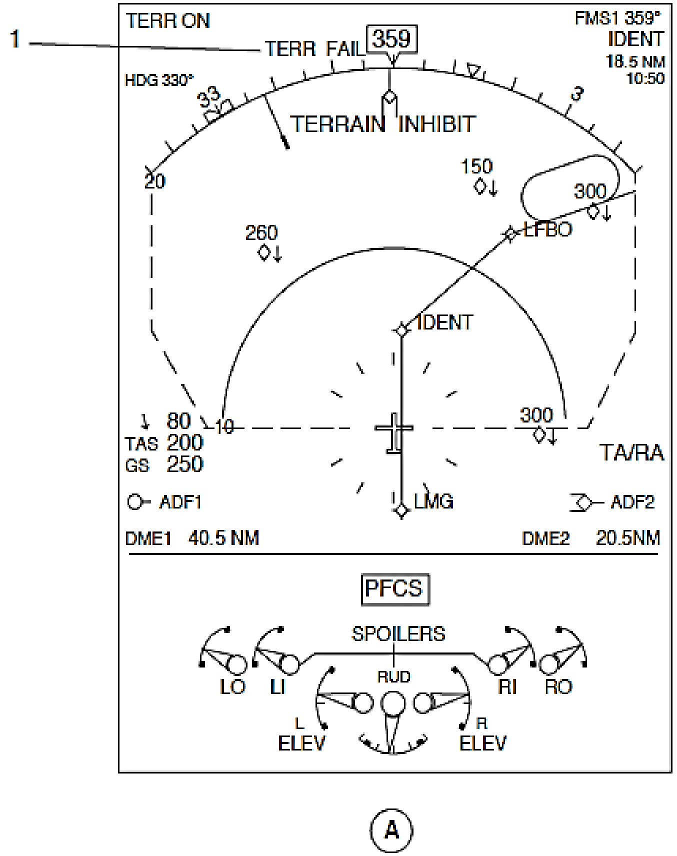
- TERRAIN INHIBIT
- TERR UNAVAILABLE.

TERRAIN INHIBIT Terrain Message: A white TERRAIN INHIBIT terrain message shows the TERRAIN INHIBIT annunciator switch selection.

TERR UNAVAILABLE Terrain Message: A white TERR UNAVAILABLE terrain message is shown when the EGPWS is not installed.

Peaks Annunciation: When the terrain image is shown, two three−number indications in the upper right part of the terrain mask show the altitude of the highest and lowest points of the terrain image. The peaks annunciation color is same as its related indication. The peaks annunciation goes out of view if the EGPWC malfunctions.





LEGEND

1. TERRAIN FAIL Message.

Figure 34- 151 MULTI−FUNCTION DISPLAY (MFD1, MFD2), EGPWS MALFUNCTION INDICATIONS

Figure 34- 151 MULTI–FUNCTION DISPLAY (MFD1, MFD2), EGPWS MALFUNCTION INDICATIONS

If the EGPWS malfunctions, the MFD navigation page (arc or plan format) will show a yellow TERR FAIL message. The terrain mode message (TERR ON or TERR TEST) stays in view but the terrain images do not.

A TERR FAIL message is shown during a condition that follow:

- EGPWS senses an internal malfunction in the terrain circuitry or unreliable external sensor data
- IOP senses a malfunction from the EGPWS outputs
- Terrain data bus has malfunctioned while the terrain display is not set off
- Range received from the terrain data bus output is not the same as the calculation made by the MFD while not in terrain alert mode
- MFD senses a malfunction of the data bus to the related IOP
- Terrain image is not supplied by the EGPWS to the MFD
- FMS is not energized.

Figure 34- 152 CAUTION AND WARNING PANEL, GPWS CAUTION INDICATION

The GPWS caution light comes for the malfunction conditions that follow:

- EGPWS computer
- IOM1, IOP1 (FDPS1) and IOM2, IOP2 (FDPS2).

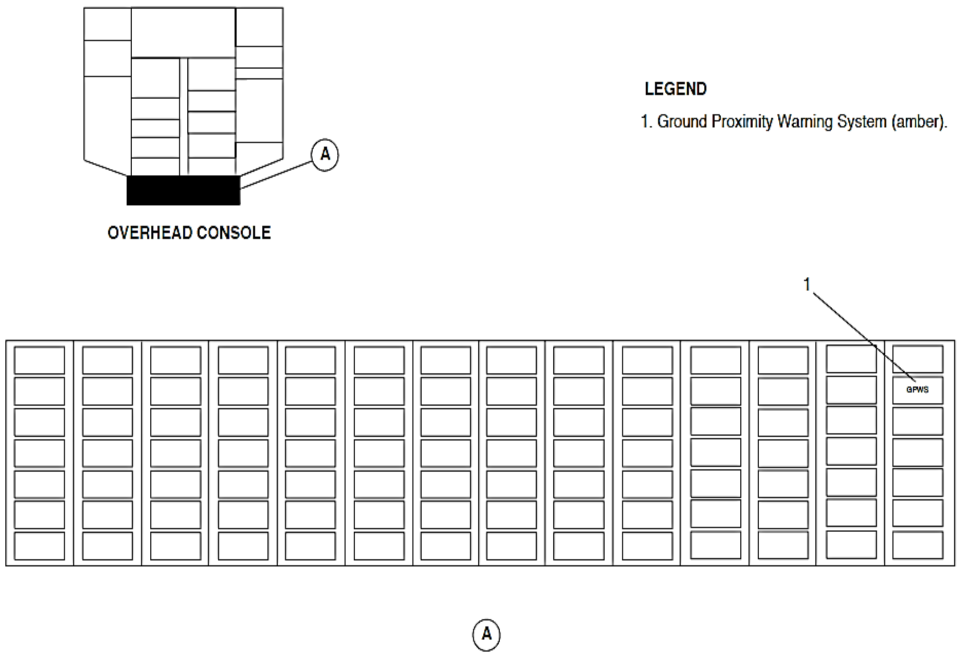


Figure 34- 152 CAUTION AND WARNING PANEL, GPWS CAUTION INDICATION

Interfaces

The left main bus supplies the Ground Proximity Warning Computer (GPWC) with 28 Vdc electrical power through a 5 Ampere circuit breaker. The circuit breaker is located on the 28 Vdc avionics circuit breaker panel at position A1..

The left main bus supplies 28 Vdc electrical power through a 1 Ampere circuit breaker to the lights in the annunciator switches that follow:

- PULL UP
- BELOW G/S
- FLAP OVERRIDE
- INHIBIT.

The circuit breaker is located on the 28 Vdc avionics circuit breaker panel in position B1.

The Advisory Control Unit (ACU) supplies discrete inputs to make the lights come on.

The Enhance Ground Proximity Warning Computer (EGPWC) receives discrete inputs from the sources that follow:

DATA	SOURCE
Landing Flaps Switch 10, 15, 35 Degree Position	GPWS LANDING FLAPS SELECTOR SWITCH
Audio Suppression	SPM1, SPM2
EGPWC Flap Override	GPWS FLAP OVERRIDE Annunciator Switch
Self Test	The PULL UP, GPWS TEST Annunciator Switch
Glide Slope Cancel	BELOW G/S Annunciator Switch
Terrain Inhibit	INHIBIT Annunciator Switch
Mode 6 Callout Low Volume	This discrete is not enabled in the DHC–8 Series 400
Steep Approach Enable	Rear connector top plug pin TP4C (Steep Approach Enabled) is permanently grounded.

DATA	SOURCE
Steep Approach Select	Steep Approach Switch
Mode 6 Altitude Callouts Enable	Pin 8C (Altitude Callouts Enable) of P1B–A1 is permanently grounded.

The Enhance Ground Proximity Warning Computer (EGPWC) receives digital inputs through the related IOP from sources that follow:

DATA	SOURCE
GPS Altitude	FMS
GPS Track Angle	FMS
GPS Latitude	FMS
GPS Longitude	FMS
GPS Ground Speed	FMS
GPS Vertical Speed	FMS
GPS Status	FMS
GPS Sensor Status	FMS
GPS Horizontal Integrity Limit	FMS
GPS Vertical Figure of Merit	FMS
GPS Horizontal Figure of Merit	FMS
FMS Destination Latitude	FMS
FMS Destination Longitude	FMS
FMS Track Angle	FMS
FMS Latitude	FMS
FMS Longitude	FMS
FMS Ground Speed	FMS
FMS True Heading	FMS
FMS Discrete Word	FMS
AGL Altitude	RA
Decision Height	ICP1, ICP2
Localiser Deviation	VHF NAV1, VHF NAV2
Glideslope Deviation	VHF NAV1, VHF NAV2
MLS Azimuth Deviation	MLS1, MLS2

DATA	SOURCE
Flap Lever Position	FCU
Range Rings	EFCP1, EFCP2
Landing Gear Position	PSEU
Terrain Display Request	INHIBIT annunciator switch
MLS Azimuth Deviation	MLS1, MLS2
MLS Elevation Deviation	MLS1, MLS2
Standard Altitude	ADC1, ADC2
Baro Corrected Altitude	ADC1, ADC2
Calibrated Airspeed	ADC1, ADC2
Altitude Rate	ADC1, ADC2
Static Air Temperature	ADC1, ADC2
Navigation Source	FGCP
Backcourse Mode	FGCP
Magnetic Heading	AHRS1, AHRS2
Pitch Attitude	AHRS1, AHRS2
Roll Attitude	AHRS1, AHRS2

The Enhance Ground Proximity Warning Computer (EGPWC) receives digital inputs from the FMS as follows:

DATA
FMS Track Angle
FMS Latitude
FMS Longitude
FMS Ground Speed
FMS True Heading
FMS Discrete Word

The Enhance Ground Proximity Warning Computer (EGPWC) receives digital inputs from the GPS as follows:

DATA
GPS Altitude
GPS Track Angle
GPS Latitude
GPS Longitude
GPS Ground Speed
GPS Vertical Speed
GPS Sensor Status
GPS Hor Integ Limit
GPS Vert Figure of Merit
GPS Horizontal Figure of Merit
Latitude fine
Longitude fine
Horizontal Dilution of Precision (HDOP)
Vertical Dilution of Precision (VDOP)

The Enhance Ground Proximity Warning Computer (EGPWC) supplies the discrete outputs that follow:

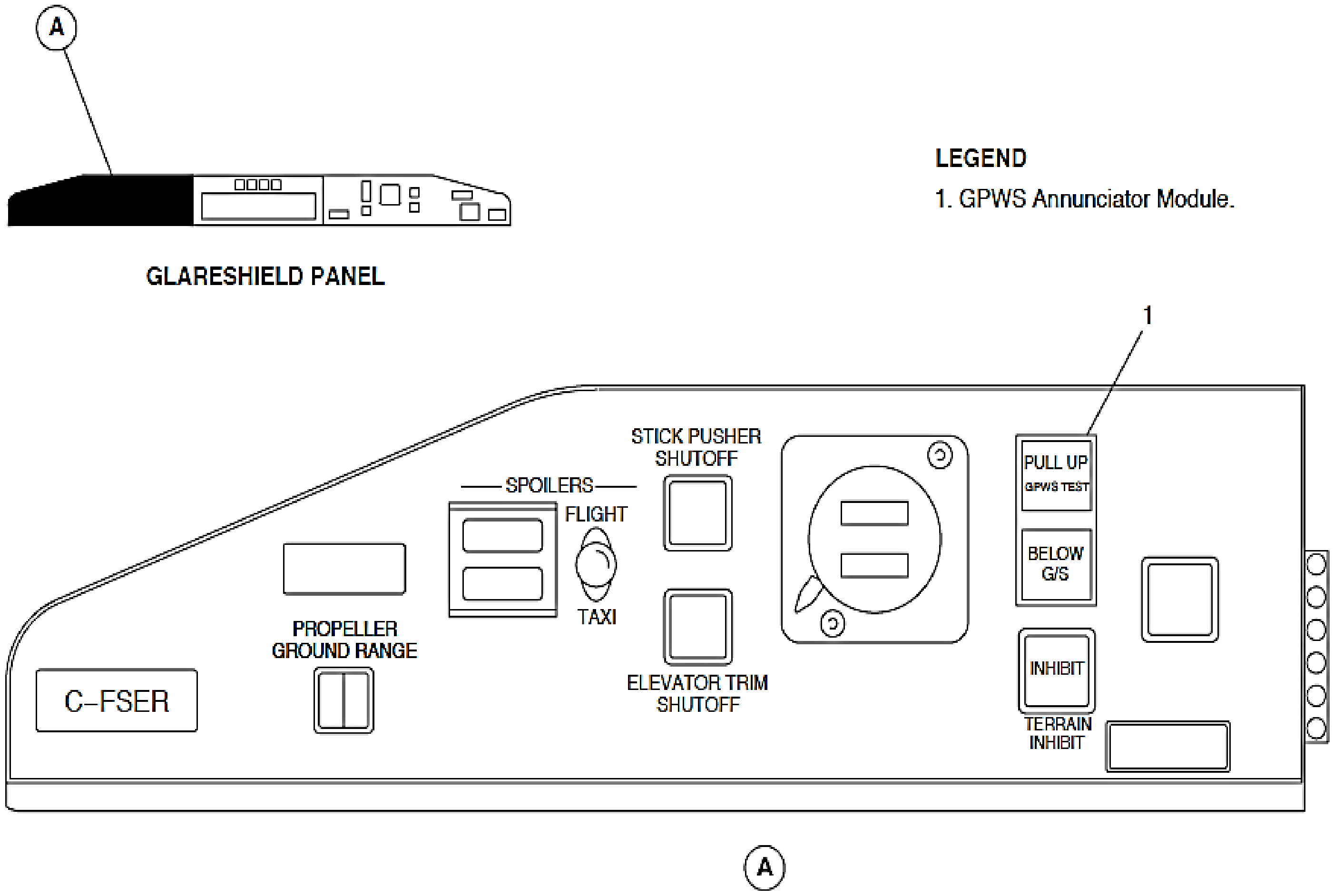
DATA	TERMINATION
EGPWC Audio On	IOM1, IOM2
Glideslope Alert light	BELOW G/S Annunciator Switch
EGPWS Alert/Warning Light	The PULL UP, GPWS TEST Annunciator Switch
EGPWC System Status	Caution and Warning Panel
Pilot, Copilot Terrain Select	MFD1 Display relay, MFD2 Display relay

NOTE

The FLAP OVERRIDE annunciator switch output is supplied through the Advisory Control Unit (ACU) back to the annunciator switch to make the cross hatching lights come on. The INHIBIT annunciator switch output is supplied through the ACU back to the annunciator switch to make the INHIBIT lights come on.

The Enhance Ground Proximity Warning Computer (EGPWC) supplies the digital outputs that follow:

DATA	TERMINATION
TERRAIN FAIL Message	Through FDPS1, FDPS2 MFDs
TERRAIN NOT AVAILABLE Message	Through FDPS1, FDPS2 to MFDs
Terrain Caution	Through FDPS1, FDPS2 to FDR
Terrain warning	Through FDPS1, FDPS2 to FDR
Pilot Terrain Pop up	Through FDPS1, FDPS2 to FDR
Copilot Terrain Pop up	Through FDPS1, FDPS2 to FDR
TERRAIN INHIBIT Message	Through FDPS1, FDPS2 to MFD1 MFD2
Peaks Values	MFD1, MFD2
Terrain Images (ARINC 453)	Through MFD Display relays to related MFD



LEGEND

1. GPWS Annunciator Module.

Figure 34- 153 EGPWS TEST SELECTION

Tests

Figure 34- 153 EGPWS TEST SELECTION

The EGPWC has a manual self−test sequence to show the status of the system. The self−test sounds the EGPWS status, makes the lights in the annunciator switches come on, and shows an MFD terrain test indication.

The EGPWS has six different levels of self−tests identify conditions that follow:

- Level 1 – Status of each of the primary functions of the system
- Level 2 – All current internal and external malfunctions in the system
- Level 3 – Configuration status of the EGPWC and the installation
- Level 4 – Faults sensed during last flight
- Level 5 – Alerts sensed during last flight
- Level 6 – Changes of the discrete inputs.

Before the EGPWS test is started, check the conditions that follow:

- Aircraft is on the ground
- On the EGPWC front panel, check that the green COMPUTER OK status indicator is on
- GPWS Landing Flap selector switch is set to the 35 degrees position
- FMS is initialized
- Course pointer is set less than 105 degrees of the aircraft heading
- The two TERRAIN INHIBIT annunciator switches are not set
- The two MFD are set to show navigation pages
- Terrain indication is shown.

Level 1 Self-Test: The pilot PULL UP, GPWS TEST annunciator switch is pushed and held for more than three seconds to start a level 1 self-test sequence. The EGPWS level 1 self-test gives the indications that follow:

- GPWS caution light comes on
- The two BELOW G/S lights come on
- “GLIDESLOPE” voice is heard
- The two BELOW G/S lights go off
- The two PULL UP lights come on.
- “PULL UP” voice is heard
- The two PULL UP lights go off
- TERRAIN TEST mode continues for 12 seconds on the two MFDs
- “TERRAIN, TERRAIN, PULL UP” voice is heard.

On aircraft with ModSum 4−309203 or 4−410300 (without ModSum 4−429519 or SB84−34−102 or SB84−34−47 or ModSum 4−457043 or SB84−34−115 or SB84−34−36 or SB84−34−83 or ModSum 4−457062 or SB84−34−160) incorporated, if the PULL UP, GPWS TEST annunciator switch is pushed and held until the “GLIDESLOPE” voice is heard, the test continues to give the indications that follow:

- “GLIDESLOPE”
- “PULL UP”
- “TERRAIN, TERRAIN”
- “PULL UP”
- “SINK RATE”
- “PULL UP”
- “TERRAIN”
- “PULL UP”
- “DON’T SINK, DON’T SINK”

- “TOO LOW TERRAIN”
- “TOO LOW GEAR”
- “TOO LOW FLAPS”
- “TOO LOW TERRAIN”
- “GLIDESLOPE”
- “BANK ANGLE, BANK ANGLE”
- “MINIMUMS”
- “MINIMUMS”
- “TOO LOW TERRAIN”
- “CAUTION TERRAIN”
- “CAUTION TERRAIN”
- “TERRAIN”
- “TERRAIN”
- “PULL UP”
- “CAUTION OBSTACLE”
- “CAUTION OBSTACLE”
- “OBSTACLE”
- “OBSTACLE”
- “PULL UP”.

Level 2 Self-Test: The pilot PULL UP, GPWS TEST annunciator switch is pushed again less than 3 seconds after the completion of the level 1 self-test to listen to currently sensed faults. The EGPWS level 2 self-test gives the indications that follow:

- “Current Faults”
- “No Faults”.

Level 3 Self-Test: The pilot PULL UP, GPWS TEST annunciator switch is pushed again less than 3 seconds after the completion of the level 2 self-test to listen to system configuration data. The EGPWS level 3 self-test gives the indications that follow:

- Part Number
- Modification Status
- Serial Number
- Application Software Version
- Configuration Software Version
- Terrain Database Version
- Envelope Modulation Database Version
- Boot Code Version
- Aircraft Type
- Audio Menu
- Altitude Call out Menu
- Configuration Option (831SO90180 only)
- Dual FMS (834SO00418−01 or 834SO90214 only)
- Dual RA (834SO00417 or 834SO90212 only)
- Dual MLS (834CH00398 only)
- Bank Angle
- Peaks
- GPS Altitude Reference (Aircraft with UNS FMS)
- Obstacle Awareness
- Terrain Display.

Level 4 Self-Test: The pilot PULL UP, GPWS TEST annunciator switch is pushed again less than 3 seconds after the completion of the level 3 self-test to listen to faults that were sensed during the last 10 flight legs. The previous flight faults are not heard if the pilot PULL UP, GPWS TEST annunciator switch is pushed less than 3 seconds after the completion of the level 3 self-test and held until the “PRESS TO CONTINUE” is heard to start a level 4 self-test sequence.

Level 5 Self-Test: The pilot PULL UP, GPWS TEST annunciator switch is pushed again less than 3 seconds after the completion of the level 4 self-test to listen to caution and warning alerts that were sensed during the last 10 flight legs. The previous flight alerts are not heard if the pilot PULL UP, GPWS TEST annunciator switch is pushed less than 3 seconds after the completion of the level 4 self-test and held until the “PRESS TO CONTINUE” is heard to start a level 5 self-test sequence.

Level 6 Self-Test: The pilot PULL UP, GPWS TEST annunciator switch is pushed again less than 3 seconds after the completion of the level 5 self-test to listen to changes in the discrete inputs. The EGPWS level 6 self-test gives the voice indications that follow:

SELECTION	INDICATION
BELOW G/S pushed	“GLIDESLOPE CANCELLED”
BELOW G/S released	“GLIDESLOPE ENABLED”
GPWS LANDING FLAPS switch to 15	“FLAP SWITCH 35 SELECTED FALSE”, “FLAP SWITCH 15 SELECTED TRUE”
GPWS LANDING FLAPS switch to 10	“FLAP SWITCH 15 SELECTED FALSE”, “FLAP SWITCH 10 SELECTED TRUE”
FLAP OVERRIDE	“NOT LANDING FLAPS”
TERRAIN INHIBIT set	“TERRAIN OFF”
TERRAIN INHIBIT released	“TERRAIN ON”

A “DISCRETE INPUT TEST, PRESS TO CANCEL” sounds every 60 seconds during Level 6 Self-Test.

The pilot PULL UP, GPWS TEST annunciator switch is pushed again to stop the self-test sequence.

NOTE

The copilot PULL UP, GPWS TEST annunciator switch selection causes a short level 1 self-test only.

NOTE

Let thirty seconds pass before starting another self-test.

Figure 34- 154 MULTI–FUNCTIONAL DISPLAY, EGPWS TEST INDICATION

During an EGPWS self-test the MFD navigation page shows TERR TEST mode message and a test indication.

On aircraft with SB84–34–123 or SB84–34–127 or ModSum 4–902388 or ModSum 4–902371 incorporated, the MFD’s terrain display area shows the terrain database version number.

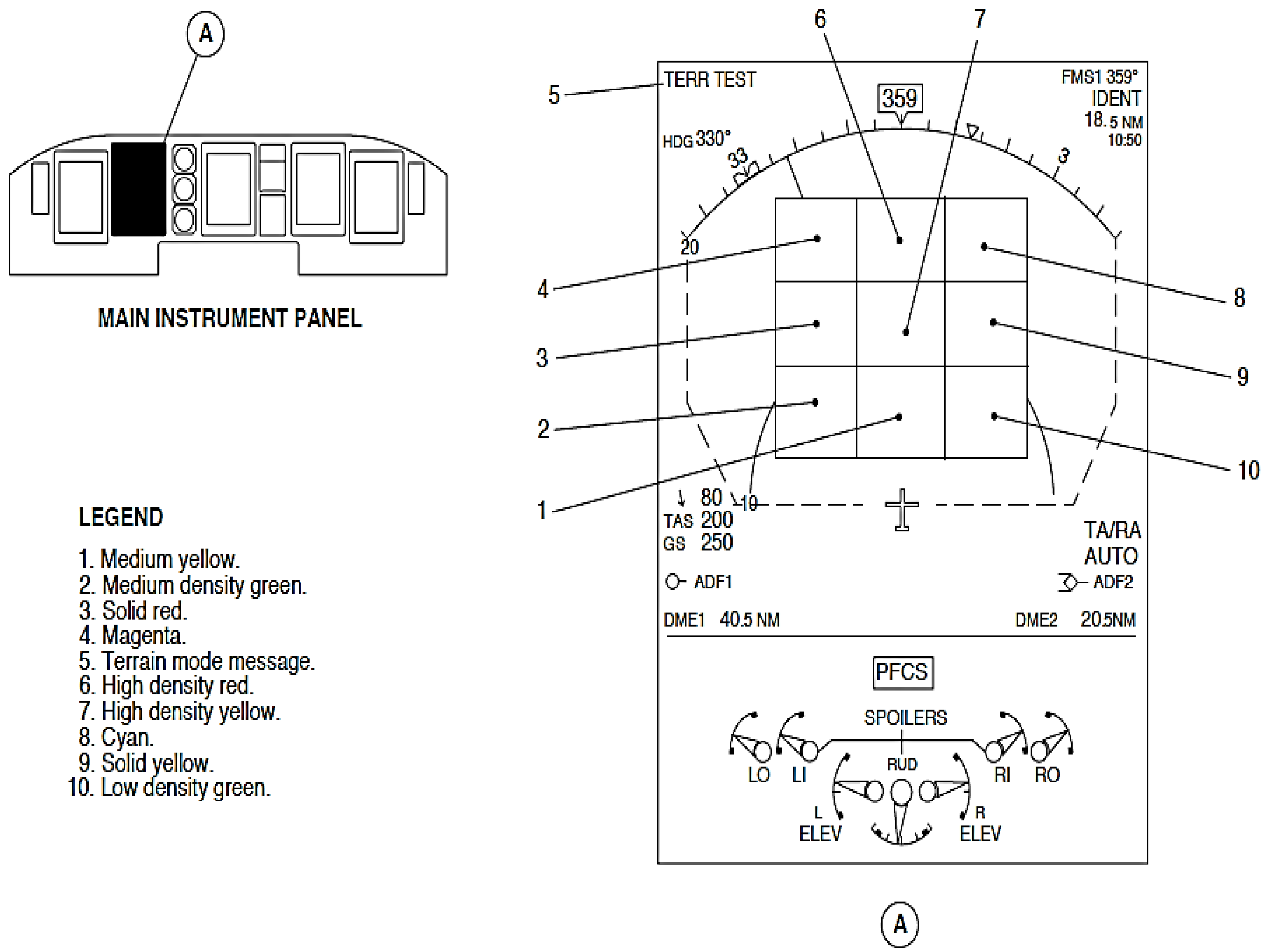


Figure 34- 154 MULTI−FUNCTIONAL DISPLAY, EGPWS TEST INDICATION

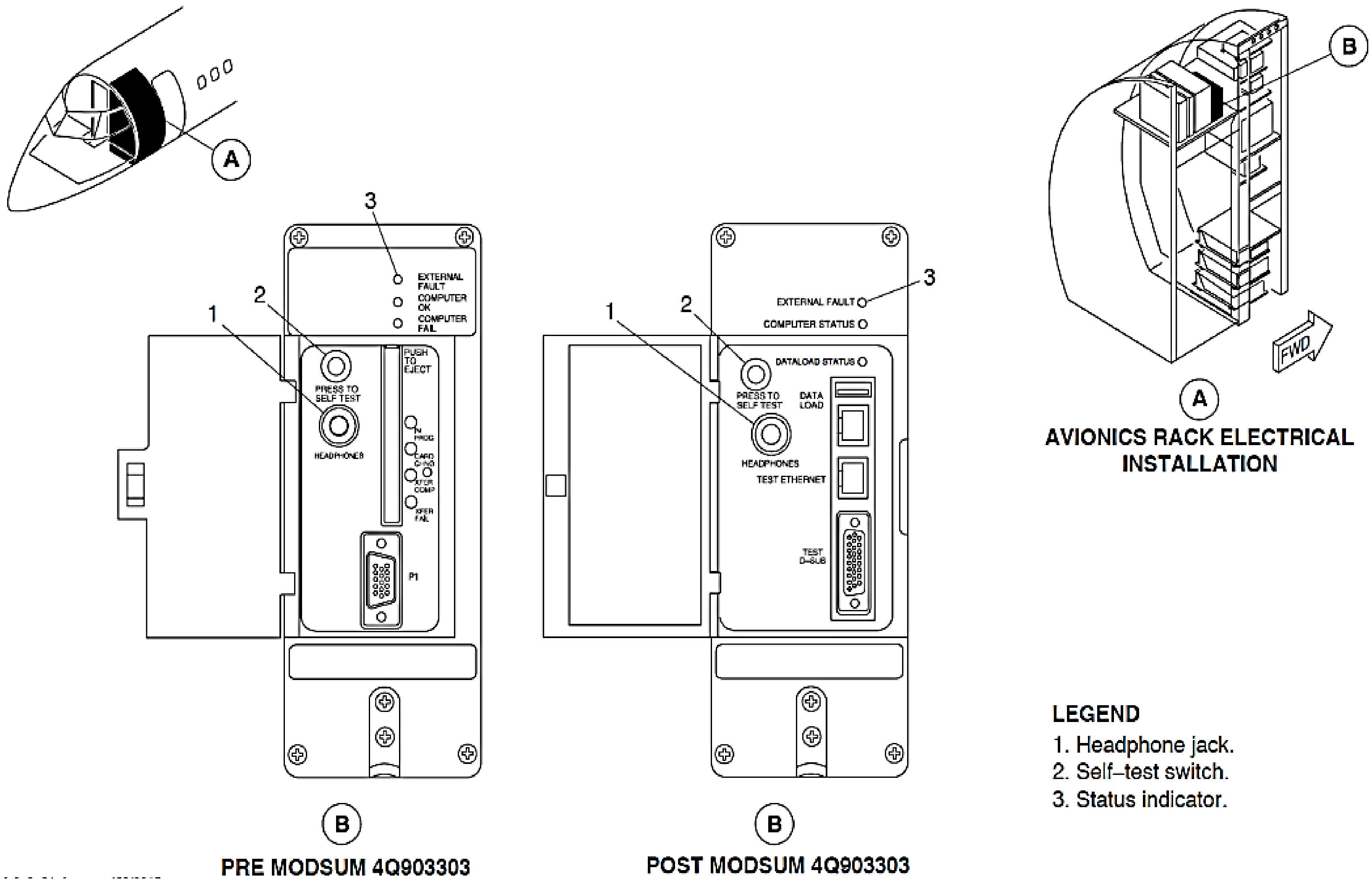


Figure 34- 155 EGPWC TEST SELECTION

Figure 34- 155 EGPWC TEST SELECTION

On aircraft with the Modsum 4Q903303=YES, the EGPWC front panel LEDs gives the status indications that follow:

EXTERNAL FAULT	COMPUTER STATUS	CONDITION	CORRECTIVE ACTION
Off	Off	EGPWC Power off	Ensure that the EGPWC is installed in the rack correctly and power to the EGPWC is turned ON.
Off	On (GREEN)	Normal	None
Off	On (RED)	EGPWC internal fault exists	Record the level 2 self test, if possible. Replace the EGPWC if required.
Off	Flashing GREEN	EGPWC is powering up	None
On (YELLOW)	Off	Invalid condition	Troubleshoot external faults using the level 2 self test. Do not replace the EGPWC.
On (YELLOW)	On (GREEN)	EGPWS external fault exists	Troubleshoot external faults using the level 2 self test. Do not replace the EGPWC.
On (YELLOW)	On (RED)	Both EGPWC internal and EGPWS external faults exist	Troubleshoot external faults using the level 2 self test. Replace the EGPWC, if required.

NOTE

The COMPUTER FAIL indication can come on during a self-test.

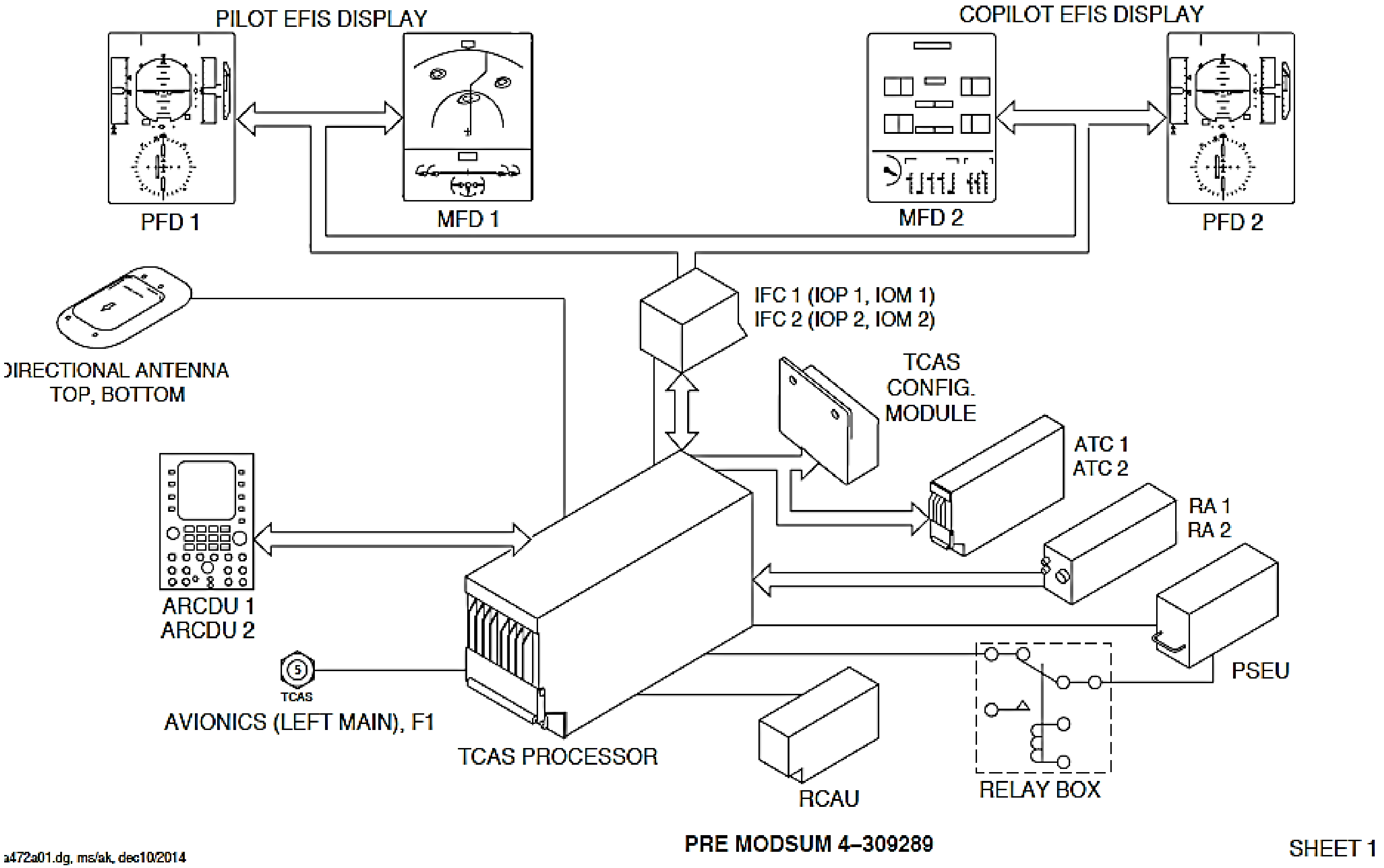


Figure 34- 156 TCAS II BLOCK DIAGRAM (SHEET 1 OF 2)

TRAFFIC ALERTING AND COLLISION AVOIDANCE SYSTEM (TCAS)

Introduction

The Traffic Collision Avoidance System II (TCAS II) gives aural and visual indications when the aircraft's position and flight path relative to an intruder aircraft may cause a potentially dangerous condition.

General Description

Figure 34- 156 TCAS II BLOCK DIAGRAM (SHEET 1 OF 2)
Figure 34- 157 TCAS II INTRUDER AIRCRAFT
Figure 34- 158 TCAS II BLOCK DIAGRAM, SUPPRESSION

The TCAS II interrogates other aircraft's transponders and uses their replies to calculate the parameters of the intruder aircraft that follow:

- Range
- Relative bearing
- Altitude and vertical speed
- Closure rate.

The TCAS II compares this data with its own position to calculate a potential collision condition, and give an audio and visual indication for vertical avoidance maneuvers.

NOTE
TCAS II is unable to detect other aircraft without an operating Air Traffic Control Radar Beacon System (ATCRBS) transponder (operating in Mode A and C) or a Mode S transponder.

The TCAS II calculates the time to, and the separation at the intruder aircraft's Closest Point of Approach (CPA).

The TCAS II will show four different symbols on the Electronic Instrument System (EIS). The type of symbol shown is based on the intruder aircraft's location and closure rate.

The symbols change shape and color to show more important levels of urgency.

The traffic symbols can also have a related altitude indication and a trend arrow to show that the intruding aircraft is vertical speed is more than 500 ft. /min.

The EIS shows the TCAS II symbols that follow:

- Non−threat traffic
- Proximity intruder traffic
- Traffic Advisory (TA)
- Resolution Advisory (RA).

Non−Threat Traffic: The hollow cyan diamond shows that an intruder aircraft's relative altitude is more than ±1200 ft., or its range is more than 6 nm.

On aircraft with ModSum 4Q309289 installed, the hollow cyan diamond shows that an intruder aircraft's relative altitude is between ±1200 to 2700 ft., or its range is more than 6 nm.

It is not considered a threat.

Proximity Intruder Traffic: The filled cyan diamond shows that an intruder aircraft's relative altitude is less than ±1200 ft., or its range is less than 6 nm.

It is not considered a threat.

TA: The traffic symbol changes to a filled yellow circle to show a possible dangerous condition. A TA is shown when the CPA is between 20 to 48 seconds.

RA: If the intruding aircraft continues to close, the TA indication changes to a filled red square. The CPA is between 15 to 35 seconds.

The RA traffic symbol is shown with an Inertial Vertical Speed Indicator (IVSI) RA to maintain a safe vertical separation. The IVSI RA indication has red and green arcs on the IVSI scale to show the required rate of climb or descent to prevent a possible collision.

The TCAS II calculates the RA indication on a five second crew reaction time to get sufficient separation. A two and one half seconds reaction time is given when there is a change in the RA indication.

NOTE

Any aircraft that is the surveillance range of the TCAS II is an intruder. If the intruder aircraft causes an RA, it then referred to as a threat aircraft.

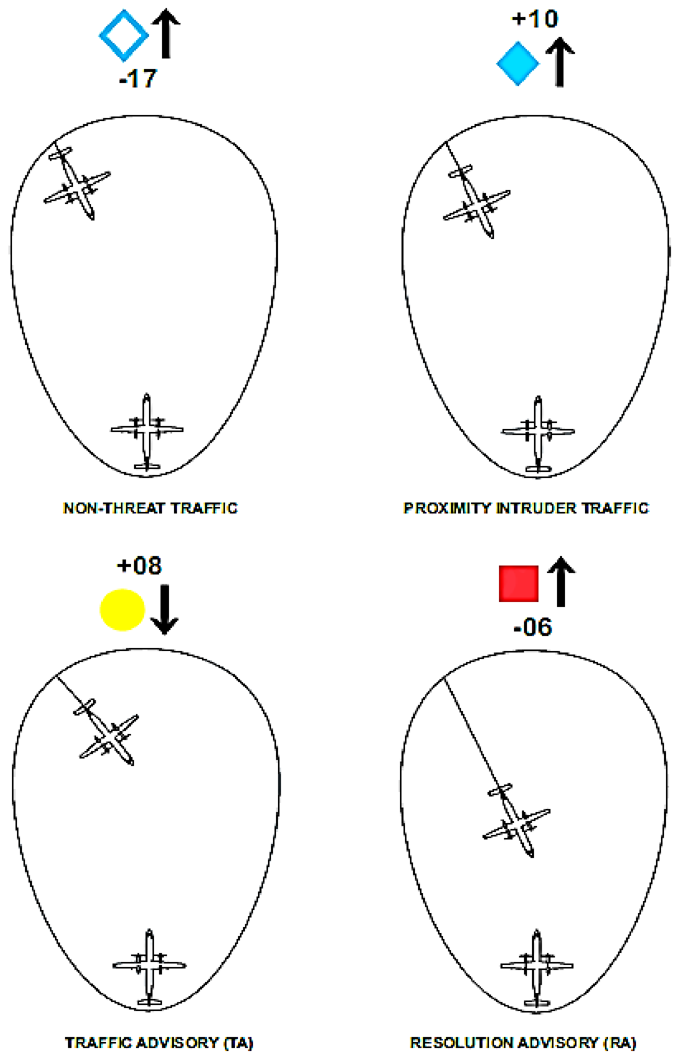


Figure 34- 157 TCAS II INTRUDER AIRCRAFT

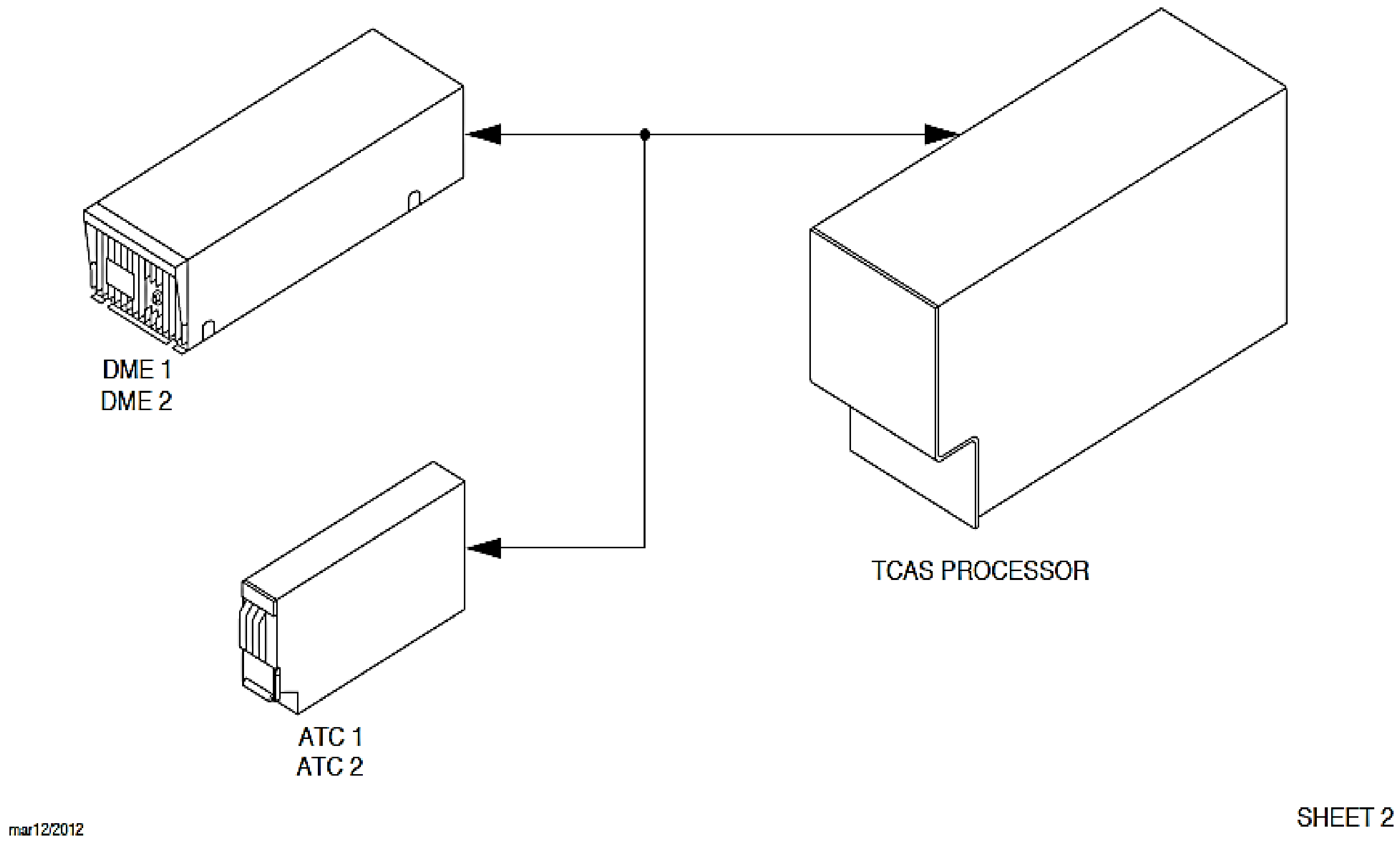


Figure 34- 158 TCAS II BLOCK DIAGRAM, SUPPRESSION

The TCAS II has different levels of protections as shown in the table that follow:

Table 34- 14 PROTECTION FROM THE INTRUDER AIRCRAFT EQUIPMENTS

INTRUDER AIRCRAFT EQUIPMENT	PROTECTION
Mode A Transponder	TA
Mode C Transponder	TA, Vertical RA
Mode S Transponder	TA, Vertical RA
TCAS I	TA, Vertical RA
TCAS II	TA, Vertical RA, TCAS – TCAS Coordination
TCAS III	TA, Vertical RA, TCAS – TCAS Coordination

The TCAS II System has the components that follow:

- Processor, TCAS
- Antenna, Directional.

The TCAS II interfaces with the systems that follow:

- Audio Integration System (AIS)
- Unit, Audio and Radio Control Display (ARCDU)
- Module, Input/Output Processor
- Module, Input/Output
- Module, Aircraft Configuration
- Unit, Proximity Sensor Electronic
- Radio Altimeter
- Air Traffic Control Transponder
- Central Diagnostic System.

The ARCDU1 and ARCDU2 are used to make the TCAS II mode selections.

Detailed Description

The mode S transponder gives airborne to airborne altitude reporting data between TCAS equipped aircraft.

The TCAS Processor Unit (TPU) does the functions that follow:

- Surveillance
- Collision avoidance calculations
- Monitor intruder aircraft and own aircraft.

Surveillance: The TCAS II monitors a 14 nautical mile (26 kilometer) area around the aircraft independently of the Air Traffic Control (ATC) system. The TCAS II monitors for squitter transmissions from other mode S transponder equipped aircraft. The squitter transmission gives the aircraft's unique address. After the squitter is received, the TCAS II then interrogates that mode S address, and from the reply, it calculates the range, bearing, and if available, the altitude of the other aircraft.

To minimize interference with other aircraft and ATC, the interrogation rate changes with the range, closing speed, and when it calculates a TA.

The range and altitude of each mode S transponder equipped aircraft is monitored for collision avoidance. A relative bearing track is also calculated for a traffic advisory indication only.

The TCAS II uses a mode C all−call to interrogate the ATCRBS transponder equipped aircraft.

The mode A transponder equipped aircraft reply with no altitude data. The TCAS II uses their framing pulses to calculate their range and bearing. This data is shown on the EIS navigation pages as non−threat, proximity intruder, or TA indications.

The mode C transponder equipped aircraft reply with altitude data. The TCAS II uses their replies to calculate their range, bearing, and altitude. This data is shown on the EIS navigation pages as non−threat, proximity intruder, TA, or RA indications.

Figure 34- 159 TCAS ON GROUND CALCULATION

When the aircraft is less than 1700 feet (518 meters) Above Ground Level (AGL), its TCAS II uses the difference between its aircraft pressure altitude and radio altitude to estimate the elevation of the ground. It then subtracts the estimated elevation of the ground value from the intruder aircraft's pressure altitude value to calculate its AGL altitude. If it is less than 180 feet (55 meters), the intruder aircraft is considered to be on the ground and an indication is not shown.

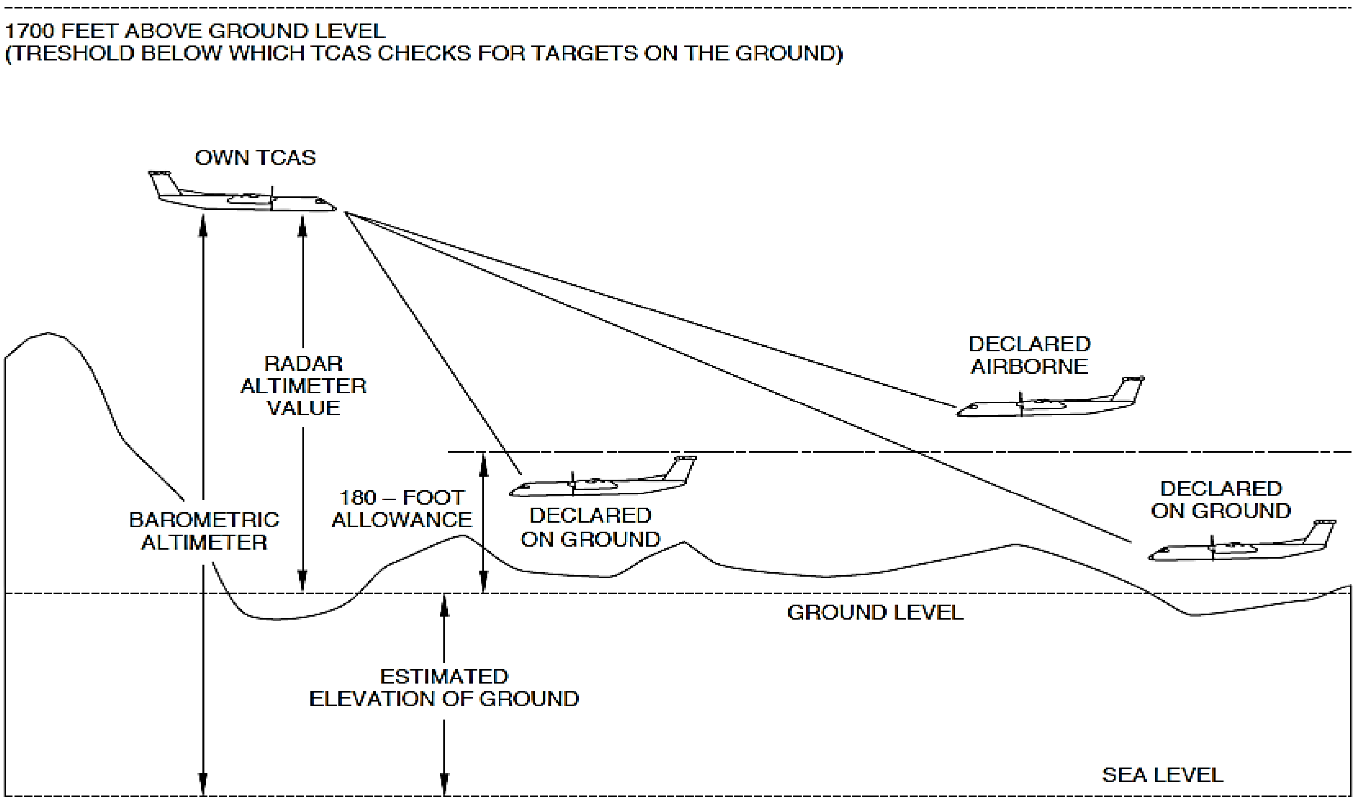


Figure 34- 159 TCAS ON GROUND CALCULATION

Figure 34- 160 TCAS RESOLUTION ADVISORY (RA) CALCULATION

If the a threat condition is calculated, a upward or downward RA is calculated based on a 1500 ft./sec (457 m/sec) vertical speed that gives more vertical separation at the CPA. The vertical separation changes between 400 and 740 feet (122 and 225.5 meters) depending the own aircraft's altitude.

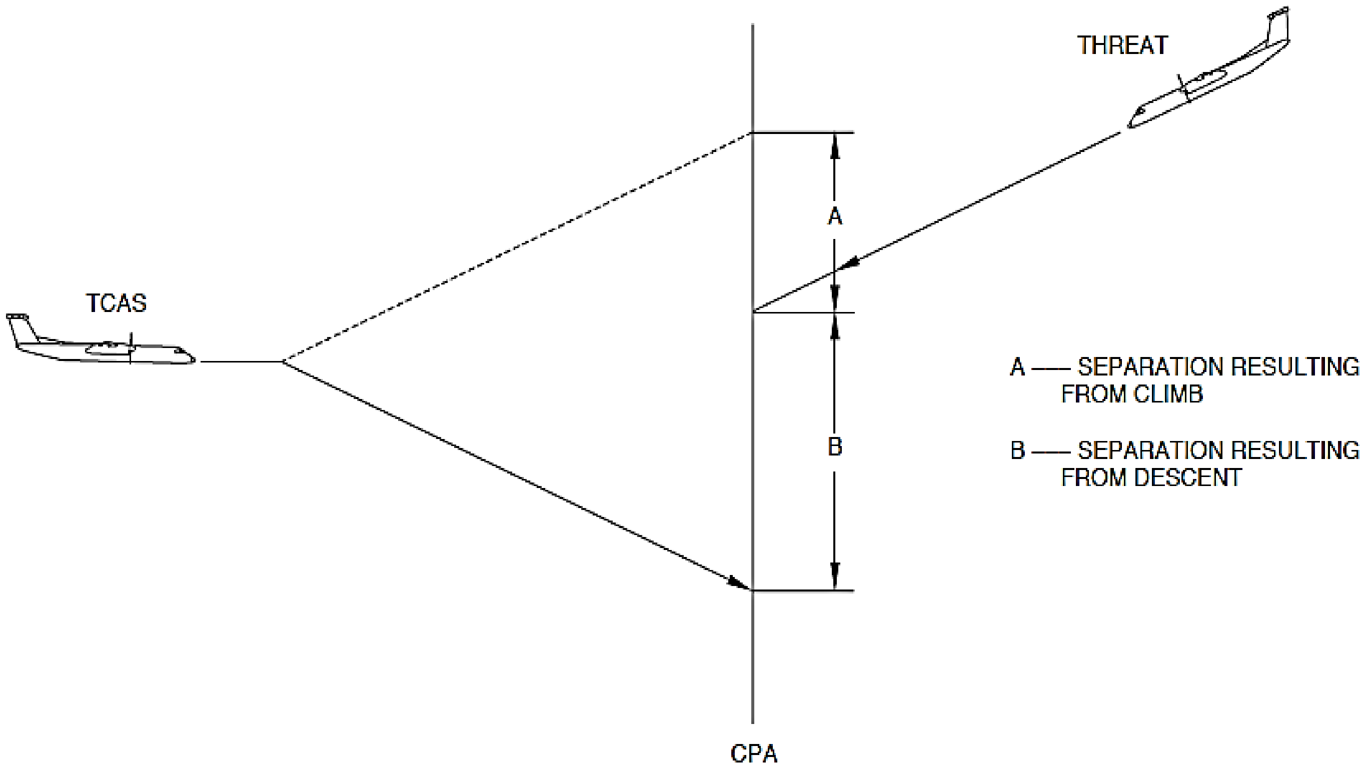


Figure 34- 160 TCAS RESOLUTION ADVISORY (RA) CALCULATION

Figure 34- 161 TCAS NON−CROSSING RA CALCULATION

If the calculated vertical separation is not between 400 and 740 feet (122 and 225.5 meters) at the CPA, a crossover RA is then calculated.

The RA shows a safe vertical speed to climb or descent throughout the altitude of the threat aircraft.

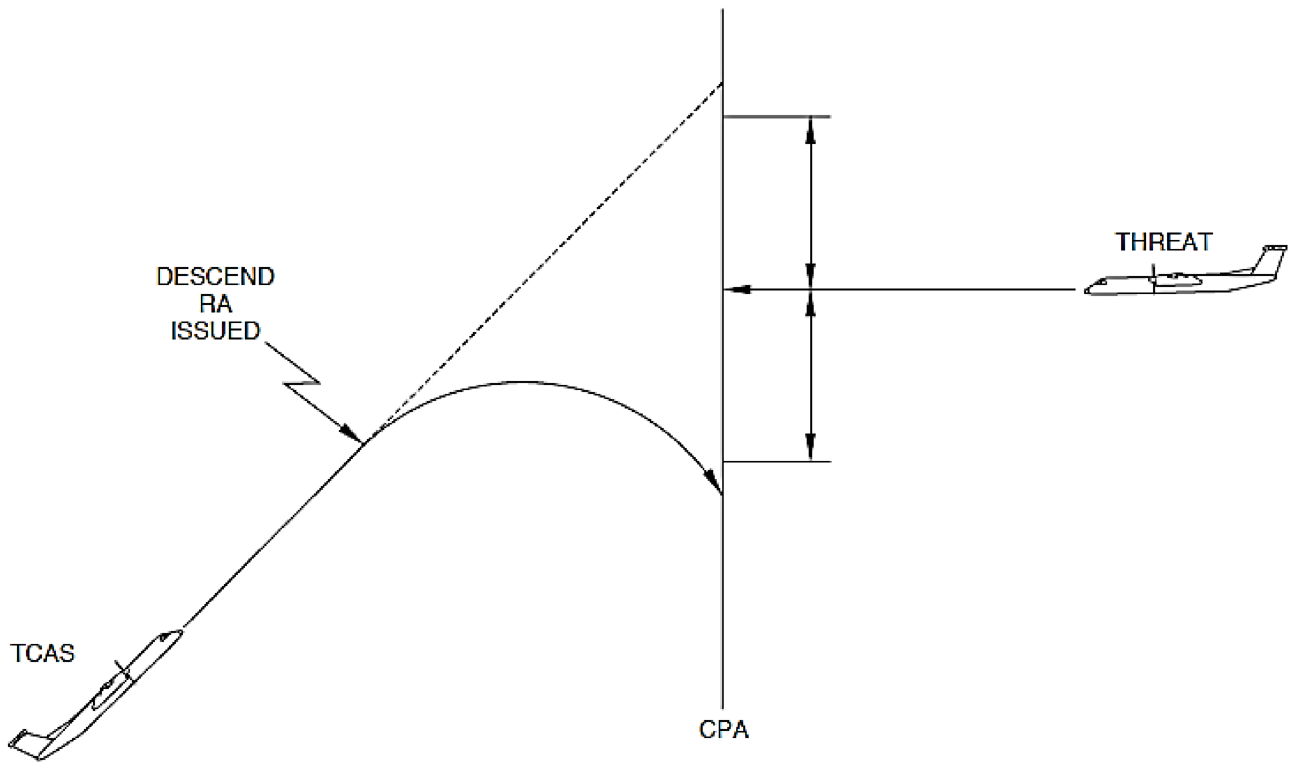


Figure 34- 161 TCAS NON−CROSSING RA CALCULATION

Figure 34- 162 TCAS INCREASE VERTICAL RATE CALCULATION

A change in the intruder aircraft's vertical speed has an effect on the calculated RA. If the threat aircraft does not have a TCAS installed, an increase vertical speed (climb or descent) or sense−reversal RA is calculated.

If the intruder aircraft has TCAS installed, an increase vertical rate (climb or descent) RA is calculated. It will not calculate a sense−reversal RA.

When the aircraft altitude is less than 1450 AGL, the increase descent RAs are not calculated and when the aircraft altitude is less than 1000 AGL, the descent RAs are not calculated.

NOTE

A climb or increase climb RA is not shown during a high altitude or landing condition.

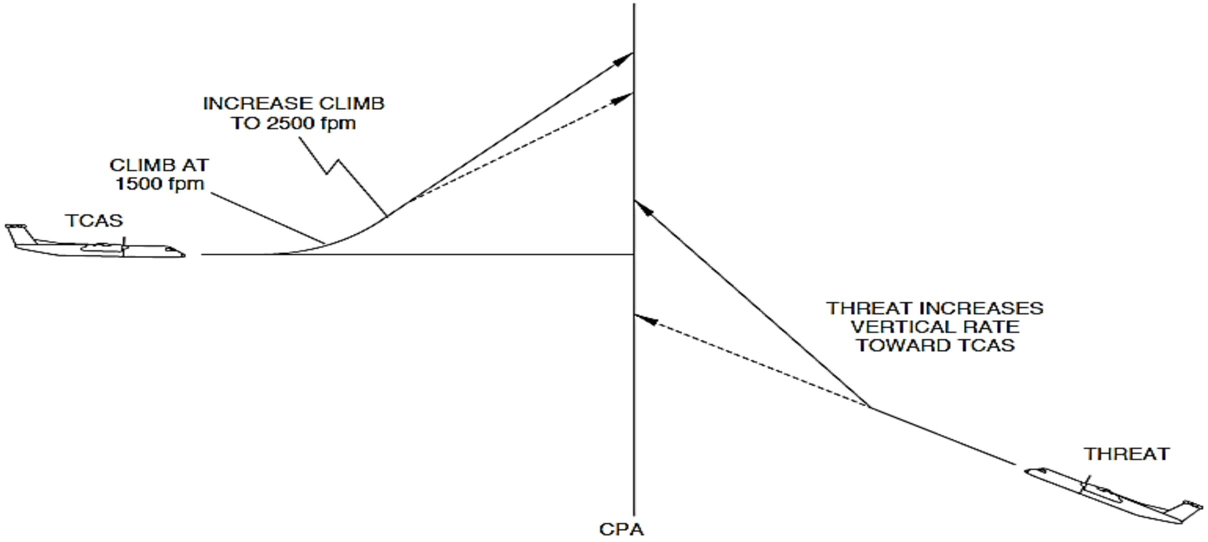


Figure 34- 162 TCAS INCREASE VERTICAL RATE CALCULATION

Figure 34- 163 TCAS RA REVERSAL CALCULATION

If the intruder aircraft does not have a TCAS II or TCAS III system, an RA reversal is calculated when the intruder aircraft reverses its vertical speed to show an opposite RA indication.

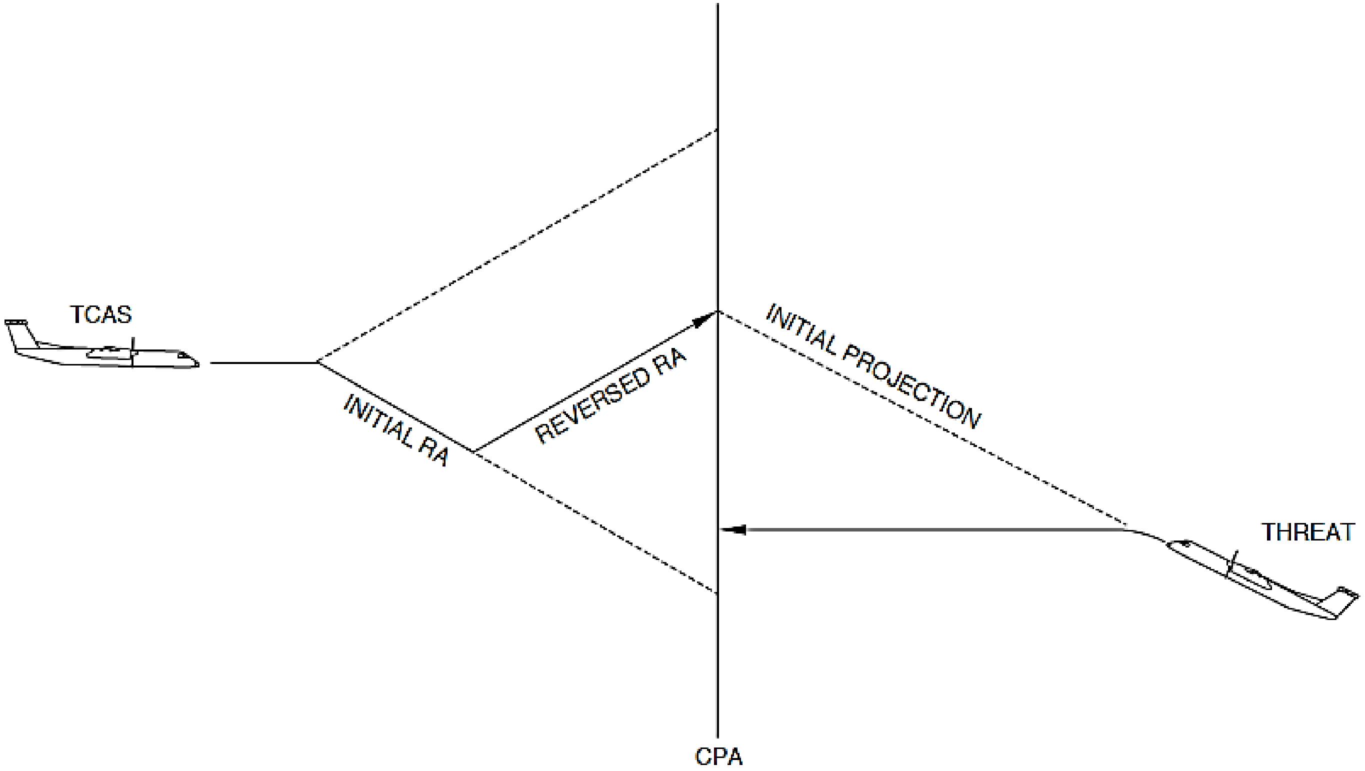


Figure 34- 163 TCAS RA REVERSAL CALCULATION

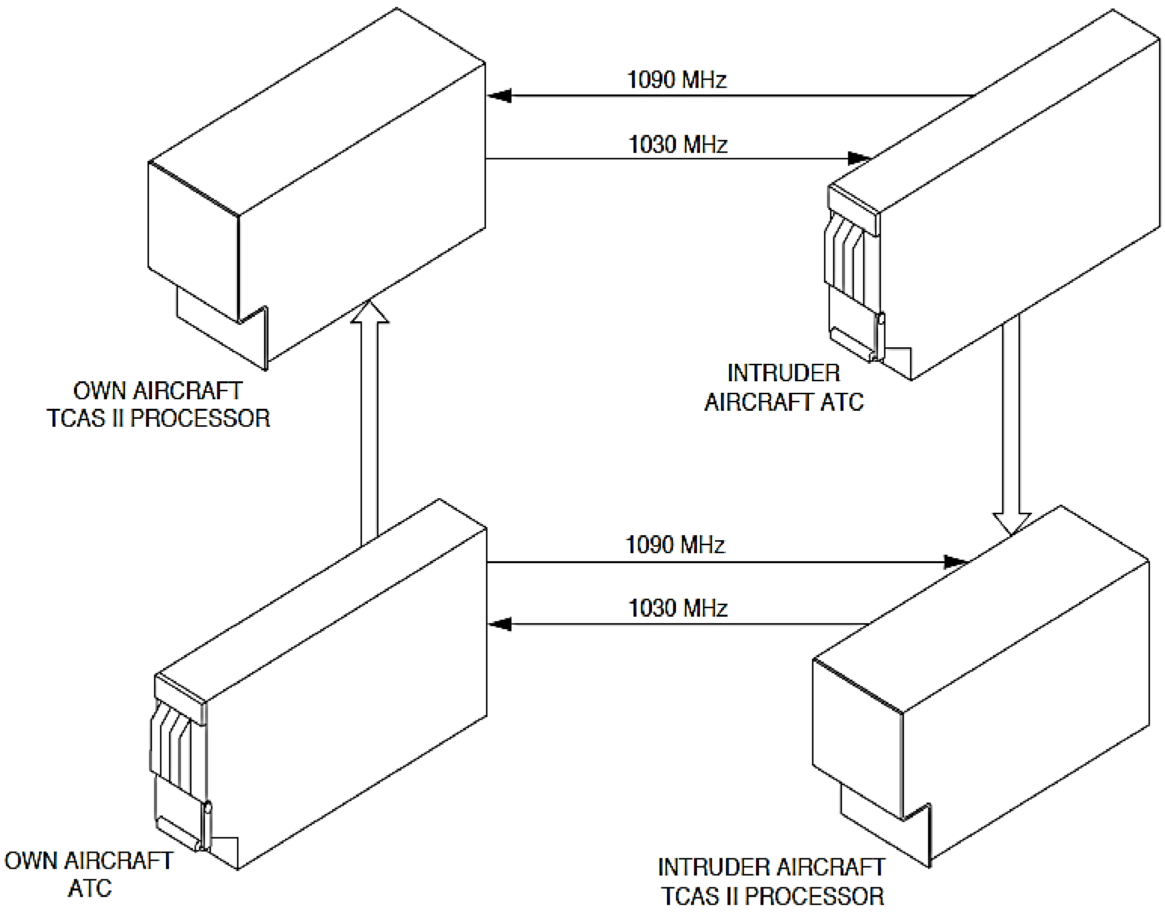


Figure 34- 164 TCAS TO TCAS COMMUNICATION

Figure 34- 164 TCAS TO TCAS COMMUNICATION

The TCAS II can monitor many aircraft situations. One RA is calculated that maintains a safe separation from each threat aircraft.

When the aircraft senses another aircraft that is equipped with TCAS, each aircraft interrogates the other's mode S transponder to make sure that complementary RA are calculated.

Coordination interrogations are transmitted once per second during the RA condition. The coordination interrogations have data that is related to its RA intention when the other aircraft causes the RA. If one aircraft has calculated a climb RA intention, it will transmit a message in the coordination interrogation to the intruder aircraft to make it calculate a descent RA intention. The amount of the descent RA depends on the geometry of the situation. The RA intentions are checked by each TCAS and opposite RA are calculated.

Each aircraft usually senses the other as a threat at different times. The first aircraft transmits its RA intention and, a small time later, the other aircraft transmits the opposite intension but if the two aircraft sense each other as threats and the same time, a RA is calculated based on geometry. If the two aircraft are flying level at the same altitude, the two systems will calculate the same RA. The aircraft with the higher mode S address will reverse its RA.

The TCAS system operates in the following modes:

- Active mode
- Standby Mode
- Test Mode
- System Mode.

Active Mode: When a mode S ATC Transponder (ATC1, ATC2) is on and the TCAS II system mode is not set to standby, the TCAS processor unit operates in an active mode, TA/RA mode or TA ONLY mode. The TA/RA active mode only operates when the aircraft is airborne. The TA ONLY active mode operates when the aircraft is on the ground and when it is airborne.

During the active mode, the TCAS Processor transmits mode S and ATCRBS ATC system interrogations to other aircraft at 1030 MHz through the top and bottom antennas. The TCAS processor unit decodes and calculates 1090 MHz replies received from ATC system equipped aircraft in ATCRBS or mode S formats.

The TCAS II system operates in collision avoidance environments that follow:

- ATCRBS
- Mode S.

ATCRBS Collision Avoidance Environment: The TCAS transmits an all call interrogation to other aircraft equipped with ATCRBS transponders. The intruder aircraft's pressure altitude (mode C data) is received, and the direction and range of the ATCRBS transponder equipped intruder aircraft in the surveillance area are calculated.

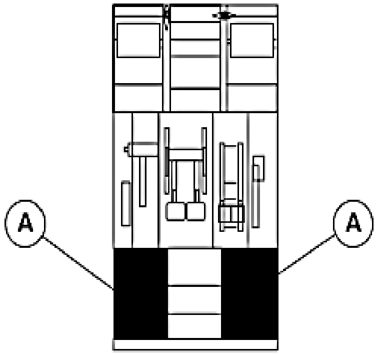
Mode S Collision Avoidance Environment: The intruder aircraft equipped with a mode S transponder continuously transmits squitter at one−second intervals. These signals alternate between the top and bottom antennas. The transmissions have data that specify its unique Mode−S, 24 bit address vertical status and equipment functions. The TCAS II equipped aircraft system monitors the squitter signals to indicate that a Mode−S ATC Transponder equipped aircraft has entered the surveillance area.

The TCAS II system then interrogates the intruder aircraft. The intruder aircraft answers to valid mode S interrogations with mode S surveillance signals. The TCAS II system uses this data to calculate the range and direction of each mode S transponder equipped intruder aircraft in its surveillance area. The TCAS II system individually interrogates and tracks each intruder aircraft using the unique Mode−S address of each aircraft.

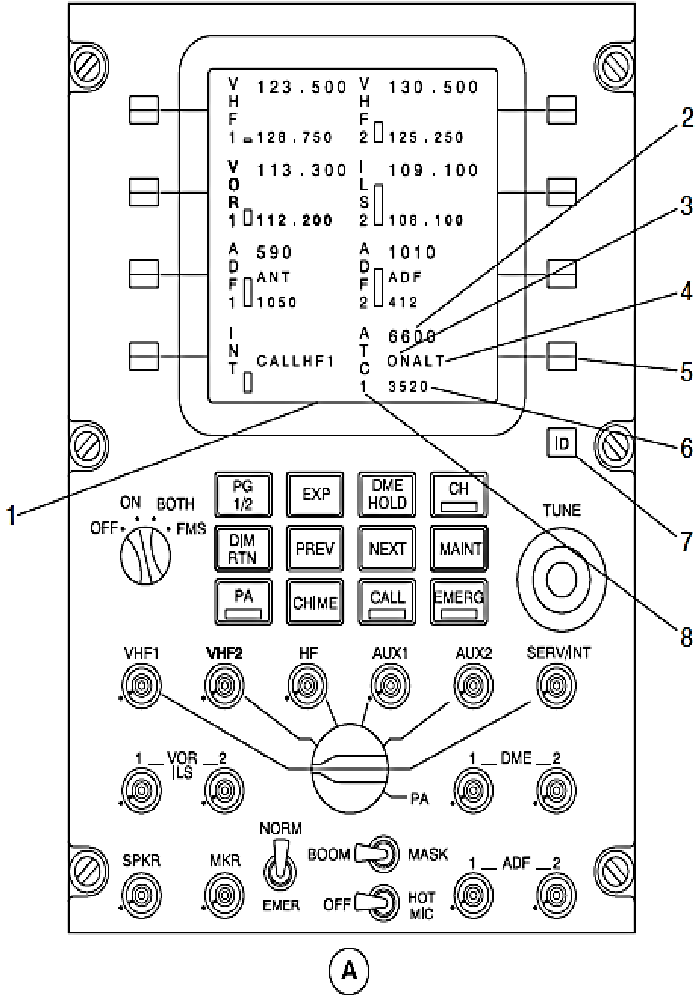
If the intruder aircraft is equipped with TCAS, the Mode−S communications that includes coordination maneuvers, between two TCAS equipped aircraft occur from the TCAS antenna of one aircraft to the mode S transponder antenna of the other aircraft. Aboard the TCAS equipped aircraft, data received by the Mode−S transponder is supplied to the TCAS Processor and the processor determines the content of the reply messages.

Standby Mode: During the standby mode, the TCAS processor unit stays in an energized state and its Built In Test (BIT) does continuous tests. The TCAS II does not transmit interrogation in the standby mode.

Test Mode: When the aircraft is on the ground and the TCAS II and ATC systems are set to the standby mode a test selection starts a self-test to determine the system’s operational status. The TCAS processor unit also sends a status signal directly to the Central Diagnostic System (CDS).



CENTER CONSOLE



LEGEND

1. Display area.
2. Active code.
3. Transponder mode.
4. Altitude reporting.
5. Side key.
6. Preset code.
7. Specific Purpose Identifier (SPI) key.
8. Label.

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Figure 34- 165 ARCDU, TCAS/ATC CONTROLS AND INDICATIONS

Figure 34- 165 ARCDU, TCAS/ATC CONTROLS AND INDICATIONS

The ARCDU1 and ARCDU2 selections control the TCAS II modes to be used.

Table 34- 15 ARCDU SELECTION

SELECTION	DESCRIPTION
OFF	The ARCDU is not energized
ON	The ARCDU is energized and it controls and tunes its related radio system
BOTH	The ARCDU is energized and it controls and tunes its related and opposite radio systems
FMS	The FMS controls and tunes its related and opposite radio systems

Figure 34- 166 ARCDU, TCAS/ATC PARTICULAR PAGE SELECTION

The side key adjacent to the ATC display area is pushed and then the EXP key to show the TCAS/ATC particular page.

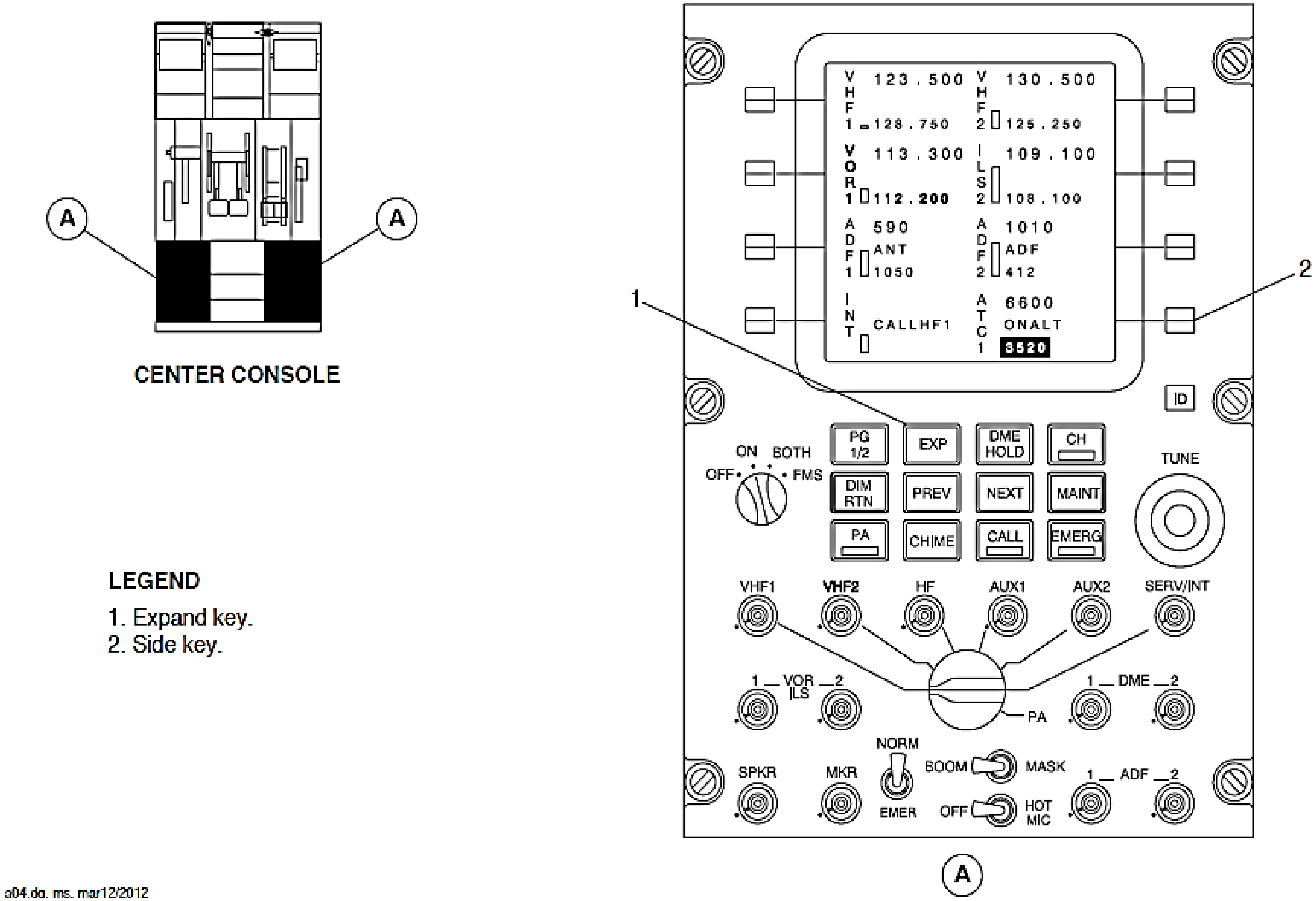


Figure 34- 166 ARCDU, TCAS/ATC PARTICULAR PAGE SELECTION

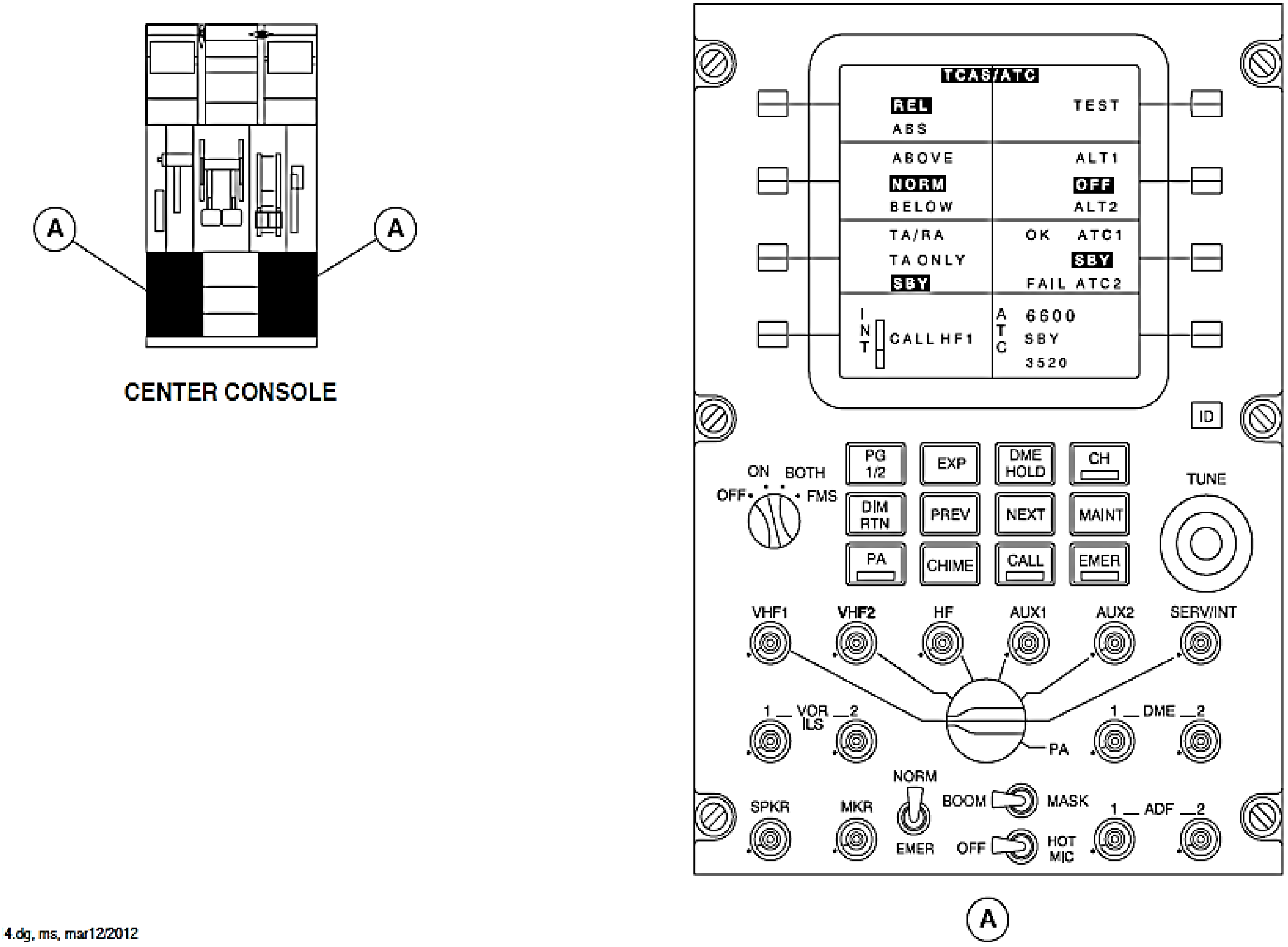


Figure 34- 167 ARCDU, TCAS/ATC PARTICULAR PAGE

Figure 34- 167 ARCDU, TCAS/ATC PARTICULAR PAGE

A TCAS/ATC label is shown in black letters on a white background in the center of the top line. The TCAS/ATC particular page is used to control the TCAS II functions that follow:

- Intruder aircraft altitude, absolute or relative to own aircraft
- Surveillance area, normal, above, or below
- System mode, TA and RA, TA only, or standby
- Test.

Intruder Aircraft Altitude: The side key adjacent to the REL, ABS display area is pushed to change the highlighted selection. It moves in a wrap round manner from REL to ABS. The selection changes to black letters on a green background from the usually white on black background. The Multi-Function Display (MFD) navigation page shows only a TCAS ABS message.

Surveillance Area: The side key adjacent to the ABOVE, NORM, BELOW display area is pushed to select the surveillance area. The selection highlights and moves in a wrap round manner from NORM, BELOW, to ABOVE. It changes to black letters on a green background from the usually white on black background. The MFD navigation pages show a TCAS ABV or TCAS BLW message.

The surveillance area is limited to the altitude that follows:

Table 34- 16 SURVEILLANCE AREA SELECTION

SELECTION	ALTITUDE ABOVE	ALTITUDE BELOW
ABOVE	9000 feet	2700 feet
NORM	2700 feet	2700 feet
BELOW	2700 feet	9000 feet

System Mode: The side key adjacent to the TA/RA, TA ONLY, and SBY display area is pushed to set the system mode. The indication in the display area changes to reverse video to show the selection. The selection changes from SBY, TA/RA, to TA ONLY with each side key selection. The MFD navigation pages show a TCAS STBY or TCAS TA ONLY message.

TA/RA: The TA/RA mode shows traffic and resolution advisory intruders. The TA ONLY mode shows traffic advisory intruders only. The standby mode sets the TCAS processor unit to the standby mode.

TA ONLY: The TCAS processor operates in the TA only mode when the radio altitude is less than 1000 feet (305 meters) AGL. When the AGL altitude is more than 1000 feet (305 meters), the TCAS II mode changes to TA/RA. When the AGL altitude is less than 100 feet (30.5 meters), the mode changes back to TA.

SBY: The TCAS processor unit and Mode S transponder do not transmit when the mode is set to STB.

NOTE

The ATC display area also shows a SBY indication.

Test Mode: The side key adjacent to the TEST display area is pushed to cause the TCAS processor unit to supply the test indications that follow:

- Intruder symbol test pattern and TCAS TEST are shown in white and yellow text on the MFDs and IVSI part of the PFD
- A climb/descent symbol test pattern is shown on the IVSI part of each PFD
- The “TCAS System Test OK" message sounds if a malfunction is not sensed and a "TCAS System Test Fail" if a malfunction is sensed.

The test mode shows the MFD indications that follow:

- RA symbol that is at the 3 o’clock position, 2 mile (3.22 kilometer) range, 200 feet (61 meters) above, and flying level
- TA symbol that is at a 9 o’clock position, 2 mile range (3.22 kilometer), 200 feet below, and climbing
- Proximity intruder traffic symbol that is at 1 o’clock position, 3.6 mile (5.79 kilometer) range, 1000 feet (305 meters) below, and descending
- Non−threat traffic that is at 11 o’clock position, 3.6 mile (5.79 kilometer) range, 1000 feet (305 meters) above, and flying level.

The test mode also shows the PFD IVSI indications that follow:

- Green arc from 0 to 250 ft./min
- Red arc from 0 to −6,000 ft./min
- Red arc from +2,000 to +6,000 ft./min.

Figure 34- 168 EFCP, TCAS POP–UP OR CONTINUOUS INDICATION SELECTION

The MFD navigation displays shows TCAS automatic (pop–up) or continuous indications. When the navigation page is set to arc or map mode and the range is set to 40 nautical miles or less, the TCAS push–button switch on the Electronic Flight Control Panel (EFCP) is pushed to show TCAS traffic continuously. If the TCAS push–button switch is pushed again, the automatic mode starts again.

If the EFCP malfunctions, the automatic mode is set.

The automatic mode only shows TA and RA indications. When a TCAS TA or RA condition is sensed, the ARC or MAP format and the 10 nmi range is automatically shown. An AUTO RANGE message below the outer range digital mark shows that the range indication does not relate to the range selection on the EFCP.

NOTE

A TA or RA condition prevents the EFCP switch selections that follow:

- TCAS push–button
- RANGE rotary switch
- FULL/ARC push–button.

After the TA or RA condition, the 10 nmi range indications stays in view. The TCAS push–button switch is pushed to reset the indication.

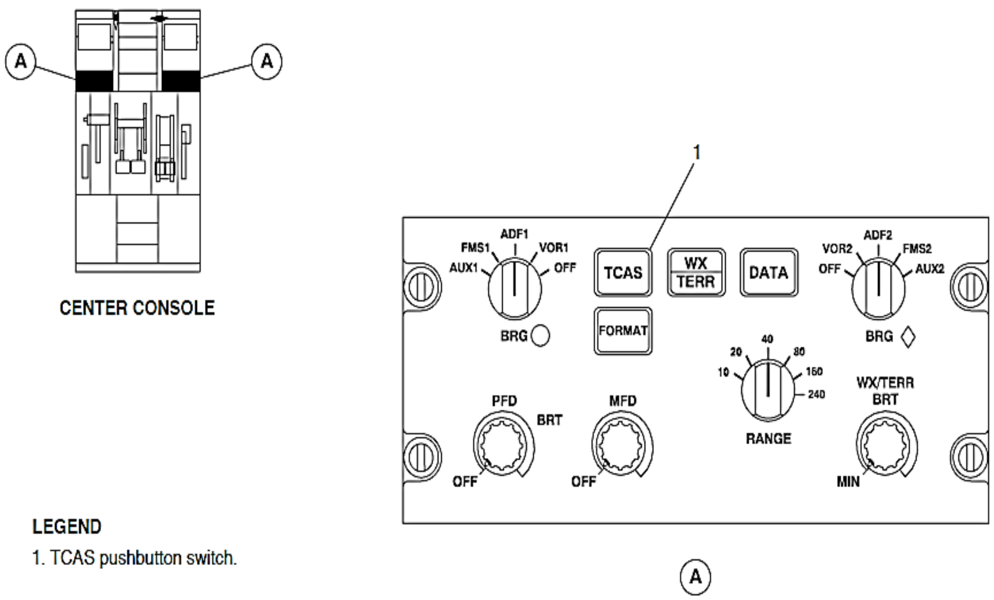
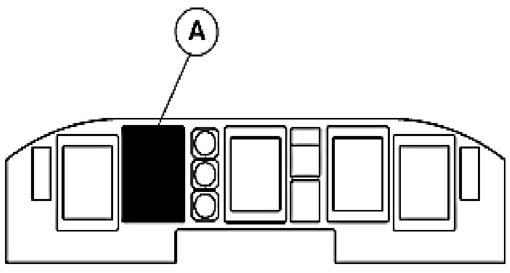
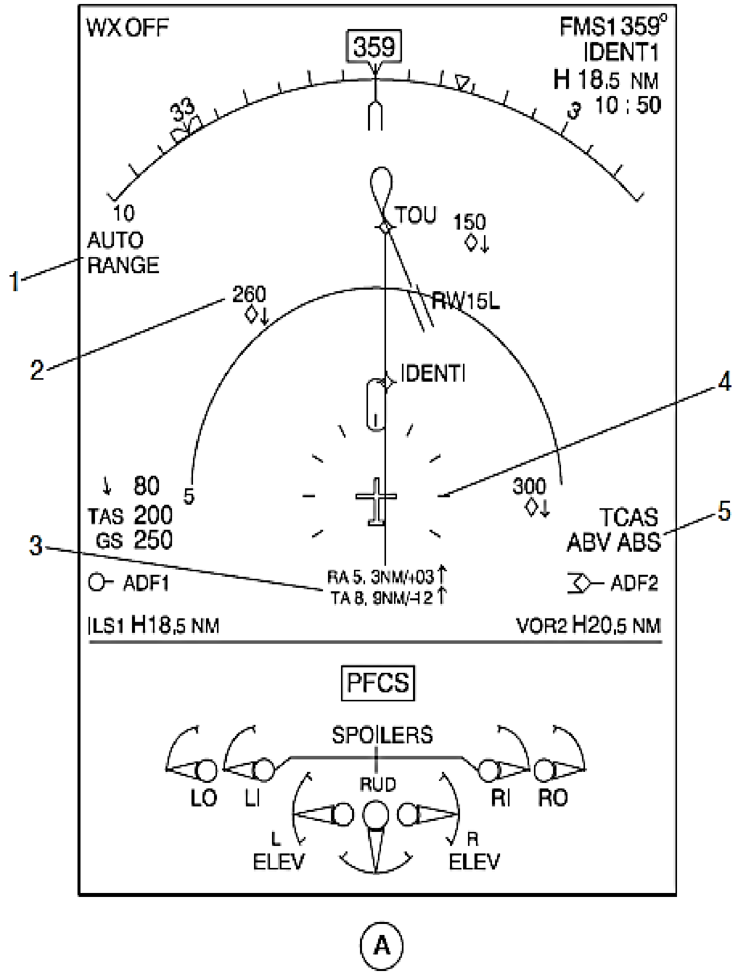


Figure 34- 168 EFCP, TCAS POP–UP OR CONTINUOUS INDICATION SELECTION





LEGEND

1. AUTO RANGE message.
2. TCAS traffic indication.
3. No–bearing advisory.
4. TCAS range ring.
5. Mode messages.

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Figure 34- 169 MFD, TCAS INDICATIONS

Figure 34- 169 MFD, TCAS INDICATIONS

The MFD navigation page shows the TCAS parameters that follow:

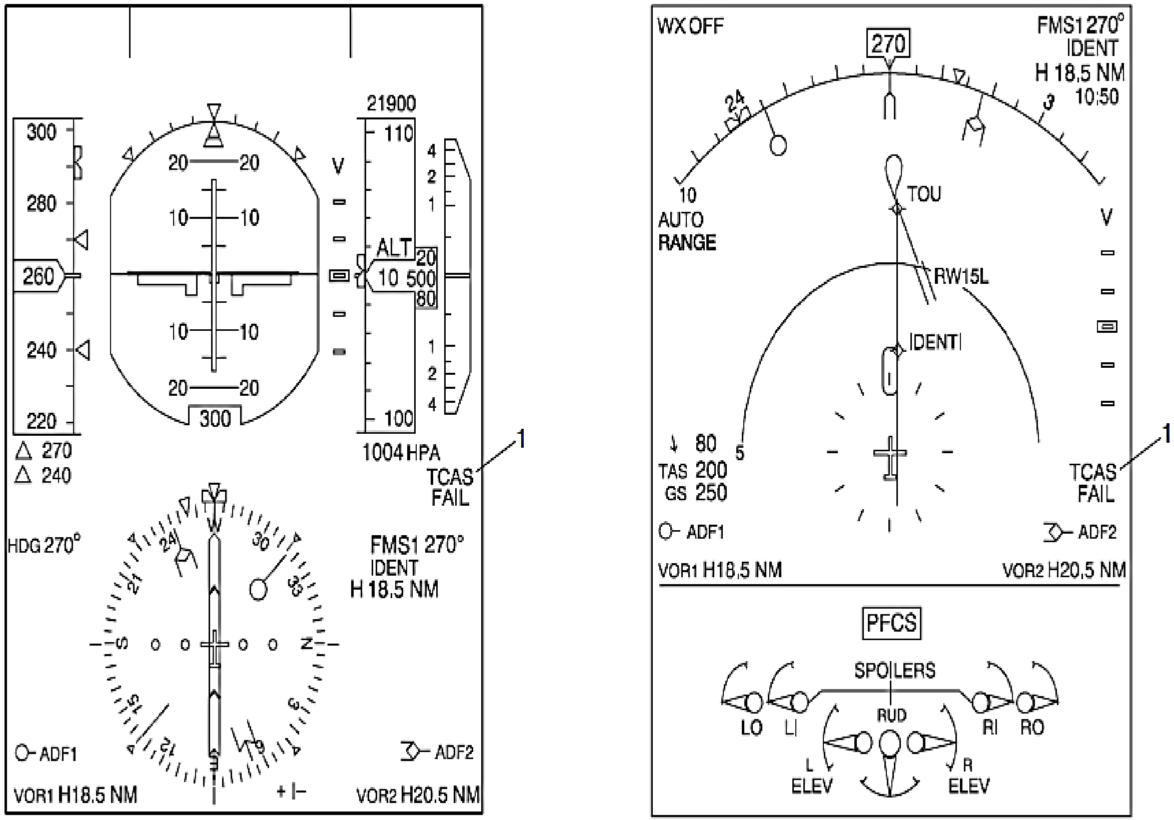
- Mode messages
- Range ring
- Intruder symbols
- No−bearing advisory.

Mode Messages: The TCAS mode selections are shown with the indication priority that follows:

- TCAS FAIL
- TCAS STBY
- TCAS TEST
- TCAS TA ONLY
- TCAS ABV or BLW, ABS.

NOTE

When the operational mode is set to TA/RA, no message is shown. Also, no message is shown for the threat aircraft altitude relative (REL) and surveillance area normal (NORM) selection.



PFD MFD

LEGEND

1. TCAS FAIL message.

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Figure 34- 170 MFD, TCAS FAIL MESSAGE

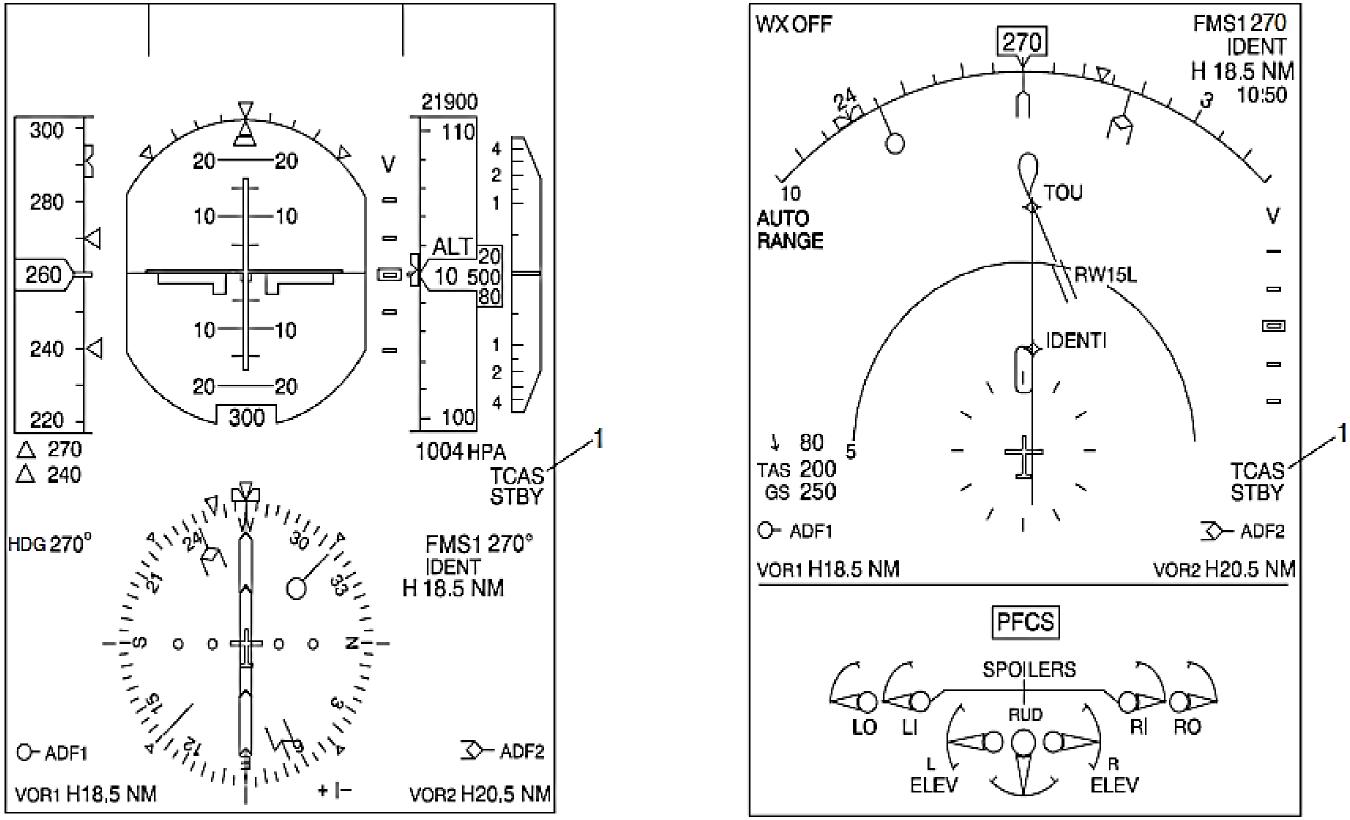
Figure 34- 170 MFD, TCAS FAIL MESSAGE

TCAS FAIL Mode Message: The TCAS FAIL mode message is shown in yellow letters when the TCAS processor unit malfunctions or when the display unit does not receive valid data from the TCAS processor unit through the IOPs.

On aircraft with ARCDU P/N CDU3933AE04, the TCAS FAIL message is an inherent nuisance message and it must be ignored.

On aircraft with ARCDU P/N CDU3933AE05, the TCAS FAIL message is removed.

When TCAS FAIL mode message is shown, no other TCAS symbols are shown.



PFD MFD

LEGEND

1. TCAS STBY message.

Figure 34- 171 MFD, TCAS STBY MESSAGE

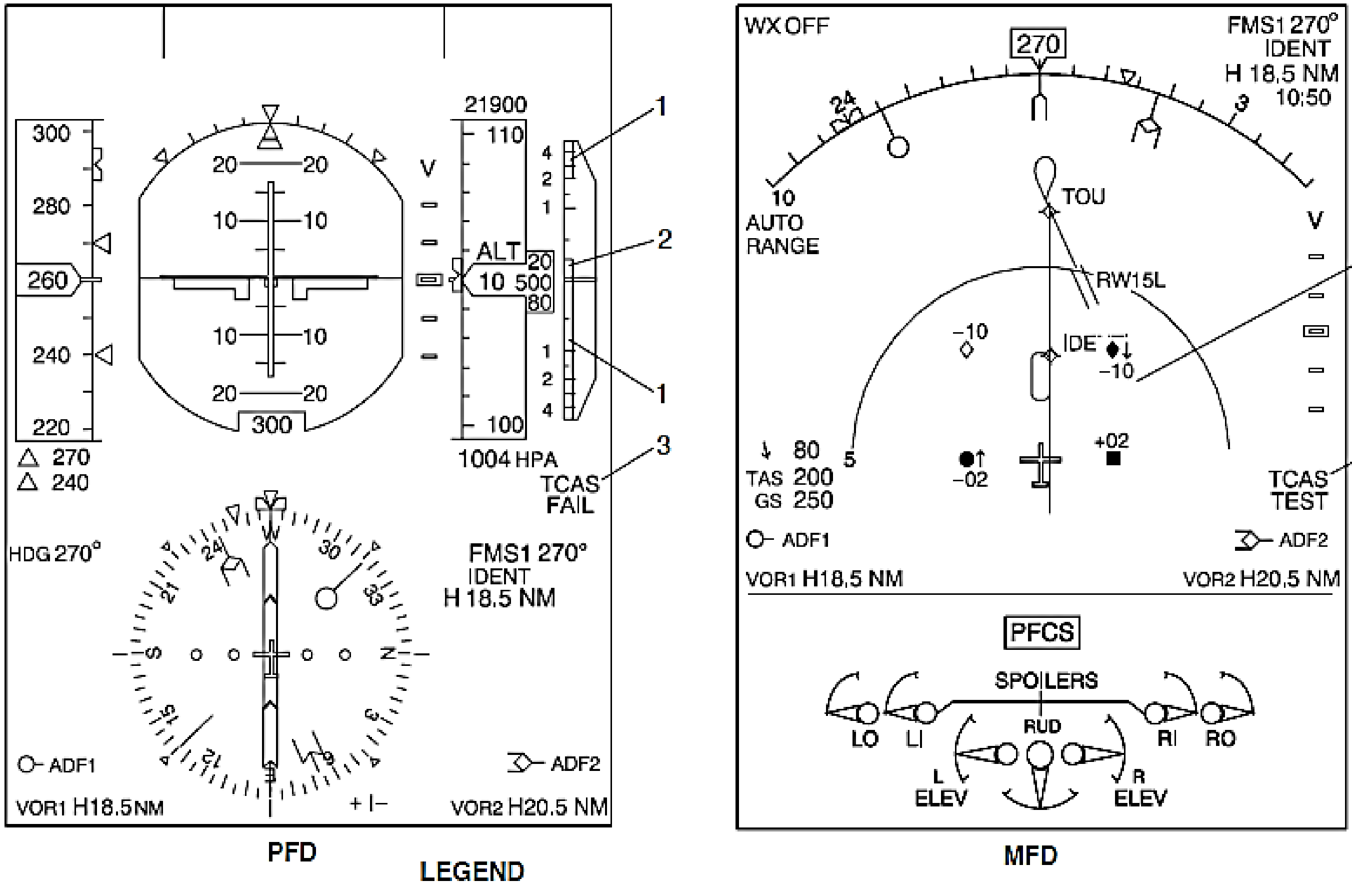
Figure 34- 171 MFD, TCAS STBY MESSAGE

TCAS STBY Mode Message: The TCAS STBY mode is shown in white letters when the TCAS processor units is in the standby mode.

NOTE

When the mode S transponder is set to the standby mode, the TCAS will also automatically operate in the standby mode.

When TCAS STBY mode message is shown, no other TCAS symbols are shown.



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LEGEND

1. Red RA band.
2. Green fly–to band.
3. TCAS TEST message.
4. TCAS test indication.

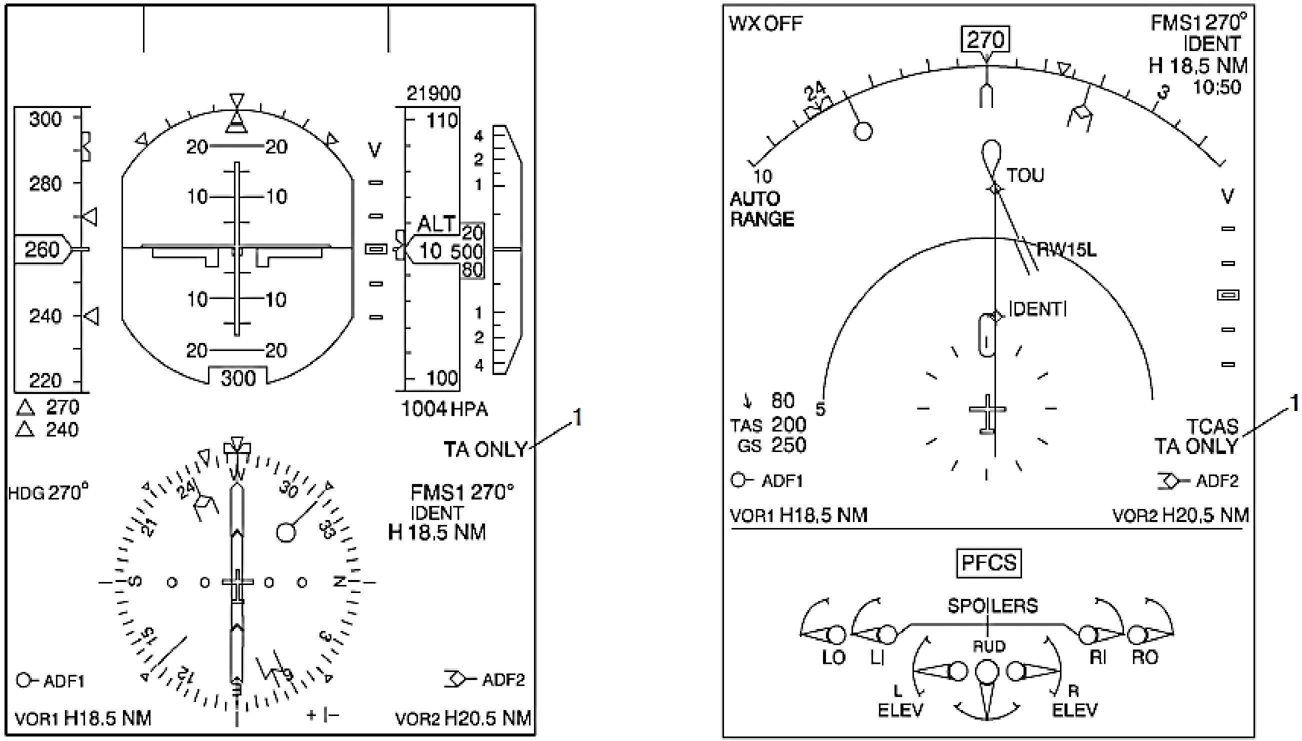
Figure 34- 172 MFD, TCAS TEST MESSAGE

Figure 34- 172 MFD, TCAS TEST MESSAGE

TCAS TEST Mode Message: The TCAS TEST mode message is shown in white and yellow letters when the TCAS processor unit does a self-test. TCAS is shown white and TEST is shown in yellow.

Figure 34- 173 MFD, TA ONLY MESSAGE

TCAS TA ONLY Mode Message: The TA ONLY message shows the TA only operational mode selection. No RAs will not be shown.

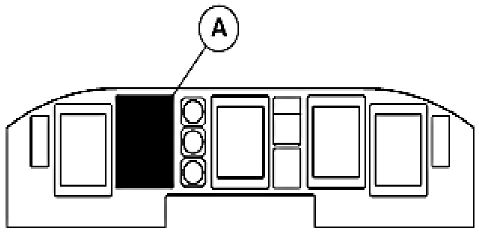


PFD MFD

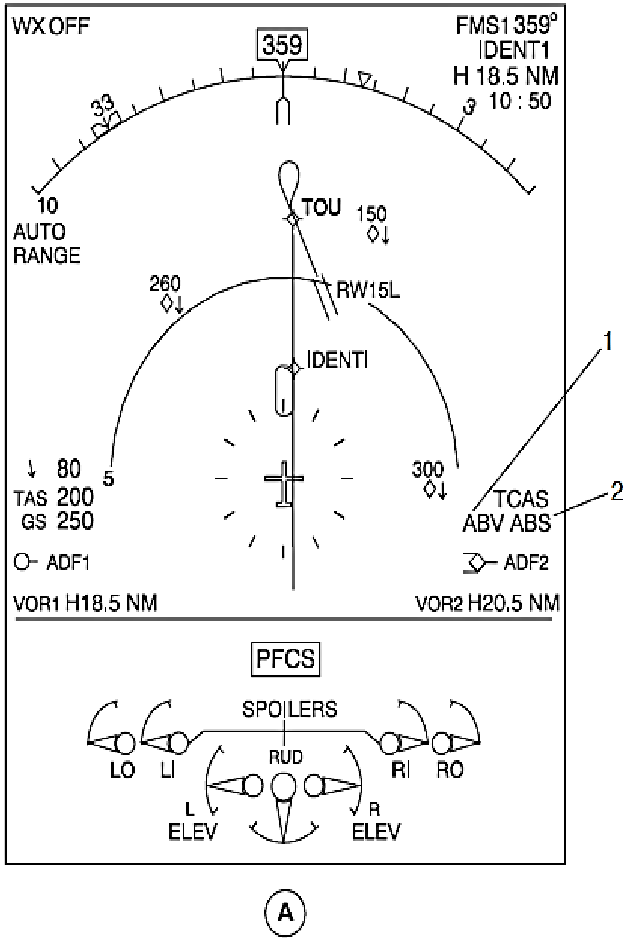
LEGEND

1. TA ONLY message.

Figure 34- 173 MFD, TA ONLY MESSAGE



MAIN INSTRUMENT PANEL



LEGEND

1. Traffic above message.
2. Traffic absolute message.

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Figure 34- 174 MFD, TCAS ABV OR BLW AND ABS MESSAGE

Figure 34- 174 MFD, TCAS ABV OR BLW AND ABS MESSAGE

TCAC ABV or BLW and ABS Mode Message: The ABV or BLW message shows the above or below mode selection.

The ABS mode message shows absolute altitude mode selection. The absolute altitude is shown in hundreds of feet using two digits with the intruder symbol.

The relative altitude mode does not show a mode message.

Range Ring: The TCAS range ring shows a 2 nmi (1.85 km) circular area around the aircraft symbol. It is shown as white short line segments or dots when the RANGE rotary switch on the EFCP is set to the 10, 20, or 40 nmi range. When the RANGE rotary switch is set to a range that is more than 40 nmi it is out of view. A TA or RA condition will cause the 10 nmi range and TCAS range ring indications to come into view with a white AUTO RANGE message below the outer range digital mark.

In the automatic mode, the range ring comes into view only when a TA or RA condition occurs. During a TA or RA condition, another range selection cannot be made. The auto range mode can be cancelled after the alert has ended.

The MFD shows the eight most dangerous intruder aircraft, as calculated by the TCAS. These intruder indications are shown on the MFD navigation pages to identify surrounding intruder aircraft that are equipped with mode S or ATCRB mode A or C transponders.

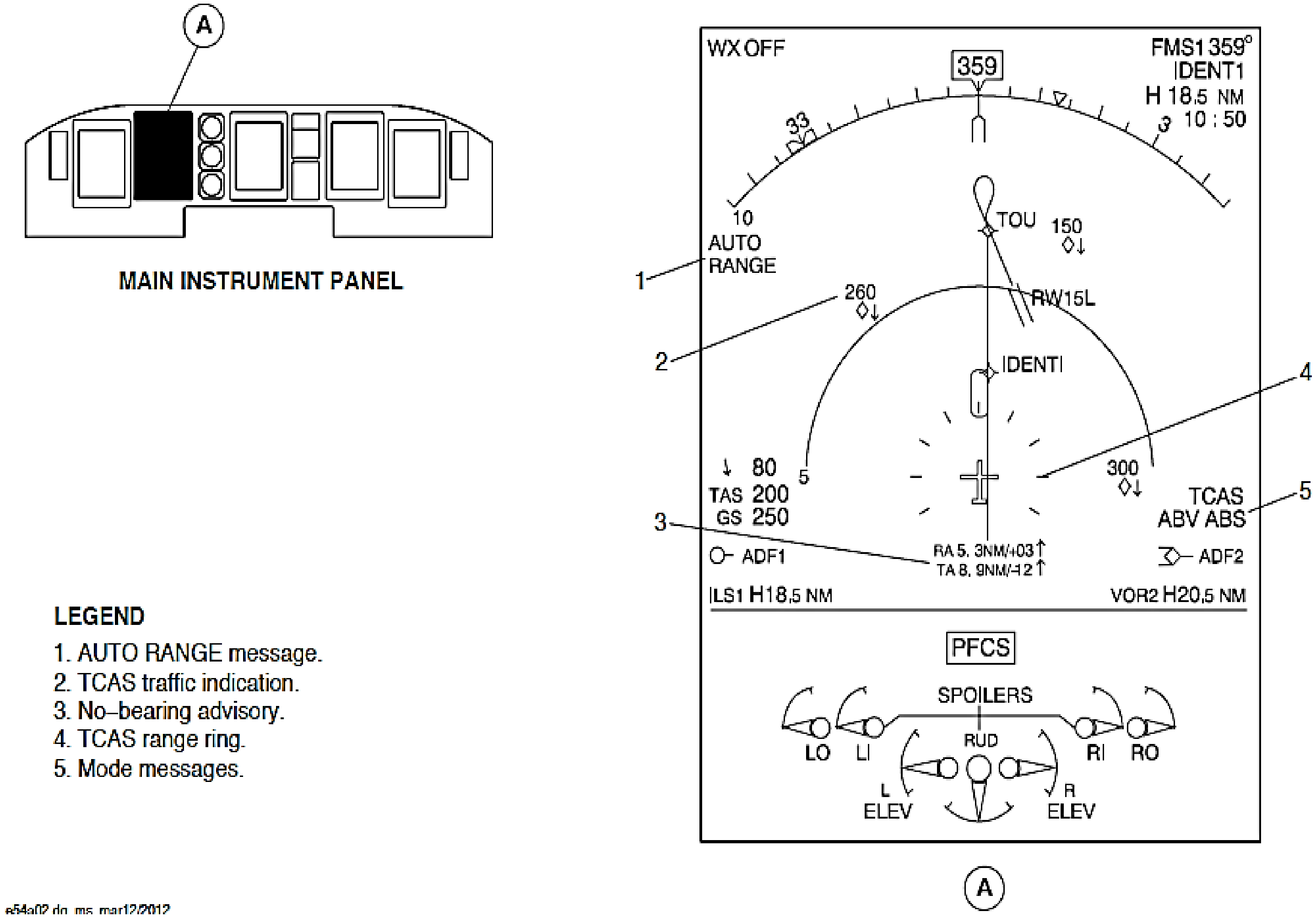

Figure 34- 175 MFD, TCAS TRAFFIC INDICATIONS

Figure 34- 175 MFD, TCAS TRAFFIC INDICATIONS

The MFD shows the intruder symbols that follows:

- Non−threat traffic
- Proximity intruder traffic
- TA
- RA
- No−bearing advisory
- Off Scale traffic.

Non−Threat Traffic (Intruder): The MFD navigation page shows non−threat traffic as an open cyan diamond symbols with their relative altitude.

Proximity Intruder Traffic (Intruder): The MFD navigation page shows proximity intruder traffic as filled blue diamond symbols with their relative altitude.

Traffic Advisory (Intruder): The MFD navigation page shows traffic advisories as filled yellow circle symbols with their relative altitude.

Resolution Advisory (Threat): The MFD navigation page shows resolution advisories as filled red circle symbols with their relative altitude.

NOTE
For better viewing, the full intruder symbol is shown on a rectangular black background.

No−Bearing Advisory: The no−bearing advisory; indication shows the two most dangerous TA or RA intruder aircraft that do not have bearing data. It is shown below the aircraft symbol when the range is set to 10, 20, or 40 nmi.

The indication is shown in red for an RA and yellow for a TA. When an intruder aircraft is not reporting altitude, the altitude and trend arrow are not shown.

Off Scale Traffic: If the TA or RA aircraft range is more than the range selection, one half of the symbol is shown at the edge of the indication. Its position gives the bearing of the intruder.

The relative altitude mode shows the relative altitude between the own aircraft and the intruder is shown as follows:

- With + symbol if the intruder is at a higher altitude
- With − symbol if the intruder is at a lower altitude.

The relative altitude is shown in hundreds of feet using two digits. It has a range of −9900 feet to +9900 feet with 1000 feet accuracy. If the relative altitude is less than 1000 feet, a leading zero is used. A 00 indication shows the other aircraft at the same altitude. This indication is shown in the same color as the intruder symbol.

Their relative or absolute altitude is shown with the intruder indication.

A vertical arrow is also shown to the right of the symbol if the vertical speed of the intruder is more than 500 ft/min. An up−pointing arrow shows climb condition and a down−pointing arrow shows a descent condition. The arrow is shown in the same color as the symbol.

When an intruder aircraft is not reporting altitude, the altitude and trend arrow is not shown.

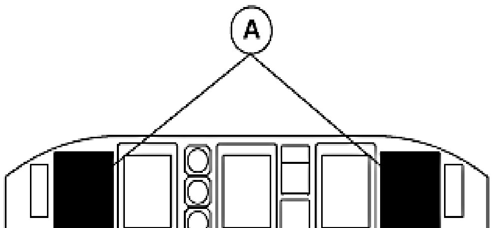

Figure 34- 176 TCAS, PFD INDICATIONS

The RAs are grouped as corrective advisories or preventive advisories. The vertical flight path of the aircraft must be changed when a corrective advisory is shown and maintained for a preventive advisory.

A pointer and vertical speed scale, which are part of the IVSI, shows the present vertical speed of the own aircraft. For TCAS II guidance, the IVSI scales show red and green command indications.

The red and green RA command indications give RA maneuvering guidance to the pilot.

If the pointer is in a red command area, the aircraft's vertical speed must be changed to put the pointer in the green command area. This is a corrective resolution advisory.

If the pointer is not in a red command area, the aircraft's vertical speed must be maintained to keep the pointer out of the red command area. This is a preventive RA.

The TCAS RA indications also sound the messages that follow:

Table 34- 17 RA MESSAGES

RA Category	Corrective	Preventive
Climb	Climb, Climb, Climb	Monitor Vertical Speed, Monitor Vertical Speed
Descent	Descend, Descend, Descend	Monitor Vertical Speed, Monitor Vertical Speed
Crossover Climb	Climb, Crossing Climb, Climb, Crossing Climb	Monitor Vertical Speed, Monitor Vertical Speed
Crossover Descent	Descend, Crossing Descend, Descend, Crossing Descend	Monitor Vertical Speed, Monitor Vertical Speed
Vertical Speed Restricted (Climb)	Reduce Climb, Reduce Climb	Monitor Vertical Speed, Monitor Vertical Speed
Vertical Speed Restricted (Descend)	Reduce Descent, Reduce Descent	Monitor Vertical Speed, Monitor Vertical Speed

NOTE

The “Monitor Vertical Speed” message sounds once if downgraded from a previous corrective advisory.

The RA increases and reversal from the data shown in Table 34-17 are shown in the table that follows:

TABLE 34- 18 RA MESSAGES

RA Category	Corrective	Preventive
Change from Climb to Descent	Descend, Descend NOW, Descend, Descend NOW	Not Applicable
Change from Descent to Climb	Climb, Climb NOW, Climb, Climb NOW	Not Applicable
Increase Climb Rate	Increase Climb, Increase Climb	Not Applicable
Increase Descent Rate	Increase Descent, Increase Descent	Not Applicable

TCAS II indications are shown in the table that follows:

TABLE 34- 19 TCAS II INDICATIONS

Aural	Visual
CLIMB, CLIMB, CLIMB	VSI RED from –6000 to +1500 ft/min, and GREEN from +1500 to +2000 ft/min.
DESCEND, DESCEND, DESCEND	VSI RED from +6000 to –1500 ft/min, and GREEN from –1500 to –2000 ft/min.
MONITOR VERTICAL SPEED, MONITOR VERTICAL SPEED	Current vertical speed is outside the RED arc as shown on the VSI
REDUCE CLIMB, REDUCE CLIMB	VSI shows prohibited vertical speed in RED. Goal is vertical speed in GREEN.
CLIMB, CROSSING CLIMB, CLIMB, CROSSING CLIMB	Same as ’CLIMB", and further shows that own flight path will cross that of intruder.
DESCEND, CROSSING DESCEND, DESCEND, CROSSING DESCEND	Same as "DESCEND", and further shows that own flight path will cross that of intruder.
CLEAR OF CONFLICT	VSI RED and GREEN arcs removed. Range is increasing and sufficient.

Aural	Visual
INCREASE CLIMB, INCREASE CLIMB	Follows a "CLIMB" advisory. VSI RED from –6000 to +2500 ft/min, and GREEN from +2500 to +3500 ft/min.
INCREASE DESCENT, INCREASE DESCENT	Follows a "DESCEND " advisory. VSI RED from +6000 to –2500 ft/min, and GREEN from –2500 to –3500 ft/min.
CLIMB, CLIMB NOW, CLIMB, CLIMB NOW	Follows a "DESCEND" advisory when it has been found that a reversal of vertical speed (direction) is needed to give sufficient separation.
DESCEND, DESCEND NOW, DESCEND, DESCEND NOW	Follows a "CLIMB" advisory when it has been found that a reversal of vertical speed (direction) is needed to give sufficient separation.

The aircraft electrical system left main bus supplies 28 Vdc electrical power through a 5 A circuit breakers to the TCAS processor unit. The circuit breaker is located in position F1 on the 28 Vdc avionics circuit breaker panel.

The Distance Measuring Equipment (DME) and the ATC Mode S Transponders (ATC1, ATC2) operate in the same frequency range. Each sends a suppression signal to the other systems before transmitting.

The TCAS parameters that follow are controlled using discrete signals and ARINC 429 low speed busses (12.0 to 14.5 kilobits/sec):

The TCAS system interfaces with the following:

- Input Output Module (IOM)
- Input−Output Processor (IOP)
- Proximity Sensing Electronics Unit (PSEU)
- Radio Altimeter (RA)
- Mode−S ATC Transponders.

IOM: The IFCs have the TCAS interfaces that follow:

- The TCAS TA and RA event discrete signals are sent to the IOMs and are monitored by the Flight Data Recorder (FDR) system
- A TCAS valid discrete is sent to the IOMs and is monitored by the CDS to know the status of the TCAS Processor

- The climb inhibit and increase climb inhibit discrete signals are sent from the IOMs and are used by the TCAS processor unit
- A TA aural traffic discrete is sent to the IOMs and is monitored by the Warning Tone Generator (WTG) to calculate the prioritizing of all the aircraft warnings
- The TA/RA indications are supplied to the PFDs and MFDs through ARINC 429 data busses and the IOPs
- The heading and attitude data is supplied to the TCAS processor unit from IOP2 through an ARINC 429 data bus.

PSEU: The PSEU supplies the TCAS processor unit with the discrete signals that follow:

- A Landing gear status discrete output is supplied to indicate that the landing gears of the aircraft are extended
- An air/ground status discrete output is supplied to indicate that the aircraft is on the ground or airborne.

RA: The RA supply AGL data to the TCAS processor unit through an ARINC 429 data bus.

Mode S ATC Transponders: The phase shift modulated signal received from the TCAS processor unit. The transponder decodes the last 24 bits of the signal, and then the remaining received message. The last 24 bits of the signal are the strapped address of the transmitting aircraft.

Audio Integrating System: The TCAS processor unit supplies an analog audio announcement output to the Remote Control Audio Unit (RCAU) part of the AIS with audio, when enabled by the WTG.

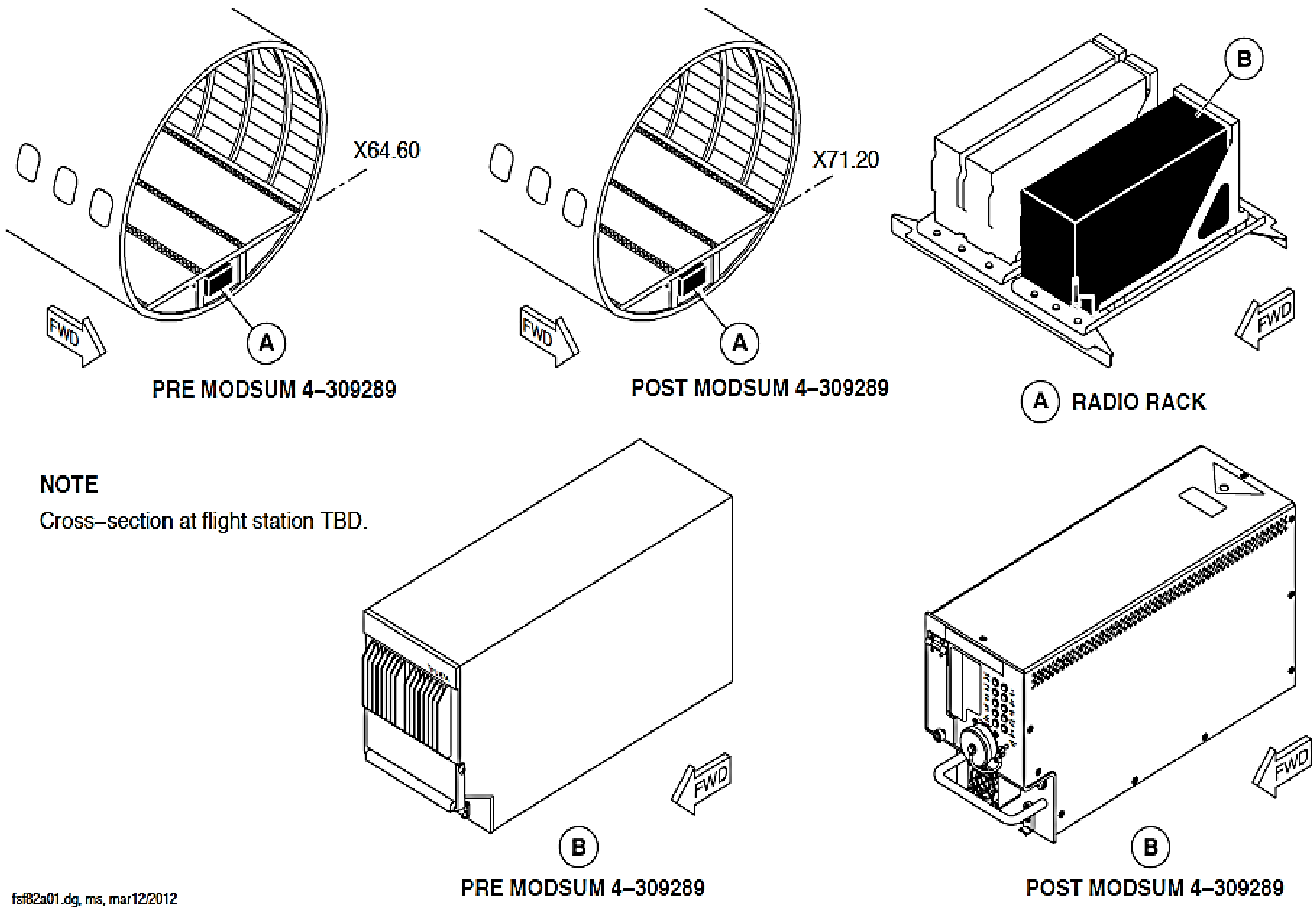

Figure 34- 177 TCAS PROCESSOR UNIT LOCATOR

Figure 34- 177 TCAS PROCESSOR UNIT LOCATOR

The TCAS Processor supplies intruder position and status information to the PFDs and MFDs through the IFCs and directly to the AIS.

The TPU has the following main assemblies:

- RF front assembly
- Receiver assembly
- Transmitter/modulator assembly
- Digital computer I/O assembly
- Power supply assembly.

On aircraft with ModSum 4Q309289 installed, the TPU mounting trays are installed below the floor in the forward portion of the center fuselage. The TCAS processor unit mounting tray is located at station X+71.2, Y0.0, and Z+86.35.

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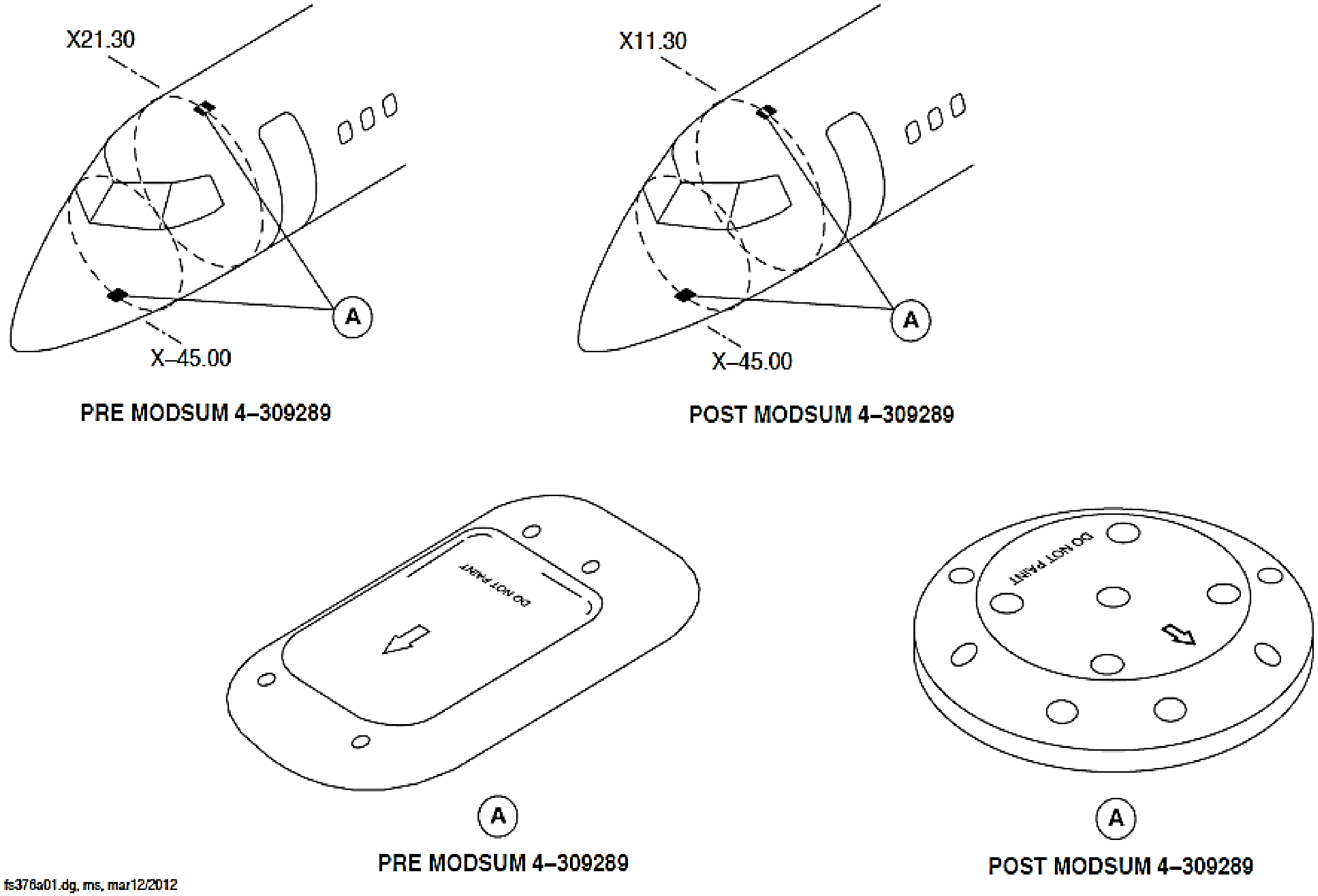


Figure 34- 178 TCAS ANTENNA LOCATOR

Figure 34- 178 TCAS ANTENNA LOCATOR

The directional antenna lets the bearing of intruder aircraft to be calculated by giving a means for the TCAS processor unit to transmit interrogations and receive reply receptions on one of four antenna beams. By selecting the beam, the TCAS processor electronically points the antenna in a surveillance direction during TCAS transmit and receive operations.

The directional antenna is an electronically steerable phased array consisting of four top loaded monopole elements.

The antenna pattern is pointed in one of four directions without physically moving the antenna. This is done by independently changing the phase to each of the four antenna elements contained in the directional antenna. The phasing of these signals directly affect the beam, therefore, the phase match of the coaxial cables is very important.

The four beam forming elements are located on the perimeter of the antenna, and the whole assembly is contained in an aerodynamic plastic enclosure.

During TCAS receptions, each of the four directional antenna elements receives any 1090 MHz signal that passes by the element. The phasing of these received signals calculated by the direction from which the rf energy is received. These signals are directed onto the same four cables that connect transmit signals between the TCAS processor unit and the directional antenna.

Each part of the four ports of the directional antenna has a resistor located across the antenna element to ground. Each of the four resistors is the same value. The TCAS processor unit periodically does a continuity check on the antenna ports and should sense the correct resistance value through an analogue to digital converter if the port is not shorted or not open.

The directional antenna is a passive device and does not require aircraft electrical power to function.

An aluminum foil gasket with an elastomeric sealant makes sure consistent distribution of conducting contacts of the gasket to the airframe. The pressure applied during installation gives a contouring feature. It lets the gasket conform to the two mating surfaces.

The top antenna is located at station X+21.3, Y+11.0, and Z+182.6. The bottom antenna is located at station X−45.0, Y0.0 and Z+89.0. Each antenna is attached to the aircraft with four mounting screws.

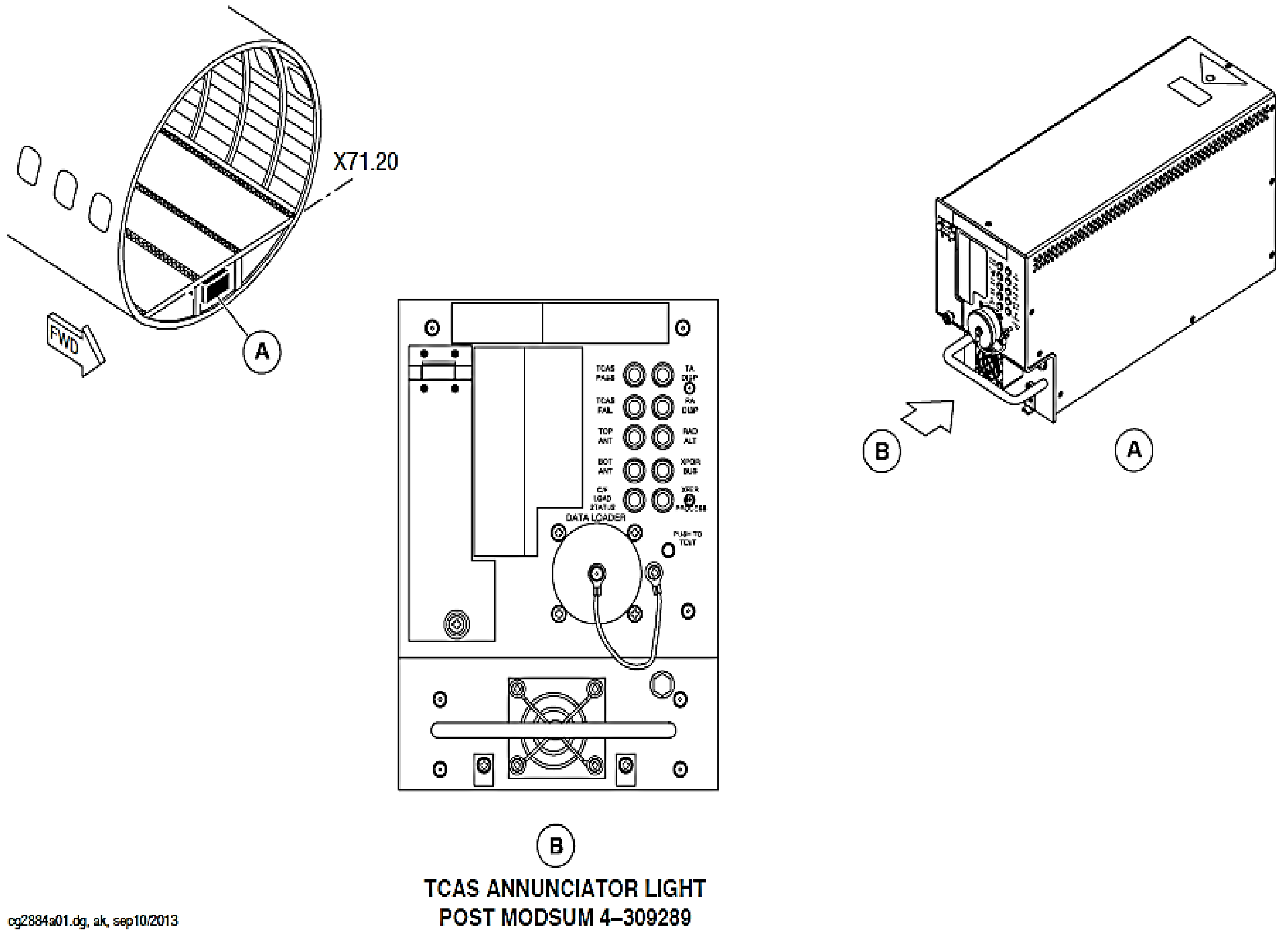


Figure 34- 179 TCAS ANNUNCIATOR LIGHT

Figure 34- 179 TCAS ANNUNCIATOR LIGHT

On aircraft with ModSum 4−309289 installed, the annunciator lights installed on the front face of the TCAS processor give the indications of the failure as follows:

TCAS PASS: This Light Emitting Diode (LED) comes on if the unit passes its own internal Built In Test Equipment (BITE) test.

TCAS FAIL: This LED comes on if the unit has not passed its own internal BITE test.

TOP ANT: This LED comes on when there is a failure in the DC resistance test of the top antenna.

BOT ANT: This LED comes on when there is a failure in the DC resistance test of the bottom antenna.

C/F LOAD STATUS: This LED comes on red if there is an error in software upload through the compact flash drive. The light C/F LOAD STATUS comes on green, if the software upload is successful.

TA DISP: This LED comes on if the traffic advisory display discrete signals from the No. 1 or the No. 2 TCAS system has failed.

RA DISP: This LED comes on if the resolution advisory display discrete signals from the No. 1 or the No. 2 TCAS system has failed.

RAD ALT: This LED comes on if the radio altitude source from the No. 1 or No. 2 radio altimeter system is invalid or has failed.

XPDR BUS: This LED comes on if the mode S transponder signals from the Air Traffic Control (ATC) No. 1 or the No. 2 system is invalid or has failed.

XFER IN PROCESS: This LED comes on to show the software upload status through the compact flash drive. When the TCAS processor reads the compact flash drive, the LED blinks. If there is an error in the software upload through the compact flash drive, the LED turns to red. The LED turns to green, if the software upload is successful.

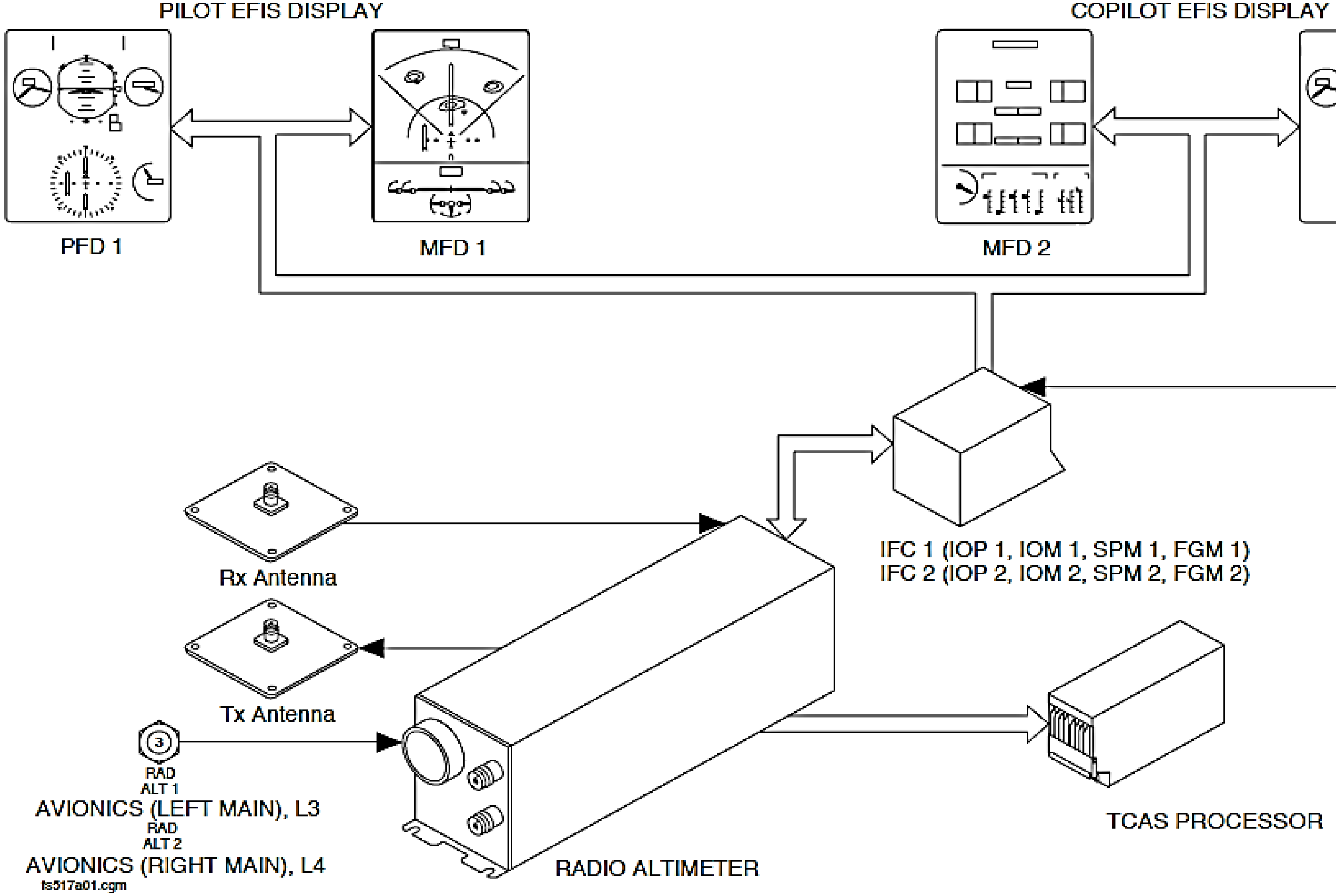


Figure 34- 180 RADIO ALTIMETER BLOCK DIAGRAM

RADIO ALTIMETER (HONEYWELL)

Introduction

The Radio Altimeter (RA) System calculates the low level Above Ground Level (AGL) altitude for display and for use by other aircraft systems.

The System allows the Decision Height (DH) altitude to be set for use during the approach phase of a flight.

General Description

The RA system transmits a signal to the ground and then it receives the reflected signal to calculate the AGL altitude.

The aircraft Primary Flight Displays (PFD1, PFD2) show the RA system parameters that follow:

- Above Ground Level (AGL) altitude
- RA mismatch
- Rising runway symbol
- DH status.

The RA System is used to make DH calculations. The AGL altitude data is compared to the DH value set by the flight crew. The Primary Flight Displays (PFD1, PFD2) shows the selected value. The Electronic Instrument System (EFIS) and the Audio and Radio Management System (ARMS) are used to alert the flight crew both visually and aurally when the aircraft reaches the set value.

In addition, the RA system supplies a signal to the Stall Protection System to prevent it from operating when the altitude is less than 200 ft. (60.96 m) AGL.

The RA system has the components that follow:

- Transmitter/receiver
- Transmitter/receiver Antenna.

The RA system supplies data directly to the Traffic Collision Avoidance System (TCAS) and through both Integrated Flight Cabinets (IFC1, IFC2) to the systems that follow:

- Auto Flight Control System (AFCS)
- Stall Protection Module (SPM1, SPM2)
- Primary Flight Displays (PFD1, PFD2)
- Solid State Flight Data Recorder (SSFDR)
- Ground Proximity Warning System (GPWS)
- Central Diagnostic System (CDS).

Detailed Description

A Frequency Modulated Carrier Wave (FMCW) is transmitted by the RA transmitter/receivers (RA1, RA2). The Frequency Modulated Carrier Wave (FMCW) continuously changes its frequency and compares the time it takes for the transmitted signal to go to the ground and back to the new frequency.

The received signal is mixed with the transmitted signal to make an Intermediate Frequency (IF). The IF is directly proportional to the time that it takes the signal to go to the ground and back. The Intermediate Frequency (IF) is changed into ARINC 429 and analogue outputs.

The RA transmitter/receiver gives continuous AGL altitude data from 2500 ft. (762 m) until landing. The center operating frequency of the transmitter that is routed through the transmit antenna is 4300 ±15 MHz It is continuously modulated at a 100 Hz rate, with a FM deviation of 100 ±5 MHz The nominal transmitted power of the signal is 160 mW.

When two RA systems are installed, both transmitter/receivers will continuously function at the same time. An external hardware strap, which is part of the aircraft wiring, is connected to the second RA transmitter/receiver interface connector. When the strap connection is sensed, the RA transmitter/receiver will use a different modulation frequency, 105 Hz.

During power−up, each RA transmitter/receiver does a comprehensive continuous self-test routine. The status of the transmitter/receiver is sent to the Central Diagnostic System (CDS).

If a single RA system is installed and the transmitter/receiver fails, then its failure status will be shown.

If a dual RA system is installed and one transmitter/receiver fails, then its failure status will not be shown to the flight crew or to the TCAS.

Each Integrated Flight Cabinet IFC1, IFC2, receives the same AGL data. If both transmitter/receivers fail, then its failure status will be shown on the Primary Flight Displays (PFD1, PFD2).

Figure 34- 181 RADIO ALTIMETER INDICATIONS

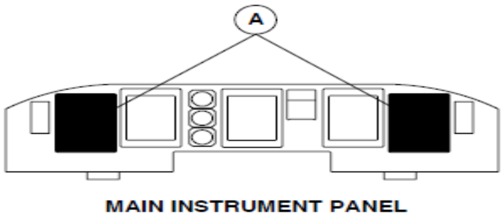
The PFD1 and PFD2 display altitude AGL as four white numbers followed by a RA label. It shows the altitude from zero to 2500 ft. (762 m) AGL.

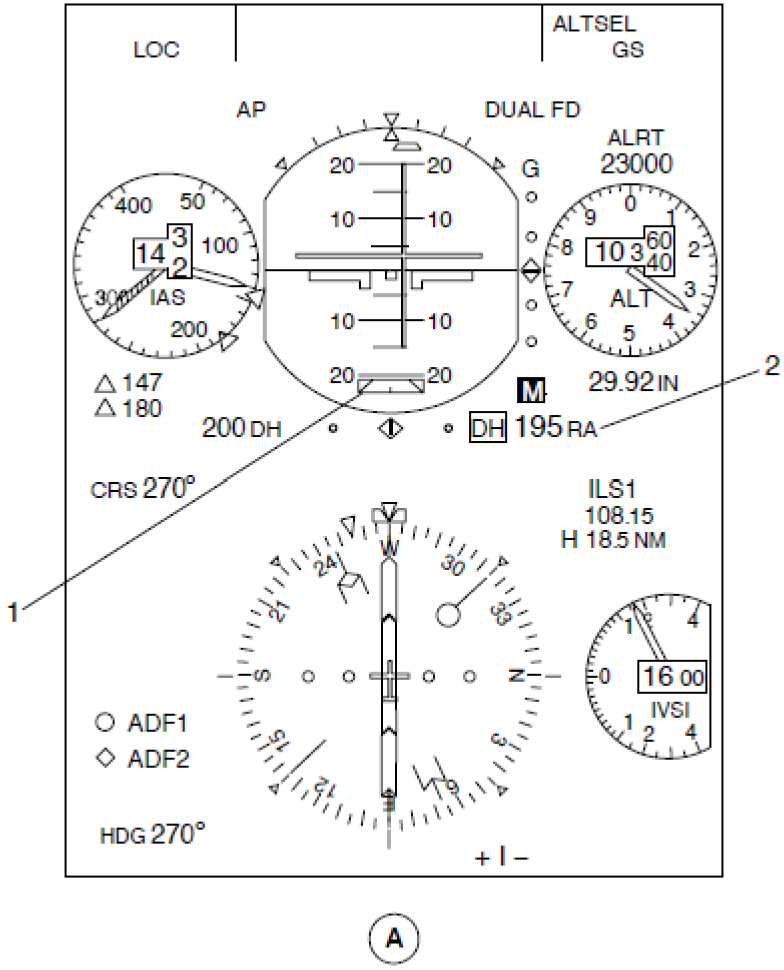
The altitude increases and decreases in 5 ft. (1.52 m) increments below 200 ft. (61 m) AGL and increases and decreases in 10 ft. increments above 200 ft. (61 m) AGL.

A rising runway symbol is used to give an analogue indication of AGL altitude. It moves vertically, linearly and correspondingly to the AGL altitude.

It comes into view at the bottom of the attitude sphere when the aircraft is at 200 ft. (61 m) AGL altitude. It moves towards the airplane symbol as the altitude AGL decreases and touches the bottom of the airplane symbol at touchdown.

The rising runway symbol goes out of view when radio altitude is not available or the aircraft is above 200 ft. (61 m) AGL altitude.





LEGEND
1. Rising Runway Indication.
2. Radio Altitude Display.

Figure 34- 181 RADIO ALTIMETER INDICATIONS

Figure 34- 182 RADIO ALTIMETER MISMATCH INDICATION

The Flight Data Processing System (FDPS) senses AGL differences between the RA transmitter/receivers (RA1, RA2), or between an output and shown parameter. If there is a mismatch condition, the FDPS causes the RA label to change from white to yellow and flash initially and then stay on steady.

In addition, the advisory display areas of the Primary Flight Displays (PFD1, PFD2) show a RAD ALT MISMATCH message. It comes into view and flashes in yellow text and then stays on steady.

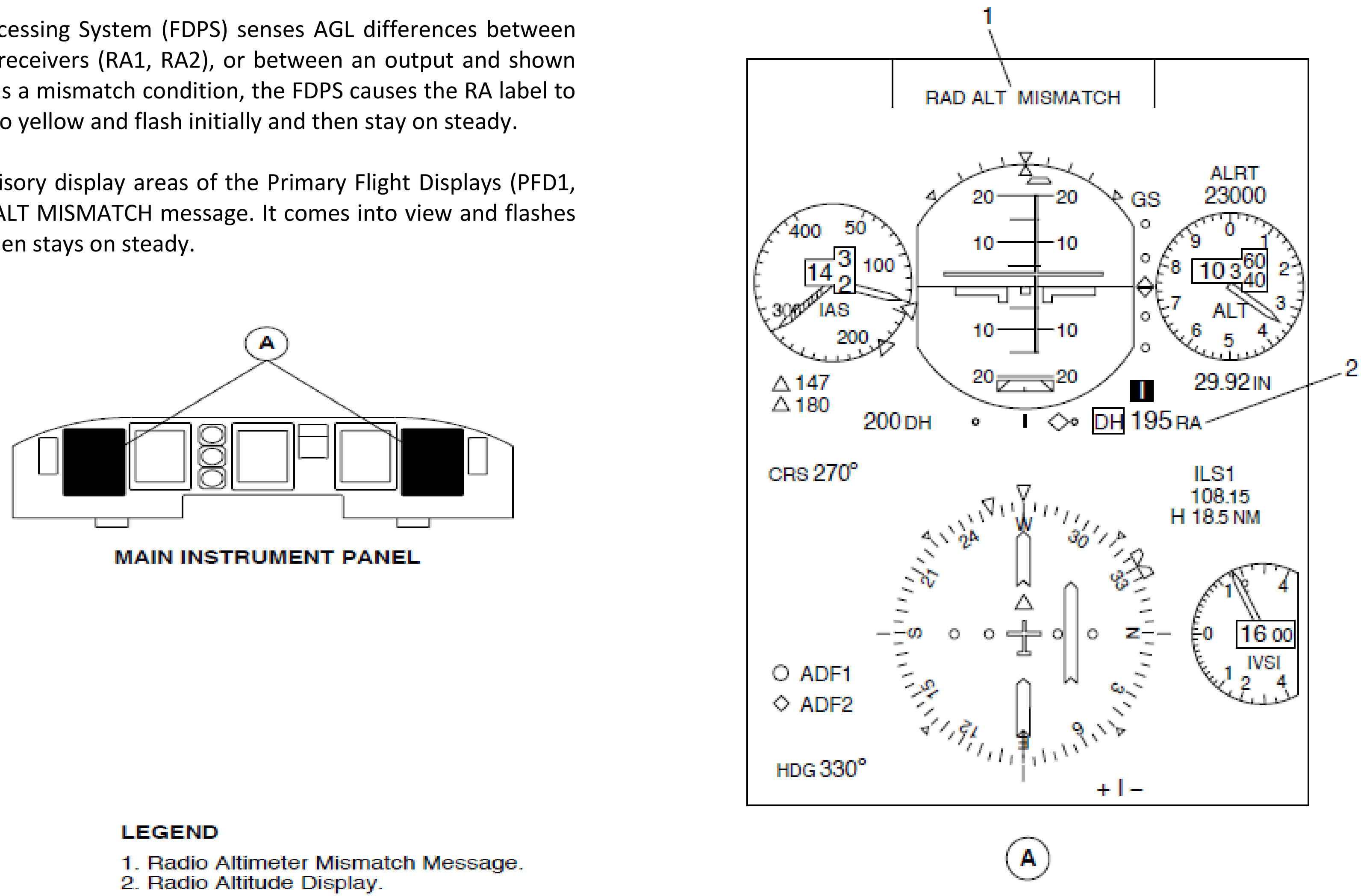
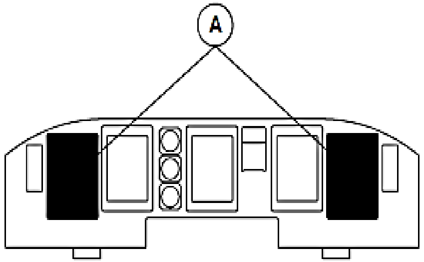


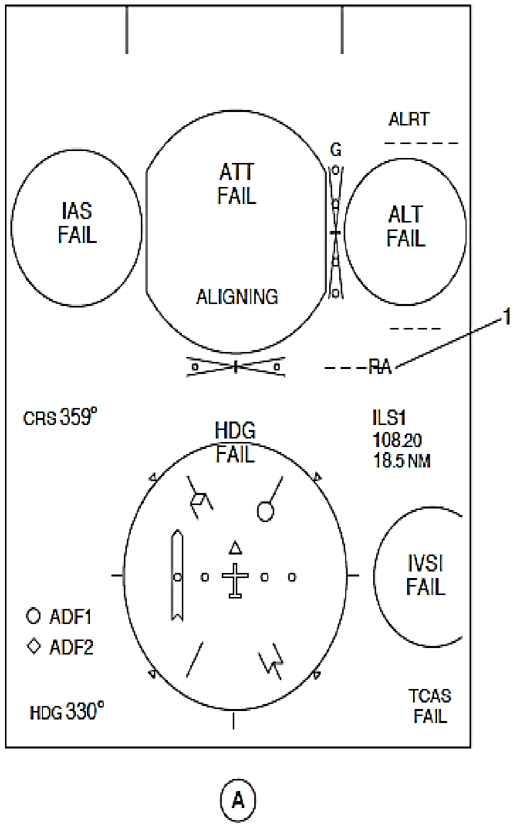
Figure 34- 182 RADIO ALTIMETER MISMATCH INDICATION

Figure 34- 183 RADIO ALTIMETER FAIL ANNUNCIATION

A failure annunciation shows four red dashes in place of the AGL altitude on the PFD1 and PFD2. If a TCAS is installed, a TCAS FAIL message will also be shown in yellow text on the PFD1 and PFD2.



MAIN INSTRUMENT PANEL



LEGEND

1. Radio Altimeter Fail Annunciation.

Figure 34- 183 RADIO ALTIMETER FAIL ANNUNCIATION

Figure 34- 184 INDEX CONTROL PANEL, DH KNOB

For DH calculations, the DH knob located on the Index Control Panel (ICP1, ICP2) is turned to set a DH altitude.

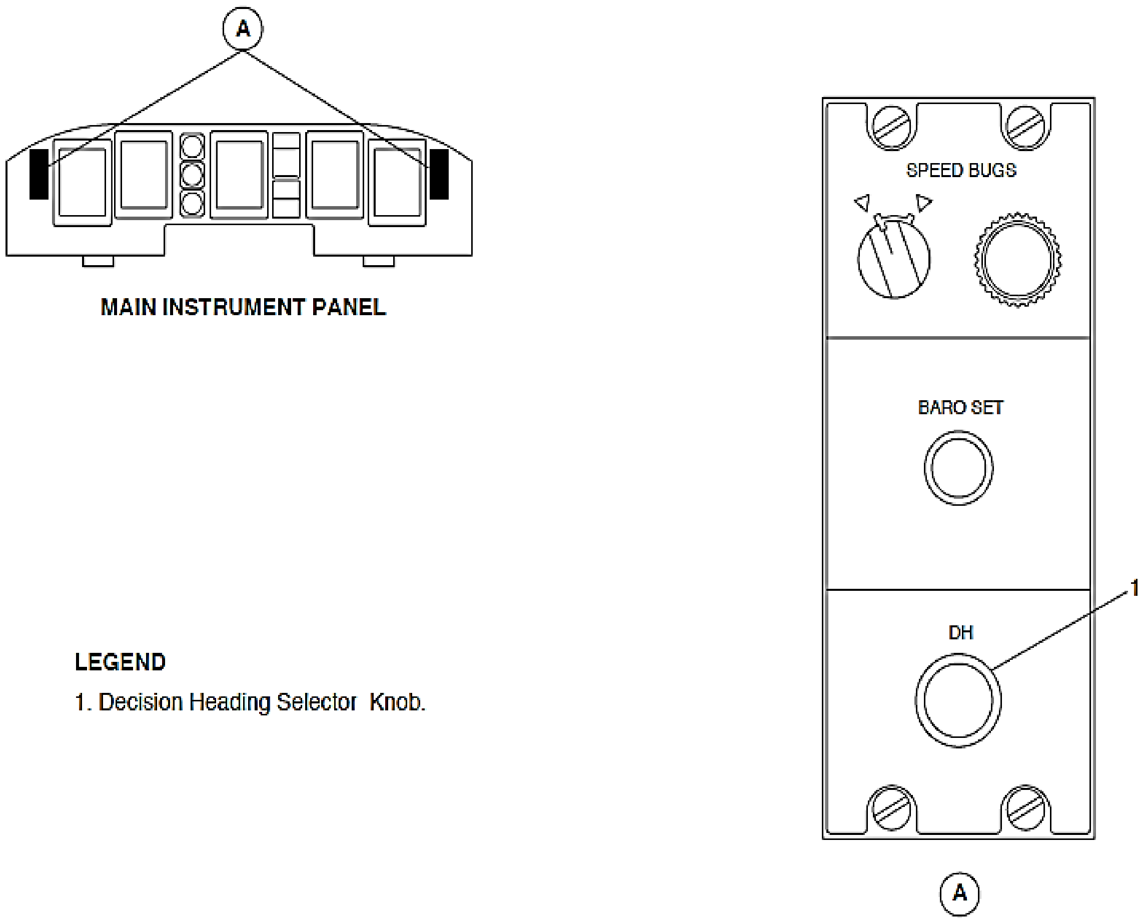
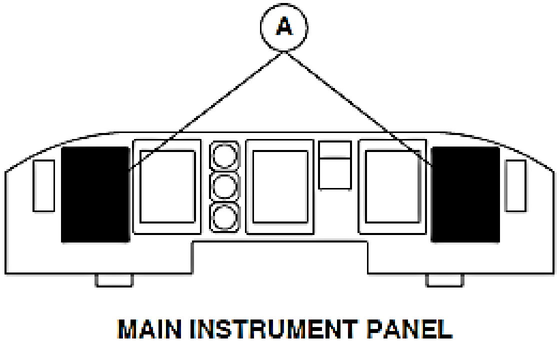
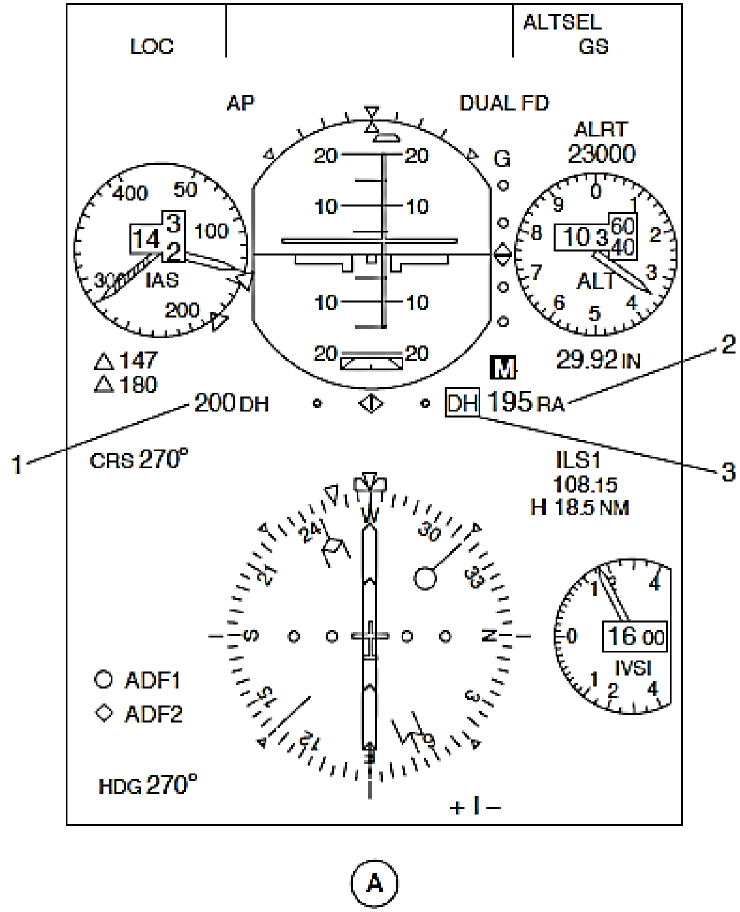


Figure 34- 184 INDEX CONTROL PANEL, DH KNOB





LEGEND

1. Decision Height Indication.
2. Radio Altitude Display.
3. Decision Height Annunciation.

Figure 34- 185 RADIO ALTIMETER DECISION HEIGHT PROCESSING INDICATIONS

Figure 34- 185 RADIO ALTIMETER DECISION HEIGHT PROCESSING INDICATIONS

The related PFD1 and PFD2 shows the selection as three digits in cyan followed by a white DH label. It shows the altitude selection in one−foot increments from zero through 999 ft. (304.49).

Three white dashes replace the digits when PFD1 and PFD2 do not receive valid data from the Integrated Flight Cabinets (IFC1, IFC2).

The DH label shows that the aircraft is at the selected height.

When aircraft is at or less than 100 ft. (30.5 m) AGL above the selected DH, the Primary Flight Displays (PFD1, PFD2) will show a white box. When the aircraft goes below the DH, a yellow DH label comes into view inside the white box.

The DH indication is out of view when the altitude AGL is greater than the DH setting.

The 28 Vdc left and right main buses supply power to the RA transmitter/receivers (RA1, RA2) through 3 ampere circuit breakers. The RA transmitter/receivers (RA1, RA2) system circuit breakers are located in positions L3 and L4 on the 28 Vdc avionics circuit breaker panel.

When two RA transmitter/receivers are installed, each unit has a related transmit and receive antenna.

In this configuration both RA transmitter/receivers (RA1, RA2) supply data to both Integrated Flight Cabinets (IFC1, IFC2) and the TCAS. When one RA transmitter/receiver (RA1, RA2) fails, the Integrated Flight Cabinets (IFC 1, IFC2) use data from the other RA transmitter/receiver (RA1, RA2). No indication of the status of the failed RA transmitter/receiver is shown.

NOTE

The status of the failed RA transmitter/receiver (RA1, RA2) is transmitted and stored in the Central Diagnostic System (CDS).

Both RA transmitter/receivers (RA1, RA2) supply a discrete health status output to each IOM1 and IOM2 for CDS monitoring purposes.

Both RA transmitter/receivers (RA1, RA2) supply radio height data to each Input Output Processors (IOP1, IOP2). A FDPS within each IOP controls the supply of data to the systems that follow:

- Ground Proximity Warning System GPWS)
- Auto Flight Control System (AFCS)
- Stall Protection System (SPS)
- Flight Data Recorder (FDR)
- Primary Flight Displays (PFD1, PFD2).

Each FDPS does a discrepancy check between the RA AGL altitude data that is received by the opposite IOP and the RA AGL altitude data that is shown on the PFD. If a difference is sensed between these inputs, then a RAD AL MISMATCH message comes into view on the opposite PFD.

The FDPS monitors the status of its related DH setting. The IOP1 sends the DH status to the GPWS. When the higher DH setting is reached the IOP1 FDPS sends the DH data to the GPWS. The "MINIMUMS, MINIMUMS" aural warning comes on if the highest DH setting is more than 50 ft. (15.2 m).

Each Stall Protection Module (SPM1, SPM2) supplies a discrete RA Test #1 Enabled discrete input that is used to test the SPS to RA system interface. The SPS test switch selection supplies a discrete RA Test #2 Enabled discrete input that is also used to test the SPS to RA system interface. Each Stall Protection Module (SPM1, SPM2) supplies an analogue test voltage input that is used to test the SPS to RA system interface.

Each Stall Protection Module (SPM1, SPM2) receives a discrete 200 ft. (60.96 m) altitude trip output that is used to prevent SPS stick pusher operation when the AGL altitude is below 200 ft. (60.96 m).

Each RA transmitter/receiver sends data to the TCAS through a single data bus.

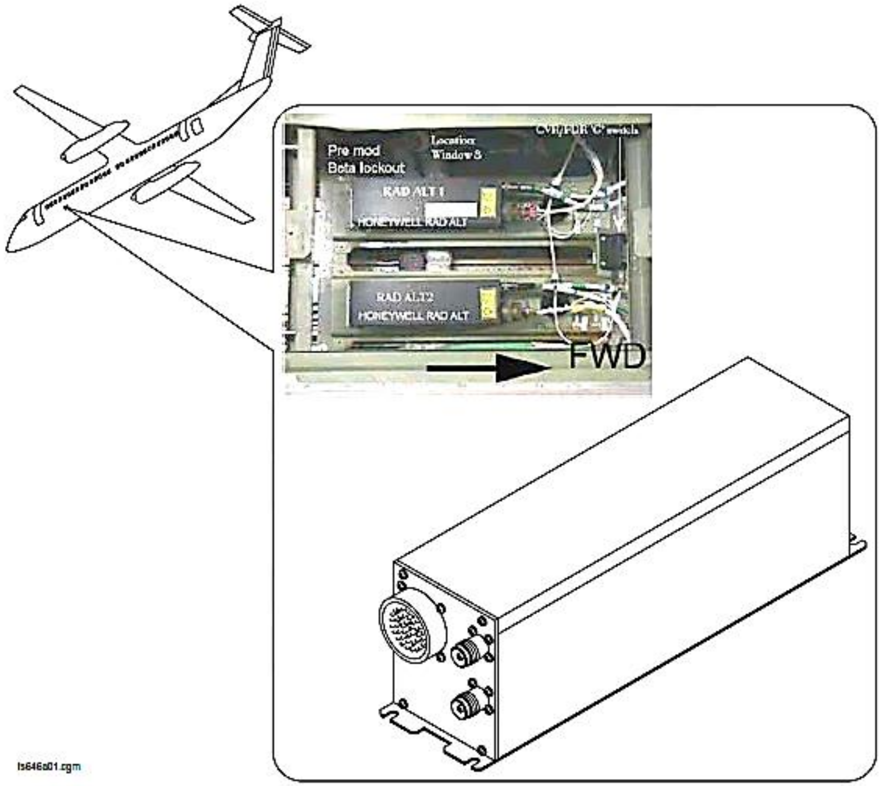


Figure 34- 186 TRANSMITTER/RECEIVER

Transmitter/Receiver

Figure 34- 186 TRANSMITTER/RECEIVER

The RA transmitter/receiver is a digital computing unit that has the modules that follow:

- An RF Board
- Power Supply and Logic Board.

Digital controlled circuitry is used to meet the functional requirements as controlled by the embedded software.

All of the transmitter/receiver parts are attached to a one−piece chassis.

The aircraft interface connectors and the zero ft. calibration switch are located on the front face of the unit.

Each transmitter/receiver is installed in a mounting rack that is located in the forward area of the center fuselage of the aircraft, below the floor. The RA 1 mounting rack is located at station X 208.6, Y−3.5 and Z 87.5, and the RA 2 mounting rack is located at station X 208.6, Y+3.5 and Z 87.5.

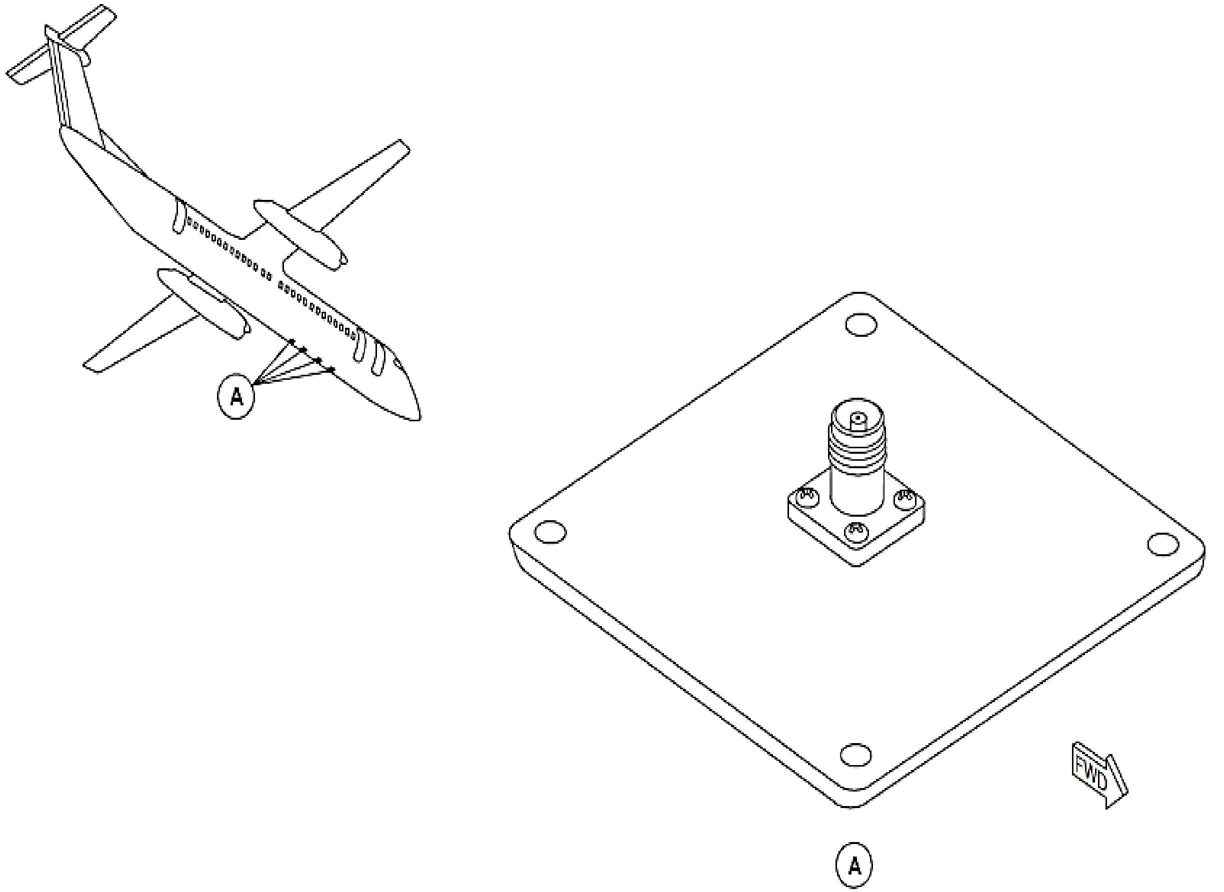


Figure 34- 187 RADIO ALTIMETER ANTENNAS LOCATOR

Antenna

Figure 34- 187 RADIO ALTIMETER ANTENNAS LOCATOR

The RA microstrip antenna operates in the RA frequency band of 4200 through 4400 MHz the antennas are linear polarized and have a beam width coverage more than 45 degrees in the roll plane and more than 40 degrees in the pitch plane.

The antenna is made from lightweight, durable Teflon and glass material with an aluminum base and TNC coaxial connector.

Each antenna can send or receive transmissions.

Each antenna is installed to the aircraft structure with four mounting screws located at each corner. A "dry fit" sealing gasket electrically bonds each antenna to the aircraft structure.

The gasket is an aluminum foil gasket with an elastomeric sealant. The pressure applied during installation gives a contouring feature that lets the gasket form itself to the two surfaces. The moisture free, contoured RF seal requires no maintenance.

The RA 1 and RA 2 transmit and receive antennas are located inline and spaced 20 in. (508 mm) apart on the bottom of the center fuselage section.

On aircraft with ModSum 4−201612 and without ModSum 4−902453 or SB 84−34−151 incorporated, the arrangements of the antenna are as follows:

TX1 (X126.0, Y2.6 and Z81.0), RX1 (X146.0, Y2.6 and Z81.0), TX2
(X166.0, Y2.6 and Z81.0) and RX2 (X186.0, Y2.6 and Z81.0)

On aircraft with ModSum 4−300800 or 4−309215 or 4−902453 or SB 84−34−151 incorporated, the arrangements of the antenna are as follows:

TX1 (X126.0, Y2.6 and Z81.0), RX1 (X146.0, Y2.6 and Z81.0), RX2
(X166.0, Y2.6 and Z81.0) and TX2 (X186.0, Y2.6 and Z81.0)

Antenna Cables

The RA system uses coaxial cables to connect the antennas to the transmitter/receiver. The coaxial cable is a transmission line with two conductors. A dielectric insulator separates the center conductor and the braided outer conductor.

The RA transmitter/receiver can use different lengths of coaxial cables. It is calibrated to 0 ft. during installation when a recessed calibration switch located on the front faceplate of the Transmitter/Receiver is pushed. The 0 ft. calibration of the RA system is done when the aircraft is static and in the normal Weight On Wheels (WOW) position.

The specified installation length is between 20 and 55 ft. (6.1 to 16.8 m).

Beta Lockout

The delayed WOW signal during landing causes a delay in the release of the propeller beta lockout function, which prevents the crew to select the propeller ground beta range and increases the rollout and stopping distance.

The occurrences of the delayed WOW signal is estimated by the addition of the two radio altimeter discrete signals with the WOW signals to the Propeller Electronic Controls (PECs). This is done by the addition of a configuration module to each radio altimeter transceiver, which provides the 25 ft. (7.62 m) trip point discrete signal input to the PECs. This allows the beta lockout system to be disabled before the main landing gear touchdown at 25 ft. (7.62 m)
AGL.

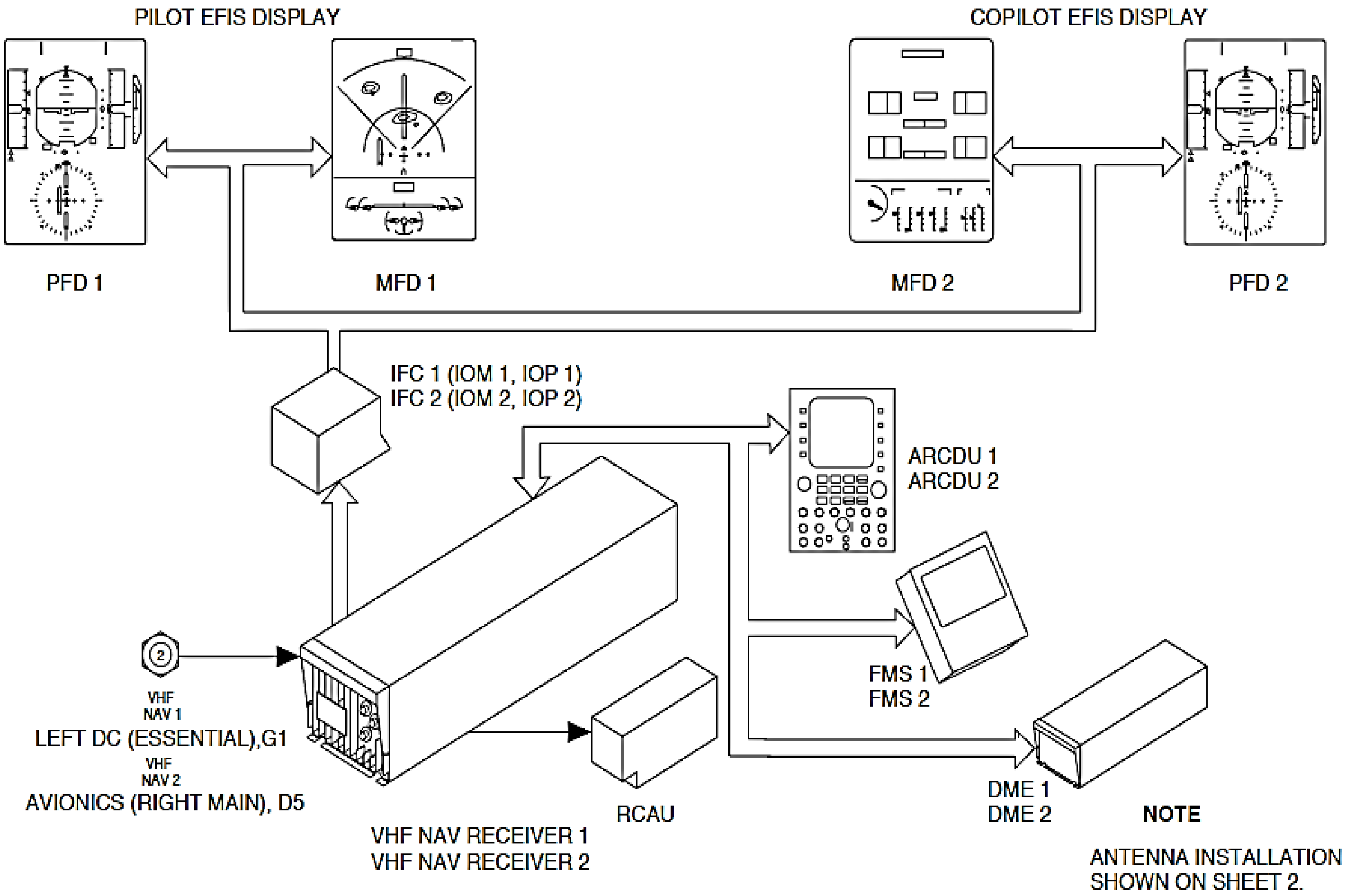


Figure 34- 188 VHF NAVIGATION SYSTEM BLOCK DIAGRAM

VHF NAVIGATION SYSTEMS

Introduction

The Very High Frequency Navigation (VHF NAV) system receives radio signals to give the navigation data that follows:

- VHF Omni–range (VOR)
- Instrument Landing System (ILS).

VHF Omni–range (VOR): The Very High Frequency Omni–range (VOR) ground stations form part of the Victor airways navigation system. The VOR gives guidance on any track to or from the station.

Instrument Landing System (ILS): The Instrument Landing System (ILS) is an integrated system that gives the approach flight path to landing on a runway.

General Description

Figure 34- 188 VHF NAVIGATION SYSTEM BLOCK DIAGRAM
Figure 34- 189 VHF NAVIGATION ANTENNAS BLOCK DIAGRAM

The two VHF Navigation System operates in VOR or ILS mode Depending on frequency selection.

In the VOR mode, the navigation receiver measures the deviation from the selected radial of the Very High Frequency Omni–range (VOR) ground transmitting station.

In the Instrument Landing System (ILS) mode, the Navigation Receiver uses the following landing aids:

- Localizer (LOC)
- Glideslope (GS)
- Marker beacon (MB).

The Localizer (LOC) system helps to guide the aircraft to the centerline approach path to the runway. The Glideslope (GS) system gives the approach angle of an aircraft. It helps to guide the aircraft down an approximate 3 degree glideslope. The Marker Beacons (MB) designated and mark an area above themselves. Up to three MBs are located near the centerline of the approach to the runway. They use the following designations:

- Outer
- Middle
- Inner.

The Very High Frequency Omni–Range (VOR) and Localizer (LOC) Ground stations periodically transmit an identification signal. The navigation receiver supplies the audio to the Audio and Radio Management System (ARMS). The Marker Beacon (MB) continuously transmits its identification signal. The navigation receiver, receives the Marker Beacon (MB) audio when directly above the transmitter. The audio is supplied to the Audio and Radio Management System (ARMS).

The VHF receivers are controlled by the components that follow:

- Audio and Radio Control Display Units (ARCDU1, ARCDU2)
- Flight Management System (FMS).

The ARCDU manually tunes the VOR and ILS (localizer and paired glideslope) frequencies. The FMS provides automatic tuning.

The ARCDU or FMS is used to self–test the marker beacons.

The Electronic Instrument System (EIS) shows the information that follows:

- VOR bearing course
- VOR lateral deviation
- Localizer lateral deviation
- Glideslope vertical deviation
- Marker beacon passage.

The Very High Frequency (VHF) Navigation System has the components that follow:

- Receiver, VHF Navigation
- Antenna, Glideslope
- Antenna, VOR/LOC
- Coupler, Antenna VOR/LOC
- Splitter, Antenna VOR/LOC
- Antenna, Marker
- Splitter, Marker.

The VHF Navigation Receivers (VHF NAV1, VHF NAV2) supply Navigation data to the systems that follow:

- Flight Guidance Modules (FGM1, FGM2)
- Audio and Radio System (ARMS)
- Input/output Processors (IOP1, IOP2)
- Distance Measuring Equipment (DME)
- Flight Management System (FMS).

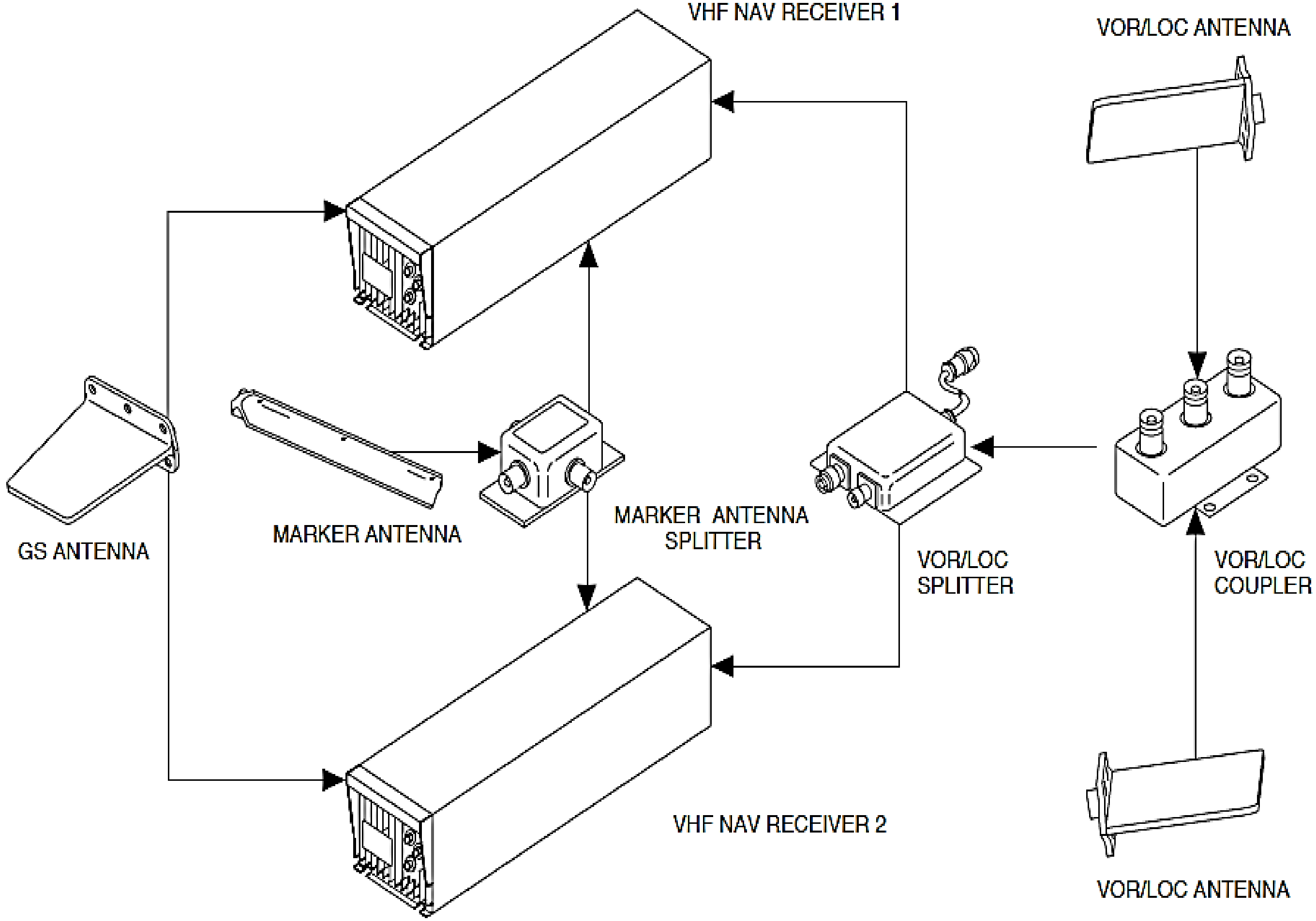


Figure 34- 189 VHF NAVIGATION ANTENNAS BLOCK DIAGRAM

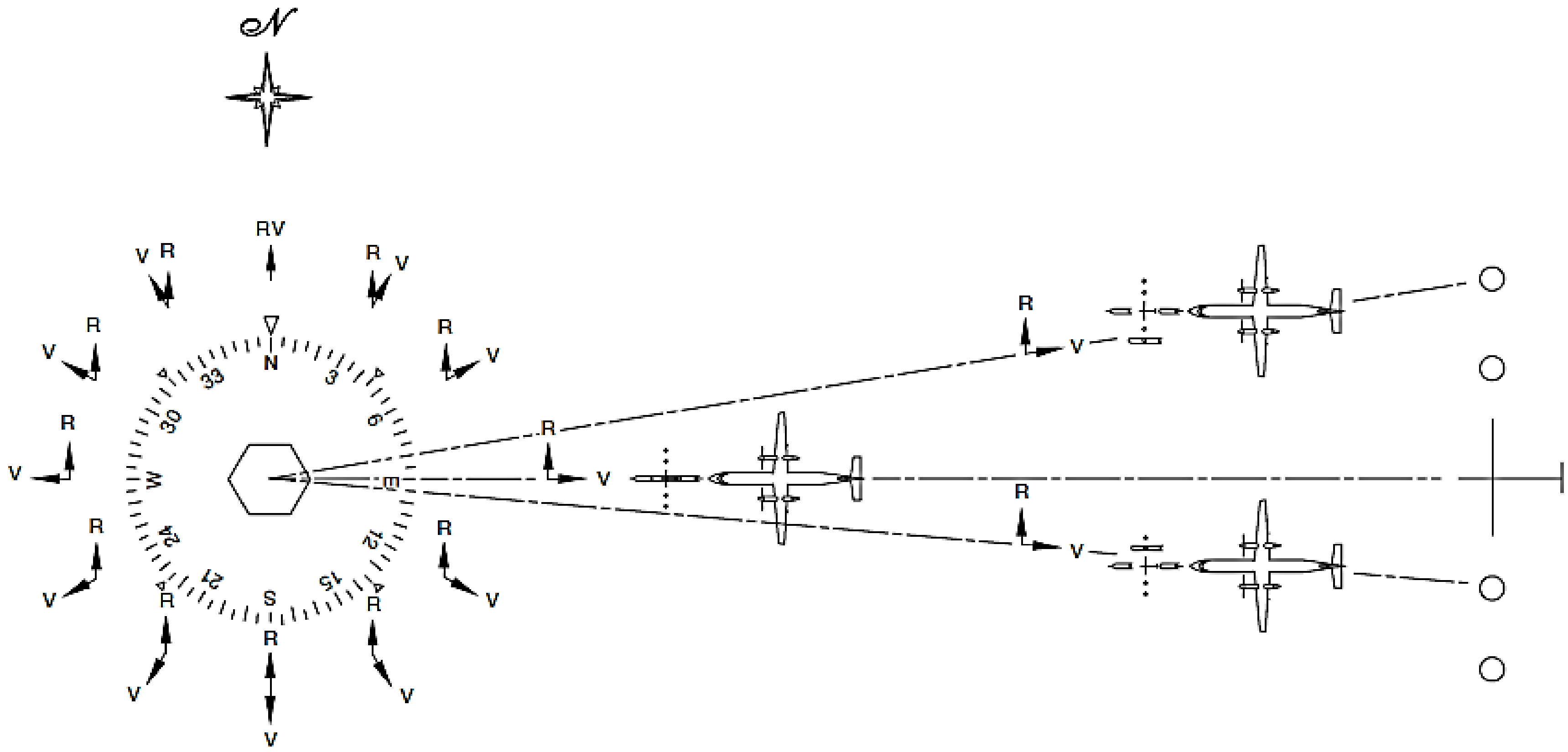


Figure 34- 190 VOR NAVIGATION PRINCIPLES

Detailed Description

Figure 34- 190 VOR NAVIGATION PRINCIPLES

The VHF Navigation Receivers (VHF NAV1, VHF NAV2) receive radio signals and data from the navigation ground stations that follow:

- VOR/DME
- VORTAC
- ILS.

VHF Omni−range (VOR): The VHF Omni−range (VOR) calculates the data that follows:

- The aircraft's position with reference to a VOR ground Station radial
- A flight path toward or away from the station.

The VOR ground station uses two antennas. The VOR ground station uses the same carrier frequency to transmit the signals to the aircraft that follow:

- Reference Phase, Frequency Modulated (FM)
- Variable phase, Amplitude Modulated (AM).

Reference Phase: A non−directional (omni−directional) antenna transmits the Reference Phase as a Frequency Modulated (FM) signal. It has a 9960 Hz sub−carrier that deviates 480 Hz 30 times a second. Its phase is the same through 360 degrees of heading.

Variable Phase: A directional antenna that turns at 30 revolutions per second and transmits a horizontally polarized signal. The phase of the received signal varies at a rate that is the same as the antenna rotation. The aircraft receives a Radio Frequency (RF) signal that appears as 30 Hz Amplitude Modulation (AM). Its phase varies through 360 degrees of heading.

The two signals align exactly in phase on magnetic north. There is Phase differential at any other point of azimuth around the station. The variable phase signal lags the reference signal. The VHF navigation receiver separates the 30 Hz frequency reference and variable phase signals from the RF carrier. It measures the difference in phase and shows the result in terms of a selected omni bearing line and the amount of deviation from it. It does not calculate direction.

A VOR deviation pointer will centre on either of two course selector settings that are 180 degrees apart to show the magnetic bearing to or from the VOR ground station. Bearing pointers are used to show the magnetic bearing to the VOR ground station.

The VOR ground stations transmits the audio identifiers that follow:

- Three−letter code group
- Voice.

The VOR ground stations continue to identify the code groups during weather and voice broadcasts. The relative audio levels of the broadcasts are set so that the code or voice transmission is easy to hear without interference from the other.

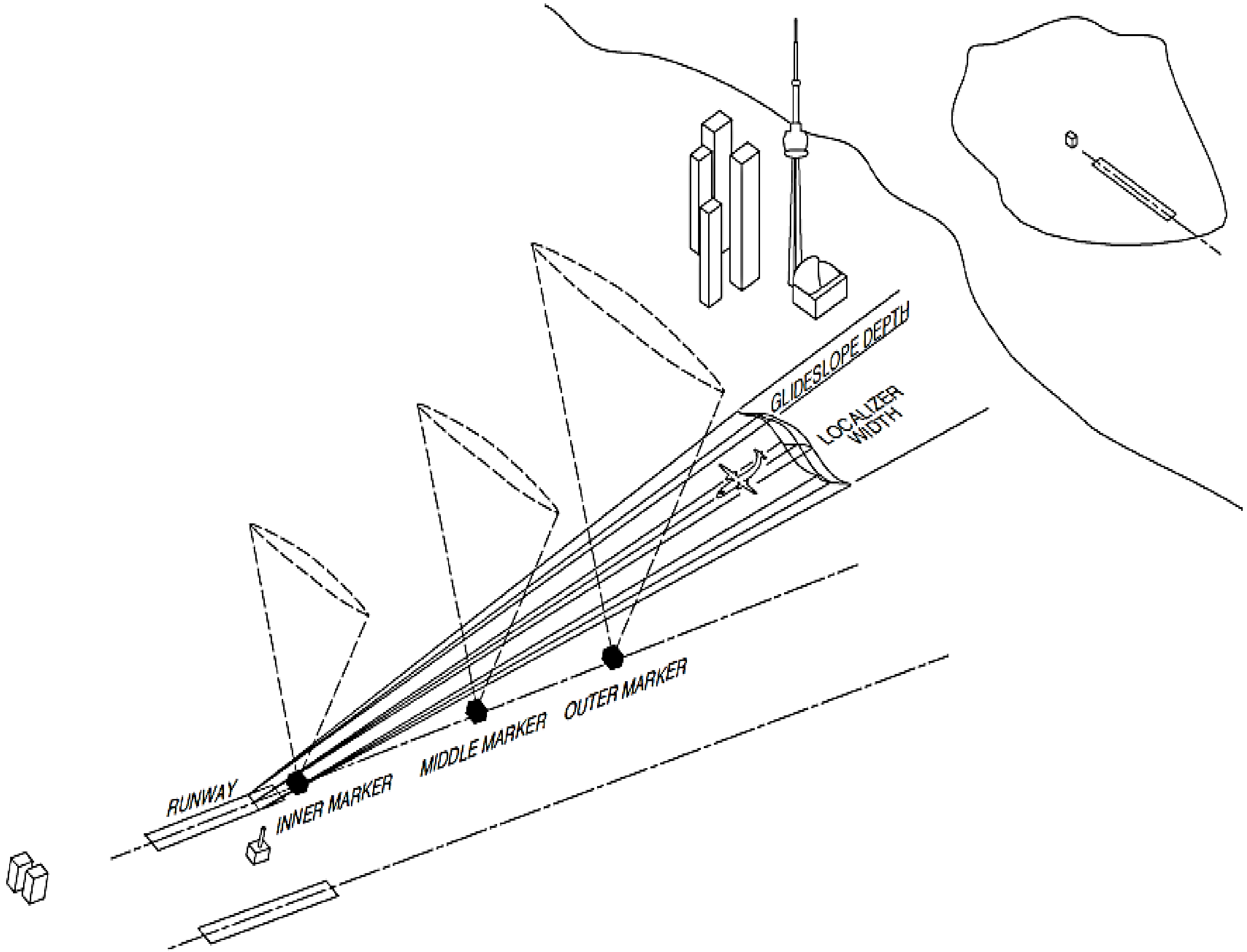


Figure 34- 191 INSTRUMENT LANDING SYSTEM (ILS) PRINCIPLES

Figure 34- 191 INSTRUMENT LANDING SYSTEM (ILS) PRINCIPLES

Instrument Landing System (ILS): The Instrument Landing System (ILS) is a precision instrument approach system. It guides the aircraft to a position from which the flight crew can make a landing.

The Instrument Landing System (ILS) supplies the information that follows to the VHF navigation system:

- Localizer
- Glideslope
- Marker Beacon.

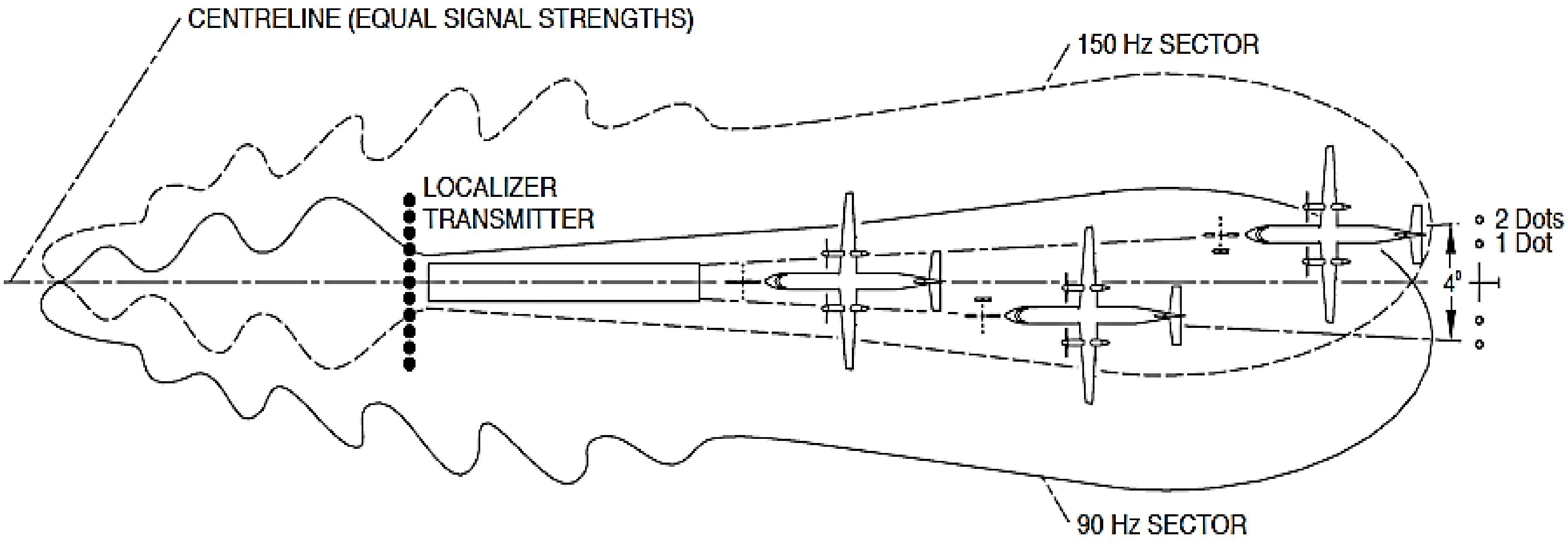


Figure 34- 192 LOCALIZER INDICATION PRINCIPLES

Figure 34- 192 LOCALIZER INDICATION PRINCIPLES

Localizer: The localizer system is used to give the position of the aircraft relative to a straight approach line to the runway. The ground based localizer antenna system sends two transmission patterns at the same frequency. One modulates at 90 Hz and the other at 150 Hz. The radio beams overlap to give a 5 degree angle to the centerline of the runway.

There are two approaches to a runway centerline:

- Front course (Localizer or ILS)
- Back course.

Front Course: A front course approach occurs when the aircraft is flown towards the departure end of the of the runway. The 90 Hz area is to the left and the 150 Hz is to the right.

Back Course: A back course approach occurs when the aircraft is flown away from the departure end of the runway. The areas reverse. The 150 Hz areas to the left and the 90 Hz is to the right.

The back courses are classified into one of the three categories as follows:

- Usable
- Restricted
- Unusable.

Usable: The usable category is suitable for back course approaches.

Restricted: It is not safe for back course approaches. The flight crew may use the back course for transitions, intersections, holding patterns, or missed approaches.

Unusable: The back course is not suitable or usable for any purpose.

The localizer projects at least 25 mi (40.2 km) from the airport and at an altitude of at least 2000 ft. (610 m).

When directly on the localizer course, the aircraft receives the 90and 150 Hz signals equal in strength. If the aircraft is flying to the left of the centerline, the 90 Hz signal is relatively stronger than the 150 Hz signal. Deviation bars move to right to show that the position of the aircrafts to the left of the runway centerline. If the aircraft is flying to the right of the centerline, the 150 Hz signal is stronger than the 90 Hz signal. The deviation bar moves to the left. The sensing reverses when flying a back course. On the back course, the aircraft is navigated away from the deviation bar to get back on course.

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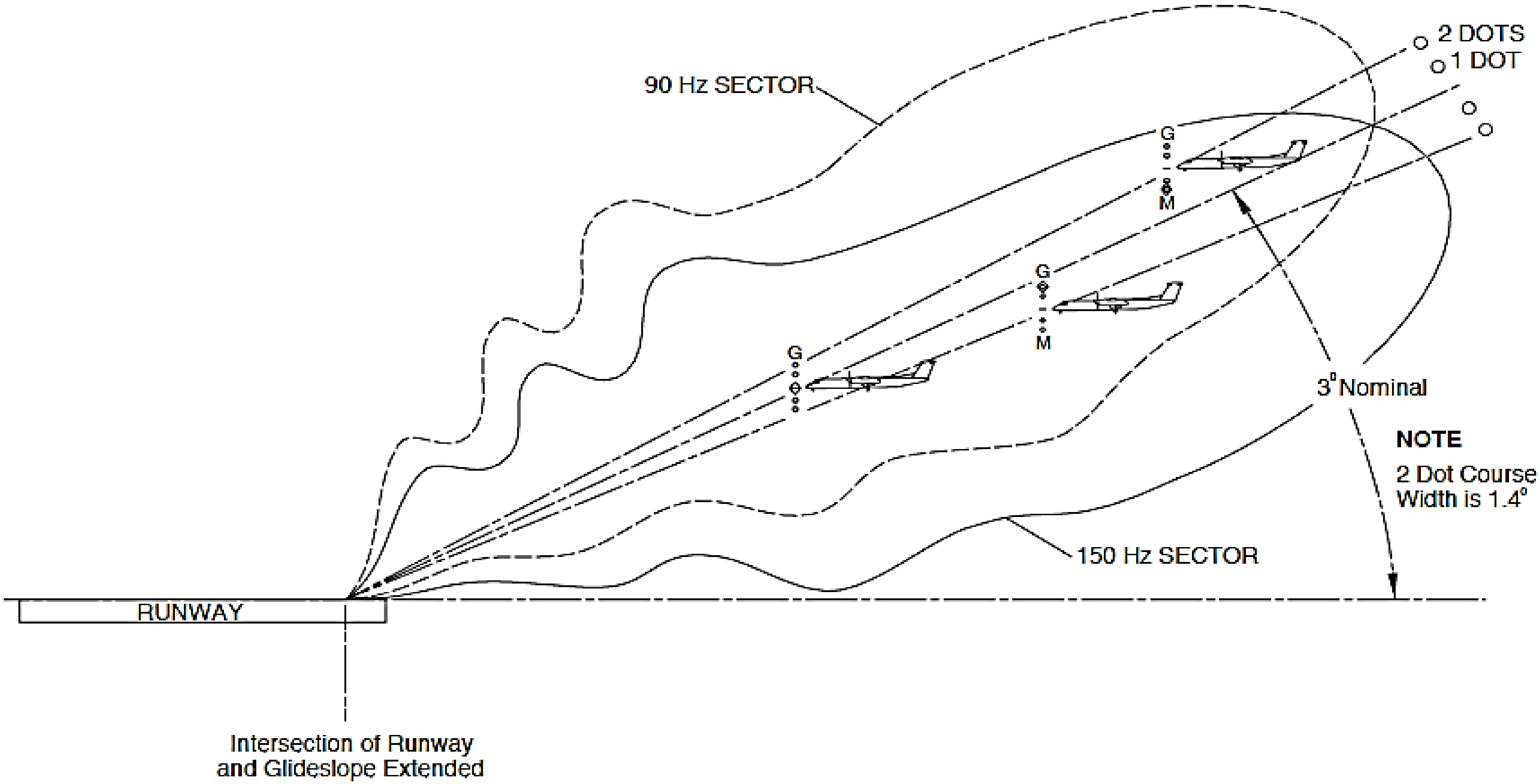


Figure 34- 193 GLIDESLOPE INDICATION PRINCIPLES

Figure 34- 193 GLIDESLOPE INDICATION PRINCIPLES

Glideslope: The glideslope system uses a transmitter and aircraft receiver to give the aircraft approach altitude. It gives a 3 degree glideslope angle from the ground horizontal. The glideslope extends out from the runway for 7 mi (11.3 km).

A directional antenna array located near the approach end of the runway transmits the glideslope signal. The signal has two transmission patterns of Radio Frequency (RF) energy that overlap. The upper transmission pattern contains 90 Hz modulation and the bottom transmitting pattern contains 150Hz modulation. The equal amplitude intersection of the two transmission patterns form the glideslope. A typical glideslope angle is between 2.25 and 3.0 degrees. When the aircraft is on the glideslope, it receives the 90 and 150 Hz signals equal in strength. If the aircraft is flying above the glideslope, the 90Hz signal is relatively stronger than the 150 Hz signal. Deviation bars move down to show that the position of the aircraft is above the glideslope. If the aircraft is flying below the glideslope, the 150 Hz signal is relatively stronger than the 90 Hz signal. The deviation bar moves up. Typically, full−scale deflection may vary between 0.4 to 0.75 degrees depending on the installation.

The distance between the outer marker and the runway is 4 to 7 mi (6.4 to 11.3km) The distance between the middle marker and the runway changes between 3250 to 3750 ft. (991 to 1143m). The glideslope intersects the runway directly opposite to the transmitter. It is between 750 and 1250 ft. (229 and 381 m) beyond the threshold.

The Back course may have one or more false glideslope. They are unusable and the flight crew must ignore their indications.

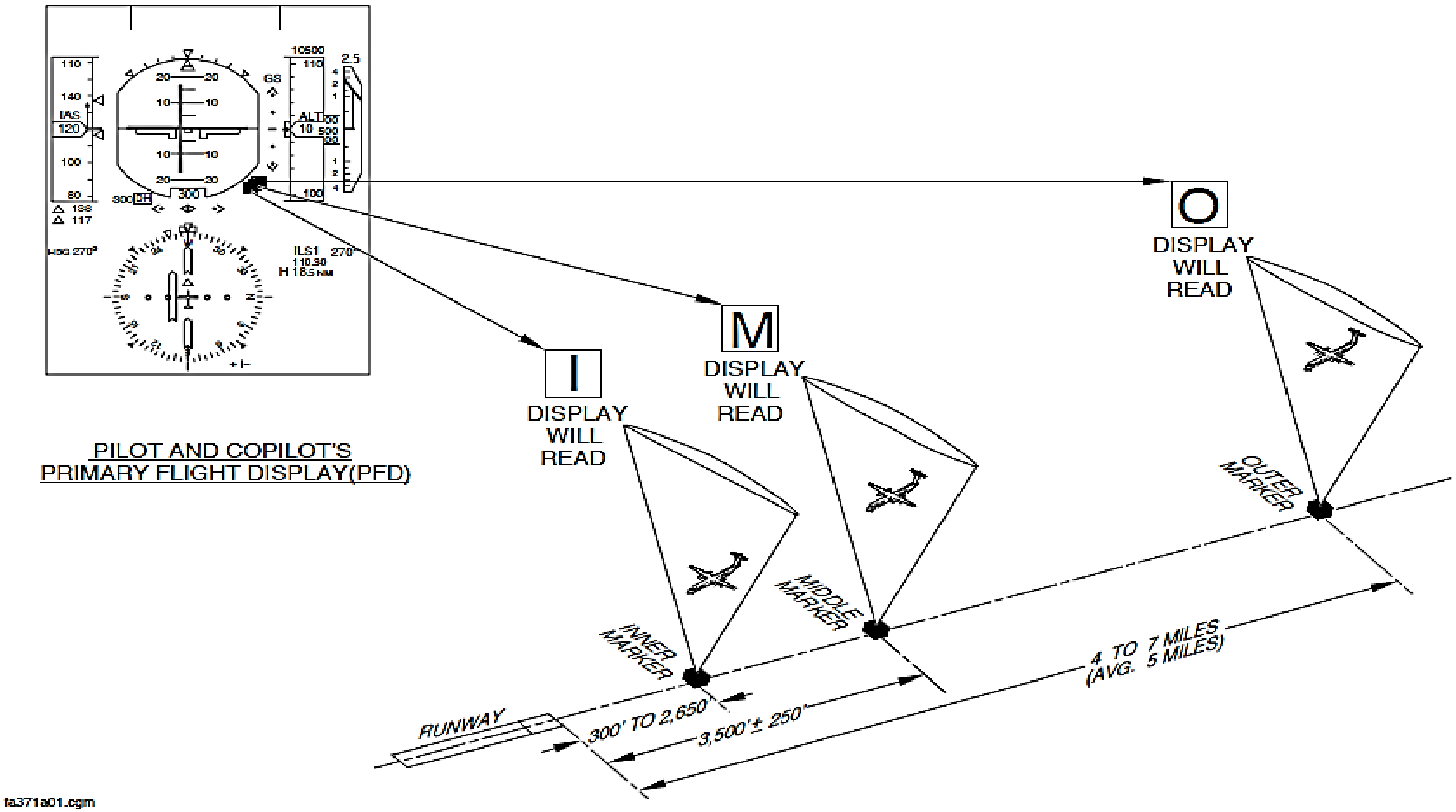


Figure 34- 194 MARKER BEACON INDICATION PRINCIPLES

Refer Figure 34- 194 MARKER BEACON INDICATION **PRINCIPLES**

Marker Beacons: The Marker Beacons are navigation aids that transmit a low power signal to mark an area above themselves. They are located along the centerline of the runway. The markers are designated from the runway as:

- Outer, 4 to 7 mi (6.4 to 11.3 km)
- Middle, 3250 to 3750 ft. (991 to 1143 m)
- Inner, 300 to 2650 ft. (91.4 to 807.7 m).

A large number of Instrument Landing Systems (ILS) have middle and outer marker beacons. The inner marker is rarely used.

Each marker has the following identifying signal and audio tone:

- Outer marker, two dashes modulated by 400 Hz
- Middle marker, alternating dot and dashes modulated by 1300Hz
- Inner marker, six dots per second modulated by 3000 Hz.

As the aircraft approaches the runway, each successive marker has higher tone. The length of time that the aircraft receives the marker beacon information varies with the speed and altitude of the aircraft. The reception time of the middle marker is about half that of the outer marker. The reception time of the inner marker is about half that of the middle marker.

NOTE

It is short enough to be missed if the flight crew is not alert.

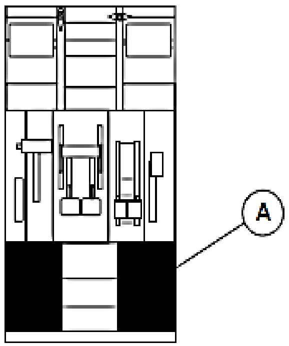
The Marker Beacon outputs the information that follows:

MARKER	AUDIO TONE (HZ)	INDICATION
Outer	400	Blue O
Middle	1300	Amber M
Inner or Airway	3000	White I

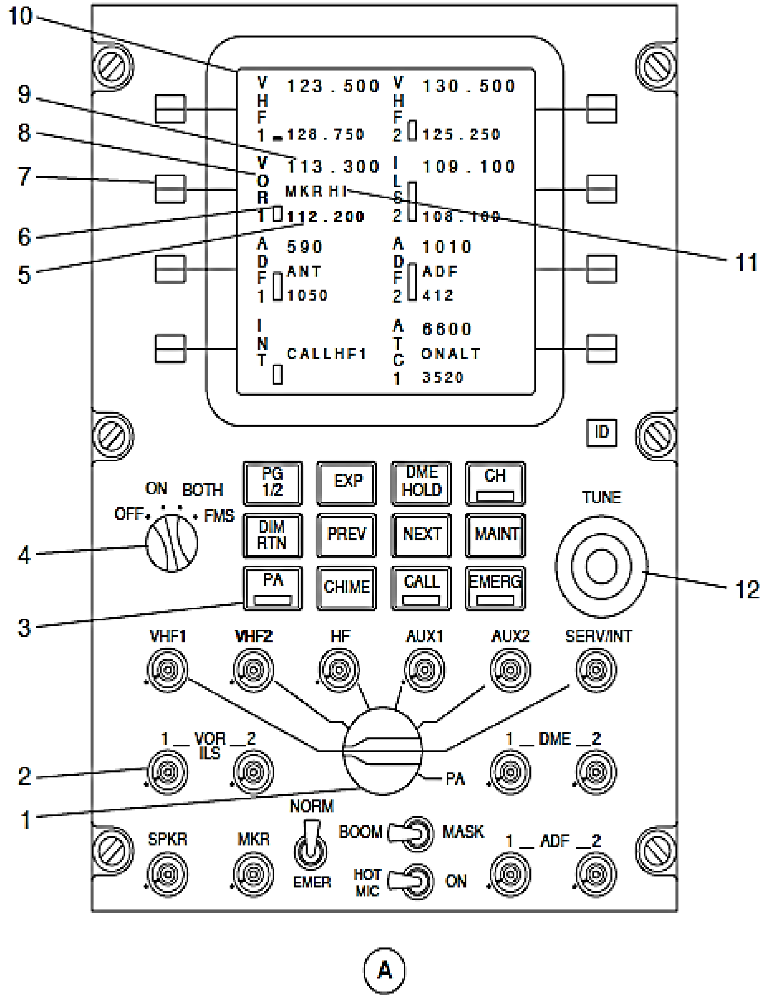
The Electronic Instrument System (EIS) shows a positive visual identification when a marker beacon signal is received and the Audio and Radio Management System (ARMS) gives an audio indication.

The VHF Navigation Receivers (VHF NAV1, VHF NAV2) receive VOR and localizer signals in the same frequency band. Their range is from 108.000 to 117.950MHz in 50 kHz increments. This gives 160 VOR channels and 40 localizer channels. As internationally accepted, localizer channels are allocated the odd–tenth frequencies between 108.00 and 111.950 MHz. For example, 108.100 MHz is a localizer frequency. These channels automatically tune a paired UHF glideslope frequency. The glideslope portion of the VHF Navigation Receivers (VHF NAV1, VHF NAV2) receives glideslope signals in the frequency range 329.15 to 335.00MHz spaced 150 kHz apart. There are 40 localizer and glideslope pairs. The VOR channels in the 108 to 111.95 MHz band are assigned "even tenth" frequencies. For example, 108.000 MHz is a VOR frequency.

When a localizer frequency selection is made, the navigation receiver automatically tunes its paired glideslope frequency. he marker beacon receiver demodulates the amplitude modulated, 75 MHz marker beacon signal.



CENTRE CONSOLE



LEGEND

1. Microphone/Interphone Selector Switch.
2. VOR Potentiometer / Push-Button Switches.
3. Keys.
4. Rotary Switch.
5. Preset Frequency.
6. Level Adjust Bar Graph.
7. Side Key.
8. Label.
9. Active Frequency.
10. Display Area.
11. VHF Navigation Mode.
12. Tuning Knob.

fs713a01.cgm

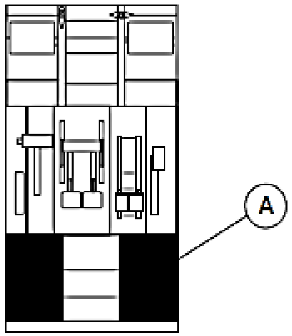
Figure - ARCDU, VHF NAVIGATION CONTROLS AND INDICATIONS

Refer Figure - ARCDU, VHF NAVIGATION CONTROLS AND INDICATIONS

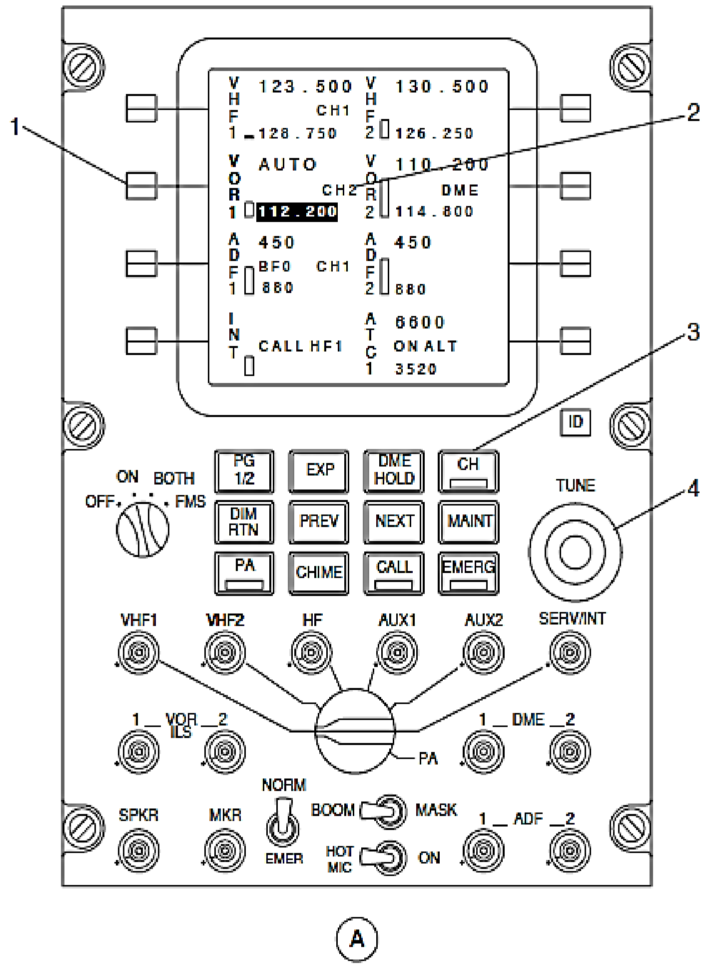
The Audio and Radio Control Display Units (ARCDU1, ARCDU2) are used to set the VHF navigation modes and to manually tune the VHF Navigation Receivers (VHF NAV1, VHF NAV2) frequencies. The Flight Management System (FMS) will also tune the navigation receivers.

The ARCDU selections control the VHF Navigation frequencies and the VOR or ILS mode to be used.

SELECTION	DESCRIPTION
OFF	The ARCDU is not energized
ON	The ARCDU is energized and it controls and tunes its related radio system
BOTH	The ARCDU is energized and it controls and tunes its related and oppositeradio systems
FMS	The FMS controls and tunes its related and opposite radio systems



CENTRE CONSOLE



LEGEND

1. Side Key.
2. Channel Memory Annunciation.
3. Channel Key.
4. Tuning Knob.

Figure - ARCDU, VHF NAVIGATION FREQUENCY CONTROL

Refer Figure - ARCDU, VHF NAVIGATION FREQUENCY CONTROL

The side key adjacent to the VHF navigation display area is pushed to highlight the preset indication. It is pushed again to make the preset frequency active and the frequency that was active, the new preset frequency. If a VOR frequency is set, the display area will show a VOR1 or VOR2 label. If an ILS frequency is set, the display area will show an ILS1 or ILS2 label.

When the side key adjacent to the VHF navigation display area is pushed, the preset indication in the display area changes to reverse video to show the selection. The TUNE double rotary knobs located at the middle right part of the ARCDU are turned to change the preset indication.

The side key adjacent to the VHF navigation display area is pushed and then the CH key to set the channel memory mode for VHF navigation. The VHF navigation channel memory mode lets the preset frequency area show one of eight different programmed channel.

If the preset frequency shown relates to a channel memory, a channel memory annunciation is shown on the second data line that relates to the channel number.

The TUNE double rotary knobs are turned to change the preset frequency indication from eight programmed channels. The channel memory annunciation changes with the preset (channel memory) frequency indication.

NOTE
If the preset frequency does not relate to a channel memory when the mode is set, no channel memory annunciation is shown. A TUNE double rotary knobs selection causes the display area to start with CH1 when turned clockwise or CH8 when turned counterclockwise.

The CH key is pushed again to change from the channel memory mode back to the frequency selection mode. The channel memory annunciator will go out of view.

NOTE
The TUNE double rotary knobs selection can roll over.

Refer Figure - ARCDU, VHF NAVIGATION PARTICULAR PAGE SELECTION

The side key adjacent to the VHF navigation display area is pushed and then the EXP key to show the VHF navigation particular page.

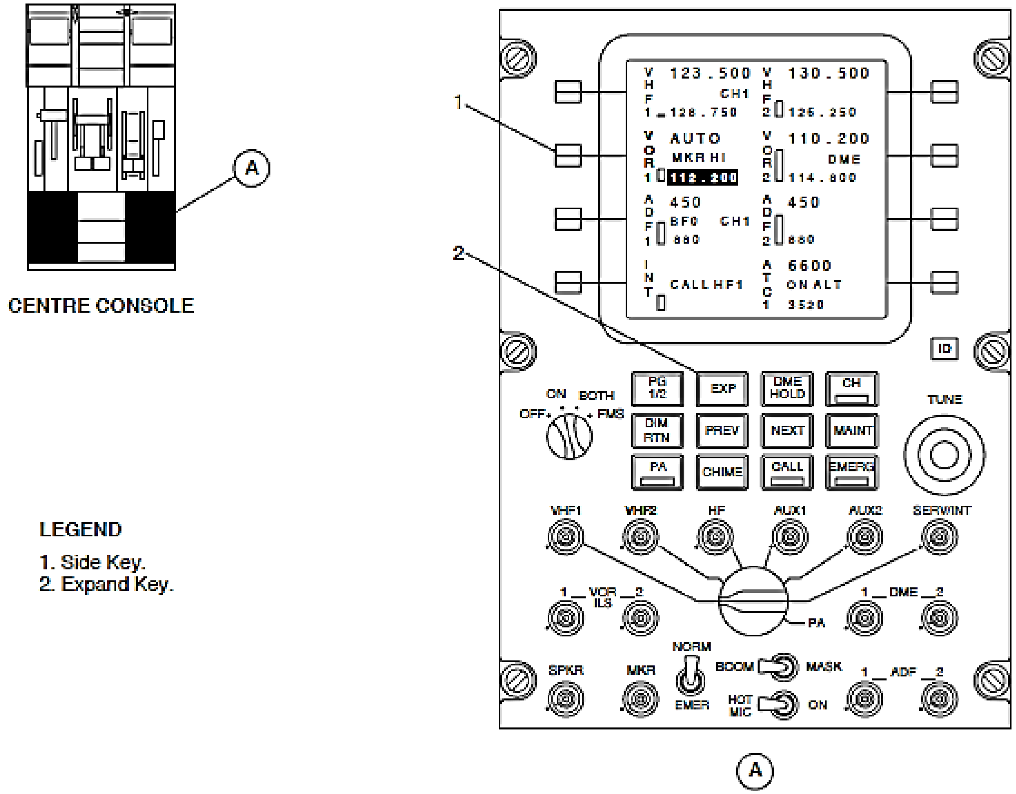


Figure - ARCDU, VHF NAVIGATION PARTICULAR PAGE SELECTION

Refer Figure - ARCDU, VHF NAVIGATION PARTICULAR PAGE

The particular page is used to change the VHF Navigation Receiver (VHF NAV1, VHF NAV2) modes and functions that follow:

- FMS automatic tuning
- Channel programming
- Marker Beacon (MKR) test
- Marker Beacon (MKR) sensitivity.

FMS Automatic Tuning: The side key adjacent to the FMS AUTO TUNE legend is pushed to set the automatic tuning mode. When the Flight Management System (FMS) is automatically tuning, the indication in the display area changes to reverse video to show the automatic tuning mode selection. The FMS AUTOTUNE legend changes from a white letters to black letters on a green background. The VOR active frequency indication changes to show a green AUTO label legend. A new navigation frequency selection causes the FMS automatic tune mode to automatically stop.

Channel Programming: Eight more preset channels are set on a different page for VHF navigation tuning.

Marker Beacon (MKR) Test: The side key adjacent to the MKR TEST legend is pushed to start a three−second test of the marker receiver. The two Primary Flight Displays (PFDs) show the Inner (I), Outer (O) and Middle (M) marker indications the marker receiver sounds the three marker audio tones through the Audio Integration System (AIS).

The white MKR TEST legend changes to reverse video during the test.

Marker Beacon (MKR) Sensitivity: The side key adjacent the MKR SENS legend is pushed to toggle between LO and HI sensitivities. The indication changes to reverse video to show the selection

NOTE

When the HI sensitivity is set, the VHF navigation display area will show a green MKR HI indication on the second data line.

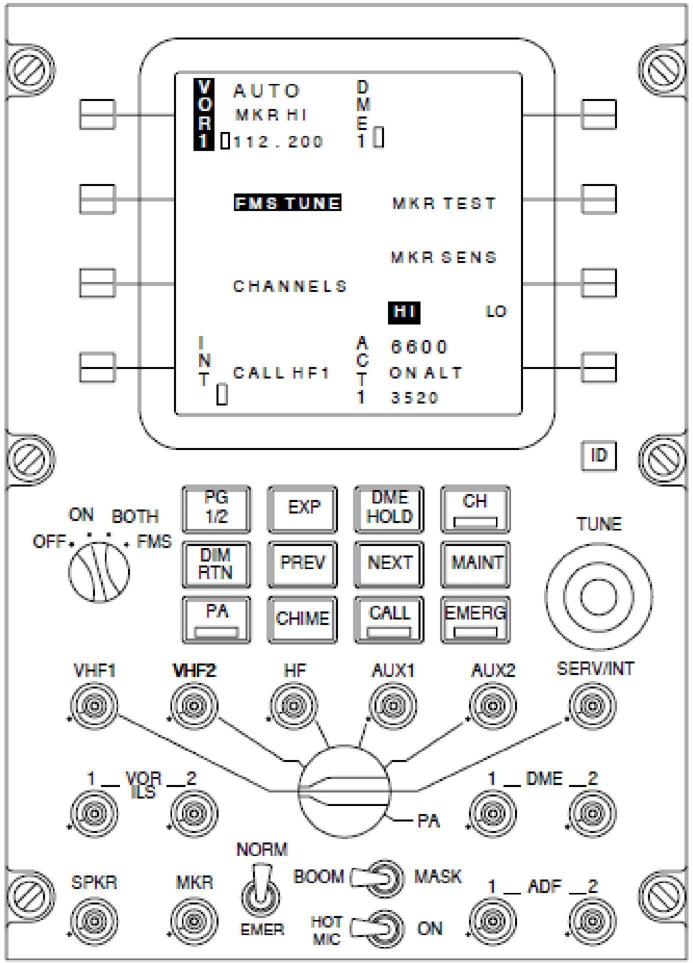


Figure - ARCDU, VHF NAVIGATION PARTICULAR PAGE

Refer Figure - ARCDU, VHF NAVIGATION CHANNELS SELECTION

The side key adjacent to the CHANNELS legend is pushed to show the channel programming page.

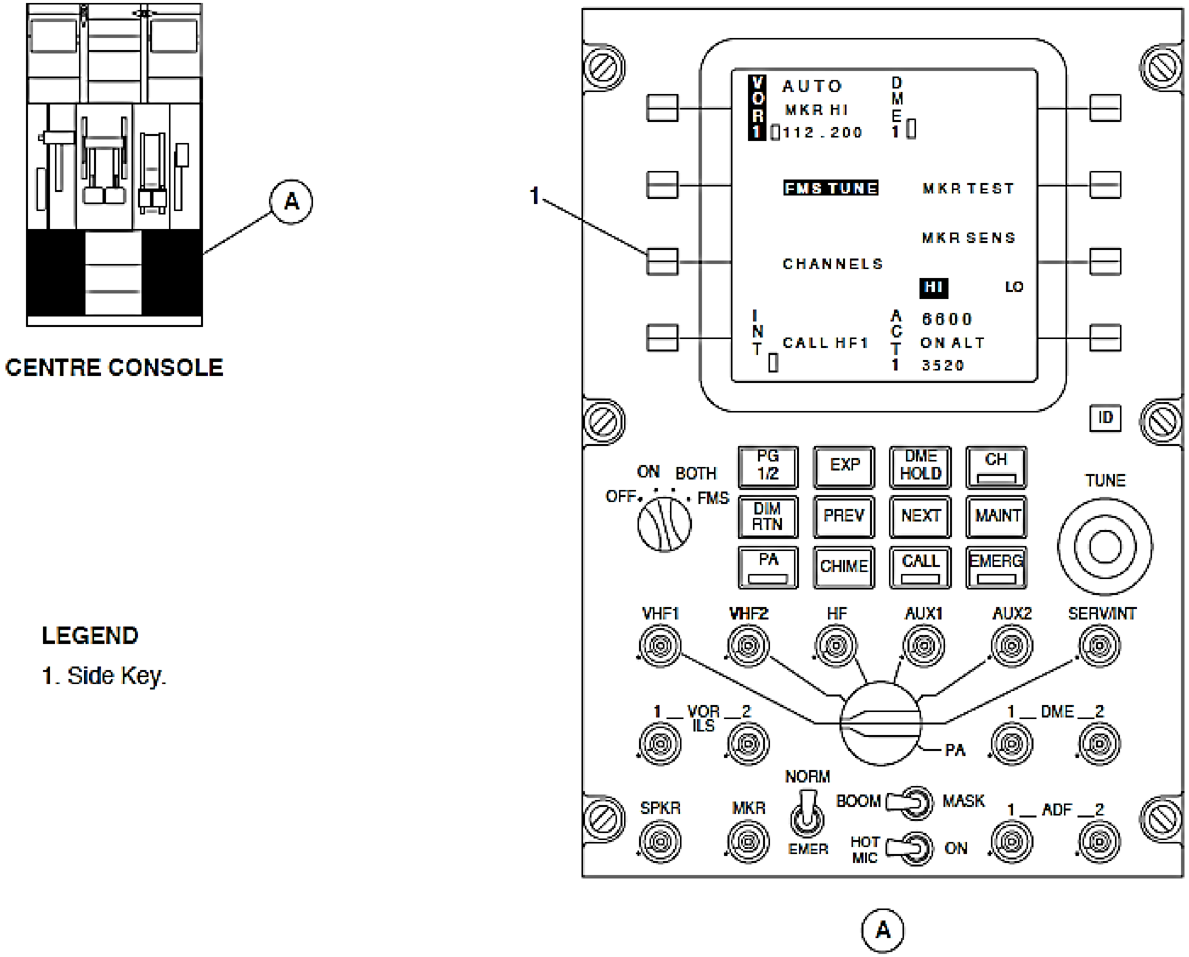


Figure - ARCDU, VHF NAVIGATION CHANNELS SELECTION

Refer Figure - ARCDU, VHF NAVIGATION CHANNELS PAGE

The channel programming page has the indications that follow:

- A index of all radios with channel programming
- Channel memory annunciators (CH1 through CH)
- Programmed frequency values for the radio selection.

The two display areas at the top of the screen show the radios that can be channel programmed. The left display area shows the radios that are related to the pilot's side and the right display area shows the radios that are related to the copilot's side.

One radio is always set and is shown in reverse video.

NOTE

The radio for the related particular page is automatically set. The side key adjacent to display area is pushed to show the channel memory frequencies for a different radio.

Four other display areas each show two channel memory annunciators with their channel memory frequency values that relate to the radio selection.

NOTE

When the ARCDU is set to the ON position, only the related radios can be channel programmed.

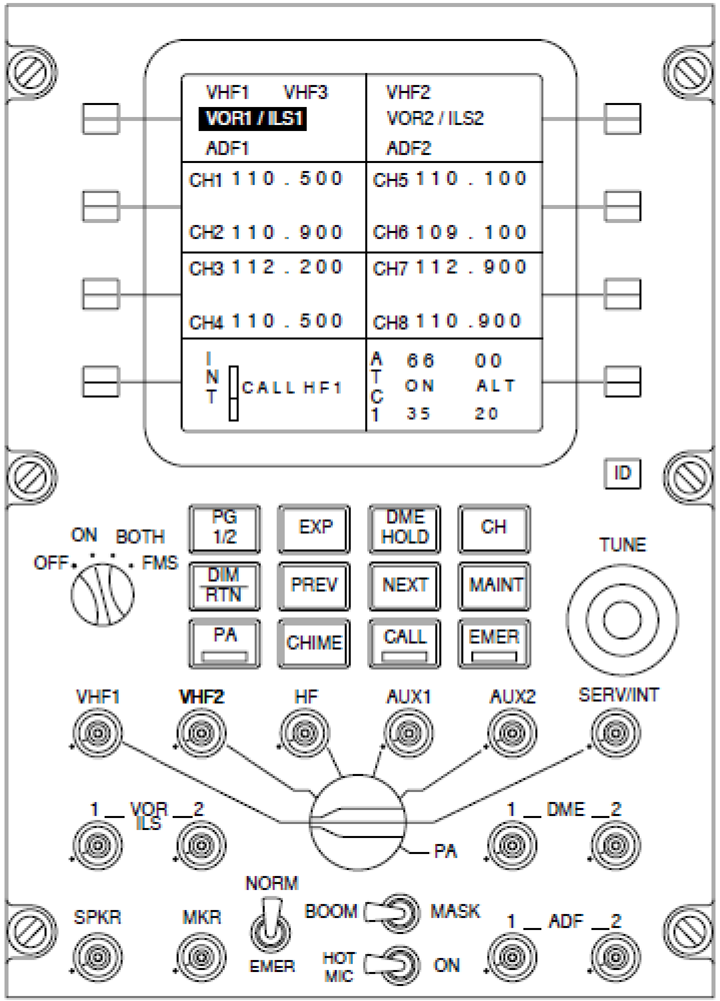


Figure - ARCDU, VHF NAVIGATION CHANNELS PAGE

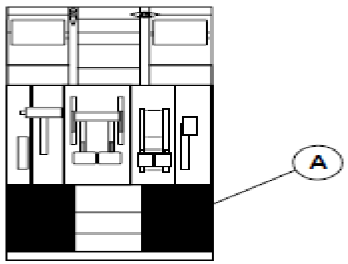
Refer Figure - ARCDU, VHF NAVIGATION CHANNEL MEMORY PROGRAMMING

The side key adjacent to the channel memory display area is pushed to highlight the top memory frequency indication. It is pushed two times to highlight the bottom memory frequency indication.

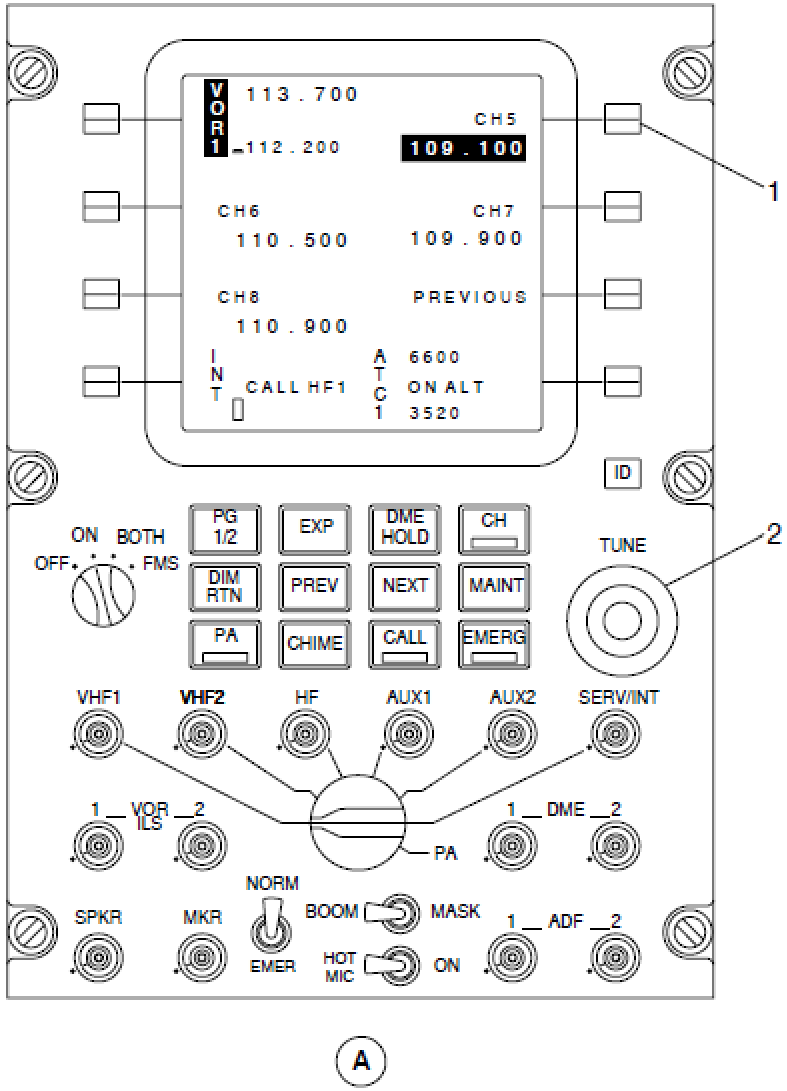
When the side key adjacent to the channel memory display area is pushed, the indication in the display area changes to reverse video to show the selection. The TUNE double rotary knobs located at the middle right part of the ARCDU are turned to change the preset indication.

The new memory frequency is set by any condition that follows:

- A side key is pushed
- Five seconds after a TUNE double rotary knobs selection
- The EXP or PG 1/2 key is pushed to show a different page.



CENTRE CONSOLE



LEGEND
1. Side Key.
2. Tuning Knob.

Figure - ARCDU, VHF NAVIGATION CHANNEL MEMORY PROGRAMMING

Refer Figure - ARCDU, VHF NAVIGATION AUDIO CONTROL

The VOR/ILS Potentiometer push−button switches control the audio parameters that follow:

- On
- Off
- Level.

The top part of the potentiometer is pushed to turn the related audio signal to the headphone on or off. The VOR ILS1, VOR ILS2, or MKR push−button switch is pushed on the ARCDU to make the audio available. When the SPKR pushbutton potentiometer switch is pushed, the VHF Navigation audio is routed to its related flight compartment speaker.

The VHF navigation display area shows the volume level as a vertical bar graph. The height of the bar graph shows the volume selection level. When the audio is set to the off position, the bar graph indication changes to white. It changes to green when the audio is set to the on position.

The VHF navigation audio is available at the observer's headphone jack when the VOR/ILS1, VOR/ILS2 is selected on the observer's audio control panel. An MLS/VOR toggle switch on the headset panel switches the audio to the headset between the VOR/LOC and MLS.

LEGEND

1. Level Adjust Bar Graph.
2. VOR Potentiometer / Push–Button Swithes.
3. Speaker Potentiometer / Push–Button Switch.

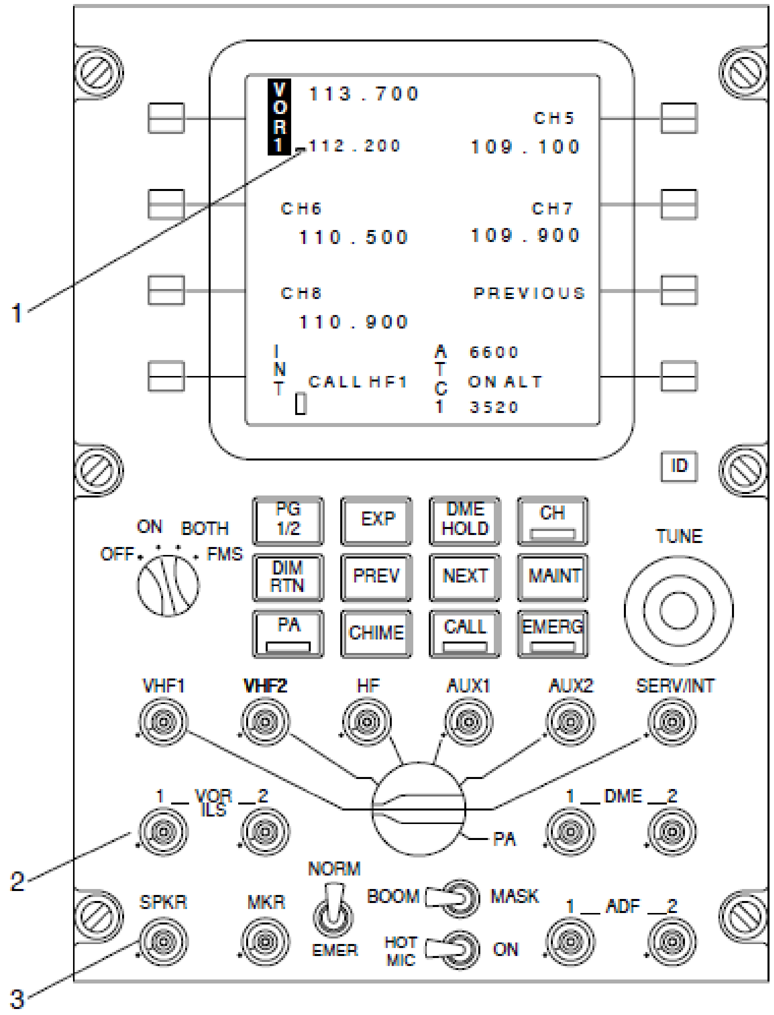


Figure - ARCDU, VHF NAVIGATION AUDIO CONTROL

Refer Figure - EFCP, VOR BEARING SELECTION

When a VOR1 or VOR2 bearing is selected on the EFIS Control Panels (EFCP1, EFCP2), a distinctive pointer comes into view on the Primary Flight Display.

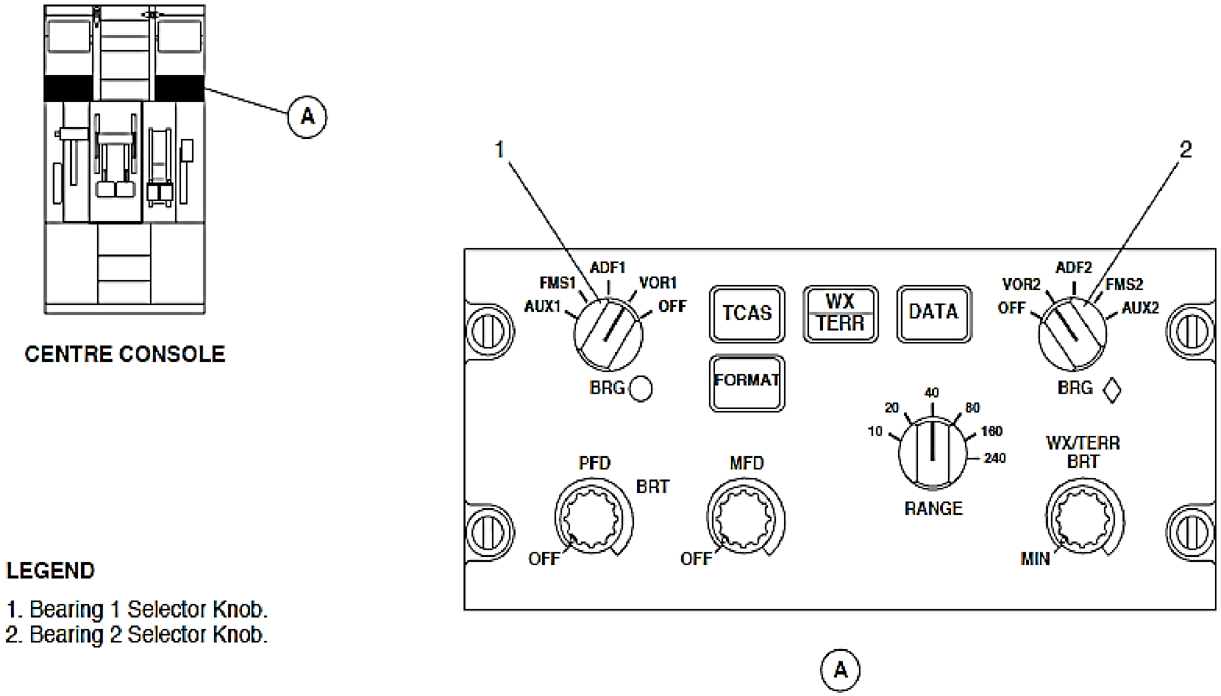


Figure - EFCP, VOR BEARING SELECTION

Refer Figure - EIS, VOR BEARING INDICATION

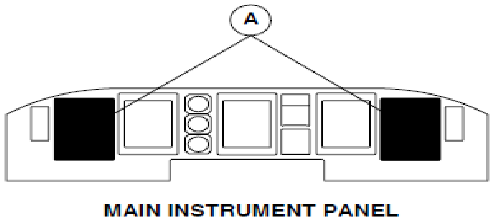
The bearing pointers in the Electronic Horizontal Situation Indicators (EHSI1, EHSI2) show the lateral direction to the navigation aid as follows:

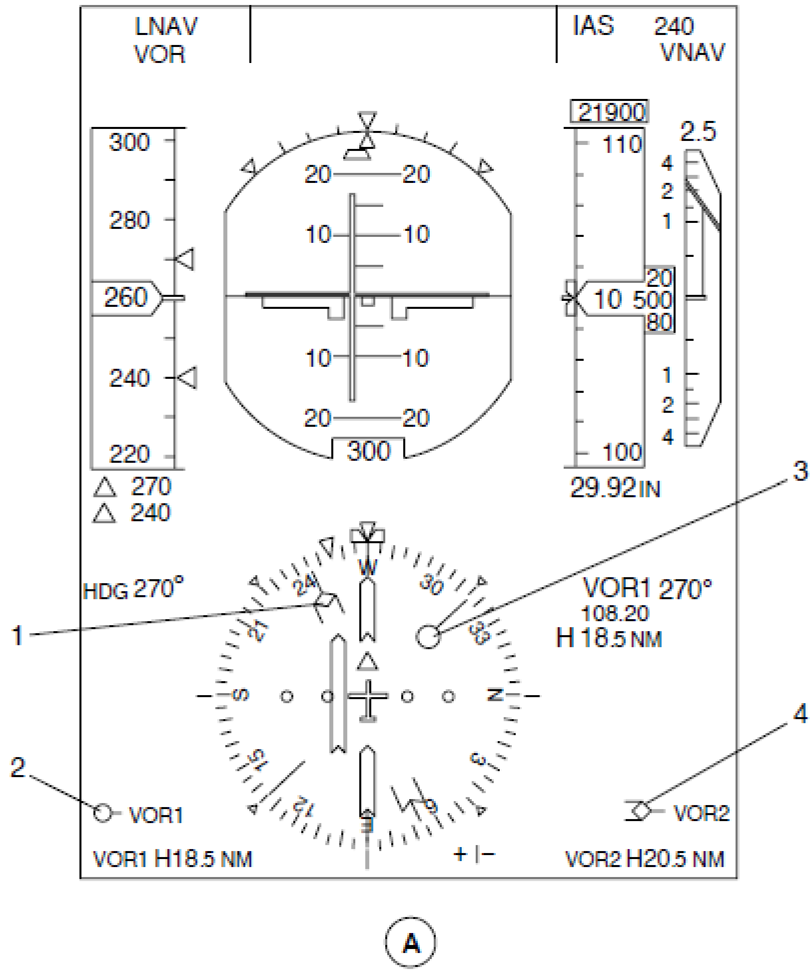
- A white, single bar pointer with a white circle to identifyVOR1
- A green, wide two−bar pointer with a green diamond to identifyVOR2.

The bottom left part of the Primary Flight Displays (PFD1, PFD2) shows a reminder label for VOR1 and the bottom right part shows a reminder label for VOR2. Each reminder label is related to the EFIS Control Panel selection. Each indication has a symbol that is similar to its bearing pointer with aVOR1 or VOR2 legend.

NOTE

The reminder bearing symbol and legend are the same color, white or green.





LEGEND

1. No. 2 Bearing Pointer.
2. No. 1 Bearing Reminder Label.
3. No. 1 Bearing Pointer.
4. No. 2 Bearing Reminder Label.

Figure - EIS, VOR BEARING INDICATION

Refer Figure - EIS, VOR BEARING MALFUNCTION INDICATION

When a VOR signal has malfunctioned, its bearing pointer goes out of view, but its reminder label stays in view to show the bearing pointer selection.

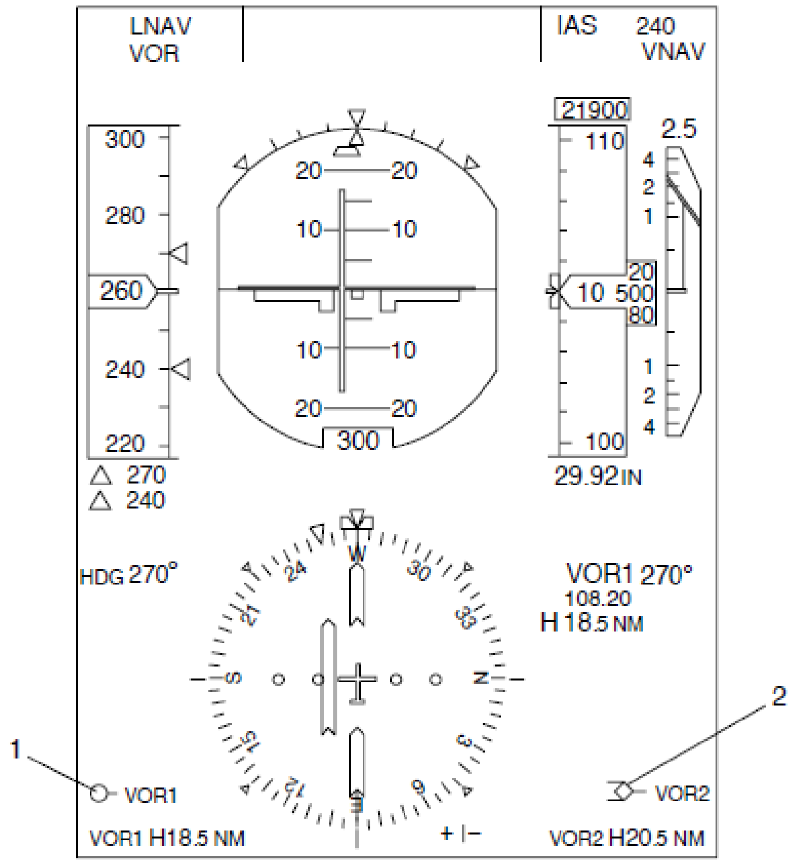
If the pilot's EFIS Control Panel (EFCP1) malfunctions, the bearing pointers will automatically show the information that follows:

- The single bar indicates Automatic Direction Finder (ADF1)
- The double bar indicates Very High Frequency Omni–range (VOR2).

If the copilot's EFIS Control Panel (EFCP2) fails, the bearing pointers will automatically show the information that follows:

- The single bar indicates Very High Frequency Omni–range (VOR1)
- The double bar indicates Automatic Direction Finder (ADF2).

The Audio and Radio Control Display Units (ARCDU1, ARCDU2) monitor the ARINC 429 data bus through the Flight Data Processing System (FDPS) for malfunction of the VHF Navigation system. The ARCDUs show a red FAIL message in the frequency preset area of the VHF navigation display area when a system malfunctions.



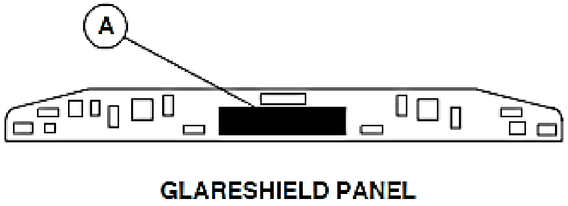
LEGEND

1. No. 1 Bearing Reminder Label.
2. No. 2 Bearing Reminder Label.

Figure - EIS, VOR BEARING MALFUNCTION INDICATION

Refer Figure - AFCP, NAVIGATION SOURCE SELECTION

The NAV SOURCE rotary switch located on the Flight Guidance Control Panel (FGCP) is used to set the VHF navigation source.



LEGEND

1. Course Selection Knob.
2. Navigation Source Selection Knob.

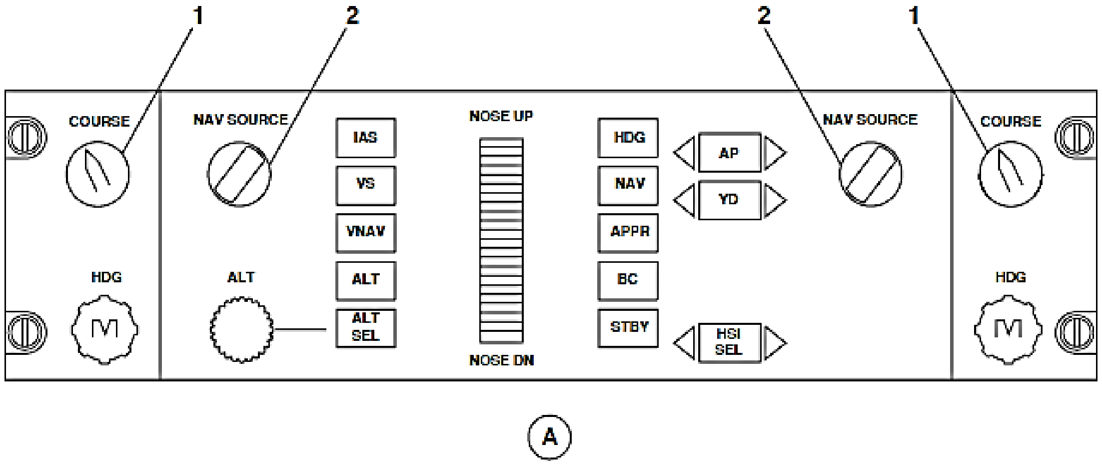
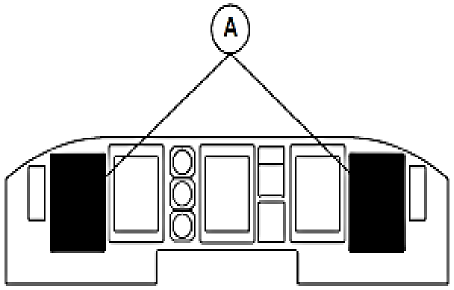


Figure - AFCP, NAVIGATION SOURCE SELECTION



LEGEND

1. Selected Course Pointer.
2. Course Deviation Scale & Bar.
3. Aircraft Symbol.
4. Heading Scale.
5. Navigation Source Annunciation.
6. Selected Course Digital Value.
7. VHF Navigation Frequency.
8. TO/FROM Pointer.

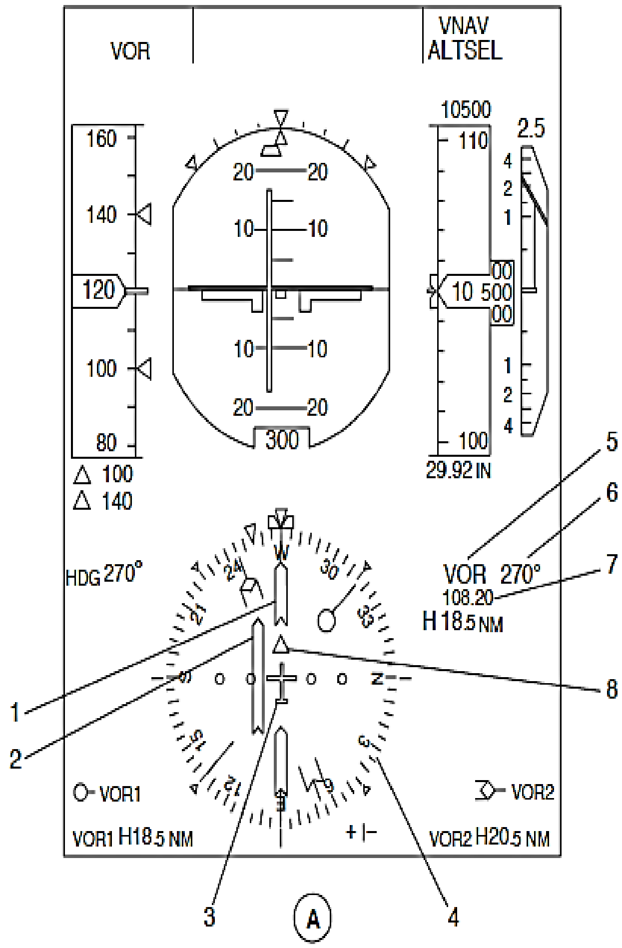


Figure - PFD, VOR COURSE INDICATION

Refer Figure - PFD, VOR COURSE INDICATION

The Primary Flight Display (PFD) shows the navigation source selection in the upper right part of the Electronic Horizontal Situation Indicators (EHSI1, EHSI2). The PFDs show the navigation source annunciations that follow:

- VOR1, VOR2
- ILS1, ILS2
- FMS1, FMS2
- MLS1, MLS2.

The VOR or ILS indication depends on the VHF Navigation frequency selection. A VOR frequency selection will cause the EHSI to show VOR deviation.

The Primary Flight Display will show the VOR parameters that follow:

- Navigation Source Annunciation
- VHF Navigation Frequency
- Selected Course Digital Value
- Selected Course Pointer
- Course Deviation Scale and Bar
- TO/FROM pointer
- Course Deviation Failure Flag.

Navigation Source Annunciation: The navigation source annunciation shows which navigation source is selected by the related NAV SOURCE rotary switch on the Flight Guidance Control Panel (FGCP). When the related source is selected, the annunciation is shown in cyan characters. If an opposite navigation source is selected, the annunciation is shown in yellow fonts.

If the section data has malfunctioned, four white dashes replace the source annunciation.

VHF Navigation Frequency: The frequency of the navigation station is shown in cyan numbers from 108.00 to 117.95.

Selected Course Digital Value: The selected course is controlled by its related COURSE knob on the Flight Guidance Control Panel (FGCP). The selected course digital value is shown by 3 cyan numbers from 1 through 360. A degree symbol is shown in white adjacent to the digital value.

If the selection data has malfunctioned, three white dashes replace the source annunciation.

Selected Course Pointer: The COURSE knob is turned to set the VOR course. It turns a cyan pointer around the 360 degree heading scale to show the selected course and supply the navigation receiver with the desired course. Each COURSE knob controls its related course pointer.

When no heading data is received or not correct, the selected course pointer goes out of view.

Course Deviation Scale and Bar: The course deviation scale and bar is part of the course pointer. It shows lateral deviations from the selected VOR course. The course deviation scale and bar moves with the course pointer and the bar moves independently left or right of the course pointer.

The bar is shown to the right of the aircraft symbol when the aircraft's flying to the left of the desired course and is shown to the left of the aircraft symbol when the aircraft is flying to the right of the desired course. The lateral deviation bar centers in the course pointer when the aircraft is on course. Two dots of deviation shows that the aircraft is 10 degrees from the desired course.

TO/FROM pointer: The TO/FROM pointer symbol is part of the course pointer indication. It turns with the course pointer to show the direction of navigation relative to a ground VOR transmitter. The TO/FROM pointer symbol is also shown in cyan.

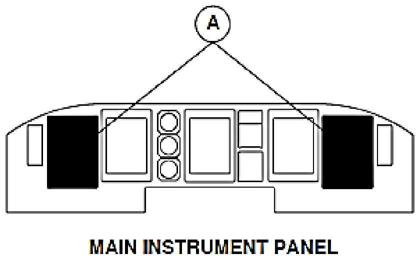
The conditions that follows cause the TO/FROM pointer symbol to go out of view:

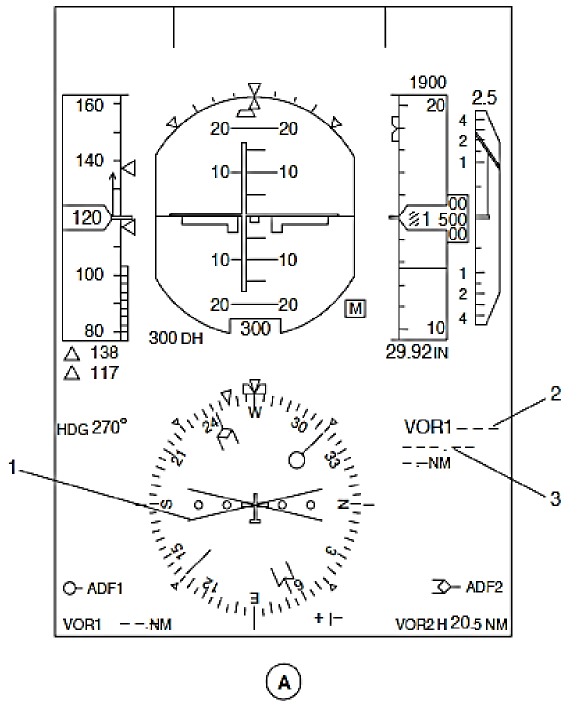
- The VOR ground station is out of range
- Navigation receiver malfunctions
- No VOR course data is available
- No VOR navigation source selection data is available.

Refer Figure - PFD, VOR MALFUNCTION INDICATION

Course Deviation Failure Flag: The course deviation failure flag is a red cross shown with the course deviation scale. The conditions that cause the course deviation failure flag to come in view are as follows:

- The VOR ground station is out of range
- Navigation Receiver malfunctions
- No VOR course data available
- No VOR navigation source selection data available.





LEGEND

1. Course Deviation Failure Flag.
2. Selected Course Fail Annunciation.
3. VOR Navigation Frequency Fail Annunciation.

Figure - PFD, VOR MALFUNCTION INDICATION

Refer Figure - ESCP, NAVIGATION PAGE SELECTION

The MFD1 and MFD2 reversion switches located on the ESID Control Panel (ESCP) are used to set the navigation page. The navigation page comes into view in the map mode format.

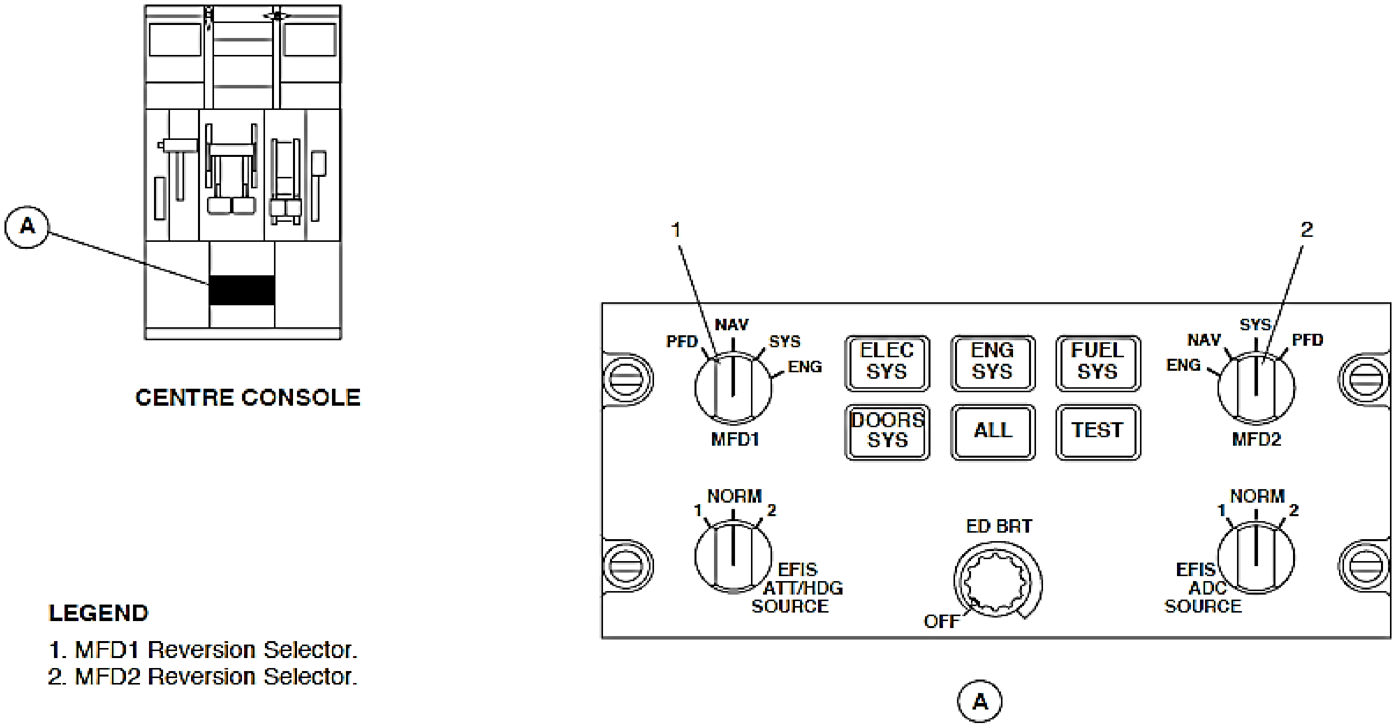
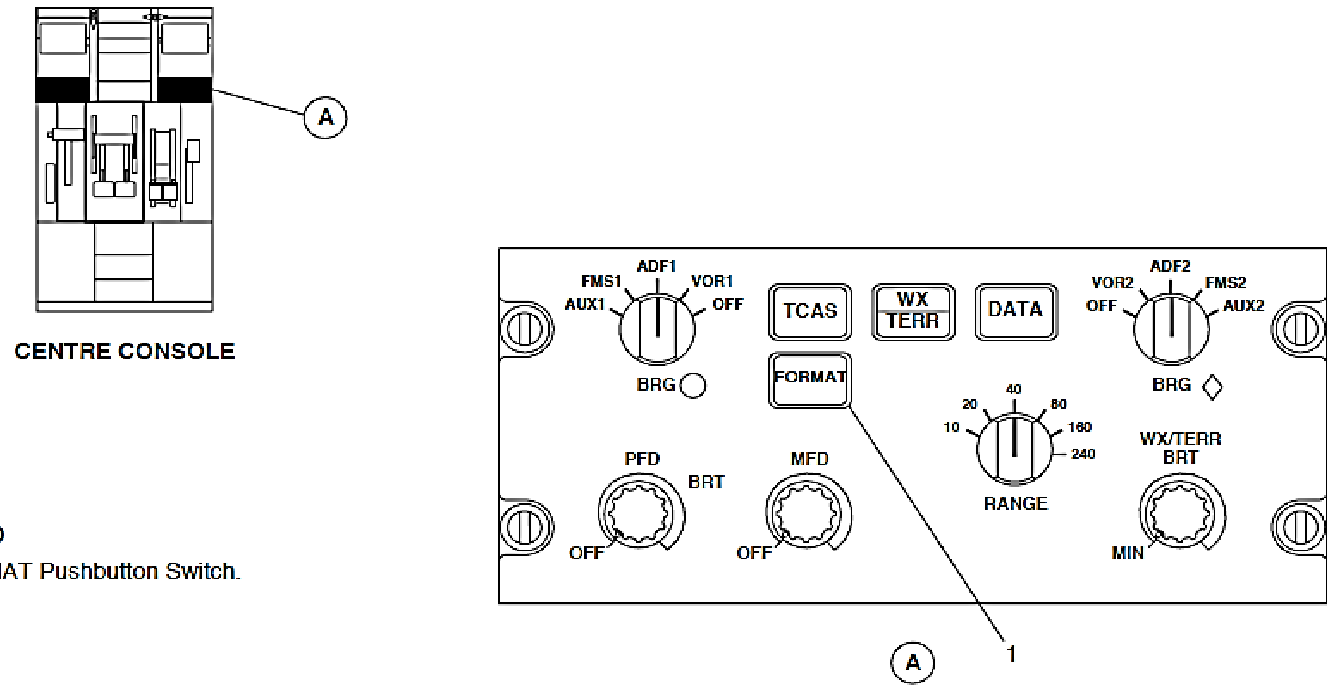


Figure - ESCP, NAVIGATION PAGE SELECTION

Refer Figure - EFCP, ARC FORMAT SELECTION

The FORMAT pushbutton switch on EFCP is pushed to change to the navigation page indication from a map (FMS) to arc (VOR/ILS) indication independently of the PFD. It is pushed again to show the map indication.



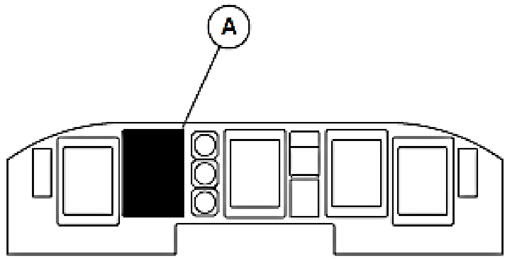
LEGEND

1. FORMAT Pushbutton Switch.

Figure - EFCP, ARC FORMAT SELECTION

Refer Figure - NAVIGATION PAGE, ARC FORMAT VOR INDICATION

The navigation page is shown on the upper part of MFD1 and permanent data is shown on the bottom. The arc mode shows the same VOR navigation parameters as the Primary Flight Display (PFD1, PFD2) indication on a 90 degree heading arc.



MAIN INSTRUMENT PANEL

LEGEND

1. Aircraft Symbol.
2. Heading Scale in Arc Format.
3. Navigation Source Annunciation.
4. Selected Course Digital Value.
5. Radio Navigation Frequency Identifier.
6. Selected Course Pointer.
7. TO/FROM Pointer.
8. Course Deviation Scale and Bar.
9. Permanent System Data Area.

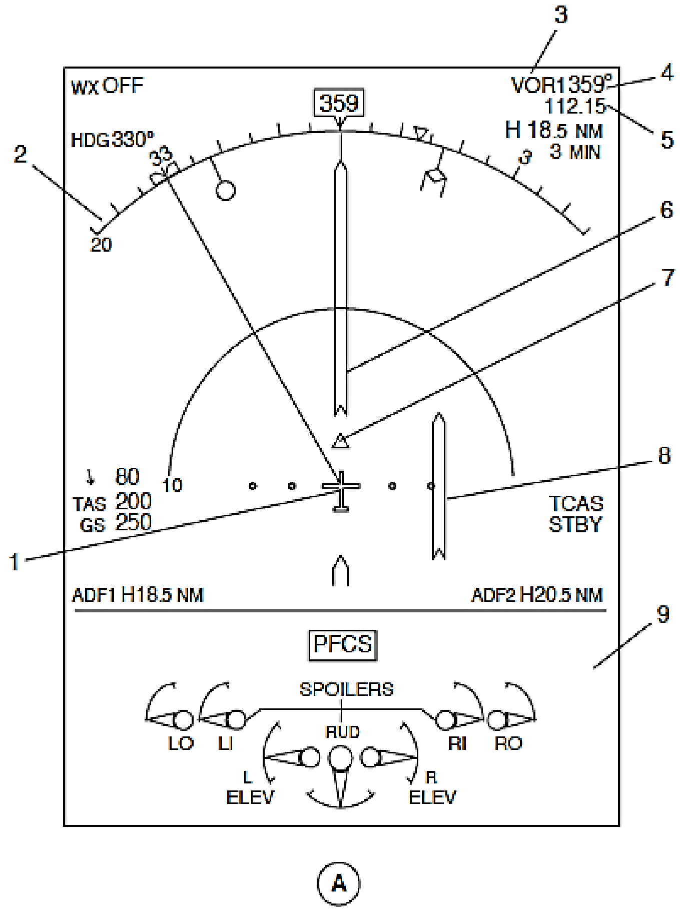
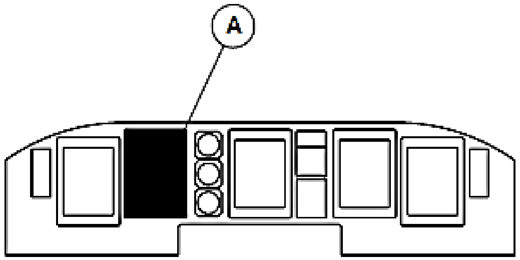


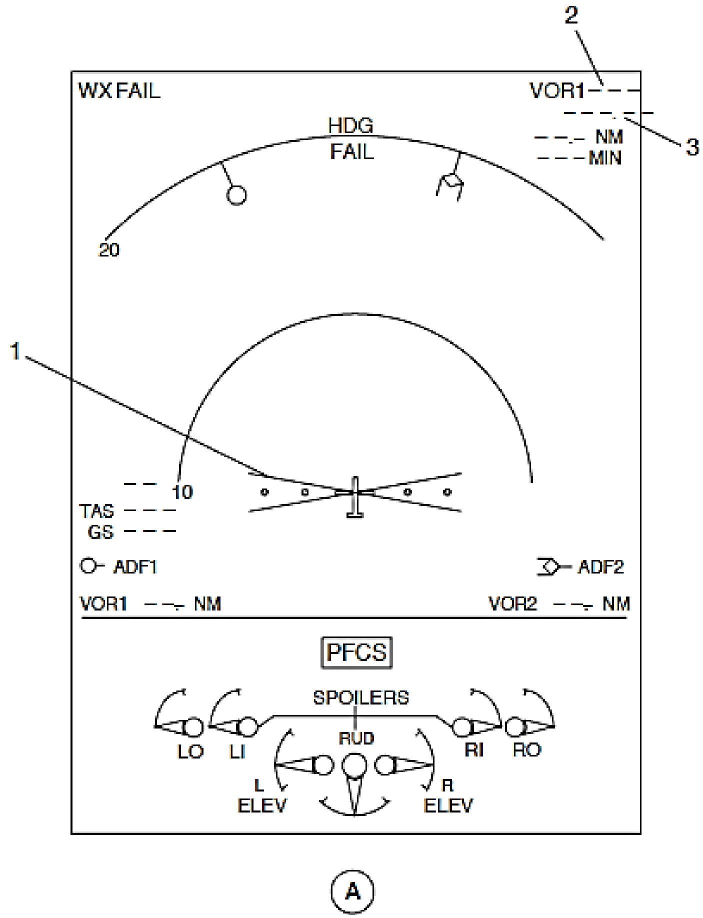
Figure - NAVIGATION PAGE, ARC FORMAT VOR INDICATION

Refer Figure - NAVIGATION PAGE, ARC FORMAT VOR MALFUNCTION INDICATION

The navigation page, arc format mode shows a course deviation failure flag for the same conditions as for the Primary Flight Displays (PFD1, PFD2). It shows a red cross on top of the course deviation scale.



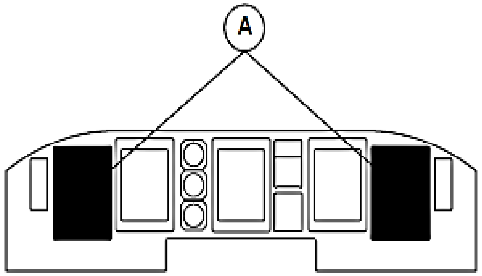
MAIN INSTRUMENT PANEL



LEGEND

1. Course Deviation Failure Flag.
2. Selected Course Fail Annunciation.
3. VHF Navigation Frequency Fail Annunciation.

Figure - NAVIGATION PAGE, ARC FORMAT VOR MALFUNCTION INDICATION



MAIN INSTRUMENT PANEL

LEGEND

1. Selected Course Pointer.
2. Expanded Localizer Pointer.
3. Glide Slope Scale.
4. Vertical Deviation Source.
5. Vertical Deviation Pointer.
6. Marker Beacon Annunciator.
7. Expanded Localizer Scale.
8. Navigation Source Annunciation.
9. Selected Course Digital Value.
10. VHF Navigation Frequency Identifier.
11. Course Deviation Scale.
12. Aircraft Symbol.
13. Heading Scale.
14. Course Deviation Bar.

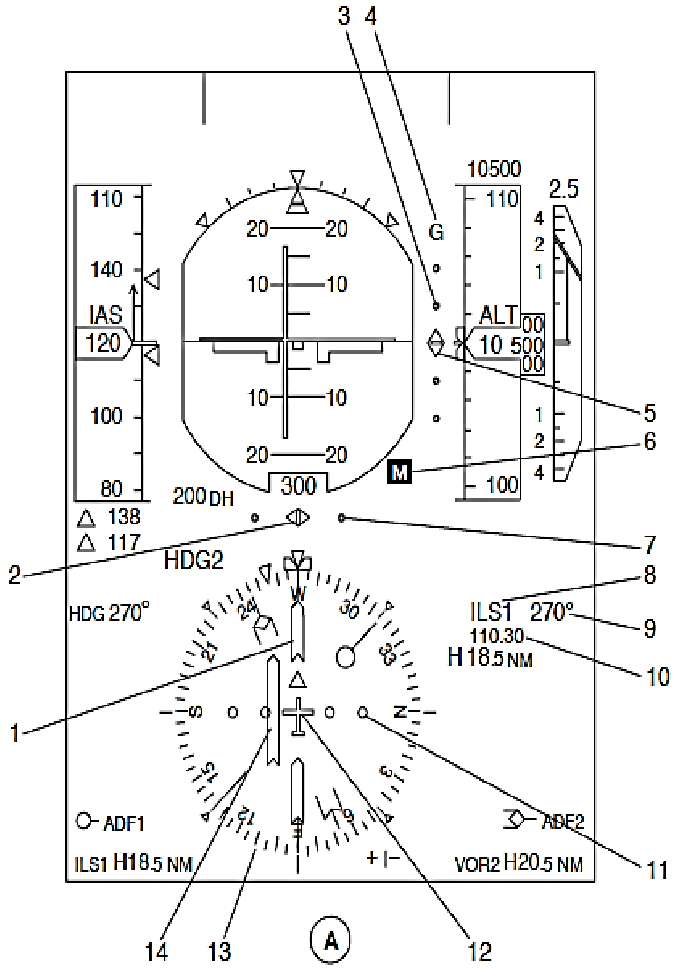


Figure - PFD, ILS COURSE INDICATION

Refer Figure - PFD, ILS COURSE INDICATION

An ILS frequency selection will cause the related PFD to show the ILS parameters that follow:

- Localizer deviation
- Glideslope deviation
- Marker beacon passage.

Localizer Deviation: The Primary Flight Display will show the localizer parameters that follow:

- Navigation Source Annunciation
- VHF Navigation Frequency
- Selected Course Digital Value
- Selected Course Pointer
- Course Deviation Scale and Bar
- Course Deviation Failure Flag
- Expanded Localizer.

Navigation Source Annunciation: The navigation source annunciation shows which navigation source is selected by the related NAV SOURCE rotary switch on the Flight Guidance Control Panel (FGCP). When the related source is selected, the annunciation is shown in cyan characters. If an opposite navigation source is selected, the annunciation is shown in yellow fonts.

If the section data is not correct, four white dashes replace the source annunciation.

VHF Navigation Frequency: The frequency of the navigation station is shown in cyan numbers from 108.00 to 117.95.

Selected Course Digital Value: The selected course is controlled by its related COURSE knob on the Flight Guidance Control Panel (FGCP). The selected course digital value is shown by 3 numbers from 1 through 360.

A degree symbol is shown in white after the digital value.

If the section data has malfunctioned, three white dashes replace the source annunciation.

Selected Course Pointer: The COURSE knob is turned to set the inbound or back course. It turns a cyan pointer around the 360 degree heading scale to show the selected course. Each COURSE knob controls its related course pointer.

NOTE

When no heading data is received or if it is not correct, the localizer selected course pointer does not go out of view.

Course Deviation Scale and Bar: The course deviation scale and bar is part of the course pointer. It shows lateral deviations from the localizer. The course deviation scale and bar moves with the course pointer and the bar moves independently left or right of the course pointer.

The bar is shown to the right of the aircraft symbol when the aircraft is flying to the left of the localizer. The bar is shown to the left of the aircraft symbol when the aircraft is flying to the right of the localizer. The lateral deviation bar centers in the course pointer when the aircraft is on the localizer. Two dots of deviation show the aircraft 2 1/2 degrees from the desired course.

Expanded Localizer: An expanded localizer scale is shown below the Electronic Attitude Direction Indicator (EADI). It is used during a Category II precision approach. The expanded localizer scale is 6 times as sensitive as the localizer display.

If the AGL altitude is less than 1200 ft. (366 m) and the localizer pointer goes off the scale, the pointer and scale will flash for 5 seconds and then go steady. The Flight Mode Annunciators (FMA) will show an LOC EXCESSDEV message.

Glideslope Deviation: The PFD will show the glideslope parameters that follow:

- Vertical deviation pointer and scale
- Vertical Deviation Source.

Vertical Deviation Pointer and Scale: The Primary Flight Displays (PFD1, PFD2) have a filled cyan pointer and white scale to show deviation from the glideslope. The vertical deviation pointer and scale is shown to the right of the attitude sphere.

If the AGL altitude is less than 1200 ft. (366 m) and the glideslope pointer goes off the scale, the pointer and scale will flash for 5 seconds and then go steady. The Flight Mode Annunciators (FMA) will show a GS EXCESSDEV message.

Vertical Deviation Source: The ILS vertical deviation source indication shows a cyan G above the scale to identify the scale.

Marker Beacon: The marker beacon reception is automatic and its indications always available. It is not a function of an ILS frequency selection.

The PFDs show the Marker Beacon (MB) passage near the attitude sphere. An O, M, or I indication is shown in as a black letter on a blue, yellow, or white rectangle background.

NOTE

When the marker beacon receiver is tested, the PFD shows all marker beacon indications at the same time.

Refer Figure - PFD, ILS MISMATCH INDICATIONS

The Flight Data Processing System (FDPS) senses signal differences between the ILS1 and the ILS2. If the difference is excessive, a yellow LOC MISMATCH message in the centre row of the centre column. The indications flash for five seconds when they come into view and then go steady.

A GS MISMATCH message is shown at the same location as the LOC MISMATCH message. If a glideslope and localizer mismatch condition occur at the same time, the glideslope mismatch message is shown. It is a more important mismatch message.

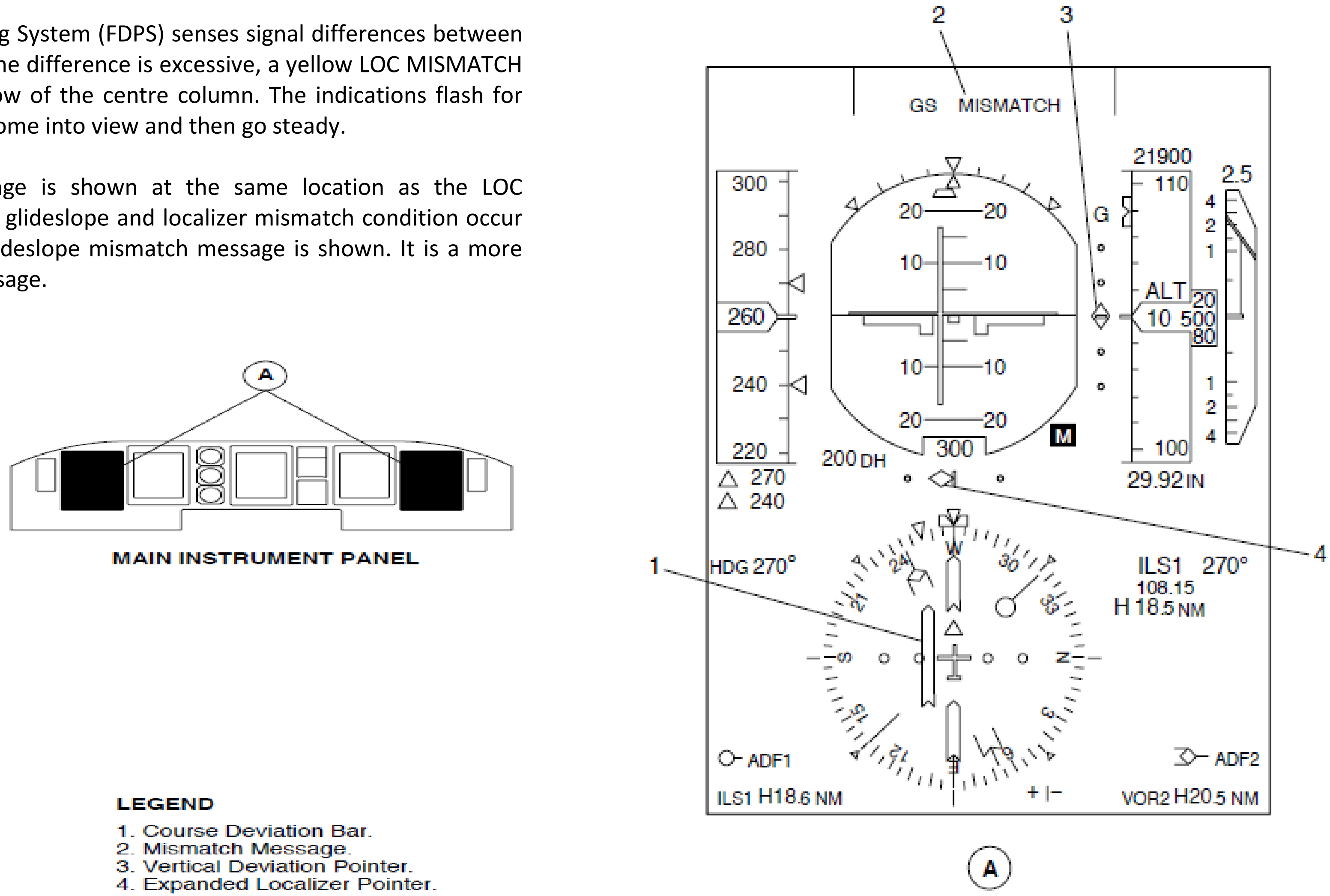


Figure - PFD, ILS MISMATCH INDICATIONS

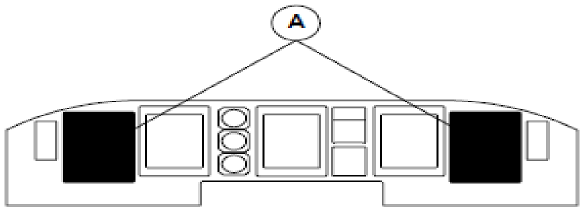
Refer Figure - PFD, ILS MALFUNCTION INDICATIONS

Course Deviation Failure Flag: The course deviation failure flag is a red cross shown with the course deviation scale. The conditions that cause the course deviation failure flag to come in view are as follows:

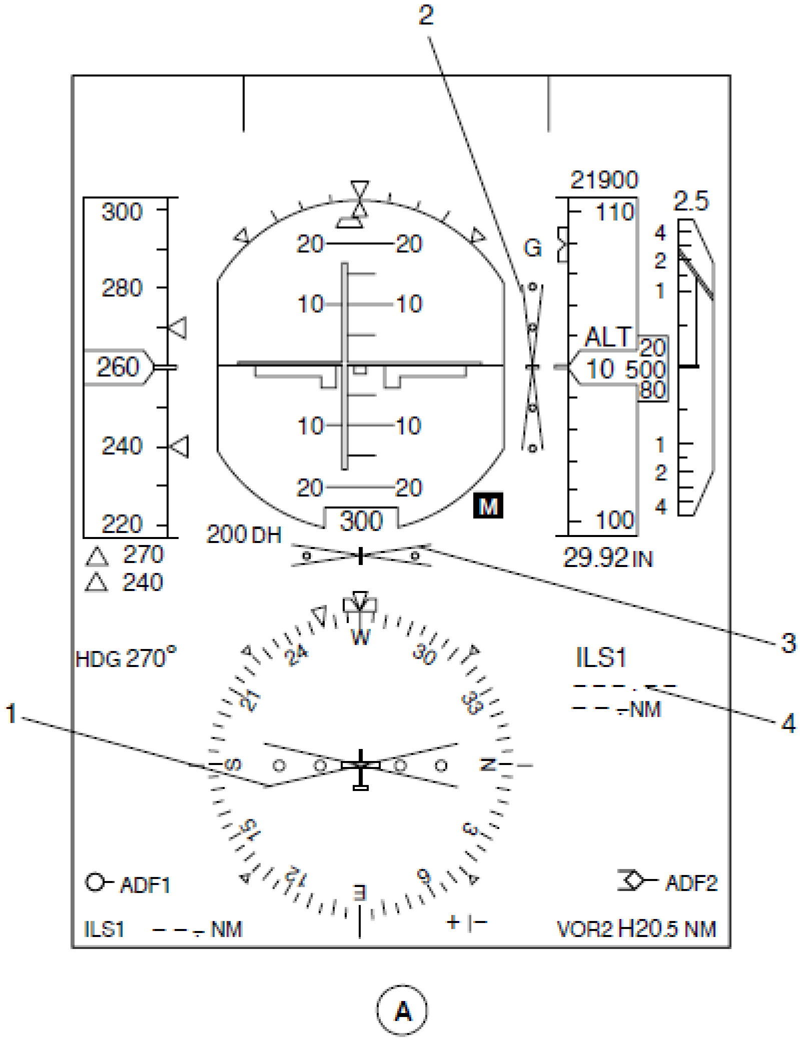
- The navigation ground station is out of range
- Navigation receiver malfunctions
- No data is available
- No navigation source selection data is available.

The PFDs show the Failure Flags that follow:

- Course Deviation
- Expanded Localizer
- Glideslope.



MAIN INSTRUMENT PANEL



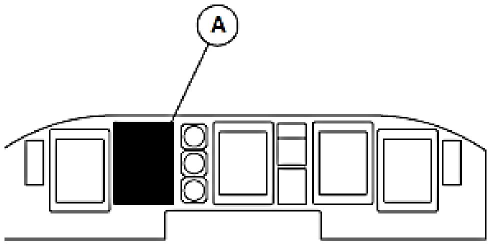
LEGEND

1. Course Deviation Failure Flag.
2. Vertical Deviation Failure Flag.
3. Expanded Localizer Failure Flag.
4. VHF Navigation Frequency Fail Annunciation

Figure - PFD, ILS MALFUNCTION INDICATIONS

Refer Figure - NAVIGATION PAGE, ARC FORMAT ILS INDICATION

The MFD1 and MFD2 Reversion switches located on the ESID Control Panel (ESCP) are used to set the navigation page. The navigation page comes into view in the map mode format. The FORMAT pushbutton switch on EFCP is pushed to change to the navigation page indication from a map (FMS) to arc (VOR/ILS) indication independently of the PFD. The arc mode shows localizer and glideslope parameters as its related PFD indication on a 90 degree heading arc.



MAIN INSTRUMENT PANEL

LEGEND

1. Selected Course Digital Value.
2. Navigation Source Annunciation.
3. Radio Navigation Frequency Identifier.
4. Selected Course Pointer.
5. Course Deviation Scale and Bar.
6. Permanent System Data Area.
7. Aircraft Symbol.
8. Heading Scale in Arc Format.

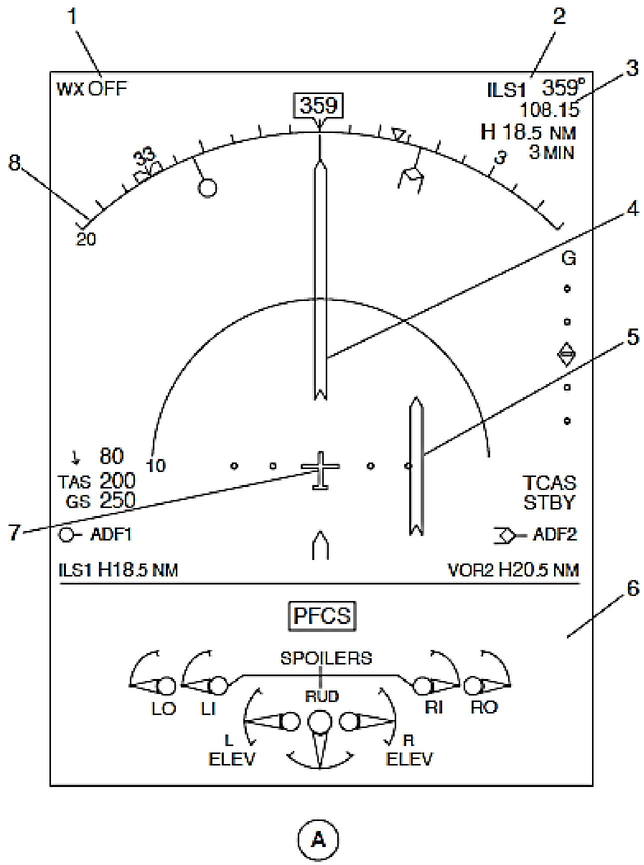
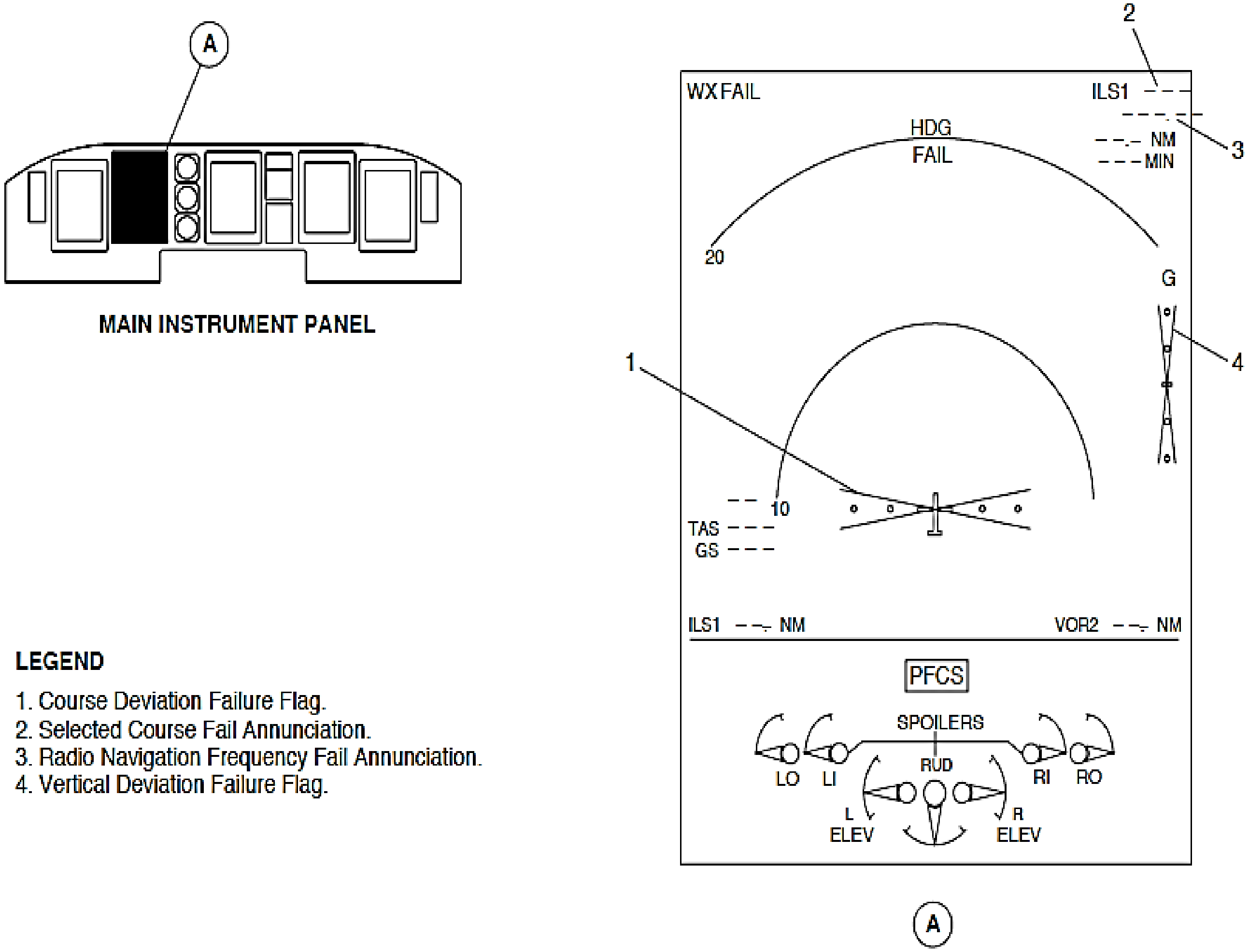


Figure - NAVIGATION PAGE, ARC FORMAT ILS INDICATION



LEGEND

1. Course Deviation Failure Flag.
2. Selected Course Fail Annunciation.
3. Radio Navigation Frequency Fail Annunciation.
4. Vertical Deviation Failure Flag.

Figure - NAVIGATION PAGE, ARC FORMAT ILS MALFUNCTION INDICATION

Refer Figure - NAVIGATION PAGE, ARC FORMAT ILS MALFUNCTION INDICATION

The navigation page, arc format shows a Course Deviation Failure Flag for the same conditions as for the PFDs. The course deviation failure flag is a red cross shown with the course deviation scale.

The Audio and Radio Control Display Units (ARCDU 1, ARCDU 2) monitor the ARINC 429 data bus through the Flight Data Processing System (FDPS) for normal functions. The ARCDUs show a red FAIL message in the frequency preset area of the VHF navigation display area for system malfunctions.

The left essential and right main busses supply 28 Vdc electrical power through 2 A VHF Navigation Receivers (VHF NAV1, VHF NAV2) circuit breakers. The circuit breakers are located in position G1 on the Left Essential andD5 on the 28 Vdc avionics circuit breaker panel.

The VHF Navigation (VHF NAV1, VHF NAV2) system has two navigation receivers with common VOR/LOC, glideslope, and marker antennas.

The VOR/Localizer antenna has two low profile blades attached to each side of the vertical stabilizer. It has phase−matched coaxial cables to connect the two antenna blades to a coupler located in the empennage and adjacent to the blades. From the coupler, a single coaxial cable is routed to a VOR/LOC antenna splitter and then to each VHF navigation radio.

Each navigation receiver connects to a single glideslope antenna with two RF connectors. Two separate coaxial cables connect the VHF navigation receivers to the antenna.

Each VHF navigation receiver connects to a single marker beacon antenna. A single coaxial cable connects the antenna to a splitter attached to the avionics rack. The splitter connects the two coaxial cables to the VHF navigation receivers.

The VHF Navigation Receivers (VHF NAV1, VHF NAV2) supply navigation information for display through the Input/output Processors (IOP1, IOP2).In addition, it supplies VOR and ILS information to the Flight Guidance Modules(FGM1, FGM2) for autopilot and flight director processing.

The Input/output Processor (IOP1, IOP2) and Flight Guidance Modules (FGM1, FGM2) are located inside its related Integrated Flight Cabinet (IFC1, IFC2).

The VHF Navigation parameters that follow are controlled using ARINC429 Low speed busses (12.0 to 14.5 kilobits per second):

Parameter	Source
Frequency	ARCDU
Outer Marker	IOP 1, IOP 2
Middle Marker	IOP 1, IOP 2
Inner Marker	IOP 1, IOP 2
VOR bearing	IOP 1, IOP 2
Left/Right Deviation	IOP 1, IOP 2
Glideslope Deviation	IOP 1, IOP 2

The Distance Measuring Equipment (DME1, DME2) is tuned by the VHF Navigation Receivers (VHF NAV1, VHF NAV2)

Part of the Flight Guidance Control Panel (FGCP) navigation source selection sends ARINC 429 tuning information to the DME. The navigation source selection causes an Input/output Module (IOM1, IOM2) discrete to control a relay that routes the tuning information to the DME. The Input/output Module (IOM1, IOM2) is part of the Flight Data Processing System (FDPS).

The Flight Management System (FMS) uses up to two sources of VOR information for the VORTAC Position Unit (VPU) mode. The Flight Management System (FMS) uses VOR information to calculate Position−fixing algorithms.

Receiver, VHF Navigation

Refer Figure - NAVIGATION RECEIVER LOCATOR

The VHF Navigation Receiver has three separate receivers. They receive and process VHF omni range, localizer, glideslope, and marker beacon data. It supplies left/right and up/down data for indication and Auto Flight Control System (AFCS) guidance.

The VHF Navigation Receiver has the sub−assemblies that follow:

- VOR/Localizer Receiver
- Glideslope Receiver
- Marker Beacon Receiver
- Navigation Converter
- Module Navigation Processor Module
- Power Supply.

The interconnections between assemblies are made using ribbon cables. Maintenance is simplified through modular construction. Modules are hinged and attached to a rigid main frame to allow easy access. The circuit boards plug into connectors to allow troubleshooting by substitution. The case has a front panel handle for easy removal and installation.

A single recessed DPX rack−mounted rear−panel connector and three front−panel coaxial RF connectors interconnect the aircraft wire harness and antenna installation.

The VHF Navigation Receivers are installed one above the other on the first and second shelves of the avionics rack, at station point X−34.00.

The VHF Navigation Receivers are located in racks that have forced−air-cooling.

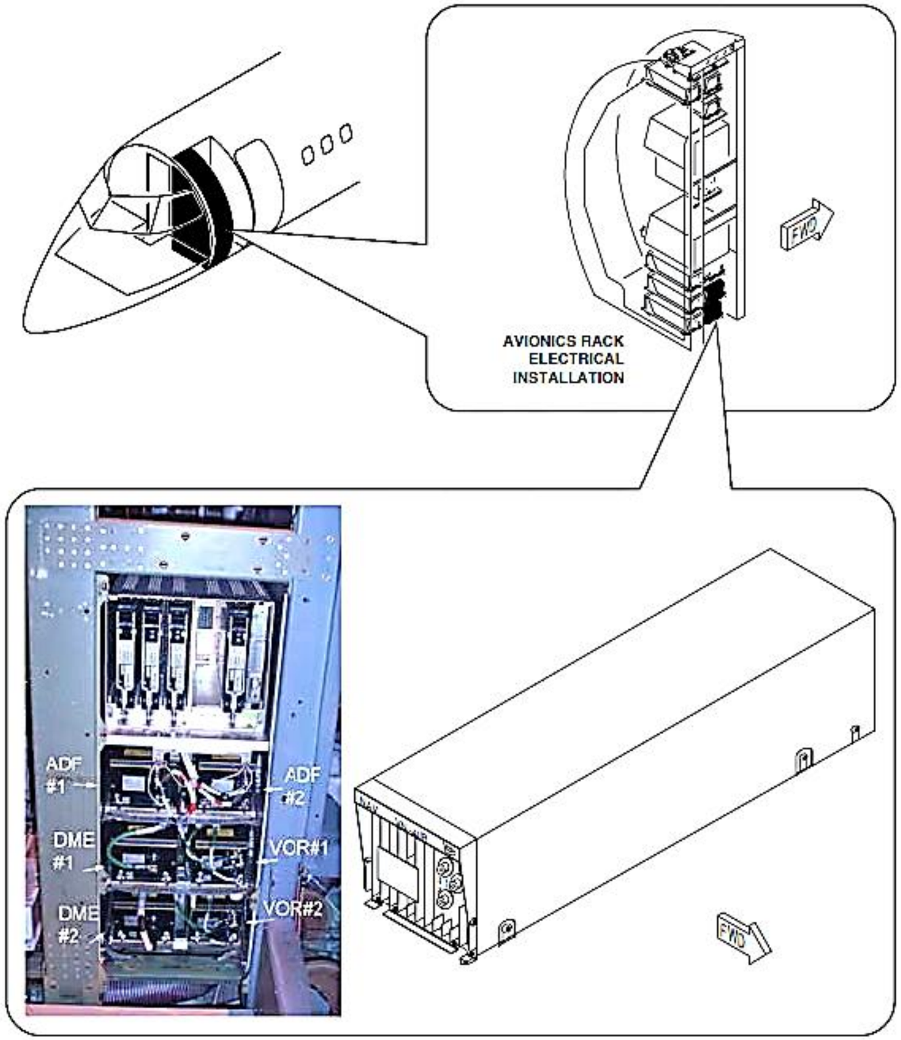


Figure - NAVIGATION RECEIVER LOCATOR

Antenna, Glideslope

Refer Figure - GLIDESLOPE ANTENNA LOCATOR

The Glideslope Antenna receives radio frequencies between 328 and 336 MHz.

The antenna encases a preformed radiating element in thermoplastic within a single assembly. Two Radio Frequency (RF) TNC female connectors are attached to an aluminum base.

An aluminum foil gasket with an elastomeric sealant makes sure consistent distribution of conducting contacts of the gasket to the airframe structure. The pressure applied during installation gives a contouring feature and lets the gasket conform to the two mating surfaces.

The glideslope antenna is located on the forward pressure bulkhead at station points X −158.00, Y 0.00, Z 115.04. The antenna is attached to the airframe structure with six mounting bolts.

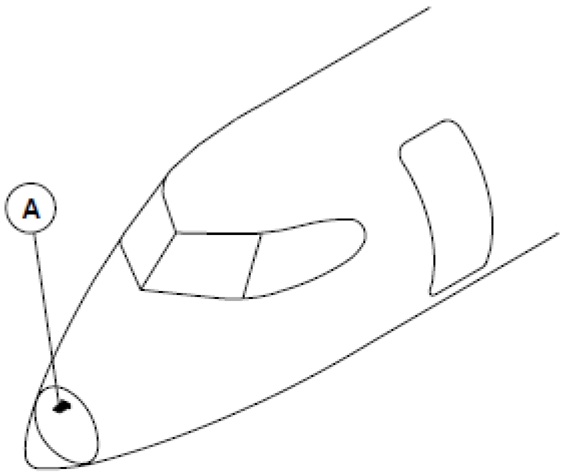


Figure - GLIDESLOPE ANTENNA LOCATOR

Antenna, VOR/LOC

Refer Figure - VOR/LOC ANTENNA LOCATOR

The VOR/Localizer Antenna receives radio frequencies between 108 and 118 MHz. The antennas have the features that follow:

- Directional
- Horizontally polarized
- Loop
- End fed.

The VOR/Localizer antenna has two low profile blades installed to each side of the vertical stabilizer. Each antenna blade has a preformed radiating element in a thermoplastic assembly. A Radio Frequency (RF) BNC female connector is attached to the aluminum base. Phase−matched coaxial cables connect the two antenna blades to a coupler housed within the empennage and adjacent to the blades.

An aluminum foil gasket with an elastomeric sealant makes electrical contact between the gasket and airframe structure. The pressure applied during installation gives a contouring feature to let the gasket contact the two mating surfaces.

The antenna has two low profile blades installed on each side of the vertical stabilizer at station points X.994.00, Z.284.00. The antenna is attached to the airframe structure with four mounting bolts.

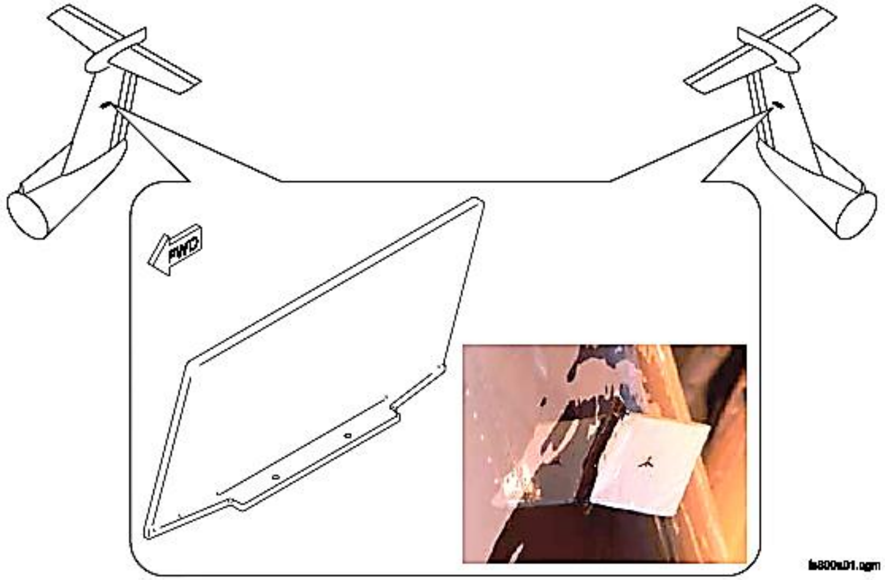


Figure - VOR/LOC ANTENNA LOCATOR

Coupler, Antenna VOR/LOC

Refer Figure - VOR/LOC ANTENNA COUPLER LOCATOR

The VOR/Localizer antenna couple is used to connect the two VOR/Localizer antenna blades.

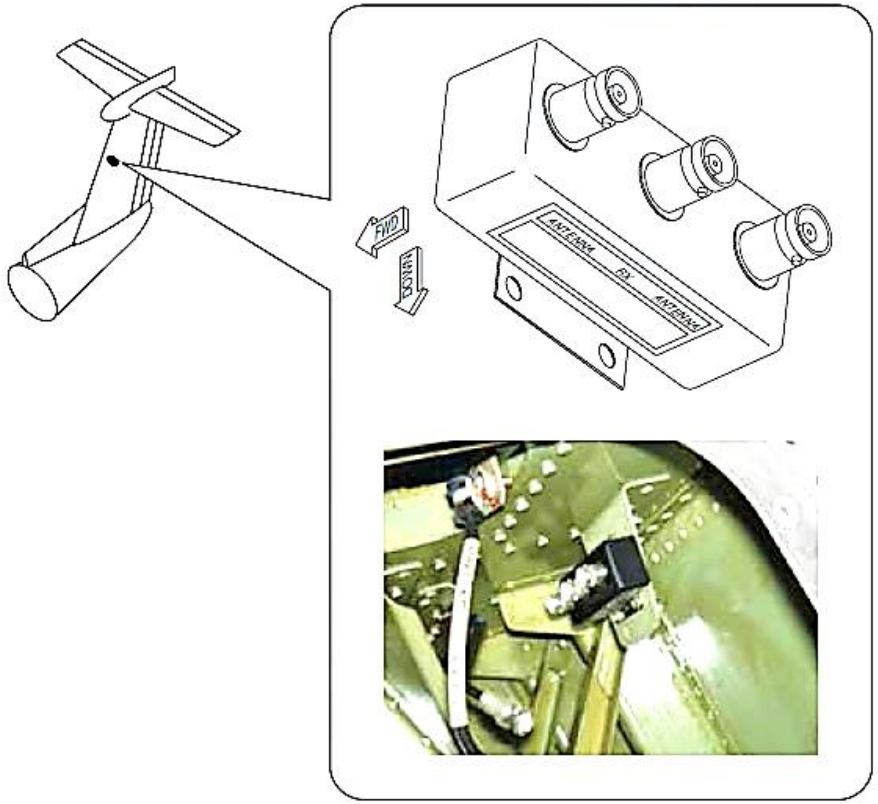


Figure - VOR/LOC ANTENNA COUPLER LOCATOR

Splitter, Antenna VOR/LOC

Refer Figure - VOR/LOC ANTENNA SPLITTER LOCATOR

The VOR/LOC antenna splitter connects the antenna to each VHF navigation radio.

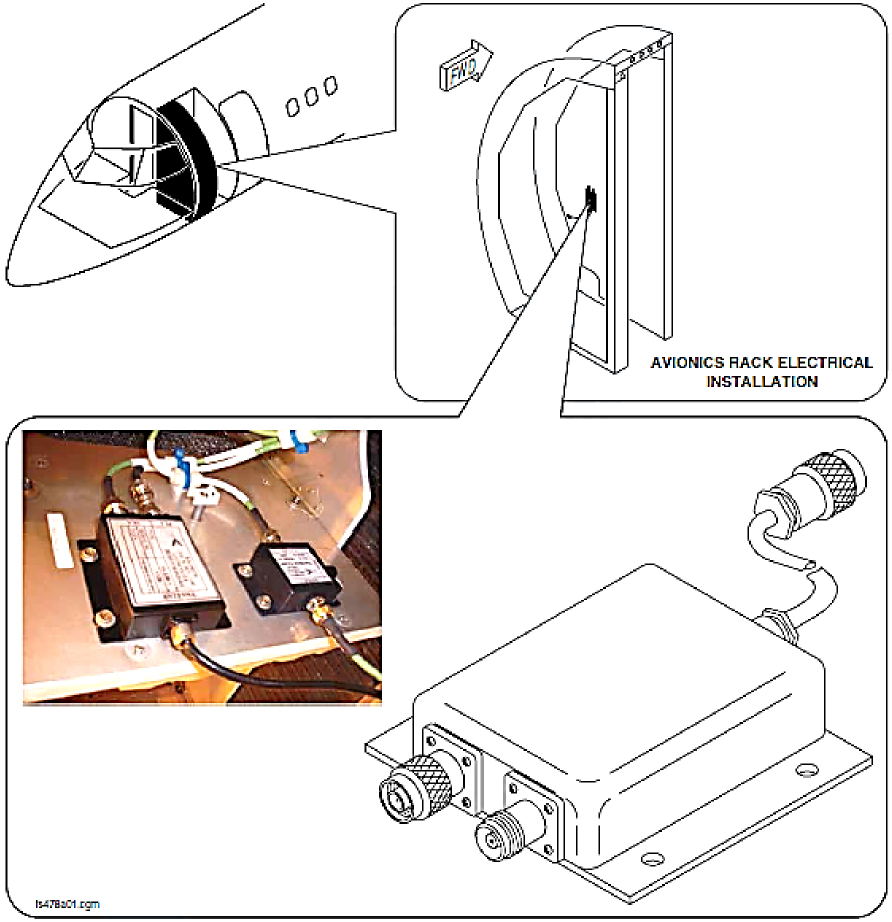


Figure - VOR/LOC ANTENNA SPLITTER LOCATOR

Antenna, Marker

Refer Figure - MARKER ANTENNA LOCATOR

The Marker Antenna receives radio frequencies at 75 MHz.

The antenna blade has a preformed radiating element in a thermoplastic assembly. A Radio Frequency (RF) BNC female connector is attached to the aluminum base.

An aluminum foil gasket with an elastomeric sealant makes electrical contact between the gasket and airframe structure. The pressure applied during installation gives a contouring feature to let the gasket contact the two mating surfaces.

The antenna is attached to the airframe structure with six mounting bolts at station point X 513.7.

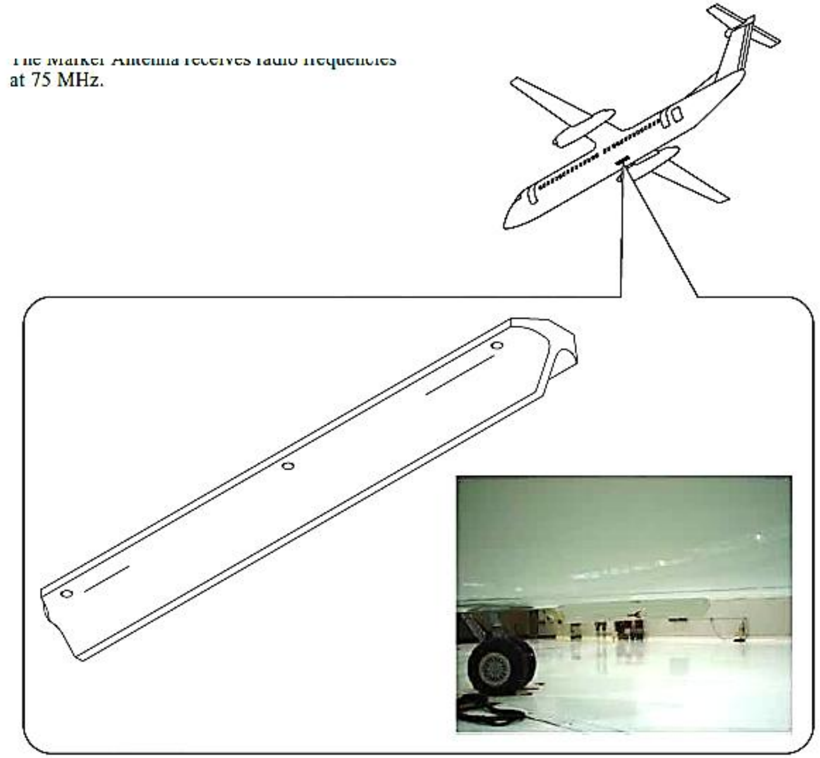


Figure - MARKER ANTENNA LOCATOR

Splitter, Marker

Refer Figure - MARKER ANTENNA SPLITTER LOCATOR

The Marker Splitter connects the Marker Antenna to each VHF Navigation radio.

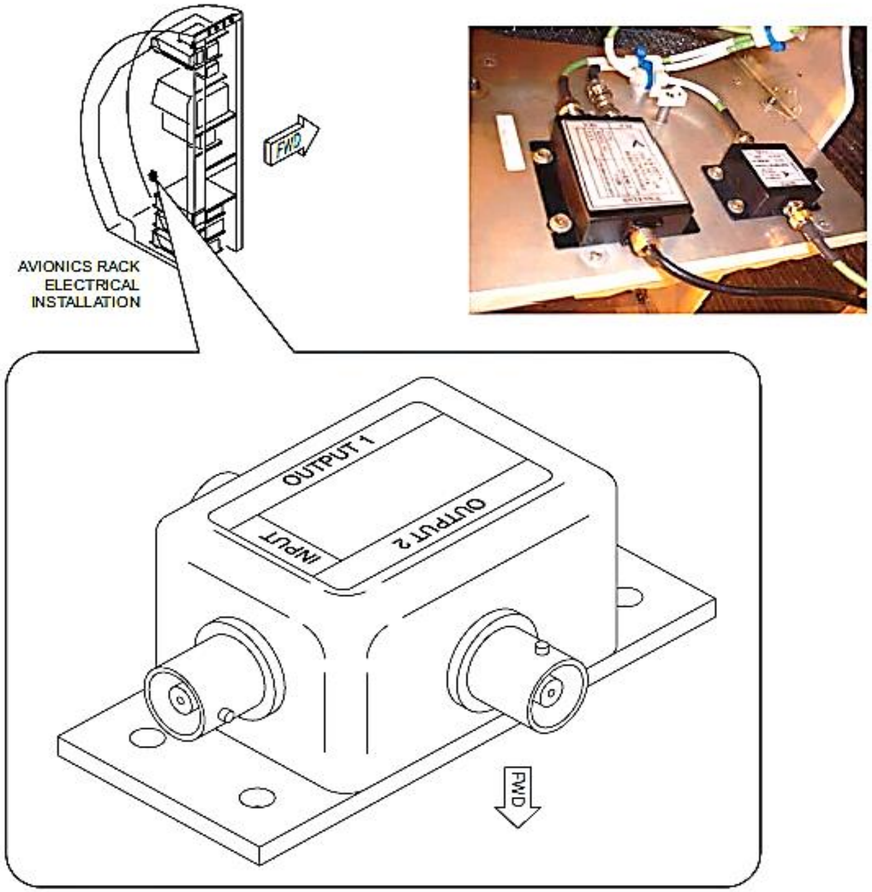


Figure - MARKER ANTENNA SPLITTER LOCATOR

Antenna Cable

Coaxial cables connect the navigation receivers to the antennas that follow:

- VOR/Localizer
- Glideslope
- Marker Beacon.

The coaxial cables are transmission lines with one conductor surrounding another. A dielectric insulator separates the two conductors. The centre conductor is seven strand silver−plated copper wire and the outer conductor is a tinned copper wire braid.

NOTE

The coaxial cables that connect the two blades of the VOR/Localizer antenna to the coupler are phase−matched within 2 degrees.

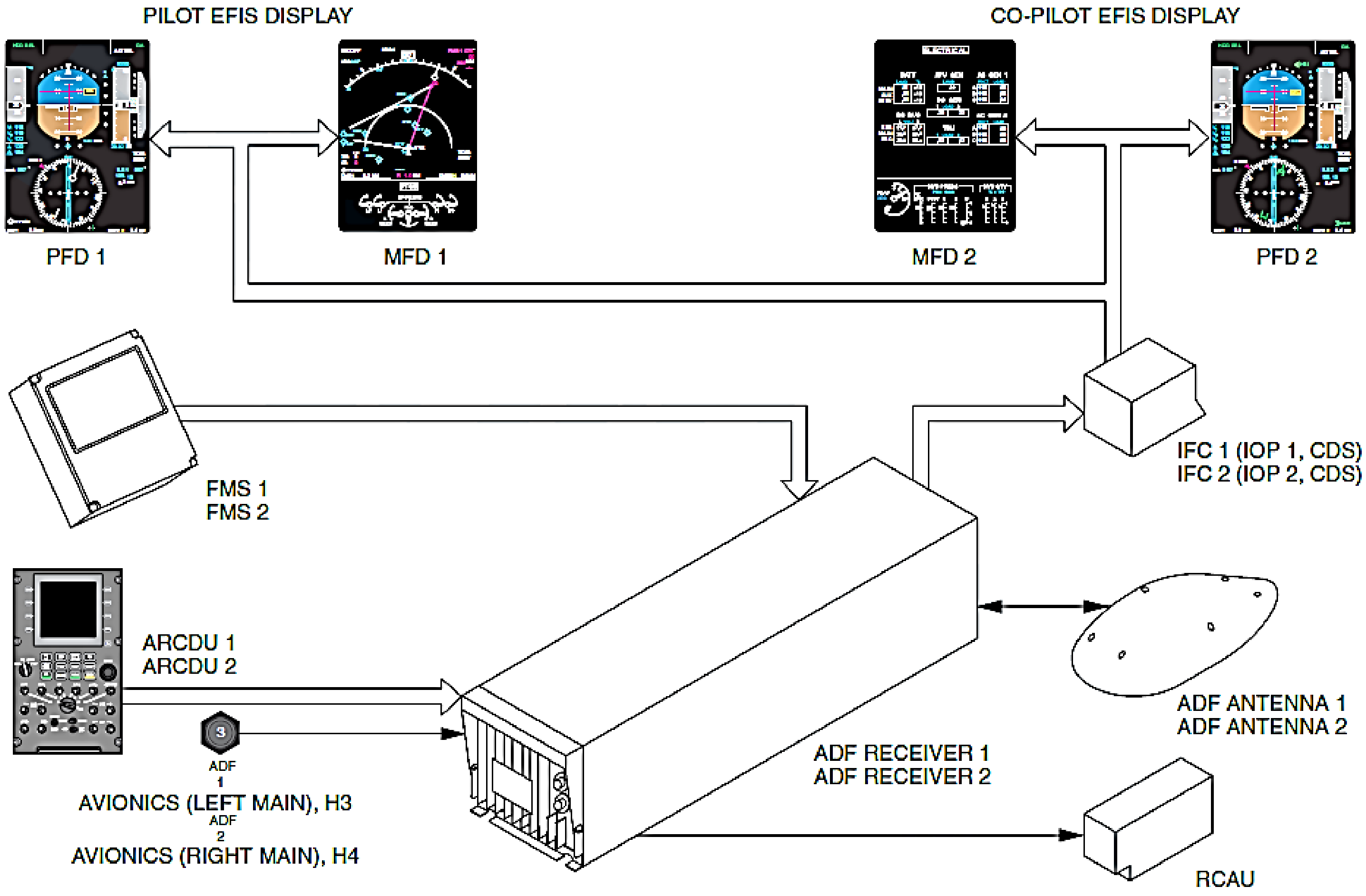


Figure - ADF BLOCK DIAGRAM

AUTOMATIC DIRECTION FINDING (ADF)

Introduction

The Automatic Direction Finding (ADF) system shows the magnetic bearing to a fixed Non−Directional Beacon (NDB).

General Description

Figure - ADF BLOCK DIAGRAM

Two Automatic Direction Finding (ADF 1, ADF 2) systems function independently to continuously show the magnetic bearing to the selected ground stations.

The Automatic Direction Finding (ADF 1, ADF 2) system removes the audio identification from the transmitting signal and sends it to the Audio and Radio Management System (ARMS).

The ADF system is controlled by the components that follow:

- Audio and Radio Control Display Units (ARCDU1, ARCDU 2)
- Flight Management System (FMS 1, FMS 2).

The Audio and Radio Control Display Units (ARCDU1, ARCDU 2) are used to manually tune the ADF system frequencies and to control the audio level.

The Flight Management System (FMS) is used to tune the ADF frequencies when the Audio and Radio Control Display Unit (ARCDU1, ARCDU 2) malfunctions. The observer uses an audio control panel to control audio level only.

The Audio and Radio Control Display Unit (ARCDU 1, ARCDU 2) or the Flight Management System (FMS 1, FMS 2) are used to manually start the Automatic Direction Finding (ADF) system self−tests.

The Electronic Instrument System (EIS) shows the Automatic Direction Finding (ADF) system bearing information.

The Automatic Direction Finding (ADF) system has the components that follow:

- Receiver, ADF
- Antenna.

The ADF Receivers (ADF 1, ADF 2) supply data to the systems that follow:

- Audio and Radio Management System (ARMS)
- Input/Output Processors (IOP 1, IOP 2)
- Flight Management System (FMS).

Detailed Description

Figure - MAGNETIC BEARING INDICATION

The Automatic Direction Finding (ADF) system gives a means to point to the relative bearing of the selected ground station.

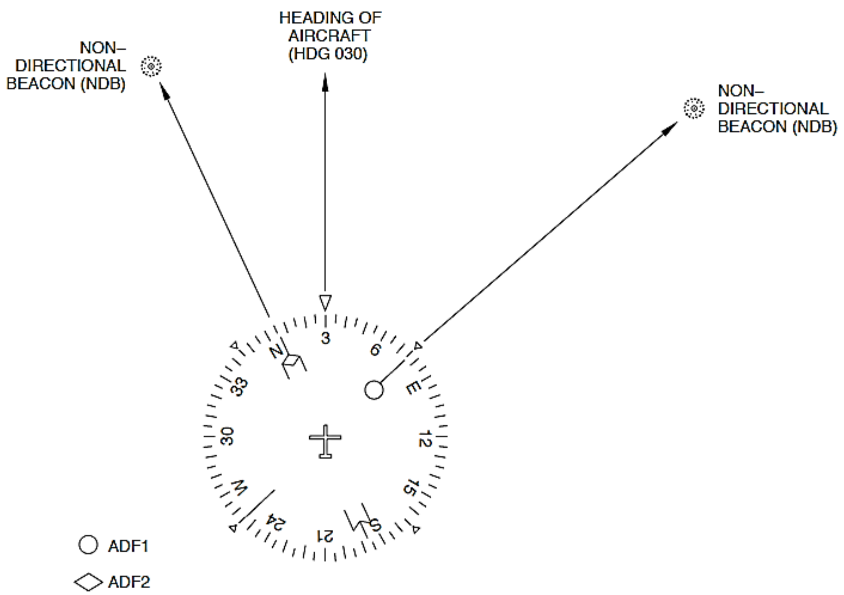


Figure - MAGNETIC BEARING INDICATION

The ground stations transmit radio waves that induce voltages in the loop windings of the ADF antenna. The loop antenna has two mutually perpendicular windings. One axis of one winding is aligned with the longitudinal axis of the aircraft and the other winding is aligned with the lateral axis.

The voltages induced are proportional to the angle between the heading of the aircraft and the track to the station.

The loop antenna has two positions 180 degrees apart where the voltages induced are the same. A sense antenna signal gives a reference phase to which the loop signals are combined to prevent bearing ambiguity.

The Automatic Direction Finding (ADF) system receives Low Frequency (LF) Amplitude Modulated (AM) Radio Frequency (RF) signals from the stations that follow:

- Non−directional beacons
- Broadcast stations
- Maritime distress.

The Non−directional beacons and the AM broadcast stations function between 190 and 1860 kHz. The maritime distress frequency range is between the 2181 and 2183 kHz.

The Automatic Direction Finding (ADF 1, ADF 2) systems function in the modes that follow:

- ADF, Automatic Direction Finding
- ANT, Antenna
- BFO, Beat Frequency Oscillator
- Test.

ADF Mode: In the ADF mode, the Electronic Instrument System (EIS) shows the bearing of the aircraft relative to the selected ground station.

ANT Mode: In the ANT mode, the receiver functions as an audio receiver to identify stations. The loop antenna is disabled to give a higher audio sensitivity than in ADF mode.

The ADF bearing pointer parks at 90 degrees relative bearing.

Beat Frequency Oscillator (BFO) Mode: Some ground stations operate with an interrupted continuous wave (CW) carrier. They do not use a modulated RF carrier for audio identification. The ADF receiver supplies an audio frequency that beats with the incoming intermittent carrier wave to give an audio output that is the same as the ground station identifying transmissions. In the BFO mode, an intermittent 1000 Hz audio tone is heard when the ADF receives a valid transmission that identifies the station.

The ADF system continues to show the bearing of the aircraft relative to the selected ground station in the BFO mode when ADF mode is selected.

Test mode: The TEST mode is a confidence test. It causes the bearing pointer of the ADF receiver under test to park at a relative bearing of 90 degrees.

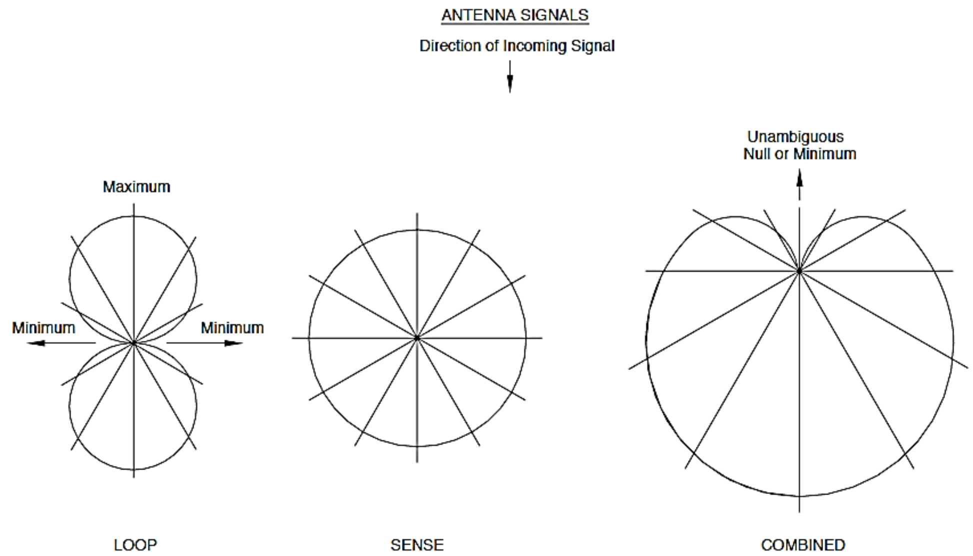
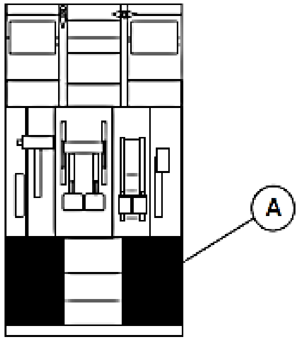
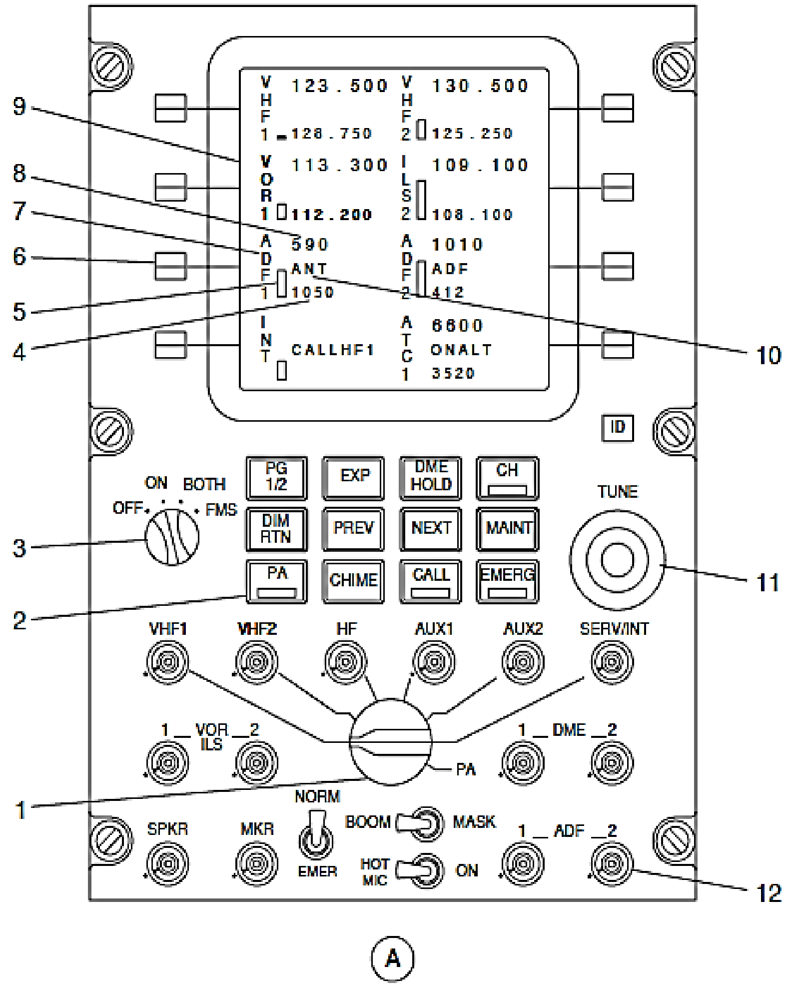


Figure - NDB ANTENNA SIGNALS



CENTRE CONSOLE



LEGEND

1. Microphone/Interphone Selector Switch.
2. Keys.
3. Rotary Switch.
4. Preset Frequency.
5. Level Adjust Bar Graph.
6. Side Key.
7. Label.
8. Active Frequency.
9. Display Area.
10. Automatic Direction Finding Mode.
11. Tuning Knob.
12. Potentiometer/Push–Button Switches.

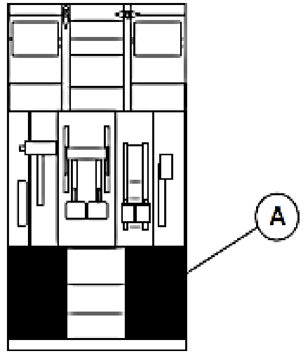
20a01.cgm

Figure - ADF, ARCDU

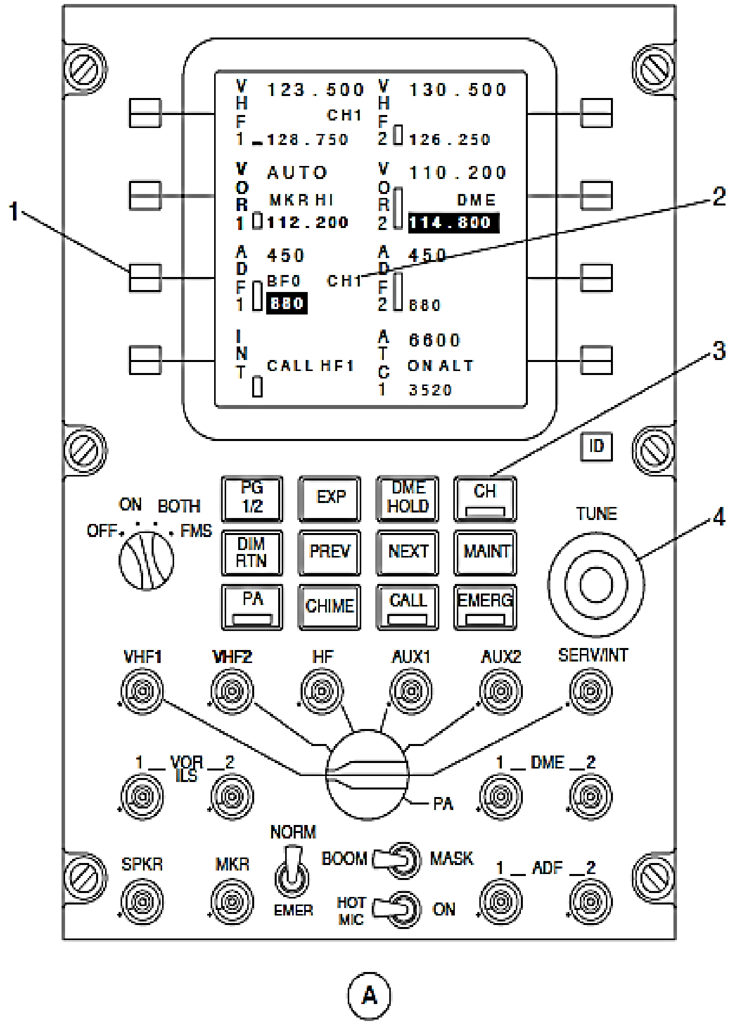
Refer Figure - ADF, ARCDU

The Radio Control Display Units (ARCDU 1, ARCDU 2) selections control the Automatic Direction Finding (ADF) frequencies and modes to be used.

Selection	Description
OFF	The ARCDU is not powered. The related FMS controls and tunes as a backup
ON	The ARCDU controls and tunes its related radio system
BOTH	The ARCDU controls and tunes its related and opposite radio systems
FMS	The FMS controls and tunes its related and opposite radio systems



CENTRE CONSOLE



LEGEND

1. Side Key.
2. Channel Memory Annunciation.
3. Channel Key.
4. Tuning Knob.

Figure - ARCDU ADF FREQUENCY CONTROL

Refer Figure - ARCDU ADF FREQUENCY CONTROL

A minimum of two steps are required to change the ADF active frequency.

The side key adjacent to the ADF display area is pushed to highlight the preset frequency. It is pushed again to make the preset frequency active. The frequency that was active now becomes the new preset frequency.

When the side key adjacent to the ADF display area is pushed, the preset frequency in the display area highlights. The TUNE double rotary knobs located at the lower right side of the ARCDU are turned to change the preset frequency. The Side key is pressed again to set the new active frequency. The frequency that was active now becomes the new preset frequency.

The CH key is pushed to set the channel memory mode from eight preset frequencies using a channel memory selection. A green bar in the CH key comes on and a channel memory annunciator comes into view in green characters below the active frequency indication. In addition, the channel memory number is shown adjacent to the preset frequency.

The side key adjacent to the ADF display area is pushed to change the preset frequency. When the TUNE double rotary knobs are turned, the channel memory number shown adjacent to the preset frequency changes. If a channel memory number is not shown in the display area, it starts with CH1. The programmed frequency for the channel memory is shown in place of the preset value as black characters on a cyan background. The double rotary knobs are turned to select the different preset channel memories. The rotary knob has a rollover capability. It shows the first preset channel again after it shows the last channel when the rotary knob is continuously turned in one direction or the other. The ADF side key is pushed again to validate the channel memory selection. Two successive pushes of the ADF side key causes the preset frequency to replace the active frequency. The active frequency now becomes the preset frequency.

The channel memory number appears in green characters below the active frequency. This shows that the active frequency is associated with a channel memory number. When the CH key is pushed again, the channel memory mode changes to the normal frequency selection. All channel memory annunciators are removed and the green light in the CH key goes out.

If the selections that follow are not made within five seconds, the previous preset frequency is restored:

- Turn one or the other TUNE double rotary knob
- Push the same side key again.

Refer Figure - ARCDU ADF PARTICULAR PAGE ACCESS

The ADF Particular Page comes into view when the ADF side key is pushed followed by the EXP key on the keyboard.

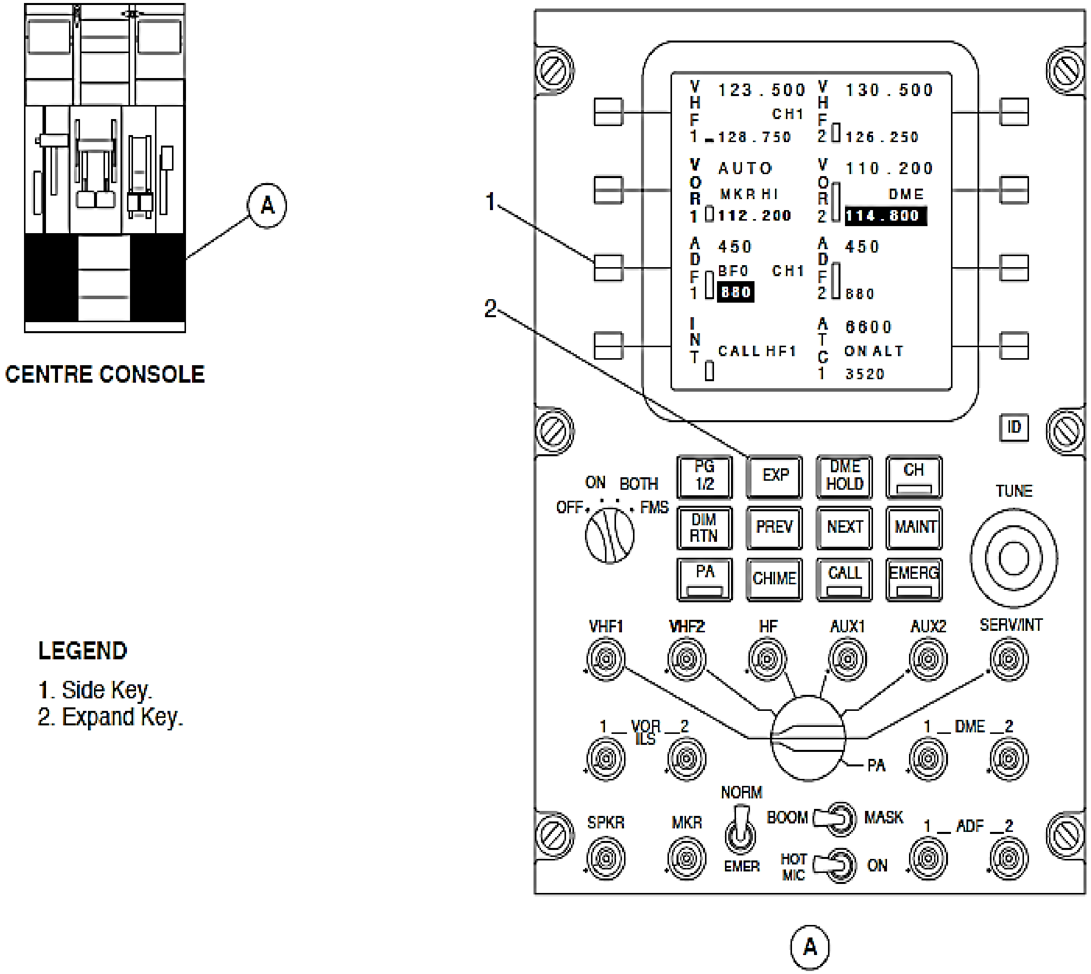


Figure - ARCDU ADF PARTICULAR PAGE ACCESS

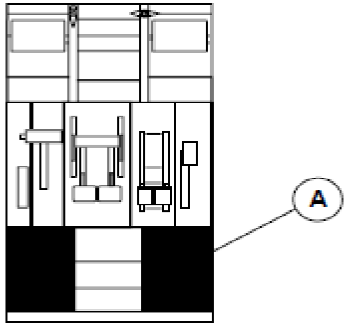
Refer Figure - ARCDU ADF PARTICULAR PAGE

The particular page can be used to change the ADF Receiver (ADF 1, ADF 2) modes and functions that follow:

- 0.5 or 1 kHz tuning
- CHANNELS, Channel programming
- ADF, Automatic Direction Finding or ANT, Antenna mode
- TEST, Test
- BFO, Beat Frequency Oscillator on or off.

Tuning Features: The side key adjacent the 0.5, 1.0 kHz legend is pushed to alternately switch between the 0.5 and 1.0 kHz tuning feature. The selected text is black letters on a green background. The non−selected text is in white text on the black background. If 0.5 kHz is selected, the ADF receiver is tuned in 0.5 kHz increments. If 1 kHz is selected, the ADF receiver is tuned in 1 kHz steps.

Channel Programming: Eight preset channels are shown on two pages for ADF tuning. The preset and active frequencies can be tuned from both channel pages.



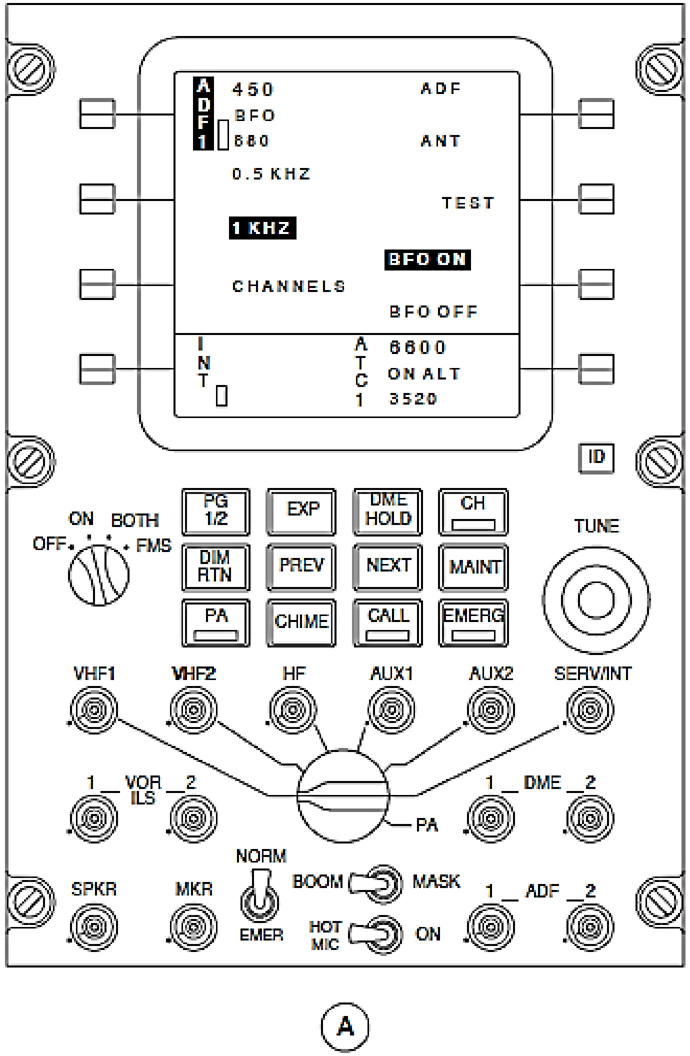
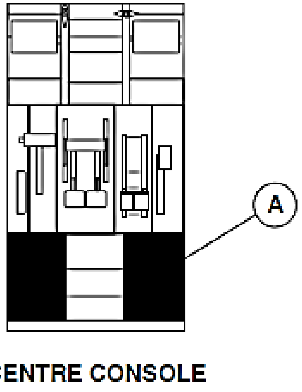
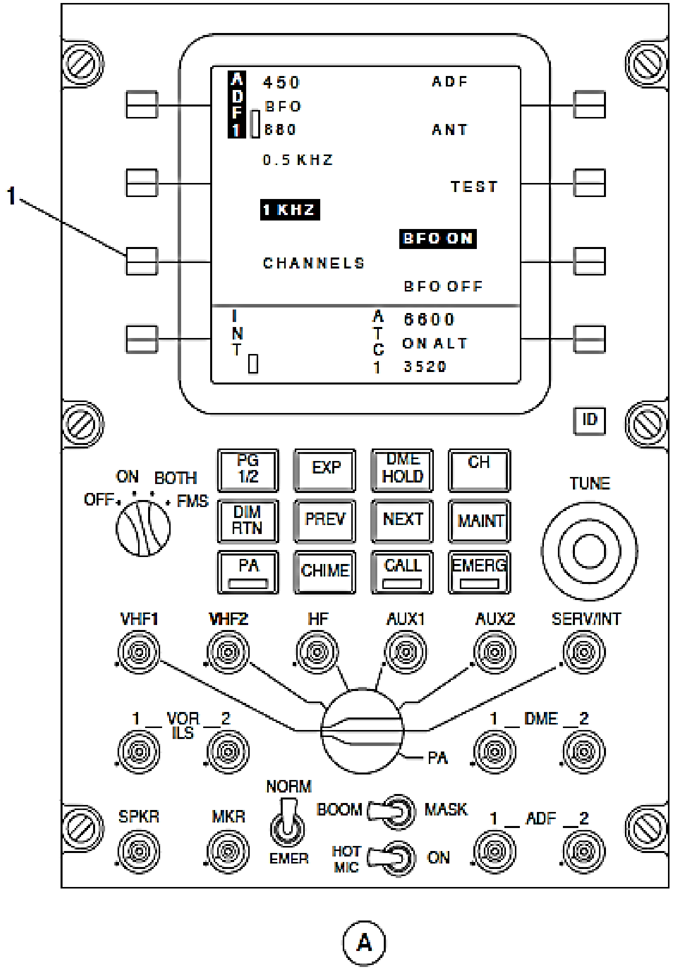


Figure - ARCDU ADF PARTICULAR PAGE

Figure - ARCDU ADF CHANNELS ACCESS PAGE

The side key adjacent to the CHANNEL legend is pushed to access the Channel Programming mode.





LEGEND

1. Side Key.

Figure - ARCDU ADF CHANNELS ACCESS PAGE

Refer Figure - ARCDU ADF FIRST CHANNELS PAGE

Channel presets are labeled as CH1 through CH8 in white characters on two pages. Each channel window shows the current preset channel number in cyan fonts below its preset labels.

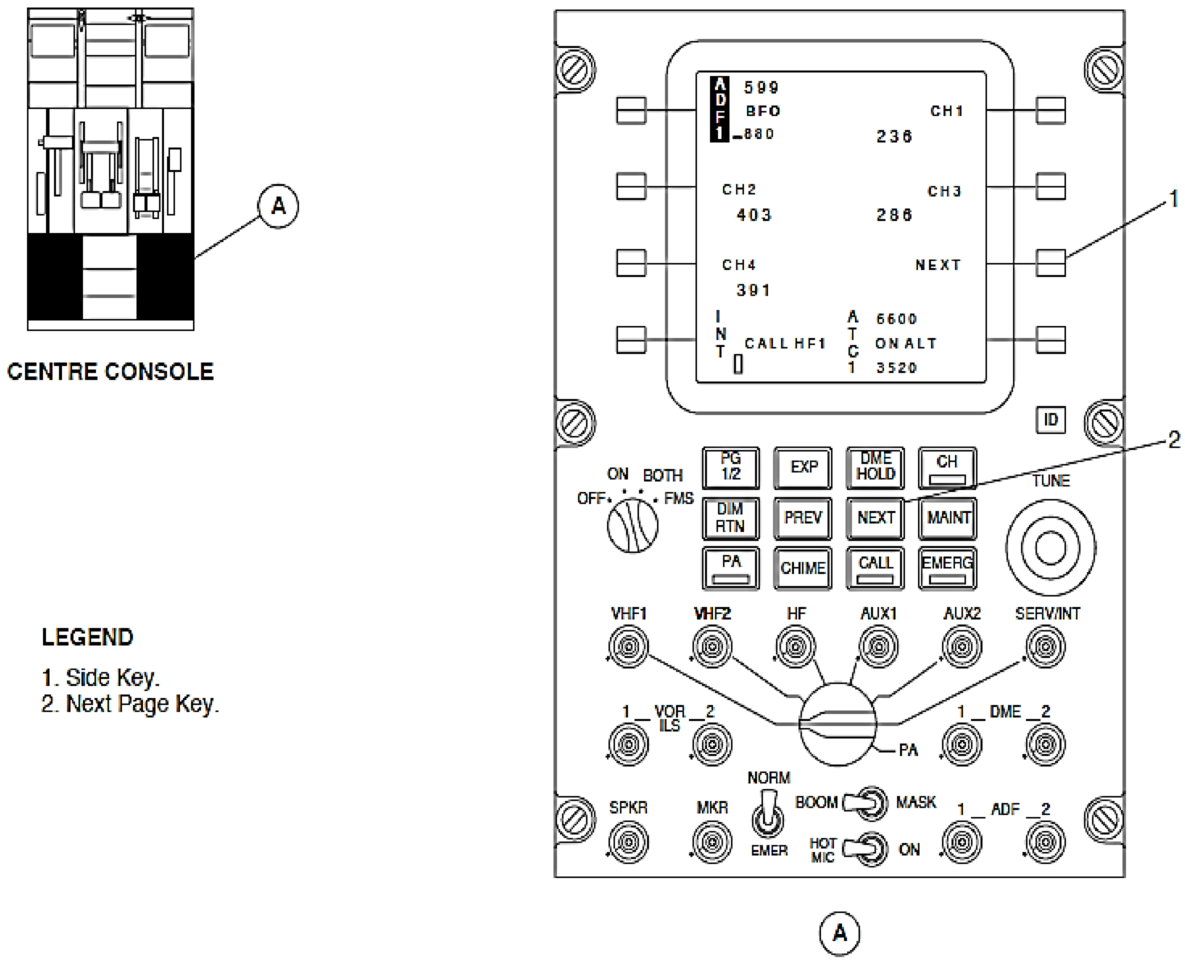


Figure - ARCDU ADF FIRST CHANNELS PAGE

Refer Figure - ARCDU ADF SECOND CHANNELS PAGE

The second channel memory page comes into view when the side key adjacent to the NEXT legend is pushed or the NEXT key is pushed. The side key adjacent to the PREVIOUS legend is pushed or the PREV key is pushed to see the first channel memory page. The PG 1/2 key is pushed to show the first main page.

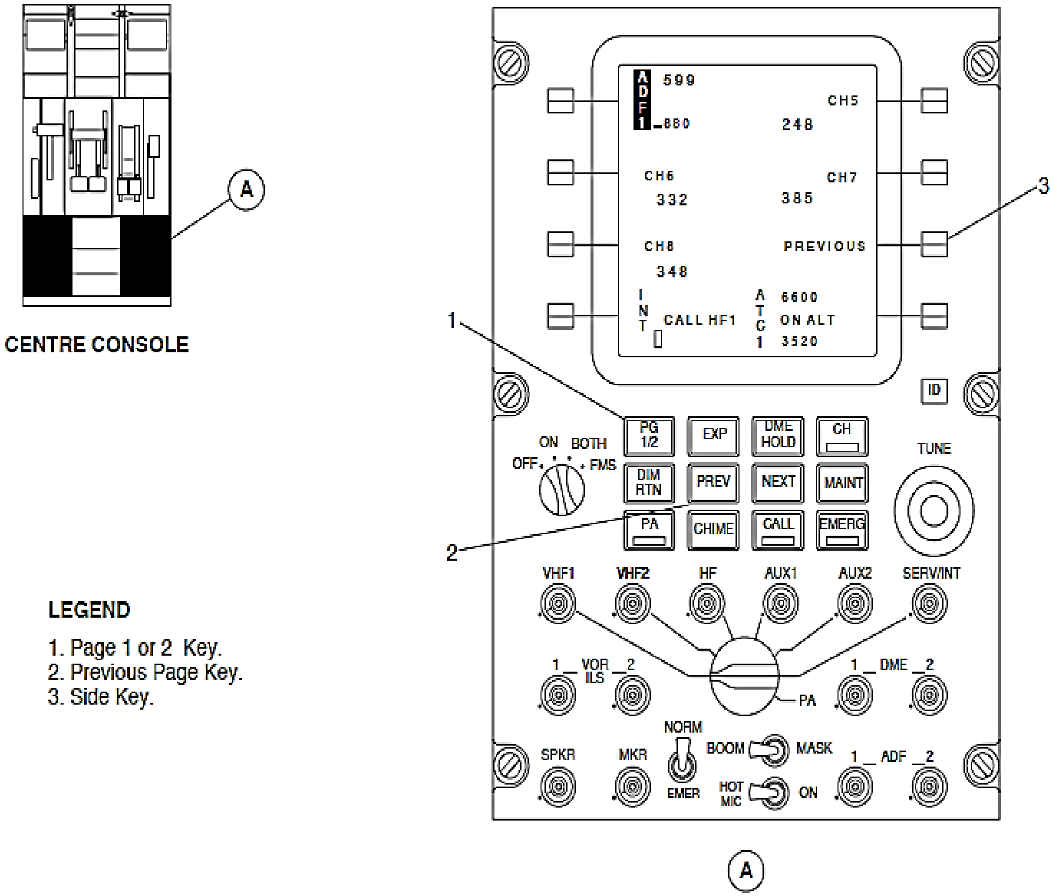
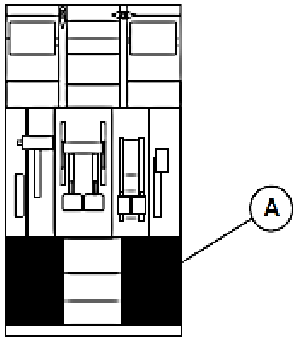


Figure - ARCDU ADF SECOND CHANNELS PAGE



CENTRE CONSOLE

LEGEND

1. Side Key.
2. Tuning Knob.

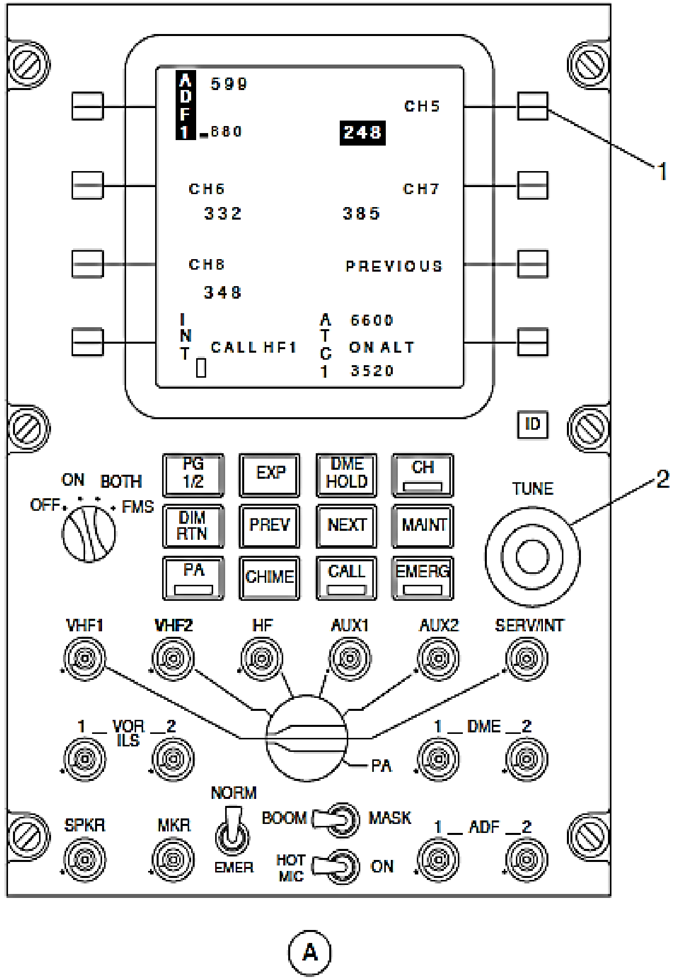


Figure - ARCDU ADF PRESET CHANNEL PROGRAMMING

Refer Figure - ARCDU ADF PRESET CHANNEL PROGRAMMING

When the side key related to the channel being changed is pushed, the current frequency value changes to black characters on a cyan background. The two TUNE double rotary knobs located at the lower right side of the ARCDU is turned to change the frequency. The side key is pushed again to validate the new frequency. The channel window shows the new frequency in cyan characters on the black display. If the new selected channel is associated to a preset radio frequency, both values change together. If the modified channel is the active radio frequency, the label will flash for five seconds at a 1 hertz rate. This shows that it is not possible to directly change an active frequency.

If the frequency selections are not made within five seconds, the previous preset frequency restores.

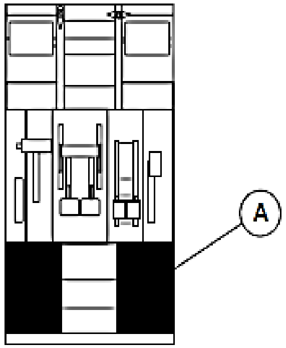
Refer Figure - ARCDU ADF PARTICULAR PAGE ACCESS

ADF and ANT mode: The side key adjacent to the ADF, ANT legend is pushed to switch between the ADF and ANT modes. The selected text is black letters on a green background. The non−selected text is in white text on the black background.

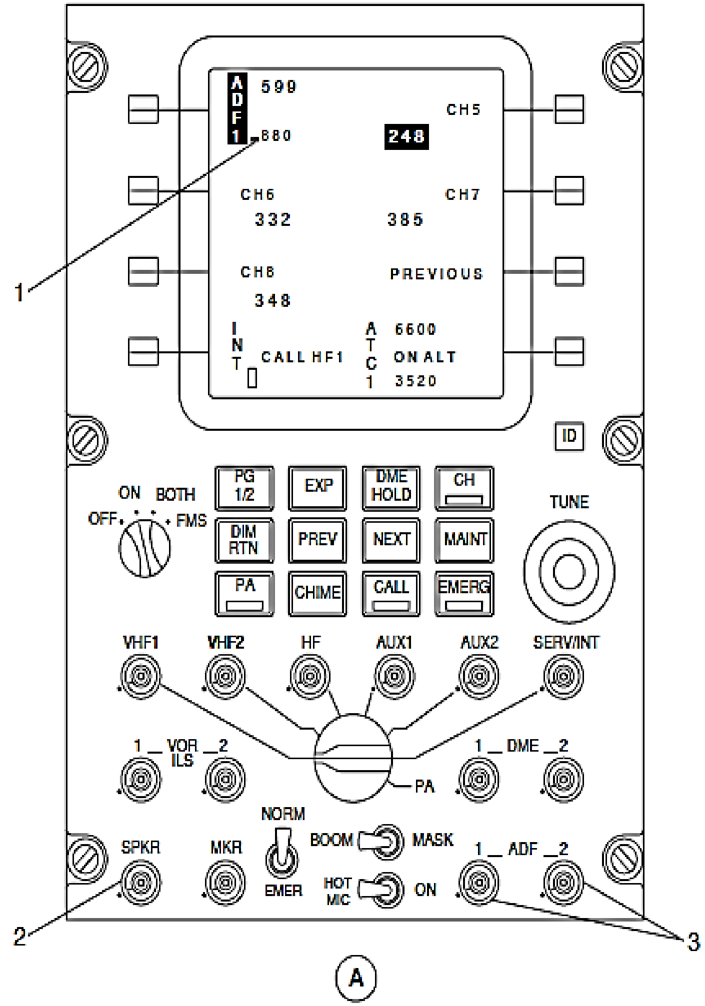
Test Mode: Push the side key adjacent to the TEST legend to test the ADF receiver. The TEST legend changes to black letters on a green background from the usually white on black background. The bearing pointer of the ADF receiver under test is parked at a relative bearing of 90 degrees. The display area shows OK in green letters for five seconds, if the test result is correct. It shows FAIL in red letters when the test result is not correct.

BFO ON or BFO OFF: The side key adjacent to the BFO ON, BFO OFF legend is pushed to alternately switch the BFO on or off.

The selected text changes to black letters on a green background and the non−selected text is in white text on the black background.



CENTRE CONSOLE



LEGEND

1. Level Adjust Bar Graph.
2. Speaker Potentiometer/Push Button Switch.
3. ADF Potentiometer/Push Button Switches.

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Figure - ARCDU ADF AUDIO CONTROL

Refer Figure - ARCDU ADF AUDIO CONTROL

Audio Control: The ADF Potentiometer push−button switches control the audio parameters that follow:

- On
- Off
- Level.

The top part of the potentiometer is pushed to set the specific audio signal to the headphone on or off. The ADF 1, or ADF 2 switch is pushed on the Audio and Radio Control Display Unit (ARCDU 1, ARCDU 2) to make its audio available to that flight crew member. When the SPKR push−button potentiometer switch is pushed, the selected ADF audio is routed to the related flight compartment speaker.

The ADF receiver display areas shows the volume level as a vertical bar graph. The height of the bar graph shows the volume selection level. When the audio is selected off, the bar graph displays in white. The bar graph changes to green when the audio is selected on. The ADF audio is available at the observer’s headphone jack when the ADF 1 or ADF 2 is selected on the observer audio control panel.

Refer Figure - EFCP ADF BEARING SELECTION

When an ADF 1 or ADF 2 bearing is selected at the EFIS Control Panels (EFCP 1, EFCP 2), a distinctive pointer comes into view on the Primary Flight Display.

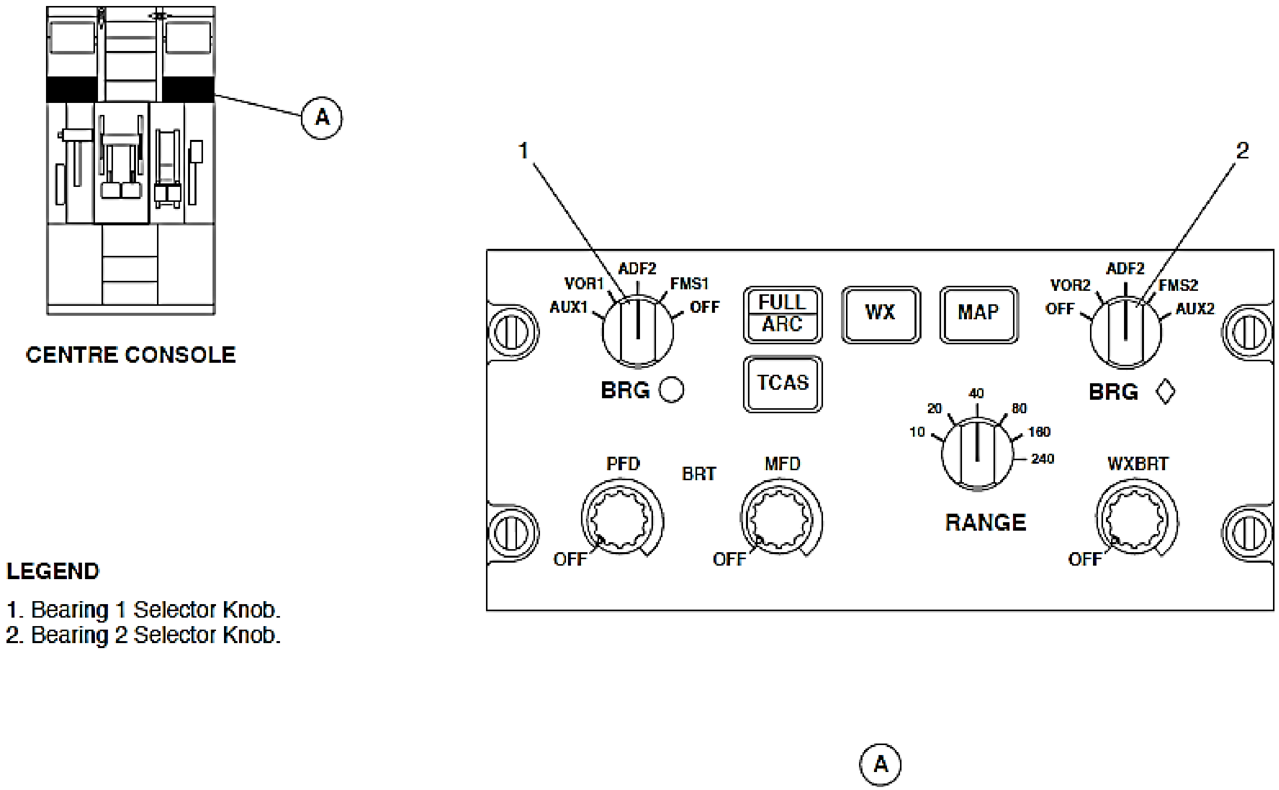


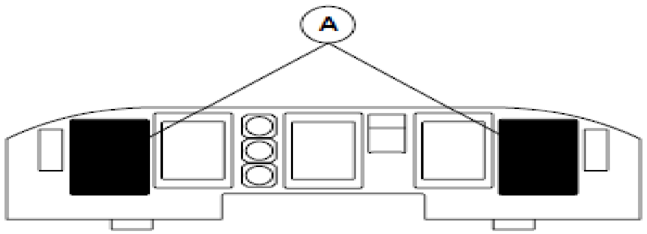
Figure - EFCP ADF BEARING SELECTION

Refer Figure - EIS ADF NORMAL BEARING INDICATION

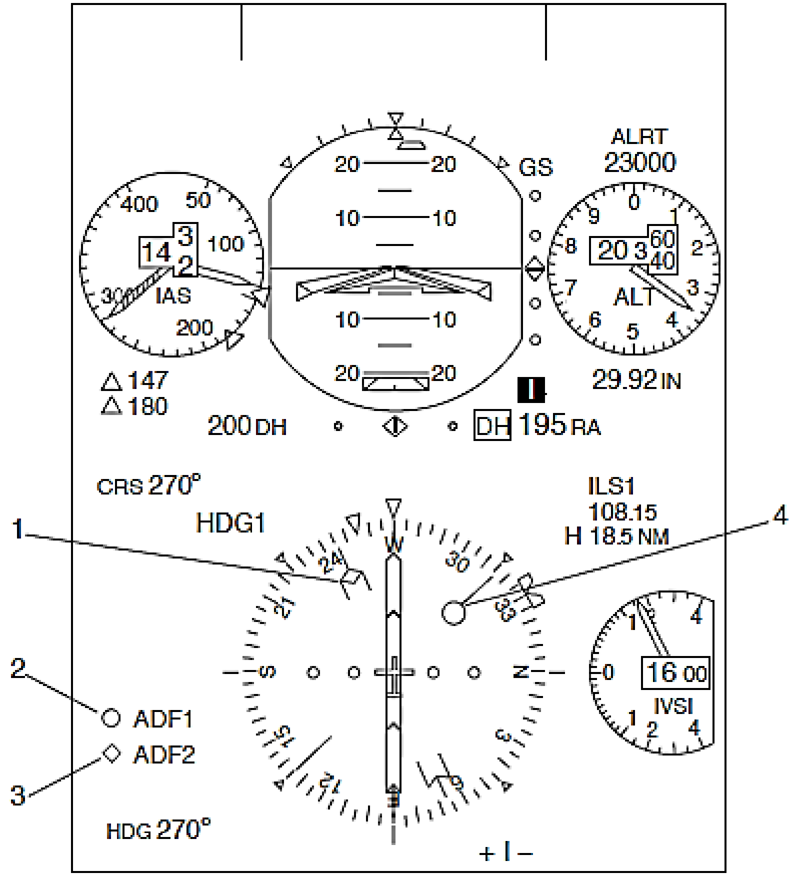
The bearing pointers in the Electronic Horizontal Situation Indicators (EHSI 1, EHSI 2) show the lateral direction to the navigation aid as follows:

- A white, single bar pointer with a white circle to identify the ADF 1
- A green, wide double–bar pointer with a green diamond to identify the ADF 2.

The lower left portion of the Primary Flight Displays (PFD 1, PFD 2) shows a reminder label. The reminder label is related to the EFIS Control Panel selection. It has the same two symbols in white along with ADF 1 and ADF 2 legends.



MAIN INSTRUMENT PANEL

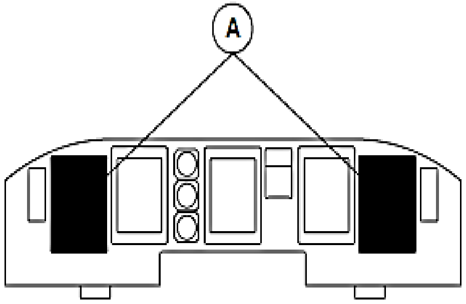


A

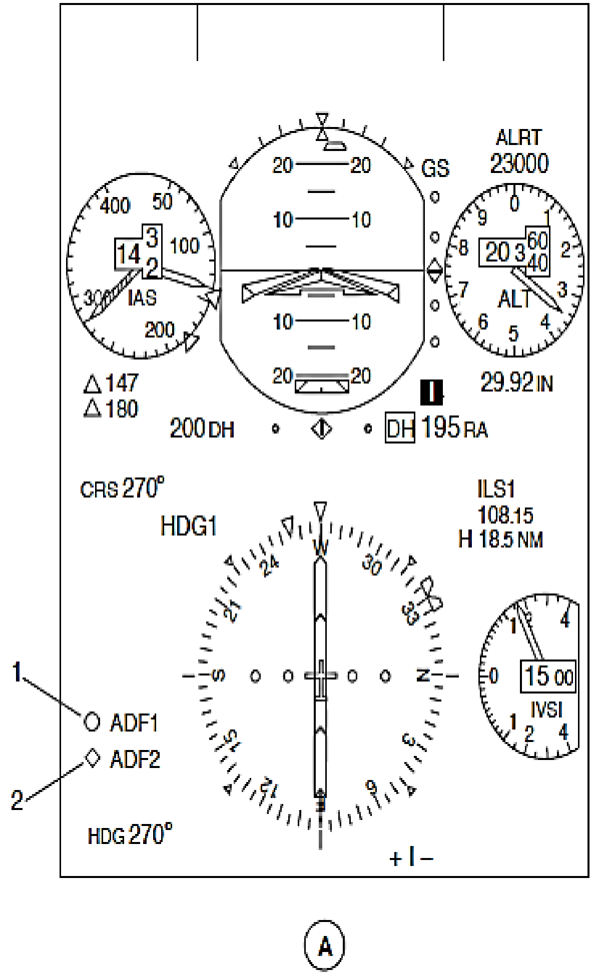
LEGEND

1. #2 Bearing Pointer.
2. #1 Bearing Reminder Label.
3. #2 BearingReminder Label.
4. #1 Bearing Pointer.

Figure - EIS ADF NORMAL BEARING INDICATION



MAIN INSTRUMENT PANEL



LEGEND

1. #1 Bearing Reminder Label.
2. #2 Bearing Reminder Label.

Figure - EIS ADF MALFUNCTION INDICATION

Refer Figure - EIS ADF MALFUNCTION INDICATION

When the ADF signal is invalid or failed, the pointers are removed from the Primary Flight Displays (PFD 1, PFD 2). It continues to display the reminder as the bearing selection.

If the pilot's EFIS Control Panel (EFCP 1) fails, the bearing pointers default to show the information that follows:

- The single bar indicates Automatic Direction Finder (ADF 1)
- The double bar indicates Very High Frequency Omni–range (VOR 2).

If the copilot's EFIS Control Panel (EFCP 1) fails, the bearing pointers default to indicate the following:

- The single bar indicates Very High Frequency Omni–range (VOR 1)
- The double bar indicates Automatic Direction Finder (ADF 2).

The Audio and Radio Control Display Units (ARCDU 1, ARCDU 2) monitor the ARINC 429 data bus through the Flight Data Processing System (FDPS) for a malfunction of the Automatic Direction Finding (ADF) system. The Audio and Radio Control Display Units (ARCDU 1, ARCDU 2) show a red FAIL message. It comes into view in the frequency–preset area of the ADF display area for system failures.

The aircraft Left Essential and Right Main buses supply 28 Vdc electrical power through 2 ampere circuit breakers to the ADF Receivers (ADF 1, ADF 2). The circuit breakers are located in position H3 and H4 on the 28 Vdc Avionics circuit breaker panel.

The Automatic Direction Finding (ADF1, ADF2) systems have two ADF receivers with related ADF antennas.

The ADF Receivers (ADF 1, ADF 2) supply bearing information for display through both Input/Output Processor (IOP 1, IOP 2). The Input/Output Processor (IOP 1, IOP 2) is located inside its related Integrated Flight Cabinet (IFC 1, IFC 2).

The ADF parameters that follows are controlled using ARINC 429 Low speed buses (12.0 to 14.5 kilobits/sec):

Parameter	Source
Frequency	ARCDU
Failure Warning	IOP 1, IOP 2
No Computed data	IOP 1, IOP 2
Normal Operation	IOP 1, IOP 2

Receiver, ADF

Refer Figure - ADF RECEIVER
Refer Figure - ADF RECEIVER LOCATOR

The ADF Receiver receives a combined sense and loop relative bearing signal from the ADF Antenna to calculate the bearing to the station. It uses a self-calibration mode that gives better bearing accuracy and digital filtering to give better bearing stability in severe environmental noise conditions.

The ADF receiver has the sub−assemblies that follow:

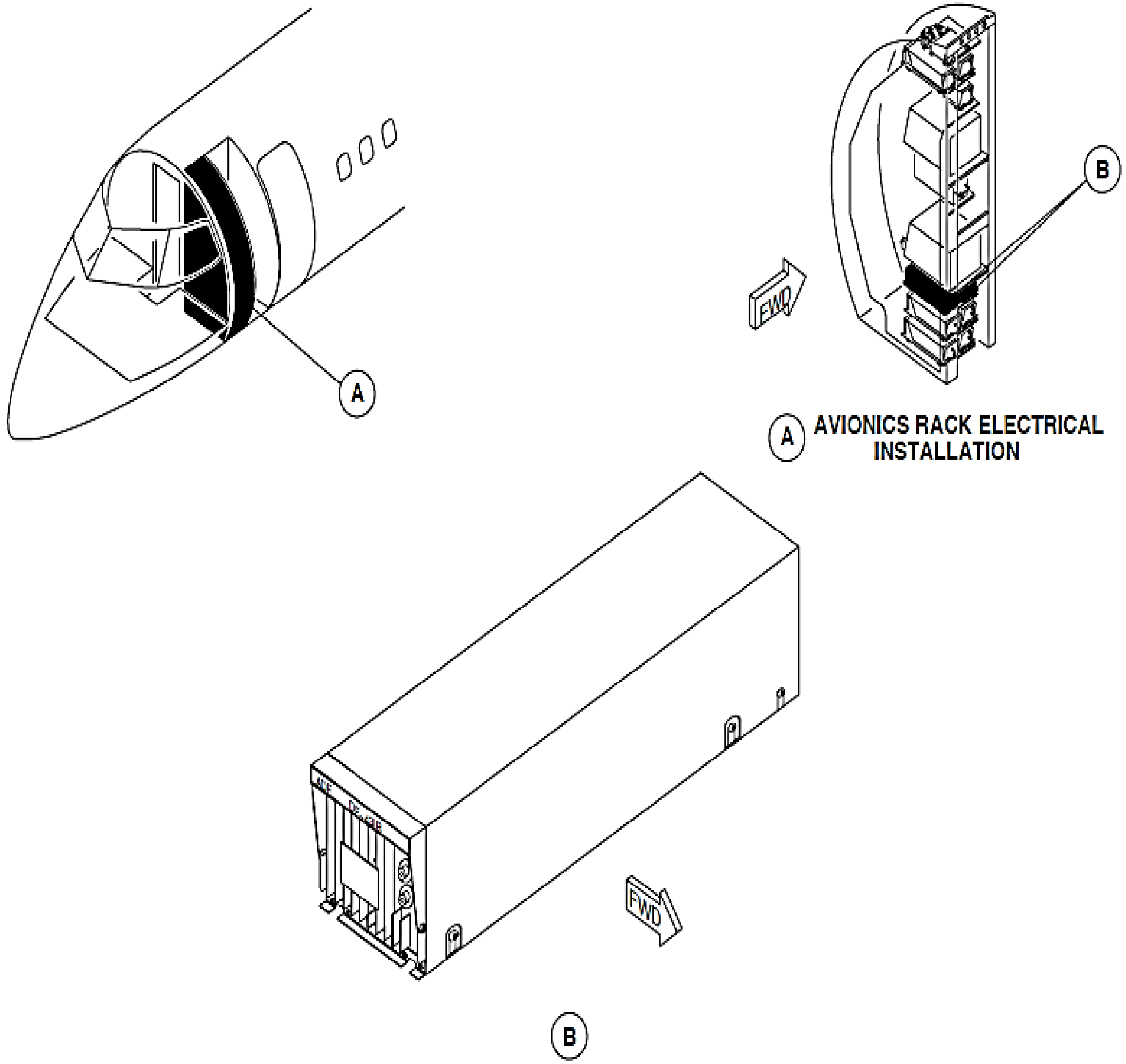
- Receiver Module
- Analog Module
- Input/Output (I/O) Module
- Processor Module
- Power Supply.

The interconnections between assemblies are made using ribbon cables. Maintenance is simplified through modular construction. Modules are hinged and attached to a rigid main frame to allow easy access. The circuit boards plug into connector to allow troubleshooting by substitution. The case has a front panel handle for easy removal and installation.

The ADF receiver weighs 5.82 lb. (2.64 kg). It has subassemblies mechanically packaged into a 4 in. (102 mm) high, 4.1 in. (104 mm) wide, by 13.33 in. (338.6 mm) long case. A single recessed DPX rack−mounted rear−panel connector and two front−panel coaxial RF connectors are installed for connection to the aircraft wire harness and antenna installation.

The ADF receivers are installed in mounting racks side by side on the third shelf of the avionics rack at station points X −28.00 and X −34.00.

Figure - ADF RECEIVER



ta01.cgm

Figure - ADF RECEIVER LOCATOR

Antenna, ADF

Refer Figure - ADF ANTENNA LOCATOR

The ADF antenna receives radio frequencies between 190 and 1860 kHz and between 2181 and 2183 kHz. The antenna has electrical circuits that sum the loop and sense signals to make an Intermediate Frequency signal. The Intermediate Frequency signal is sent to the ADF receiver.

The ADF antenna weighs five pounds. It has a combined loop and sense element with a solid−state amplifier. The antenna enclosure has a low profile to reduce aerodynamic drag. It is 2.85 in. (72.4 mm) high, 6.70 in. (170.2 mm) wide, by 13.80 in. (350.5 mm) long.

The ADF antennas are attached with six mounting bolts to the bottom of the aircraft fuselage along the centerline at station points X 246.00 and X 291.00.

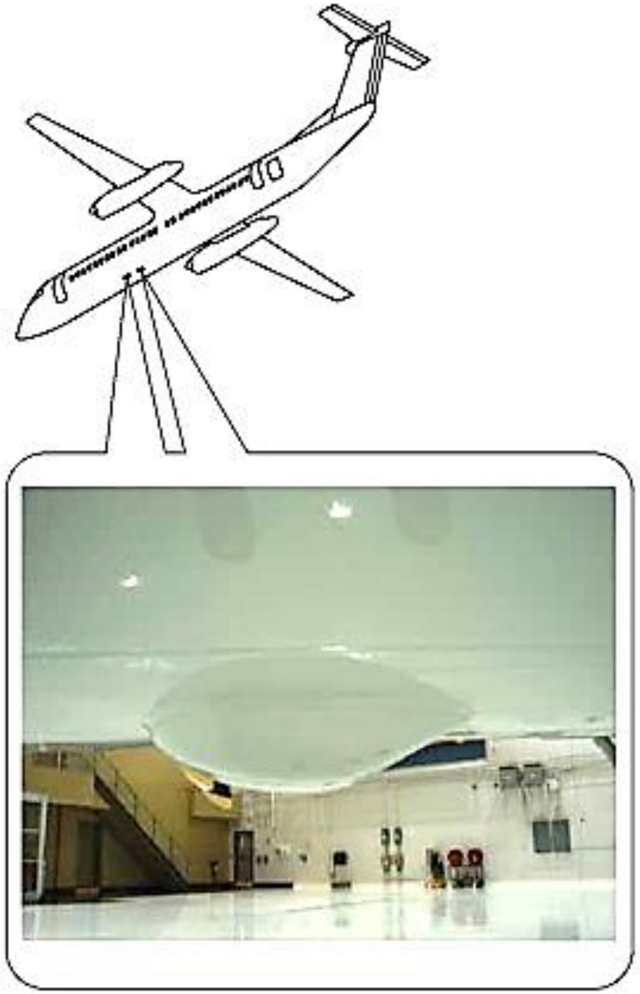


Figure - ADF ANTENNA LOCATOR

Antenna Cable

The Automatic Direction Finding (ADF) systems use twinax antenna cables to connect the ADF receiver to its respective antenna. The twinax cable is a transmission line with two conductors twisted together as a balanced wire line with a copper shielding braid around both wires. Two insulated braids are connected to the system ground and to the airframe ground. The two balanced signal−carrying wires are twisted to cancel any random induced voltage pickup. The antenna cable length is not critical.

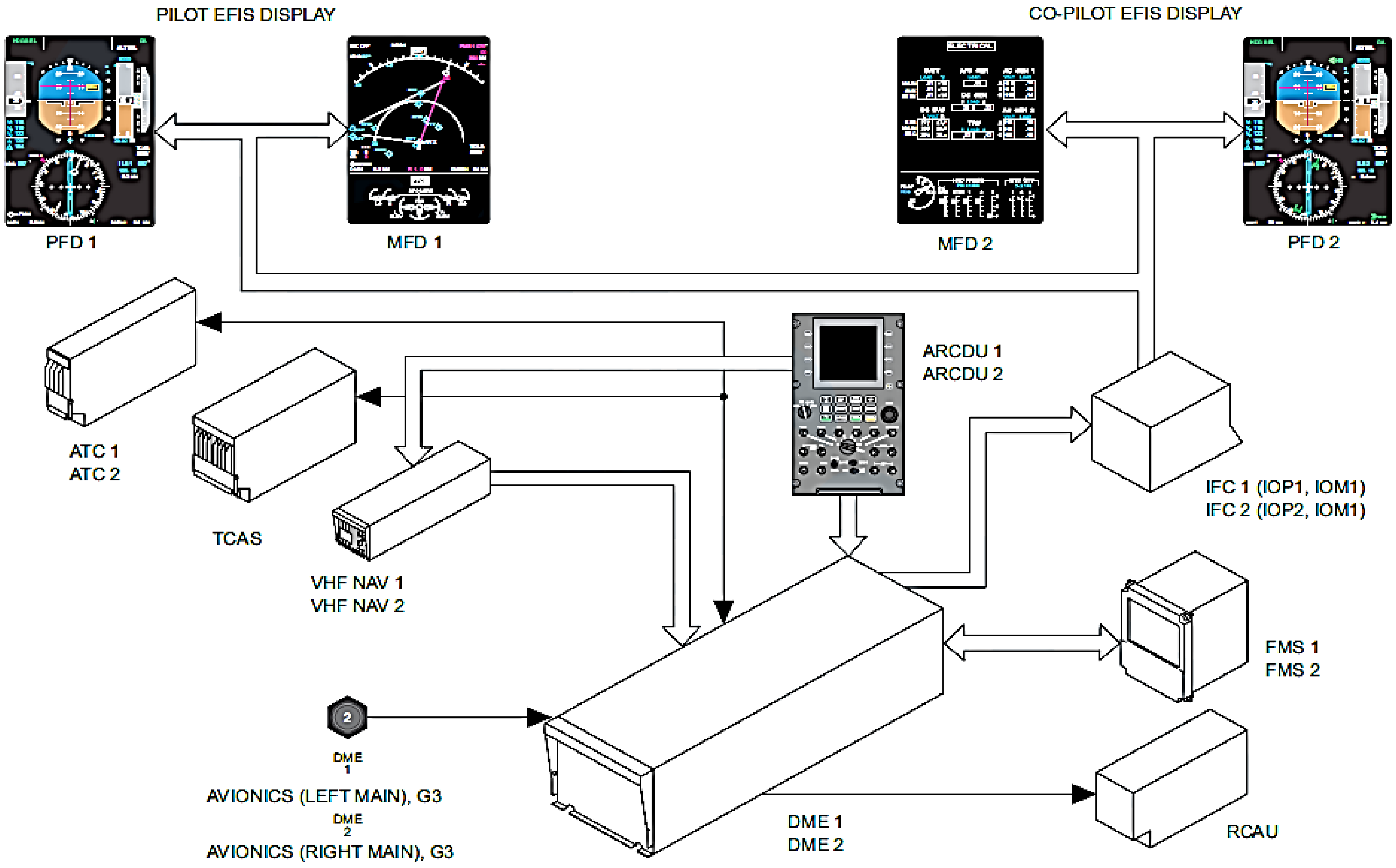


Figure - DME BLOCK DIAGRAM

DISTANCE MEASURING EQUIPMENT (DME)

Introduction

The airborne Distance Measuring Equipment (DME) measures the line of sight slant range from the aircraft to the DME ground station. It uses the slant distance data to calculate the ground speed and the time−to−go parameters.

General Description

Refer Figure - DME BLOCK DIAGRAM

The Distance Measuring Equipment (DME) system transmits Radio Frequency (RF) pulse−pairs to a ground station.

It measures the time it takes for the signal to travel to the station and back to make the calculations that follow:

- Distance to station
- Ground Speed
- Time−to−go.

The VHF Navigation Receivers (VHF NAV1, VHF NAV2) tune the Distance Measuring Equipment (DME1, DME2) from an Audio and Radio Control Display Units (ARCDU1, ARCDU2) selection. A DME HOLD selection is used to keep the current DME station active when a new frequency is selected. The Microwave Landing System Receivers (MLS1, MLS2) tune the Distance Measuring Equipment (DME1, DME2) to its paired channel selection when the Microwave Landing System is used.

The Audio and Radio Control Display Units (ARCDU1, ARCDU2) are used to make the DME audio selections.

The observer uses an Audio Control Panel (ACP) to set the DME audio.

The Electronic Flight Instrument System (EFIS1, EFIS2) shows the Distance Measuring Equipment (DME) parameters that follow:

- Slant Range
- Time To Station
- Ground Speed
- DME Hold.

The Distance Measuring Equipment (DME) system supplies data from multiple ground stations to the Flight Management System (FMS) to calculate its position−fixing algorithm.

The Distance Measuring Equipment (DME1, DME2) has the components that follow.

- Transceiver, DME
- Antenna, DME.

The Distance Measuring Equipment (DME1, DME2) supplies data to the systems that follow:

- Audio and Radio System (ARMS)
- Input/Output Processors (IOP1, IOP2)
- Traffic Collision Avoidance System (TCAS)
- ATC Transponder (ATC1, ATC2)
- Flight Management System (FMS).

Detailed Description

The airborne Distance Measuring Equipment (DME) interrogates the ground station with pulse pair transmissions spaced either 12 or 36 microseconds apart. The ground station replies with pulse pairs transmissions spaced either 12 or 30 microseconds apart. The ground station has a built in 50 microsecond delay between reception of an interrogation and transmission of the reply to allow operation at close range.

The airborne DME senses the time interval between transmission of an interrogation and reception of the reply to calculate the slant range.

The Distance Measuring Equipment (DME) ground station transmits at a constant Pulse Repetition Frequency (PRF) of 2700 pulse pairs per second. It consists of replies to interrogations and random pulses called squitter. If the DME ground station receives more than 2700 interrogations per second, it will reply to the stronger interrogation rather than increase its Pulse Repetition Frequency (PRF) rate. The airborne DME contains circuitry that lets it identify replies to its own interrogations from squitter or replies to other Distance Measuring Equipment (DME).

The airborne Distance Measuring Equipment (DME) operates in the search mode when its transmissions are sent and replies are not received. In the search mode, the DME transmits about 100 pulse pairs per second.

Once the Distance Measuring Equipment (DME) receives replies from the ground station, it operates in the tracking mode and reduces the interrogations to about 30 pulse pairs per second.

The Distance Measuring Equipment (DME) must receive replies from the selected station within 40 seconds. It will not transmit again until it receives squitter from the selected ground station.

The DME transceivers (DME1, DME2) operate within the frequency range of 962 to 1213 MHz each transceiver uses two hundred DME channels that are paired with VHF Navigation frequencies between 108.00 and 117.95 MHz The channels are divided into 100 X and 100 Y channels. The X channel spacing of the transmitted pulse−pairs is 12 microseconds for both the interrogation and reply signals. The Y channel spacing is 30 microseconds for the interrogation and 36 microseconds for the reply.

The DME ground stations transmits a three−letter code group as an identifier.

When a VHF Omni−range (VOR) or Instrument Landing System (ILS) frequency selection is made, the navigation receiver automatically tunes a paired DME frequency.

The Radio Control Display Units (ARCDU1, ARCDU2) are used to control the VHF Navigation and paired DME frequencies.

Refer Figure - ARCDU DME HOLD FUNCTION

The DME HOLD key is pushed on the Audio and Radio Control Display Unit (ARCDU1, ARCDU2) to hold the DME channel active while a new VHF Navigation frequency is selected.

The ARCDU shows the held frequency adjacent to a DME hold legend. The held frequency is shown in a green font on a white background below the active VHF NAV frequency.

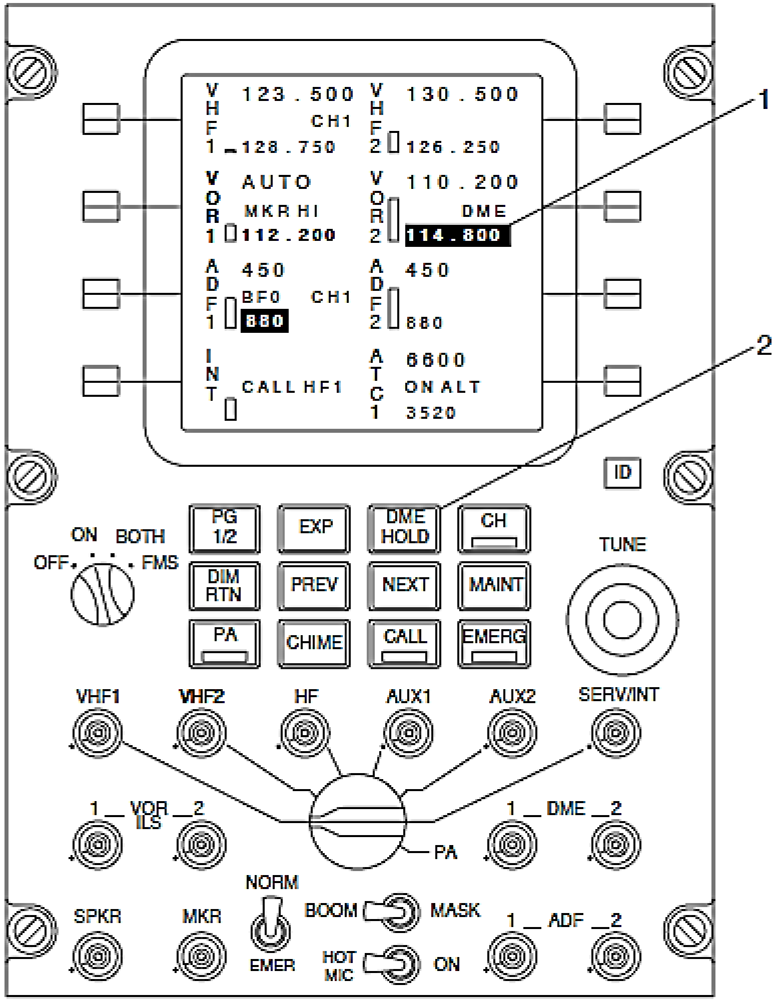
The Primary Flight Display (PFD1, PFD2) and related Navigation Page show a yellow H to the left of the displayed distance. It shows that the Distance Measuring Equipment (DME) is not tuned to the station frequency that is shown directly above it.

When the DME Hold key is pushed again it cancels the DME hold function.

NOTE

The conditions that follow also cancel the DME hold function:

- The Microwave Landing System (MLS) is selected as the navigation source
- The Flight Management System (FMS) is autotuning



LEGEND

1. DME Hold Indication.
2. DME Hold Key.

Figure - ARCDU DME HOLD FUNCTION

Refer Figure - ARCDU DME AUDIO CONTROL

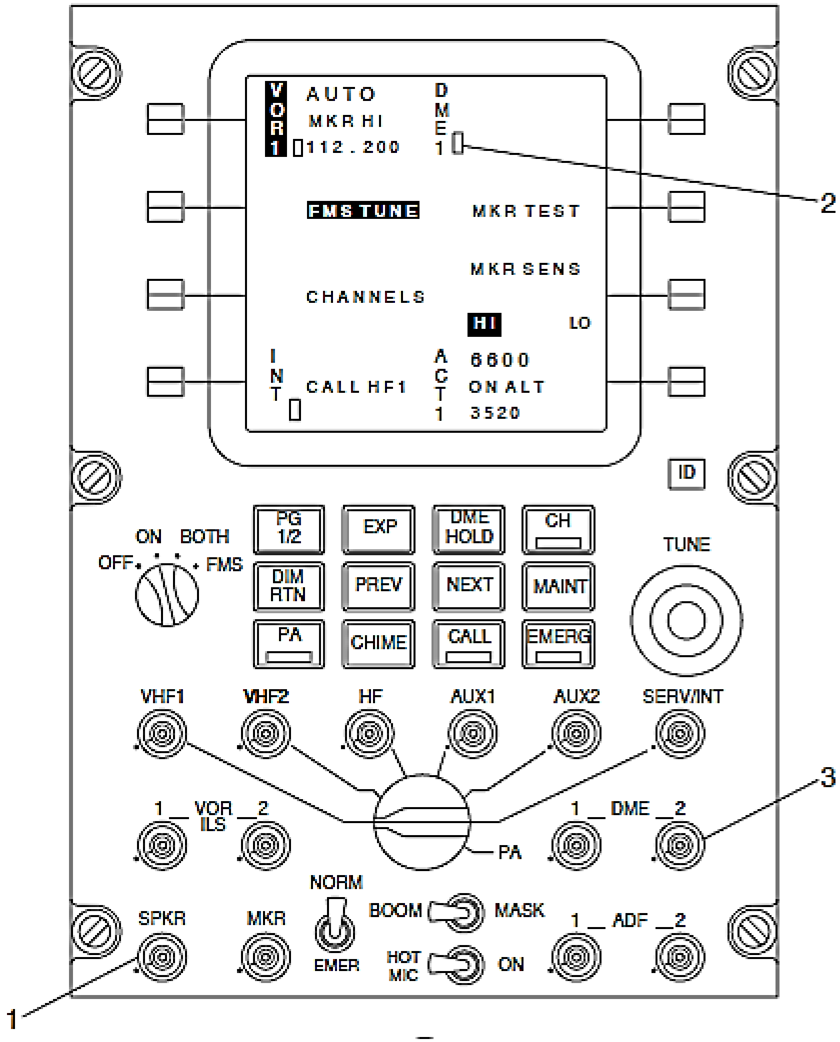
The DME Potentiometer push–button switches give the audio control functions that follow:

- On
- Off
- Level.

The top part of the potentiometer is pushed to set the specific audio signal to the headphone on or off. The DME1 or DME2 push–button switch is pushed on the Audio and Radio Control Display Unit (ARCDU1, ARCDU2) to make its audio available to that flight crew member. When the SPKR push–button potentiometer switch is pushed, the DME audio is routed to its related flight compartment speaker.

The DME display areas shows the volume level as a vertical bar graph. The height of the bar graph shows the volume selection level. When the audio is selected off, the bar graph displays in white. The bar graph changes to green when the audio is selected on.

The DME audio is available at the observer's headphone jack when the DME 1or DME 2 is selected on the observer's Audio Control Panel (ACP).



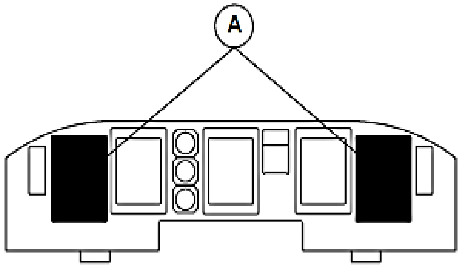
LEGEND

1. Speaker Potentiometer / Push–Button Switch.
2. Level Adjust Bar Graph.
3. DME Potentiometer / Push–Button Switches.

Figure - ARCDU DME AUDIO CONTROL

Refer Figure - PFD DME NORMAL INDICATION

The upper right part of the Electronic Horizontal Situation Indicator (EHSI) area of the Primary Flight Display (PFD1, PFD2) shows the DME distance in a blue font. It shows the distance to station information from 0 to 300 nmi. The range resolution is 0.1 nmi for distances to 99.9 nmi and 1 nmi for distances greater than 99.9 nmi.



MAIN INSTRUMENT PANEL

LEGEND

1. DME Distance.
2. DME Hold Annunciation.

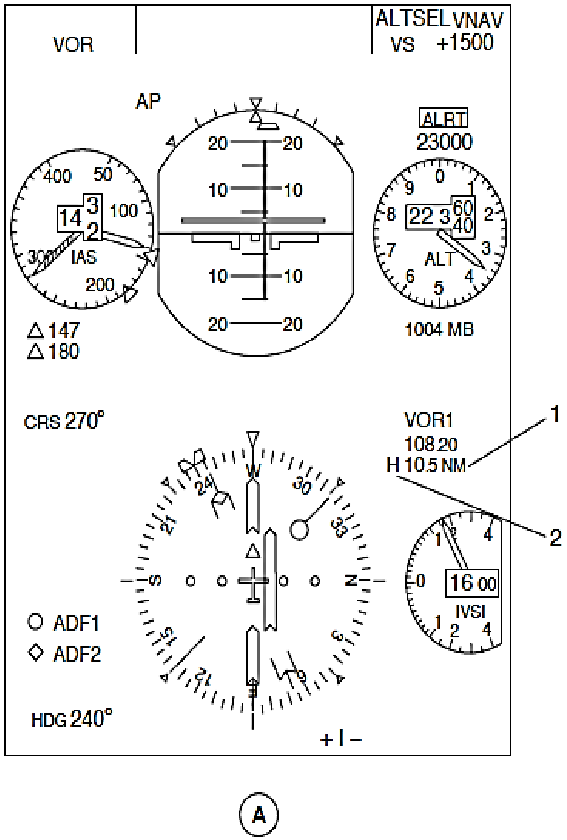


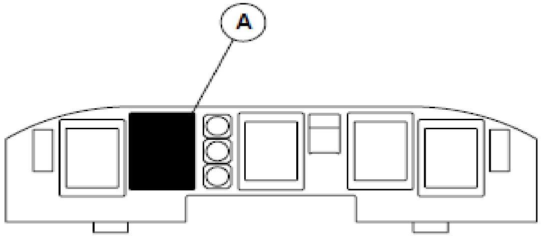
Figure - PFD DME NORMAL INDICATION

Refer Figure - MFD DME NORMAL INDICATION

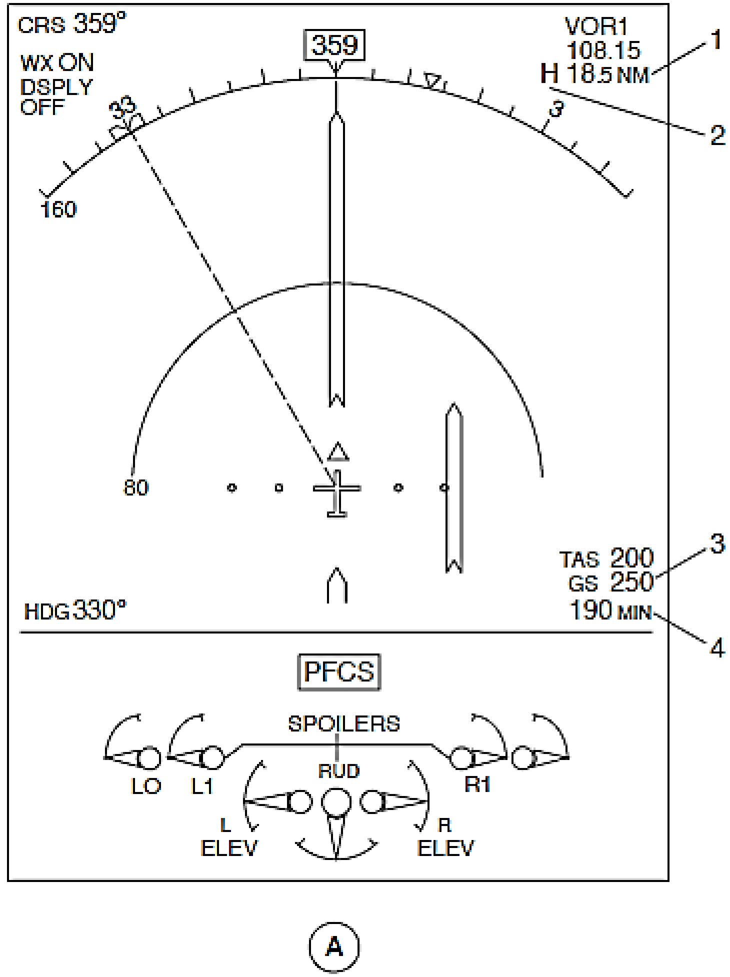
The Navigation page of the Multi−Function Displays (MFD1, MFD2) shows the DME functions that follow:

- Distance
- GS, Ground Speed
- Time to station.

The distance is shown in the same location on the MFD as on the PFD. The ground speed and time to station are shown in a blue font in the lower right area of the Navigation Page.



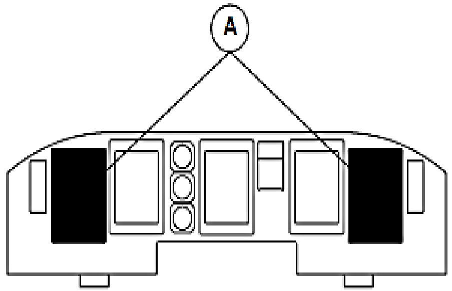
MAIN INSTRUMENT PANEL



LEGEND

1. DME Distance.
2. DME Hold Annunciation.
3. Ground Speed Digital Value.
4. Time To Go Digital Value.

Figure - MFD DME NORMAL INDICATION



LEGEND

1. DME Fail Annunciation.

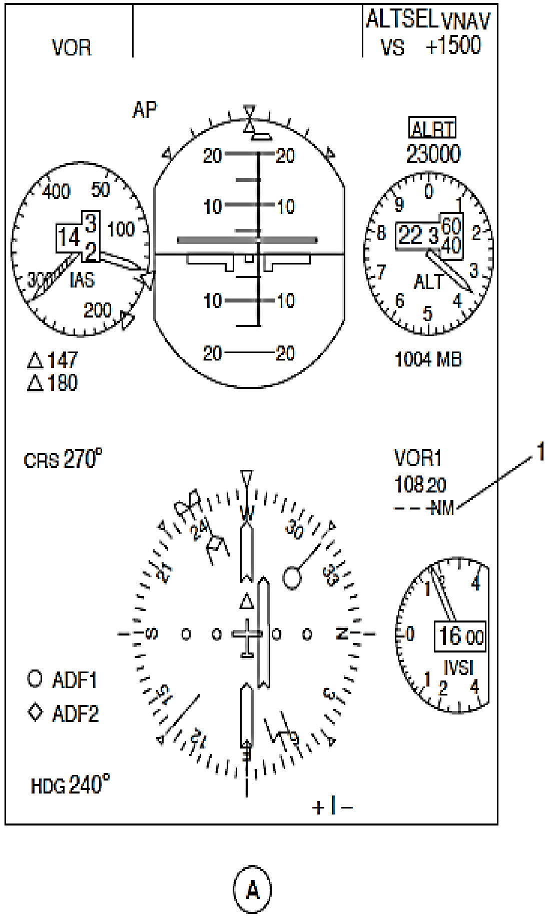
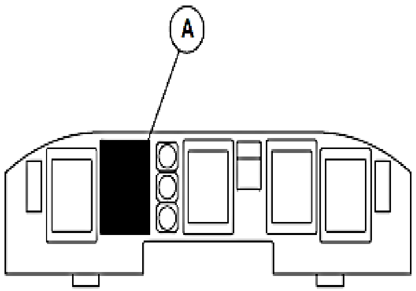
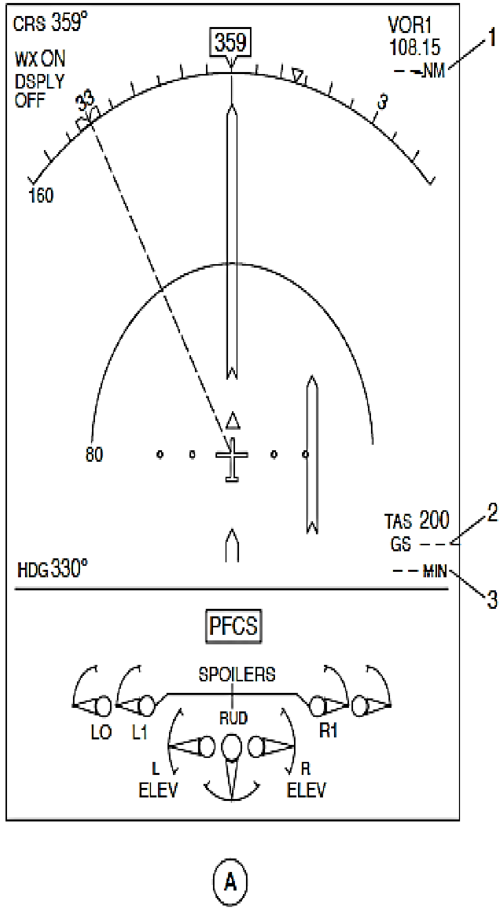


Figure - PFD DME FAIL INDICATION



MAIN INSTRUMENT PANEL



LEGEND

1. DME Fail Annunciation.
2. Ground Speed Fail Annunciation.
3. Time To Go Fail Annunciation.

Figure - MFD DME FAIL INDICATION

Refer Figure - PFD DME FAIL INDICATION
Refer Figure - MFD DME FAIL INDICATION

The numerical values for all indications change to three white dashes when there is a source failure or the distance is out of range.

If the DME fails, the Audio and Radio Control Display Unit (ARCDU1, ARCDU2) shows a FAIL legend in red characters below the active frequency.

The Left and Right Main DC buses supply 28 Vdc electrical power to the Distance Measuring Equipment (DME1, DME2) through 2 A circuit breakers. The circuit breakers are located in positions G3 and G4 on the 28 Vdc avionics circuit breaker panel.

The DME, ATC transponder and the Traffic Collision Avoidance System (TCAS) operate in the same frequency range. Suppression pulses are output to momentarily suppress the transmissions of the other L–band equipment.

Each Distance Measuring Equipment (DME1, DME2) transceiver interfaces with the systems that follow so that only one system transmits at a time:

- The other DME transceiver
- Traffic Collision Avoidance System (TCAS)
- Transponders (ATC1, ATC2).

Each Distance Measuring Equipment (DME1, DME2) systems has a transceiver with a related antenna.

The Distance Measuring Equipment (DME1, DME2) supplies navigation information for display through the Input/Output Processors (IOP1, IOP2). The Input/Output Processors (IOP1, IOP2) and are located inside its related Integrated Flight Cabinet (IFC1, IFC2).

The DME sends the parameters that follow on an ARINC 429 Low speed data bus:

Parameter	Source
DME Frequency	ARCDU
DME Distance (Range)	IOP1, IOP2
DME Time–to–Station	IOP1, IOP2
DME Ground Speed	IOP1, IOP2

The Distance Measuring Equipment (DME1, DME2) are tuned by the navigation sources that follow:

- VHF Navigation Receivers (VHF NAV1, VHF NAV 2)
- Microwave Landing System (MLS1, MLS2) receivers.

Part of the Flight Guidance Control Panel (FGCP) navigation source selection routes ARINC 429 tuning information to the DME. The navigation source selection causes an Input/Output Module (IOM1, IOM2) discrete to control a relay to route tuning information to the DME. The Input/Output Modules (IOM1, IOM2) are located inside of its related Integrated Flight Cabinet.

The Distance Measuring Equipment (DME1, DME2) supplies data to the Flight Management System (FMS). It uses Distance Measuring Equipment (DME) data for the VORTAC Position Unit (VPU) mode to calculate its position–fixing algorithms

Transceiver, DME

Refer Figure - DME TRANSCEIVER LOCATOR
Refer Figure - DME TRANSCEIVER

The DME Transceiver gives slant range distance data from DME ground stations. The transceiver transmits pulse pair interrogation signals to a ground station which in turn sends back coded reply signals for each interrogation signal received. The DME Transceiver measures the two way transmission delay from the aircraft to the ground station and changes it to a distance value in nautical miles. Microprocessor circuits use the distance data to calculate ground speed and time−to−go data.

Each DME ground station periodically transmits an audio identification signal which is received, decoded, and then sent to the audio system.

The DME Transmitter is tuned by external sources using the ARINC 429 format. The system is capable of tracking three DME ground stations for the Flight Management System (FMS) VORTAC Position Unit (VPU) mode to calculate its position−fixing algorithms.

The power supply module is activated by a hard−wired low−level enable signal at the rear connector.

A single recessed DPX rack−mounted rear−panel connector and a front−panel TNC coaxial connector is used for connection to the aircraft wire harness and the antenna cable.

The DME transceivers are installed in mounting trays one above the other on the two lower shelves of the avionics rack, at approximately station point X −28.0.

The DME transceivers are installed in the avionics rack that is supplied with forced air cooling.

Figure - DME TRANSCEIVER

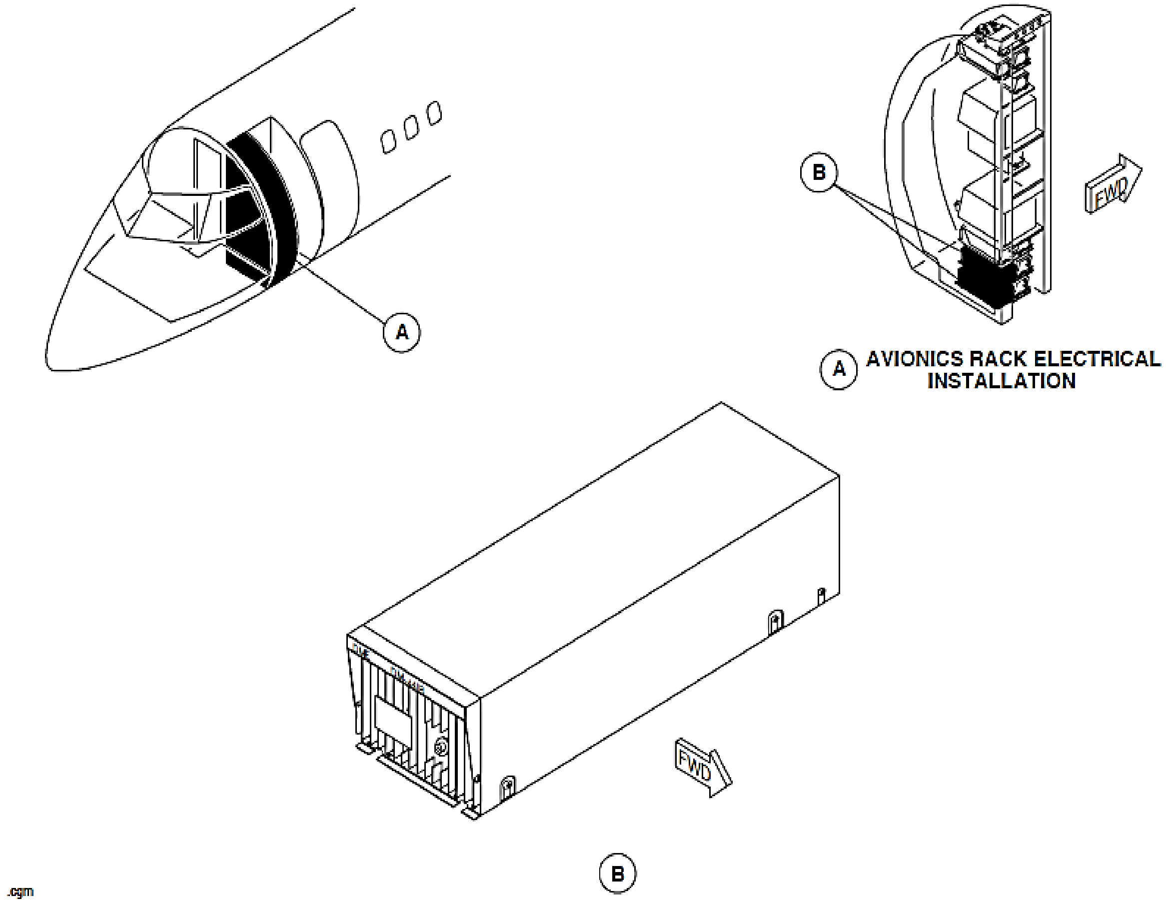


Figure - DME TRANSCEIVER LOCATOR

Antenna, DME

Refer Figure - DME ANTENNA **LOCATOR**
Refer Figure - NO.1 AND NO.2 DME **ANTENNA**

The L−band DME antenna transmits and receives radio frequencies in the 950 to 1250 MHz frequency bands. The antenna has the features that follow:

- Omni−directional
- Quarter wavelength
- Vertically polarized
- Monopole
- End fed.

The L−band antenna is an advanced lightweight durable thermoplastic blade antenna. The antenna encases the pre−formed radiating element within a single assembly. The antenna’s Radio Frequency (RF) coaxial type HN female connector is attached to an aluminum base plate.

The L−band antenna weighs less than 0.25 pounds (0.11 kilograms). The DME ANT 1 is installed in the forward, left−side area below the forward fuselage section at station point’s X −94.41, Y −15.41. DME No.2 antenna is installed on the aircraft centerline at X 479.5. Each antenna is installed to the aircraft with five mounting screws. The antennas are bonded to the aircraft structure using a "wet fit" sealing gasket.

The gasket consists of an aluminum foil gasket with an elastomeric sealant. The pressure applied during installation gives a contour feature. It lets the gasket form itself precisely to the two metallic surfaces to give good conduction between the antenna and airframe. The moisture free, contoured seal requires no maintenance.

Antenna Cable

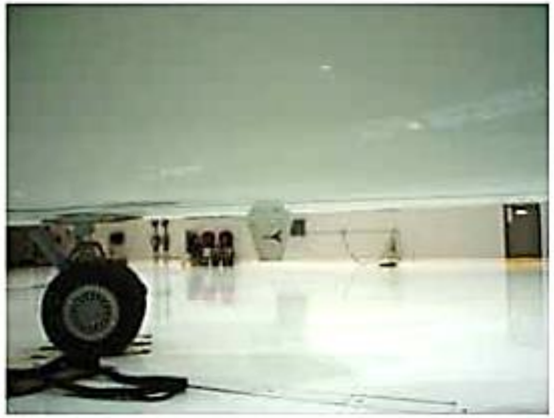
Coaxial cables connect the antennas to the DME transceivers. The coaxial cable is a transmission line with one conductor surrounding another. A dielectric insulator separates the two conductors. The center conductor is seven strand silver−plated copper wire and the outer conductor is a tinned copper wire braid.

Two types of dc resistance coaxial cables are installed because of different cable lengths in the aircraft. The DME2 has a lower loss cable because of the longer run between the transceiver and the antenna.

The table that follows show the difference between the two co−axial cables.

Cable	Attenuation	DC Resistance
DME 1	7dB / 100 ft	1.8 Ohms / 100 ft
DME 2	4dB / 100 ft	0.95 Ohms / 100 ft

No.1 DME Antenna



No.2 DME Antenna

Figure - NO.1 AND NO.2 DME ANTENNA

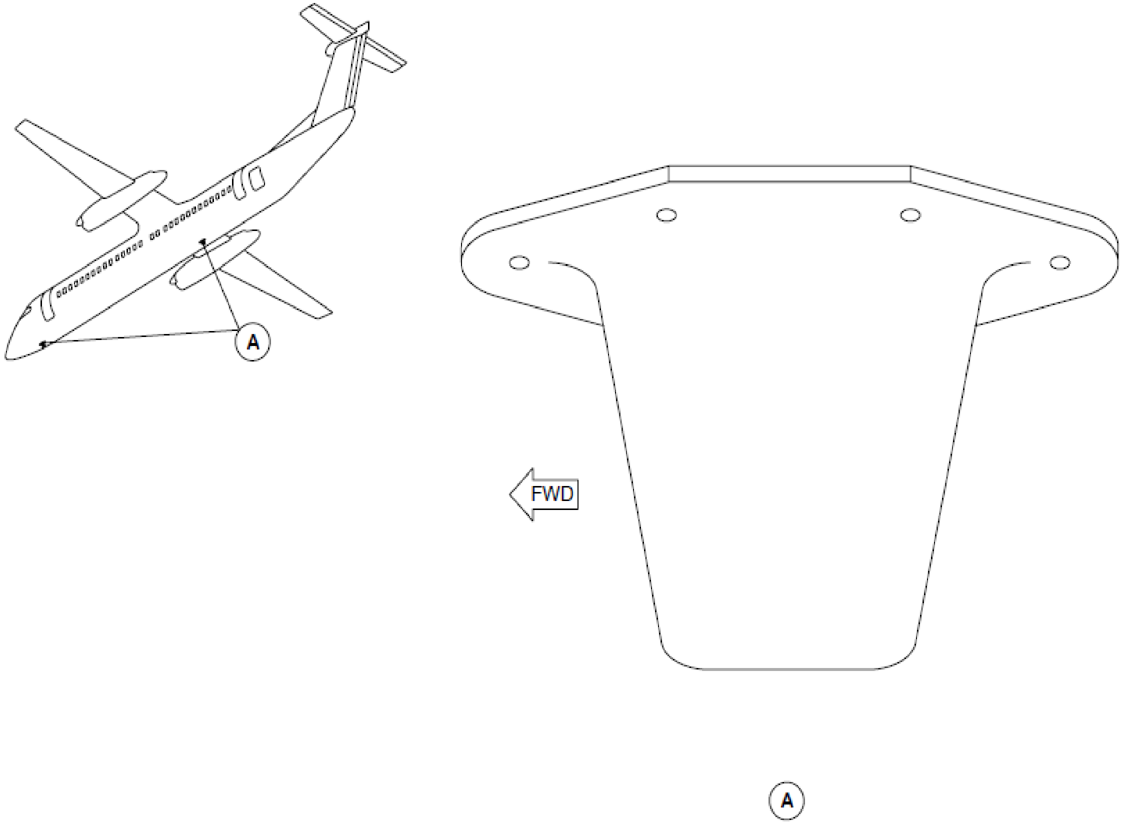


Figure - DME ANTENNA LOCATOR

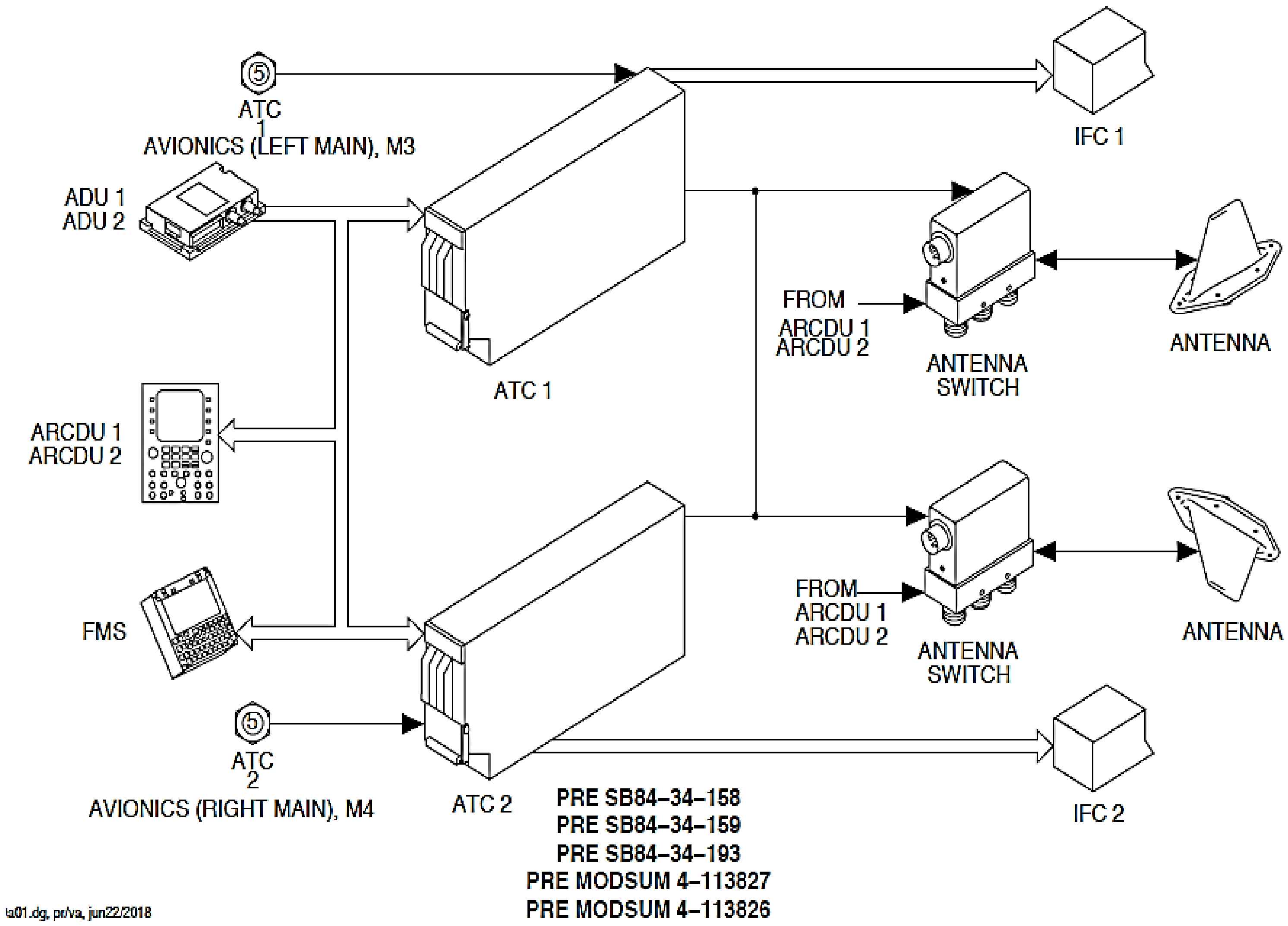


Figure - ATC BLOCK DIAGRAM

AIR TRAFFIC CONTROL TRANSPONDER

Introduction

The Mode S transponder system transmits a signal to identify the aircraft and its altitude parameters to Air Traffic Control (ATC) ground stations and to other aircraft equipped with Traffic Alert and Collision Avoidance System (TCAS).

General Description

Refer Figure - ATC BLOCK DIAGRAM

The Mode S transponder replies to ATC ground station interrogations and other TCAS equipped aircraft. It transmits the information that follows:

- Altitude
- Maximum airspeed
- TCAS
- Mode S Identifier of the aircraft.

In addition, the interrogating ATC ground station or other aircraft fitted with TCAS determines aircraft's range and azimuth.

The Mode S transponder system has two transponders (ATC1, ATC2).

One transponder operates in the active mode and the other will automatically operate in the standby mode.

The active Mode S transponder is selected to operate in one of the sub−modes that follow:

- Air Traffic Control Radar Beacon System (ATCRBS), Mode A and Mode C
- Mode S Surveillance
- Test.

In addition, the pilots can select the altitude reporting data source.

The Mode S transponder system has the components that follow:

- Receiver, Mode S Transponder
- L−Band Antenna, Transponder
- Coaxial Cable Switch, Antenna
- Switch, ATC ID.

A coaxial cable switch connects each transponder to an upper and lower fitted L−band omni−directional antennae.

The Mode S transponder system interfaces with both of the Audio and Radio Control Display Units (ARCDU1, ARCDU2) and the Flight Management System (FMS). Pilot inputs to the ARCDU and the FMS control the operating modes of the transponder. In addition to mode selections, the ARCDU1 and the FMS sends a pilot selected four−digit octal code to the transponders.

The Mode S transponder system interfaces with the Air Data System (ADS). It converts altitude information from the ADS into formats required for Mode C and Mode S replies.

Detailed Description

The Mode S transponder functions in air traffic alert and collision avoidance environments.

The Mode S transponder receives 1030 MHz interrogations on the aircraft top and bottom antennas from ground ATC ground stations and from TCAS equipped aircraft. The transponder decodes and processes the interrogations and if required, transmits its replies at 1090 MHz in one of the modes that follow:

- ATCRBS
- Mode S Surveillance
- TCAS Environment
- Squitter Transmission (Unsolicited transmissions).

The replies have data that relates to the altitude, maximum airspeed, TCAS, and Mode S Identifier of the aircraft. The replies will be transmitted from the antenna that had the stronger interrogation signal strength to more accurately determine the aircraft position.

ATCRBS Environment: The airborne Mode S transponder system is part of the ATCRBS. It transmits a coded response to a coded interrogation from an ATC ground station.

There are three types of radar at each the ATC ground stations:

- Primary Surveillance Radar (PSR)
- Secondary Surveillance Radar (SSR)
- Omni−directional.

The PSR determines the relative bearing and distance of the aircraft. It operates on the normal radar principle to process energy reflected from the aircraft under surveillance.

The SSR determines the aircraft parameters that follows:

- Identity
- Distance
- Altitude.

The SSR transmits two pulses on a 1030 MHz carrier to interrogate the airborne transponder. The airborne transponder replies

- Mode A, Code
- Mode C, Altitude.

The transponder must reply to ATCRBS interrogations from the transmitting antenna main beam and not from the side lobes

.

The transponder replies to Mode A interrogations with a pilot selected four−digit octal code. After the Mode A reply, an 18 second Special Purpose Identifier (SPI) can be transmitted when one of the ATC ID switches is pushed.

The transponder replies to Mode C interrogations. It transmits aircraft barometric altitude information.

The SPI pulse is transmitted when the ID key is pushed on the ARCDU to give a positive aircraft identification to ATC.

Mode S Air Traffic Control Environment: When an aircraft equipped with a transponder enters the airspace that is monitored by ground ATC facilities or TCAS equipped aircraft, it is interrogated by an ATCRBS/Mode S All−Call interrogation. The ATCBRS transponders reply with a Mode A or Mode C format and the Mode S transponders as installed on this aircraft reply with a Mode S format that includes a unique address. The address with the aircraft location is entered into a roll call file. On the next interrogation, that Mode S transponder is interrogated. The azimuth and range are known for all aircraft on roll−call. Each aircraft is interrogated according to a precomputed schedule. The schedule has ATCRBS/Mode S All−Calls that provide tracking of known ATCRBS aircraft and the acquisition of more ATCBRS and Mode S aircraft that enter the airspace.

TCAS Environment: The Mode S transponder is part of the TCAS. It supplies an air−to−air data link for altitude reporting information for the TCAS.

The Mode S transponder transmits a squitter signal to identify itself. It is the initial indication that a Mode S equipped aircraft has entered a surveillance area. ATC ground stations and TCAS equipped aircraft monitors the 1090 MHz frequency for squitter. After a valid identification squitter is received, the aircraft's unique Mode S address is added to a list of aircraft the other TCAS equipped aircraft will interrogate.

The aircraft also receives squitter signals from other Mode S equipped aircraft. Its TCAS compiles a list of Mode S equipped aircraft in the area to interrogate. In addition, the TCAS processor transmits interrogations to any transponder Mode A or Mode C equipped aircraft in the area. These aircraft respond to the ATCRBS type interrogation.

Squitter: The Mode S transponder transmits its unique address at a periodic rate, once every second known as "squitter". The transmissions alternate between the upper and lower antennas. Other aircraft equipped with TCAS receive the squitter signal and then interrogate the Mode S transponder equipped aircraft based on the address received from the squitter. These interrogations are used for Mode S target tracking and collision threat assessment.

The squitter transmissions have the data that follow:

- A unique Mode S address
- Altitude
- Equipment capabilities of the sending aircraft

The ARCDU1, ARCDU2 or the FMS control the Mode S transponders (ATC1, ATC2).

If the ARCDU1 fails and switched to OFF, then the ARCDU2 or the FMS maintains and gives full transponder control from pilot commands.

The transponder system defaults to ATC 1 the conditions when both ARCDU malfunction, and both ARCDUs transfer switches are set to the OFF position.

The transponder uses default ARCDU control logic and pilot commands through the FMS. The opposite Mode S transponder,
ATC2 operates in standby.

The Mode S transponders (ATC1, ATC2) functions in the modes that follow:

- Active (ATC1 or ATC2)
- Standby (SBY)
- Initiated Test (TEST).

Active: The selected Mode S transponder receives interrogations through the upper and lower antennae from ATC ground stations and from TCAS equipped aircraft. The transponder decodes and processes the interrogations and if required, replies in Mode A, C, or S format. The antenna that receives the stronger interrogation signal strength is used to transmit replies. The transponder operates in the L radio Frequency band. It receives interrogations at 1030 MHz and transmits replies at 1090 MHz.

Standby Mode: The Mode S transponder functions in the standby mode when it is not selected. The Mode S transponder stays in a powered state in the standby mode. Its Built In Test (BIT) does a Radio Frequency (RF) loop−around checkout in addition to the normal continuous test routine. The Mode S transponder will not respond to any ATCRBS or Mode S interrogations when in the standby operation mode.

Initiated Test Mode: The selected Mode S transponder will not respond to any ATCRBS or Mode S interrogations when in the test operation mode.

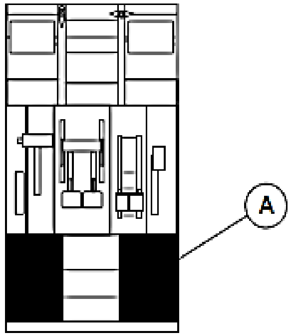
The Mode S transponder continuously carries out system fault isolation. It finds the source of a failure at the component or Aircraft wiring level.

The transponder sends a failure message to the ARCDUs through a data bus. A Light Emitting Diode (LED) code inside the transponder indicates a control data failure. When a failure that does not affect the ATC surveillance function is detected, the transponder will not send a failure message to ARCDU.

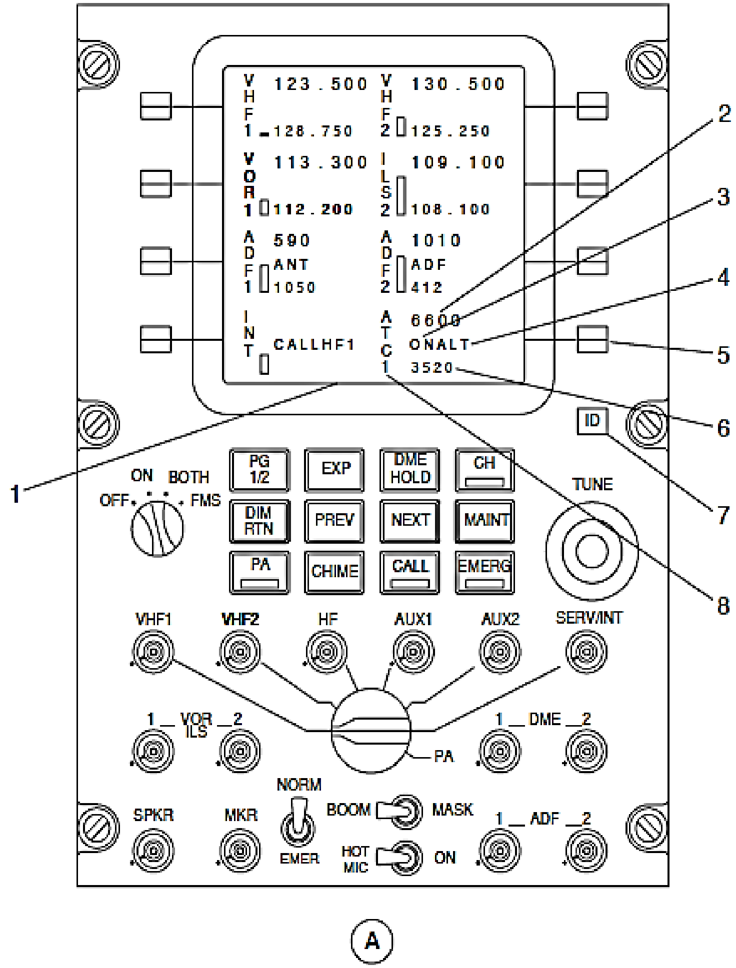
Each Mode S transponder has an initiated and continuous Built In Test Equipment (BITE) capability. It includes an RF loop−around test that is only executed when a transponder is in the standby operation mode. Each transponder reports its overall status and its interfaces with other systems to each ARCDU through a discrete health status interface. Each ARCDU shows the status of each Mode S transponder.

The Central Diagnosis System (CDS) continuously monitors the discrete health status output from each Mode S transponder.

If the Mode S transponder determines a failure that causes the system not to function for ATC purposes, the ARCDUs show "FAIL" in the code area of the display.



CENTRE CONSOLE



LEGEND

1. Display Area.
2. Active Code.
3. Transponder Mode.
4. Altitude Reporting.
5. Side Key.
6. Preset Code.
7. Specific Purpose Idendifier (SPI) Key.
8. Label.

cgm

Figure - AIR TRAFFIC CONTROL TRANSPONDER, ARCDU

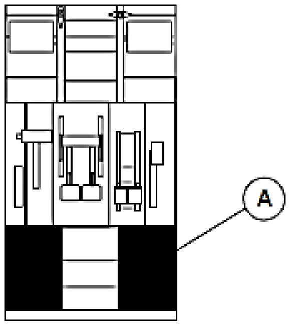
Refer Figure - AIR TRAFFIC CONTROL TRANSPONDER, ARCDU

The ARCDUs selections control the ATC transponder mode and codes to be used.

SELECTION	DESCRIPTION
OFF	The ARCDU is not powered. The related FMS controls and tunes as a backup
ON	The ARCDU controls and tunes its related radio system
BOTH	The ARCDU controls and tunes its related and opposite radio systems
FMS	The FMS controls and tunes its related and opposite radio systems

The ATC display area is shown on both main pages and all particular pages. A double horizontal line separates the ATC display area from the particular pages. The main page shows the information that follows in the ATC display area.

- Active ATC label
- Active Code
- Mode, On or Standby
- Altitude, On or Off.
- Specific Purpose Identifier (ID).



CENTRE CONSOLE

LEGEND

1. Side Key.
2. Tuning Knob.

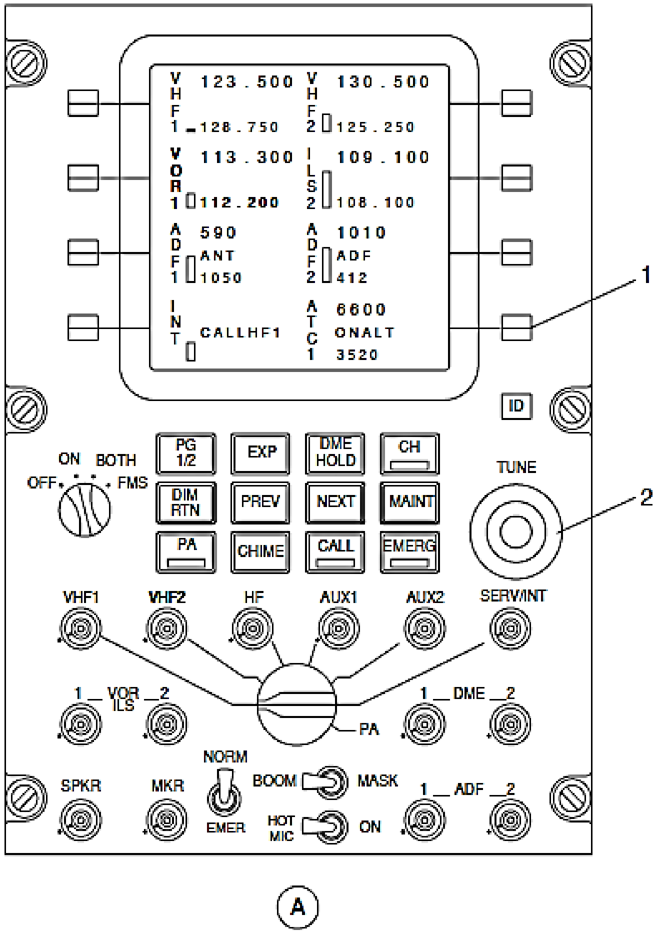


Figure - ARCDU ATC MODE A CONTROL

Refer Figure - ARCDU ATC MODE A CONTROL

A minimum of two steps are required to change the active code.

The side key adjacent to the ATC display area is pushed to highlight the preset code. It is pushed again to make the preset code active. The code that was active now becomes the new preset code.

When the side key adjacent to the ATC display area is pushed, the preset code in the ATC transponder display area highlights. The TUNE double rotary knobs located at the lower right side of the ARCDU are turned to change the preset code. The Side key is pressed again to set the new active code. The code that was active now becomes the new preset code.

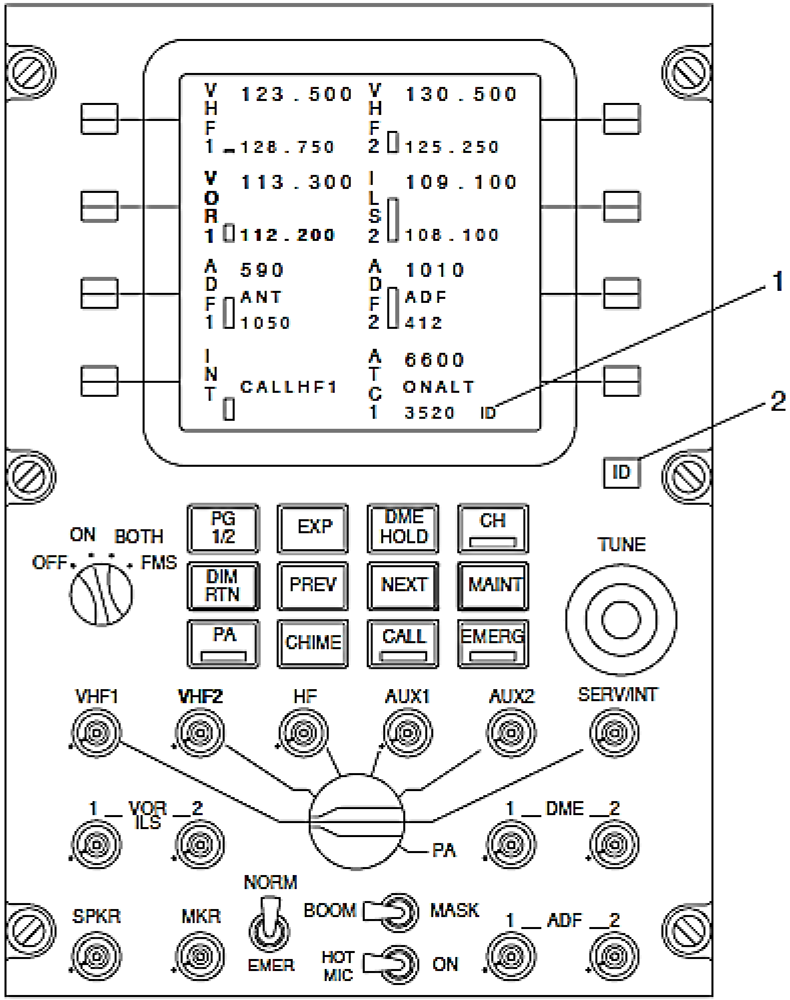
When a side key is selected, the flight crew must follow up that selection with one of the following actions within five seconds:

- Rotate either TUNE double rotary switch
- Push the same side key again

If not, the previous preset data to restores.

Refer Figure - ARCDU ATC IDENTIFIER CONTROL

One ARCDU or handwheel located ATC IDENT push−button switch is pushed to cause the transponder to transmit an identifier for 17 seconds. The ATC display area shows the Specific Purpose Identifier (SPI) as an ID legend in green fonts for the duration of the identifier.



LEGEND

1. Specific Purpose Identifier.
2. Specific Purpose Identifier (SPI) Key.

Figure - ARCDU ATC IDENTIFIER CONTROL

Refer Figure - ARCDU ATC PARTICULAR PAGE ACCESS

The ATC Particular Page comes into view when the ATC side key is pushed followed by the EXP key on the keyboard.

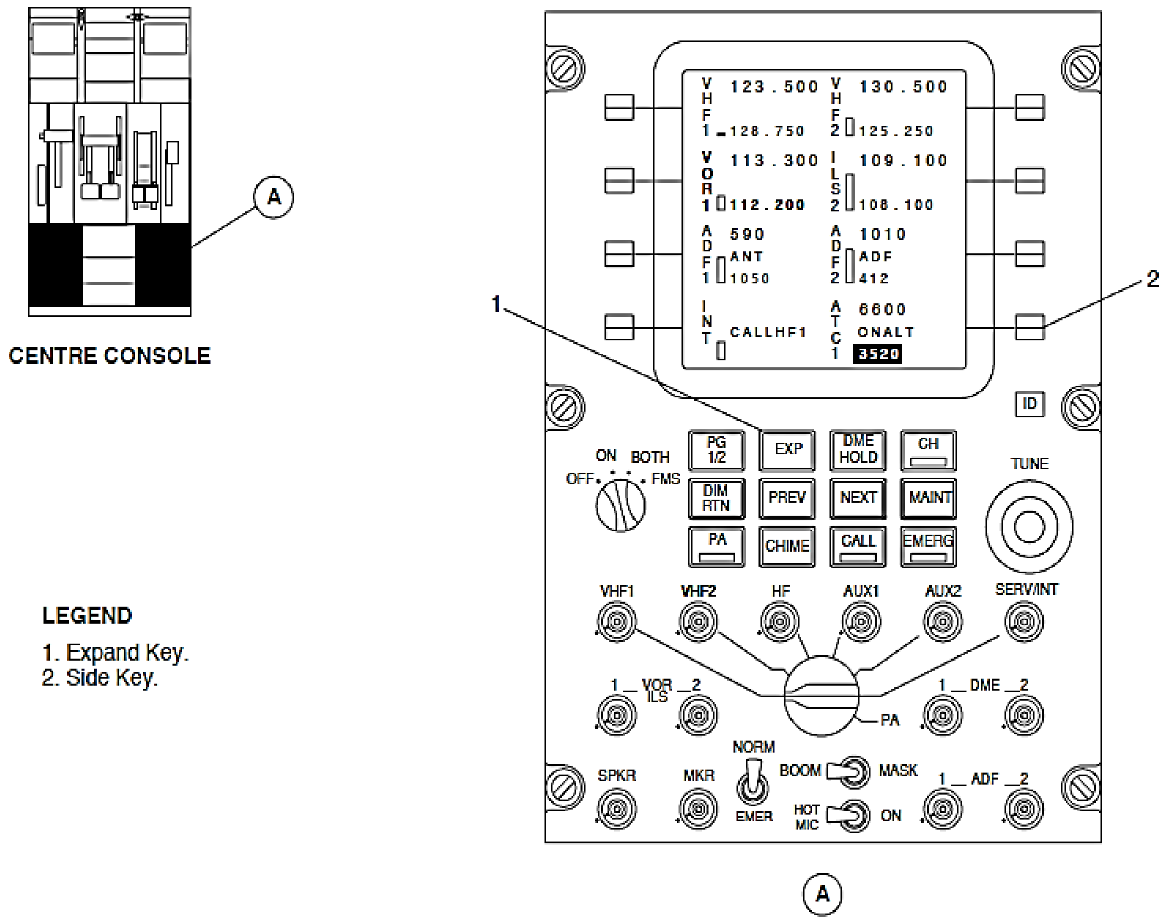
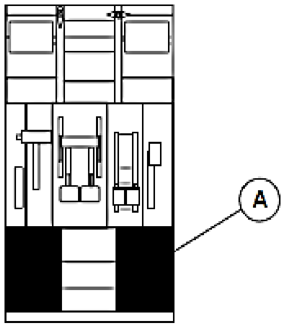


Figure - ARCDU ATC PARTICULAR PAGE ACCESS



CENTRE CONSOLE

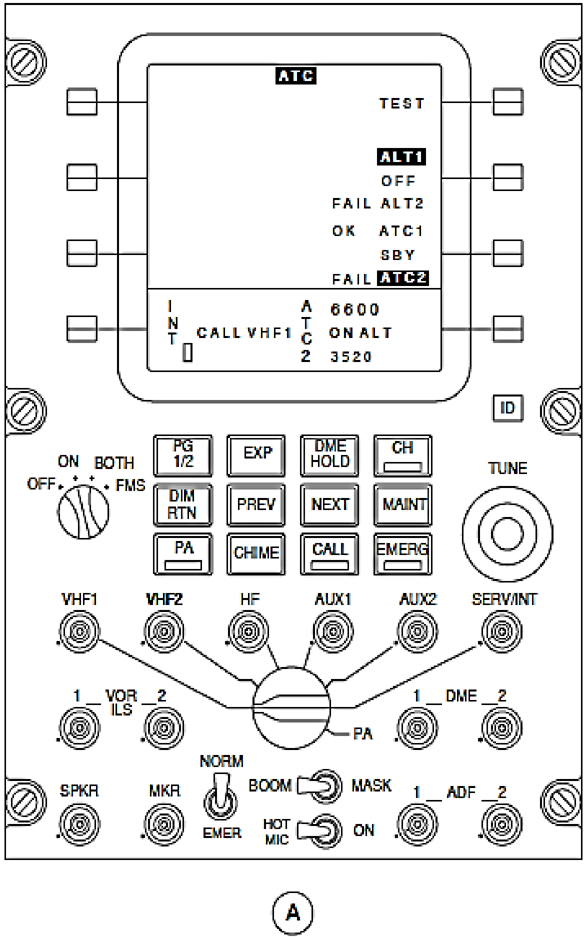


Figure - ARCDU ATC PARTICULAR PAGE

Refer Figure - ARCDU ATC PARTICULAR PAGE

An ATC label is shown in black letters on a white background in the center of the first line. The ATC particular page is used to change the following Mode S transponder (ATC 1, ATC 2) modes and functions:

- Standby, Active
- Test
- Altitude source selection.

Standby and Active Modes: The side key adjacent to the ATC1, SBY, and ATC2 display area is pushed to change the highlighted selection. It moves in a wrap round manner from ATC1, SBY, to ATC2. The selection changes to black letters on a green background from the usually white on black background. The first main page shows a SBY or ON legend in green letters in the ATC display area.

If ATC1 is selected the ATC display area will show ATC1 and ATC2 when ATC2 is selected.

Test Mode: The side key adjacent to the TEST legend is pushed to test the transponder that is operating in the standby mode. It is possible to test both transponders at the same time by setting both transponders to the standby operating mode. The TEST legend changes to black letters on a green background from the usually white on black background. The ATC1, SBY, ATC2 display area shows the test results adjacent to the ATC1 and ATC2 legends. The display area shows OK in green fonts for five seconds if the test result is successfully completed. It shows FAIL in red fonts when the test result shows a malfunction. The test feature is not available when the aircraft is in the air. An attempt to test the transponders while the aircraft is in the air will causes the ATC label to flash for five seconds at a one hertz rate.

The side key adjacent to the ALT1, OFF, ALT2 display area is pushed to select the encoding altitude source. The selection highlights and moves in a wrap round manner from ALT1, OFF, to ALT2. It changes to black letters on a green background from the usually white on black background.

The main pages show an ALT legend in the ATC display area. It does not come into view if both transponders are operating in the standby mode. If the selected ADU malfunctions, a FAIL label comes into view next the ALT1 or ALT2 legend.

The aircraft electrical system left main and right main buses supply 28 Vdc power through 5 ampere circuit breakers to the ATC Mode S transponders (ATC1, ATC2) and to the two coaxial cable switches. The circuit breakers are located in positions M3 and M4 on the 28 Vdc Avionics circuit breaker panel. Each Mode S transponder consumes less than 45 W and the coaxial cable switches use less than 7.5 W.

The onboard Distance Measuring Equipment (DME) and the TCAS operate in the same frequency range. Each sends a suppression signal to the other systems before transmitting. The transponder receives the suppression signal and stops transponder operations while the DME is transmitting. Prior to transmitting, the transponder sends a suppression signal to momentarily inhibit operation of the DME.

.

Receiver, Mode S Transponder

Refer Figure - MODE S TRANSPONDER RECEIVER
Refer Figure - MODE S TRANSPONDER RECEIVER LOCATOR

The Mode S transponder decodes the received phase shift modulated signal from the TCAS or ATC ground station transmitter. The last 24 bits of the signal represents the strapped address of the transmitting aircraft.

The transponder mounting trays are installed below the floor in the forward portion of the center fuselage.

The ATC 1 and ATC 2 mode S transponder mounting trays are located between station X+49.50 and X+71.20, between Y+0.0 and Y+9.25 and Z+89.6.

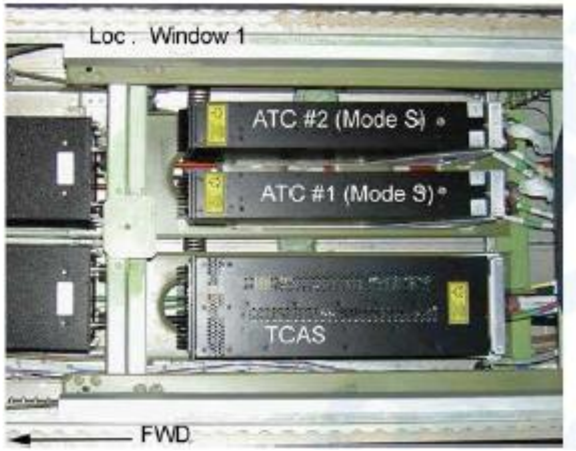


Figure - MODE S TRANSPONDER RECEIVER

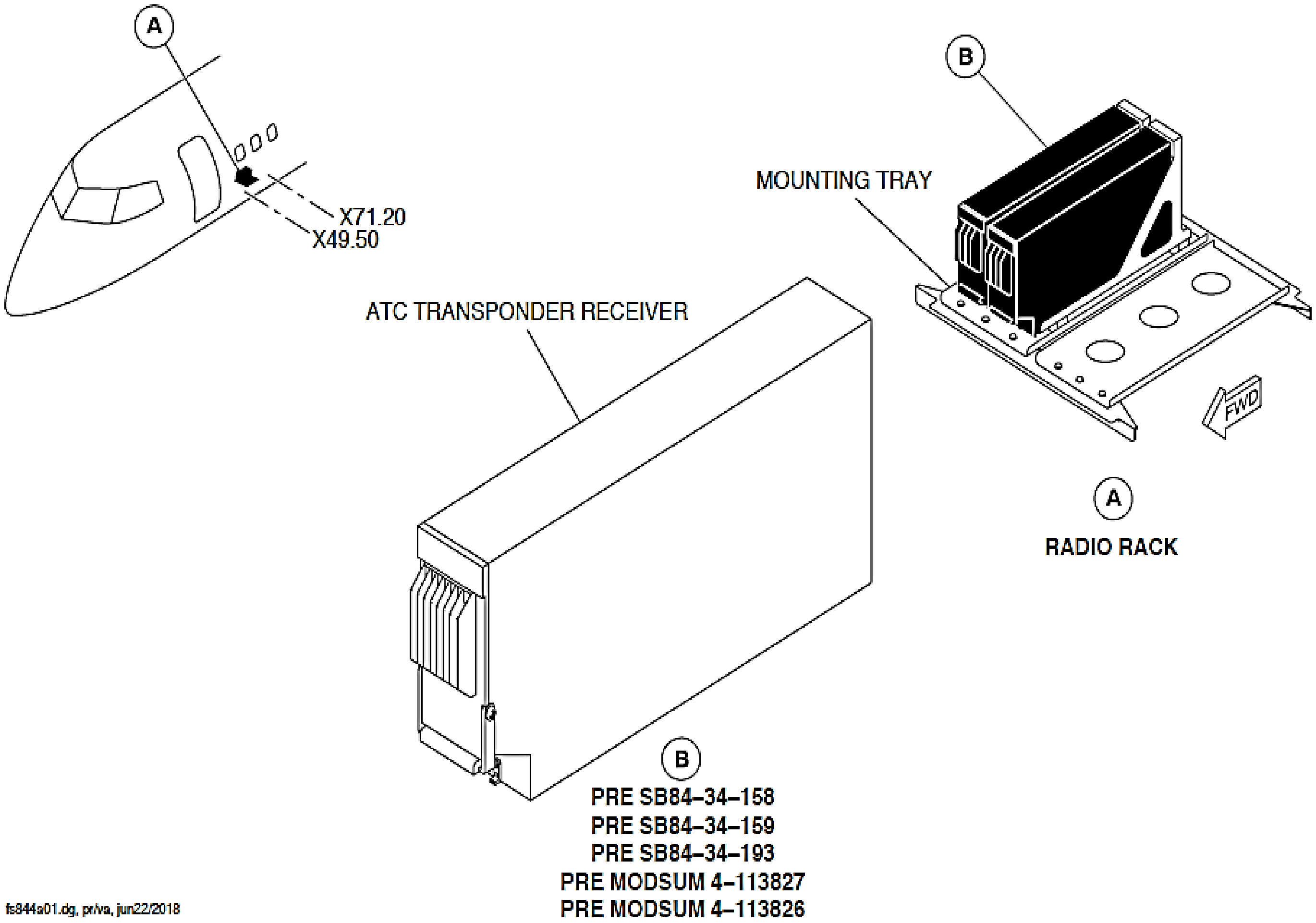


Figure - MODE S TRANSPONDER RECEIVER LOCATOR

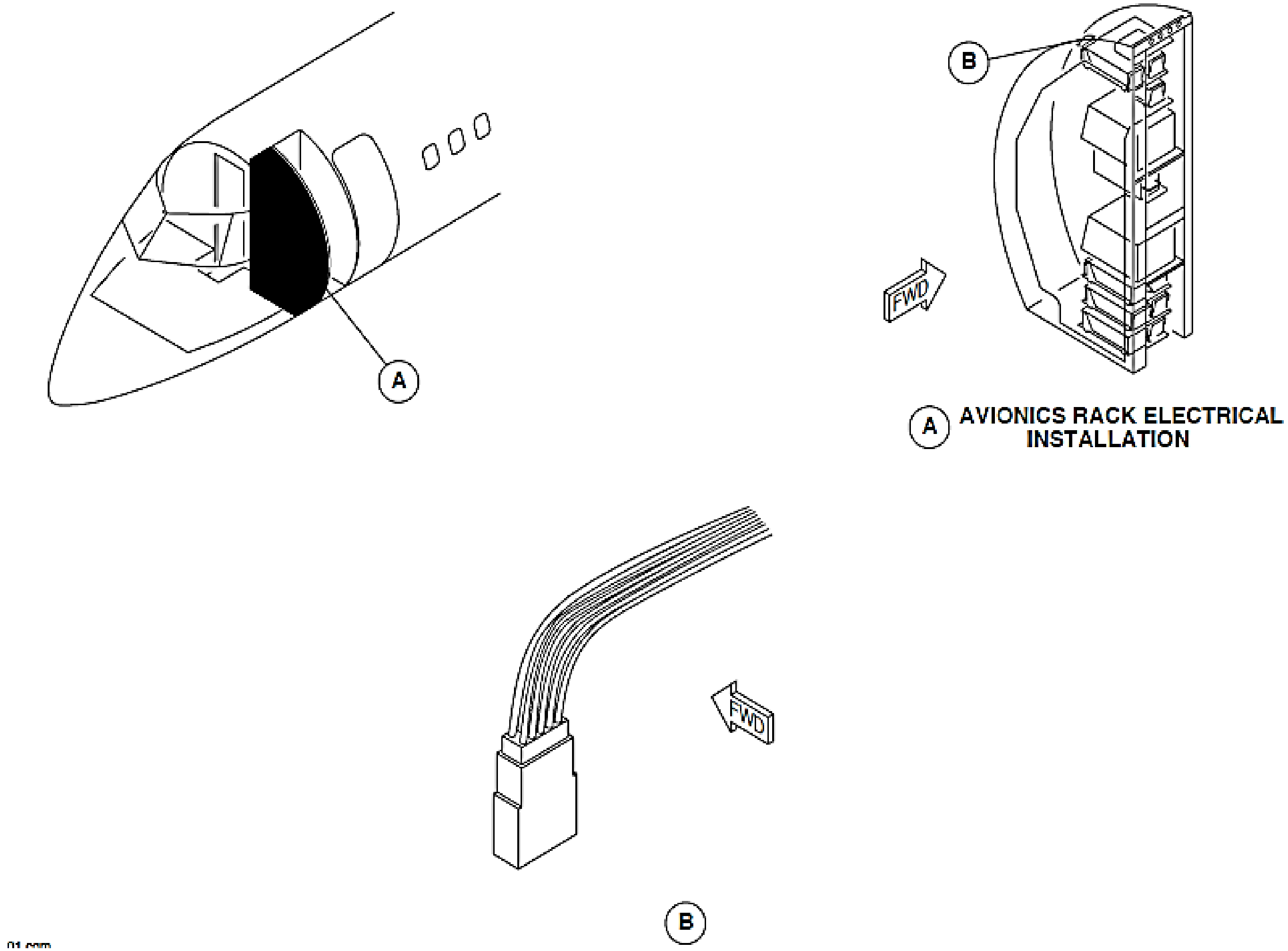


Figure - MODE S TRANSPONDER ADDRESS CONNECTOR LOCATOR

Refer Figure - MODE S TRANSPONDER ADDRESS CONNECTOR LOCATOR

The unique Mode S strapping is part of the data in the Mode S squitter and reply messages. Specific rear mating pin connections to ground potential determines the system's wire strapping. The transponder uses software to read and identify the configuration of the wire strap. Unique Mode S strapping determines the following aircraft parameters:

- Mode S address
- Maximum airspeed
- Antenna configuration
- Altitude source type.

The Mode S strap gives the aircraft a 24 bit unique Mode S address. Some Mode S replies from the transponder contain information related to the designated maximum cruising speed of the aircraft. The airspeed program strap sets the maximum airspeed between 150 and 300 knots (278 and 556 km/h).

The ATC transponder system provides a weight−on−wheels signal input (ground to open) to the transponder on connection J1A, pin 79 described as the Air/Ground ATCRBS Inhibit discrete input (not). The discrete is in the active state when a logic 0 (aircraft electrical ground) is sensed. Ground is applied when the aircraft is on ground. As per Mode S requirements, the transponder will only transmit on the ground when interrogated by its unique 24 bit address and if the airspeed is less than 100 kts (185.2 km/h).

For the minimum operation of the ATCRBS/Mode S system, the airspeed of 100 kts (185.2 km/h) will allow the Mode S transponder to report an airborne state and takes precedence over pin 79 status (the Air/Ground Inhibit discrete input status of the transponder). If the discrete input at pin 79 remains stuck in the on−ground condition and airspeed is more than 100 kts (185.2 km/h), this will cause the transponder to transmit the airborne status.

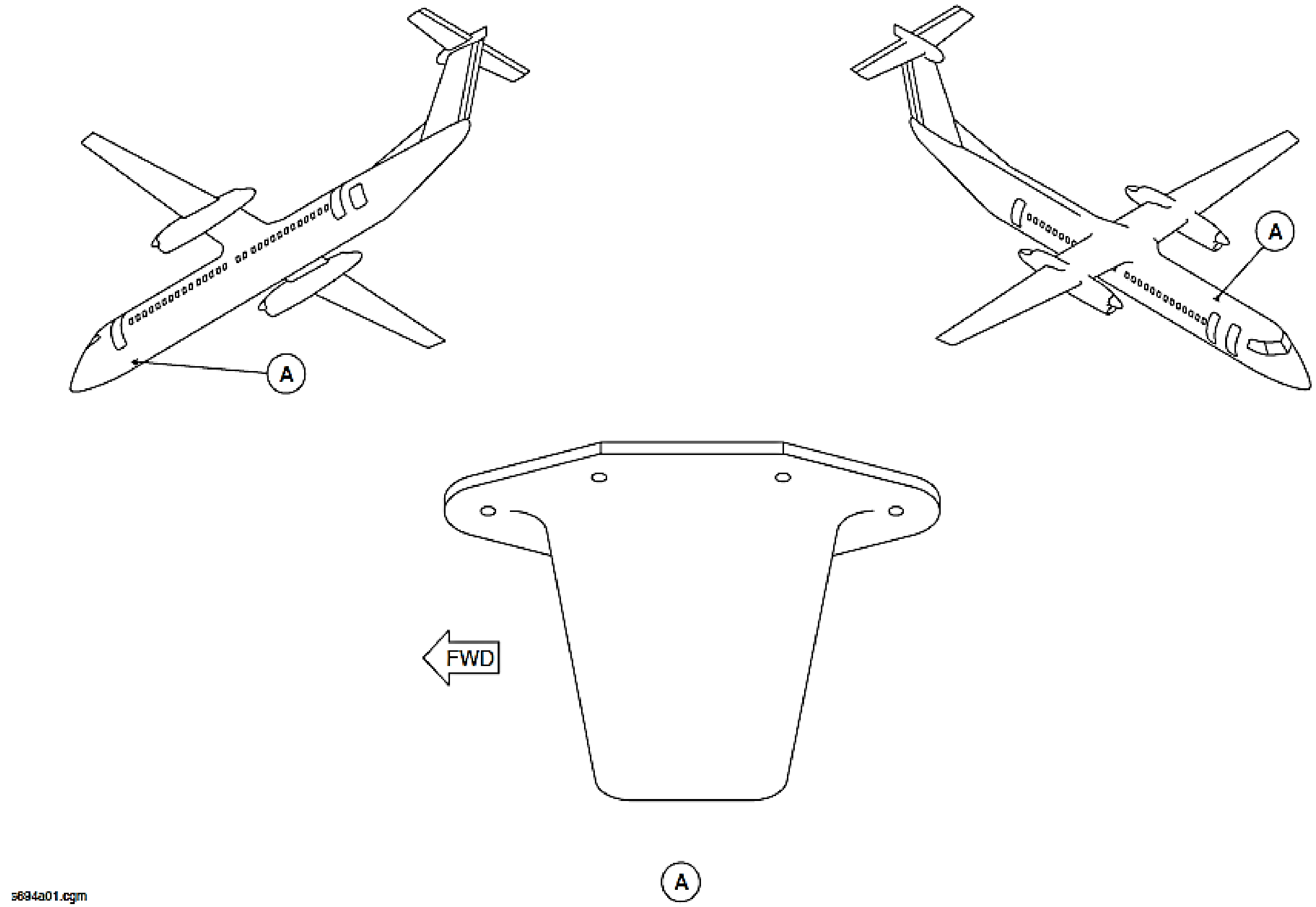


Figure - TRANSPONDER L−BAND ANTENNA LOCATOR

L–Band Antenna, Transponder

Refer Figure - TRANSPONDER L–BAND ANTENNA LOCATOR

The L–band antenna transmits and receives radio frequencies in the 950 to 1250 MHz frequency bands. The L–band antenna has the features that follow:

- Omni–directional
- Quarter wavelength
- Vertically polarized
- Monopole
- End fed.

The L–band blade antennae are compatible with the Mode S transponder.

The L–band antenna encases a preformed radiating element in thermoplastic within a single assembly. A Radio Frequency (RF) connector is attached to an aluminum base.

An aluminum foil gasket with an elastomeric sealant makes sure consistent distribution of conducting contacts of the gasket to the airframe. The pressure applied during installation gives a contouring feature. It lets the gasket conform to the two mating surfaces.

The Mode S transponder L–band antenna weighs less than 0.25 pounds (0.11 kilograms). The upper antenna is attached to the fuselage at station X+146, Y−2.0, and Z+182.7. The lower antenna is attached to the fuselage at X+11.9, Y+2.6, and Z+80.3. Each antenna is installed to the aircraft with five mounting screws.

Coaxial Cable Switch, Antenna

Refer Figure - TRANSPONDER COAXIAL CABLE ANTENNA SWITCH LOCATOR
Refer Figure - COAXIAL CABLE ANTENNA SWITCH

Two antenna coaxial switches connect the shared upper and lower L–band antennae to the active Mode S transponder. The antenna switch defaults to connect upper and lower L–band antennas to the ATC1 transponder.

The antenna coaxial switches are attached to the fuselage at station X+71.20, Y+9.35. Each antenna coaxial switch is installed to the aircraft using mounting screws. It electrically bonds to the aircraft structure.

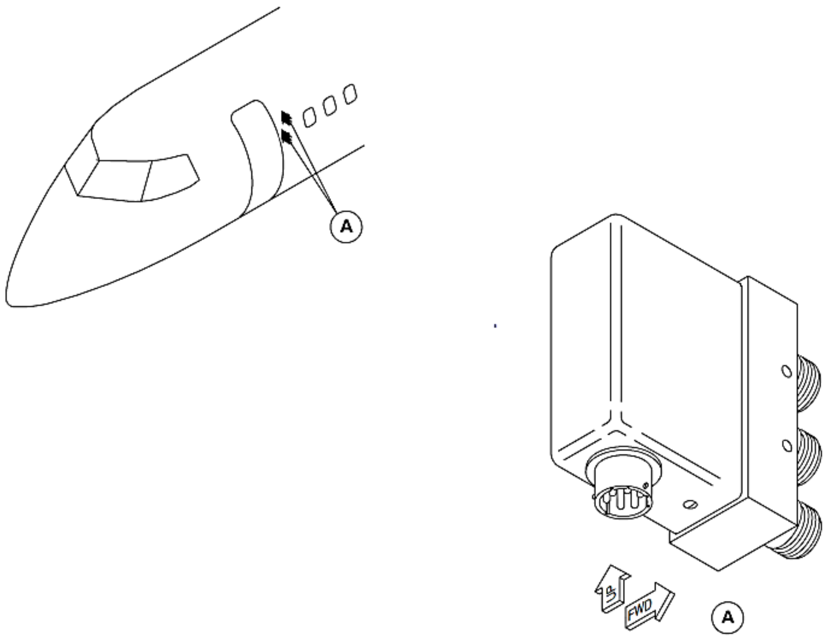


Figure - TRANSPONDER COAXIAL CABLE ANTENNA SWITCH LOCATOR

Figure - COAXIAL CABLE ANTENNA SWITCH

Switch, ATC ID

Refer Figure - ATC ID SWITCH

Two ATC identifier switches gives a remote selection of the SPI. The switch selection supplies a discrete input to its related ARCDU.

The ATC identifier switch is a device that opens or closes the electrical circuit. It is a usually open and pushed momentarily for close. Electrical wires are soldered to the switches.

The ATC identifier switches attach to the housing covers. In turn, the housing covers are attached to the pilot and copilot hand−wheel assemblies with set screws.

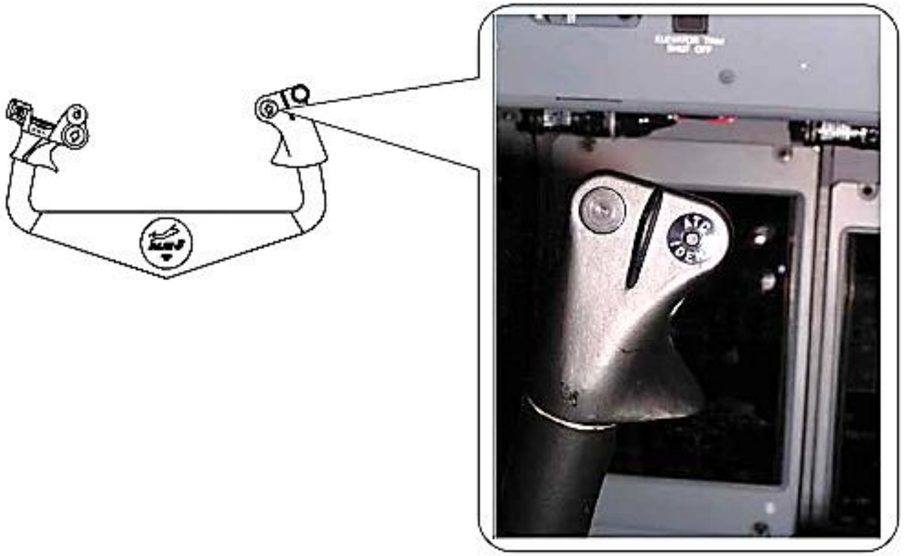


Figure - ATC ID SWITCH

Training Information Points

Do not operate the Mode S transponder when the ARCDUs show the codes that follow:

- 0000, reserved
- 7500, hijack
- 7600, radio failure
- 7700, emergency.

FLIGHT MANAGEMENT SYSTEM – UNS–1C/1E/1EW

INTRODUCTION

The Flight Management Systems (FMS) are integrated navigation management systems providing the flight crew with centralized control for:

- Navigation
- Flight planning
- Fuel management data
- Communication and navigation radio management.

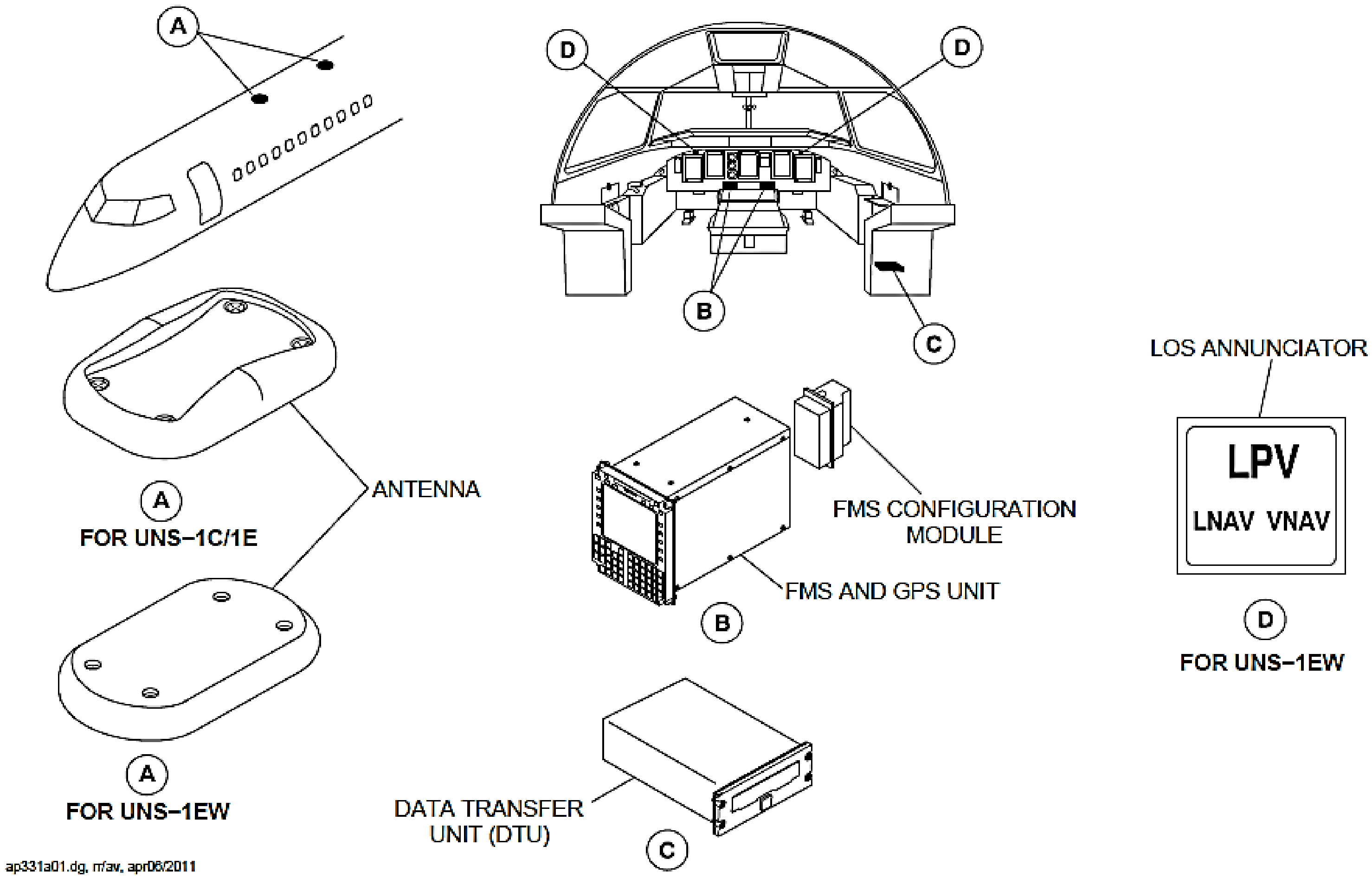


Figure - FMS SYSTEM − COMPONENT LOCATIONS

COMPONENT LOCATION

Refer Figure - FMS SYSTEM – COMPONENT LOCATIONS

The flight management system consists of a Flight Management Computer (FMC) located in the center console and an FMS configuration module that is attached to the back of the FMC. The Data Transfer Unit (DTU) or Solid State Data Transfer Unit (SSDTU) is installed under the lower shelf in the copilot's side panel. The system antennas are mounted on top of the fuselage.

GENERAL DESCRIPTION

Refer Figure - UNS–1C/1E/1EW FMS

The UNS FMS is a cockpit–mounted single Line Replaceable Unit (LRU) consisting of:

- Multi–functional Control Display Unit (MCDU)
- Flight Management Computer (FMC)
- Global Positioning System (GPS).

Position information is derived from VOR#1, DME#1 and GPS#1 for the pilot's side FMS, and VOR#2, DME#2 and GPS#2 for the optional co–pilot's side FMS. True airspeed, altitude, static air temperature data, fuel flow data, etc., are routed to each FMS from the Flight Data Processing System (FDPS).

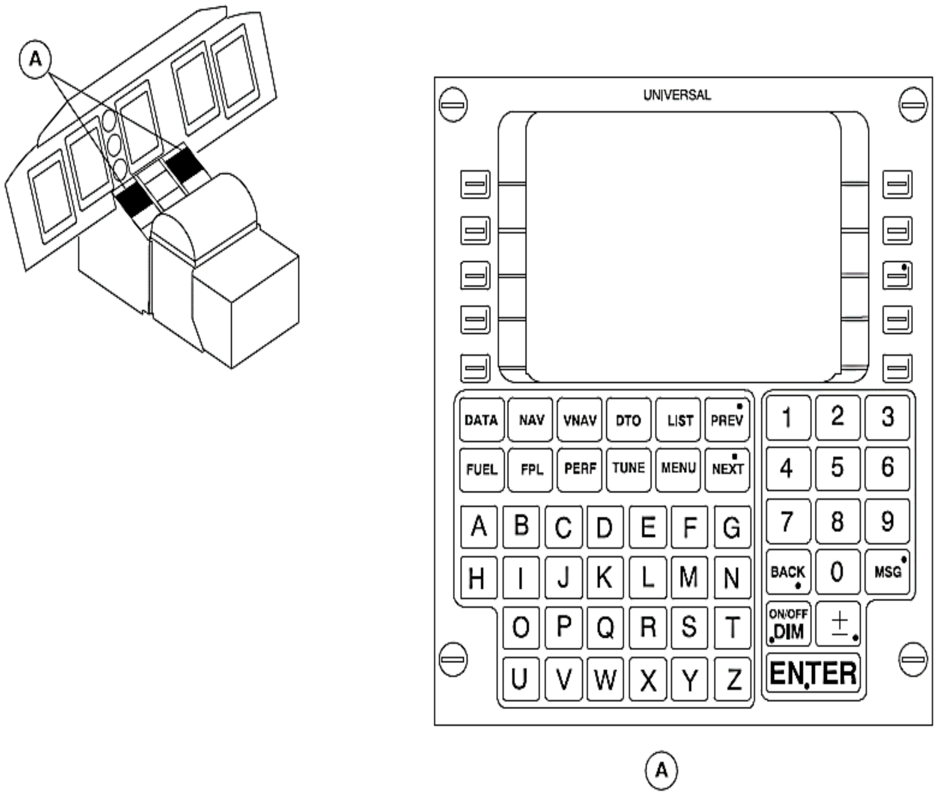


Figure - UNS–1C/1E/1EW FMS

The Q400 Series UNS−1E FMS configuration requires the following minimum equipment to be functional and operating:

Description	Part Number	Model Number	FMS #1 Quantity	FMS #2 Quantity
FMS Hardware P/N	2017−41−221	UNS−1E	1	1
FMS Software P/N	SCN 801.5 or SCN 802 or SCN 802.2 or SCN 802.8 or SCN 803.0 SCN 802.8 or SCN 801.5 with MOD 23 World Magnetic Model (WMM) (SB84−34−183 or SB84−34−184)	−	1	1
Antenna	10705	Universal Avionics	1	1
Data Transfer Unit	1406−01−1 or 1408−00−1	Universal Avionics	1	(part of FMS #1)

System capabilities for UNS−1E include:

- Full EFIS interface including GAMA 429 Map display
- Full compatibility with all ARINC 424 leg types
- Remote navigation tuning of VOR/DME for approach navigation as well as third DME channel
- GPS navigation
- Full IFR/IMC, supplementary means navigation capability
- Enroute, Terminal and Approach LNAV operations with AFCS coupling
- Enroute, Terminal and Approach VNAV operations with AFCS coupling
- TSO C129A Class A1/B1/C1 GPS navigation capability.

The FMS provides Lateral Navigation (LNAV) guidance and Vertical Navigation (VNAV) (approved for UNS−1E and UNS−1Ew on Q400) guidance based on data from the navigation receivers and air data sensors. An FMS approach mode allows the operator to fly flight director or autopilot non−precision approaches. The FMS provides the information for:

- Course
- Course Deviation
- Bearing−to−waypoint
- Distance−to−waypoint.

This information is displayed on the pilot's and co−pilot's PFD and MFD and is coupled to the AFCS for flight director and autopilot functions. During flight the FMS automatically updates the fuel on board and gross weight as well as predicted fuel requirements and estimated time of arrival based on:

- Fuel flow
- Ground speed
- Enroute flight plan winds (if entered).

A self−contained database stored in non−volatile memory provides the FMS with information on over 100,000 waypoints, navaids and airports. Data base updates are provided by Jeppesen.

DETAILED DESCRIPTION

SENSORS

The FMS accepts primary position information from short and long range navigation sensor(s) which include:

- VOR
- DME
- GPS.

The FMS also receives true airspeed and altitude information from the air data computer and fuel flow data from the aircraft's fuel flow sensors.

BEST COMPUTED POSITION

Primary position data received from each sensor is combined in a Kalman Filter contained within the FMS to derive a Best Computed Position (BCP). Utilizing this BCP, the FMS navigates the aircraft along the pilot−programmed flight path

The FMS best computed position is determined by using position inputs from all available navigation sensors and DME distance information from the scanning DME.

This DME derived position is then integrated with position information from the GPS, radial from the VOR, TAS from the air data computer and heading information to derive the best computed position as a weighted average of the various sensor inputs. Once the best computed position is obtained, secondary navigational functions are computed for display and include:

- Waypoint
- ETA
- Distance to waypoint
- Wind
- Ground speed.

Estimated winds may be defined at various waypoints in the flight plan to give a more accurate ETA.

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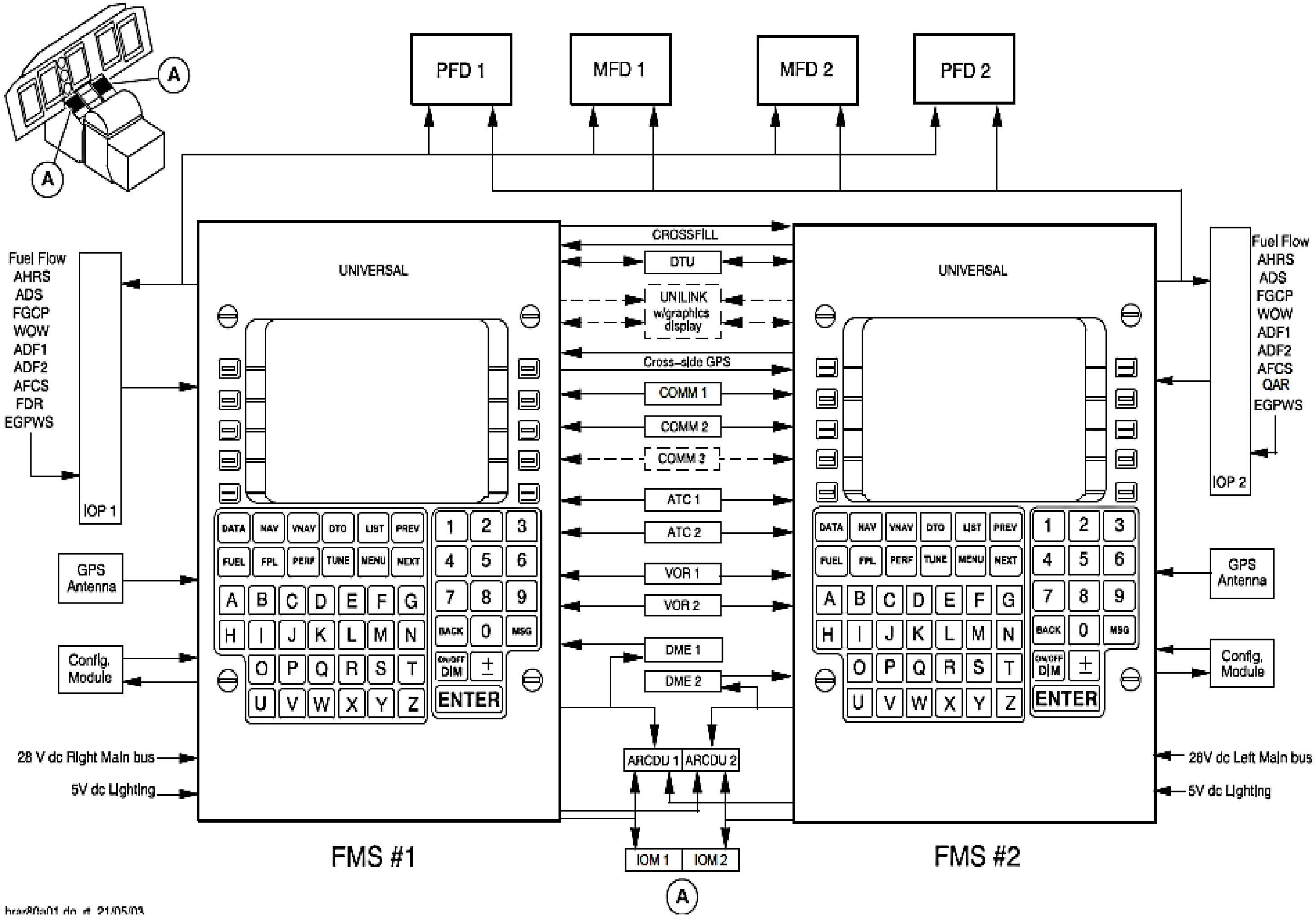


Figure - DUAL UNS−1C/1E FMS CONFIGURATION

MULTIPLE FMS'S

Refer Figure - DUAL UNS–1C/1E FMS CONFIGURATION

In dual installations, each FMS provides system initialization and flight plan information through an interconnecting bus to other systems. While all long range navigation sensors are interfaced with both units, each FMS acts as the "master" to the associated sensor. Each FMS computes an independent position using all available navigation sensors and provides selected information to the associated flight instruments. Each FMS also shares its best computed position with the other system and continuously monitors the other system's position output.

RADIO TUNING

Incorporated into the FMS, an optional TUNE mode provides the operator with a centralized means to tune the aircraft's radios, select and store preset frequencies for each radio and view the selected (preset or active) frequencies for each radio.

ANNUNCIATIONS

MESSAGES

The FMS contains an array of messages that alert the operator to system status and flight plan sequencing. When a message is active, "MSG" will appear on the display and the message can be accessed by pressing the MSG key.

The Message function can also be utilized for AFIS and UniLink communications in applicable installations.

REMOTE ANNUNCIATORS

Outputs are provided to cause the six external annunciators to alert the pilot of system status or flight plan sequencing. These annunciations may be incorporated into the EFIS display:

MSG	A new message has been generated.
WPT	(Steady) Lateral waypoint alert. (Flashing) Vertical waypoint Flight Path Angle alert.
SXTK	FMS is in selected crosstrack mode.
FMS HDG	FMS is in Heading Mode.
FMS APPR	FMS is in Approach Mode.
GPS INTEG	RAIM is not available or a fault is detected. DME may be still in use.

SYSTEM COMPONENTS

The basic FMS installation consists of the FMS (containing the display, keyboard, navigation computer and a GPS sensor), a configuration module, a DTU or a SSDTU, a GPS antenna/WAAS antenna and a Radio Tuning Unit.

FMS DISPLAY

The FMS utilizes an active matrix flat panel color liquid crystal display (LCD). The high resolution display provides alphanumeric characters, icons, and graphics capabilities to facilitate on screen data recognition. The display allows for data display on eleven lines with 24 characters each in two different character sizes.

FMS KEY BOARD

The full alphanumeric keyboard contains dedicated functions keys that when used in conjunction with the ten line elect keys, provides the operator with all the controls necessary to communicate with the internal navigation computer and all associated sensors. The keyboard allows manual data entry, system mode selection and control (selection and de−selection) of navigation sensors.

FMS NAVIGATION COMPUTER

The FMS has a powerful central processing unit (navigation computer), navigation sensor interface circuits, flight guidance system interface and the Navigation Data Base.

The unit processes data from DME, VOR, Air Data, Fuel Flow and up to five long range navigation sensors. A radar joystick input can also be accommodated. At installation, the system is configured with the number and type of long range navigation sensors, which will optimize the FMS system to the aircraft's specific requirements. The long range navigation sensors act in addition to the multi−channel scanning DME and VOR sensing capability integrated into the basic circuit.

SHORT RANGE SENSORS

DME

The FMS unique auto−scanning DME−DME−DME position function uses the internal Navigation Data Base to continually map the stations surrounding the aircraft (VORTACs, DMEs, ILS−DMSs, LOC−DMEs, TACANs). The FMS then automatically tunes and reads these, one every four seconds, utilizing the blind channel of an external digital DME or Universal's Radio Reference Sensor (VOR, DME, TACAN). Range information from all responding DMEs within approximately 300 nautical miles of the FMS position are corrected for slant range error then compared with those computed from the Navigation Data Base to verify reasonableness and to ensure DME position integrity is maintained.

Flights that never leave areas of multiple DME coverage can expect exceptional position accuracy. Over−water flights will retain DME accuracy long into the flight, reestablishing this same accuracy well before landfall as DMEs are again received.

VOR

The aircraft's digital VOR receiver or Universal's RRS provides the FMS with azimuth information from the referenced VOR navaid. This azimuth data combined with the scanning DME distances provides accurate navigation information for the FMS system throughout the enroute, terminal and approach modes of operation in accordance with the criteria set forth in AC 20−130A when within range of the required ground based navaid.

RADIO REFERENCE SENSOR (RRS)

Universal's Radio Reference Sensor (RRS) functions as a multi−channel DME−TACAN receiver/transmitter and VOR receiver, able to provide DME, VOR and TACAN data simultaneously to two FMS systems. The RRS provides the required navigation information via an ARINC 429 data bus without interfering with the aircraft's existing radios. RRS operation is fully automatic required no pilot intervention.

LONG RANGE SENSORS

GPS SENSOR

The FMS CDU presents the following GPS information:

- Latitude and longitude position
- A radial difference in nautical miles between the FMS and the GPS position
- GPS horizontal position error
- GPS position uncertainty
- GPS time in Universal Time Coordinated (UTC)
- Using altitude annunciation
- GPS operating in Dead Reckoning (DR) mode annunciation
- Receiver Autonomous Integrity Monitoring (RAIM) availability at the current position
- FDE availability at the current position
- When RAIM will not be available at the Active Flight Plan Destination annunciation
- GPS integrity mode
- GPS accuracy or integrity is not adequate annunciation
- Satellite status for up to eight satellites.

For the UNS−1C and UNS−1E, the GPS sensor is the source of three−dimensional position and velocity, time and status information. The unit comprises an embedded 12−channel GPS receiver which receives signals transmitted by the U.S. GPS satellite network to process measurements from all satellites in view simultaneously.

For the UNS−1C and UNS−1E, the associated GPS active antenna is powered by 5 VDC from the FMS and it is installed on the top of the aircraft forward fuselage.

The GPS sensor uses data in satellite navigation messages to calculate:

- Latitude
- Longitude
- Altitude
- Horizontal velocity
- Vertical velocity
- Current Universal Time Coordinated (UTC).

The Primary Flight Display (PFD) presents a yellow GPS INTEG message to the pilots if:

- The satellite geometry is not adequate to support at least RAIM error detection
- An uncorrectable satellite error occurs.

CONFIGURATION MODULE

A configuration module is installed on the FMS rear connector. It is programmed at the time of installation, and defines all the aircraft interfaces unique to the particular installation.

AIR DATA COMPUTER (ADC)

The FMS accepts digital or analog air data from a remote air data computer. True Airspeed, barometrically corrected or non−corrected altitude, and Static Air Temperature are utilized. Baro−corrected altitude is required for vertical guidance display during approaches and for VNAV (Approved for UNS−1E and UNS−1Ew on Q400) below 18,000 feet.

DATA TRANSFER UNIT (DTU)

The Data Transfer Unit (DTU) is designed to be mounted in the aircraft and used to update the FMS navigation data base within minutes. A portable unit is also available for multi−aircraft operations.

The DTU is connected directly to the FMS such that the data is transmitted directly to the FMS during the update process. The update data will refresh the entire navigation data base with each update.

A subscription service is available through Universal, which provides navigation data base updates on a 28 day cycle.

The DTU is also used to load aircraft specific performance data into the FMS for computation and display of aircraft performance data. The DTU may also load pilot−defined data created with off−line flight planning software loaded into an IBM compatible computer. FMS in−flight data parameters can be written to a disk utilizing the DTU. Maintenance log diagnostic history is also downloaded to a disk using the DTU. FMS software updates can also be loaded via the DTU.

Solid State Data Transfer Unit (SSDTU)

The fixed Solid State Data Transfer Unit (SSDTU) or portable SSDTU replaces the ZIP disk, floppy disk and the CD/DVD−ROM technology with the flash memory technology. The SSDTU is capable of supporting all the products previously supported by the DTU−100 (e.g. UNS−1C or UNS−1C+ or UNS−1E or UNS−1Ew).

Two data storage device ports are available on the faceplate of SSDTU. These ports support the Universal Serial Bus (USB), Secure Digital (SD) data storage devices and USB CD/DVD ROM disk drives. The unit has a high speed USB and a SD media port on the faceplate. These media ports function as disk drives.

RADIO TUNING UNIT (RTU)

An optional Radio Tuning Unit (RTU) allows the operator to use the FMS control display unit to tune a selected radio and store up to four preset frequencies for later use by each selected radio. COMMs, NAVs, ATC, ADFs and TACAN are selectable through the RTU, depending upon the specific aircraft configuration. Both 25 kHz and 8.33 kHz spacing is supported.

CONFIGURATION

Single or Dual UNS FMS UNS−1C or UNS−1E or UNS−1Ew installation can be used on the Q400.

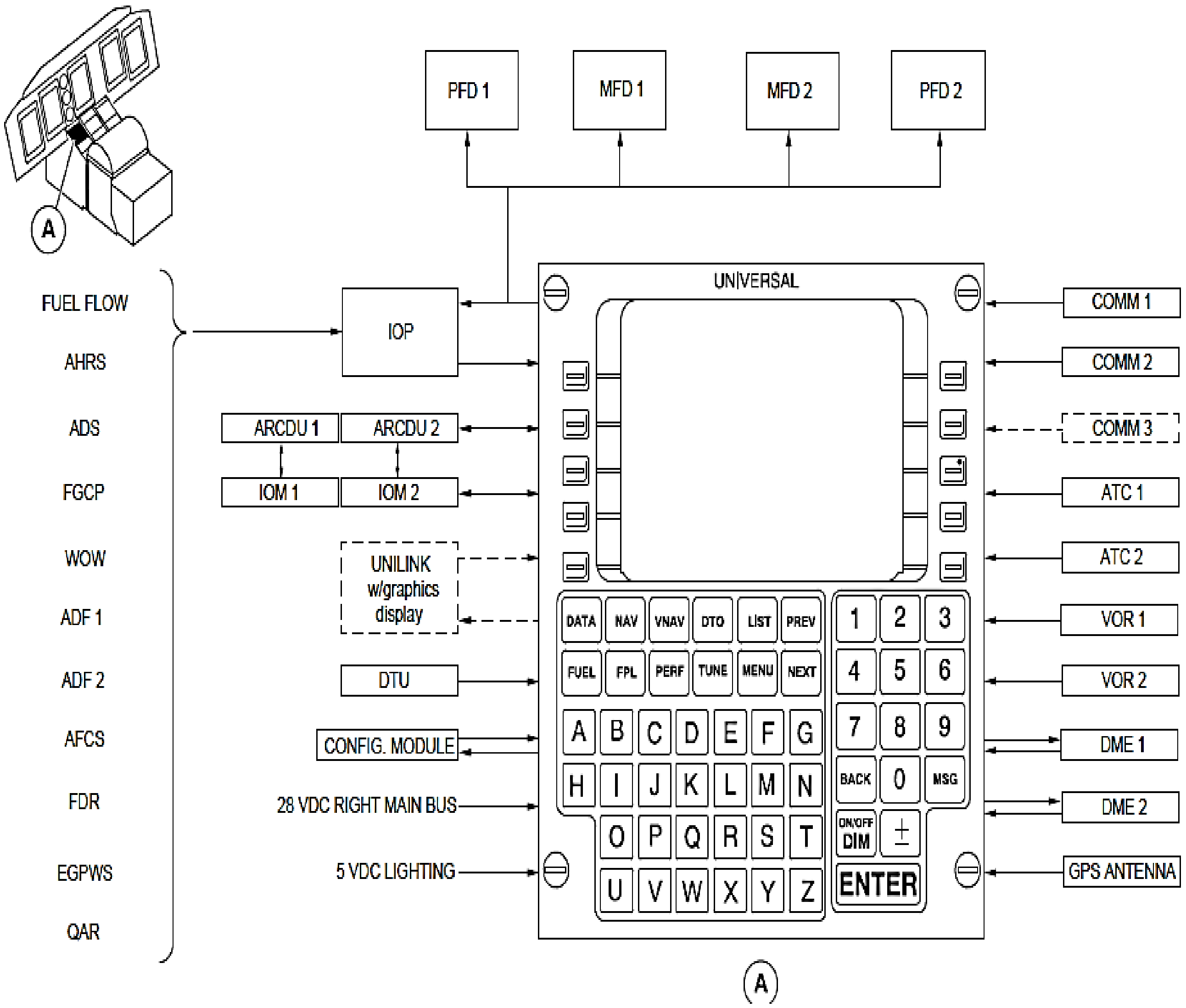


Figure - SINGLE UNS−1C/1E FMS CONFIGURATION

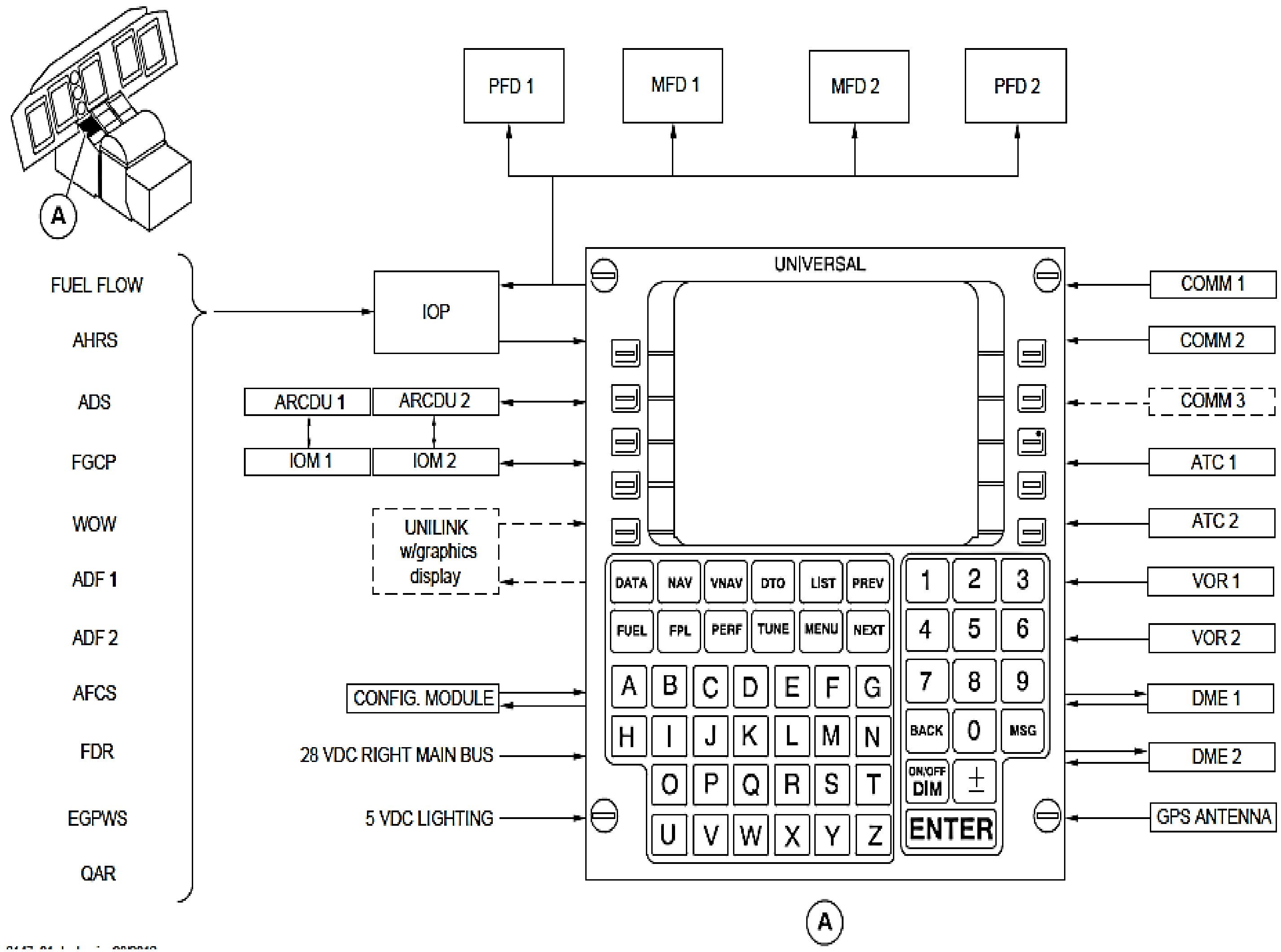


Figure - SINGLE UNS−1EW FMS CONFIGURATION

SINGLE SYSTEMS

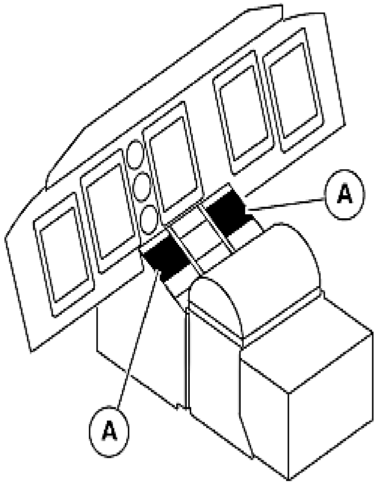
In a single FMS UNS−1C or UNS−1E or UNS−1Ew installation, the FMS simultaneously accepts inputs from either digital DME and VOR receivers or DME/VOR/TACAN from an RRS, multiple long−range navigational sensors, the Air data Computer and engine fuel flow sensors.

MULTIPLE SYSTEMS

Refer Figure - DUAL UNS−1C/1E FMS CONFIGURATION

In multiple system installations, communication between FMSs is accomplished through an interconnection bus. There is no “Master/Slave” type relationship between the FMS’s. Each system makes data available (Initialization, Flight Plan, Fuel) to other interfaced system(s). Crossfill of this data can be initiated by the sending or receiving system. Each system acts as “MASTER” of its associated sensors, even though that sensor may also be providing data to another system. Typically, the on−side compass and air data computer will supply heading and air data, while fuel flow data will be shared by the systems.

Each FMS computes its own independent position using all available navigation sensors and provides selected information to the associated flight instruments. Each FMS also shares its best computed position with the other system and continuously monitors the other system’s position output.



LEGEND

1. ON/OFF – DIM key.
2. ENTER key.
3. ± key.
4. BACK key.
5. MSG key (message).
6. NEXT key.
7. PREV key (previous).
8. LSK key (line select key).

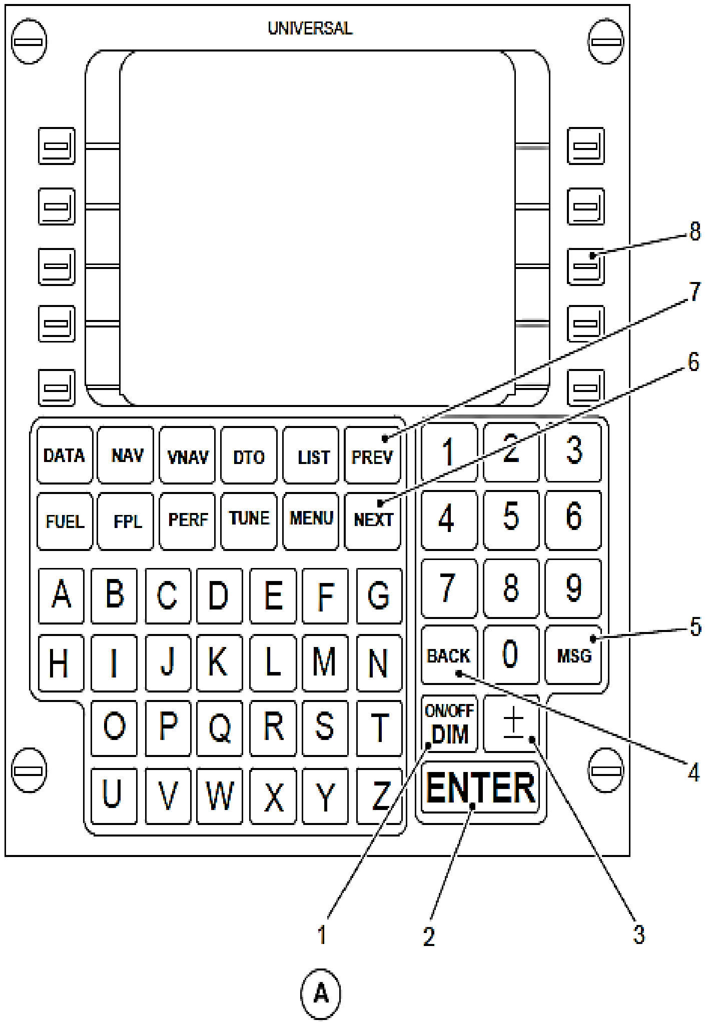


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Figure - UNS FMS MCDU – CONTROL KEYS

FMS MCDU CONTROL KEYS

Refer Figure - UNS FMS MCDU – CONTROL KEYS

- ON/OFF – DIM key (momentary action) PUSH – provides power, display dimming and unit shutdown functions.
- ENTER key (momentary action) PUSH – stores input data to memory. The cursor marks data that can be changed using the normal input processes (dark letters on a light background). When variable data is marked by the cursor, pressing the ENTER key stores data to memory.
- ± key (momentary action) PUSH – used in conjunction with the alphanumeric keys to enter data. It reverses the selected entry data field from + to −, N to S, and L to R. In the alpha field, it is used as a dash or period.
- BACK key (momentary action) PUSH – When the cursor is over a data entry field, the BACK key functions as a delete or backspace key.
- MSG key (momentary action) PUSH – displays an active message when the 'MSG' annunciation appears on the screen. The message alerts the operator to system status and flight plan sequencing. Selecting the UniLink option will access the UniLink menu page.
- NEXT key (momentary action) PUSH – used to cycle forward, one page at a time, through multiple pages of the same mode.
- PREV key (momentary action) PUSH – used to cycle backwards, one page at a time, through multiple pages of the same mode.
- LSK key (momentary action) PUSH – Line Select Keys (LSK's) are used to enter data utilizing the alphanumeric keys. Pressing the ENTER key completes the entry. 10 LSK's (5 down left side and five down right side of display) align with associated data lines.

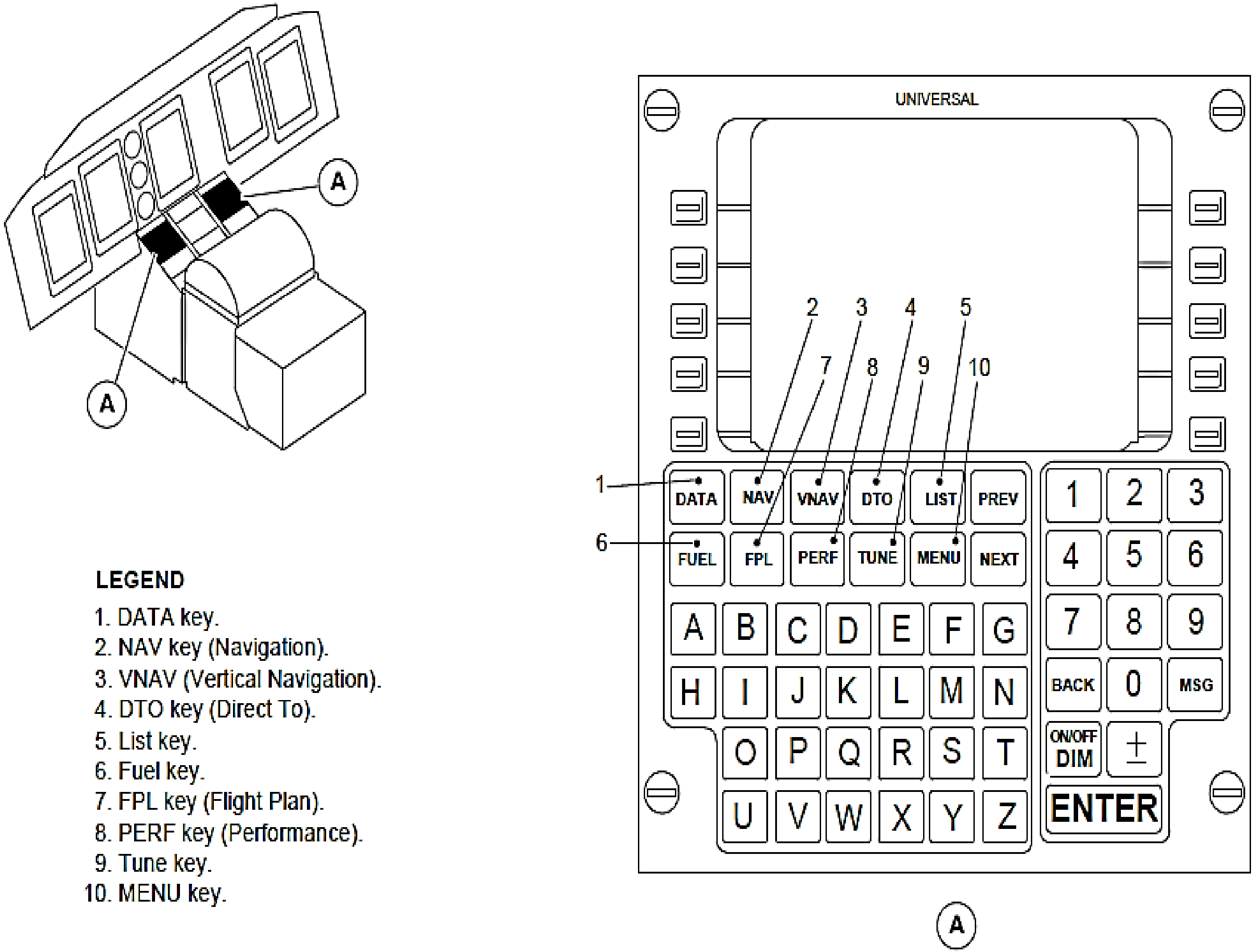

Figure - UNS FMS MCDU – FUNCTION KEYS

FMS MCDU FUNCTION KEYS

Refer Figure - UNS FMS MCDU – FUNCTION KEYS

- DATA key (momentary action) PUSH – to obtain information and status about the FMS, Navigation Data Base and sensors which operate with the FMS. The function is also used to select/deselect individual sensors and to add/delete or change pilot defined locations.

- NAV key (momentary action) PUSH – to display all navigation data normally required by the flight crew. There are normally two Navigation pages. Additional pages are added when APPROACH or HEADING is selected. Pressing the PREV or NEXT function keys cycles through the associated pages.

- VNAV key (momentary action) PUSH – to define a desired vertical flight profiles along the flight plan route. It computes the aircraft deviation from that profiles for display.

- DTO key (momentary action) PUSH – accesses the ’DIRECT To’ page and is specifically dedicated to changing the flight plan in response to the ’direct to’ clearances. If the ’direct to’ location is off the flight plan, provisions are made to link the location into the flight plan.

- LIST key (momentary action) PUSH – to display a list of options during data entry appropriate to the entry being made.

- FUEL key (momentary action) PUSH – to provide access to all fuel management functions. These include fuel and weight entry, range and endurance estimates, fuel requirement summary, projected landing weight, fuel flow and fuel consumption.

- FPL key (momentary action) PUSH – to access the Flight Plan pages or to access stored arrivals and routes. The FPL function is also used to create a new flight plan, alter current flight plan, and insert a SID, STAR and approach into the flight plan.

- PERF key (momentary action) PUSH – to view a synopsis of in–flight performance information.

- TUNE key (momentary action) PUSH – to tune, select and store pre–selected frequencies and view selected frequencies (active and preset) for each radio.

- MENU key (momentary action) PUSH – to display a list of alternate formats or options for the FUEL, FPL, NAV, VNAV or TUNE modes. The letter ’M’ will appear in a box on the title line of any page in which the MENU key is active.